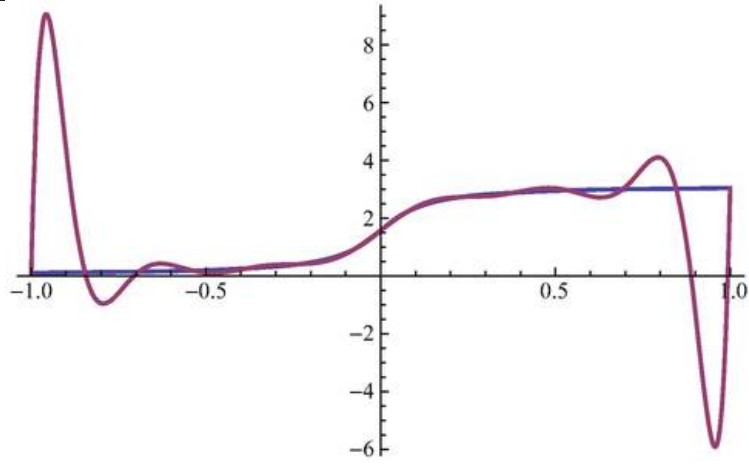
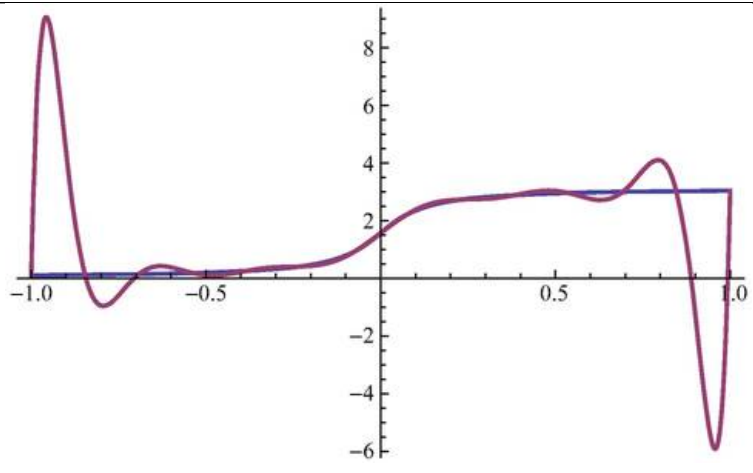
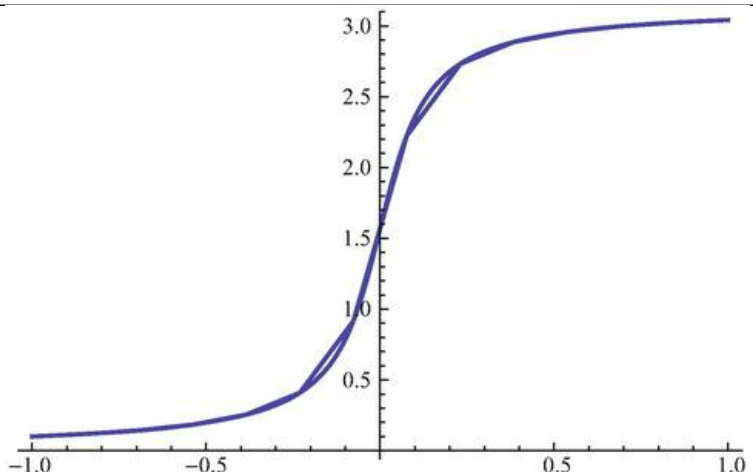


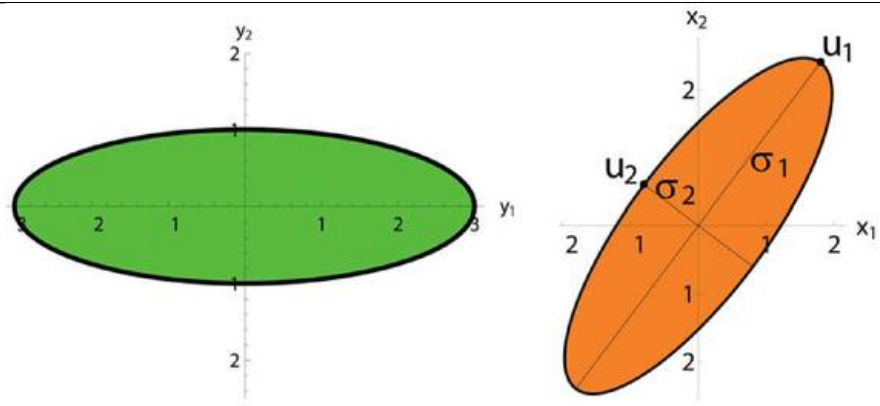
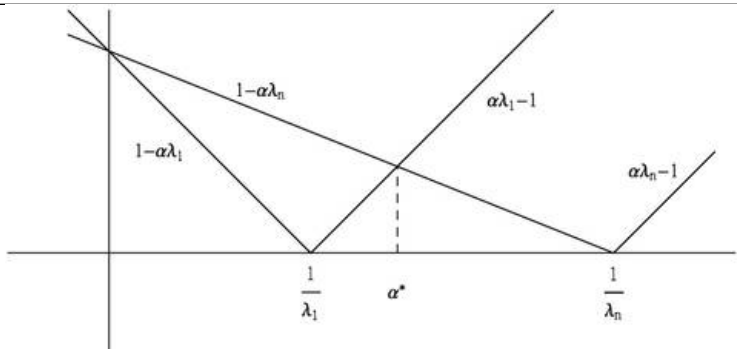
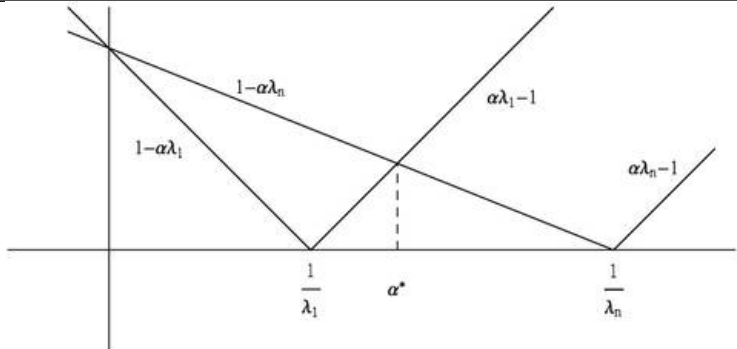
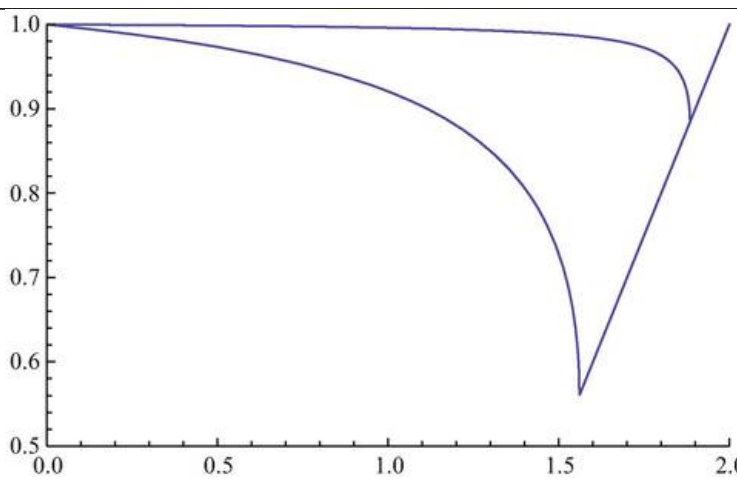
lyche-numerical-linear-algebra: formula evidence

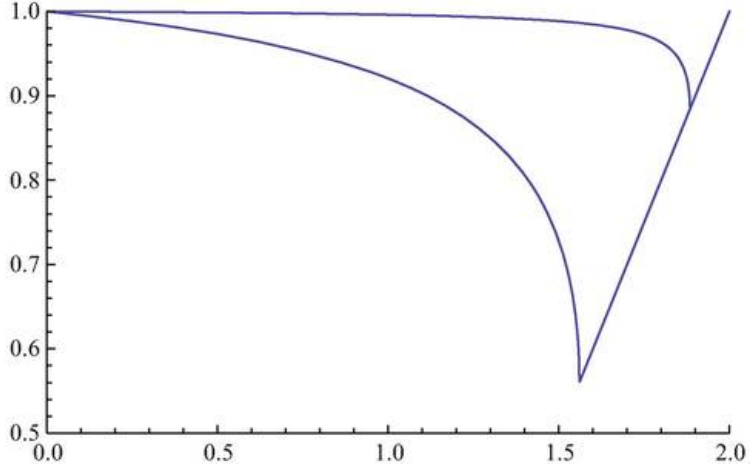
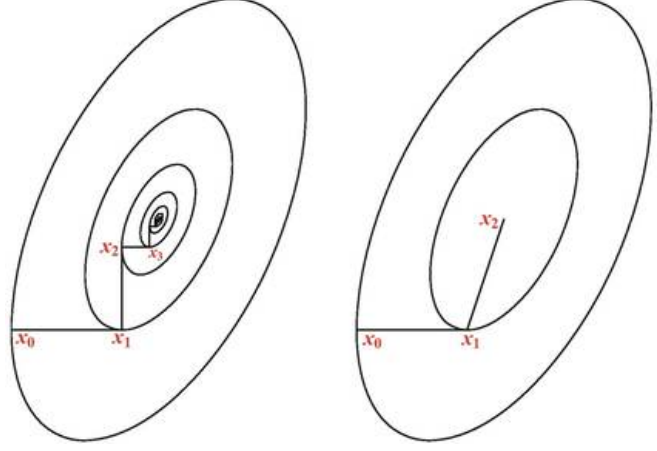
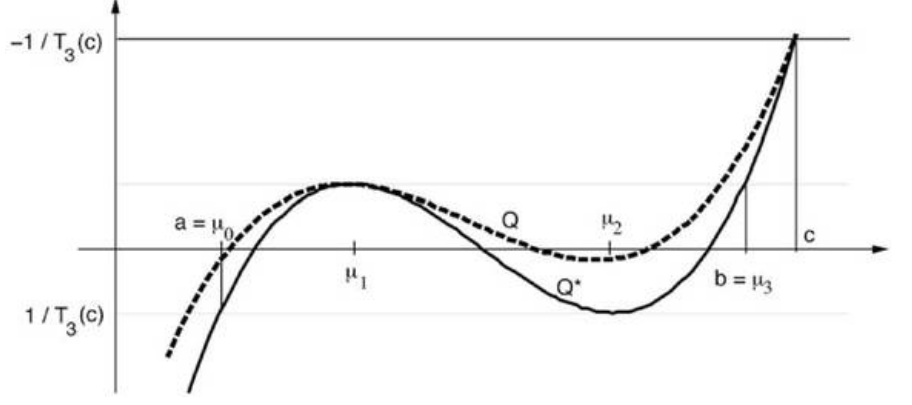
The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00001	7	■ 1.000	\boldsymbol{A}	$\boldsymbol{A}$	• <i>System of linear equations.</i> Given a (square) matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$. Find a
lyche-numerical-linear-algebra_F00002	7	■ 1.000	\boldsymbol{b}	$\boldsymbol{b}$	• <i>System of linear equations.</i> Given a (square) matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$. Find a
lyche-numerical-linear-algebra_F00003	7	■ 0.993	\boldsymbol{x}	$\boldsymbol{x}$	vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{b}$.
lyche-numerical-linear-algebra_F00004	7	■ 0.993	\boldsymbol{A} \ \boldsymbol{x}=\boldsymbol{b}	$\boldsymbol{Ax} = \boldsymbol{b}$	vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{b}$.
lyche-numerical-linear-algebra_F00005	7	■ 1.000	\boldsymbol{b}-\boldsymbol{A} \ \boldsymbol{x}	$\boldsymbol{b} - \boldsymbol{Ax}$	such that the sum of squares of the components of $\boldsymbol{b} - \boldsymbol{Ax}$ is as small as possible.
lyche-numerical-linear-algebra_F00006	7	■ 1.000	\lambda	λ	• <i>Eigenvalues and eigenvectors.</i> Given a (square) matrix $\boldsymbol{A}$. Find a number λ
lyche-numerical-linear-algebra_F00007	7	■ 0.997	\boldsymbol{A} \ \boldsymbol{x}=\lambda \ \boldsymbol{x}	$\boldsymbol{Ax} = \lambda \boldsymbol{x}$	and/or a nonzero vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \lambda \boldsymbol{x}$.
lyche-numerical-linear-algebra_F00008	21	■ 1.000	\mathbb{R}^n	$\mathbb{R}^n$	emphasis on $\mathbb{R}^n$ and $\mathbb{C}^n$. For a more elementary introduction, see for example the
lyche-numerical-linear-algebra_F00009	21	■ 1.000	\mathbb{C}^n	$\mathbb{C}^n$	emphasis on $\mathbb{R}^n$ and $\mathbb{C}^n$. For a more elementary introduction, see for example the
lyche-numerical-linear-algebra_F00010	179	■ 0.987	\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^* \boldsymbol{A}^* \boldsymbol{A}	$\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^*$	Exercise 7.6 (The Matrices $\boldsymbol{A}^* \boldsymbol{A}$, $\boldsymbol{A} \boldsymbol{A}^*$ and SVD) Show the following: If $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F00011	26	■ 1.000	p	p	where the coefficients $a_0, \dots, a_n$ are real or complex numbers. p is called the
lyche-numerical-linear-algebra_F00012	267	■ 1.000	\omega	ω	parameter ω in the range $1 \leq \omega < 2$ and then $\boldsymbol{x}_{k+1}$ is computed by linear
lyche-numerical-linear-algebra_F00013	335	■ 1.000	\lambda_m	λ_m	we consider a method to compute an approximation to the m th eigenvalue λ_m for
lyche-numerical-linear-algebra_F00014	43	■ 1.000	T	T	Fig. 1.1 The triangle T
lyche-numerical-linear-algebra_F00015	43	■ 1.000	P_{1}, P_{2}	P_1, P_2	P_1, P_2 and P_3
lyche-numerical-linear-algebra_F00016	43	■ 1.000	P_{3}	P_3	P_1, P_2 and P_3
lyche-numerical-linear-algebra_F00017	48	—	f(x)=\arctan (10 x)+\pi / 2	$f(x) = \arctan(10x) + \pi/2$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00018	48	—	$[-1,1]$	$[-1,1]$	
lyche-numerical-linear-algebra_F00019	48	—	$n=14$	$n = 14$	
lyche-numerical-linear-algebra_F00020	48	■ 1.000	$f(x)=x^4$	$f(x) = x^4$	function $f : [0, 2] \rightarrow \mathbb{R}$ given by $f(x) = x^4$. Thus we consider the D_2 -spline
lyche-numerical-linear-algebra_F00024	120	■ 1.000	$\boldsymbol{v}_1$	$\boldsymbol{v}_1$	$\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in Gram-Schmidt.
lyche-numerical-linear-algebra_F00025	120	■ 1.000	$\boldsymbol{v}_2$	$\boldsymbol{v}_2$	$\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in Gram-Schmidt.
lyche-numerical-linear-algebra_F00026	27	■ 1.000	c	c	not all the c 's are equal to zero. Pick a nonzero c , say c_{i_1} . Then $\boldsymbol{v}_{i_1}$ can be
lyche-numerical-linear-algebra_F00027	120	■ 0.990	$c:=\left\langle \boldsymbol{s}_2, \boldsymbol{v}_1 \right\rangle / \left\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \right\rangle$	$c := \langle \boldsymbol{s}_2, \boldsymbol{v}_1 \rangle / \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle$	$c := \langle \boldsymbol{s}_2, \boldsymbol{v}_1 \rangle / \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle$
lyche-numerical-linear-algebra_F00028	122	■ 1.000	$\boldsymbol{s}+\boldsymbol{t}$	$\boldsymbol{s} + \boldsymbol{t}$	projections of $\boldsymbol{s} + \boldsymbol{t}$ into $\mathcal{S}$
lyche-numerical-linear-algebra_F00029	28	■ 1.000	$\mathcal{S}$	$\mathcal{S}$	Definition 1.5 (Subspace) A nonempty subset $\mathcal{S}$ of a real or complex vector space
lyche-numerical-linear-algebra_F00030	28	■ 1.000	$\mathcal{T}$	$\mathcal{T}$	4. The sum of two subspaces $\mathcal{S}$ and $\mathcal{T}$ of a vector space $\mathcal{V}$ is defined by

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F00031	175	—	$y_{\{1\}^{\{2\}} / 9 + y_{\{2\}^{\{2\}} = 1$	$y_1^2/9 + y_2^2 = 1$	
lyche-numerical-linear-algebra-F00033	251	■ 1.000	$n=100$	$n = 100$	The nonzero elements in L are shown in Fig. 11.1 for $n = 100$. Note that most
lyche-numerical-linear-algebra-F00034	274	—	$\alpha \rightarrow 1 - \alpha \lambda_1 $	$\alpha \rightarrow 1 - \alpha \lambda_1 $	
lyche-numerical-linear-algebra-F00035	274	—	$\alpha \rightarrow 1 - \alpha \lambda_n $	$\alpha \rightarrow 1 - \alpha \lambda_n $	
lyche-numerical-linear-algebra-F00036	278	—	$12.2 \ \rho(\boldsymbol{G}_{\omega})$	$12.2\rho(\boldsymbol{G}_{\omega})$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00037	278	—	$\omega \in [0, 2]$	$\omega \in [0, 2]$	
lyche-numerical-linear-algebra_F00038	277	■ 1.000	$n=2500$	$n = 2500$	(lower curve) and $n = 2500$ (upper curve). As ω increases the spectral radius of G_ω
lyche-numerical-linear-algebra_F00039	292	—	$Q(x, y)$	$Q(x, y)$	
lyche-numerical-linear-algebra_F00040	291	■ 1.000	Q	Q	are shown in Fig. 13.1. The level curves are ellipses and the graph of Q is a
lyche-numerical-linear-algebra_F00041	301	■ 1.000	$\boldsymbol{x} - \boldsymbol{x}_0$	$\boldsymbol{x} - \boldsymbol{x}_0$	$(\boldsymbol{x}_k - \boldsymbol{x}_0)$ and $\boldsymbol{x}_k - \boldsymbol{x}_0$ is the element in $\mathbb{W}_k$ that is closest to $\boldsymbol{x} - \boldsymbol{x}_0$ in the A -norm.
lyche-numerical-linear-algebra_F00042	301	■ 1.000	$\mathbb{W}_k$	$\mathbb{W}_k$	If $\boldsymbol{x}_0 = \mathbf{0}$ then (13.24) says that $\boldsymbol{x}_k$ is the element in $\mathbb{W}_k$ that is closest to the
lyche-numerical-linear-algebra_F00043	306	—	$k=3 . f \equiv Q-Q^*$	$k = 3.f \equiv Q - Q^*$	
lyche-numerical-linear-algebra_F00044	51	■ 1.000	μ_1	μ_1	determined from (2.9). For $n \geq 3$ and with μ_1 and μ_{n+1} given (2.9) can be written

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F00045	306	—	$\mu_{\{2\}}$	μ_2	
lyche-numerical-linear-algebra-F00046	306	—	$\mu_{\{3\}}$	μ_3	
lyche-numerical-linear-algebra-F00047	328	■ 0.989	$R_{\{i\}}$	R_i	disk R_i
lyche-numerical-linear-algebra-F00048	268	■ 1.000	$k_{\{n\}}$	k_n	Table 12.1 The number of iterations k_n to solve the discrete Poisson problem with n unknowns
lyche-numerical-linear-algebra-F00049	21	■ 1.000	n	n	3. $\mathbb{R}^n$ is the set of n -tuples of real numbers which we will represent as bold face
lyche-numerical-linear-algebra-F00050	268	■ 1.000	$10^{\{-8\}}$	10^{-8}	using the methods of Jacobi, Gauss-Seidel, and SOR (see text) with a tolerance 10^{-8}
lyche-numerical-linear-algebra-F00053	97	■ 1.000	K	K	LU1 factorization $\boldsymbol{H} = \boldsymbol{L}\boldsymbol{U}$ with $\ \boldsymbol{L}\ _\infty\ \boldsymbol{U}\ _\infty \leq K\ \boldsymbol{H}\ _\infty$ for a constant K not
lyche-numerical-linear-algebra-F00056	21	■ 1.000	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	complex numbers are denoted by $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, respectively.
lyche-numerical-linear-algebra-F00057	21	■ 1.000	$v:=e$	$v := e$	2. We use the “colon equal” symbol  to indicate that the symbol v is defined
lyche-numerical-linear-algebra-F00058	21	■ 1.000	v	v	2. We use the “colon equal” symbol $v := e$ to indicate that the symbol v is defined
lyche-numerical-linear-algebra-F00059	21	■ 1.000	e	e	by the expression e .
lyche-numerical-linear-algebra-F00060	21	■ 1.000	$\boldsymbol{x} \in \mathbb{R}^n$	$\boldsymbol{x} \in \mathbb{R}^n$	column vectors. Thus $\boldsymbol{x} \in \mathbb{R}^n$ means
lyche-numerical-linear-algebra-F00061	21	■ 1.000	$x_{\{i\}} \in \mathbb{R}$	$x_i \in \mathbb{R}$	where $x_i \in \mathbb{R}$ for $i = 1, \dots, n$. Row vectors are normally identified using the
lyche-numerical-linear-algebra-F00062	21	■ 1.000	$i=1, \ldots, n$	$i = 1, \dots, n$	where $x_i \in \mathbb{R}$ for $i = 1, \dots, n$. Row vectors are normally identified using the
lyche-numerical-linear-algebra-F00063	21	■ 1.000	$\boldsymbol{x}^T$	$\boldsymbol{x}^T$	transpose operation. Thus if $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{x}$ is a column vector and $\boldsymbol{x}^T$ is a row
lyche-numerical-linear-algebra-F00064	22	■ 1.000	$\mathbb{R}^{m \times n}$	$\mathbb{R}^{m \times n}$	5. $\mathbb{R}^{m \times n}$ is the set of matrices $\boldsymbol{A}$ with real elements. The integers m and n are the
lyche-numerical-linear-algebra-F00065	22	■ 1.000	m	m	5. $\mathbb{R}^{m \times n}$ is the set of matrices $\boldsymbol{A}$ with real elements. The integers m and n are the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00066	22	■ 0.998	i	i	The element in the i th row and j th column of A will be denoted by $a_{i,j}$, a_{ij} ,
lyche-numerical-linear-algebra_F00067	22	■ 0.998	j	j	The element in the i th row and j th column of A will be denoted by $a_{i,j}$, a_{ij} ,
lyche-numerical-linear-algebra_F00068	22	■ 0.998	a_{i, j}, a_{i j}	$a_{i,j}, a_{ij}$	The element in the i th row and j th column of A will be denoted by $a_{i,j}$, a_{ij} ,
lyche-numerical-linear-algebra_F00069	22	■ 1.000	\boldsymbol{A}(i, j)	$\boldsymbol{A}(i, j)$	$\boldsymbol{A}(i, j)$ or $(\boldsymbol{A})_{i,j}$. We use the notations
lyche-numerical-linear-algebra_F00070	22	■ 1.000	(\boldsymbol{A})_{i, j}	$(\boldsymbol{A})_{i,j}$	$\boldsymbol{A}(i, j)$ or $(\boldsymbol{A})_{i,j}$. We use the notations
lyche-numerical-linear-algebra_F00071	22	■ 1.000	\boldsymbol{a}_{: j}	$\boldsymbol{a}_{:j}$	for the columns $\boldsymbol{a}_{:j}$ and rows $\boldsymbol{a}_{i:}^T$ of A . We often drop the colon and write $\boldsymbol{a}_j$
lyche-numerical-linear-algebra_F00072	22	■ 1.000	\boldsymbol{a}_{i:}^T	$\boldsymbol{a}_{i:}^T$	for the columns $\boldsymbol{a}_{:j}$ and rows $\boldsymbol{a}_{i:}^T$ of A . We often drop the colon and write $\boldsymbol{a}_j$
lyche-numerical-linear-algebra_F00073	22	■ 1.000	\boldsymbol{a}_{j}	$\boldsymbol{a}_j$	for the columns $\boldsymbol{a}_{:j}$ and rows $\boldsymbol{a}_{i:}^T$ of A . We often drop the colon and write $\boldsymbol{a}_j$
lyche-numerical-linear-algebra_F00074	22	■ 1.000	\boldsymbol{a}_{i}^T	$\boldsymbol{a}_i^T$	and $\boldsymbol{a}_i^T$ with the risk of some confusion. If $m = 1$ then A is a row vector, if
lyche-numerical-linear-algebra_F00075	22	■ 1.000	m=1	$m = 1$	and $\boldsymbol{a}_i^T$ with the risk of some confusion. If $m = 1$ then A is a row vector, if
lyche-numerical-linear-algebra_F00076	22	■ 1.000	n=1	$n = 1$	$n = 1$ then A is a column vector, while if $m = n$ then A is a square matrix.
lyche-numerical-linear-algebra_F00077	22	■ 1.000	m=n	$m = n$	$n = 1$ then A is a column vector, while if $m = n$ then A is a square matrix.
lyche-numerical-linear-algebra_F00078	22	■ 0.999	\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \cdots	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \cdots$	In this text we will denote matrices by boldface capital letters $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \cdots$ and
lyche-numerical-linear-algebra_F00079	22	■ 1.000	\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \cdots	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \cdots$	vectors most often by boldface lower case letters x, y, z,
lyche-numerical-linear-algebra_F00080	22	■ 1.000	x=a+i b	$x = a + ib$	6. A complex number is a number written in the form $x = a + ib$, where a, b
lyche-numerical-linear-algebra_F00081	22	■ 1.000	a, b	a, b	6. A complex number is a number written in the form $x = a + ib$, where a, b
lyche-numerical-linear-algebra_F00082	22	■ 1.000	i^2=-1	$i^2 = -1$	are real numbers and i , the imaginary unit , satisfies $i^2 = -1$. The set of all
lyche-numerical-linear-algebra_F00083	22	■ 1.000	\mathbb{C}	$\mathbb{C}$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra_F00084	22	■ 1.000	a=\operatorname{Re} x	$a = \operatorname{Re} x$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra_F00085	22	■ 1.000	b=\operatorname{Im} x	$b = \operatorname{Im} x$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra_F00086	22	■ 1.000	x	x	real and imaginary part of x . The number $\bar{x} := a - ib$ is called the complex
lyche-numerical-linear-algebra_F00087	22	■ 1.000	\bar{x}:=a-i b	$\bar{x} := a - ib$	real and imaginary part of x . The number $\bar{x} := a - ib$ is called the complex
lyche-numerical-linear-algebra_F00088	22	■ 1.000	x :=\sqrt{\bar{x} x}=\sqrt{a^2+b^2}	$ x := \sqrt{\bar{x}x} = \sqrt{a^2 + b^2}$	conjugate of $x = a + ib$, and x := \sqrt{\bar{x}x} = \sqrt{a^2 + b^2} the absolute value or
lyche-numerical-linear-algebra_F00089	23	■ 1.000	e^{x+y}=e^x e^y	$e^{x+y} = e^x e^y$	We have e^{x+y} = e^x e^y for all $x, y \in \mathbb{C}$. The polar form of a complex number is
lyche-numerical-linear-algebra_F00090	23	■ 1.000	x, y \in \mathbb{C}	$x, y \in \mathbb{C}$	We have $e^{x+y} = e^x e^y$ for all x, y \in \mathbb{C} . The polar form of a complex number is
lyche-numerical-linear-algebra_F00091	23	■ 1.000	\boldsymbol{A} \in \mathbb{C}^{m \times n}	$\boldsymbol{A} \in \mathbb{C}^{m \times n}$	7. For matrices and vectors with complex elements we use the notation $\boldsymbol{A} \in \mathbb{C}^{m \times n}$
lyche-numerical-linear-algebra_F00092	23	■ 1.000	\boldsymbol{x} \in \mathbb{C}^n	$\boldsymbol{x} \in \mathbb{C}^n$	and $\boldsymbol{x} \in \mathbb{C}^n$. We define complex row vectors using either the transpose $\boldsymbol{x}^T$ or

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lyche-numerical-linear-algebra-F00093	23	■ 1.000	$\overline{\boldsymbol{x}} := \overline{\boldsymbol{x}^T} = [\overline{x_1}, \dots, \overline{x_n}]$	$\boldsymbol{x}^* := \overline{\boldsymbol{x}}^T = [\overline{x_1}, \dots, \overline{x_n}]$	the conjugate transpose operation $\boldsymbol{x}^* := \overline{\boldsymbol{x}}^T = [\overline{x_1}, \dots, \overline{x_n}]$. If $\boldsymbol{x} \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra-F00094	23	■ 0.809	$\boldsymbol{x}^* = \boldsymbol{x}^T$	$\boldsymbol{x}^* = \boldsymbol{x}^T$	$\boldsymbol{x}^* = \boldsymbol{x}^T$.
lyche-numerical-linear-algebra-F00095	23	■ 1.000	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	8. For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ and $a \in \mathbb{C}$ the operations of vector addition and scalar
lyche-numerical-linear-algebra-F00096	23	■ 1.000	$a \in \mathbb{C}$	$a \in \mathbb{C}$	8. For $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ and $a \in \mathbb{C}$ the operations of vector addition and scalar
lyche-numerical-linear-algebra-F00097	23	■ 1.000	$\boldsymbol{C} := \boldsymbol{A} + \boldsymbol{B}$	$\boldsymbol{C} := \boldsymbol{A} + \boldsymbol{B}$	• matrix addition $\boldsymbol{C} := \boldsymbol{A} + \boldsymbol{B}$ if $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are matrices of the same size, i.e.,
lyche-numerical-linear-algebra-F00098	23	■ 1.000	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$	• matrix addition $\boldsymbol{C} := \boldsymbol{A} + \boldsymbol{B}$ if $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are matrices of the same size, i.e.,
lyche-numerical-linear-algebra-F00099	23	■ 1.000	$c_{ij} := a_{ij} + b_{ij}$	$c_{ij} := a_{ij} + b_{ij}$	with the same number of rows and columns, and $c_{ij} := a_{ij} + b_{ij}$ for all i, j .
lyche-numerical-linear-algebra-F00100	23	■ 1.000	i, j	i, j	with the same number of rows and columns, and $c_{ij} := a_{ij} + b_{ij}$ for all i, j .
lyche-numerical-linear-algebra-F00101	23	■ 1.000	$\boldsymbol{C} := \alpha \boldsymbol{A}$	$\boldsymbol{C} := \alpha \boldsymbol{A}$	• multiplication by a scalar $\boldsymbol{C} := \alpha \boldsymbol{A}$, where $c_{ij} := \alpha a_{ij}$ for all i, j .
lyche-numerical-linear-algebra-F00102	23	■ 1.000	$c_{ij} := \alpha a_{ij}$	$c_{ij} := \alpha a_{ij}$	• multiplication by a scalar $\boldsymbol{C} := \alpha \boldsymbol{A}$, where $c_{ij} := \alpha a_{ij}$ for all i, j .
lyche-numerical-linear-algebra-F00103	23	■ 1.000	$\boldsymbol{C} := \boldsymbol{A} \boldsymbol{B}, \boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$	$\boldsymbol{C} := \boldsymbol{A} \boldsymbol{B}, \boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$	• matrix multiplication $\boldsymbol{C} := \boldsymbol{A} \boldsymbol{B}, \boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$ or $\boldsymbol{C} = \boldsymbol{A} * \boldsymbol{B}$, where $\boldsymbol{A} \in$
lyche-numerical-linear-algebra-F00104	23	■ 1.000	$\boldsymbol{C} := \boldsymbol{A} * \boldsymbol{B}$	$\boldsymbol{C} = \boldsymbol{A} * \boldsymbol{B}$	• matrix multiplication $\boldsymbol{C} := \boldsymbol{A} \boldsymbol{B}, \boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$ or $\boldsymbol{C} = \boldsymbol{A} * \boldsymbol{B}$, where $\boldsymbol{A} \in$
lyche-numerical-linear-algebra-F00105	23	■ 1.000	$\boldsymbol{A} \in$	$\boldsymbol{A} \in$	• matrix multiplication $\boldsymbol{C} := \boldsymbol{A} \boldsymbol{B}, \boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$ or $\boldsymbol{C} = \boldsymbol{A} * \boldsymbol{B}$, where $\boldsymbol{A} \in$
lyche-numerical-linear-algebra-F00106	23	■ 1.000	$\boldsymbol{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}, \boldsymbol{C} \in \mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}, \boldsymbol{C} \in \mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}, \boldsymbol{C} \in \mathbb{C}^{m \times n}$, and $c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$ for $i = 1, \dots, m$,
lyche-numerical-linear-algebra-F00107	23	■ 1.000	$c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$	$c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$	$\mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}, \boldsymbol{C} \in \mathbb{C}^{m \times n}$, and $c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$ for $i = 1, \dots, m$,
lyche-numerical-linear-algebra-F00108	23	■ 1.000	$i=1, \dots, m$	$i = 1, \dots, m$	$\mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}, \boldsymbol{C} \in \mathbb{C}^{m \times n}$, and $c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$ for $i = 1, \dots, m$,
lyche-numerical-linear-algebra-F00109	23	■ 1.000	$j=1, \dots, n$	$j = 1, \dots, n$	$j = 1, \dots, n$.
lyche-numerical-linear-algebra-F00110	23	■ 1.000	$\boldsymbol{C} := \boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D} := \boldsymbol{A} / \boldsymbol{B}$	$\boldsymbol{C} := \boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D} := \boldsymbol{A} / \boldsymbol{B}$	• element-by-element matrix operations $\boldsymbol{C} := \boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D} := \boldsymbol{A} / \boldsymbol{B}$, and $\boldsymbol{E} :=$
lyche-numerical-linear-algebra-F00111	23	■ 1.000	$\boldsymbol{E} :=$	$\boldsymbol{E} :=$	• element-by-element matrix operations $\boldsymbol{C} := \boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D} := \boldsymbol{A} / \boldsymbol{B}$, and $\boldsymbol{E} :=$
lyche-numerical-linear-algebra-F00112	23	■ 1.000	$\boldsymbol{A} \wedge \boldsymbol{r}$	$\boldsymbol{A} \wedge \boldsymbol{r}$	$\boldsymbol{A} \wedge \boldsymbol{r}$ where all matrices are of the same size and $c_{ij} := a_{ij} b_{ij}, d_{ij} := a_{ij} / b_{ij}$
lyche-numerical-linear-algebra-F00113	23	■ 1.000	$c_{ij} := a_{ij} b_{ij}, d_{ij} := a_{ij} / b_{ij}$	$c_{ij} := a_{ij} b_{ij}, d_{ij} := a_{ij} / b_{ij}$	$\boldsymbol{A} \wedge \boldsymbol{r}$ where all matrices are of the same size and $c_{ij} := a_{ij} b_{ij}, d_{ij} := a_{ij} / b_{ij}$
lyche-numerical-linear-algebra-F00114	23	■ 1.000	$e_{ij} := a_{ij}^r$	$e_{ij} := a_{ij}^r$	and $e_{ij} := a_{ij}^r$ for all i, j and suitable r . For the division $\boldsymbol{A} / \boldsymbol{B}$ we assume that
lyche-numerical-linear-algebra-F00115	23	■ 1.000	r	r	and $e_{ij} := a_{ij}^r$ for all i, j and suitable r . For the division $\boldsymbol{A} / \boldsymbol{B}$ we assume that
lyche-numerical-linear-algebra-F00116	23	■ 1.000	$\boldsymbol{A} / \boldsymbol{B}$	$\boldsymbol{A} / \boldsymbol{B}$	and $e_{ij} := a_{ij}^r$ for all i, j and suitable r . For the division $\boldsymbol{A} / \boldsymbol{B}$ we assume that
lyche-numerical-linear-algebra-F00117	23	■ 1.000	$\boldsymbol{B}$	$\boldsymbol{B}$	all elements of $\boldsymbol{B}$ are nonzero. The element-by-element product $\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B}$
lyche-numerical-linear-algebra-F00118	23	■ 1.000	$\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B}$	$\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B}$	all elements of $\boldsymbol{B}$ are nonzero. The element-by-element product $\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B}$
lyche-numerical-linear-algebra-F00119	23	■ 1.000	$\boldsymbol{A} \in \mathbb{R}^{m \times n}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}$	10. Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ or $\boldsymbol{A} \in \mathbb{C}^{m \times n}$. The transpose $\boldsymbol{A}^T$ and conjugate transpose

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lyche-numerical-linear-algebra-F00120	23	■ 1.000	$\boldsymbol{A}^{\mathrm{T}}$	$\boldsymbol{A}^T$	10. Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ or $\boldsymbol{A} \in \mathbb{C}^{m \times n}$. The transpose $\boldsymbol{A}^T$ and conjugate transpose
lyche-numerical-linear-algebra-F00121	23	■ 1.000	$\boldsymbol{A}^*$	$\boldsymbol{A}^*$	$\boldsymbol{A}^*$ are $n \times m$ matrices with elements $a_{ij}^T := a_{ji}$ and $a_{ij}^* := \bar{a}_{ji}$, respectively.
lyche-numerical-linear-algebra-F00122	23	■ 1.000	$n \times m$	$n \times m$	$\boldsymbol{A}^*$ are $n \times m$ matrices with elements $a_{ij}^T := a_{ji}$ and $a_{ij}^* := \bar{a}_{ji}$, respectively.
lyche-numerical-linear-algebra-F00123	23	■ 1.000	$a_{ij}^{\mathrm{T}} := a_{ji}$	$a_{ij}^T := a_{ji}$	$\boldsymbol{A}^*$ are $n \times m$ matrices with elements $a_{ij}^T := a_{ji}$ and $a_{ij}^* := \bar{a}_{ji}$, respectively.
lyche-numerical-linear-algebra-F00124	23	■ 1.000	$a_{ij}^* := \bar{a}_{ji}$	$a_{ij}^* := \bar{a}_{ji}$	$\boldsymbol{A}^*$ are $n \times m$ matrices with elements $a_{ij}^T := a_{ji}$ and $a_{ij}^* := \bar{a}_{ji}$, respectively.
lyche-numerical-linear-algebra-F00125	23	■ 1.000	n, p	n, p	If $\boldsymbol{B}$ is an n, p matrix then $(\boldsymbol{AB})^T = \boldsymbol{B}^T \boldsymbol{A}^T$ and $(\boldsymbol{AB})^* = \boldsymbol{B}^* \boldsymbol{A}^*$. A matrix
lyche-numerical-linear-algebra-F00126	23	■ 1.000	$(\boldsymbol{AB})^{\mathrm{T}} = \boldsymbol{B}^{\mathrm{T}} \boldsymbol{A}^{\mathrm{T}}$	$(\boldsymbol{AB})^T = \boldsymbol{B}^T \boldsymbol{A}^T$	If $\boldsymbol{B}$ is an n, p matrix then $(\boldsymbol{AB})^T = \boldsymbol{B}^T \boldsymbol{A}^T$ and $(\boldsymbol{AB})^* = \boldsymbol{B}^* \boldsymbol{A}^*$. A matrix
lyche-numerical-linear-algebra-F00127	23	■ 1.000	$(\boldsymbol{AB})^* = \boldsymbol{B}^* \boldsymbol{A}^*$	$(\boldsymbol{AB})^* = \boldsymbol{B}^* \boldsymbol{A}^*$	If $\boldsymbol{B}$ is an n, p matrix then $(\boldsymbol{AB})^T = \boldsymbol{B}^T \boldsymbol{A}^T$ and $(\boldsymbol{AB})^* = \boldsymbol{B}^* \boldsymbol{A}^*$. A matrix
lyche-numerical-linear-algebra-F00128	23	■ 1.000	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is symmetric if $\boldsymbol{A}^T = \boldsymbol{A}$ and Hermitian if $\boldsymbol{A}^* = \boldsymbol{A}$.
lyche-numerical-linear-algebra-F00129	23	■ 1.000	$\boldsymbol{A}^{\mathrm{T}} = \boldsymbol{A}$	$\boldsymbol{A}^T = \boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is symmetric if $\boldsymbol{A}^T = \boldsymbol{A}$ and Hermitian if $\boldsymbol{A}^* = \boldsymbol{A}$.
lyche-numerical-linear-algebra-F00130	23	■ 1.000	$\boldsymbol{A}^* = \boldsymbol{A}$	$\boldsymbol{A}^* = \boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is symmetric if $\boldsymbol{A}^T = \boldsymbol{A}$ and Hermitian if $\boldsymbol{A}^* = \boldsymbol{A}$.
lyche-numerical-linear-algebra-F00131	23	■ 1.000	$\boldsymbol{I}_n = \boldsymbol{I} := [\delta_{ij}]_{i,j=1}^n$	$\boldsymbol{I}_n = \boldsymbol{I} := [\delta_{ij}]_{i,j=1}^n$	while $\boldsymbol{I}_n = \boldsymbol{I} := [\delta_{ij}]_{i,j=1}^n$, where
lyche-numerical-linear-algebra-F00132	23	■ 1.000	$\boldsymbol{I}$	$\boldsymbol{I}$	rows of $\boldsymbol{I}$ are the unit vectors $\boldsymbol{e}_1, \boldsymbol{e}_2, \dots, \boldsymbol{e}_n$.
lyche-numerical-linear-algebra-F00133	23	■ 1.000	$\boldsymbol{e}_1, \boldsymbol{e}_2, \dots, \boldsymbol{e}_n$	$\boldsymbol{e}_1, \boldsymbol{e}_2, \dots, \boldsymbol{e}_n$	rows of $\boldsymbol{I}$ are the unit vectors $\boldsymbol{e}_1, \boldsymbol{e}_2, \dots, \boldsymbol{e}_n$.
lyche-numerical-linear-algebra-F00134	24	■ 1.000	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$ or $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. Then $\boldsymbol{A}$ is
lyche-numerical-linear-algebra-F00135	24	■ 1.000	$a_{ij} = 0$	$a_{ij} = 0$	• diagonal if $a_{ij} = 0$ for $i \neq j$.
lyche-numerical-linear-algebra-F00136	24	■ 1.000	$i \neq j$	$i \neq j$	• diagonal if $a_{ij} = 0$ for $i \neq j$.
lyche-numerical-linear-algebra-F00137	24	■ 1.000	$i > j$	$i > j$	• upper triangular or right triangular if $a_{ij} = 0$ for $i > j$.
lyche-numerical-linear-algebra-F00138	24	■ 1.000	$i < j$	$i < j$	• lower triangular or left triangular if $a_{ij} = 0$ for $i < j$.
lyche-numerical-linear-algebra-F00139	24	■ 1.000	$i > j + 1$	$i > j + 1$	• upper Hessenberg if $a_{ij} = 0$ for $i > j + 1$.
lyche-numerical-linear-algebra-F00140	24	■ 1.000	$i < j + 1$	$i < j + 1$	• lower Hessenberg if $a_{ij} = 0$ for $i < j + 1$.
lyche-numerical-linear-algebra-F00141	24	■ 1.000	$ i - j > 1$	$ i - j > 1$	• tridiagonal if $a_{ij} = 0$ for $ i - j > 1$.
lyche-numerical-linear-algebra-F00142	24	■ 0.999	d	d	• d-banded if $a_{ij} = 0$ for $ i - j > d$.
lyche-numerical-linear-algebra-F00143	24	■ 0.999	$ i - j > d$	$ i - j > d$	• d-banded if $a_{ij} = 0$ for $ i - j > d$.
lyche-numerical-linear-algebra-F00144	24	■ 1.000	$n \times n$	$n \times n$	13. We use the following notations for diagonal- and tridiagonal $n \times n$ matrices
lyche-numerical-linear-algebra-F00145	24	■ 1.000	$b_{ii} := d_i$	$b_{ii} := d_i$	Here $b_{ii} := d_i$ for $i = 1, \dots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$ for $i = 1, \dots, n - 1$,
lyche-numerical-linear-algebra-F00146	24	■ 1.000	$i = 1, \dots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$	$i = 1, \dots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$	Here $b_{ii} := d_i$ for $i = 1, \dots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$ for $i = 1, \dots, n - 1$,

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lyche-numerical-linear-algebra-F00147	24	1.000	$i=1, \ldots, n-1$	$i = 1, \dots, n - 1$	Here $b_{ii} := d_i$ for $i = 1, \dots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$ for $i = 1, \dots, n - 1$,
lyche-numerical-linear-algebra-F00148	24	1.000	$b_{ij}:=0$	$b_{ij} := 0$	and $b_{ij} := 0$ otherwise.
lyche-numerical-linear-algebra-F00149	24	1.000	$1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j_1 < j_2 < \cdots <$	$1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j_1 < j_2 < \cdots <$	14. Suppose $A \in \mathbb{C}^{m \times n}$ and $1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j_1 < j_2 < \cdots <$
lyche-numerical-linear-algebra-F00150	24	1.000	$j_{\{c\}} \leq n$	$j_c \leq n$	$j_c \leq n$. The matrix $A(i, j) \in \mathbb{C}^{r \times c}$ is the submatrix of A consisting of rows
lyche-numerical-linear-algebra-F00151	24	1.000	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r \times c}$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r \times c}$	$j_c \leq n$. The matrix $\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r \times c}$ is the submatrix of $\boldsymbol{A}$ consisting of rows
lyche-numerical-linear-algebra-F00152	24	1.000	$\boldsymbol{i} := [i_1, \dots, i_r]$	$\boldsymbol{i} := [i_1, \dots, i_r]$	$\boldsymbol{i} := [i_1, \dots, i_r]$ and columns $\boldsymbol{j} := [j_1, \dots, j_c]$
lyche-numerical-linear-algebra-F00153	24	1.000	$\boldsymbol{j} := [j_1, \dots, j_c]$	$\boldsymbol{j} := [j_1, \dots, j_c]$	$\boldsymbol{i} := [i_1, \dots, i_r]$ and columns $\boldsymbol{j} := [j_1, \dots, j_c]$
lyche-numerical-linear-algebra-F00154	25	1.000	$\mathbb{R}^2$	$\mathbb{R}^2$	Many mathematical systems have analogous properties to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$.
lyche-numerical-linear-algebra-F00155	25	1.000	$\mathbb{R}^3$	$\mathbb{R}^3$	Many mathematical systems have analogous properties to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$.
lyche-numerical-linear-algebra-F00156	25	1.000	$\mathcal{V}$	$\mathcal{V}$	Definition 1.1 (Real Vector Space) A real vector space is a nonempty set $\mathcal{V}$,
lyche-numerical-linear-algebra-F00157	25	1.000	$+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$	$+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$	whose objects are called vectors , together with two operations $+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$
lyche-numerical-linear-algebra-F00158	25	0.992	$\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$	$\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$	and $\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$, called addition and scalar multiplication , satisfying the
lyche-numerical-linear-algebra-F00159	25	1.000	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$	following axioms for all vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.
lyche-numerical-linear-algebra-F00160	25	1.000	c, d	c, d	following axioms for all vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.
lyche-numerical-linear-algebra-F00161	25	1.000	$\mathbb{R}$	$\mathbb{R}$	following axioms for all vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.
lyche-numerical-linear-algebra-F00162	25	1.000	$\boldsymbol{u} + \boldsymbol{v}$	$\boldsymbol{u} + \boldsymbol{v}$	(V1) The sum $\boldsymbol{u} + \boldsymbol{v}$ is in $\mathcal{V}$,
lyche-numerical-linear-algebra-F00163	25	0.960	$\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$	$\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$	(V2) $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$,
lyche-numerical-linear-algebra-F00164	25	0.994	$\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$	$\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$	(V3) $\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$,
lyche-numerical-linear-algebra-F00165	25	0.999	$\mathbf{0}$	$\mathbf{0}$	(V4) There is a zero vector $\mathbf{0}$ such that $\boldsymbol{u} + \mathbf{0} = \boldsymbol{u}$,
lyche-numerical-linear-algebra-F00166	25	0.999	$\boldsymbol{u} + \mathbf{0} = \boldsymbol{u}$	$\boldsymbol{u} + \mathbf{0} = \boldsymbol{u}$	(V4) There is a zero vector $\mathbf{0}$ such that $\boldsymbol{u} + \mathbf{0} = \boldsymbol{u}$,
lyche-numerical-linear-algebra-F00167	25	1.000	$\boldsymbol{u}$	$\boldsymbol{u}$	(V5) For each $\boldsymbol{u}$ in $\mathcal{V}$ there is a vector $-\boldsymbol{u}$ in $\mathcal{V}$ such that $\boldsymbol{u} + (-\boldsymbol{u}) = \mathbf{0}$,
lyche-numerical-linear-algebra-F00168	25	1.000	$-\boldsymbol{u}$	$-\boldsymbol{u}$	(V5) For each $\boldsymbol{u}$ in $\mathcal{V}$ there is a vector $-\boldsymbol{u}$ in $\mathcal{V}$ such that $\boldsymbol{u} + (-\boldsymbol{u}) = \mathbf{0}$,
lyche-numerical-linear-algebra-F00169	25	1.000	$\boldsymbol{u} + (-\boldsymbol{u}) = \mathbf{0}$	$\boldsymbol{u} + (-\boldsymbol{u}) = \mathbf{0}$	(V5) For each $\boldsymbol{u}$ in $\mathcal{V}$ there is a vector $-\boldsymbol{u}$ in $\mathcal{V}$ such that $\boldsymbol{u} + (-\boldsymbol{u}) = \mathbf{0}$,
lyche-numerical-linear-algebra-F00170	25	1.000	$c \cdot \boldsymbol{u}$	$c \cdot \boldsymbol{u}$	(S1) The scalar multiple $c \cdot \boldsymbol{u}$ is in $\mathcal{V}$,
lyche-numerical-linear-algebra-F00171	25	1.000	$c \cdot (\boldsymbol{u} + \boldsymbol{v}) = c \cdot \boldsymbol{u} + c \cdot \boldsymbol{v}$	$c \cdot (\boldsymbol{u} + \boldsymbol{v}) = c \cdot \boldsymbol{u} + c \cdot \boldsymbol{v}$	(S2) $c \cdot (\boldsymbol{u} + \boldsymbol{v}) = c \cdot \boldsymbol{u} + c \cdot \boldsymbol{v}$,
lyche-numerical-linear-algebra-F00172	25	1.000	$(c + d) \cdot \boldsymbol{u} = c \cdot \boldsymbol{u} + d \cdot \boldsymbol{u}$	$(c + d) \cdot \boldsymbol{u} = c \cdot \boldsymbol{u} + d \cdot \boldsymbol{u}$	(S3) $(c + d) \cdot \boldsymbol{u} = c \cdot \boldsymbol{u} + d \cdot \boldsymbol{u}$,
lyche-numerical-linear-algebra-F00173	25	1.000	$c \cdot (d \cdot \boldsymbol{u}) = (cd) \cdot \boldsymbol{u}$	$c \cdot (d \cdot \boldsymbol{u}) = (cd) \cdot \boldsymbol{u}$	(S4) $c \cdot (d \cdot \boldsymbol{u}) = (cd) \cdot \boldsymbol{u}$,
lyche-numerical-linear-algebra-F00174	25	1.000	$1 \cdot \boldsymbol{u} = \boldsymbol{u}$	$1 \cdot \boldsymbol{u} = \boldsymbol{u}$	(S5) $1 \cdot \boldsymbol{u} = \boldsymbol{u}$.

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lyche-numerical-linear-algebra-F00175	25	0.589	$c \, \boldsymbol{v}$	$c\boldsymbol{v}$	The scalar multiplication symbol $\cdot$ is often omitted, writing $c\boldsymbol{v}$ instead of $c \cdot \boldsymbol{v}$. We
lyche-numerical-linear-algebra-F00176	25	0.589	$c \, \boldsymbol{\cdot} \, \boldsymbol{v}$	$c \cdot \boldsymbol{v}$	The scalar multiplication symbol $\cdot$ is often omitted, writing $c\boldsymbol{v}$ instead of $c \cdot \boldsymbol{v}$. We
lyche-numerical-linear-algebra-F00177	25	1.000	$\boldsymbol{u} - \boldsymbol{v} := \boldsymbol{u} + (-\boldsymbol{v})$	$\boldsymbol{u} - \boldsymbol{v} := \boldsymbol{u} + (-\boldsymbol{v})$	define $\boldsymbol{u} - \boldsymbol{v} := \boldsymbol{u} + (-\boldsymbol{v})$. We call $\mathcal{V}$ a complex vector space if the scalars consist
lyche-numerical-linear-algebra-F00178	25	0.974	$\boldsymbol{u} \in \mathcal{V}$	$\boldsymbol{u} \in \mathcal{V}$	2. For each $\boldsymbol{u} \in \mathcal{V}$ the negative $-\boldsymbol{u}$ of $\boldsymbol{u}$ is unique.
lyche-numerical-linear-algebra-F00179	25	1.000	$0 \, \boldsymbol{u} = \boldsymbol{0}, c \, \boldsymbol{0} = \boldsymbol{0}$	$0\boldsymbol{u} = \boldsymbol{0}, c\boldsymbol{0} = \boldsymbol{0}$	3. $0\boldsymbol{u} = \boldsymbol{0}, c\boldsymbol{0} = \boldsymbol{0}$, and $-\boldsymbol{u} = (-1)\boldsymbol{u}$.
lyche-numerical-linear-algebra-F00180	25	1.000	$-\boldsymbol{u} = (-1) \, \boldsymbol{u}$	$-\boldsymbol{u} = (-1)\boldsymbol{u}$	3. $0\boldsymbol{u} = \boldsymbol{0}, c\boldsymbol{0} = \boldsymbol{0}$, and $-\boldsymbol{u} = (-1)\boldsymbol{u}$.
lyche-numerical-linear-algebra-F00181	25	1.000	$n \in \mathbb{N}$	$n \in \mathbb{N}$	1. The spaces $\mathbb{R}^n$ and $\mathbb{C}^n$, where $n \in \mathbb{N}$, are real and complex vector spaces,
lyche-numerical-linear-algebra-F00182	25	1.000	$\mathcal{D}$	$\mathcal{D}$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\boldsymbol{f}, \boldsymbol{g} : \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra-F00183	25	1.000	$d \in \mathbb{N}$	$d \in \mathbb{N}$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\boldsymbol{f}, \boldsymbol{g} : \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra-F00184	25	1.000	$\boldsymbol{f}, \boldsymbol{g} : \mathcal{D} \rightarrow \mathbb{R}^d$	$\boldsymbol{f}, \boldsymbol{g} : \mathcal{D} \rightarrow \mathbb{R}^d$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\boldsymbol{f}, \boldsymbol{g} : \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra-F00185	25	1.000	$\boldsymbol{f}, \boldsymbol{g}$	$\boldsymbol{f}, \boldsymbol{g}$	Two functions $\boldsymbol{f}, \boldsymbol{g}$ in $\mathcal{V}$ are equal if $\boldsymbol{f}(t) = \boldsymbol{g}(t)$ for all $t \in \mathcal{D}$. The zero
lyche-numerical-linear-algebra-F00186	25	1.000	$\boldsymbol{f}(t) = \boldsymbol{g}(t)$	$\boldsymbol{f}(t) = \boldsymbol{g}(t)$	Two functions $\boldsymbol{f}, \boldsymbol{g}$ in $\mathcal{V}$ are equal if $\boldsymbol{f}(t) = \boldsymbol{g}(t)$ for all $t \in \mathcal{D}$. The zero
lyche-numerical-linear-algebra-F00187	25	1.000	$t \in \mathcal{D}$	$t \in \mathcal{D}$	Two functions $\boldsymbol{f}, \boldsymbol{g}$ in $\mathcal{V}$ are equal if $\boldsymbol{f}(t) = \boldsymbol{g}(t)$ for all $t \in \mathcal{D}$. The zero
lyche-numerical-linear-algebra-F00188	25	0.998	$\boldsymbol{f}(t) = \boldsymbol{0}$	$\boldsymbol{f}(t) = \boldsymbol{0}$	element is the zero function given by $\boldsymbol{f}(t) = \boldsymbol{0}$ for all $t \in \mathcal{D}$ and the negative
lyche-numerical-linear-algebra-F00189	25	1.000	$\boldsymbol{f}$	$\boldsymbol{f}$	of $\boldsymbol{f}$ is given by $-\boldsymbol{f} = (-1)\boldsymbol{f}$. In the following we will use boldface letters for
lyche-numerical-linear-algebra-F00190	25	1.000	$-\boldsymbol{f} = (-1) \, \boldsymbol{f}$	$-\boldsymbol{f} = (-1)\boldsymbol{f}$	of $\boldsymbol{f}$ is given by $-\boldsymbol{f} = (-1)\boldsymbol{f}$. In the following we will use boldface letters for
lyche-numerical-linear-algebra-F00191	25	1.000	$d > 1$	$d > 1$	functions only if $d > 1$.
lyche-numerical-linear-algebra-F00192	25	0.997	$n \geq 0$	$n \geq 0$	3. For $n \geq 0$ the space Π_n of polynomials of degree at most n consists of all
lyche-numerical-linear-algebra-F00193	25	0.997	Π_n	Π_n	3. For $n \geq 0$ the space Π_n of polynomials of degree at most n consists of all
lyche-numerical-linear-algebra-F00194	25	1.000	$p : \mathbb{R} \rightarrow \mathbb{R}, p : \mathbb{R} \rightarrow \mathbb{C}$	$p : \mathbb{R} \rightarrow \mathbb{R}, p : \mathbb{R} \rightarrow \mathbb{C}$	polynomials $p : \mathbb{R} \rightarrow \mathbb{R}, p : \mathbb{R} \rightarrow \mathbb{C}$, or $p : \mathbb{C} \rightarrow \mathbb{C}$ of the form
lyche-numerical-linear-algebra-F00195	25	1.000	$p : \mathbb{C} \rightarrow \mathbb{C}$	$p : \mathbb{C} \rightarrow \mathbb{C}$	polynomials $p : \mathbb{R} \rightarrow \mathbb{R}, p : \mathbb{R} \rightarrow \mathbb{C}$, or $p : \mathbb{C} \rightarrow \mathbb{C}$ of the form
lyche-numerical-linear-algebra-F00196	26	1.000	$a_0, \dots, a_n$	$a_0, \dots, a_n$	where the coefficients $a_0, \dots, a_n$ are real or complex numbers. p is called the
lyche-numerical-linear-algebra-F00197	26	1.000	$0 \leq k \leq n$	$0 \leq k \leq n$	integer $0 \leq k \leq n$ such that $p(t) = a_0 + \dots + a_k t^k$ with $a_k \neq 0$. The degree of
lyche-numerical-linear-algebra-F00198	26	1.000	$p(t) = a_0 + \dots + a_k t^k$	$p(t) = a_0 + \dots + a_k t^k$	integer $0 \leq k \leq n$ such that $p(t) = a_0 + \dots + a_k t^k$ with $a_k \neq 0$. The degree of
lyche-numerical-linear-algebra-F00199	26	1.000	$a_k \neq 0$	$a_k \neq 0$	integer $0 \leq k \leq n$ such that $p(t) = a_0 + \dots + a_k t^k$ with $a_k \neq 0$. The degree of
lyche-numerical-linear-algebra-F00200	26	1.000	$n \geq 1$	$n \geq 1$	Definition 1.2 (Linear Combination) For $n \geq 1$ let $\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ be a set of
lyche-numerical-linear-algebra-F00201	26	1.000	$\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$	$\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$	Definition 1.2 (Linear Combination) For $n \geq 1$ let $\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ be a set of

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lyche-numerical-linear-algebra-F00202	26	■ 1.000	$c_{\{1\}}, \ldots, c_{\{n\}}$	$c_1, \dots, c_n$	vectors in a vector space $\mathcal{V}$ and let $c_1, \dots, c_n$ be scalars.
lyche-numerical-linear-algebra-F00203	26	■ 1.000	$c_{\{1\}} \boldsymbol{x}_{\{1\}} + \cdots + c_{\{n\}} \boldsymbol{x}_{\{n\}}$	$c_1 \boldsymbol{x}_1 + \cdots + c_n \boldsymbol{x}_n$	1. The sum $c_1 \boldsymbol{x}_1 + \cdots + c_n \boldsymbol{x}_n$ is called a linear combination of $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$.
lyche-numerical-linear-algebra-F00204	26	■ 1.000	$\boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{n\}}$	$\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$	1. The sum $c_1 \boldsymbol{x}_1 + \cdots + c_n \boldsymbol{x}_n$ is called a linear combination of $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$.
lyche-numerical-linear-algebra-F00205	26	■ 1.000	$c_{\{j\}} \boldsymbol{x}_{\{j\}} \neq \mathbf{0}$	$c_j \boldsymbol{x}_j \neq \mathbf{0}$	2. The linear combination is nontrivial if $c_j \boldsymbol{x}_j \neq \mathbf{0}$ for at least one j .
lyche-numerical-linear-algebra-F00206	26	■ 0.999	$\mathcal{X}$	$\mathcal{X}$	3. The set of all linear combinations of elements in $\mathcal{X}$ is denoted $\text{span}(\mathcal{X})$.
lyche-numerical-linear-algebra-F00207	26	■ 0.999	$\operatorname{span}(\mathcal{X})$	$\text{span}(\mathcal{X})$	3. The set of all linear combinations of elements in $\mathcal{X}$ is denoted $\text{span}(\mathcal{X})$.
lyche-numerical-linear-algebra-F00208	26	■ 1.000	$\left\{ \boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{n\}} \right\}$	$\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$	exists $n \in \mathbb{N}$ and $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ in $\mathcal{V}$ such that $\mathcal{V} = \text{span}(\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\})$.
lyche-numerical-linear-algebra-F00209	26	■ 1.000	$\mathcal{V} = \operatorname{span}\left(\left\{ \boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{n\}} \right\}\right)$	$\mathcal{V} = \text{span}(\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\})$	exists $n \in \mathbb{N}$ and $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ in $\mathcal{V}$ such that $\mathcal{V} = \text{span}(\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\})$.
lyche-numerical-linear-algebra-F00210	26	■ 1.000	$\boldsymbol{x} = \left[\boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{m\}} \right]^T$	$\boldsymbol{x} = [x_1, \dots, x_m]^T$	1. Any $\boldsymbol{x} = [x_1, \dots, x_m]^T$ in $\mathbb{C}^m$ can be written as a linear combination of the unit
lyche-numerical-linear-algebra-F00211	26	■ 1.000	$\mathbb{C}^{\wedge m}$	$\mathbb{C}^m$	1. Any $\boldsymbol{x} = [x_1, \dots, x_m]^T$ in $\mathbb{C}^m$ can be written as a linear combination of the unit
lyche-numerical-linear-algebra-F00212	26	■ 0.991	$\boldsymbol{x} = \boldsymbol{x}_{\{1\}} \boldsymbol{e}_{\{1\}} + \boldsymbol{x}_{\{2\}} \boldsymbol{e}_{\{2\}} + \cdots + \boldsymbol{x}_{\{m\}} \boldsymbol{e}_{\{m\}}$	$\boldsymbol{x} = x_1 \boldsymbol{e}_1 + x_2 \boldsymbol{e}_2 + \cdots + x_m \boldsymbol{e}_m$	vectors as $\boldsymbol{x} = x_1 \boldsymbol{e}_1 + x_2 \boldsymbol{e}_2 + \cdots + x_m \boldsymbol{e}_m$. Thus, $\mathbb{C}^m = \text{span}(\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_m\})$ and
lyche-numerical-linear-algebra-F00213	26	■ 0.991	$\mathbb{C}^{\wedge m} = \operatorname{span}\left(\left\{ \boldsymbol{e}_{\{1\}}, \ldots, \boldsymbol{e}_{\{m\}} \right\}\right)$	$\mathbb{C}^m = \text{span}(\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_m\})$	vectors as $\boldsymbol{x} = x_1 \boldsymbol{e}_1 + x_2 \boldsymbol{e}_2 + \cdots + x_m \boldsymbol{e}_m$. Thus, $\mathbb{C}^m = \text{span}(\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_m\})$ and
lyche-numerical-linear-algebra-F00214	26	■ 1.000	$\mathbb{R}^{\wedge m}$	$\mathbb{R}^m$	$\mathbb{C}^m$ is finite dimensional. Similarly $\mathbb{R}^m$ is finite dimensional.
lyche-numerical-linear-algebra-F00215	26	■ 1.000	$\Pi = \cup_{\{n\}} \Pi_{\{n\}}$	$\Pi = \cup_n \Pi_n$	2. Let $\Pi = \cup_n \Pi_n$ be the space of all polynomials. Π is a vector space that
lyche-numerical-linear-algebra-F00216	26	■ 1.000	Π	Π	2. Let $\Pi = \cup_n \Pi_n$ be the space of all polynomials. Π is a vector space that
lyche-numerical-linear-algebra-F00217	26	■ 1.000	$\Pi =$	$\Pi =$	is not finite dimensional. For suppose Π is finite dimensional. Then $\Pi =$
lyche-numerical-linear-algebra-F00218	26	■ 0.986	$\operatorname{span}\left(\left\{ \boldsymbol{p}_{\{1\}}, \ldots, \boldsymbol{p}_{\{m\}} \right\}\right)$	$\text{span}(\{p_1, \dots, p_m\})$	$\text{span}(\{p_1, \dots, p_m\})$ for some polynomials $p_1, \dots, p_m$. Let d be an integer such
lyche-numerical-linear-algebra-F00219	26	■ 0.986	$\boldsymbol{p}_{\{1\}}, \ldots, \boldsymbol{p}_{\{m\}}$	$p_1, \dots, p_m$	$\text{span}(\{p_1, \dots, p_m\})$ for some polynomials $p_1, \dots, p_m$. Let d be an integer such
lyche-numerical-linear-algebra-F00220	26	■ 1.000	$\boldsymbol{p}_{\{j\}}$	p_j	that the degree of p_j is less than d for $j = 1, \dots, m$. A polynomial of degree d
lyche-numerical-linear-algebra-F00221	26	■ 1.000	$j=1, \ldots, m$	$j = 1, \dots, m$	that the degree of p_j is less than d for $j = 1, \dots, m$. A polynomial of degree d
lyche-numerical-linear-algebra-F00222	26	■ 0.958	$\mathcal{X} = \left\{ \boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{n\}} \right\}$	$\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$	Definition 1.3 (Linear Independence) A set $\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ of nonzero
lyche-numerical-linear-algebra-F00223	26	■ 1.000	$\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$	$\boldsymbol{x} \in \text{span}(\mathcal{X})$	1. If $\boldsymbol{x} \in \text{span}(\mathcal{X})$ then the scalars $c_1, \dots, c_n$ in the representation $\boldsymbol{x} = c_1 \boldsymbol{x}_1 + \cdots +$
lyche-numerical-linear-algebra-F00224	26	■ 1.000	$\boldsymbol{x} = c_{\{1\}} \boldsymbol{x}_{\{1\}} + \cdots +$	$\boldsymbol{x} = c_1 \boldsymbol{x}_1 + \cdots +$	1. If $\boldsymbol{x} \in \text{span}(\mathcal{X})$ then the scalars $c_1, \dots, c_n$ in the representation $\boldsymbol{x} = c_1 \boldsymbol{x}_1 + \cdots +$
lyche-numerical-linear-algebra-F00225	26	■ 0.995	$c_{\{n\}} \boldsymbol{x}_{\{n\}}$	$c_n \boldsymbol{x}_n$	$c_n \boldsymbol{x}_n$ are unique.
lyche-numerical-linear-algebra-F00226	26	■ 0.995	$\boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{v}_{\{n\}}$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$	Lemma 1.1 (Linear Independence and Span) Suppose $\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$ span a vector
lyche-numerical-linear-algebra-F00227	26	■ 1.000	$\boldsymbol{w}_{\{1\}}, \ldots, \boldsymbol{w}_{\{k\}}$	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$	space $\mathcal{V}$ and that $\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$ are linearly independent vectors in $\mathcal{V}$. Then $k \leq n$.

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lyche-numerical-linear-algebra-F00228	26	■ 1.000	$k \leq n$	$k \leq n$	space $\mathcal{V}$ and that $\mathbf{w}_1, \dots, \mathbf{w}_k$ are linearly independent vectors in $\mathcal{V}$. Then $k \leq n$.
lyche-numerical-linear-algebra-F00229	27	■ 1.000	$k > n$	$k > n$	Proof Suppose $k > n$. Write $\mathbf{w}_1$ as a linear combination of elements from
lyche-numerical-linear-algebra-F00230	27	■ 1.000	$\boldsymbol{w}_1$	$\mathbf{w}_1$	Proof Suppose $k > n$. Write $\mathbf{w}_1$ as a linear combination of elements from
lyche-numerical-linear-algebra-F00231	27	■ 1.000	$\mathcal{X}_0 := \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	$\mathcal{X}_0 := \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$	the set $\mathcal{X}_0 := \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, say $\mathbf{w}_1 = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n$. Since $\mathbf{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra-F00232	27	■ 1.000	$\boldsymbol{w}_1 = c_1 \boldsymbol{v}_1 + \dots + c_n \boldsymbol{v}_n$	$\mathbf{w}_1 = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n$	the set $\mathcal{X}_0 := \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, say $\mathbf{w}_1 = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n$. Since $\mathbf{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra-F00233	27	■ 1.000	$\boldsymbol{w}_1 \neq \mathbf{0}$	$\mathbf{w}_1 \neq \mathbf{0}$	the set $\mathcal{X}_0 := \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, say $\mathbf{w}_1 = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n$. Since $\mathbf{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra-F00234	27	■ 1.000	c_{i_1}	c_{i_1}	not all the c 's are equal to zero. Pick a nonzero c , say c_{i_1} . Then $\mathbf{v}_{i_1}$ can be
lyche-numerical-linear-algebra-F00235	27	■ 1.000	$\boldsymbol{v}_{i_1}$	$\mathbf{v}_{i_1}$	not all the c 's are equal to zero. Pick a nonzero c , say c_{i_1} . Then $\mathbf{v}_{i_1}$ can be
lyche-numerical-linear-algebra-F00236	27	■ 1.000	$\boldsymbol{v}$	$\mathbf{v}$	expressed as a linear combination of $\mathbf{w}_1$ and the remaining $\mathbf{v}$'s. So the set $\mathcal{X}_1 :=$
lyche-numerical-linear-algebra-F00237	27	■ 1.000	$\mathcal{X}_1 :=$	$\mathcal{X}_1 :=$	expressed as a linear combination of $\mathbf{w}_1$ and the remaining $\mathbf{v}$'s. So the set $\mathcal{X}_1 :=$
lyche-numerical-linear-algebra-F00238	27	■ 1.000	$\{\boldsymbol{w}_1, \boldsymbol{v}_1, \dots, \boldsymbol{v}_{i_1-1}, \boldsymbol{v}_{i_1+1}, \dots, \boldsymbol{v}_n\}$	$\{\mathbf{w}_1, \mathbf{v}_1, \dots, \mathbf{v}_{i_1-1}, \mathbf{v}_{i_1+1}, \dots, \mathbf{v}_n\}$	$\{\mathbf{w}_1, \mathbf{v}_1, \dots, \mathbf{v}_{i_1-1}, \mathbf{v}_{i_1+1}, \dots, \mathbf{v}_n\}$ must also be a spanning set for $\mathcal{V}$. We repeat this
lyche-numerical-linear-algebra-F00239	27	■ 1.000	$\boldsymbol{w}_2$	$\mathbf{w}_2$	for $\mathbf{w}_2$ and $\mathcal{X}_1$. In the linear combination $\mathbf{w}_2 = d_{i_1} \mathbf{w}_1 + \sum_{j \neq i_1} d_j \mathbf{v}_j$, we must
lyche-numerical-linear-algebra-F00240	27	■ 1.000	$\mathcal{X}_1$	$\mathcal{X}_1$	for $\mathbf{w}_2$ and $\mathcal{X}_1$. In the linear combination $\mathbf{w}_2 = d_{i_1} \mathbf{w}_1 + \sum_{j \neq i_1} d_j \mathbf{v}_j$, we must
lyche-numerical-linear-algebra-F00241	27	■ 1.000	$\boldsymbol{w}_2 = d_{i_1} \boldsymbol{w}_1 + \sum_{j \neq i_1} d_j \boldsymbol{v}_j$	$\mathbf{w}_2 = d_{i_1} \mathbf{w}_1 + \sum_{j \neq i_1} d_j \mathbf{v}_j$	for $\mathbf{w}_2$ and $\mathcal{X}_1$. In the linear combination $\mathbf{w}_2 = d_{i_1} \mathbf{w}_1 + \sum_{j \neq i_1} d_j \mathbf{v}_j$, we must
lyche-numerical-linear-algebra-F00242	27	■ 0.900	$d_{i_2} \neq 0$	$d_{i_2} \neq 0$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\mathbf{w}_2 = d_{i_1} \mathbf{w}_1$ contradicting
lyche-numerical-linear-algebra-F00243	27	■ 0.900	i_2	i_2	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\mathbf{w}_2 = d_{i_1} \mathbf{w}_1$ contradicting
lyche-numerical-linear-algebra-F00244	27	■ 0.900	$i_2 \neq i_1$	$i_2 \neq i_1$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\mathbf{w}_2 = d_{i_1} \mathbf{w}_1$ contradicting
lyche-numerical-linear-algebra-F00245	27	■ 0.900	$\boldsymbol{w}_2 = d_1 \boldsymbol{w}_1$	$\mathbf{w}_2 = d_1 \mathbf{w}_1$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\mathbf{w}_2 = d_{i_1} \mathbf{w}_1$ contradicting
lyche-numerical-linear-algebra-F00246	27	■ 1.000	$\boldsymbol{w}$	$\mathbf{w}$	the linear independence of the $\mathbf{w}$'s. So the set $\mathcal{X}_2$ consisting of the $\mathbf{v}$'s with $\mathbf{v}_{i_1}$
lyche-numerical-linear-algebra-F00247	27	■ 1.000	$\mathcal{X}_2$	$\mathcal{X}_2$	the linear independence of the $\mathbf{w}$'s. So the set $\mathcal{X}_2$ consisting of the $\mathbf{v}$'s with $\mathbf{v}_{i_1}$
lyche-numerical-linear-algebra-F00248	27	■ 0.998	$\boldsymbol{v}_{i_2}$	$\mathbf{v}_{i_2}$	replaced by $\mathbf{w}_1$ and $\mathbf{v}_{i_2}$ replaced by $\mathbf{w}_2$ is again a spanning set for $\mathcal{V}$. Repeating this
lyche-numerical-linear-algebra-F00249	27	■ 1.000	$n - 2$	$n - 2$	process $n - 2$ more times we obtain a spanning set $\mathcal{X}_n$ where $\mathbf{v}_1, \dots, \mathbf{v}_n$ have been
lyche-numerical-linear-algebra-F00250	27	■ 1.000	$\mathcal{X}_n$	$\mathcal{X}_n$	process $n - 2$ more times we obtain a spanning set $\mathcal{X}_n$ where $\mathbf{v}_1, \dots, \mathbf{v}_n$ have been
lyche-numerical-linear-algebra-F00251	27	■ 1.000	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$	$\mathbf{w}_1, \dots, \mathbf{w}_n$	replaced by $\mathbf{w}_1, \dots, \mathbf{w}_n$. Since $k > n$ we can then write $\mathbf{w}_k$ as a linear combination
lyche-numerical-linear-algebra-F00252	27	■ 1.000	$\boldsymbol{w}_k$	$\mathbf{w}_k$	replaced by $\mathbf{w}_1, \dots, \mathbf{w}_n$. Since $k > n$ we can then write $\mathbf{w}_k$ as a linear combination
lyche-numerical-linear-algebra-F00253	27	—	$\square$	$\square$	—
lyche-numerical-linear-algebra-F00254	27	■ 1.000	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	$\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$	Definition 1.4 (Basis) A finite set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space $\mathcal{V}$ is a

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lyche-numerical-linear-algebra-F00255	27	■ 1.000	$\operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\} = \mathcal{V}$	$\operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\} = \mathcal{V}$	1. $\operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\} = \mathcal{V}$.
lyche-numerical-linear-algebra-F00256	27	■ 0.999	$\{\boldsymbol{v}_{i_1}, \ldots, \boldsymbol{v}_{i_k}\}$	$\{\boldsymbol{v}_{i_1}, \ldots, \boldsymbol{v}_{i_k}\}$	that $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ is a spanning set for $\mathcal{V}$. Then we can find a subset $\{\boldsymbol{v}_{i_1}, \ldots, \boldsymbol{v}_{i_k}\}$
lyche-numerical-linear-algebra-F00257	27	■ 1.000	$\mathcal{V} = \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$	$\mathcal{V} = \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$	Proof Let $\mathcal{V} = \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ be a finite dimensional vector space. By Theo-
lyche-numerical-linear-algebra-F00258	27	■ 1.000	$\operatorname{dim} \mathcal{V}$	$\dim \mathcal{V}$	space and denoted $\dim \mathcal{V}$.
lyche-numerical-linear-algebra-F00259	27	■ 1.000	$\mathcal{X} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$	$\mathcal{X} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$	Proof Suppose $\mathcal{X} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ and $\mathcal{Y} = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k\}$ are two bases for $\mathcal{V}$. By
lyche-numerical-linear-algebra-F00260	27	■ 1.000	$\mathcal{Y} = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k\}$	$\mathcal{Y} = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k\}$	Proof Suppose $\mathcal{X} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ and $\mathcal{Y} = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k\}$ are two bases for $\mathcal{V}$. By
lyche-numerical-linear-algebra-F00261	27	■ 1.000	$\mathcal{Y}$	$\mathcal{Y}$	Lemma 1.1 we have $k \leq n$. Using the same Lemma with $\mathcal{X}$ and $\mathcal{Y}$ switched we
lyche-numerical-linear-algebra-F00262	27	■ 1.000	$n \leq k$	$n \leq k$	obtain $n \leq k$. We conclude that $n = k$.
lyche-numerical-linear-algebra-F00263	27	■ 1.000	$n = k$	$n = k$	obtain $n \leq k$. We conclude that $n = k$.
lyche-numerical-linear-algebra-F00264	27	■ 1.000	$\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$	$\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$	The set of unit vectors $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$ form a basis for both $\mathbb{R}^n$ and $\mathbb{C}^n$.
lyche-numerical-linear-algebra-F00265	27	■ 0.996	$\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$	$\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$	vectors $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ in a finite dimensional vector space $\mathcal{V}$ can be enlarged to a
lyche-numerical-linear-algebra-F00266	28	■ 1.000	$\boldsymbol{v}_{k+1}$	$\boldsymbol{v}_{k+1}$	Proof If $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ does not span $\mathcal{V}$ we can enlarge the set by one vector $\boldsymbol{v}_{k+1}$
lyche-numerical-linear-algebra-F00267	28	■ 1.000	$\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$	$\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$	(V1) The sum $\boldsymbol{u} + \boldsymbol{v}$ is in $\mathcal{S}$ for any $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$.
lyche-numerical-linear-algebra-F00268	28	■ 1.000	$c \boldsymbol{u}$	$c\boldsymbol{u}$	(S1) The scalar multiple $c\boldsymbol{u}$ is in $\mathcal{S}$ for any scalar c and any $\boldsymbol{u} \in \mathcal{S}$.
lyche-numerical-linear-algebra-F00269	28	■ 1.000	$\boldsymbol{u} \in \mathcal{S}$	$\boldsymbol{u} \in \mathcal{S}$	(S1) The scalar multiple $c\boldsymbol{u}$ is in $\mathcal{S}$ for any scalar c and any $\boldsymbol{u} \in \mathcal{S}$.
lyche-numerical-linear-algebra-F00270	28	■ 1.000	$V1 - V5$	$V1 - V5$	axioms $V1 - V5$ and $S1 - S5$ are satisfied for $\mathcal{S}$. In particular, $\mathcal{S}$ must contain the
lyche-numerical-linear-algebra-F00271	28	■ 1.000	$S1 - S5$	$S1 - S5$	axioms $V1 - V5$ and $S1 - S5$ are satisfied for $\mathcal{S}$. In particular, $\mathcal{S}$ must contain the
lyche-numerical-linear-algebra-F00272	28	■ 1.000	$\{\mathbf{0}\}$	$\{\mathbf{0}\}$	1. $\{\mathbf{0}\}$, where $\mathbf{0}$ is the zero vector is a subspace, the trivial subspace . The dimension
lyche-numerical-linear-algebra-F00273	28	■ 1.000	$\mathcal{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\} \subseteq \mathcal{V}$	$\mathcal{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\} \subseteq \mathcal{V}$	3. $\operatorname{span}(\mathcal{X})$ is a subspace of $\mathcal{V}$ for any $\mathcal{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\} \subseteq \mathcal{V}$. Indeed, it is easy to
lyche-numerical-linear-algebra-F00274	28	■ 1.000	$\mathcal{S} \oplus \mathcal{T}$	$\mathcal{S} \oplus \mathcal{T}$	denoted $\mathcal{S} \oplus \mathcal{T}$ if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$.
lyche-numerical-linear-algebra-F00275	28	■ 1.000	$\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$	$\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$	denoted $\mathcal{S} \oplus \mathcal{T}$ if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$.
lyche-numerical-linear-algebra-F00276	29	■ 1.000	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$	Proof Let $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra-F00277	29	■ 1.000	$\mathcal{S} \cap \mathcal{T}$	$\mathcal{S} \cap \mathcal{T}$	Proof Let $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra-F00278	29	■ 1.000	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} = \emptyset$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} = \emptyset$	Proof Let $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra-F00279	29	■ 1.000	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q\}$ for $\mathcal{S}$ and a basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$ for $\mathcal{T}$. Every
lyche-numerical-linear-algebra-F00280	29	■ 1.000	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q\}$ for $\mathcal{S}$ and a basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$ for $\mathcal{T}$. Every
lyche-numerical-linear-algebra-F00281	29	■ 1.000	$\boldsymbol{x} \in \mathcal{S} + \mathcal{T}$	$\boldsymbol{x} \in \mathcal{S} + \mathcal{T}$	$\boldsymbol{x} \in \mathcal{S} + \mathcal{T}$ can be written as a linear combination of
lyche-numerical-linear-algebra-F00282	29	■ 1.000	$\mathcal{S} + \mathcal{T}$	$\mathcal{S} + \mathcal{T}$	so these vectors span $\mathcal{S} + \mathcal{T}$. We show that they are linearly independent and hence

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lyche-numerical-linear-algebra-F00283	29	■ 1.000	$\boldsymbol{u}+\boldsymbol{s}+\boldsymbol{t}=\mathbf{0}$	$\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$	a basis. Suppose $\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$, where $\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$, and
lyche-numerical-linear-algebra-F00284	29	■ 1.000	$\boldsymbol{u}:=\sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s}:=\sum_{j=1}^q \rho_j \boldsymbol{s}_j$	$\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j, \boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$	a basis. Suppose $\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$, where $\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$, and
lyche-numerical-linear-algebra-F00285	29	■ 1.000	$\boldsymbol{t}:=\sum_{j=1}^r \sigma_j \boldsymbol{t}_j$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra-F00286	29	■ 1.000	$\boldsymbol{s}=-\left(\boldsymbol{u}+\boldsymbol{t}\right)$	$\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra-F00287	29	■ 1.000	$\boldsymbol{s} \in$	$\boldsymbol{s} \in$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra-F00288	29	■ 1.000	$\boldsymbol{s}$	$\boldsymbol{s}$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra-F00289	29	■ 1.000	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra-F00290	29	■ 1.000	$\boldsymbol{s}:=$	$\boldsymbol{s} :=$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra-F00291	29	■ 1.000	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$. But then $\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$ and since
lyche-numerical-linear-algebra-F00292	29	■ 1.000	$\mathbf{0}=\sum_{j=1}^p \beta_j \boldsymbol{u}_j-\sum_{j=1}^q \rho_j \boldsymbol{s}_j$	$\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$. But then $\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$ and since
lyche-numerical-linear-algebra-F00293	29	■ 1.000	$\beta_1=\cdots=\beta_p=\rho_1=\cdots=\rho_q=0$	$\beta_1 = \cdots = \beta_p = \rho_1 = \cdots = \rho_q = 0$	is linearly independent we must have $\beta_1 = \cdots = \beta_p = \rho_1 = \cdots = \rho_q = 0$
lyche-numerical-linear-algebra-F00294	29	■ 1.000	$\boldsymbol{s}=\mathbf{0}$	$\boldsymbol{s} = \mathbf{0}$	and hence $\boldsymbol{s} = \mathbf{0}$. We then have $\boldsymbol{u} + \boldsymbol{t} = \mathbf{0}$ and by linear independence of
lyche-numerical-linear-algebra-F00295	29	■ 1.000	$\boldsymbol{u}+\boldsymbol{t}=\mathbf{0}$	$\boldsymbol{u} + \boldsymbol{t} = \mathbf{0}$	and hence $\boldsymbol{s} = \mathbf{0}$. We then have $\boldsymbol{u} + \boldsymbol{t} = \mathbf{0}$ and by linear independence of
lyche-numerical-linear-algebra-F00296	29	■ 1.000	$\alpha_1=\cdots=\alpha_p=\sigma_1=\cdots=\sigma_r=0$	$\alpha_1 = \cdots = \alpha_p = \sigma_1 = \cdots = \sigma_r = 0$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$ we obtain $\alpha_1 = \cdots = \alpha_p = \sigma_1 = \cdots = \sigma_r = 0$. We
lyche-numerical-linear-algebra-F00297	29	■ 1.000	$\left\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$	have shown that the vectors $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$ constitute a basis
lyche-numerical-linear-algebra-F00298	29	■ 1.000	$\operatorname{dim}\{\mathbf{0}\}=0$	$\dim\{\mathbf{0}\} = 0$	and (1.7) follows. Equation (1.7) implies (1.8) since $\dim\{\mathbf{0}\} = 0$.
lyche-numerical-linear-algebra-F00299	29	■ 1.000	$\left\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\right\}$	$\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$	n -dimensional vector space $\mathcal{V}$ and let $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$ be a basis for $\mathcal{S}$ and $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$
lyche-numerical-linear-algebra-F00300	29	■ 1.000	$\left\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\right\}$	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$	m -dimensional vector space $\mathcal{V}$ and let $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$ be a basis for $\mathcal{S}$ and $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$
lyche-numerical-linear-algebra-F00301	29	■ 1.000	$\boldsymbol{s}_j$	$\boldsymbol{s}_j$	a basis for $\mathcal{V}$. Then each $\boldsymbol{s}_j$ can be expressed as a linear combination of $\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$,
lyche-numerical-linear-algebra-F00302	29	■ 1.000	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$	a basis for $\mathcal{V}$. Then each $\boldsymbol{s}_j$ can be expressed as a linear combination of $\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$.
lyche-numerical-linear-algebra-F00303	30	■ 1.000	$\boldsymbol{x} \in \mathcal{S}$	$\boldsymbol{x} \in \mathcal{S}$	If $\boldsymbol{x} \in \mathcal{S}$ then $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some coefficients $\boldsymbol{b} :=$
lyche-numerical-linear-algebra-F00304	30	■ 1.000	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{s}_j=\sum_{i=1}^m b_i \boldsymbol{v}_i$	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$	If $\boldsymbol{x} \in \mathcal{S}$ then $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some coefficients $\boldsymbol{b} :=$
lyche-numerical-linear-algebra-F00305	30	■ 1.000	$\boldsymbol{b}:=$	$\boldsymbol{b} :=$	If $\boldsymbol{x} \in \mathcal{S}$ then $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some coefficients $\boldsymbol{b} :=$
lyche-numerical-linear-algebra-F00306	30	■ 1.000	$\left[\boldsymbol{b}_1, \dots, \boldsymbol{b}_m\right]^T, \boldsymbol{c}:=\left[c_1, \dots, c_n\right]^T$	$[b_1, \dots, b_m]^T, \boldsymbol{c} := [c_1, \dots, c_n]^T$	$[b_1, \dots, b_m]^T, \boldsymbol{c} := [c_1, \dots, c_n]^T$. Moreover $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$, where $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m \times n}$ is

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lyche-numerical-linear-algebra_F00307	30	1.000	$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{c}$	$\boldsymbol{b} = \boldsymbol{A} \boldsymbol{c}$	$[b_1, \dots, b_m]^T$, $\boldsymbol{c} := [c_1, \dots, c_n]^T$. Moreover $\boldsymbol{b} = \boldsymbol{A} \boldsymbol{c}$, where $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m \times n}$ is
lyche-numerical-linear-algebra_F00308	30	1.000	$\boldsymbol{A}=\left[a_{i j}\right] \in \mathbb{C}^{m \times n}$	$\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m \times n}$	$[b_1, \dots, b_m]^T$, $\boldsymbol{c} := [c_1, \dots, c_n]^T$. Moreover $\boldsymbol{b} = \boldsymbol{A} \boldsymbol{c}$, where $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m \times n}$ is
lyche-numerical-linear-algebra_F00309	30	0.999	$a_{i j}$	a_{ij}	Proof Equation (1.9) holds for some a_{ij} since $\boldsymbol{s}_j \in \mathcal{V}$ and $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$ spans $\mathcal{V}$.
lyche-numerical-linear-algebra_F00310	30	0.999	$\boldsymbol{s}_{j} \in \mathcal{V}$	$\boldsymbol{s}_j \in \mathcal{V}$	Proof Equation (1.9) holds for some a_{ij} since $\boldsymbol{s}_j \in \mathcal{V}$ and $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$ spans $\mathcal{V}$.
lyche-numerical-linear-algebra_F00311	30	0.980	$\left(c_{j}\right)$	(c_j)	be written $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some scalars (c_j) and (b_i) . But then
lyche-numerical-linear-algebra_F00312	30	0.980	$\left(b_{i}\right)$	(b_i)	be written $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some scalars (c_j) and (b_i) . But then
lyche-numerical-linear-algebra_F00313	30	1.000	$b_{i}=\sum_{j=1}^n a_{i j} c_{j}$	$b_i = \sum_{j=1}^n a_{ij} c_j$	Since $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$ is linearly independent it follows that $b_i = \sum_{j=1}^n a_{ij} c_j$ for $i =$
lyche-numerical-linear-algebra_F00314	30	1.000	$i=$	$i =$	Since $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$ is linearly independent it follows that $b_i = \sum_{j=1}^n a_{ij} c_j$ for $i =$
lyche-numerical-linear-algebra_F00315	30	1.000	$1, \ldots, m$	$1, \dots, m$	$1, \dots, m$ or $\boldsymbol{b} = \boldsymbol{A} \boldsymbol{c}$. Finally, to show that $\boldsymbol{A}$ has linearly independent columns
lyche-numerical-linear-algebra_F00316	30	1.000	$\boldsymbol{b}:=\boldsymbol{A} \boldsymbol{c}=\mathbf{0}$	$\boldsymbol{b} := \boldsymbol{A} \boldsymbol{c} = \mathbf{0}$	suppose $\boldsymbol{b} := \boldsymbol{A} \boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \dots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_F00317	30	1.000	$\boldsymbol{c}=\left[c_{1}, \ldots, c_{n}\right]^T$	$\boldsymbol{c} = [c_1, \dots, c_n]^T$	suppose $\boldsymbol{b} := \boldsymbol{A} \boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \dots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_F00318	30	1.000	$\boldsymbol{x}:=\sum_{j=1}^n c_{j} \boldsymbol{s}_{j}$	$\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$	suppose $\boldsymbol{b} := \boldsymbol{A} \boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \dots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_F00319	30	0.995	$\boldsymbol{x}=\sum_{i=1}^m b_{i} \boldsymbol{v}_{i}$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$ is linearly
lyche-numerical-linear-algebra_F00320	30	0.995	$\boldsymbol{b}=\mathbf{0}$	$\boldsymbol{b} = \mathbf{0}$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$ is linearly
lyche-numerical-linear-algebra_F00321	30	0.995	$\boldsymbol{x}=\mathbf{0}$	$\boldsymbol{x} = \mathbf{0}$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$ is linearly
lyche-numerical-linear-algebra_F00322	30	1.000	$\boldsymbol{c}=\mathbf{0}$	$\boldsymbol{c} = \mathbf{0}$	independent it follows that $\boldsymbol{c} = \mathbf{0}$.
lyche-numerical-linear-algebra_F00323	30	1.000	$\mathcal{V}=\mathbb{R}^m$	$\mathcal{V} = \mathbb{R}^m$	When $\mathcal{V} = \mathbb{R}^m$ or $\mathbb{C}^m$ we can think of n vectors in $\mathcal{V}$, say $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$, as a set
lyche-numerical-linear-algebra_F00324	30	1.000	$m \times n$	$m \times n$	$\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ or as the columns of an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n]$. A
lyche-numerical-linear-algebra_F00325	30	1.000	$\boldsymbol{X}=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right]$	$\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n]$	$\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ or as the columns of an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n]$. A
lyche-numerical-linear-algebra_F00326	30	0.998	$\boldsymbol{X} \boldsymbol{c}$	$\boldsymbol{X} \boldsymbol{c}$	linear combination can then be written as a matrix times vector $\boldsymbol{X} \boldsymbol{c}$, where $\boldsymbol{c} =$
lyche-numerical-linear-algebra_F00327	30	0.998	$\boldsymbol{c} =$	$\boldsymbol{c} =$	linear combination can then be written as a matrix times vector $\boldsymbol{X} \boldsymbol{c}$, where $\boldsymbol{c} =$
lyche-numerical-linear-algebra_F00328	30	1.000	$\left[c_{1}, \ldots, c_{n}\right]^T$	$[c_1, \dots, c_n]^T$	$[c_1, \dots, c_n]^T$ is the vector of scalars. Thus
lyche-numerical-linear-algebra_F00329	30	1.000	$\boldsymbol{x}_{j} \in \mathcal{V}, j=1, \ldots, n$	$\boldsymbol{x}_j \in \mathcal{V}, j = 1, \dots, n$	ated with an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n]$, where $\boldsymbol{x}_j \in \mathcal{V}, j = 1, \dots, n$ are the
lyche-numerical-linear-algebra_F00330	30	1.000	$\mathcal{R}(\boldsymbol{X})$	$\mathcal{R}(\boldsymbol{X})$	1. The subspace $\mathcal{R}(\boldsymbol{X})$ is called the column space of $\boldsymbol{X}$. It is the smallest subspace
lyche-numerical-linear-algebra_F00331	30	1.000	$\boldsymbol{X}$	$\boldsymbol{X}$	1. The subspace $\mathcal{R}(\boldsymbol{X})$ is called the column space of $\boldsymbol{X}$. It is the smallest subspace
lyche-numerical-linear-algebra_F00332	30	1.000	$\mathcal{R}\left(\boldsymbol{X}^T\right)$	$\mathcal{R}\left(\boldsymbol{X}^T\right)$	2. $\mathcal{R}(\boldsymbol{X}^T)$ is called the row space of $\boldsymbol{X}$. It is generated by the rows of $\boldsymbol{X}$ written as

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lyche-numerical-linear-algebra-F00333	30	1.000	$\mathcal{N}(\boldsymbol{X}) := \left\{ \boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{X} \boldsymbol{y} = \mathbf{0} \right\}$	$\mathcal{N}(\boldsymbol{X}) := \{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{X} \boldsymbol{y} = \mathbf{0}\}$	3. The subspace $\mathcal{N}(\boldsymbol{X}) := \{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{X} \boldsymbol{y} = \mathbf{0}\}$ is called the null space or kernel
lyche-numerical-linear-algebra-F00334	30	1.000	$\mathcal{N}(\boldsymbol{X})$	$\mathcal{N}(\boldsymbol{X})$	space of $\boldsymbol{X}$. The dimension of $\mathcal{N}(\boldsymbol{X})$ is called the nullity of $\boldsymbol{X}$ and denoted
lyche-numerical-linear-algebra-F00335	30	1.000	$\operatorname{null}(\boldsymbol{X})$	$\operatorname{null}(\boldsymbol{X})$	$\operatorname{null}(\boldsymbol{X})$.
lyche-numerical-linear-algebra-F00336	31	0.922	$\boldsymbol{X} \in$	$\boldsymbol{X} \in$	Theorem 1.5 (Counting Dimensions of Fundamental Subspaces) Suppose $\boldsymbol{X} \in$
lyche-numerical-linear-algebra-F00337	31	1.000	$\mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times n}$. Then
lyche-numerical-linear-algebra-F00338	31	1.000	$\operatorname{rank}(\boldsymbol{X}) = \operatorname{rank}(\boldsymbol{X}^*)$	$\operatorname{rank}(\boldsymbol{X}) = \operatorname{rank}(\boldsymbol{X}^*)$	1. $\operatorname{rank}(\boldsymbol{X}) = \operatorname{rank}(\boldsymbol{X}^*)$.
lyche-numerical-linear-algebra-F00339	31	1.000	$\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}) = n$	$\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}) = n$	2. $\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}) = n$.
lyche-numerical-linear-algebra-F00340	31	1.000	$\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}^*) = m$	$\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}^*) = m$	3. $\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}^*) = m$.
lyche-numerical-linear-algebra-F00341	31	1.000	x_j	x_j	x_j , and the components b_i of the right hand side are real or complex numbers. The
lyche-numerical-linear-algebra-F00342	31	1.000	b_i	b_i	x_j , and the components b_i of the right hand side are real or complex numbers. The
lyche-numerical-linear-algebra-F00343	32	0.992	$\boldsymbol{a}_j = [a_{1j}, \dots, a_{mj}]^T \in \mathbb{C}^m$	$\boldsymbol{a}_j = [a_{1j}, \dots, a_{mj}]^T \in \mathbb{C}^m$	where $\boldsymbol{a}_j = [a_{1j}, \dots, a_{mj}]^T \in \mathbb{C}^m$ for $j = 1, \dots, n$ and $\boldsymbol{b} = [b_1, \dots, b_m]^T \in \mathbb{C}^m$.
lyche-numerical-linear-algebra-F00344	32	0.992	$\boldsymbol{b} = [b_1, \dots, b_m]^T \in \mathbb{C}^m$	$\boldsymbol{b} = [b_1, \dots, b_m]^T \in \mathbb{C}^m$	where $\boldsymbol{a}_j = [a_{1j}, \dots, a_{mj}]^T \in \mathbb{C}^m$ for $j = 1, \dots, n$ and $\boldsymbol{b} = [b_1, \dots, b_m]^T \in \mathbb{C}^m$.
lyche-numerical-linear-algebra-F00345	32	1.000	$m < n, m = n$	$m < n, m = n$	square , or overdetermined if $m < n$, $m = n$, or $m > n$, respectively.
lyche-numerical-linear-algebra-F00346	32	1.000	$m > n$	$m > n$	square , or overdetermined if $m < n$, $m = n$, or $m > n$, respectively.
lyche-numerical-linear-algebra-F00347	32	1.000	$m < n$	$m < n$	Lemma 1.3 (Underdetermined System) Suppose $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ with $m < n$. Then
lyche-numerical-linear-algebra-F00348	32	0.570	$\boldsymbol{A} \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} = \mathbf{0}$	there is a nonzero $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{A} \boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra-F00349	32	1.000	$\boldsymbol{A} \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} = \mathbf{0}$	nonzero $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{A} \boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra-F00350	32	1.000	$\mathbb{R}^{n \times n}$	$\mathbb{R}^{n \times n}$	$\mathbb{R}^{n \times n}$ is said to be nonsingular if the only real solution of the homogeneous system
lyche-numerical-linear-algebra-F00351	32	0.658	$\boldsymbol{b} \in \mathbb{R}^n$	$\boldsymbol{b} \in \mathbb{R}^n$	The linear system $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ has a unique solution $\boldsymbol{x} \in \mathbb{R}^n$ for any $\boldsymbol{b} \in \mathbb{R}^n$ if and only
lyche-numerical-linear-algebra-F00352	32	1.000	$\boldsymbol{B} = [\boldsymbol{A} \boldsymbol{b}] \in \mathbb{R}^{n \times (n+1)}$	$\boldsymbol{B} = [\boldsymbol{A} \boldsymbol{b}] \in \mathbb{R}^{n \times (n+1)}$	Proof Suppose $\boldsymbol{A}$ is nonsingular. We define $\boldsymbol{B} = [\boldsymbol{A} \boldsymbol{b}] \in \mathbb{R}^{n \times (n+1)}$ by adding a
lyche-numerical-linear-algebra-F00353	32	0.997	$\boldsymbol{z} \in \mathbb{R}^{n+1}$	$\boldsymbol{z} \in \mathbb{R}^{n+1}$	column to $\boldsymbol{A}$. By Lemma 1.3 there is a nonzero $\boldsymbol{z} \in \mathbb{R}^{n+1}$ such that $\boldsymbol{B} \boldsymbol{z} = \mathbf{0}$. If we
lyche-numerical-linear-algebra-F00354	32	0.997	$\boldsymbol{B} \boldsymbol{z} = \mathbf{0}$	$\boldsymbol{B} \boldsymbol{z} = \mathbf{0}$	column to $\boldsymbol{A}$. By Lemma 1.3 there is a nonzero $\boldsymbol{z} \in \mathbb{R}^{n+1}$ such that $\boldsymbol{B} \boldsymbol{z} = \mathbf{0}$. If we
lyche-numerical-linear-algebra-F00355	32	0.896	$\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$	$\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$	write $\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$ where $\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then
lyche-numerical-linear-algebra-F00356	32	0.896	$\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$	$\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$	write $\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$ where $\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then
lyche-numerical-linear-algebra-F00357	32	0.896	$z_{n+1} \in \mathbb{R}$	$z_{n+1} \in \mathbb{R}$	write $\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$ where $\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then

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lyche-numerical-linear-algebra-F00358	33	0.737	$z_{n+1}=0$	$z_{n+1} = 0$	We cannot have $z_{n+1} = 0$ for then $A\tilde{z} = \mathbf{0}$ for a nonzero $\tilde{z}$, contradicting the
lyche-numerical-linear-algebra-F00359	33	0.737	$\boldsymbol{A} \ \tilde{\boldsymbol{z}}=\mathbf{0}$	$A\tilde{z} = \mathbf{0}$	We cannot have $z_{n+1} = 0$ for then $A\tilde{z} = \mathbf{0}$ for a nonzero $\tilde{z}$, contradicting the
lyche-numerical-linear-algebra-F00360	33	0.737	$\tilde{\boldsymbol{z}}$	$\tilde{z}$	We cannot have $z_{n+1} = 0$ for then $A\tilde{z} = \mathbf{0}$ for a nonzero $\tilde{z}$, contradicting the
lyche-numerical-linear-algebra-F00361	33	0.997	$\boldsymbol{x}:-\tilde{\boldsymbol{z}}/z_{n+1}$	$x := -\tilde{z}/z_{n+1}$	nonsingularity of A . Define $x := -\tilde{z}/z_{n+1}$. Then
lyche-numerical-linear-algebra-F00362	33	1.000	$\boldsymbol{A} \ \boldsymbol{y}=\boldsymbol{b}$	$Ay = b$	Suppose $Ax = b$ and $Ay = b$ for $x, y \in \mathbb{R}^n$. Then $A(x - y) = \mathbf{0}$ and since A is
lyche-numerical-linear-algebra-F00363	33	1.000	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$	$x, y \in \mathbb{R}^n$	Suppose $Ax = b$ and $Ay = b$ for $x, y \in \mathbb{R}^n$. Then $A(x - y) = \mathbf{0}$ and since A is
lyche-numerical-linear-algebra-F00364	33	1.000	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\mathbf{0}$	$A(x - y) = \mathbf{0}$	Suppose $Ax = b$ and $Ay = b$ for $x, y \in \mathbb{R}^n$. Then $A(x - y) = \mathbf{0}$ and since A is
lyche-numerical-linear-algebra-F00365	33	1.000	$\boldsymbol{x}-\boldsymbol{y}=\mathbf{0}$	$x - y = \mathbf{0}$	nonsingular we conclude that $x - y = \mathbf{0}$ or $x = y$. Thus the solution is unique.
lyche-numerical-linear-algebra-F00366	33	1.000	$\boldsymbol{x}=\boldsymbol{y}$	$x = y$	nonsingular we conclude that $x - y = \mathbf{0}$ or $x = y$. Thus the solution is unique.
lyche-numerical-linear-algebra-F00367	33	1.000	$m<$	$m <$	Lemma 1.4 (Complex Underdetermined System) Suppose $A \in \mathbb{C}^{m \times n}$ with $m <$
lyche-numerical-linear-algebra-F00368	33	1.000	$\boldsymbol{b} \in \mathbb{C}^n$	$b \in \mathbb{C}^n$	$b \in \mathbb{C}^n$ if and only if the matrix A is nonsingular.
lyche-numerical-linear-algebra-F00369	33	1.000	$\boldsymbol{B} \in \mathbb{C}^{n \times n}$	$B \in \mathbb{C}^{n \times n}$	Suppose $A \in \mathbb{C}^{n \times n}$ is a square matrix. A matrix $B \in \mathbb{C}^{n \times n}$ is called a right inverse
lyche-numerical-linear-algebra-F00370	33	1.000	$\boldsymbol{A} \ \boldsymbol{B}=\boldsymbol{I}$	$AB = I$	of A if $AB = I$. A matrix $C \in \mathbb{C}^{n \times n}$ is said to be a left inverse of A if $CA = I$.
lyche-numerical-linear-algebra-F00371	33	1.000	$\boldsymbol{C} \in \mathbb{C}^{n \times n}$	$C \in \mathbb{C}^{n \times n}$	of A if $AB = I$. A matrix $C \in \mathbb{C}^{n \times n}$ is said to be a left inverse of A if $CA = I$.
lyche-numerical-linear-algebra-F00372	33	1.000	$\boldsymbol{C} \ \boldsymbol{A}=\boldsymbol{I}$	$CA = I$	of A if $AB = I$. A matrix $C \in \mathbb{C}^{n \times n}$ is said to be a left inverse of A if $CA = I$.
lyche-numerical-linear-algebra-F00373	33	1.000	$\boldsymbol{C}$	C	inverse B and a left inverse C then
lyche-numerical-linear-algebra-F00374	33	1.000	$\boldsymbol{A}^{-1}$	A^{-1}	and this common inverse is called the inverse of A and denoted by A^{-1} . Thus the
lyche-numerical-linear-algebra-F00375	33	1.000	$\boldsymbol{A}^{-1} \ \boldsymbol{A}=\boldsymbol{A} \ \boldsymbol{A}^{-1}=\boldsymbol{I}$	$A^{-1}A = AA^{-1} = I$	inverse satisfies $A^{-1}A = AA^{-1} = I$.
lyche-numerical-linear-algebra-F00376	33	0.998	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	$A, B, C \in \mathbb{C}^{n \times n}$	Theorem 1.8 (Product of Nonsingular Matrices) If $A, B, C \in \mathbb{C}^{n \times n}$ with $AB =$
lyche-numerical-linear-algebra-F00377	33	0.998	$\boldsymbol{A} \ \boldsymbol{B}=\boldsymbol{A}$	$AB =$	Theorem 1.8 (Product of Nonsingular Matrices) If $A, B, C \in \mathbb{C}^{n \times n}$ with $AB =$
lyche-numerical-linear-algebra-F00378	33	1.000	$\boldsymbol{B} \ \boldsymbol{A}=\boldsymbol{I}$	$BA = I$	if either $AB = I$ or $BA = I$ then A is nonsingular and $A^{-1} = B$.
lyche-numerical-linear-algebra-F00379	33	1.000	$\boldsymbol{A}^{-1}=\boldsymbol{B}$	$A^{-1} = B$	if either $AB = I$ or $BA = I$ then A is nonsingular and $A^{-1} = B$.
lyche-numerical-linear-algebra-F00380	33	1.000	$\boldsymbol{C} \ \boldsymbol{x}=\mathbf{0}$	$Cx = \mathbf{0}$	Proof Suppose both A and B are nonsingular and let $Cx = \mathbf{0}$. Then $ABx = \mathbf{0}$ and
lyche-numerical-linear-algebra-F00381	33	1.000	$\boldsymbol{A} \ \boldsymbol{B} \ \boldsymbol{x}=\mathbf{0}$	$ABx = \mathbf{0}$	Proof Suppose both A and B are nonsingular and let $Cx = \mathbf{0}$. Then $ABx = \mathbf{0}$ and
lyche-numerical-linear-algebra-F00382	33	1.000	$\boldsymbol{B} \ \boldsymbol{x}=\mathbf{0}$	$Bx = \mathbf{0}$	since A is nonsingular we see that $Bx = \mathbf{0}$. Since B is nonsingular we have $x = \mathbf{0}$.
lyche-numerical-linear-algebra-F00383	34	1.000	$\boldsymbol{C} \ \boldsymbol{x}=(\boldsymbol{A} \ \boldsymbol{B}) \ \boldsymbol{x}=\boldsymbol{A} \ (\boldsymbol{B} \ \boldsymbol{x})=\boldsymbol{A} \ \mathbf{0}=\mathbf{0}$	$Cx = (AB)x = A(Bx) = A\mathbf{0} = \mathbf{0}$	vector so that $Bx = \mathbf{0}$. But then $Cx = (AB)x = A(Bx) = A\mathbf{0} = \mathbf{0}$ so C is

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lyche-numerical-linear-algebra_F00384	34	■ 1.000	$\tilde{\boldsymbol{x}}$	$\tilde{x}$	singular. Finally suppose B is nonsingular, but A is singular. Let $\tilde{x}$ be a nonzero
lyche-numerical-linear-algebra_F00385	34	■ 1.000	$\boldsymbol{A} \tilde{\boldsymbol{x}} = \mathbf{0}$	$A\tilde{x} = 0$	vector such that $A\tilde{x} = 0$. By Theorem 1.7 there is a vector x such that $Bx = \tilde{x}$ and
lyche-numerical-linear-algebra_F00386	34	■ 1.000	$\boldsymbol{B} \boldsymbol{x} = \tilde{\boldsymbol{x}}$	$Bx = \tilde{x}$	vector such that $A\tilde{x} = 0$. By Theorem 1.7 there is a vector x such that $Bx = \tilde{x}$ and
lyche-numerical-linear-algebra_F00387	34	■ 1.000	$\boldsymbol{C} \boldsymbol{x} = (\boldsymbol{A} \boldsymbol{B}) \boldsymbol{x} = \boldsymbol{A} (\boldsymbol{B} \boldsymbol{x}) = \boldsymbol{A} \tilde{\boldsymbol{x}} = \mathbf{0}$	$Cx = (AB)x = A(Bx) = A\tilde{x} = 0$	x is nonzero since $\tilde{x}$ is nonzero. But then $Cx = (AB)x = A(Bx) = Ax = 0$ for a
lyche-numerical-linear-algebra_F00388	34	■ 1.000	$\boldsymbol{A} \boldsymbol{b}_i = \boldsymbol{e}_i$	$Ab_i = e_i$	systems $Ab_i = e_i$ has a unique solution b_i for $i = 1, \dots, n$. Let $B = [b_1, \dots, b_n]$.
lyche-numerical-linear-algebra_F00389	34	■ 1.000	$\boldsymbol{b}_i$	b_i	systems $Ab_i = e_i$ has a unique solution b_i for $i = 1, \dots, n$. Let $B = [b_1, \dots, b_n]$.
lyche-numerical-linear-algebra_F00390	34	■ 1.000	$\boldsymbol{B} = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$	$B = [b_1, \dots, b_n]$	systems $Ab_i = e_i$ has a unique solution b_i for $i = 1, \dots, n$. Let $B = [b_1, \dots, b_n]$.
lyche-numerical-linear-algebra_F00391	34	■ 0.968	$\boldsymbol{A} \boldsymbol{B} = [\boldsymbol{A} \boldsymbol{b}_1, \dots, \boldsymbol{A} \boldsymbol{b}_n] = [\boldsymbol{e}_1, \dots, \boldsymbol{e}_n] = \boldsymbol{I}$	$AB = [Ab_1, \dots, Ab_n] = [e_1, \dots, e_n] = I$	Then $AB = [Ab_1, \dots, Ab_n] = [e_1, \dots, e_n] = I$ so that A has a right inverse B .
lyche-numerical-linear-algebra_F00392	34	■ 1.000	1.8 $\boldsymbol{B}$	$1.8B$	By Theorem 1.8 B is nonsingular since I is nonsingular and $AB = I$. Since B is
lyche-numerical-linear-algebra_F00393	34	■ 1.000	$\boldsymbol{B} \boldsymbol{C} = \boldsymbol{I}$	$BC = I$	inverse C , i.e. $BC = I$. But then $AB = BC = I$ so B has both a right inverse and
lyche-numerical-linear-algebra_F00394	34	■ 1.000	$\boldsymbol{A} \boldsymbol{B} = \boldsymbol{B} \boldsymbol{C} = \boldsymbol{I}$	$AB = BC = I$	inverse C , i.e. $BC = I$. But then $AB = BC = I$ so B has both a right inverse and
lyche-numerical-linear-algebra_F00395	34	■ 1.000	$\boldsymbol{A} = \boldsymbol{C}$	$A = C$	a left inverse which must be equal so $A = C$. Since $BC = I$ we have $BA = I$, so
lyche-numerical-linear-algebra_F00396	34	■ 0.995	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$	$A, B \in \mathbb{C}^{n \times n}$	Corollary 1.2 (Basic Properties of the Inverse Matrix) Suppose $A, B \in \mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra_F00397	34	■ 1.000	$(\boldsymbol{A}^{-1})^{-1} = \boldsymbol{A}$	$(A^{-1})^{-1} = A$	1. A^{-1} is nonsingular and $(A^{-1})^{-1} = A$.
lyche-numerical-linear-algebra_F00398	34	■ 1.000	$\boldsymbol{C} = \boldsymbol{A} \boldsymbol{B}$	$C = AB$	2. $C = AB$ is nonsingular and $C^{-1} = B^{-1}A^{-1}$.
lyche-numerical-linear-algebra_F00399	34	■ 1.000	$\boldsymbol{C}^{-1} = \boldsymbol{B}^{-1} \boldsymbol{A}^{-1}$	$C^{-1} = B^{-1}A^{-1}$	2. $C = AB$ is nonsingular and $C^{-1} = B^{-1}A^{-1}$.
lyche-numerical-linear-algebra_F00400	34	■ 1.000	$(\boldsymbol{A}^T)^{-1} = (\boldsymbol{A}^{-1})^T =: \boldsymbol{A}^{-T}$	$(A^T)^{-1} = (A^{-1})^T =: A^{-T}$	3. A^T is nonsingular and $(A^T)^{-1} = (A^{-1})^T =: A^{-T}$.
lyche-numerical-linear-algebra_F00401	34	■ 1.000	$(\boldsymbol{A}^*)^{-1} = (\boldsymbol{A}^{-1})^* =: \boldsymbol{A}^{-*}$	$(A^*)^{-1} = (A^{-1})^* =: A^{-*}$	4. A^* is nonsingular and $(A^*)^{-1} = (A^{-1})^* =: A^{-*}$.
lyche-numerical-linear-algebra_F00402	34	■ 1.000	$c \boldsymbol{A}$	cA	5. cA is nonsingular and $(cA)^{-1} = \frac{1}{c}A^{-1}$.
lyche-numerical-linear-algebra_F00403	34	■ 1.000	$(c \boldsymbol{A})^{-1} = \frac{1}{c} \boldsymbol{A}^{-1}$	$(cA)^{-1} = \frac{1}{c}A^{-1}$	5. cA is nonsingular and $(cA)^{-1} = \frac{1}{c}A^{-1}$.
lyche-numerical-linear-algebra_F00404	34	■ 1.000	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{I}$	$A^{-1}A = I$	1. Since $A^{-1}A = I$ the matrix A is a right inverse of A^{-1} . Thus A^{-1} is nonsingular
lyche-numerical-linear-algebra_F00405	34	■ 1.000	$(\boldsymbol{B}^{-1} \boldsymbol{A}^{-1}) (\boldsymbol{A} \boldsymbol{B}) = \boldsymbol{B}^{-1} (\boldsymbol{A}^{-1} \boldsymbol{A}) \boldsymbol{B} = \boldsymbol{B}^{-1} \boldsymbol{B} = \boldsymbol{I}$	$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$	2. We note that $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$. Thus AB is
lyche-numerical-linear-algebra_F00406	34	■ 1.000	$\boldsymbol{A} \boldsymbol{B}$	AB	2. We note that $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$. Thus AB is

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lyche-numerical-linear-algebra_F00407	34	1.000	$\boldsymbol{I}=\boldsymbol{I}^T=\left(\boldsymbol{A}^{-1}\boldsymbol{A}\right)^T=\boldsymbol{A}^T\left(\boldsymbol{A}^{-1}\right)^T$	$\boldsymbol{I}=\boldsymbol{I}^T=\left(\boldsymbol{A}^{-1}\boldsymbol{A}\right)^T=\boldsymbol{A}^T\left(\boldsymbol{A}^{-1}\right)^T$	3. Now $\boldsymbol{I}=\boldsymbol{I}^T=\left(\boldsymbol{A}^{-1}\boldsymbol{A}\right)^T=\boldsymbol{A}^T\left(\boldsymbol{A}^{-1}\right)^T$ showing that $\left(\boldsymbol{A}^{-1}\right)^T$ is a right inverse
lyche-numerical-linear-algebra_F00408	34	1.000	$\left(\boldsymbol{A}^{-1}\right)^T$	$\left(\boldsymbol{A}^{-1}\right)^T$	3. Now $\boldsymbol{I}=\boldsymbol{I}^T=\left(\boldsymbol{A}^{-1}\boldsymbol{A}\right)^T=\boldsymbol{A}^T\left(\boldsymbol{A}^{-1}\right)^T$ showing that $\left(\boldsymbol{A}^{-1}\right)^T$ is a right inverse
lyche-numerical-linear-algebra_F00409	34	1.000	$\frac{1}{c}\boldsymbol{A}^{-1}$	$\frac{1}{c}\boldsymbol{A}^{-1}$	4. The matrix $\frac{1}{c}\boldsymbol{A}^{-1}$ is a one sided inverse of $c\boldsymbol{A}$.
lyche-numerical-linear-algebra_F00410	35	1.000	$n!$	$n!$	This sum ranges of all $n!$ permutations of $\{1,2,\dots,n\}$. Moreover, $\text{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00411	35	1.000	$\{1,2,\ldots,n\}$	$\{1,2,\dots,n\}$	This sum ranges of all $n!$ permutations of $\{1,2,\dots,n\}$. Moreover, $\text{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00412	35	1.000	$\operatorname{sign}(\sigma)$	$\text{sign}(\sigma)$	This sum ranges of all $n!$ permutations of $\{1,2,\dots,n\}$. Moreover, $\text{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00413	35	1.000	σ	σ	the number of times a bigger integer precedes a smaller one in σ . We also denote
lyche-numerical-linear-algebra_F00414	35	1.000	ϵ	ϵ	The first term on the right corresponds to the identity permutation ϵ given by $\epsilon(i)=$
lyche-numerical-linear-algebra_F00415	35	1.000	$\epsilon(i)=$	$\epsilon(i)=$	The first term on the right corresponds to the identity permutation ϵ given by $\epsilon(i)=$
lyche-numerical-linear-algebra_F00416	35	1.000	$i, i=1,2$	$i, i=1,2$	$i, i=1,2$. The second term comes from the permutation $\sigma=\{2,1\}$. For $n=3$
lyche-numerical-linear-algebra_F00417	35	1.000	$\sigma=\{2,1\}$	$\sigma=\{2,1\}$	$i, i=1,2$. The second term comes from the permutation $\sigma=\{2,1\}$. For $n=3$
lyche-numerical-linear-algebra_F00418	35	1.000	$n=3$	$n=3$	$i, i=1,2$. The second term comes from the permutation $\sigma=\{2,1\}$. For $n=3$
lyche-numerical-linear-algebra_F00419	35	0.719	$\{1,2,3\}$	$\{1,2,3\}$	there are six permutations of $\{1,2,3\}$. Then
lyche-numerical-linear-algebra_F00420	35	0.999	$\operatorname{sign}(\{1,2,3\})=\operatorname{sign}(\{2,3,1\})=\operatorname{sign}(\{3,1,2\})=1$	$\text{sign}(\{1,2,3\})=\text{sign}(\{2,3,1\})=\text{sign}(\{3,1,2\})=1$	This follows since $\text{sign}(\{1,2,3\})=\text{sign}(\{2,3,1\})=\text{sign}(\{3,1,2\})=1$, and
lyche-numerical-linear-algebra_F00421	35	0.749	$\operatorname{sign}(\{2,1,3\})=\operatorname{sign}(\{3,2,1\})=\operatorname{sign}(\{1,3,2\})=-1$	$\text{sign}(\{2,1,3\})=\text{sign}(\{3,2,1\})=\text{sign}(\{1,3,2\})=-1$	$\text{sign}(\{2,1,3\})=\text{sign}(\{3,2,1\})=\text{sign}(\{1,3,2\})=-1$.
lyche-numerical-linear-algebra_F00422	35	1.000	$\det(\boldsymbol{A})=$	$\det(\boldsymbol{A})=$	to reduce it to a simpler form. For example, if $\boldsymbol{A}$ is triangular then $\det(\boldsymbol{A})=$
lyche-numerical-linear-algebra_F00423	36	1.000	$a_{11}a_{22}\cdots a_{nn}$	$a_{11}a_{22}\cdots a_{nn}$	$a_{11}a_{22}\cdots a_{nn}$, the product of the diagonal elements. In particular, for the identity
lyche-numerical-linear-algebra_F00424	36	1.000	$\det(\boldsymbol{I})=1$	$\det(\boldsymbol{I})=1$	matrix $\det(\boldsymbol{I})=1$. The elementary operations using either rows or columns are
lyche-numerical-linear-algebra_F00425	36	1.000	$\det(\boldsymbol{B})=-\det(\boldsymbol{A})$	$\det(\boldsymbol{B})=-\det(\boldsymbol{A})$	1. Interchanging two rows(columns): $\det(\boldsymbol{B})=-\det(\boldsymbol{A})$,
lyche-numerical-linear-algebra_F00426	36	1.000	$\alpha,\det(\boldsymbol{B})=\alpha\det(\boldsymbol{A})$	$\alpha,\det(\boldsymbol{B})=\alpha\det(\boldsymbol{A})$	2. Multiply a row(column) by a scalar: $\alpha,\det(\boldsymbol{B})=\alpha\det(\boldsymbol{A})$,
lyche-numerical-linear-algebra_F00427	36	1.000	$\boldsymbol{A}_{i,j}$	$\boldsymbol{A}_{i,j}$	Here $\boldsymbol{A}_{i,j}$ denotes the submatrix of $\boldsymbol{A}$ obtained by deleting the i th row and j th
lyche-numerical-linear-algebra_F00428	36	1.000	$1\leq i,j\leq n$	$1\leq i,j\leq n$	column of $\boldsymbol{A}$. For $\boldsymbol{A}\in\mathbb{C}^{n\times n}$ and $1\leq i,j\leq n$ the determinant $\det(\boldsymbol{A}_{ij})$ is called the
lyche-numerical-linear-algebra_F00429	36	1.000	$\det(\boldsymbol{A}_{ij})$	$\det(\boldsymbol{A}_{ij})$	column of $\boldsymbol{A}$. For $\boldsymbol{A}\in\mathbb{C}^{n\times n}$ and $1\leq i,j\leq n$ the determinant $\det(\boldsymbol{A}_{ij})$ is called the
lyche-numerical-linear-algebra_F00430	36	0.969	(x_1,y_1)	(x_1,y_1)	line through two points (x_1,y_1) and (x_2,y_2) in the plane can be written as the
lyche-numerical-linear-algebra_F00431	36	0.969	(x_2,y_2)	(x_2,y_2)	line through two points (x_1,y_1) and (x_2,y_2) in the plane can be written as the
lyche-numerical-linear-algebra_F00432	36	1.000	$\det(\boldsymbol{A})=0$	$\det(\boldsymbol{A})=0$	Rearranging the equation $\det(\boldsymbol{A})=0$ we obtain
lyche-numerical-linear-algebra_F00433	37	1.000	$\boldsymbol{A},\boldsymbol{B}$	$\boldsymbol{A},\boldsymbol{B}$	$\boldsymbol{A},\boldsymbol{B}$ are square matrices of order n with real or complex elements, then

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lyche-numerical-linear-algebra-F00434	37	■ 0.984	$\operatorname{det}(\boldsymbol{A} \boldsymbol{B})=\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{B})$	$\det(\boldsymbol{A}\boldsymbol{B}) = \det(\boldsymbol{A}) \det(\boldsymbol{B})$	1. $\det(\boldsymbol{A}\boldsymbol{B}) = \det(\boldsymbol{A}) \det(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F00435	37	■ 1.000	$\operatorname{det}\left(\boldsymbol{A}^T\right)=\operatorname{det}(\boldsymbol{A})$	$\det\left(\boldsymbol{A}^T\right) = \det(\boldsymbol{A})$	2. $\det(\boldsymbol{A}^T) = \det(\boldsymbol{A})$, and $\det(\boldsymbol{A}^*) = \overline{\det(\boldsymbol{A})}$, (complex conjugate).
lyche-numerical-linear-algebra-F00436	37	■ 1.000	$\operatorname{det}\left(\boldsymbol{A}^*\right)=\overline{\operatorname{det}(\boldsymbol{A})}$	$\det\left(\boldsymbol{A}^*\right) = \overline{\det(\boldsymbol{A})}$	2. $\det(\boldsymbol{A}^T) = \det(\boldsymbol{A})$, and $\det(\boldsymbol{A}^*) = \overline{\det(\boldsymbol{A})}$, (complex conjugate).
lyche-numerical-linear-algebra-F00437	37	■ 1.000	$\operatorname{det}(a \boldsymbol{A})=a^n \operatorname{det}(\boldsymbol{A})$	$\det(a\boldsymbol{A}) = a^n \det(\boldsymbol{A})$	3. $\det(a\boldsymbol{A}) = a^n \det(\boldsymbol{A})$, for $a \in \mathbb{C}$.
lyche-numerical-linear-algebra-F00438	37	■ 0.943	$\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E}\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E} \end{bmatrix}$	5. If $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E} \end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$.
lyche-numerical-linear-algebra-F00439	37	■ 0.943	$\boldsymbol{C}, \boldsymbol{E}$	$\boldsymbol{C}, \boldsymbol{E}$	5. If $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E} \end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$.
lyche-numerical-linear-algebra-F00440	37	■ 0.943	$\operatorname{det}(\boldsymbol{A})=\operatorname{det}(\boldsymbol{C}) \operatorname{det}(\boldsymbol{E})$	$\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$	5. If $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E} \end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$.
lyche-numerical-linear-algebra-F00441	37	■ 1.000	$\boldsymbol{x} =$	$\boldsymbol{x} =$	6. Cramer's rule Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is nonsingular and $\boldsymbol{b} \in \mathbb{C}^n$. Let $\boldsymbol{x} =$
lyche-numerical-linear-algebra-F00442	37	■ 0.982	$\left[x_1, x_2, \ldots, x_n\right]^T$	$[x_1, x_2, \dots, x_n]^T$	$[x_1, x_2, \dots, x_n]^T$ be the unique solution of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. Then
lyche-numerical-linear-algebra-F00443	37	■ 1.000	$\boldsymbol{A}_j(\boldsymbol{b})$	$\boldsymbol{A}_j(\boldsymbol{b})$	where $\boldsymbol{A}_j(\boldsymbol{b})$ denote the matrix obtained from $\boldsymbol{A}$ by replacing the j th column of
lyche-numerical-linear-algebra-F00444	37	■ 0.999	$\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	$\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	where the matrix $\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$ with elements $\operatorname{adj}(\boldsymbol{A})_{i, j} = (-1)^{i+j} \det(\boldsymbol{A}_{j, i})$
lyche-numerical-linear-algebra-F00445	37	■ 0.999	$\operatorname{adj}(\boldsymbol{A})_{i, j}=(-1)^{i+j} \det\left(\boldsymbol{A}_{j, i}\right)$	$\operatorname{adj}(\boldsymbol{A})_{i, j} = (-1)^{i+j} \det\left(\boldsymbol{A}_{j, i}\right)$	where the matrix $\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$ with elements $\operatorname{adj}(\boldsymbol{A})_{i, j} = (-1)^{i+j} \det(\boldsymbol{A}_{j, i})$
lyche-numerical-linear-algebra-F00446	37	■ 1.000	$\boldsymbol{A}_{j, i}$	$\boldsymbol{A}_{j, i}$	is called the adjoint of $\boldsymbol{A}$. Moreover, $\boldsymbol{A}_{j, i}$ denotes the submatrix of $\boldsymbol{A}$ obtained
lyche-numerical-linear-algebra-F00447	37	■ 1.000	$\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}$	$\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}$	8. Cauchy-Binet formula: Let $\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}$ and $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$. Suppose
lyche-numerical-linear-algebra-F00448	37	■ 1.000	$1 \leq r \leq \min \{m, n, p\}$	$1 \leq r \leq \min \{m, n, p\}$	$1 \leq r \leq \min \{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \dots, i_r\}$ and $\boldsymbol{j} = \{j_1, \dots, j_r\}$ be integers
lyche-numerical-linear-algebra-F00449	37	■ 1.000	$\boldsymbol{i}=\left\{i_1, \ldots, i_r\right\}$	$\boldsymbol{i} = \{i_1, \dots, i_r\}$	$1 \leq r \leq \min \{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \dots, i_r\}$ and $\boldsymbol{j} = \{j_1, \dots, j_r\}$ be integers
lyche-numerical-linear-algebra-F00450	37	■ 1.000	$\boldsymbol{j}=\left\{j_1, \ldots, j_r\right\}$	$\boldsymbol{j} = \{j_1, \dots, j_r\}$	$1 \leq r \leq \min \{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \dots, i_r\}$ and $\boldsymbol{j} = \{j_1, \dots, j_r\}$ be integers
lyche-numerical-linear-algebra-F00451	37	■ 1.000	$1 \leq i_1 < i_2 < \cdots < i_r \leq m$	$1 \leq i_1 < i_2 < \cdots < i_r \leq m$	with $1 \leq i_1 < i_2 < \cdots < i_r \leq m$ and $1 \leq j_1 < j_2 < \cdots < j_r \leq n$. Then
lyche-numerical-linear-algebra-F00452	37	■ 1.000	$1 \leq j_1 < j_2 < \cdots < j_r \leq n$	$1 \leq j_1 < j_2 < \cdots < j_r \leq n$	with $1 \leq i_1 < i_2 < \cdots < i_r \leq m$ and $1 \leq j_1 < j_2 < \cdots < j_r \leq n$. Then
lyche-numerical-linear-algebra-F00453	37	■ 0.999	$\boldsymbol{k}=\left\{k_1, \ldots, k_r\right\}$	$\boldsymbol{k} = \{k_1, \dots, k_r\}$	where we sum over all $\boldsymbol{k} = \{k_1, \dots, k_r\}$ with $1 \leq k_1 < k_2 < \cdots < k_r \leq p$.
lyche-numerical-linear-algebra-F00454	37	■ 0.999	$1 \leq k_1 < k_2 < \cdots < k_r \leq p$	$1 \leq k_1 < k_2 < \cdots < k_r \leq p$	where we sum over all $\boldsymbol{k} = \{k_1, \dots, k_r\}$ with $1 \leq k_1 < k_2 < \cdots < k_r \leq p$.
lyche-numerical-linear-algebra-F00455	38	■ 0.937	$\lambda \in \mathbb{C}$	$\lambda \in \mathbb{C}$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is a square matrix, $\lambda \in \mathbb{C}$ and $\boldsymbol{x} \in \mathbb{C}^n$. We say that $(\lambda, \boldsymbol{x})$ is an
lyche-numerical-linear-algebra-F00456	38	■ 0.937	$\lambda, \boldsymbol{x}$	$\lambda, \boldsymbol{x}$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is a square matrix, $\lambda \in \mathbb{C}$ and $\boldsymbol{x} \in \mathbb{C}^n$. We say that $(\lambda, \boldsymbol{x})$ is an
lyche-numerical-linear-algebra-F00457	38	■ 1.000	$\sigma(\boldsymbol{A})$	$\sigma(\boldsymbol{A})$	of $\boldsymbol{A}$ and is denoted by $\sigma(\boldsymbol{A})$. For example, $\sigma(\boldsymbol{I}) = \{1, \dots, 1\} = \{1\}$.
lyche-numerical-linear-algebra-F00458	38	■ 1.000	$\sigma(\boldsymbol{I})=\{1, \ldots, 1\}=\{1\}$	$\sigma(\boldsymbol{I}) = \{1, \dots, 1\} = \{1\}$	of $\boldsymbol{A}$ and is denoted by $\sigma(\boldsymbol{A})$. For example, $\sigma(\boldsymbol{I}) = \{1, \dots, 1\} = \{1\}$

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lyche-numerical-linear-algebra_F00459	38	■ 1.000	$\lambda \in$	$\lambda \in$	Lemma 1.5 (Characteristic Equation) For any $A \in \mathbb{C}^{n \times n}$ we have $\lambda \in$
lyche-numerical-linear-algebra_F00460	38	■ 0.995	$\sigma(\mathbf{A}) \Longleftrightarrow \det(\mathbf{A} - \lambda \mathbf{I}) = 0$	$\sigma(\mathbf{A}) \Longleftrightarrow \det(\mathbf{A} - \lambda \mathbf{I}) = 0$	$\sigma(\mathbf{A}) \Longleftrightarrow \det(\mathbf{A} - \lambda \mathbf{I}) = 0$.
lyche-numerical-linear-algebra_F00461	38	■ 0.699	$(\lambda, \mathbf{x})$	$(\lambda, \mathbf{x})$	Proof Suppose $(\lambda, \mathbf{x})$ is an eigenpair for A . The equation $A\mathbf{x} = \lambda\mathbf{x}$ can be written
lyche-numerical-linear-algebra_F00462	38	■ 1.000	$(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0}$	$(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0}$	$(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0}$. Since $\mathbf{x}$ is nonzero the matrix $\mathbf{A} - \lambda \mathbf{I}$ must be singular with a
lyche-numerical-linear-algebra_F00463	38	■ 1.000	$\mathbf{A} - \lambda \mathbf{I}$	$\mathbf{A} - \lambda \mathbf{I}$	$(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0}$. Since $\mathbf{x}$ is nonzero the matrix $\mathbf{A} - \lambda \mathbf{I}$ must be singular with a
lyche-numerical-linear-algebra_F00464	38	■ 1.000	$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$	$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$	zero determinant. Conversely, if $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$ then $\mathbf{A} - \lambda \mathbf{I}$ is singular and
lyche-numerical-linear-algebra_F00465	38	■ 1.000	$\det(\mathbf{A} - \lambda \mathbf{I})$	$\det(\mathbf{A} - \lambda \mathbf{I})$	The expression $\det(\mathbf{A} - \lambda \mathbf{I})$ is a polynomial of exact degree n in λ . For $n = 3$
lyche-numerical-linear-algebra_F00466	38	■ 1.000	$r(\lambda)$	$r(\lambda)$	where each term in $r(\lambda)$ has at most $n - 2$ factors containing λ . It follows that r is
lyche-numerical-linear-algebra_F00467	38	■ 0.998	$n - 2, \det(\mathbf{A} - \lambda \mathbf{I})$	$n - 2, \det(\mathbf{A} - \lambda \mathbf{I})$	a polynomial of degree at most $n - 2$. $\det(\mathbf{A} - \lambda \mathbf{I})$ is a polynomial of exact degree
lyche-numerical-linear-algebra_F00468	38	■ 1.000	$\det(\mathbf{A} - \lambda \mathbf{I}) = (-1)^n \det(\lambda \mathbf{I} - \mathbf{A})$	$\det(\mathbf{A} - \lambda \mathbf{I}) = (-1)^n \det(\lambda \mathbf{I} - \mathbf{A})$	We observe that $\det(\mathbf{A} - \lambda \mathbf{I}) = (-1)^n \det(\lambda \mathbf{I} - \mathbf{A})$ so $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$ if and
lyche-numerical-linear-algebra_F00469	38	■ 1.000	$\det(\lambda \mathbf{I} - \mathbf{A}) = 0$	$\det(\lambda \mathbf{I} - \mathbf{A}) = 0$	only if $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$.
lyche-numerical-linear-algebra_F00470	39	■ 0.958	$\pi_{\mathbf{A}}: \mathbb{C} \rightarrow$	$\pi_{\mathbf{A}}: \mathbb{C} \rightarrow$	Definition 1.9 (Characteristic Polynomial of a Matrix) The function $\pi_{\mathbf{A}}: \mathbb{C} \rightarrow$
lyche-numerical-linear-algebra_F00471	39	■ 1.000	$\pi_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I})$	$\pi_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I})$	$\mathbb{C}$ given by $\pi_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I})$ is called the characteristic polynomial of A .
lyche-numerical-linear-algebra_F00472	39	■ 0.994	$\lambda_1, \dots, \lambda_n$	$\lambda_1, \dots, \lambda_n$	ties, precisely n eigenvalues $\lambda_1, \dots, \lambda_n$ some of which might be complex even if A
lyche-numerical-linear-algebra_F00473	39	■ 1.000	$A\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}$	$A\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}$	A real gives $A\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}$.
lyche-numerical-linear-algebra_F00474	39	■ 0.945	$\pi_{\mathbf{A}}$	$\pi_{\mathbf{A}}$	Proof We compare two different expansions of $\pi_{\mathbf{A}}$. On the one hand from (1.16)
lyche-numerical-linear-algebra_F00475	39	■ 1.000	$c_{n-1} = (-1)^{n-1} \text{trace}(\mathbf{A})$	$c_{n-1} = (-1)^{n-1} \text{trace}(\mathbf{A})$	where $c_{n-1} = (-1)^{n-1} \text{trace}(\mathbf{A})$ and $c_0 = \pi_{\mathbf{A}}(0) = \det(\mathbf{A})$. On the other hand
lyche-numerical-linear-algebra_F00476	39	■ 1.000	$c_0 = \pi_{\mathbf{A}}(0) = \det(\mathbf{A})$	$c_0 = \pi_{\mathbf{A}}(0) = \det(\mathbf{A})$	where $c_{n-1} = (-1)^{n-1} \text{trace}(\mathbf{A})$ and $c_0 = \pi_{\mathbf{A}}(0) = \det(\mathbf{A})$. On the other hand
lyche-numerical-linear-algebra_F00477	39	■ 1.000	$d_{n-1} = (-1)^{n-1} (\lambda_1 + \dots + \lambda_n)$	$d_{n-1} = (-1)^{n-1} (\lambda_1 + \dots + \lambda_n)$	where $d_{n-1} = (-1)^{n-1} (\lambda_1 + \dots + \lambda_n)$ and $d_0 = \lambda_1 \cdots \lambda_n$. Since $c_j = d_j$ for all j
lyche-numerical-linear-algebra_F00478	39	■ 1.000	$d_0 = \lambda_1 \cdots \lambda_n$	$d_0 = \lambda_1 \cdots \lambda_n$	where $d_{n-1} = (-1)^{n-1} (\lambda_1 + \dots + \lambda_n)$ and $d_0 = \lambda_1 \cdots \lambda_n$. Since $c_j = d_j$ for all j
lyche-numerical-linear-algebra_F00479	39	■ 1.000	$c_j = d_j$	$c_j = d_j$	where $d_{n-1} = (-1)^{n-1} (\lambda_1 + \dots + \lambda_n)$ and $d_0 = \lambda_1 \cdots \lambda_n$. Since $c_j = d_j$ for all j
lyche-numerical-linear-algebra_F00480	39	■ 1.000	$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	Thus, if $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ then $\text{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$ so that $\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$.
lyche-numerical-linear-algebra_F00481	39	■ 1.000	$\text{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$	$\text{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$	Thus, if $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ then $\text{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$ so that $\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$.
lyche-numerical-linear-algebra_F00482	39	■ 1.000	$\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$	$\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$	Thus, if $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ then $\text{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$ so that $\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$.
lyche-numerical-linear-algebra_F00483	39	■ 1.000	$\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$	$\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$	Since $\mathbf{A}$ is singular $\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$, some $\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$, some $\mathbf{x} \neq$
lyche-numerical-linear-algebra_F00484	39	■ 1.000	$\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$	$\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$	Since $\mathbf{A}$ is singular $\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$, some $\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$, some $\mathbf{x} \neq$

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lyche-numerical-linear-algebra_F00485	39	■ 1.000	$\boldsymbol{x} \neq$	$x \neq$	Since A is singular $\iff Ax = \mathbf{0}$, some $x \neq 0 \iff Ax = \mathbf{0}x$, some $x \neq$
lyche-numerical-linear-algebra_F00486	39	■ 0.981	$\emptyset \Longleftarrow \longrightarrow$	$0 \iff$	$0 \iff$ zero is an eigenvalue of A , we obtain
lyche-numerical-linear-algebra_F00487	40	■ 0.848	$\boldsymbol{A} = \left[\begin{array}{cc} \boldsymbol{B} & \boldsymbol{D} \\ 0 & \boldsymbol{C} \end{array} \right]$	$A = \begin{bmatrix} B & D \\ 0 & C \end{bmatrix}$	Theorem 1.12 (Eigenvalues of a Block Triangular Matrix) If $A = \begin{bmatrix} B & D \\ 0 & C \end{bmatrix}$ is
lyche-numerical-linear-algebra_F00488	40	■ 0.998	$\pi_{\boldsymbol{A}} = \pi_{\boldsymbol{B}} \cdot \pi_{\boldsymbol{C}}$	$\pi_A = \pi_B \cdot \pi_C$	block triangular then $\pi_A = \pi_B \cdot \pi_C$.
lyche-numerical-linear-algebra_F00489	40	■ 1.000	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n \times n}$	$A, B \in \mathbb{R}^{n \times n}$	a) With $A, B \in \mathbb{R}^{n \times n}$, how many arithmetic operations are required to form the
lyche-numerical-linear-algebra_F00490	40	■ 1.000	$2n \times 2n$	$2n \times 2n$	b) Consider the $2n \times 2n$ block matrix
lyche-numerical-linear-algebra_F00491	40	■ 1.000	$\boldsymbol{A}, \dots, \boldsymbol{Z}$	$A, \dots, Z$	where all matrices $A, \dots, Z$ are in $\mathbb{R}^{n \times n}$. How many operations does it take to
lyche-numerical-linear-algebra_F00492	40	■ 1.000	$\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}$	W, X, Y	compute W, X, Y and Z by the obvious algorithm?
lyche-numerical-linear-algebra_F00493	40	■ 1.000	$\boldsymbol{Z}$	Z	compute W, X, Y and Z by the obvious algorithm?
lyche-numerical-linear-algebra_F00494	40	■ 1.000	$m \times m$	$m \times m$	matrices A and B of size $m \times m$, with $m = 2^k$ for some $k \geq 0$, calculates the
lyche-numerical-linear-algebra_F00495	40	■ 1.000	$m = 2^k$	$m = 2^k$	matrices A and B of size $m \times m$, with $m = 2^k$ for some $k \geq 0$, calculates the
lyche-numerical-linear-algebra_F00496	40	■ 1.000	$k \geq 0$	$k \geq 0$	matrices A and B of size $m \times m$, with $m = 2^k$ for some $k \geq 0$, calculates the
lyche-numerical-linear-algebra_F00497	40	■ 1.000	$\mathcal{O}(m^{\log_2(7)})$	$\mathcal{O}(m^{\log_2(7)})$	e) Show that the operation count of the recursive algorithm is $\mathcal{O}(m^{\log_2(7)})$. Note
lyche-numerical-linear-algebra_F00498	40	■ 1.000	$\log_2(7) \approx 2.8 < 3$	$\log_2(7) \approx 2.8 < 3$	that $\log_2(7) \approx 2.8 < 3$, so this is less costly than straightforward matrix
lyche-numerical-linear-algebra_F00499	41	■ 1.000	2×2	2×2	Exercise 1.2 (The Inverse of a General 2×2 Matrix) Show that
lyche-numerical-linear-algebra_F00500	41	■ 1.000	a, b, c, d	a, b, c, d	for any a, b, c, d such that $ad - bc \neq 0$.
lyche-numerical-linear-algebra_F00501	41	■ 1.000	$a d - b c \neq 0$	$ad - bc \neq 0$	for any a, b, c, d such that $ad - bc \neq 0$.
lyche-numerical-linear-algebra_F00502	41	■ 1.000	$\boldsymbol{B}, \boldsymbol{C} \in$	$B, C \in$	Exercise 1.4 (Sherman-Morrison Formula) Suppose $A \in \mathbb{C}^{n \times n}$, and $B, C \in$
lyche-numerical-linear-algebra_F00503	41	■ 1.000	$\mathbb{R}^{n \times m}$	$\mathbb{R}^{n \times m}$	$\mathbb{R}^{n \times m}$ for some $n, m \in \mathbb{N}$. If $(I + C^T A^{-1} B)^{-1}$ exists then
lyche-numerical-linear-algebra_F00504	41	■ 1.000	$n, m \in \mathbb{N}$	$n, m \in \mathbb{N}$	$\mathbb{R}^{n \times m}$ for some $n, m \in \mathbb{N}$. If $(I + C^T A^{-1} B)^{-1}$ exists then
lyche-numerical-linear-algebra_F00505	41	■ 1.000	$\left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B} \right)^{-1}$	$(I + C^T A^{-1} B)^{-1}$	$\mathbb{R}^{n \times m}$ for some $n, m \in \mathbb{N}$. If $(I + C^T A^{-1} B)^{-1}$ exists then
lyche-numerical-linear-algebra_F00506	41	■ 1.000	$\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$	$u, v \in \mathbb{R}^n$	a) Let $u, v \in \mathbb{R}^n$ and suppose $v^T u \neq 1$. Show that $I - uv^T$ has an inverse given
lyche-numerical-linear-algebra_F00507	41	■ 1.000	$\boldsymbol{v}^T \boldsymbol{u} \neq 1$	$v^T u \neq 1$	a) Let $u, v \in \mathbb{R}^n$ and suppose $v^T u \neq 1$. Show that $I - uv^T$ has an inverse given
lyche-numerical-linear-algebra_F00508	41	■ 1.000	$\boldsymbol{I} - \boldsymbol{u} \boldsymbol{v}^T$	$I - uv^T$	a) Let $u, v \in \mathbb{R}^n$ and suppose $v^T u \neq 1$. Show that $I - uv^T$ has an inverse given
lyche-numerical-linear-algebra_F00509	41	■ 0.813	$\boldsymbol{I} - \tau \boldsymbol{u} \boldsymbol{v}^T$	$I - \tau uv^T$	by $I - \tau uv^T$, where $\tau = 1/(v^T u - 1)$.
lyche-numerical-linear-algebra_F00510	41	■ 0.813	$\tau = 1 / (v^T u - 1)$	$\tau = 1 / (v^T u - 1)$	by $I - \tau uv^T$, where $\tau = 1/(v^T u - 1)$.
lyche-numerical-linear-algebra_F00511	41	■ 1.000	$\boldsymbol{C} := \boldsymbol{A}^{-1}$	$C := A^{-1}$	b) Let $A \in \mathbb{R}^{n \times n}$ be nonsingular with inverse $C := A^{-1}$, and let $a \in \mathbb{R}^n$. Let $\bar{A}$ be

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lyche-numerical-linear-atgdra_F00512	41	1.000	$\boldsymbol{a} \in \mathbb{R}^n$	$\boldsymbol{a} \in \mathbb{R}^n$	b) Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be nonsingular with inverse $\boldsymbol{C} := \boldsymbol{A}^{-1}$, and let $\boldsymbol{a} \in \mathbb{R}^n$. Let $\overline{\boldsymbol{A}}$ be
lyche-numerical-linear-atgdra_F00513	41	1.000	$\overline{\boldsymbol{A}}$	$\overline{\boldsymbol{A}}$	b) Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be nonsingular with inverse $\boldsymbol{C} := \boldsymbol{A}^{-1}$, and let $\boldsymbol{a} \in \mathbb{R}^n$. Let $\overline{\boldsymbol{A}}$ be
lyche-numerical-linear-atgdra_F00514	41	1.000	$\boldsymbol{a}^T$	$\boldsymbol{a}^T$	the matrix which differs from $\boldsymbol{A}$ by exchanging the i th row of $\boldsymbol{A}$ with $\boldsymbol{a}^T$, i.e.,
lyche-numerical-linear-atgdra_F00515	41	0.995	$\overline{\boldsymbol{A}} = \boldsymbol{A} - \boldsymbol{e}_i(\boldsymbol{e}_i^T \boldsymbol{A} - \boldsymbol{a}^T)$	$\overline{\boldsymbol{A}} = \boldsymbol{A} - \boldsymbol{e}_i(\boldsymbol{e}_i^T \boldsymbol{A} - \boldsymbol{a}^T)$	$\overline{\boldsymbol{A}} = \boldsymbol{A} - \boldsymbol{e}_i(\boldsymbol{e}_i^T \boldsymbol{A} - \boldsymbol{a}^T)$, where $\boldsymbol{e}_i$ is the i th column in the identity matrix $\boldsymbol{I}$.
lyche-numerical-linear-atgdra_F00516	41	0.995	$\boldsymbol{e}_i$	$\boldsymbol{e}_i$	$\overline{\boldsymbol{A}} = \boldsymbol{A} - \boldsymbol{e}_i(\boldsymbol{e}_i^T \boldsymbol{A} - \boldsymbol{a}^T)$, where $\boldsymbol{e}_i$ is the i th column in the identity matrix $\boldsymbol{I}$.
lyche-numerical-linear-atgdra_F00517	41	1.000	$\overline{\boldsymbol{C}} = \overline{\boldsymbol{A}}^{-1}$	$\overline{\boldsymbol{C}} = \overline{\boldsymbol{A}}^{-1}$	then $\overline{\boldsymbol{A}}$ has an inverse $\overline{\boldsymbol{C}} = \overline{\boldsymbol{A}}^{-1}$ given by
lyche-numerical-linear-atgdra_F00518	41	1.000	$\boldsymbol{a}$	$\boldsymbol{a}$	c) Write an algorithm which to given $\boldsymbol{C}$ and $\boldsymbol{a}$ checks if (1.20) holds and computes
lyche-numerical-linear-atgdra_F00519	41	0.999	$\overline{\boldsymbol{C}}$	$\overline{\boldsymbol{C}}$	$\overline{\boldsymbol{C}}$ provided $\lambda \neq 0$. (hint: Use (1.21) to find formulas for computing each column
lyche-numerical-linear-atgdra_F00520	41	0.999	$\lambda \neq 0$	$\lambda \neq 0$	$\overline{\boldsymbol{C}}$ provided $\lambda \neq 0$. (hint: Use (1.21) to find formulas for computing each column
lyche-numerical-linear-atgdra_F00521	41	1.000	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in$	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in$	Exercise 1.6 (Matrix Products (Exam Exercise 2009-1)) Let $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in$
lyche-numerical-linear-atgdra_F00522	41	1.000	$\boldsymbol{B} \boldsymbol{C}$	$\boldsymbol{B} \boldsymbol{C}$	a) How many operations are required to compute the matrix product $\boldsymbol{B} \boldsymbol{C}$? How
lyche-numerical-linear-atgdra_F00523	41	1.000	$\boldsymbol{L} \in \mathbb{R}^{n \times n}$	$\boldsymbol{L} \in \mathbb{R}^{n \times n}$	b) Show that there exists a lower triangular matrix $\boldsymbol{L} \in \mathbb{R}^{n \times n}$ such that $\boldsymbol{A} = \boldsymbol{L} + \boldsymbol{L}^T$.
lyche-numerical-linear-atgdra_F00524	41	1.000	$\boldsymbol{A} = \boldsymbol{L} + \boldsymbol{L}^T$	$\boldsymbol{A} = \boldsymbol{L} + \boldsymbol{L}^T$	b) Show that there exists a lower triangular matrix $\boldsymbol{L} \in \mathbb{R}^{n \times n}$ such that $\boldsymbol{A} = \boldsymbol{L} + \boldsymbol{L}^T$.
lyche-numerical-linear-atgdra_F00525	42	1.000	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \boldsymbol{S} + \boldsymbol{S}^T$	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \boldsymbol{S} + \boldsymbol{S}^T$	c) We have $\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \boldsymbol{S} + \boldsymbol{S}^T$ where $\boldsymbol{S} = \boldsymbol{E}^T \boldsymbol{L} \boldsymbol{E}$. How many operations are
lyche-numerical-linear-atgdra_F00526	42	1.000	$\boldsymbol{S} = \boldsymbol{E}^T \boldsymbol{L} \boldsymbol{E}$	$\boldsymbol{S} = \boldsymbol{E}^T \boldsymbol{L} \boldsymbol{E}$	c) We have $\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \boldsymbol{S} + \boldsymbol{S}^T$ where $\boldsymbol{S} = \boldsymbol{E}^T \boldsymbol{L} \boldsymbol{E}$. How many operations are
lyche-numerical-linear-atgdra_F00527	42	1.000	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E}$	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E}$	required to compute $\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E}$ in this way?
lyche-numerical-linear-atgdra_F00528	42	1.000	$(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$	$(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$	is the equation for a plane through three points $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$
lyche-numerical-linear-atgdra_F00529	43	1.000	$P_i = (x_i, y_i), i = 1, 2, 3$	$P_i = (x_i, y_i), i = 1, 2, 3$	Exercise 1.10 (Signed Area of a Triangle) Let $P_i = (x_i, y_i), i = 1, 2, 3$, be three
lyche-numerical-linear-atgdra_F00530	43	0.816	T is ²	T is ²	points in the plane defining a triangle T . Show that the area of T is ²
lyche-numerical-linear-atgdra_F00531	43	0.999	$\prod_{i>j} (x_i - x_j) = \prod_{i=2}^n (x_i - x_1)(x_i - x_2) \cdots (x_i - x_{i-1})$	$\prod_{i>j} (x_i - x_j) = \prod_{i=2}^n (x_i - x_1)(x_i - x_2) \cdots (x_i - x_{i-1})$	where $\prod_{i>j} (x_i - x_j) = \prod_{i=2}^n (x_i - x_1)(x_i - x_2) \cdots (x_i - x_{i-1})$. This determinant
lyche-numerical-linear-atgdra_F00532	43	1.000	$\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_n]^T, \boldsymbol{\beta} =$	$\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_n]^T, \boldsymbol{\beta} =$	Exercise 1.12 (Cauchy Determinant (1842)) Let $\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_n]^T, \boldsymbol{\beta} =$
lyche-numerical-linear-atgdra_F00533	43	1.000	$[\beta_1, \dots, \beta_n]^T$	$[\beta_1, \dots, \beta_n]^T$	$[\beta_1, \dots, \beta_n]^T$ be in $\mathbb{R}^n$.
lyche-numerical-linear-atgdra_F00534	43	0.996	$a_{i,j} = 1/(\alpha_i + \beta_j), i, j =$	$a_{i,j} = 1/(\alpha_i + \beta_j), i, j =$	a) Consider the matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ with elements $a_{i,j} = 1/(\alpha_i + \beta_j), i, j =$
lyche-numerical-linear-atgdra_F00535	43	1.000	$1, 2, \dots, n$	$1, 2, \dots, n$	$1, 2, \dots, n$. Show that
lyche-numerical-linear-atgdra_F00536	43	0.997	$A(T) = A(\boldsymbol{A} \boldsymbol{B} \boldsymbol{P}_3 \boldsymbol{P}_1) + A(\boldsymbol{P}_3 \boldsymbol{B} \boldsymbol{C} \boldsymbol{P}_2) - A(\boldsymbol{P}_1 \boldsymbol{A} \boldsymbol{C} \boldsymbol{P}_2)$	$A(T) = A(\boldsymbol{A} \boldsymbol{B} \boldsymbol{P}_3 \boldsymbol{P}_1) + A(\boldsymbol{P}_3 \boldsymbol{B} \boldsymbol{C} \boldsymbol{P}_2) - A(\boldsymbol{P}_1 \boldsymbol{A} \boldsymbol{C} \boldsymbol{P}_2)$	² Hint: $A(T) = A(\boldsymbol{A} \boldsymbol{B} \boldsymbol{P}_3 \boldsymbol{P}_1) + A(\boldsymbol{P}_3 \boldsymbol{B} \boldsymbol{C} \boldsymbol{P}_2) - A(\boldsymbol{P}_1 \boldsymbol{A} \boldsymbol{C} \boldsymbol{P}_2)$ c.f. Fig. 1.1.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00537	43	0.997	x_n^k	x_n^k	³ Hint: subtract x_n^k times column k from column $k+1$ for $k = n-1, n-2, \dots, 1$.
lyche-numerical-linear-algebra_F00538	43	0.997	k	k	³ Hint: subtract x_n^k times column k from column $k+1$ for $k = n-1, n-2, \dots, 1$.
lyche-numerical-linear-algebra_F00539	43	0.997	$k+1$	$k+1$	³ Hint: subtract x_n^k times column k from column $k+1$ for $k = n-1, n-2, \dots, 1$.
lyche-numerical-linear-algebra_F00540	43	0.997	$k=n-1, n-2, \ldots, 1$	$k = n-1, n-2, \dots, 1$	³ Hint: subtract x_n^k times column k from column $k+1$ for $k = n-1, n-2, \dots, 1$.
lyche-numerical-linear-algebra_F00541	44	1.000	$P = \prod_{i=1}^n \prod_{j=1}^n a_{ij}$	$P = \prod_{i=1}^n \prod_{j=1}^n a_{ij}$	where $P = \prod_{i=1}^n \prod_{j=1}^n a_{ij}$, and for $\boldsymbol{\gamma} = [\gamma_1, \dots, \gamma_n]^T$
lyche-numerical-linear-algebra_F00542	44	1.000	$\boldsymbol{\gamma} = [\gamma_1, \dots, \gamma_n]^T$	$\boldsymbol{\gamma} = [\gamma_1, \dots, \gamma_n]^T$	where $P = \prod_{i=1}^n \prod_{j=1}^n a_{ij}$, and for $\boldsymbol{\gamma} = [\gamma_1, \dots, \gamma_n]^T$
lyche-numerical-linear-algebra_F00543	44	1.000	$\prod_{j=1}^n (\alpha_i + \beta_j)$	$\prod_{j=1}^n (\alpha_i + \beta_j)$	Hint: Multiply the i th row of A by $\prod_{j=1}^n (\alpha_i + \beta_j)$ for $i = 1, 2, \dots, n$. Call the
lyche-numerical-linear-algebra_F00544	44	1.000	$i=1, 2, \ldots, n$	$i = 1, 2, \dots, n$	Hint: Multiply the i th row of A by $\prod_{j=1}^n (\alpha_i + \beta_j)$ for $i = 1, 2, \dots, n$. Call the
lyche-numerical-linear-algebra_F00545	44	0.999	$n-1$	$n-1$	resulting matrix C . Each element of C is a product of $n-1$ factors $\alpha_r + \beta_s$. Hence
lyche-numerical-linear-algebra_F00546	44	0.999	$\alpha_r + \beta_s$	$\alpha_r + \beta_s$	resulting matrix C . Each element of C is a product of $n-1$ factors $\alpha_r + \beta_s$. Hence
lyche-numerical-linear-algebra_F00547	44	1.000	$\det(C)$	$\det(C)$	$\det(C)$ is a sum of terms where each term contain precisely $n(n-1)$ factors
lyche-numerical-linear-algebra_F00548	44	1.000	$n(n-1)$	$n(n-1)$	$\det(C)$ is a sum of terms where each term contain precisely $n(n-1)$ factors
lyche-numerical-linear-algebra_F00549	44	0.997	$\det(C) = q(\alpha, \beta)$	$\det(C) = q(\alpha, \beta)$	$\alpha_r + \beta_s$. Thus $\det(C) = q(\alpha, \beta)$ where q is a polynomial of degree at most
lyche-numerical-linear-algebra_F00550	44	0.997	q	q	$\alpha_r + \beta_s$. Thus $\det(C) = q(\alpha, \beta)$ where q is a polynomial of degree at most
lyche-numerical-linear-algebra_F00551	44	1.000	α_i	α_i	$n(n-1)$ in α_i and β_j . Since $\det(A)$ and therefore $\det(C)$ vanishes if $\alpha_i = \alpha_j$ for
lyche-numerical-linear-algebra_F00552	44	1.000	β_j	β_j	$n(n-1)$ in α_i and β_j . Since $\det(A)$ and therefore $\det(C)$ vanishes if $\alpha_i = \alpha_j$ for
lyche-numerical-linear-algebra_F00553	44	1.000	$\det(A)$	$\det(A)$	$n(n-1)$ in α_i and β_j . Since $\det(A)$ and therefore $\det(C)$ vanishes if $\alpha_i = \alpha_j$ for
lyche-numerical-linear-algebra_F00554	44	1.000	$\alpha_i = \alpha_j$	$\alpha_i = \alpha_j$	$n(n-1)$ in α_i and β_j . Since $\det(A)$ and therefore $\det(C)$ vanishes if $\alpha_i = \alpha_j$ for
lyche-numerical-linear-algebra_F00555	44	1.000	$\beta_r = \beta_s$	$\beta_r = \beta_s$	some $i \neq j$ or $\beta_r = \beta_s$ for some $r \neq s$, we have that $q(\boldsymbol{\alpha}, \boldsymbol{\beta})$ must be divisible
lyche-numerical-linear-algebra_F00556	44	1.000	$r \neq s$	$r \neq s$	some $i \neq j$ or $\beta_r = \beta_s$ for some $r \neq s$, we have that $q(\boldsymbol{\alpha}, \boldsymbol{\beta})$ must be divisible
lyche-numerical-linear-algebra_F00557	44	1.000	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})$	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})$	some $i \neq j$ or $\beta_r = \beta_s$ for some $r \neq s$, we have that $q(\boldsymbol{\alpha}, \boldsymbol{\beta})$ must be divisible
lyche-numerical-linear-algebra_F00558	44	1.000	$g(\boldsymbol{\alpha})$	$g(\boldsymbol{\alpha})$	by each factor in $g(\boldsymbol{\alpha})$ and $g(\boldsymbol{\beta})$. Since $g(\boldsymbol{\alpha})$ and $g(\boldsymbol{\beta})$ is a polynomial of degree
lyche-numerical-linear-algebra_F00559	44	1.000	$g(\boldsymbol{\beta})$	$g(\boldsymbol{\beta})$	by each factor in $g(\boldsymbol{\alpha})$ and $g(\boldsymbol{\beta})$. Since $g(\boldsymbol{\alpha})$ and $g(\boldsymbol{\beta})$ is a polynomial of degree
lyche-numerical-linear-algebra_F00560	44	1.000	$\boldsymbol{\alpha}$	$\boldsymbol{\alpha}$	for some constant k independent of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Show that $k = 1$ by choosing
lyche-numerical-linear-algebra_F00561	44	1.000	$\boldsymbol{\beta}$	$\boldsymbol{\beta}$	for some constant k independent of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Show that $k = 1$ by choosing
lyche-numerical-linear-algebra_F00562	44	1.000	$k=1$	$k = 1$	for some constant k independent of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Show that $k = 1$ by choosing
lyche-numerical-linear-algebra_F00563	44	1.000	$\beta_i + \alpha_i = 0, i=1, 2, \ldots, n$	$\beta_i + \alpha_i = 0, i = 1, 2, \dots, n$	$\beta_i + \alpha_i = 0, i = 1, 2, \dots, n$.

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lyche-numerical-linear-algebra_F00564	44	1.000	$\boldsymbol{A}^{-1}=\left(b_{j, k}\right)$	$\boldsymbol{A}^{-1}=\left(b_{j, k}\right)$	the elements of $\boldsymbol{A}^{-1}=\left(b_{i, k}\right)$. Answer:
lyche-numerical-linear-algebra_F00565	44	1.000	$\boldsymbol{H}_{n}=\left(h_{i, j}\right)$	$\boldsymbol{H}_n=\left(h_{i, j}\right)$	Exercise 1.13 (Inverse of the Hilbert Matrix) Let $\boldsymbol{H}_n=\left(h_{i, j}\right)$ be the $n \times n$
lyche-numerical-linear-algebra_F00566	44	1.000	$h_{i, j}=1 /\left(i+j-1\right)$	$h_{i, j}=1 /\left(i+j-1\right)$	matrix with elements $h_{i, j}=1 /\left(i+j-1\right)$. Use Exercise 1.12 to show that the elements
lyche-numerical-linear-algebra_F00567	44	1.000	$t_{i, j}^n$	$t_{i, j}^n$	$t_{i, j}^n$ in $\boldsymbol{T}_n=\boldsymbol{H}_n^{-1}$ are given by
lyche-numerical-linear-algebra_F00568	44	1.000	$\boldsymbol{T}_{n}=\boldsymbol{H}_{n}^{-1}$	$\boldsymbol{T}_n=\boldsymbol{H}_n^{-1}$	$t_{i, j}^n$ in $\boldsymbol{T}_n=\boldsymbol{H}_n^{-1}$ are given by
lyche-numerical-linear-algebra_F00569	45	1.000	$\boldsymbol{L}$	$\boldsymbol{L}$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F00570	45	1.000	$\boldsymbol{U}$	$\boldsymbol{U}$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F00571	45	1.000	$\boldsymbol{A} =$	$\boldsymbol{A} =$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F00572	45	1.000	$\boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{L} \boldsymbol{U}$. We also consider the factorization $\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}$, where $\boldsymbol{L}$ is lower triangular,
lyche-numerical-linear-algebra_F00573	45	1.000	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}$	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}$	$\boldsymbol{L} \boldsymbol{U}$. We also consider the factorization $\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}$, where $\boldsymbol{L}$ is lower triangular,
lyche-numerical-linear-algebra_F00574	45	1.000	$\boldsymbol{D}$	$\boldsymbol{D}$	$\boldsymbol{D}$ is diagonal and $\boldsymbol{U}$ is upper triangular. Moreover, $\boldsymbol{L}$ and $\boldsymbol{U}$ have ones on their
lyche-numerical-linear-algebra_F00575	45	1.000	$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$	$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$	on QR factorization. Here $\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^* \boldsymbol{Q}=\boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra_F00576	45	1.000	$\boldsymbol{Q}$	$\boldsymbol{Q}$	on QR factorization. Here $\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^* \boldsymbol{Q}=\boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra_F00577	45	1.000	$\boldsymbol{Q}^* \boldsymbol{Q}=\boldsymbol{I}$	$\boldsymbol{Q}^* \boldsymbol{Q}=\boldsymbol{I}$	on QR factorization. Here $\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^* \boldsymbol{Q}=\boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra_F00578	45	1.000	$\boldsymbol{R}$	$\boldsymbol{R}$	on QR factorization. Here $\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^* \boldsymbol{Q}=\boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra_F00579	46	0.996	$[a, b], n+1 \geq 2$	$[a, b], n+1 \geq 2$	Given an interval $[a, b], n+1 \geq 2$ equidistant sites in $[a, b]$
lyche-numerical-linear-algebra_F00580	46	0.996	$[a, b]$	$[a, b]$	Given an interval $[a, b], n+1 \geq 2$ equidistant sites in $[a, b]$
lyche-numerical-linear-algebra_F00581	46	1.000	y	y	and y values $\boldsymbol{y}:=\left[y_1, \ldots, y_{n+1}\right]^T \in \mathbb{R}^{n+1}$. We seek a function $g:[a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_F00582	46	1.000	$\boldsymbol{y}:=\left[y_1, \ldots, y_{n+1}\right]^T \in \mathbb{R}^{n+1}$	$\boldsymbol{y}:=\left[y_1, \ldots, y_{n+1}\right]^T \in \mathbb{R}^{n+1}$	and y values $\boldsymbol{y}:=\left[y_1, \ldots, y_{n+1}\right]^T \in \mathbb{R}^{n+1}$. We seek a function $g:[a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_F00583	46	1.000	$g:[a, b] \rightarrow \mathbb{R}$	$g:[a, b] \rightarrow \mathbb{R}$	and y values $\boldsymbol{y}:=\left[y_1, \ldots, y_{n+1}\right]^T \in \mathbb{R}^{n+1}$. We seek a function $g:[a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_F00584	47	1.000	$a \leq x_1 < x_2 < \cdots < x_{n+1} \leq b$	$a \leq x_1 < x_2 < \cdots < x_{n+1} \leq b$	$a < x_1 < x_2 < \cdots < x_{n+1} < b$.
lyche-numerical-linear-algebra_F00585	47	1.000	$n+1$	$n+1$	Since there are $n+1$ interpolation conditions in (2.2) a natural choice for a function
lyche-numerical-linear-algebra_F00586	47	1.000	g	g	g is a polynomial of degree n . As shown in most books on numerical methods such
lyche-numerical-linear-algebra_F00587	47	1.000	$n=1, g$	$n=1, g$	when $n=1, g$ is the straight line
lyche-numerical-linear-algebra_F00588	47	1.000	f	f	The function f and the polynomial g of degree at most 13 satisfying (2.2) with
lyche-numerical-linear-algebra_F00589	47	1.000	$[a, b]=[-1, 1]$	$[a, b]=[-1, 1]$	$[a, b]=[-1, 1]$ and $y_i=f\left(x_i\right), i=1, \ldots, 14$ is shown in Fig. 2.1. The interpolant
lyche-numerical-linear-algebra_F00590	47	1.000	$y_i=f\left(x_i\right), i=1, \ldots, 14$	$y_i=f\left(x_i\right), i=1, \ldots, 14$	$[a, b]=[-1, 1]$ and $y_i=f\left(x_i\right), i=1, \ldots, 14$ is shown in Fig. 2.1. The interpolant

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lyche-numerical-linear-algebra-F00591	48	■ 1.000	$p_{\{i\}}$	p_i	where each p_i is a polynomial of degree ≤ 1 . In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra-F00592	48	■ 1.000	≤ 1	≤ 1	where each p_i is a polynomial of degree ≤ 1 . In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra-F00593	48	■ 1.000	$p_{\{1\}}$	p_1	where each p_i is a polynomial of degree ≤ 1 . In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra-F00594	48	■ 0.656	≤ 2	≤ 2	continuous derivatives of order ≤ 2 (C^2). We consider here the following functions
lyche-numerical-linear-algebra-F00595	48	■ 0.656	$C^{\{2\}}$	C^2	continuous derivatives of order ≤ 2 (C^2). We consider here the following functions
lyche-numerical-linear-algebra-F00596	49	■ 0.950	$D_{\{2\}}$	D_2	Definition 2.1 (The D_2-Spline Problem) Given $n \in \mathbb{N}$, an interval $[a, b]$, $y \in$
lyche-numerical-linear-algebra-F00597	49	■ 0.950	$\boldsymbol{y} \in$	$\boldsymbol{y} \in$	Definition 2.1 (The D_2-Spline Problem) Given $n \in \mathbb{N}$, an interval $[a, b]$, $y \in$
lyche-numerical-linear-algebra-F00598	49	■ 1.000	$\mathbb{R}^{\{n+1\}}$	$\mathbb{R}^{n+1}$	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \dots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1} . The problem
lyche-numerical-linear-algebra-F00599	49	■ 1.000	$x_{\{1\}}, \ldots, x_{\{n+1\}}$	$x_1, \dots, x_{n+1}$	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \dots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1} . The problem
lyche-numerical-linear-algebra-F00600	49	■ 1.000	$\mu_{\{1\}}, \mu_{\{n+1\}}$	μ_1, μ_{n+1}	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \dots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1} . The problem
lyche-numerical-linear-algebra-F00601	49	■ 1.000	$g \in C^{\{2\}}[a, b]$	$g \in C^2[a, b]$	• smoothness: $g \in C^2[a, b]$, i.e., derivatives of order ≤ 2 are continuous on $\mathbb{R}$,
lyche-numerical-linear-algebra-F00602	49	■ 1.000	$g(\text{left}(x_{\{i\}}\text{right})=y_{\{i\}}, \quad \text{quad } i=1,2, \ldots, n+1$	$g(x_i) = y_i, \quad i = 1, 2, \dots, n+1$	• interpolation: $g(x_i) = y_i, \quad i = 1, 2, \dots, n+1$,
lyche-numerical-linear-algebra-F00603	49	■ 1.000	$\boldsymbol{D}_{\{\mathbf{2}\}}$	$\boldsymbol{D}_2$	• D_2 boundary conditions: $g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$.
lyche-numerical-linear-algebra-F00604	49	■ 1.000	$g^{\{\prime \prime\}}(a)=\mu_{\{1\}}, \quad \text{quad } g^{\{\prime \prime\}}(b)=\mu_{\{n+1\}}$	$g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$	• D_2 boundary conditions: $g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$.
lyche-numerical-linear-algebra-F00605	49	■ 1.000	N	N	We call g a D_2 -spline. It is called an N -spline or natural spline if $\mu_1 = \mu_{n+1} = 0$.
lyche-numerical-linear-algebra-F00606	49	■ 1.000	$\mu_{\{1\}}=\mu_{\{n+1\}}=0$	$\mu_1 = \mu_{n+1} = 0$	We call g a D_2 -spline. It is called an N -spline or natural spline if $\mu_1 = \mu_{n+1} = 0$.
lyche-numerical-linear-algebra-F00607	49	■ 1.000	$n=2$	$n = 2$	<i>Example 2.1 (A D_2-Spline)</i> Suppose we choose $n = 2$ and sample data from the
lyche-numerical-linear-algebra-F00608	49	■ 1.000	$f:[0,2] \rightarrow \mathbb{R}$	$f : [0, 2] \rightarrow \mathbb{R}$	function $f : [0, 2] \rightarrow \mathbb{R}$ given by $f(x) = x^4$. Thus we consider the D_2 -spline
lyche-numerical-linear-algebra-F00609	49	■ 1.000	$[a, b]=[0,2], \quad \boldsymbol{y}:=[0,1,16]^{\{T\}}$	$[a, b] = [0, 2], \boldsymbol{y} := [0, 1, 16]^T$	problem with $[a, b] = [0, 2], \boldsymbol{y} := [0, 1, 16]^T$ and $\mu_1 = g''(0) = 0, \mu_3 = g''(2) =$
lyche-numerical-linear-algebra-F00610	49	■ 1.000	$\mu_{\{1\}}=g^{\{\prime \prime\}}(0)=0, \quad \mu_{\{3\}}=g^{\{\prime \prime\}}(2)=$	$\mu_1 = g''(0) = 0, \mu_3 = g''(2) =$	problem with $[a, b] = [0, 2], \boldsymbol{y} := [0, 1, 16]^T$ and $\mu_1 = g''(0) = 0, \mu_3 = g''(2) =$
lyche-numerical-linear-algebra-F00611	49	■ 1.000	$x_{\{1\}}=0, \quad x_{\{2\}}=1$	$x_1 = 0, x_2 = 1$	48. The knots are $x_1 = 0, x_2 = 1$ and $x_3 = 2$. The function g given by
lyche-numerical-linear-algebra-F00612	49	■ 1.000	$x_{\{3\}}=2$	$x_3 = 2$	48. The knots are $x_1 = 0, x_2 = 1$ and $x_3 = 2$. The function g given by
lyche-numerical-linear-algebra-F00613	49	■ 1.000	$p_{\{2\}}$	p_2	is a D_2 -spline solving this problem. Indeed, p_1 and p_2 are cubic polynomials. For
lyche-numerical-linear-algebra-F00614	49	■ 1.000	$p_{\{1\}}(1)=p_{\{2\}}(1)=1, \quad p_{\{1\}}^{\{\prime\}}(1)=p_{\{2\}}^{\{\prime\}}(1)=4, \quad p_{\{1\}}^{\{\prime \prime\}}(1)=p_{\{2\}}^{\{\prime \prime\}}(1)=$	$p_1(1) = p_2(1) = 1, p_1'(1) = p_2'(1) = 4, p_1''(1) = p_2''(1) =$	smoothness we find $p_1(1) = p_2(1) = 1, p_1'(1) = p_2'(1) = 4, p_1''(1) = p_2''(1) =$
lyche-numerical-linear-algebra-F00615	49	■ 1.000	$g \in C^{\{2\}}[0,2]$	$g \in C^2[0, 2]$	9 which implies that $g \in C^2[0, 2]$. Finally we check that the interpolation and
lyche-numerical-linear-algebra-F00616	49	■ 1.000	$g(0)=p_{\{1\}}(0)=0, \quad g(1)=p_{\{2\}}(1)=1, \quad g(2)=$	$g(0) = p_1(0) = 0, g(1) = p_2(1) = 1, g(2) =$	boundary conditions hold. Indeed, $g(0) = p_1(0) = 0, g(1) = p_2(1) = 1, g(2) =$
lyche-numerical-linear-algebra-F00617	49	■ 1.000	$p_{\{2\}}(2)=16, \quad g^{\{\prime \prime\}}(0)=p_{\{1\}}^{\{\prime \prime\}}(0)=0$	$p_2(2) = 16, g''(0) = p_1''(0) = 0$	$p_2(2) = 16, g''(0) = p_1''(0) = 0$ and $g''(2) = p_2''(2) = 48$. Note that $p_1'''(x) = 9 \neq$

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lyche-numerical-linear-algebra_F00637	51	1.000	$c_{i, 4}$	$c_{i,4}$	$2c_{i,3} = \mu_i$ (a moment), and then $c_{i,4}$ from $p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$. Finally
lyche-numerical-linear-algebra_F00638	51	1.000	$p_{i+1}'(x_{i+1}) = \mu_{i+1}$	$p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$	$2c_{i,3} = \mu_i$ (a moment), and then $c_{i,4}$ from $p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$. Finally
lyche-numerical-linear-algebra_F00639	51	1.000	$c_{i, 2}$	$c_{i,2}$	we find $c_{i,2}$ by solving $p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1}-\mu_i}{6h}h^3 = y_{i+1}$. For
lyche-numerical-linear-algebra_F00640	51	1.000	$p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1}-\mu_i}{6h}h^3 = y_{i+1}$	$p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1}-\mu_i}{6h}h^3 = y_{i+1}$	we find $c_{i,2}$ by solving $p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1}-\mu_i}{6h}h^3 = y_{i+1}$. For
lyche-numerical-linear-algebra_F00641	51	1.000	$j=0,1,2,3$	$j = 0, 1, 2, 3$	$j = 0, 1, 2, 3$ the shifted powers $(x - x_i)^j$ constitute a basis for cubic polynomials
lyche-numerical-linear-algebra_F00642	51	1.000	$(x - x_i)^j$	$(x - x_i)^j$	$j = 0, 1, 2, 3$ the shifted powers $(x - x_i)^j$ constitute a basis for cubic polynomials
lyche-numerical-linear-algebra_F00643	51	0.999	$\mu_1, \dots$	$\mu_1, \dots$	Theorem 2.1 (Constructing a D_2-Spline) Suppose for some moments $\mu_1, \dots$,
lyche-numerical-linear-algebra_F00644	51	1.000	μ_{n+1}	μ_{n+1}	μ_{n+1} that each p_i is given as in Lemma 2.1 for $i = 1, \dots, n$. If in addition
lyche-numerical-linear-algebra_F00645	51	0.973	$\mu_1, \dots, \mu_{n+1}$	$\mu_1, \dots, \mu_{n+1}$	$\mu_1, \dots, \mu_{n+1}$. Consider the C^2 requirement. Since $p_{i-1}(x_i) = p_i(x_i) = y_i$ and
lyche-numerical-linear-algebra_F00646	51	0.973	$p_{i-1}(x_i) = p_i(x_i) = y_i$	$p_{i-1}(x_i) = p_i(x_i) = y_i$	$\mu_1, \dots, \mu_{n+1}$. Consider the C^2 requirement. Since $p_{i-1}(x_i) = p_i(x_i) = y_i$ and
lyche-numerical-linear-algebra_F00647	51	1.000	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$ for $i = 2, \dots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_F00648	51	1.000	$i=2, \dots, n$	$i = 2, \dots, n$	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$ for $i = 2, \dots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_F00649	51	1.000	$g \in C^2$	$g \in C^2$	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$ for $i = 2, \dots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_F00650	51	1.000	$p_{i-1}'(x_i) = p_i'(x_i)$	$p_{i-1}'(x_i) = p_i'(x_i)$	$p_{i-1}'(x_i) = p_i'(x_i)$ for $i = 2, \dots, n$. By (2.7)
lyche-numerical-linear-algebra_F00651	51	1.000	$g(x_i) = y_i, i = 1, \dots, n+1, g'(a) = p_1'(x_1) = \mu_1$	$g(x_i) = y_i, i = 1, \dots, n+1, g''(a) = p_1''(x_1) = \mu_1$	By construction $g(x_i) = y_i, i = 1, \dots, n+1, g''(a) = p_1''(x_1) = \mu_1$ and $g''(b) =$
lyche-numerical-linear-algebra_F00652	51	1.000	$g'(b) =$	$g''(b) =$	By construction $g(x_i) = y_i, i = 1, \dots, n+1, g''(a) = p_1''(x_1) = \mu_1$ and $g''(b) =$
lyche-numerical-linear-algebra_F00653	51	1.000	$p_n''(x_{n+1}) = \mu_{n+1}$	$p_n''(x_{n+1}) = \mu_{n+1}$	$p_n''(x_{n+1}) = \mu_{n+1}$. It follows that g solves the D_2 -spline problem.
lyche-numerical-linear-algebra_F00654	51	1.000	$\mu_2, \dots, \mu_n$	$\mu_2, \dots, \mu_n$	In order for the D_2 -spline to exist we need to show that $\mu_2, \dots, \mu_n$ always can be
lyche-numerical-linear-algebra_F00655	51	1.000	$n \geq 3$	$n \geq 3$	determined from (2.9). For $n \geq 3$ and with μ_1 and μ_{n+1} given (2.9) can be written
lyche-numerical-linear-algebra_F00656	52	1.000	$A = [a_{ij}] \in \mathbb{C}^{n \times n}$	$A = [a_{ij}] \in \mathbb{C}^{n \times n}$	Definition 2.2 (Strict Diagonal Dominance) The matrix $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ is
lyche-numerical-linear-algebra_F00657	52	1.000	$ x_k = \max_i x_i $	$ x_k = \max_i x_i $	that $ x_k = \max_i x_i $. Then
lyche-numerical-linear-algebra_F00658	52	0.999	$\max_{1 \leq i \leq n} x_i = x_k \leq \frac{ b_k }{\sigma_k} \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right)$	$\max_{1 \leq i \leq n} x_i = x_k \leq \frac{ b_k }{\sigma_k} \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right)$	and this implies $\max_{1 \leq i \leq n} x_i = x_k \leq \frac{ b_k }{\sigma_k} \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right)$. For the
lyche-numerical-linear-algebra_F00659	52	0.999	$\max_{1 \leq i \leq n} x_i \leq 0$	$\max_{1 \leq i \leq n} x_i \leq 0$	nonsingularity, if $Ax = 0$, then $\max_{1 \leq i \leq n} x_i \leq 0$ by (2.13), and so $x = 0$.
lyche-numerical-linear-algebra_F00660	52	1.000	g_1	g_1	uniqueness suppose g_1 and g_2 are two D_2 -splines interpolating the same data. Then
lyche-numerical-linear-algebra_F00661	52	1.000	g_2	g_2	uniqueness suppose g_1 and g_2 are two D_2 -splines interpolating the same data. Then

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lyche-numerical-linear-algebra_F00662	52	■ 1.000	$g:=g_{\{1\}}-g_{\{2\}}$	$g := g_1 - g_2$	$g := g_1 - g_2$ is an N -spline satisfying (2.2) with $\mathbf{y} = \mathbf{0}$. The solution $[\mu_2, \dots, \mu_n]^T$
lyche-numerical-linear-algebra_F00663	52	■ 1.000	$\boldsymbol{y}=\mathbf{0}$	$\mathbf{y} = \mathbf{0}$	$g := g_1 - g_2$ is an N -spline satisfying (2.2) with $\mathbf{y} = \mathbf{0}$. The solution $[\mu_2, \dots, \mu_n]^T$
lyche-numerical-linear-algebra_F00664	52	■ 1.000	$\left[\mu_{\{2\}}, \ldots, \mu_{\{n\}}\right]^T$	$[\mu_2, \dots, \mu_n]^T$	$g := g_1 - g_2$ is an N -spline satisfying (2.2) with $\mathbf{y} = \mathbf{0}$. The solution $[\mu_2, \dots, \mu_n]^T$
lyche-numerical-linear-algebra_F00665	52	■ 1.000	$\mu_{\{1\}}=\mu_{\{n+1\}}$	$\mu_1 = \mu_{n+1}$	of (2.11) and also $\mu_1 = \mu_{n+1}$ are zero. It follows from (2.8) that all coefficients $c_{i,j}$
lyche-numerical-linear-algebra_F00666	52	■ 1.000	$c_{\{i, j\}}$	$c_{i,j}$	of (2.11) and also $\mu_1 = \mu_{n+1}$ are zero. It follows from (2.8) that all coefficients $c_{i,j}$
lyche-numerical-linear-algebra_F00667	52	■ 1.000	$g=0$	$g = 0$	are zero. We conclude that $g = 0$ and $g_1 = g_2$.
lyche-numerical-linear-algebra_F00668	52	■ 1.000	$g_{\{1\}}=g_{\{2\}}$	$g_1 = g_2$	are zero. We conclude that $g = 0$ and $g_1 = g_2$.
lyche-numerical-linear-algebra_F00669	52	■ 1.000	B	B	<i>Example 2.2 (Cubic B-Spline)</i> For the N -spline with $[a, b] = [0, 4]$ and $\mathbf{y} =$
lyche-numerical-linear-algebra_F00670	52	■ 1.000	$[a, b]=[0, 4]$	$[a, b] = [0, 4]$	<i>Example 2.2 (Cubic B-Spline)</i> For the N -spline with $[a, b] = [0, 4]$ and $\mathbf{y} =$
lyche-numerical-linear-algebra_F00671	52	■ 1.000	$\boldsymbol{y}=$	$\mathbf{y} =$	<i>Example 2.2 (Cubic B-Spline)</i> For the N -spline with $[a, b] = [0, 4]$ and $\mathbf{y} =$
lyche-numerical-linear-algebra_F00672	52	■ 0.997	$\left[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0\right]$	$\left[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0\right]$	$[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0]$ the linear system (2.9) takes the form
lyche-numerical-linear-algebra_F00673	53	■ 1.000	$\mu_{\{2\}}=\mu_{\{4\}}=1, \mu_{\{3\}}=-2$	$\mu_2 = \mu_4 = 1, \mu_3 = -2$	The solution is $\mu_2 = \mu_4 = 1, \mu_3 = -2$. The knotset is $\{0, 1, 2, 3, 4\}$. Using (2.8)
lyche-numerical-linear-algebra_F00674	53	■ 1.000	$\{0, 1, 2, 3, 4\}$	$\{0, 1, 2, 3, 4\}$	The solution is $\mu_2 = \mu_4 = 1, \mu_3 = -2$. The knotset is $\{0, 1, 2, 3, 4\}$. Using (2.8)
lyche-numerical-linear-algebra_F00675	53	■ 1.000	$(0, 4)$	$(0, 4)$	A plot of this spline is shown in Fig. 2.4. On $(0, 4)$ the function g equals the nonzero
lyche-numerical-linear-algebra_F00676	53	■ 1.000	$D^{\{2\}}$	D^2	To find the D^2 -spline g we have to solve the triangular system (2.11). Consider
lyche-numerical-linear-algebra_F00677	53	■ 1.000	$\boldsymbol{A}=\operatorname{tridiag}\left(\mathbf{a}_{\{i\}}, \mathbf{d}_{\{i\}}, \mathbf{c}_{\{i\}}\right) \in$	$\mathbf{A} = \operatorname{tridiag}(a_i, d_i, c_i) \in$	solving a general tridiagonal linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ where $\mathbf{A} = \operatorname{tridiag}(a_i, d_i, c_i) \in$
lyche-numerical-linear-algebra_F00678	54	■ 1.000	$\mathbb{C}^{n \times n}$	$\mathbb{C}^{n \times n}$	$\mathbb{C}^{n \times n}$. Instead of using Gaussian elimination directly, we can construct two matrices
lyche-numerical-linear-algebra_F00679	54	■ 1.000	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}$	$\mathbf{A} = \mathbf{L}\mathbf{U}$	$\mathbf{L}$ and $\mathbf{U}$ such that $\mathbf{A} = \mathbf{L}\mathbf{U}$. Since $\mathbf{A}\mathbf{x} = \mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b}$ we can find $\mathbf{x}$ by solving
lyche-numerical-linear-algebra_F00680	54	■ 1.000	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{L} \boldsymbol{U} \boldsymbol{x}=\boldsymbol{b}$	$\mathbf{A}\mathbf{x} = \mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b}$	$\mathbf{L}$ and $\mathbf{U}$ such that $\mathbf{A} = \mathbf{L}\mathbf{U}$. Since $\mathbf{A}\mathbf{x} = \mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b}$ we can find $\mathbf{x}$ by solving
lyche-numerical-linear-algebra_F00681	54	■ 1.000	$\boldsymbol{L} \boldsymbol{z}=\boldsymbol{b}$	$\mathbf{L}\mathbf{z} = \mathbf{b}$	two systems $\mathbf{L}\mathbf{z} = \mathbf{b}$ and $\mathbf{U}\mathbf{x} = \mathbf{z}$. Moreover $\mathbf{L}$ and $\mathbf{U}$ are both triangular and
lyche-numerical-linear-algebra_F00682	54	■ 1.000	$\boldsymbol{U} \boldsymbol{x}=\boldsymbol{z}$	$\mathbf{U}\mathbf{x} = \mathbf{z}$	two systems $\mathbf{L}\mathbf{z} = \mathbf{b}$ and $\mathbf{U}\mathbf{x} = \mathbf{z}$. Moreover $\mathbf{L}$ and $\mathbf{U}$ are both triangular and
lyche-numerical-linear-algebra_F00683	54	■ 1.000	$u_{\{1\}}, u_{\{2\}}, \ldots, u_{\{n-1\}}$	$u_1, u_2, \dots, u_{n-1}$	then $\mathbf{A} = \mathbf{L}\mathbf{U}$. If $u_1, u_2, \dots, u_{n-1}$ are nonzero then (2.16) is well defined. If in
lyche-numerical-linear-algebra_F00684	54	■ 0.999	$u_{\{n\}} \neq 0$	$u_n \neq 0$	addition $u_n \neq 0$ then we can solve $\mathbf{L}\mathbf{z} = \mathbf{b}$ and $\mathbf{U}\mathbf{x} = \mathbf{z}$ for $\mathbf{z}$ and $\mathbf{x}$. We formulate
lyche-numerical-linear-algebra_F00685	54	■ 0.999	$\boldsymbol{z}$	$\mathbf{z}$	addition $u_n \neq 0$ then we can solve $\mathbf{L}\mathbf{z} = \mathbf{b}$ and $\mathbf{U}\mathbf{x} = \mathbf{z}$ for $\mathbf{z}$ and $\mathbf{x}$. We formulate
lyche-numerical-linear-algebra_F00686	54	■ 1.000	$\boldsymbol{l} \in \mathbb{C}^{n-1}, \boldsymbol{u} \in \mathbb{C}^n$	$\boldsymbol{l} \in \mathbb{C}^{n-1}, \boldsymbol{u} \in \mathbb{C}^n$	this as two algorithms. In <code>trifactor</code> , vectors $\boldsymbol{l} \in \mathbb{C}^{n-1}, \boldsymbol{u} \in \mathbb{C}^n$ are computed
lyche-numerical-linear-algebra_F00687	54	■ 0.852	$\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}, \boldsymbol{d} \in \mathbb{C}^n$	$\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}, \boldsymbol{d} \in \mathbb{C}^n$	from $\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}, \boldsymbol{d} \in \mathbb{C}^n$. This implements the LU factorization of a tridiagonal
lyche-numerical-linear-algebra_F00688	55	■ 1.000	$\boldsymbol{l}, \in \mathbb{C}^{n-1}$	$\boldsymbol{l}, \in \mathbb{C}^{n-1}$	computed from a previous call to <code>trifactor</code> . Here $\boldsymbol{l}, \in \mathbb{C}^{n-1}$ and $\boldsymbol{u} \in \mathbb{C}^n$ are

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lyche-numerical-linear-algebra-F00689	55	■ 1.000	$\boldsymbol{u} \in \mathbb{C}^n$	$\boldsymbol{u} \in \mathbb{C}^n$	computed from a previous call to <code>trifactor</code> . Here $\boldsymbol{L} \in \mathbb{C}^{n-1}$ and $\boldsymbol{u} \in \mathbb{C}^n$ are
lyche-numerical-linear-algebra-F00690	55	■ 1.000	$\boldsymbol{b} \in \mathbb{C}^{n,r}$	$\boldsymbol{b} \in \mathbb{C}^{n,r}$	output from <code>trifactor</code> and $\boldsymbol{b} \in \mathbb{C}^{n,r}$ for some $r \in \mathbb{N}$:
lyche-numerical-linear-algebra-F00691	55	■ 1.000	$r \in \mathbb{N}$	$r \in \mathbb{N}$	output from <code>trifactor</code> and $\boldsymbol{b} \in \mathbb{C}^{n,r}$ for some $r \in \mathbb{N}$:
lyche-numerical-linear-algebra-F00692	55	■ 1.000	$u_{\{k\}}$	u_k	Proof We show that the u_k 's in (2.16) are nonzero for $k = 1, \dots, n$. For this it is
lyche-numerical-linear-algebra-F00693	55	■ 1.000	$k=1, \ldots, n$	$k = 1, \dots, n$	Proof We show that the u_k 's in (2.16) are nonzero for $k = 1, \dots, n$. For this it is
lyche-numerical-linear-algebra-F00694	55	■ 1.000	$a_{\{0\}}:=c_{\{n\}}:=0$	$a_0 := c_n := 0$	and where $a_0 := c_n := 0$. By assumption $ u_1 = d_1 = \sigma_1 + c_1 $. Suppose $ u_k \geq$
lyche-numerical-linear-algebra-F00695	55	■ 1.000	$ u_{\{1\}} = d_{\{1\}} = \sigma_{\{1\}} + c_{\{1\}} $	$ u_1 = d_1 = \sigma_1 + c_1 $	and where $a_0 := c_n := 0$. By assumption $ u_1 = d_1 = \sigma_1 + c_1 $. Suppose $ u_k \geq$
lyche-numerical-linear-algebra-F00696	55	■ 1.000	$ u_{\{k\}} \geq$	$ u_k \geq$	and where $a_0 := c_n := 0$. By assumption $ u_1 = d_1 = \sigma_1 + c_1 $. Suppose $ u_k \geq$
lyche-numerical-linear-algebra-F00697	55	■ 0.999	$\sigma_{\{k\}} + c_{\{k\}} $	$\sigma_k + c_k $	$\sigma_k + c_k $ for some $1 \leq k \leq n - 1$. Then $ c_k / u_k < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra-F00698	55	■ 0.999	$1 \leq k \leq n-1$	$1 \leq k \leq n - 1$	$\sigma_k + c_k $ for some $1 \leq k \leq n - 1$. Then $ c_k / u_k < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra-F00699	55	■ 0.999	$ c_{\{k\}} / u_{\{k\}} < 1$	$ c_k / u_k < 1$	$\sigma_k + c_k $ for some $1 \leq k \leq n - 1$. Then $ c_k / u_k < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra-F00700	56	■ 1.000	$u_{\{1\}}=d_{\{1\}}$	$u_1 = d_1$	given by (2.16). Then (2.17) holds, $u_1 = d_1$ and
lyche-numerical-linear-algebra-F00701	56	■ 1.000	$l_{\{k\}}$	l_k	we cannot have severe growth in the computed elements u_k and l_k .
lyche-numerical-linear-algebra-F00702	56	■ 1.000	$3n-3$	$3n - 3$	onal matrix of order n using (2.16) is $3n - 3$, while the number of arithmetic oper-
lyche-numerical-linear-algebra-F00703	56	■ 1.000	$r(5n-4)$	$r(5n - 4)$	ations for Algorithm <code>trisolve</code> is $r(5n - 4)$, where r is the number of right-hand
lyche-numerical-linear-algebra-F00704	56	■ 1.000	$O(n)$	$O(n)$	sides. This means that the complexity to solve a tridiagonal system is $O(n)$, or
lyche-numerical-linear-algebra-F00705	56	■ 1.000	$8n-7$	$8n - 7$	more precisely $8n - 7$ when $r = 1$, and this number only grows linearly ² with n .
lyche-numerical-linear-algebra-F00706	56	■ 1.000	$r=1$	$r = 1$	more precisely $8n - 7$ when $r = 1$, and this number only grows linearly ² with n .
lyche-numerical-linear-algebra-F00707	56	■ 0.939	$[0,1]$	$[0,1]$	where f is a given continuous function on $[0, 1]$ and u is an unknown function. This
lyche-numerical-linear-algebra-F00708	56	■ 0.939	u	u	where f is a given continuous function on $[0, 1]$ and u is an unknown function. This
lyche-numerical-linear-algebra-F00709	56	■ 1.000	$u(0)=u(1)=$	$u(0) = u(1) =$	integration constants so that the homogeneous boundary conditions $u(0) = u(1) =$
lyche-numerical-linear-algebra-F00710	56	■ 1.000	$f(x)=1$	$f(x) = 1$	0 are satisfied. For example, if $f(x) = 1$ then $u(x) = x(x - 1)/2$ is the solution.
lyche-numerical-linear-algebra-F00711	56	■ 1.000	$u(x)=x(x-1) / 2$	$u(x) = x(x - 1)/2$	0 are satisfied. For example, if $f(x) = 1$ then $u(x) = x(x - 1)/2$ is the solution.
lyche-numerical-linear-algebra-F00712	56	■ 1.000	$O(n^3)$	$O(n^3)$	² We show in Sect. 3.3.2 that Gaussian elimination on a full $n \times n$ system is an $O(n^3)$ process.
lyche-numerical-linear-algebra-F00713	57	■ 1.000	$h:=1/(m+1)$	$h := 1/(m + 1)$	positive integer m , let $h := 1/(m + 1)$ be the discretization parameter, and replace
lyche-numerical-linear-algebra-F00714	57	■ 0.816	$x_{\{j\}}:=j h$	$x_j := jh$	the interval $[0, 1]$ by grid points $x_j := jh$ for $j = 0, 1, \dots, m + 1$. We then obtain
lyche-numerical-linear-algebra-F00715	57	■ 0.816	$j=0,1, \ldots, m+1$	$j = 0, 1, \dots, m + 1$	the interval $[0, 1]$ by grid points $x_j := jh$ for $j = 0, 1, \dots, m + 1$. We then obtain

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lyche-numerical-linear-algebra-F00716	57	■ 0.958	$v_{\{j\}}$	v_j	approximations v_j to the exact solution $u(x_j)$ for $j = 1, \dots, m$ by replacing the
lyche-numerical-linear-algebra-F00717	57	■ 0.958	$u\left(x_{\{j\}}\right)$	$u\left(x_j\right)$	approximations v_j to the exact solution $u(x_j)$ for $j = 1, \dots, m$ by replacing the
lyche-numerical-linear-algebra-F00718	57	■ 1.000	$h^{\{2\}}$	h^2	Moving the h^2 factor to the right hand side this can be written as an $m \times m$ linear
lyche-numerical-linear-algebra-F00719	57	■ 1.000	$\boldsymbol{T}$	$\boldsymbol{T}$	The matrix $\boldsymbol{T}$ is called the second derivative matrix and will occur frequently in
lyche-numerical-linear-algebra-F00720	57	■ 0.999	$\boldsymbol{T}=\operatorname{tridiag}\left(a_{\{i\}}, d_{\{i\}}, c_{\{i\}}\right) \text { in }$	$\boldsymbol{T}=\operatorname{tridiag}\left(a_i, d_i, c_i\right) \in$	this book. It is our second example of a tridiagonal matrix, $\boldsymbol{T}=\operatorname{tridiag}\left(a_i, d_i, c_i\right) \in$
lyche-numerical-linear-algebra-F00721	57	■ 1.000	$\mathbb{R}^{\{m \times m\}}$	$\mathbb{R}^{m \times m}$	$\mathbb{R}^{m \times m}$, where in this case $a_i=c_i=-1$ and $d_i=2$ for all i .
lyche-numerical-linear-algebra-F00722	57	■ 1.000	$a_{\{i\}}=c_{\{i\}}=-1$	$a_i=c_i=-1$	$\mathbb{R}^{m \times m}$, where in this case $a_i=c_i=-1$ and $d_i=2$ for all i .
lyche-numerical-linear-algebra-F00723	57	■ 1.000	$d_{\{i\}}=2$	$d_i=2$	$\mathbb{R}^{m \times m}$, where in this case $a_i=c_i=-1$ and $d_i=2$ for all i .
lyche-numerical-linear-algebra-F00724	58	■ 1.000	$\boldsymbol{A}_{\{1\}}$	$\boldsymbol{A}_1$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra-F00725	58	■ 1.000	$\boldsymbol{A}_{\{2\}}$	$\boldsymbol{A}_2$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra-F00726	58	■ 1.000	$\boldsymbol{A}_{\{3\}}$	$\boldsymbol{A}_3$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra-F00727	58	■ 1.000	$\operatorname{det}\left(\boldsymbol{A}_{\{3\}}\right)=4 \neq 0$	$\det\left(\boldsymbol{A}_3\right)=4 \neq 0$	has a zero row, and $\det\left(\boldsymbol{A}_3\right)=4 \neq 0$. It follows that for the nonsingularity and
lyche-numerical-linear-algebra-F00728	58	■ 1.000	$\left d_{\{1\}}\right >\left c_{\{1\}}\right $	$\left d_1\right >\left c_1\right $	$\mathbb{C}^{n \times n}$ is tridiagonal and weakly diagonally dominant. If in addition $\left d_1\right >\left c_1\right $ and
lyche-numerical-linear-algebra-F00729	58	■ 0.574	$a_{\{i\}} \neq 0$	$a_i \neq 0$	$a_i \neq 0$ for $i=1, \dots, n-2$, then $\boldsymbol{A}$ has a unique LU factorization (2.15). If in
lyche-numerical-linear-algebra-F00730	58	■ 0.574	$i=1, \ldots, n-2$	$i=1, \dots, n-2$	$a_i \neq 0$ for $i=1, \dots, n-2$, then $\boldsymbol{A}$ has a unique LU factorization (2.15). If in
lyche-numerical-linear-algebra-F00731	58	■ 0.574	L U	LU	$a_i \neq 0$ for $i=1, \dots, n-2$, then $\boldsymbol{A}$ has a unique LU factorization (2.15). If in
lyche-numerical-linear-algebra-F00732	58	■ 1.000	$d_{\{n\}} \neq 0$	$d_n \neq 0$	addition $d_n \neq 0$, then $\boldsymbol{A}$ is nonsingular.
lyche-numerical-linear-algebra-F00733	58	■ 0.999	$k=1, \ldots, n-1$	$k=1, \dots, n-1$	factorization if the u_k 's in (2.16) are nonzero for $k=1, \dots, n-1$. For this it is
lyche-numerical-linear-algebra-F00734	58	■ 0.991	$\left u_{\{k\}}\right >\left c_{\{k\}}\right $	$\left u_k\right >\left c_k\right $	sufficient to show by induction that $\left u_k\right >\left c_k\right $ for $k=1, \dots, n-1$. By assumption
lyche-numerical-linear-algebra-F00735	58	■ 0.798	$\left u_{\{1\}}\right =\left d_{\{1\}}\right >\left c_{\{1\}}\right $	$\left u_1\right =\left d_1\right >\left c_1\right $	$\left u_1\right =\left d_1\right >\left c_1\right $. Suppose $\left u_k\right >\left c_k\right $ for some $1 \leq k \leq n-2$. Then $\left c_k\right /\left u_k\right <1$
lyche-numerical-linear-algebra-F00736	58	■ 0.798	$1 \leq k \leq n-2$	$1 \leq k \leq n-2$	$\left u_1\right =\left d_1\right >\left c_1\right $. Suppose $\left u_k\right >\left c_k\right $ for some $1 \leq k \leq n-2$. Then $\left c_k\right /\left u_k\right <1$
lyche-numerical-linear-algebra-F00737	58	■ 1.000	$k=n-1$	$k=n-1$	This also holds for $k=n-1$ if $a_{n-1} \neq 0$. By (2.23) and weak diagonal
lyche-numerical-linear-algebra-F00738	58	■ 1.000	$a_{\{n-1\}} \neq 0$	$a_{n-1} \neq 0$	This also holds for $k=n-1$ if $a_{n-1} \neq 0$. By (2.23) and weak diagonal
lyche-numerical-linear-algebra-F00739	58	■ 1.000	$\left u_{\{k+1\}}\right >\left d_{\{k+1\}}\right -\left a_{\{k\}}\right \geq\left c_{\{k+1\}}\right $	$\left u_{k+1}\right >\left d_{k+1}\right -\left a_k\right \geq\left c_{k+1}\right $	dominance $\left u_{k+1}\right >\left d_{k+1}\right -\left a_k\right \geq\left c_{k+1}\right $ and it follows by induction that an
lyche-numerical-linear-algebra-F00740	58	■ 1.000	$\left u_{\{n\}}\right >\left d_{\{n\}}\right -\left a_{\{n-1\}}\right \geq 0$	$\left u_n\right >\left d_n\right -\left a_{n-1}\right \geq 0$	$a_{n-1} \neq 0$ then by (2.16) $\left u_n\right >\left d_n\right -\left a_{n-1}\right >0$ by weak diagonal dominance,
lyche-numerical-linear-algebra-F00741	58	■ 1.000	$a_{\{n-1\}}=0$	$a_{n-1}=0$	while if $a_{n-1}=0$ then again by (2.23) $\left u_n\right \geq\left d_n\right >0$.
lyche-numerical-linear-algebra-F00742	58	■ 1.000	$\left u_{\{n\}}\right \geq\left d_{\{n\}}\right >0$	$\left u_n\right \geq\left d_n\right >0$	while if $a_{n-1}=0$ then again by (2.23) $\left u_n\right \geq\left d_n\right >0$.
lyche-numerical-linear-algebra-F00743	58	■ 1.000	$\boldsymbol{T} \boldsymbol{v}=\boldsymbol{b}$	$\boldsymbol{T} \boldsymbol{v}=\boldsymbol{b}$	Consider now the special system $\boldsymbol{T} \boldsymbol{v}=\boldsymbol{b}$ given by (2.21). The matrix $\boldsymbol{T}$ is weakly

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lyche-numerical-linear-algebra_F00744	58	■ 1.000	$\boldsymbol{v}=\boldsymbol{T}^{-1} \boldsymbol{b}$	$\boldsymbol{v} = \boldsymbol{T}^{-1}\boldsymbol{b}$	solution of $\boldsymbol{T}\boldsymbol{v} = \boldsymbol{b}$ as $\boldsymbol{v} = \boldsymbol{T}^{-1}\boldsymbol{b}$. However, this is not a good idea. In fact, all
lyche-numerical-linear-algebra_F00745	58	■ 1.000	$\boldsymbol{T}^{-1}$	$\boldsymbol{T}^{-1}$	elements in $\boldsymbol{T}^{-1}$ are nonzero and the calculation of $\boldsymbol{T}^{-1}\boldsymbol{b}$ requires $O(n^2)$ operations.
lyche-numerical-linear-algebra_F00746	58	■ 1.000	$\boldsymbol{T}^{-1} \boldsymbol{b}$	$\boldsymbol{T}^{-1}\boldsymbol{b}$	elements in $\boldsymbol{T}^{-1}$ are nonzero and the calculation of $\boldsymbol{T}^{-1}\boldsymbol{b}$ requires $O(n^2)$ operations.
lyche-numerical-linear-algebra_F00747	58	■ 1.000	$O(n^2)$	$O(n^2)$	elements in $\boldsymbol{T}^{-1}$ are nonzero and the calculation of $\boldsymbol{T}^{-1}\boldsymbol{b}$ requires $O(n^2)$ operations.
lyche-numerical-linear-algebra_F00748	59	■ 1.000	$\boldsymbol{x} \in$	$\boldsymbol{x} \in$	Recall that if $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is a square matrix and $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ for some nonzero $\boldsymbol{x} \in$
lyche-numerical-linear-algebra_F00749	59	■ 1.000	L	L	Consider a horizontal beam of length L located between 0 and L on the x -axis of
lyche-numerical-linear-algebra_F00750	59	■ 1.000	$x=0$	$x = 0$	the plane. We assume that the beam is fixed at $x = 0$ and $x = L$ and that a force F is
lyche-numerical-linear-algebra_F00751	59	■ 1.000	$x=L$	$x = L$	the plane. We assume that the beam is fixed at $x = 0$ and $x = L$ and that a force F is
lyche-numerical-linear-algebra_F00752	59	■ 1.000	F	F	the plane. We assume that the beam is fixed at $x = 0$ and $x = L$ and that a force F is
lyche-numerical-linear-algebra_F00753	59	■ 1.000	$(L, 0)$	$(L, 0)$	applied at $(L, 0)$ in the direction towards the origin. This situation can be modeled
lyche-numerical-linear-algebra_F00754	59	■ 1.000	$y(x)$	$y(x)$	where $y(x)$ is the vertical displacement of the beam at x , and R is a constant defined
lyche-numerical-linear-algebra_F00755	59	■ 1.000	R	R	where $y(x)$ is the vertical displacement of the beam at x , and R is a constant defined
lyche-numerical-linear-algebra_F00756	59	■ 1.000	$u:[0,1] \rightarrow \mathbb{R}$	$u : [0, 1] \rightarrow \mathbb{R}$	by considering the function $u : [0, 1] \rightarrow \mathbb{R}$ given by $u(t) := y(tL)$. Since $u''(t) =$
lyche-numerical-linear-algebra_F00757	59	■ 1.000	$u(t):=y(t L)$	$u(t) := y(tL)$	by considering the function $u : [0, 1] \rightarrow \mathbb{R}$ given by $u(t) := y(tL)$. Since $u''(t) =$
lyche-numerical-linear-algebra_F00758	59	■ 1.000	$u^{\prime \prime}(t)=$	$u''(t) =$	by considering the function $u : [0, 1] \rightarrow \mathbb{R}$ given by $u(t) := y(tL)$. Since $u''(t) =$
lyche-numerical-linear-algebra_F00759	59	■ 1.000	$L^2 y^{\prime \prime}(t L)$	$L^2 y''(tL)$	$L^2 y''(tL)$, the problem (2.24) then becomes
lyche-numerical-linear-algebra_F00760	59	■ 1.000	$u=0$	$u = 0$	Clearly $u = 0$ is a solution, but we can have nonzero solutions corresponding
lyche-numerical-linear-algebra_F00761	59	■ 1.000	$F=0$	$F = 0$	called an eigenfunction. If $F = 0$ then $K = 0$ and $u = 0$ is the only solution, but
lyche-numerical-linear-algebra_F00762	59	■ 1.000	$K=0$	$K = 0$	called an eigenfunction. If $F = 0$ then $K = 0$ and $u = 0$ is the only solution, but
lyche-numerical-linear-algebra_F00763	59	■ 1.000	$u(t)=\sin (\pi t)$	$u(t) = \sin(\pi t)$	With $u(t) = \sin(\pi t)$ we find $u''(t) = -\pi^2 u(t)$ and this u is a solution if $K = \pi^2$.
lyche-numerical-linear-algebra_F00764	59	■ 1.000	$u^{\prime \prime}(t)=-\pi^2 u(t)$	$u''(t) = -\pi^2 u(t)$	With $u(t) = \sin(\pi t)$ we find $u''(t) = -\pi^2 u(t)$ and this u is a solution if $K = \pi^2$.
lyche-numerical-linear-algebra_F00765	59	■ 1.000	$K=\pi^2$	$K = \pi^2$	With $u(t) = \sin(\pi t)$ we find $u''(t) = -\pi^2 u(t)$ and this u is a solution if $K = \pi^2$.
lyche-numerical-linear-algebra_F00766	59	■ 1.000	$F=\frac{\pi^2}{L^2} R$	$F = \frac{\pi^2 R}{L^2}$	find $F = \frac{\pi^2 R}{L^2}$.
lyche-numerical-linear-algebra_F00767	59	■ 1.000	$m \in \mathbb{N}, h:=1 /(m+$	$m \in \mathbb{N}, h := 1/(m+$	We can approximate this eigenvalue numerically. Choosing $m \in \mathbb{N}, h := 1/(m+$
lyche-numerical-linear-algebra_F00768	60	■ 1.000	$v_j \approx u(j h)$	$v_j \approx u(jh)$	where $v_j \approx u(jh)$ for $j = 0, \dots, m+1$. If we define $\lambda := h^2 K$ then we obtain the
lyche-numerical-linear-algebra_F00769	60	■ 1.000	$j=0, \ldots, m+1$	$j = 0, \dots, m+1$	where $v_j \approx u(jh)$ for $j = 0, \dots, m+1$. If we define $\lambda := h^2 K$ then we obtain the
lyche-numerical-linear-algebra_F00770	60	■ 1.000	$\lambda:=h^2 K$	$\lambda := h^2 K$	where $v_j \approx u(jh)$ for $j = 0, \dots, m+1$. If we define $\lambda := h^2 K$ then we obtain the

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lyche-numerical-linear-algebra-F00771	60	1.000	$\lambda=4 \sin ^{2}(\pi h / 2)$	$\lambda = 4 \sin ^2(\pi h/2)$	that the smallest eigenvalue of (2.26) is given by $\lambda = 4 \sin ^2(\pi h/2)$. Since $\lambda =$
lyche-numerical-linear-algebra-F00772	60	1.000	$\lambda=$	$\lambda =$	that the smallest eigenvalue of (2.26) is given by $\lambda = 4 \sin ^2(\pi h/2)$. Since $\lambda =$
lyche-numerical-linear-algebra-F00773	60	1.000	$h^{2} K=\frac{h^{2} F L^{2}}{R}$	$h^2 K = \frac{h^2 F L^2}{R}$	$h^2 K = \frac{h^2 F L^2}{R}$ we can solve for F to obtain
lyche-numerical-linear-algebra-F00774	60	1.000	h	h	For small h this is a good approximation to the value $\frac{\pi^2 R}{L^2}$ we computed above.
lyche-numerical-linear-algebra-F00775	60	1.000	$\frac{\pi^{2} R}{L^{2}}$	$\frac{\pi^2 R}{L^2}$	For small h this is a good approximation to the value $\frac{\pi^2 R}{L^2}$ we computed above.
lyche-numerical-linear-algebra-F00776	60	0.997	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1)$	$\boldsymbol{T} = \operatorname{tridiag}(-1, 2, -1)$	The second derivative matrix $\boldsymbol{T} = \operatorname{tridiag}(-1, 2, -1)$ is a special case of the
lyche-numerical-linear-algebra-F00777	60	0.990	$a, d \in \mathbb{R}$	$a, d \in \mathbb{R}$	where $a, d \in \mathbb{R}$. We call this the 1D test matrix . It is symmetric and strictly
lyche-numerical-linear-algebra-F00778	60	0.990	$\mathbf{1 D}$	1D	where $a, d \in \mathbb{R}$. We call this the 1D test matrix . It is symmetric and strictly
lyche-numerical-linear-algebra-F00779	60	1.000	$ d >2 a $	$ d > 2 a $	diagonally dominant if $ d > 2 a $.
lyche-numerical-linear-algebra-F00780	61	1.000	$m=3$	$m = 3$	For $m = 3$,
lyche-numerical-linear-algebra-F00781	61	1.000	$\boldsymbol{T}_{1}=\left(t_{k j}\right)_{k, j}=$	$\boldsymbol{T}_1 = (t_{kj})_{k,j} =$	Lemma 2.2 (Eigenpairs of 1D Test Matrix) Suppose $\boldsymbol{T}_1 = (t_{kj})_{k,j} =$
lyche-numerical-linear-algebra-F00782	61	0.999	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ with $m \geq 2$, $a, d \in \mathbb{R}$, and let $h = 1/(m + 1)$.
lyche-numerical-linear-algebra-F00783	61	0.999	$m \geq 2, a, d \in \mathbb{R}$	$m \geq 2, a, d \in \mathbb{R}$	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ with $m > 2, a, d \in \mathbb{R}$ and let $h = 1/(m + 1)$.
lyche-numerical-linear-algebra-F00784	61	0.999	$h=1 /(m+1)$	$h = 1/(m + 1)$	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ with $m \geq 2$, $a, d \in \mathbb{R}$, and let $h = 1/(m + 1)$.
lyche-numerical-linear-algebra-F00785	61	1.000	$\boldsymbol{T}_{1} \boldsymbol{s}_j=\lambda_j \boldsymbol{s}_j$	$\boldsymbol{T}_1 \boldsymbol{s}_j = \lambda_j \boldsymbol{s}_j$	1. We have $\boldsymbol{T}_1 \boldsymbol{s}_j = \lambda_j \boldsymbol{s}_j$ for $j = 1, \dots, m$, where
lyche-numerical-linear-algebra-F00786	61	1.000	$1 < k < m$	$1 < k < m$	Proof We find for $1 < k < m$
lyche-numerical-linear-algebra-F00787	61	1.000	$k=1, m$	$k = 1, m$	This also holds for $k = 1, m$, and part 1 follows. Since $j \pi h = j \pi/(m + 1) \in (0, \pi)$
lyche-numerical-linear-algebra-F00788	61	1.000	$j \pi h=j \pi /(m+1) \in(0, \pi)$	$j \pi h = j \pi/(m + 1) \in (0, \pi)$	This also holds for $k = 1, m$, and part 1 follows. Since $j \pi h = j \pi/(m + 1) \in (0, \pi)$
lyche-numerical-linear-algebra-F00789	61	0.988	$(0, \pi)$	$(0, \pi)$	for $j = 1, \dots, m$ and the cosine function is strictly monotone decreasing on $(0, \pi)$
lyche-numerical-linear-algebra-F00790	61	1.000	$\boldsymbol{T}_{1}$	$\boldsymbol{T}_1$	the eigenvalues are distinct, and since $\boldsymbol{T}_1$ is symmetric it follows from Lemma 2.3
lyche-numerical-linear-algebra-F00791	62	1.000	$i=\sqrt{-1}$	$i = \sqrt{-1}$	complex exponentials. With $i = \sqrt{-1}$ we find
lyche-numerical-linear-algebra-F00792	62	1.000	$\boldsymbol{A}^{*}:=\overline{\boldsymbol{A}}^T$	$\boldsymbol{A}^* := \overline{\boldsymbol{A}}^T$	Recall that the conjugate transpose of a matrix is defined by $\boldsymbol{A}^* := \overline{\boldsymbol{A}}^T$, where
lyche-numerical-linear-algebra-F00793	62	1.000	$\boldsymbol{x} \neq \mathbf{0}$	$\boldsymbol{x} \neq \mathbf{0}$	Proof Suppose $\boldsymbol{A}^* = \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ with $\boldsymbol{x} \neq \mathbf{0}$. We multiply both sides of
lyche-numerical-linear-algebra-F00794	62	0.998	$\boldsymbol{x}^{*}$	$\boldsymbol{x}^*$	$\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ from the left by $\boldsymbol{x}^*$ and divide by $\boldsymbol{x}^* \boldsymbol{x}$ to obtain $\lambda = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$. Taking
lyche-numerical-linear-algebra-F00795	62	0.998	$\boldsymbol{x}^{*} \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{x}$	$\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ from the left by $\boldsymbol{x}^*$ and divide by $\boldsymbol{x}^* \boldsymbol{x}$ to obtain $\lambda = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$. Taking
lyche-numerical-linear-algebra-F00796	62	0.998	$\lambda=\frac{\boldsymbol{x}^{*} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{*} \boldsymbol{x}}$	$\lambda = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$	$\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ from the left by $\boldsymbol{x}^*$ and divide by $\boldsymbol{x}^* \boldsymbol{x}$ to obtain $\lambda = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$. Taking

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lyche-numerical-linear-algebra-F00797	62	■ 0.996	$\overline{\lambda} = \lambda^* = \frac{(x^*Ax)^*}{(x^*x)^*} = \frac{x^*A^*x}{x^*x} = \frac{x^*Ax}{x^*x} = \lambda$	$\bar{\lambda} = \lambda^* = \frac{(x^*Ax)^*}{(x^*x)^*} = \frac{x^*A^*x}{x^*x} = \frac{x^*Ax}{x^*x} = \lambda$	complex conjugates we find $\bar{\lambda} = \lambda^* = \frac{(x^*Ax)^*}{(x^*x)^*} = \frac{x^*A^*x}{x^*x} = \frac{x^*Ax}{x^*x} = \lambda$, and λ is real.
lyche-numerical-linear-algebra-F00798	62	■ 0.725	$(\mu, \mathbf{y})$	$(\mu, \mathbf{y})$	Suppose that $(\lambda, \mathbf{x})$ and $(\mu, \mathbf{y})$ are two eigenpairs for A with $\mu \neq \lambda$. Multiplying
lyche-numerical-linear-algebra-F00799	62	■ 0.725	$\mu \neq \lambda$	$\mu \neq \lambda$	Suppose that $(\lambda, \mathbf{x})$ and $(\mu, \mathbf{y})$ are two eigenpairs for A with $\mu \neq \lambda$. Multiplying
lyche-numerical-linear-algebra-F00800	62	■ 1.000	$\mathbf{y}^*$	$\mathbf{y}^*$	$A\mathbf{x} = \lambda\mathbf{x}$ by $\mathbf{y}^*$ gives
lyche-numerical-linear-algebra-F00801	62	■ 1.000	μ	μ	using that μ is real. Since $\lambda \neq \mu$ it follows that $\mathbf{y}^*\mathbf{x} = 0$, which means that $\mathbf{x}$ and $\mathbf{y}$
lyche-numerical-linear-algebra-F00802	62	■ 1.000	$\lambda \neq \mu$	$\lambda \neq \mu$	using that μ is real. Since $\lambda \neq \mu$ it follows that $\mathbf{y}^*\mathbf{x} = 0$, which means that $\mathbf{x}$ and $\mathbf{y}$
lyche-numerical-linear-algebra-F00803	62	■ 1.000	$\mathbf{y}^*\mathbf{x} = 0$	$\mathbf{y}^*\mathbf{x} = 0$	using that μ is real. Since $\lambda \neq \mu$ it follows that $\mathbf{y}^*\mathbf{x} = 0$, which means that $\mathbf{x}$ and $\mathbf{y}$
lyche-numerical-linear-algebra-F00804	62	■ 1.000	$\mathbf{y}$	$\mathbf{y}$	using that μ is real. Since $\lambda \neq \mu$ it follows that $\mathbf{y}^*\mathbf{x} = 0$, which means that $\mathbf{x}$ and $\mathbf{y}$
lyche-numerical-linear-algebra-F00805	63	■ 0.997	$\left[\begin{array}{ll} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{array} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$\left[\begin{array}{cc cc} \mathbf{A}_{11} & \mathbf{A}_{12} & & \\ \mathbf{A}_{21} & \mathbf{A}_{22} & & \end{array} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	(i) $\left[\begin{array}{cc cc} \mathbf{A}_{11} & \mathbf{A}_{12} & & \\ \mathbf{A}_{21} & \mathbf{A}_{22} & & \end{array} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right],$
lyche-numerical-linear-algebra-F00806	63	■ 0.949	$\left[\mathbf{a}_{:1}, \mathbf{a}_{:2}, \mathbf{a}_{:3} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$[\mathbf{a}_{:1}, \mathbf{a}_{:2}, \mathbf{a}_{:3}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	(ii) $[\mathbf{a}_{:1}, \mathbf{a}_{:2}, \mathbf{a}_{:3}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right],$
lyche-numerical-linear-algebra-F00807	63	■ 0.125	$\left[\begin{array}{c} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{array} \right] = \left[\begin{array}{c} 123 \\ 45 \\ 789 \end{array} \right]$	$\left[\begin{array}{c} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{array} \right] = \left[\begin{array}{c} 123 \\ 45 \\ 789 \end{array} \right]$	(iii) $\left[\begin{array}{c} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{array} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right],$
lyche-numerical-linear-algebra-F00808	63	■ 0.673	$\left[\mathbf{A}_{11}, \mathbf{A}_{12} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$[\mathbf{A}_{11}, \mathbf{A}_{12}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	(iv) $[\mathbf{A}_{11}, \mathbf{A}_{12}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right].$
lyche-numerical-linear-algebra-F00809	63	■ 0.902	(i i)	(ii)	while in (ii) and (iii) A has been partitioned into columns and rows, respectively.
lyche-numerical-linear-algebra-F00810	63	■ 0.902	(i i i) $\mathbf{A}$	(iii) $\mathbf{A}$	while in (ii) and (iii) A has been partitioned into columns and rows, respectively.
lyche-numerical-linear-algebra-F00811	63	■ 1.000	$\mathbf{A} \in \mathbb{C}^{m \times p}$	$\mathbf{A} \in \mathbb{C}^{m \times p}$	In the following we assume that $\mathbf{A} \in \mathbb{C}^{m \times p}$ and $\mathbf{B} \in \mathbb{C}^{p \times n}$. Here are some rules
lyche-numerical-linear-algebra-F00812	63	■ 1.000	$\mathbf{B} \in \mathbb{C}^{p \times n}$	$\mathbf{B} \in \mathbb{C}^{p \times n}$	In the following we assume that $\mathbf{A} \in \mathbb{C}^{m \times p}$ and $\mathbf{B} \in \mathbb{C}^{p \times n}$. Here are some rules
lyche-numerical-linear-algebra-F00813	63	■ 1.000	$\mathbf{B} = [\mathbf{b}_{:1}, \dots, \mathbf{b}_{:n}]$	$\mathbf{B} = [\mathbf{b}_{:1}, \dots, \mathbf{b}_{:n}]$	1. If $\mathbf{B} = [\mathbf{b}_{:1}, \dots, \mathbf{b}_{:n}]$ is partitioned into columns then the partition of the product

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lyche-numerical-linear-algebra-F00814	63	1.000	$\boldsymbol{A} \boldsymbol{e}_j$	Ae_j	and we see that column j of A can be written Ae_j for $j = 1, \dots, p$.
lyche-numerical-linear-algebra-F00815	63	1.000	$j=1, \ldots, p$	$j = 1, \dots, p$	and we see that column j of A can be written Ae_j for $\boxed{j = 1, \dots, p}$.
lyche-numerical-linear-algebra-F00816	63	1.000	$\boldsymbol{A}=\boldsymbol{I}$	$A = I$	and taking $A = I$ it follows that row i of B can be written $e_i^T B$ for $i = 1, \dots, m$.
lyche-numerical-linear-algebra-F00817	63	1.000	$\boldsymbol{e}_i^T \boldsymbol{B}$	$e_i^T B$	and taking $A = I$ it follows that row i of B can be written $e_i^T B$ for $i = 1, \dots, m$.
lyche-numerical-linear-algebra-F00818	63	1.000	$\boldsymbol{A} \boldsymbol{x}$	Ax	3. It is often useful to write the matrix-vector product Ax as a linear combination
lyche-numerical-linear-algebra-F00819	64	1.000	$\boldsymbol{B}=\left[\boldsymbol{B}_1, \boldsymbol{B}_2\right]$	$B = [B_1, B_2]$	4. If $B = [B_1, B_2]$, where $B_1 \in \mathbb{C}^{p \times r}$ and $B_2 \in \mathbb{C}^{p \times (n-r)}$ then
lyche-numerical-linear-algebra-F00820	64	1.000	$\boldsymbol{B}_1 \in \mathbb{C}^{p \times r}$	$B_1 \in \mathbb{C}^{p \times r}$	4. If $B = [B_1, B_2]$, where $B_1 \in \mathbb{C}^{p \times r}$ and $B_2 \in \mathbb{C}^{p \times (n-r)}$ then
lyche-numerical-linear-algebra-F00821	64	1.000	$\boldsymbol{B}_2 \in \mathbb{C}^{p \times (n-r)}$	$B_2 \in \mathbb{C}^{p \times (n-r)}$	4. If $B = [B_1, B_2]$, where $B_1 \in \mathbb{C}^{p \times r}$ and $B_2 \in \mathbb{C}^{p \times (n-r)}$ then
lyche-numerical-linear-algebra-F00822	64	0.993	$\boldsymbol{A}=\left[\begin{array}{l} \boldsymbol{A}_1 \\ \boldsymbol{A}_2 \end{array}\right]$	$A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$	5. If $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{k \times p}$ and $A_2 \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra-F00823	64	0.993	$\boldsymbol{A}_1 \in \mathbb{C}^{k \times p}$	$A_1 \in \mathbb{C}^{k \times p}$	5. If $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{k \times p}$ and $A_2 \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra-F00824	64	0.993	$\boldsymbol{A}_2 \in \mathbb{C}^{(m-k) \times p}$	$A_2 \in \mathbb{C}^{(m-k) \times p}$	5. If $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{k \times p}$ and $A_2 \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra-F00825	64	0.985	$\boldsymbol{A}=\left[\boldsymbol{A}_1, \boldsymbol{A}_2\right]$	$A = [A_1, A_2]$	6. If $A = [A_1, A_2]$ and $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{m \times s}$, $A_2 \in \mathbb{C}^{m \times (p-s)}$, $B_1 \in$
lyche-numerical-linear-algebra-F00826	64	0.985	$\boldsymbol{B}=\left[\begin{array}{l} \boldsymbol{B}_1 \\ \boldsymbol{B}_2 \end{array}\right]$	$B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$	6. If $A = [A_1, A_2]$ and $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{m \times s}$, $A_2 \in \mathbb{C}^{m \times (p-s)}$, $B_1 \in$
lyche-numerical-linear-algebra-F00827	64	0.985	$\boldsymbol{A}_1 \in \mathbb{C}^{m \times s}, \boldsymbol{A}_2 \in \mathbb{C}^{m \times (p-s)}, \boldsymbol{B}_1 \in$	$A_1 \in \mathbb{C}^{m \times s}, A_2 \in \mathbb{C}^{m \times (p-s)}, B_1 \in$	6. If $A = [A_1, A_2]$ and $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, where $\boxed{A_1 \in \mathbb{C}^{m \times s}, A_2 \in \mathbb{C}^{m \times (p-s)}, B_1 \in}$
lyche-numerical-linear-algebra-F00828	64	1.000	$\mathbb{C}^{s \times n}$	$\mathbb{C}^{s \times n}$	$\mathbb{C}^{s \times n}$ and $B_2 \in \mathbb{C}^{(p-s) \times n}$ then
lyche-numerical-linear-algebra-F00829	64	1.000	$\boldsymbol{B}_2 \in \mathbb{C}^{(p-s) \times n}$	$B_2 \in \mathbb{C}^{(p-s) \times n}$	$\mathbb{C}^{s \times n}$ and $\boxed{B_2 \in \mathbb{C}^{(p-s) \times n}}$ then
lyche-numerical-linear-algebra-F00830	64	1.000	$\boldsymbol{A}=\left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right]$	$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$	7. If $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$ then
lyche-numerical-linear-algebra-F00831	64	1.000	$\boldsymbol{B}=\left[\begin{array}{ll} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{array}\right]$	$B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$	7. If $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$ then
lyche-numerical-linear-algebra-F00832	64	1.000	$\boldsymbol{A}_{11}$	A_{11}	number of columns in A_{11} and A_{21} equals the number of rows in B_{11} and B_{12}
lyche-numerical-linear-algebra-F00833	64	1.000	$\boldsymbol{A}_{21}$	A_{21}	number of columns in A_{11} and A_{21} equals the number of rows in B_{11} and B_{12}
lyche-numerical-linear-algebra-F00834	64	1.000	$\boldsymbol{B}_{11}$	B_{11}	number of columns in A_{11} and A_{21} equals the number of rows in B_{11} and B_{12}

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lyche-numerical-linear-algebra-F00835	64	■ 1.000	$\boldsymbol{B}_{12}$	B_{12}	number of columns in A_{11} and A_{21} equals the number of rows in B_{11} and B_{12}
lyche-numerical-linear-algebra-F00836	65	■ 1.000	$\boldsymbol{A}_{ik} \boldsymbol{B}_{kj}$	$A_{ik} B_{kj}$	8. Consider finally the general case. If all the matrix products $A_{ik} B_{kj}$ in
lyche-numerical-linear-algebra-F00837	65	■ 1.000	$\boldsymbol{A}, \boldsymbol{A}_{11}$	A, A_{11}	where A, A_{11} and A_{22} are square matrices. Then A is nonsingular if and only if
lyche-numerical-linear-algebra-F00838	65	■ 1.000	$\boldsymbol{A}_{22}$	A_{22}	where A, A_{11} and A_{22} are square matrices. Then A is nonsingular if and only if
lyche-numerical-linear-algebra-F00839	65	■ 1.000	$\boldsymbol{B} := \boldsymbol{A}^{-1}$	$B := A^{-1}$	Proof Suppose A is nonsingular. We partition $B := A^{-1}$ conformally with A and
lyche-numerical-linear-algebra-F00840	66	■ 1.000	$\boldsymbol{B}_{21} =$	$B_{21} =$	The first equation implies that A_{11} is nonsingular, this in turn implies that $B_{21} =$
lyche-numerical-linear-algebra-F00841	66	■ 0.577	$\boldsymbol{0} \boldsymbol{A}_{11}^{-1} = \boldsymbol{0}$	$0A_{11}^{-1} = 0$	$0A_{11}^{-1} = 0$ in the second equation, and then the third equation simplifies to
lyche-numerical-linear-algebra-F00842	66	■ 1.000	$\boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}$	$B_{22} A_{22} = I$	$B_{22} A_{22} = I$. We conclude that also A_{22} is nonsingular. From the fourth equation
lyche-numerical-linear-algebra-F00843	66	■ 1.000	a_{ii}	a_{ii}	$A = [a_{ij}] \in \mathbb{C}^{n \times n}$ is nonsingular if and only if the diagonal elements a_{ii} ,
lyche-numerical-linear-algebra-F00844	66	■ 1.000	$a_{ii}^{-1}, i=1, \ldots, n$	$a_{ii}^{-1}, i=1, \ldots, n$	diagonal elements $a_{ii}^{-1}, i=1, \ldots, n$.
lyche-numerical-linear-algebra-F00845	66	■ 0.785	a_{11}	a_{11}	$[a_{11}]$ is nonsingular if and only if $a_{11} \neq 0$ and in that case $A^{-1} = [a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra-F00846	66	■ 0.785	$a_{11} \neq 0$	$a_{11} \neq 0$	$[a_{11}]$ is nonsingular if and only if $a_{11} \neq 0$ and in that case $A^{-1} = [a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra-F00847	66	■ 0.785	$\boldsymbol{A}^{-1} = [a_{11}^{-1}]$	$A^{-1} = [a_{11}^{-1}]$	$[a_{11}]$ is nonsingular if and only if $a_{11} \neq 0$ and in that case $A^{-1} = [a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra-F00848	66	■ 1.000	$\boldsymbol{A} \in \mathbb{C}^{(k+1) \times (k+1)}$	$A \in \mathbb{C}^{(k+1) \times (k+1)}$	the result holds for $n = k$ and let $A \in \mathbb{C}^{(k+1) \times (k+1)}$ be upper triangular. We partition
lyche-numerical-linear-algebra-F00849	66	■ 1.000	$\boldsymbol{A}_k \in \mathbb{C}^{k \times k}$	$A_k \in \mathbb{C}^{k \times k}$	and note that $A_k \in \mathbb{C}^{k \times k}$ is upper triangular. By Lemma 2.4 A is nonsingular if and
lyche-numerical-linear-algebra-F00850	66	■ 1.000	2.4 $\boldsymbol{A}$	2.4 A	and note that $A_k \in \mathbb{C}^{k \times k}$ is upper triangular. By Lemma 2.4 A is nonsingular if and
lyche-numerical-linear-algebra-F00851	66	■ 1.000	$\boldsymbol{A}_k$	A_k	only if A_k and $(a_{k+1, k+1})$ are nonsingular and in that case
lyche-numerical-linear-algebra-F00852	66	■ 1.000	$(a_{k+1, k+1})$	$(a_{k+1, k+1})$	only if A_k and $(a_{k+1, k+1})$ are nonsingular and in that case
lyche-numerical-linear-algebra-F00853	66	■ 1.000	$\boldsymbol{c} \in \mathbb{C}^n$	$c \in \mathbb{C}^n$	for some $c \in \mathbb{C}^n$. By the induction hypothesis A_k is nonsingular if and only if the
lyche-numerical-linear-algebra-F00854	66	■ 1.000	$a_{11}, \ldots, a_{kk}$	$a_{11}, \ldots, a_{kk}$	diagonal elements $a_{11}, \ldots, a_{kk}$ of A_k are nonzero and in that case A_k^{-1} is upper
lyche-numerical-linear-algebra-F00855	66	■ 1.000	$\boldsymbol{A}_k^{-1}$	A_k^{-1}	diagonal elements $a_{11}, \ldots, a_{kk}$ of A_k are nonzero and in that case A_k^{-1} is upper
lyche-numerical-linear-algebra-F00856	66	■ 1.000	$a_{ii}^{-1}, i=1, \ldots, k$	$a_{ii}^{-1}, i=1, \ldots, k$	triangular with diagonal elements $a_{ii}^{-1}, i=1, \ldots, k$. The result for A follows.
lyche-numerical-linear-algebra-F00857	66	■ 0.515	$\boldsymbol{C} = \boldsymbol{A} \boldsymbol{B} = (c_{ij})$	$C = AB = (c_{ij})$	Lemma 2.6 (Product of Triangular Matrices) The product $C = AB = (c_{ij})$ of
lyche-numerical-linear-algebra-F00858	66	■ 1.000	$\boldsymbol{A} = (a_{ij})$	$A = (a_{ij})$	two upper (lower) triangular matrices $A = (a_{ij})$ and $B = (b_{ij})$ is upper (lower)
lyche-numerical-linear-algebra-F00859	66	■ 1.000	$\boldsymbol{B} = (b_{ij})$	$B = (b_{ij})$	two upper (lower) triangular matrices $A = (a_{ij})$ and $B = (b_{ij})$ is upper (lower)
lyche-numerical-linear-algebra-F00860	66	■ 1.000	$c_{ii} = a_{ii} b_{ii}$	$c_{ii} = a_{ii} b_{ii}$	triangular with diagonal elements $c_{ii} = a_{ii} b_{ii}$ for all i .

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lyche-numerical-linear-algebra_F00861	67	0.822	$\left\{\left(x-x_i\right)^j\right\}_{0 \leq j \leq n}$	$\left\{\left(x-x_i\right)^j\right\}_{0 \leq j \leq n}$	$\left\{\left(x-x_i\right)^j\right\}_{0 \leq j \leq n}$ is a basis for polynomials of degree n . ³
lyche-numerical-linear-algebra_F00862	67	0.822	$n .^{\{3\}}$	$n .^3$	$\left\{\left(x-x_i\right)^j\right\}_{0 \leq j \leq n}$ is a basis for polynomials of degree n . ³
lyche-numerical-linear-algebra_F00863	67	1.000	$D_{\{1\}}$	D_1	pose instead of the D_2 boundary conditions we use D_1 conditions given by $g'(a) =$
lyche-numerical-linear-algebra_F00864	67	1.000	$g^{\{\prime\}}(a)=$	$g'(a) =$	pose instead of the D_2 boundary conditions we use D_1 conditions given by $g'(a) =$
lyche-numerical-linear-algebra_F00865	67	1.000	$s_{\{1\}}$	s_1	s_1 and $g'(b) = s_{n+1}$ for some given numbers s_1 and s_{n+1} . Show that the linear
lyche-numerical-linear-algebra_F00866	67	1.000	$g^{\{\prime\}}(b)=s_{\{n+1\}}$	$g'(b) = s_{n+1}$	s_1 and $g'(b) = s_{n+1}$ for some given numbers s_1 and s_{n+1} . Show that the linear
lyche-numerical-linear-algebra_F00867	67	1.000	$s_{\{n+1\}}$	s_{n+1}	s_1 and $g'(b) = s_{n+1}$ for some given numbers s_1 and s_{n+1} . Show that the linear
lyche-numerical-linear-algebra_F00868	67	1.000	$\delta^2 y_{\{i\}}:=y_{\{i+1\}}-2 y_{\{i\}}+y_{\{i-1\}}, i=2, \ldots, n$	$\delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}, i = 2, \dots, n$	where $\delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}, i = 2, \dots, n$. Hint: Use (2.10) to compute $g'(x_1)$
lyche-numerical-linear-algebra_F00869	67	1.000	$g^{\{\prime\}}\left(x_{\{1\}}\right)$	$g'(x_1)$	where $\delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}, i = 2, \dots, n$. Hint: Use (2.10) to compute $g'(x_1)$
lyche-numerical-linear-algebra_F00870	67	1.000	$g^{\{\prime\}}\left(x_{\{n+1\}}\right)$	$g'(x_{n+1})$	and $g'(x_{n+1})$, Is g unique?
lyche-numerical-linear-algebra_F00871	67	1.000	$x_{\{i\}}$	x_i	³ Hint: consider an arbitrary polynomial of degree n and expanded it in Taylor series around x_i .
lyche-numerical-linear-algebra_F00872	68	0.999	$h \in C^2[a, b]$	$h \in C^2[a, b]$	for all $h \in C^2[a, b]$ such that $h(x_i) = g(x_i), i = 1, \dots, n+1$.
lyche-numerical-linear-algebra_F00873	68	0.999	$h\left(x_{\{i\}}\right)=g\left(x_{\{i\}}\right), i=1, \ldots, n+1$	$h\left(x_i\right)=g\left(x_i\right), i = 1, \dots, n+1$	for all $h \in C^2[a, b]$ such that $h(x_i) = g(x_i), i = 1, \dots, n+1$.
lyche-numerical-linear-algebra_F00874	68	1.000	$\int_{\{a\}}^{\{b\}} g^{\{\prime\prime\prime\}} e^{\{\prime\prime\prime\}}=\left[g^{\{\prime\prime\prime\}} e^{\{\prime\prime\prime\}}\right]_{\{a\}}^{\{b\}}-\int_{\{a\}}^{\{b\}} g^{\{\prime\prime\prime\}} e^{\{\prime\prime\prime\}}$	$\int_a^b g'' e'' = [g'' e']_a^b - \int_a^b g''' e'$	Integration by parts gives $\int_a^b g'' e'' = [g'' e']_a^b - \int_a^b g''' e'$. The first term is zero since
lyche-numerical-linear-algebra_F00875	68	1.000	$g^{\{\prime\prime\prime\}}$	g''	g'' is continuous and $g''(b) = g''(a) = 0$. For the second term, since g''' is equal to
lyche-numerical-linear-algebra_F00876	68	1.000	$g^{\{\prime\prime\prime\}}(b)=g^{\{\prime\prime\prime\}}(a)=0$	$g''(b) = g''(a) = 0$	g'' is continuous and $g''(b) = g''(a) = 0$. For the second term, since g''' is equal to
lyche-numerical-linear-algebra_F00877	68	1.000	$g^{\{\prime\prime\prime\prime\}}$	g'''	g'' is continuous and $g''(b) = g''(a) = 0$. For the second term, since g''' is equal to
lyche-numerical-linear-algebra_F00878	68	0.998	$v_{\{i\}}$	v_i	a constant v_i on each subinterval (x_i, x_{i+1}) and $e(x_i) = 0$, for $i = 1, \dots, n+1$
lyche-numerical-linear-algebra_F00879	68	0.998	$x_{\{i\}}, x_{\{i+1\}}$	x_i, x_{i+1}	a constant v_i on each subinterval (x_i, x_{i+1}) and $e(x_i) = 0$, for $i = 1, \dots, n+1$
lyche-numerical-linear-algebra_F00880	68	0.998	$e\left(x_{\{i\}}\right)=0$	$e\left(x_i\right)=0$	a constant v_i on each subinterval (x_i, x_{i+1}) and $e(x_i) = 0$, for $i = 1, \dots, n+1$
lyche-numerical-linear-algebra_F00881	68	1.000	$h=g+e$	$h = g + e$	Writing $h = g + e$ and using (2.35)
lyche-numerical-linear-algebra_F00882	68	1.000	$\boldsymbol{y} \in \mathbb{R}^{\{n+1\}}$	$\boldsymbol{y} \in \mathbb{R}^{n+1}$	an interval $[a, b]$, a vector $\boldsymbol{y} \in \mathbb{R}^{n+1}$ and μ_1, μ_{n+1} . The vector $\boldsymbol{x} = [x_1, \dots, x_n]$ and
lyche-numerical-linear-algebra_F00883	68	1.000	$\boldsymbol{x}=\left[x_{\{1\}}, \ldots, x_{\{n\}}\right]$	$\boldsymbol{x} = [x_1, \dots, x_n]$	an interval $[a, b]$, a vector $\boldsymbol{y} \in \mathbb{R}^{n+1}$ and μ_1, μ_{n+1} . The vector $\boldsymbol{x} = [x_1, \dots, x_n]$ and
lyche-numerical-linear-algebra_F00884	69	1.000	$\boldsymbol{C} \in \mathbb{R}^{\{n \times 4\}}$	$\boldsymbol{C} \in \mathbb{R}^{n \times 4}$	the coefficient matrix $\boldsymbol{C} \in \mathbb{R}^{n \times 4}$ in (2.7) are returned in the following algorithm. It
lyche-numerical-linear-algebra_F00885	69	1.000	$g\left(r_{\{j\}}\right)$	$g\left(r_j\right)$	form (2.4) we need to compute values $g(r_j)$ at a number of sites $\boldsymbol{r} = [r_1, \dots, r_m] \in$
lyche-numerical-linear-algebra_F00886	69	1.000	$\boldsymbol{r}=\left[r_{\{1\}}, \ldots, r_{\{m\}}\right] \in$	$\boldsymbol{r} = [r_1, \dots, r_m] \in$	form (2.4) we need to compute values $g(r_j)$ at a number of sites $\boldsymbol{r} = [r_1, \dots, r_m] \in$

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lyche-numerical-linear-algebra_F08887	69	■ 1.000	$i_{\{j\}}$	i_j	find an integer i_j so that $g(r_j) = p_{i_j}(r_j)$.
lyche-numerical-linear-algebra_F08888	69	■ 1.000	$g\left(r_{\{j\}}\right)=p_{\{i_{\{j\}}\}}\left(r_{\{j\}}\right)$	$g\left(r_j\right)=p_{i_j}\left(r_j\right)$	find an integer i_j so that $g(r_j) = p_{i_j}(r_j)$.
lyche-numerical-linear-algebra_F08889	69	■ 1.000	$k \in \mathbb{N}, \boldsymbol{t}=\left[t_{\{1\}}, \ldots, t_{\{k\}}\right]$	$k \in \mathbb{N}, \boldsymbol{t}=\left[t_1, \ldots, t_k\right]$	Given $k \in \mathbb{N}, \boldsymbol{t}=\left[t_1, \ldots, t_k\right]$ and a real number x . We consider the problem
lyche-numerical-linear-algebra_F08890	69	■ 1.000	$i=1$	$i=1$	of computing an integer i so that $i=1$ if $x < t_2, i=k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F08891	69	■ 1.000	$x<t_{\{2\}}, i=k$	$x < t_2, i=k$	of computing an integer i so that $i=1$ if $x < t_2, i=k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F08892	69	■ 1.000	$x \geq t_{\{k\}}$	$x \geq t_k$	of computing an integer i so that $i=1$ if $x < t_2, i=k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F08893	69	■ 1.000	$t_{\{i\}} \leq$	$t_i \leq$	of computing an integer i so that $i=1$ if $x < t_2, i=k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F08894	69	■ 1.000	$x<t_{\{i+1\}}$	$x < t_{i+1}$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m -vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F08895	69	■ 1.000	$\boldsymbol{x} \in \mathbb{R}^m$	$\boldsymbol{x} \in \mathbb{R}^m$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m -vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F08896	69	■ 1.000	$\boldsymbol{i}$	$\boldsymbol{i}$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m -vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F08897	69	■ 1.000	$\boldsymbol{i}=\left[i_{\{1\}}, \ldots, i_{\{m\}}\right]$	$\boldsymbol{i}=\left[i_1, \ldots, i_m\right]$	The following MATLAB function determines $\boldsymbol{i}=\left[i_1, \ldots, i_m\right]$. It uses the built in
lyche-numerical-linear-algebra_F08898	70	■ 1.000	$\boldsymbol{x}, \boldsymbol{C}$	$\boldsymbol{x}, \boldsymbol{C}$	Given output $\boldsymbol{x}, \boldsymbol{C}$ of <code>splineint</code> , defining a cubic spline g , and a vector $\boldsymbol{X}$,
lyche-numerical-linear-algebra_F08899	70	■ 1.000	$\boldsymbol{G}=g(\boldsymbol{X})$	$\boldsymbol{G}=g(\boldsymbol{X})$	<code>splineval</code> computes the vector $\boldsymbol{G}=g(\boldsymbol{X})$.
lyche-numerical-linear-algebra_F08900	71	■ 1.000	$f \in C^2[x-h, x+h]$	$f \in C^2[x-h, x+h]$	a) Show using Taylor expansion that if $f \in C^2[x-h, x+h]$ then for some η_2
lyche-numerical-linear-algebra_F08901	71	■ 1.000	η_2	η_2	a) Show using Taylor expansion that if $f \in C^2[x-h, x+h]$ then for some η_2
lyche-numerical-linear-algebra_F08902	71	■ 1.000	$f \in C^4[x-h, x+h]$	$f \in C^4[x-h, x+h]$	b) Show that if $f \in C^4[x-h, x+h]$ then for some η_4
lyche-numerical-linear-algebra_F08903	71	■ 1.000	η_4	η_4	b) Show that if $f \in C^4[x-h, x+h]$ then for some η_4
lyche-numerical-linear-algebra_F08904	71	■ 1.000	$\delta^2 f(x)$	$\delta^2 f(x)$	$\delta^2 f(x)$ is known as the central difference approximation to the second derivative
lyche-numerical-linear-algebra_F08905	71	■ 1.000	f, q	f, q	We assume that the given functions f, q and r are continuous on $[a, b]$ and that
lyche-numerical-linear-algebra_F08906	71	■ 1.000	$q(x) \geq 0$	$q(x) \geq 0$	$q(x) \geq 0$ for $x \in [a, b]$. It can then be shown that (2.36) has a unique solution u .
lyche-numerical-linear-algebra_F08907	71	■ 1.000	$x \in[a, b]$	$x \in[a, b]$	$q(x) \geq 0$ for $x \in[a, b]$. It can then be shown that (2.36) has a unique solution u .
lyche-numerical-linear-algebra_F08908	71	■ 1.000	$m \in \mathbb{N}, h=(b-a) /(m+1), x_{\{j\}}=a+j h$	$m \in \mathbb{N}, h=(b-a) /(m+1), x_j=a+j h$	To solve (2.36) numerically we choose $m \in \mathbb{N}, h=(b-a) /(m+1), x_j=a+j h$
lyche-numerical-linear-algebra_F08909	71	■ 0.993	$v_{\{0\}}=g_{\{0\}}$	$v_0=g_0$	with $v_0=g_0$ and $v_{m+1}=g_1$.
lyche-numerical-linear-algebra_F08910	71	■ 0.993	$v_{\{m+1\}}=g_{\{1\}}$	$v_{m+1}=g_1$	with $v_0=g_0$ and $v_{m+1}=g_1$.
lyche-numerical-linear-algebra_F08911	71	■ 1.000	$\boldsymbol{A} \boldsymbol{v}=\boldsymbol{b}$	$\boldsymbol{A} \boldsymbol{v}=\boldsymbol{b}$	a) Show that (2.37) leads to a tridiagonal linear system $\boldsymbol{A} \boldsymbol{v}=\boldsymbol{b}$, where $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F08912	71	■ 0.523	$\operatorname{tridiag}\left(a_{\{j\}}, d_{\{j\}}, c_{\{j\}}\right) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}\left(a_j, d_j, c_j\right) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}\left(a_j, d_j, c_j\right) \in \mathbb{R}^{m \times m}$ has elements
lyche-numerical-linear-algebra_F08913	72	■ 1.000	$\frac{h}{2} r(x) < 1$	$\frac{h}{2} r(x) < 1$	h is so small that $\frac{h}{2} r(x) < 1$ for all $x \in[a, b]$.

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lyche-numerical-linear-algebra-F00914	72	1.000	$v_{\{1\}}, \ldots, v_{\{m\}}$	$v_1, \dots, v_m$	(c) Propose a method to find $v_1, \dots, v_m$.
lyche-numerical-linear-algebra-F00915	72	1.000	$r=0, f=q=1$	$r = 0, f = q = 1$	a) Consider the problem (2.36) with $r = 0, f = q =$ and boundary conditions
lyche-numerical-linear-algebra-F00916	72	1.000	$u(0)=1, u(1)=0$	$u(0) = 1, u(1) = 0$	$u(0) = 1, u(1) = 0$. The exact solution is $u(x) = 1 - \sinh x / \sinh 1$. Write a
lyche-numerical-linear-algebra-F00917	72	1.000	$u(x)=1-\sinh x / \sinh 1$	$u(x) = 1 - \sinh x / \sinh 1$	$u(0) = 1, u(1) = 0$. The exact solution is $u(x) = 1 - \sinh x / \sinh 1$. Write a
lyche-numerical-linear-algebra-F00918	72	1.000	$h=0.1, 0.05, 0.025, 0.0125$	$h = 0.1, 0.05, 0.025, 0.0125$	computer program to solve (2.37) for $h = 0.1, 0.05, 0.025, 0.0125$, and compute
lyche-numerical-linear-algebra-F00919	72	1.000	$\max_{1 \leq j \leq m} u(x_j) - v_j $	$\max_{1 \leq j \leq m} u(x_j) - v_j $	the “error” $\max_{1 \leq j \leq m} u(x_j) - v_j $ for each h .
lyche-numerical-linear-algebra-F00920	72	1.000	$v_{\{j\}}, j=$	$v_j, j =$	b) Make a combined plot of the solution u and the computed points $v_j, j =$
lyche-numerical-linear-algebra-F00921	72	1.000	$0, \ldots, m+1$	$0, \dots, m + 1$	$0, \dots, m + 1$ for $h = 0.1$.
lyche-numerical-linear-algebra-F00922	72	1.000	$h=0.1$	$h = 0.1$	$0, \dots, m + 1$ for $h = 0.1$.
lyche-numerical-linear-algebra-F00923	72	1.000	$h^{\{p\}}$	h^p	c) One can show that the error is proportional to h^p for some integer p . Estimate p
lyche-numerical-linear-algebra-F00924	72	1.000	$a_{\{i, i+1\}} a_{\{i+1, i\}} > 0$	$a_{i,i+1} a_{i+1,i} > 0$	tridiagonal and suppose $a_{i,i+1} a_{i+1,i} > 0$ for $i = 1, \dots, n - 1$. Show that there
lyche-numerical-linear-algebra-F00925	72	1.000	$\boldsymbol{D} = \operatorname{diag}(d_{\{1\}}, \ldots, d_{\{n\}})$	$\boldsymbol{D} = \operatorname{diag}(d_1, \dots, d_n)$	exists a diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, \dots, d_n)$ with $d_i > 0$ for all i such that
lyche-numerical-linear-algebra-F00926	72	1.000	$d_{\{i\}} > 0$	$d_i > 0$	exists a diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, \dots, d_n)$ with $d_i > 0$ for all i such that
lyche-numerical-linear-algebra-F00927	72	1.000	$\boldsymbol{B} := \boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$	$\boldsymbol{B} := \boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$	$\boldsymbol{B} := \boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$ is symmetric.
lyche-numerical-linear-algebra-F00928	73	1.000	$\boldsymbol{T} = \boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{T} = \boldsymbol{L} \boldsymbol{U}$	Exercise 2.14 (LU Factorization of 2. Derivative Matrix) Show that $\boldsymbol{T} = \boldsymbol{L} \boldsymbol{U}$,
lyche-numerical-linear-algebra-F00929	73	0.998	$\boldsymbol{S} \in \mathbb{R}^{m \times m}$	$\boldsymbol{S} \in \mathbb{R}^{m \times m}$	Exercise 2.15 (Inverse of the 2. Derivative Matrix) Let $\boldsymbol{S} \in \mathbb{R}^{m \times m}$ have elements
lyche-numerical-linear-algebra-F00930	73	1.000	$s_{\{i j\}}$	s_{ij}	s_{ij} given by
lyche-numerical-linear-algebra-F00931	73	1.000	$\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$	$\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$	Show that $\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$ and conclude that $\boldsymbol{T}^{-1} = \boldsymbol{S}$.
lyche-numerical-linear-algebra-F00932	73	1.000	$\boldsymbol{T}^{-1} = \boldsymbol{S}$	$\boldsymbol{T}^{-1} = \boldsymbol{S}$	Show that $\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$ and conclude that $\boldsymbol{T}^{-1} = \boldsymbol{S}$.
lyche-numerical-linear-algebra-F00933	73	1.000	$a_{\{i j\}} = \boldsymbol{e}_{\{i\}}^T \boldsymbol{A} \boldsymbol{e}_{\{j\}}$	$a_{ij} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j$	that $a_{ij} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j$ for all i, j .
lyche-numerical-linear-algebra-F00934	73	1.000	$\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \boldsymbol{e}_{\{i\}} \boldsymbol{e}_{\{j\}}^T$	$\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \boldsymbol{e}_i \boldsymbol{e}_j^T$	show that $\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \boldsymbol{e}_i \boldsymbol{e}_j^T$.
lyche-numerical-linear-algebra-F00935	73	0.942	$\boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{A}^T \boldsymbol{A}$	Exercise 2.18 (The Product $\boldsymbol{A}^T \boldsymbol{A}$) Let $\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$. Explain why this product is
lyche-numerical-linear-algebra-F00936	73	0.942	$\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$	Exercise 2.18 (The Product $\boldsymbol{A}^T \boldsymbol{A}$) Let $\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$. Explain why this product is
lyche-numerical-linear-algebra-F00937	73	1.000	$b_{\{i j\}} = \boldsymbol{a}_{\{i\}}^T \boldsymbol{a}_{\{j\}}$	$b_{ij} = \boldsymbol{a}_{\cdot i}^T \boldsymbol{a}_{\cdot j}$	defined for any matrix $\boldsymbol{A}$. Show that $b_{ij} = \boldsymbol{a}_{\cdot i}^T \boldsymbol{a}_{\cdot j}$ for all i, j .
lyche-numerical-linear-algebra-F00938	73	1.000	$\boldsymbol{B} \in \mathbb{R}^{p \times n}$	$\boldsymbol{B} \in \mathbb{R}^{p \times n}$	Exercise 2.19 (Outer Product Expansion) For $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{B} \in \mathbb{R}^{p \times n}$ show
lyche-numerical-linear-algebra-F00939	73	1.000	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$, and $\boldsymbol{X} \in \mathbb{R}^{n \times p}$. Show that
lyche-numerical-linear-algebra-F00940	73	1.000	$\boldsymbol{X} \in \mathbb{R}^{n \times p}$	$\boldsymbol{X} \in \mathbb{R}^{n \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$, and $\boldsymbol{X} \in \mathbb{R}^{n \times p}$. Show that

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lyche-numerical-linear-algebra_F00941	74	■ 0.989	$\backslash\boldsymbol{B}=\backslash\boldsymbol{B}$	$B =$	Exercise 2.21 (Block Multiplication Example) Suppose $A = [A_1, A_2]$ and $B =$
lyche-numerical-linear-algebra_F00942	74	■ 1.000	$\left[\begin{array}{c}\boldsymbol{B}_{_1} \\ \boldsymbol{0}\end{array}\right]$	$\left[\begin{array}{c} B_1 \\ 0 \end{array}\right]$	$\left[\begin{array}{c} B_1 \\ 0 \end{array}\right]$. When is $AB = A_1B_1$?
lyche-numerical-linear-algebra_F00943	74	■ 1.000	$\backslash\boldsymbol{A} \ \backslash\boldsymbol{B}=\backslash\boldsymbol{A}_{_1} \ \backslash\boldsymbol{B}_{_1}$	$AB = A_1B_1$	$\left[\begin{array}{c} B_1 \\ 0 \end{array}\right]$. When is $AB = A_1B_1$?
lyche-numerical-linear-algebra_F00944	74	■ 0.999	$\backslash\boldsymbol{A}, \ \backslash\boldsymbol{B}, \ \backslash\boldsymbol{C} \ \backslash\text{in}$	$A, B, C \in$	Exercise 2.22 (Another Block Multiplication Example) Suppose $A, B, C \in$
lyche-numerical-linear-algebra_F00945	74	■ 1.000	$\backslash\boldsymbol{A}_{_1}, \ \backslash\boldsymbol{B}_{_1}, \ \backslash\boldsymbol{C}_{_1} \ \backslash\text{in} \ \mathbb{R}^{(n-1) \times (n-1)}$	$A_1, B_1, C_1 \in \mathbb{R}^{(n-1) \times (n-1)}$	where $A_1, B_1, C_1 \in \mathbb{R}^{(n-1) \times (n-1)}$. Show that
lyche-numerical-linear-algebra_F00946	74	■ 0.921	$\backslash\boldsymbol{T} \ \backslash\boldsymbol{x}=\backslash\boldsymbol{b} \ ?$	$Tx = b?$	$Tx = b?$
lyche-numerical-linear-algebra_F00947	74	■ 1.000	$\left[\begin{array}{cc} 1 & 1+i \\ 1+i & 2 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 1+i \\ 1+i & 2 \end{array}\right]$	2.7.6 Is the matrix $\left[\begin{array}{cc} 1 & 1+i \\ 1+i & 2 \end{array}\right]$ Hermitian? Symmetric?
lyche-numerical-linear-algebra_F00948	75	■ 1.000	$\backslash\boldsymbol{P}$	P	factorization, if row interchanges are introduced. Here P is a permutation matrix, L
lyche-numerical-linear-algebra_F00949	75	■ 0.951	$3 \ \times \ 3$	3×3	linear equations in n unknowns. ¹ We first recall how it works on a 3×3 system.
lyche-numerical-linear-algebra_F00950	76	■ 1.000	$a_{11}^{(1)} \neq 0$	$a_{11}^{(1)} \neq 0$	To solve this system by Gaussian elimination suppose $a_{11}^{(1)} \neq 0$. We subtract $l_{21}^{(1)} :=$
lyche-numerical-linear-algebra_F00951	76	■ 1.000	$l_{21}^{(1)} :=$	$l_{21}^{(1)} :=$	To solve this system by Gaussian elimination suppose $a_{11}^{(1)} \neq 0$. We subtract $l_{21}^{(1)} :=$
lyche-numerical-linear-algebra_F00952	76	■ 0.997	$a_{21}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)} / a_{11}^{(1)}$ times equation I from equation II and $l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$ times equation I
lyche-numerical-linear-algebra_F00953	76	■ 0.997	$l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$	$l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)} / a_{11}^{(1)}$ times equation I from equation II and $l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$ times equation I
lyche-numerical-linear-algebra_F00954	76	■ 1.000	$b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$	$b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00955	76	■ 1.000	$i=2,3$	$i = 2, 3$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00956	76	■ 1.000	$a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$	$a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00957	76	■ 1.000	$i, j=2,3$	$i, j = 2, 3$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00958	76	■ 1.000	$a_{11}^{(1)} = 0$	$a_{11}^{(1)} = 0$	If $a_{11}^{(1)} = 0$ and $a_{21}^{(1)} \neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)} =$
lyche-numerical-linear-algebra_F00959	76	■ 1.000	$a_{21}^{(1)} \neq 0$	$a_{21}^{(1)} \neq 0$	If $a_{11}^{(1)} = 0$ and $a_{21}^{(1)} \neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)} =$
lyche-numerical-linear-algebra_F00960	76	■ 1.000	$a_{11}^{(1)} =$	$a_{11}^{(1)} =$	If $a_{11}^{(1)} = 0$ and $a_{21}^{(1)} \neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)} =$
lyche-numerical-linear-algebra_F00961	76	■ 0.997	$a_{21}^{(1)} = 0$	$a_{21}^{(1)} = 0$	$a_{21}^{(1)} = 0$ we interchange equation I and III. Since the system is nonsingular the first
lyche-numerical-linear-algebra_F00962	76	■ 0.705	$a_{22}^{(2)} \neq 0$	$a_{22}^{(2)} \neq 0$	If $a_{22}^{(2)} \neq 0$ we subtract $l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$ times equation II' from equation III' to
lyche-numerical-linear-algebra_F00963	76	■ 0.705	$l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$	$l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$	If $a_{22}^{(2)} \neq 0$ we subtract $l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$ times equation II' from equation III' to
lyche-numerical-linear-algebra_F00964	76	■ 0.705	$\{ \}^{\prime}$	$'$	If $a_{22}^{(2)} \neq 0$ we subtract $l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$ times equation II' from equation III' to

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lyche-numerical-linear-algebra_F00965	76	■ 1.000	$a_{33}^{(3)}=a_{33}^{(2)}-l_{32}^{(2)} a_{23}^{(2)}$	$a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$	where $a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$ and $b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$. If $a_{22}^{(2)} = 0$ then $a_{32}^{(2)} \neq 0$
lyche-numerical-linear-algebra_F00966	76	■ 1.000	$b_{3}^{(3)}=b_{3}^{(2)}-l_{32}^{(2)} b_{2}^{(2)}$	$b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$	where $a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$ and $b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$. If $a_{22}^{(2)} = 0$ then $a_{32}^{(2)} \neq 0$
lyche-numerical-linear-algebra_F00967	76	■ 1.000	$a_{22}^{(2)}=0$	$a_{22}^{(2)} = 0$	where $a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$ and $b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$. If $a_{22}^{(2)} = 0$ then $a_{32}^{(2)} \neq 0$
lyche-numerical-linear-algebra_F00968	76	■ 1.000	$a_{32}^{(2)} \neq 0$	$a_{32}^{(2)} \neq 0$	where $a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$ and $b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$. If $a_{22}^{(2)} = 0$ then $a_{32}^{(2)} \neq 0$
lyche-numerical-linear-algebra_F00969	76	■ 1.000	$a_{k k}^{(k)} \neq 0, k=1,2$	$a_{kk}^{(k)} \neq 0, k = 1, 2$	Indeed, if $a_{kk}^{(k)} \neq 0, k = 1, 2$ then
lyche-numerical-linear-algebra_F00970	77	■ 1.000	$\boldsymbol{A}^{(1)}$	$\boldsymbol{A}^{(1)}$	$\boldsymbol{A}^{(1)}$ (cf. the proof of Theorem 3.2).
lyche-numerical-linear-algebra_F00971	77	■ 1.000	$\boldsymbol{A}^{(k)} \boldsymbol{x}=\boldsymbol{b}^{(k)}$	$\boldsymbol{A}^{(k)} \boldsymbol{x} = \boldsymbol{b}^{(k)}$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ and generate a sequence of equivalent systems $\boldsymbol{A}^{(k)} \boldsymbol{x} = \boldsymbol{b}^{(k)}$ for $k =$
lyche-numerical-linear-algebra_F00972	77	■ 1.000	$k=$	$k =$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ and generate a sequence of equivalent systems $\boldsymbol{A}^{(k)} \boldsymbol{x} = \boldsymbol{b}^{(k)}$ for $k =$
lyche-numerical-linear-algebra_F00973	77	■ 1.000	$1, \ldots, n$	$1, \dots, n$	$1, \dots, n$, where $\boldsymbol{A}^{(1)} = \boldsymbol{A}$, $\boldsymbol{b}^{(1)} = \boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00974	77	■ 1.000	$\boldsymbol{A}^{(1)}=\boldsymbol{A}, \boldsymbol{b}^{(1)}=\boldsymbol{b}$	$\boldsymbol{A}^{(1)} = \boldsymbol{A}, \boldsymbol{b}^{(1)} = \boldsymbol{b}$	$1, \dots, n$, where $\boldsymbol{A}^{(1)} = \boldsymbol{A}, \boldsymbol{b}^{(1)} = \boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00975	77	■ 1.000	$\boldsymbol{A}^{(k)}$	$\boldsymbol{A}^{(k)}$	$1, \dots, n$, where $\boldsymbol{A}^{(1)} = \boldsymbol{A}, \boldsymbol{b}^{(1)} = \boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00976	77	■ 1.000	$k-1$	$k - 1$	first $k - 1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00977	77	■ 1.000	$\boldsymbol{A}^{(n)}$	$\boldsymbol{A}^{(n)}$	first $k - 1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00978	77	■ 1.000	$\boldsymbol{A}^{(n)} \boldsymbol{x}=\boldsymbol{b}^{(n)}$	$\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$	first $k - 1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00979	77	■ 1.000	$\boldsymbol{A}^{(k+1)}$	$\boldsymbol{A}^{(k+1)}$	The process transforming $\boldsymbol{A}^{(k)}$ into $\boldsymbol{A}^{(k+1)}$ for $k = 1, \dots, n - 1$ can be described
lyche-numerical-linear-algebra_F00980	78	■ 1.000	$j=k$	$j = k$	For $j = k$ it follows from (3.2) that $a_{ik}^{(k+1)} = a_{ik}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kk}^{(k)} = 0$ for $i =$
lyche-numerical-linear-algebra_F00981	78	■ 1.000	$a_{i k}^{(k+1)}=a_{i k}^{(k)}-\frac{a_{i k}^{(k)}}{a_{k k}^{(k)}} a_{k k}^{(k)}=0$	$a_{ik}^{(k+1)} = a_{ik}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kk}^{(k)} = 0$	For $j = k$ it follows from (3.2) that $a_{ik}^{(k+1)} = a_{ik}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kk}^{(k)} = 0$ for $i =$
lyche-numerical-linear-algebra_F00982	78	■ 1.000	$k+1, \ldots, n$	$k + 1, \dots, n$	$k + 1, \dots, n$. Thus $\boldsymbol{A}^{(k+1)}$ will have zeros under the diagonal in its first k columns
lyche-numerical-linear-algebra_F00983	78	■ 1.000	$l_{i k}^{(k)}$	$l_{ik}^{(k)}$	and the elimination is carried one step further. The numbers $l_{ik}^{(k)}$ in (3.2) are called
lyche-numerical-linear-algebra_F00984	78	■ 1.000	$a_{k k}^{(k)}$	$a_{kk}^{(k)}$	$a_{kk}^{(k)}$ are nonzero for $k = 1, \dots, n - 1$. This depends on certain submatrices of $\boldsymbol{A}$
lyche-numerical-linear-algebra_F00985	78	■ 1.000	$\boldsymbol{A}_{[k]} \in \mathbb{C}^{k \times k}$	$\boldsymbol{A}_{[k]} \in \mathbb{C}^{k \times k}$	Definition 3.1 (Principal Submatrix) For $k = 1, \dots, n$ the matrices $\boldsymbol{A}_{[k]} \in \mathbb{C}^{k \times k}$
lyche-numerical-linear-algebra_F00986	78	■ 0.997	$\boldsymbol{B} \in \mathbb{C}^{k \times k}$	$\boldsymbol{B} \in \mathbb{C}^{k \times k}$	matrix $\boldsymbol{B} \in \mathbb{C}^{k \times k}$ is called a principal submatrix of $\boldsymbol{A}$ if $\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$, where
lyche-numerical-linear-algebra_F00987	78	■ 0.997	$\boldsymbol{B}=\boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$	$\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$	matrix $\boldsymbol{B} \in \mathbb{C}^{k \times k}$ is called a principal submatrix of $\boldsymbol{A}$ if $\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$, where
lyche-numerical-linear-algebra_F00988	78	■ 1.000	$\boldsymbol{r}=\left[r_1, \ldots, r_k\right]$	$\boldsymbol{r} = [r_1, \dots, r_k]$	$\boldsymbol{r} = [r_1, \dots, r_k]$ for some $1 \leq r_1 < \dots < r_k \leq n$. Thus,

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lyche-numerical-linear-algebra_F00989	78	1.000	$1 \leq r_1 < \cdots < r_k \leq n$	$1 \leq r_1 < \cdots < r_k \leq n$	$\mathbf{r} = [r_1, \dots, r_k]$ for some $\boxed{1 < r_1 < \cdots < r_k < n}$. Thus,
lyche-numerical-linear-algebra_F00990	78	1.000	$r_{\{j\}}=j$	$r_j = j$	A principal submatrix is leading if $r_j = j$ for $j = 1, \dots, k$. Also a principal
lyche-numerical-linear-algebra_F00991	78	1.000	$j=1, \ldots, k$	$j = 1, \dots, k$	A principal submatrix is leading if $r_j = j$ for $\boxed{j = 1, \dots, k}$. Also a principal
lyche-numerical-linear-algebra_F00992	78	1.000	$\boldsymbol{A}=\left[\begin{array}{lll}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]$	$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	<i>Example 3.2 (Principal Submatrices)</i> The principal submatrices of $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$
lyche-numerical-linear-algebra_F00993	78	1.000	$a_{\{k, k\}}^{\{(k)\}} \neq 0$	$a_{k,k}^{(k)} \neq 0$	Theorem 3.1 We have $a_{k,k}^{(k)} \neq 0$ for $k = 1, \dots, n - 1$ if and only if the leading
lyche-numerical-linear-algebra_F00994	78	1.000	$\boldsymbol{A}_{\{[k]\}}$	$\mathbf{A}_{[k]}$	principal submatrices $\mathbf{A}_{[k]}$ of $\mathbf{A}$ are nonsingular for $k = 1, \dots, n - 1$. Moreover
lyche-numerical-linear-algebra_F00995	78	1.000	$\boldsymbol{B}_{\{k\}}=\boldsymbol{A}_{\{k-1\}}^{\{(k)\}}$	$\mathbf{B}_k = \mathbf{A}_{k-1}^{(k)}$	Proof Let $\boxed{\mathbf{B}_k = \mathbf{A}_{k-1}^{(k)}}$ be the upper left $k - 1$ corner of $\mathbf{A}^{(k)}$ given by (3.1). Observe
lyche-numerical-linear-algebra_F00996	78	1.000	$\boldsymbol{B}_{\{k\}}$	$\mathbf{B}_k$	that the elements of $\mathbf{B}_k$ are computed from $\mathbf{A}$ by using only elements from $\mathbf{A}_{[k-1]}$.
lyche-numerical-linear-algebra_F00997	79	1.000	$\boldsymbol{A}_{\{[k-1]\}}$	$\mathbf{A}_{[k-1]}$	that the elements of $\mathbf{B}_k$ are computed from $\mathbf{A}$ by using only elements from $\mathbf{A}_{[k-1]}$.
lyche-numerical-linear-algebra_F00998	78	1.000	$\boldsymbol{B}_{\{k+1\}}$	$\mathbf{B}_{k+1}$	of diagonal elements of $\mathbf{B}_{k+1}$ and (3.3) follows. But then $a_{11}^{(1)} \cdots a_{kk}^{(k)} \neq 0$ for $k =$
lyche-numerical-linear-algebra_F00999	79	1.000	$a_{\{11\}}^{\{(1)\}} \cdots a_{\{k k\}}^{\{(k)\}} \neq 0$	$a_{11}^{(1)} \cdots a_{kk}^{(k)} \neq 0$	of diagonal elements of $\mathbf{B}_{k+1}$ and (3.3) follows. But then $\boxed{a_{11}^{(1)} \cdots a_{kk}^{(k)} \neq 0}$ for $k =$
lyche-numerical-linear-algebra_F01000	78	1.000	$1, \ldots, n-1$	$1, \dots, n - 1$	$1, \dots, n - 1$ if and only if $\det(\mathbf{A}_{[k]}) \neq 0$ for $k = 1, \dots, n - 1$, or equivalently $\mathbf{A}_{[k]}$
lyche-numerical-linear-algebra_F01001	78	1.000	$\operatorname{det}\left(\boldsymbol{A}_{\{[k]\}}\right) \neq 0$	$\det(\mathbf{A}_{[k]}) \neq 0$	$1, \dots, n - 1$ if and only if $\det(\mathbf{A}_{[k]}) \neq 0$ for $k = 1, \dots, n - 1$, or equivalently $\mathbf{A}_{[k]}$
lyche-numerical-linear-algebra_F01002	78	1.000	$l_{\{i j\}}^{\{(j)\}}$	$l_{ij}^{(j)}$	where the $l_{ij}^{(j)}$ and $a_{ij}^{(i)}$ are given by (3.2).
lyche-numerical-linear-algebra_F01003	78	1.000	$a_{\{i j\}}^{\{(i)\}}$	$a_{ij}^{(i)}$	where the $l_{ij}^{(j)}$ and $a_{ij}^{(i)}$ are given by (3.2).
lyche-numerical-linear-algebra_F01004	79	1.000	$i \leq j$	$i \leq j$	Thus for $\boxed{i < j}$ we find
lyche-numerical-linear-algebra_F01005	79	1.000	$a_{\{n n\}}^{\{(n)\}}=0$	$a_{nn}^{(n)} = 0$	matrix $\mathbf{U}$ is then singular, and we must have $a_{nn}^{(n)} = 0$ when $\mathbf{A}$ is singular.
lyche-numerical-linear-algebra_F01006	80	1.000	$\boldsymbol{L} \boldsymbol{U} \boldsymbol{x}=\boldsymbol{b}$	$\mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b}$	Since $\mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b}$ we have $\mathbf{L}\mathbf{y} = \mathbf{b}$, where $\mathbf{y} := \mathbf{U}\mathbf{x}$. We first solve $\mathbf{L}\mathbf{y} = \mathbf{b}$, for $\mathbf{y}$
lyche-numerical-linear-algebra_F01007	80	1.000	$\boldsymbol{L} \boldsymbol{y}=\boldsymbol{b}$	$\mathbf{L}\mathbf{y} = \mathbf{b}$	Since $\mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b}$ we have $\mathbf{L}\mathbf{y} = \mathbf{b}$, where $\mathbf{y} := \mathbf{U}\mathbf{x}$. We first solve $\mathbf{L}\mathbf{y} = \mathbf{b}$, for $\mathbf{y}$
lyche-numerical-linear-algebra_F01008	80	1.000	$\boldsymbol{y}:=\boldsymbol{U} \boldsymbol{x}$	$\mathbf{y} := \mathbf{U}\mathbf{x}$	Since $\mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{b}$ we have $\mathbf{L}\mathbf{y} = \mathbf{b}$, where $\mathbf{y} := \mathbf{U}\mathbf{x}$. We first solve $\mathbf{L}\mathbf{y} = \mathbf{b}$, for $\mathbf{y}$
lyche-numerical-linear-algebra_F01009	80	1.000	$\boldsymbol{U} \boldsymbol{x}=\boldsymbol{y}$	$\mathbf{U}\mathbf{x} = \mathbf{y}$	and then $\mathbf{U}\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$.
lyche-numerical-linear-algebra_F01010	80	0.999	$2.5 \boldsymbol{A}$	$2.5\mathbf{A}$	A nonsingular triangular linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is easy to solve. By Lemma 2.5 $\mathbf{A}$
lyche-numerical-linear-algebra_F01011	80	1.000	$x_{\{1\}}=b_{\{1\}} / a_{\{11\}}$	$x_1 = b_1/a_{11}$	From the first equation we find $x_1 = b_1/a_{11}$. Solving the second equation for x_2 we
lyche-numerical-linear-algebra_F01012	80	1.000	$x_{\{2\}}$	x_2	From the first equation we find $x_1 = b_1/a_{11}$. Solving the second equation for x_2 we
lyche-numerical-linear-algebra_F01013	80	0.999	$x_{\{2\}}=\left(b_{\{2\}}-a_{\{21\}} x_{\{1\}}\right) / a_{\{22\}}$	$x_2 = (b_2 - a_{21}x_1)/a_{22}$	obtain $x_2 = (b_2 - a_{21}x_1)/a_{22}$. Finally the third equation gives $x_3 = (b_3 - a_{31}x_1 -$
lyche-numerical-linear-algebra_F01014	80	0.999	$x_{\{3\}}=\left(b_{\{3\}}-a_{\{31\}} x_{\{1\}}-\right.$	$x_3 = (b_3 - a_{31}x_1 -$	obtain $x_2 = (b_2 - a_{21}x_1)/a_{22}$. Finally the third equation gives $x_3 = (b_3 - a_{31}x_1 -$

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lyche-numerical-linear-algebra-F01015	80	1.000	$\left. a_{32} x_2 \right) / a_{33}$	$a_{32} x_2) / a_{33}$	$a_{32} x_2) / a_{33}$. This process is known as forward substitution. In general
lyche-numerical-linear-algebra-F01016	80	1.000	$a_{k,j}=0$	$a_{k,j} = 0$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k + 1, \dots, k$ for $k = 1, 2, \dots, n$, and where
lyche-numerical-linear-algebra-F01017	80	1.000	$j \notin \{l_k, l_k + 1, \dots, k\}$	$j \notin \{l_k, l_k + 1, \dots, k$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k + 1, \dots, k$ for $k = 1, 2, \dots, n$, and where
lyche-numerical-linear-algebra-F01018	80	1.000	$k=1,2, \ldots, n$	$k = 1, 2, \dots, n$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k + 1, \dots, k$ for $k = 1, 2, \dots, n$, and where
lyche-numerical-linear-algebra-F01019	80	1.000	$l_k := \max(1, k - d)$	$l_k := \max(1, k - d)$	$l_k := \max(1, k - d)$, see Fig. 3.2. For a lower triangular d -band matrix the
lyche-numerical-linear-algebra-F01020	80	1.000	$d=n$	$d = n$	Note that (3.8) reduces to (3.7) if $d = n$. Letting $A(k, l_k : (k - 1)) * x(l_k : (k - 1))$
lyche-numerical-linear-algebra-F01021	80	1.000	$A \left(k, l_k : (k - 1) \right) * x \left(l_k : (k - 1) \right)$	$A(k, l_k : (k - 1)) * x(l_k : (k - 1))$	Note that (3.8) reduces to (3.7) if $d = n$. Letting $A(k, l_k : (k - 1)) * x(l_k : (k - 1))$
lyche-numerical-linear-algebra-F01022	80	0.995	$\sum_{j=l_k}^{k-1} a_{kj} x_j$	$\sum_{j=l_k}^{k-1} a_{kj} x_j$	denote the sum $\sum_{j=l_k}^{k-1} a_{kj} x_j$ we arrive at the following algorithm, where the initial
lyche-numerical-linear-algebra-F01023	81	1.000	$x_{\{n\}}$	x_n	substitution or 'bottom-up'. We first find x_n from the last equation and then move
lyche-numerical-linear-algebra-F01024	81	1.000	$\text{\textbf{A}} \in \mathbb{C}^{n \times n}, \text{\textbf{b}} \in \mathbb{C}^n$	$\text{\textbf{A}} \in \mathbb{C}^{n \times n}, \text{\textbf{b}} \in \mathbb{C}^n$	banded matrix $\text{\textbf{A}} \in \mathbb{C}^{n \times n}, \text{\textbf{b}} \in \mathbb{C}^n$ and d , as input, and returns an $\text{\textbf{x}} \in \mathbb{C}^n$ so
lyche-numerical-linear-algebra-F01025	81	1.000	$n-k+1$	$n - k + 1$	This system is of order $n - k + 1$. The unknowns are $x_k, \dots, x_n$.
lyche-numerical-linear-algebra-F01026	81	1.000	$x_{\{k\}}, \ldots, x_{\{n\}}$	$x_k, \dots, x_n$	This system is of order $n - k + 1$. The unknowns are $x_k, \dots, x_n$.
lyche-numerical-linear-algebra-F01027	82	1.000	$x_{\{k\}}=b_{\{k\}} / a_{\{k, k\}}$	$x_k = b_k / a_{k,k}$	We see that $x_k = b_k / a_{k,k}$ and eliminating x_k from the remaining equations we
lyche-numerical-linear-algebra-F01028	82	1.000	$x_{\{k\}}$	x_k	We see that $x_k = b_k / a_{k,k}$ and eliminating x_k from the remaining equations we
lyche-numerical-linear-algebra-F01029	82	1.000	$n-k$	$n - k$	obtain a system of order $n - k$ with unknowns $x_{k+1}, \dots, x_n$
lyche-numerical-linear-algebra-F01030	82	1.000	$x_{\{k+1\}}, \ldots, x_{\{n\}}$	$x_{k+1}, \dots, x_n$	obtain a system of order $n - k$ with unknowns $x_{k+1}, \dots, x_n$
lyche-numerical-linear-algebra-F01031	82	1.000	$k=1,2, \ldots n$	$k = 1, 2, \dots n$	Thus at the k th step, $k = 1, 2, \dots n$ we set $x_k = b_k / A(k, k)$ and update b as follows:
lyche-numerical-linear-algebra-F01032	82	1.000	$x_{\{k\}}=b_{\{k\}} / A(k, k)$	$x_k = b_k / A(k, k)$	Thus at the k th step, $k = 1, 2, \dots n$ we set $x_k = b_k / A(k, k)$ and update b as follows:
lyche-numerical-linear-algebra-F01033	82	1.000	b	b	Thus at the k th step, $k = 1, 2, \dots n$ we set $x_k = b_k / A(k, k)$ and update b as follows:
lyche-numerical-linear-algebra-F01034	82	1.000	M, D, A, S	M, D, A, S	operations. Let M, D, A, S be the number of (complex) multiplications, divisions,
lyche-numerical-linear-algebra-F01035	82	1.000	$a_{ij}^{k+1}=a_{ij}^{(k)}-l_{ik}^{(k)} a_{kj}^{(k)}$	$a_{ij}^{k+1} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$	calculation of $a_{ij}^{k+1} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$ which is carried out $(n - k)^2$ times. Moreover,
lyche-numerical-linear-algebra-F01036	82	1.000	$(n-k)^{2}$	$(n - k)^2$	calculation of $a_{ij}^{k+1} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$ which is carried out $(n - k)^2$ times. Moreover,
lyche-numerical-linear-algebra-F01037	83	1.000	$M+$	$M+$	each calculation involves one subtraction and one multiplication. Thus we find $M+$
lyche-numerical-linear-algebra-F01038	83	1.000	$S=2 \sum_{k=1}^{n-1} (n-k)^2=2 \sum_{m=1}^{n-1} m^2=\frac{2}{3} n(n-1) \left(n-\frac{1}{2}\right)$	$S = 2 \sum_{k=1}^{n-1} (n - k)^2 = 2 \sum_{m=1}^{n-1} m^2 = \frac{2}{3} n(n - 1) \left(n - \frac{1}{2} \right)$	$S = 2 \sum_{k=1}^{n-1} (n - k)^2 = 2 \sum_{m=1}^{n-1} m^2 = \frac{2}{3} n(n - 1) \left(n - \frac{1}{2} \right)$ For each k there are $n - k$
lyche-numerical-linear-algebra-F01039	83	1.000	$\sum_{k=1}^{n-1} (n-k)=\frac{1}{2} n(n-1)$	$\sum_{k=1}^{n-1} (n - k) = \frac{1}{2} n(n - 1)$	divisions giving a sum of $\sum_{k=1}^{n-1} (n - k) = \frac{1}{2} n(n - 1)$. Since there are no additions
lyche-numerical-linear-algebra-F01040	83	1.000	N_{LU}	N_{LU}	We are only interested in N_{LU} when n is large and for such n the term $\frac{2}{3} n^3$

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lyche-numerical-linear-algebra-F01041	83	■ 1.000	$\frac{2}{3} n^3$	$\frac{2}{3}n^3$	We are only interested in N_{LU} when n is large and for such n the term $\frac{2}{3}n^3$
lyche-numerical-linear-algebra-F01042	83	■ 1.000	$G_{\{n\}}$	G_n	the number G_n as follows:
lyche-numerical-linear-algebra-F01043	83	■ 0.997	$3.2\left(G_{\{n\}}:=\frac{2}{3} n^3\right)$	$3.2\left(G_n:=\frac{2}{3}n^3\right)$	Definition 3.2 ($G_n := \frac{2}{3}n^3$) We define $G_n := \frac{2}{3}n^3$.
lyche-numerical-linear-algebra-F01044	83	■ 0.997	$G_{\{n\}}:=\frac{2}{3} n^3$	$G_n:=\frac{2}{3}n^3$	Definition 3.2 ($G_n := \frac{2}{3}n^3$) We define $G_n := \frac{2}{3}n^3$.
lyche-numerical-linear-algebra-F01045	83	■ 1.000	$2 n^3 / 3$	$2n^3/3$	There is a quick way to arrive at the leading term $2n^3/3$. We only consider the
lyche-numerical-linear-algebra-F01046	83	■ 1.000	$M+S$	$M+S$	loop contributing to $M+S$. Then we replace sums by integrals letting the summation
lyche-numerical-linear-algebra-F01047	83	■ 1.000	$N_{\{S\}}$	N_S	Consider next N_S , the number of forward plus backward substitutions. By (3.7)
lyche-numerical-linear-algebra-F01048	83	■ 1.000	$n=10^{\{6\}}$	$n=10^6$	system requires $O(n^2)$ arithmetic operations. Thus, if $n=10^6$ and one arith-
lyche-numerical-linear-algebra-F01049	83	■ 1.000	$c=10^{\{-14\}}$	$c=10^{-14}$	metic operation requires $c=10^{-14}$ seconds of computing time then $cn^3=$
lyche-numerical-linear-algebra-F01050	83	■ 1.000	$c n^3=$	$cn^3=$	metic operation requires $c=10^{-14}$ seconds of computing time then $cn^3=$
lyche-numerical-linear-algebra-F01051	83	■ 1.000	$10^{\{4\}}$	10^4	10^4 seconds ≈ 3 hours and $cn^2=0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra-F01052	83	■ 1.000	≈ 3	≈ 3	10^4 seconds ≈ 3 hours and $cn^2=0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra-F01053	83	■ 1.000	$c n^2=0.01$	$cn^2=0.01$	10^4 seconds ≈ 3 hours and $cn^2=0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra-F01054	84	■ 1.000	$\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{l} 1 \\ 1 \end{array}\right]$	$\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{l} 1 \\ 1 \end{array}\right]$	nonsingular system. A simple example is $\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{l} 1 \\ 1 \end{array}\right]$. We show here that any
lyche-numerical-linear-algebra-F01055	84	■ 0.980	(k, k)	(k, k)	as pivoting . The element which is moved to the diagonal position (k, k) is called
lyche-numerical-linear-algebra-F01056	84	■ 1.000	$r_{\{k\}} \geq k$	$r_k \geq k$	1. Choose $r_k \geq k$ so that $a_{r_k,k}^{(k)} \neq 0$.
lyche-numerical-linear-algebra-F01057	84	■ 1.000	$a_{\{r_{\{k\}}, k\}}^{\{(k)\}} \neq 0$	$a_{r_k,k}^{(k)} \neq 0$	1. Choose $r_k \geq k$ so that $a_{r_k,k}^{(k)} \neq 0$.
lyche-numerical-linear-algebra-F01058	84	■ 1.000	$r_{\{k\}}$	r_k	2. Interchange rows r_k and k of $A^{(k)}$.
lyche-numerical-linear-algebra-F01059	84	■ 1.000	$a_{\{i j\}}^{\{(k+1)\}}$	$a_{ij}^{(k+1)}$	3. Eliminate by computing $a_{ik}^{(k)}$ and $a_{ij}^{(k+1)}$ using (3.2).
lyche-numerical-linear-algebra-F01060	84	■ 1.000	$\boldsymbol{A}^{\{1\}}=\boldsymbol{A}$	$\boldsymbol{A}^{(1)}=\boldsymbol{A}$	$\boldsymbol{A}^{(1)}=\boldsymbol{A}$ this holds for $k=1$. By Lemma 2.4 the lower right diagonal block in
lyche-numerical-linear-algebra-F01061	84	■ 1.000	$\boldsymbol{e}_{\{i_1\}}, \ldots, \boldsymbol{e}_{\{i_n\}}$	$\boldsymbol{e}_{i_1}, \ldots, \boldsymbol{e}_{i_n}$	where $\boldsymbol{e}_{i_1}, \ldots, \boldsymbol{e}_{i_n}$ is a permutation of the unit vectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n \in \mathbb{R}^n$.
lyche-numerical-linear-algebra-F01062	84	■ 1.000	$\boldsymbol{e}_{\{1\}}, \ldots, \boldsymbol{e}_{\{n\}} \in \mathbb{R}^{\{n\}}$	$\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n \in \mathbb{R}^n$	where $\boldsymbol{e}_{i_1}, \ldots, \boldsymbol{e}_{i_n}$ is a permutation of the unit vectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n \in \mathbb{R}^n$.
lyche-numerical-linear-algebra-F01063	84	■ 1.000	$\boldsymbol{p}=\left[i_{\{1\}}, \ldots, i_{\{n\}}\right]^T$	$\boldsymbol{p}=\left[i_1, \ldots, i_n\right]^T$	Every permutation $\boldsymbol{p}=\left[i_1, \ldots, i_n\right]^T$ of the integers $1, 2, \ldots, n$ gives rise to a
lyche-numerical-linear-algebra-F01064	85	■ 0.997	$\boldsymbol{A} \boldsymbol{P}=\left(\boldsymbol{A} \boldsymbol{e}_{\{i_1\}}, \ldots, \boldsymbol{A} \boldsymbol{e}_{\{i_n\}}\right)=\boldsymbol{A}(:, \boldsymbol{p})$	$\boldsymbol{A} \boldsymbol{P}=\left(\boldsymbol{A} \boldsymbol{e}_{i_1}, \ldots, \boldsymbol{A} \boldsymbol{e}_{i_n}\right)=\boldsymbol{A}(:, \boldsymbol{p})$	Indeed, $\boldsymbol{A} \boldsymbol{P}=\left(\boldsymbol{A} \boldsymbol{e}_{i_1}, \ldots, \boldsymbol{A} \boldsymbol{e}_{i_n}\right)=\boldsymbol{A}(:, \boldsymbol{p})$ and $\boldsymbol{P}^T \boldsymbol{A}=\left(\boldsymbol{A}^T \boldsymbol{P}\right)^T=\left(\boldsymbol{A}^T(:,$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01090	88	■ 0.967	$\mathbf{L}$	$\mathbf{LU}$	that $A = \mathbf{LU}$ is an LU factorization of $A \in \mathbb{C}^{n \times n}$ if $\mathbf{L} \in \mathbb{C}^{n \times n}$ is lower triangular
lyche-numerical-linear-algebra-F01091	88	■ 0.967	$\mathbf{L} \in \mathbb{C}^{n \times n}$	$\mathbf{L} \in \mathbb{C}^{n \times n}$	that $A = \mathbf{LU}$ is an LU factorization of $A \in \mathbb{C}^{n \times n}$ if $\mathbf{L} \in \mathbb{C}^{n \times n}$ is lower triangular
lyche-numerical-linear-algebra-F01092	88	■ 1.000	$\mathbf{U} \in \mathbb{C}^{n \times n}$	$\mathbf{U} \in \mathbb{C}^{n \times n}$	and $\mathbf{U} \in \mathbb{C}^{n \times n}$ is upper triangular , i.e.,
lyche-numerical-linear-algebra-F01093	88	■ 0.963	n^2	n^2	To find an LU factorization there is one equation for each of the n^2 elements in A ,
lyche-numerical-linear-algebra-F01094	88	■ 1.000	$n^2 + n$	$n^2 + n$	and $\mathbf{L}$ and $\mathbf{U}$ contain a total of $n^2 + n$ unknown elements. There are several ways to
lyche-numerical-linear-algebra-F01095	89	■ 0.701	$l_{ii} = 1$	$l_{ii} = 1$	L1U: $l_{ii} = 1$ all i ,
lyche-numerical-linear-algebra-F01096	89	■ 0.992	$u_{ii} = 1$	$u_{ii} = 1$	LU1: $u_{ii} = 1$ all i ,
lyche-numerical-linear-algebra-F01097	89	■ 0.998	$\mathbf{A} = \mathbf{LDU}, l_{ii} = u_{ii} = 1$	$\mathbf{A} = \mathbf{LDU}, l_{ii} = u_{ii} = 1$	LDU: $\mathbf{A} = \mathbf{LDU}, l_{ii} = u_{ii} = 1$ all i , $\mathbf{D} = \text{diag}(d_{11}, \dots, d_{nn})$.
lyche-numerical-linear-algebra-F01098	89	■ 0.998	$\mathbf{D} = \text{diag}(d_{11}, \dots, d_{nn})$	$\mathbf{D} = \text{diag}(d_{11}, \dots, d_{nn})$	LDU: $\mathbf{A} = \mathbf{LDU}, l_{ii} = u_{ii} = 1$ all i , $\mathbf{D} = \text{diag}(d_{11}, \dots, d_{nn})$.
lyche-numerical-linear-algebra-F01099	89	■ 0.889	$a, b, c, d \in \mathbb{C}$	$a, b, c, d \in \mathbb{C}$	<i>Example 3.5 (L1U of 2×2 Matrix)</i> Let $a, b, c, d \in \mathbb{C}$. An L1U factorization of
lyche-numerical-linear-algebra-F01100	89	■ 0.855	$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ must satisfy the equations
lyche-numerical-linear-algebra-F01101	89	■ 0.984	l_1	l_1	for the unknowns l_1 in $\mathbf{L}$ and u_1, u_2, u_3 in $\mathbf{U}$. The equations are
lyche-numerical-linear-algebra-F01102	89	■ 0.984	u_1, u_2, u_3	u_1, u_2, u_3	for the unknowns l_1 in $\mathbf{L}$ and u_1, u_2, u_3 in $\mathbf{U}$. The equations are
lyche-numerical-linear-algebra-F01103	89	■ 0.999	$al_1 = c$	$al_1 = c$	equation $al_1 = c$. There are essentially three cases
lyche-numerical-linear-algebra-F01104	89	■ 1.000	$a \neq 0$	$a \neq 0$	1. $a \neq 0$: The matrix has a unique L1U factorization.
lyche-numerical-linear-algebra-F01105	89	■ 1.000	$a = c = 0$	$a = c = 0$	2. $a = c = 0$: The L1U factorization exists, but it is not unique. Any value for l_1
lyche-numerical-linear-algebra-F01106	89	■ 1.000	$a = 0, c \neq 0$	$a = 0, c \neq 0$	3. $a = 0, c \neq 0$: No L1U factorization exists.
lyche-numerical-linear-algebra-F01107	89	■ 1.000	$\mathbf{A}_4$	$\mathbf{A}_4$	has no L1U factorization, $\mathbf{A}_3$ has an L1U factorization but it is not unique, and $\mathbf{A}_4$
lyche-numerical-linear-algebra-F01108	90	■ 1.000	$\mathbf{A}_{[k]}, \mathbf{L}_{[k]}, \mathbf{U}_{[k]}$	$\mathbf{A}_{[k]}, \mathbf{L}_{[k]}, \mathbf{U}_{[k]}$	<i>L1U factorization of $\mathbf{A} \in \mathbb{C}^{n \times n}$. For $k = 1, \dots, n$ let $\mathbf{A}_{[k]}, \mathbf{L}_{[k]}, \mathbf{U}_{[k]}$ be the leading</i>
lyche-numerical-linear-algebra-F01109	90	■ 0.986	$\mathbf{A}, \mathbf{L}, \mathbf{U}$	$\mathbf{A}, \mathbf{L}, \mathbf{U}$	<i>principal submatrices of $\mathbf{A}, \mathbf{L}, \mathbf{U}$, respectively. Then $\mathbf{A}_{[k]} = \mathbf{L}_{[k]} \mathbf{U}_{[k]}$ is an L1U</i>
lyche-numerical-linear-algebra-F01110	90	■ 0.986	$\mathbf{A}_{[k]} = \mathbf{L}_{[k]} \mathbf{U}_{[k]}$	$\mathbf{A}_{[k]} = \mathbf{L}_{[k]} \mathbf{U}_{[k]}$	<i>principal submatrices of $\mathbf{A}, \mathbf{L}, \mathbf{U}$, respectively. Then $\mathbf{A}_{[k]} = \mathbf{L}_{[k]} \mathbf{U}_{[k]}$ is an L1U</i>
lyche-numerical-linear-algebra-F01111	90	■ 1.000	$\mathbf{F}_k, \mathbf{N}_k, \mathbf{T}_k \in \mathbb{C}^{n-k, n-k}$	$\mathbf{F}_k, \mathbf{N}_k, \mathbf{T}_k \in \mathbb{C}^{n-k, n-k}$	where $\mathbf{F}_k, \mathbf{N}_k, \mathbf{T}_k \in \mathbb{C}^{n-k, n-k}$. Comparing blocks we find $\mathbf{A}_{[k]} = \mathbf{L}_{[k]} \mathbf{U}_{[k]}$. Since
lyche-numerical-linear-algebra-F01112	90	■ 1.000	$\mathbf{L}_{[k]}$	$\mathbf{L}_{[k]}$	$\mathbf{L}_{[k]}$ is unit lower triangular and $\mathbf{U}_{[k]}$ is upper triangular this is an L1U factorization
lyche-numerical-linear-algebra-F01113	90	■ 1.000	$\mathbf{U}_{[k]}$	$\mathbf{U}_{[k]}$	$\mathbf{L}_{[k]}$ is unit lower triangular and $\mathbf{U}_{[k]}$ is upper triangular this is an L1U factorization
lyche-numerical-linear-algebra-F01114	90	■ 0.597	1×1	1×1	is clearly true for $n = 1$, since the unique L1U factorization of a 1×1 matrix is
lyche-numerical-linear-algebra-F01115	90	■ 1.000	$[a_{11}] = [1][a_{11}]$	$[a_{11}] = [1][a_{11}]$	$[a_{11}] = [1][a_{11}]$. Suppose that $\mathbf{A}_{[n-1]}$ has a unique L1U factorization $\mathbf{A}_{[n-1]} =$
lyche-numerical-linear-algebra-F01116	90	■ 1.000	$\mathbf{A}_{[n-1]}$	$\mathbf{A}_{[n-1]}$	$[a_{11}] = [1][a_{11}]$. Suppose that $\mathbf{A}_{[n-1]}$ has a unique L1U factorization $\mathbf{A}_{[n-1]} =$

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lyche-numerical-linear-algebra_F01117	90	1.000	$\boldsymbol{A}_{[n-1]}=$	$\boldsymbol{A}_{[n-1]} =$	$[a_{11}] = [1][a_{11}]$. Suppose that $\boldsymbol{A}_{[n-1]}$ has a unique L1U factorization $\boldsymbol{A}_{[n-1]} =$
lyche-numerical-linear-algebra_F01118	90	0.992	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$, and that $\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By block multiplication
lyche-numerical-linear-algebra_F01119	90	0.992	$\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$	$\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$, and that $\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By block multiplication
lyche-numerical-linear-algebra_F01120	90	1.000	$\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra_F01121	90	1.000	$\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$	$\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra_F01122	90	1.000	$u_{nn} \in \mathbb{C}$	$u_{nn} \in \mathbb{C}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra_F01123	90	1.000	$\boldsymbol{L}_{n-1}$	$\boldsymbol{L}_{n-1}$	Since $\boldsymbol{A}_{[n-1]}$ is nonsingular it follows that $\boldsymbol{L}_{n-1}$ and $\boldsymbol{U}_{n-1}$ are nonsingular and
lyche-numerical-linear-algebra_F01124	90	1.000	$\boldsymbol{U}_{n-1}$	$\boldsymbol{U}_{n-1}$	Since $\boldsymbol{A}_{[n-1]}$ is nonsingular it follows that $\boldsymbol{L}_{n-1}$ and $\boldsymbol{U}_{n-1}$ are nonsingular and
lyche-numerical-linear-algebra_F01125	90	1.000	$\boldsymbol{l}_n, \boldsymbol{u}_n$	$\boldsymbol{l}_n, \boldsymbol{u}_n$	therefore $\boldsymbol{l}_n, \boldsymbol{u}_n$, and u_{nn} are uniquely given. Thus (3.17) gives a unique L1U
lyche-numerical-linear-algebra_F01126	90	1.000	u_{nn}	u_{nn}	therefore $\boldsymbol{l}_n, \boldsymbol{u}_n$, and u_{nn} are uniquely given. Thus (3.17) gives a unique L1U
lyche-numerical-linear-algebra_F01127	91	1.000	$k \leq n-1$	$k \leq n-1$	$\boldsymbol{A}_{[k]}$ is singular for some $k \leq n-1$. We will show that this leads to a contradiction.
lyche-numerical-linear-algebra_F01128	91	1.000	$\boldsymbol{A}_{[j]}$	$\boldsymbol{A}_{[j]}$	Let k be the smallest integer so that $\boldsymbol{A}_{[k]}$ is singular. Since $\boldsymbol{A}_{[j]}$ is nonsingular for
lyche-numerical-linear-algebra_F01129	91	0.916	$j \leq k-1$	$j \leq k-1$	$j < k-1$ it follows from what we have already shown that $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]}$ is the
lyche-numerical-linear-algebra_F01130	91	1.000	$\boldsymbol{U}_{[k]}^T \boldsymbol{M}_k^T = \boldsymbol{C}_k^T$	$\boldsymbol{U}_{[k]}^T \boldsymbol{M}_k^T = \boldsymbol{C}_k^T$	and $\boldsymbol{L}_{[k]}$ is nonsingular. By (3.16) we have $\boldsymbol{U}_{[k]}^T \boldsymbol{M}_k^T = \boldsymbol{C}_k^T$. This can be written as
lyche-numerical-linear-algebra_F01131	91	1.000	$\boldsymbol{M}_k^T$	$\boldsymbol{M}_k^T$	$n-k$ linear systems for the columns of $\boldsymbol{M}_k^T$. By assumption $\boldsymbol{M}_k^T$ exists, but since
lyche-numerical-linear-algebra_F01132	91	1.000	$\boldsymbol{U}_{[k]}^T$	$\boldsymbol{U}_{[k]}^T$	$\boldsymbol{U}_{[k]}^T$ is singular $\boldsymbol{M}_k$ is not unique, a contradiction.
lyche-numerical-linear-algebra_F01133	91	1.000	$\boldsymbol{M}_k$	$\boldsymbol{M}_k$	$\boldsymbol{U}_{[k]}^T$ is singular $\boldsymbol{M}_k$ is not unique, a contradiction.
lyche-numerical-linear-algebra_F01134	91	1.000	$k=n$	$k = n$	By combining the last two equations in (3.18) we obtain with $k = n$
lyche-numerical-linear-algebra_F01135	91	1.000	$k-d$	$k-d$	is d -banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$
lyche-numerical-linear-algebra_F01136	91	1.000	$\boldsymbol{r}_k$	$\boldsymbol{r}_k$	is d -banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$
lyche-numerical-linear-algebra_F01137	91	1.000	$\boldsymbol{c}_k$	$\boldsymbol{c}_k$	is d -banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$
lyche-numerical-linear-algebra_F01138	91	1.000	$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$	$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$	to solve the lower triangular system $\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$ for the k th row $\boldsymbol{l}_k^T$ in $\boldsymbol{L}$ and the
lyche-numerical-linear-algebra_F01139	91	1.000	$\boldsymbol{l}_k^T$	$\boldsymbol{l}_k^T$	to solve the lower triangular system $\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$ for the k th row $\boldsymbol{l}_k^T$ in $\boldsymbol{L}$ and the
lyche-numerical-linear-algebra_F01140	91	0.699	$\begin{bmatrix} \boldsymbol{u}_k \\ u_{kk} \end{bmatrix}$	$\begin{bmatrix} \boldsymbol{u}_k \\ u_{kk} \end{bmatrix}$	k th column $\begin{bmatrix} \boldsymbol{u}_k \\ u_{kk} \end{bmatrix}$ in $\boldsymbol{U}$ for $k = 2, \dots, n$. This leads to the following algorithm to
lyche-numerical-linear-algebra_F01141	91	0.699	$k=2, \dots, n$	$k = 2, \dots, n$	k th column $\begin{bmatrix} \boldsymbol{u}_k \\ u_{kk} \end{bmatrix}$ in $\boldsymbol{U}$ for $k = 2, \dots, n$. This leads to the following algorithm to
lyche-numerical-linear-algebra_F01142	91	1.000	$d \geq 1$	$d \geq 1$	compute the L1U factorization of a d -banded matrix $\boldsymbol{A}$ with $d \geq 1$. The algorithm

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lyche-numerical-linear-algebra-F01143	91	■ 1.000	$0\left(d^2n\right)$	$O\left(d^2n\right)$	the number of arithmetic operation for this algorithm is $O(d^2n)$.
lyche-numerical-linear-algebra-F01144	91	■ 0.996	$k<n$	$k < n$	factorization even if $A_{[k]}$ is singular for some $k < n$. By Theorem 4.1 such an LU
lyche-numerical-linear-algebra-F01145	91	■ 1.000	$\boldsymbol{A}=\boldsymbol{I}\boldsymbol{A}$	$\boldsymbol{A} = \boldsymbol{I}\boldsymbol{A}$	$\boldsymbol{A}$ is $\boldsymbol{A} = \boldsymbol{I}\boldsymbol{A}$ so it always exists even if $\boldsymbol{A}$ has zeros somewhere on the diagonal.
lyche-numerical-linear-algebra-F01146	91	■ 1.000	$a_{k\ k}$	a_{kk}	By Lemma 2.5, if some a_{kk} is zero then $A_{[k]}$ is singular and the L1U factorization
lyche-numerical-linear-algebra-F01147	92	■ 1.000	$\boldsymbol{A}_{i\ i}$	$\boldsymbol{A}_{ii}$	where each diagonal block $\boldsymbol{A}_{ii}$ is square. We call the factorization
lyche-numerical-linear-algebra-F01148	92	■ 0.434	$\mathbf{L}$	$\mathbf{L}$	a block L1U factorization of $\boldsymbol{A}$. Here the i th diagonal blocks $\boldsymbol{I}$ and $\boldsymbol{U}_{ii}$ in $\boldsymbol{L}$ and
lyche-numerical-linear-algebra-F01149	92	■ 0.434	$\boldsymbol{U}_{i\ i}$	$\boldsymbol{U}_{ii}$	a block L1U factorization of $\boldsymbol{A}$. Here the i th diagonal blocks $\boldsymbol{I}$ and $\boldsymbol{U}_{ii}$ in $\boldsymbol{L}$ and
lyche-numerical-linear-algebra-F01150	92	■ 0.999	$k=1,\ldots,m-1$	$k = 1, \dots, m - 1$	are nonsingular for $\boxed{k = 1, \dots, m - 1}$.
lyche-numerical-linear-algebra-F01151	92	■ 1.000	$\boldsymbol{A}_{\{k\}}$	$\boldsymbol{A}_{\{k\}}$	Proof Suppose $\boldsymbol{A}_{[k]}$ is nonsingular for $k = 1, \dots, m - 1$. Following the proof
lyche-numerical-linear-algebra-F01152	92	■ 0.993	$\boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{A}_{\{m-1\}}$	in Theorem 3.4 suppose $\boldsymbol{A}_{[m-1]}$ has a unique block LU factorization $\boldsymbol{A}_{[m-1]} =$
lyche-numerical-linear-algebra-F01153	92	■ 0.993	$\boldsymbol{A}_{\{m-1\}}=$	$\boldsymbol{A}_{\{m-1\}} =$	in Theorem 3.4 suppose $\boldsymbol{A}_{[m-1]}$ has a unique block LU factorization $\boldsymbol{A}_{[m-1]} =$
lyche-numerical-linear-algebra-F01154	92	■ 0.999	$\boldsymbol{L}_{\{m-1\}}\boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}}\boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{L}_{[m-1]}\boldsymbol{U}_{[m-1]}$, and that $\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{[m-1]}$ are nonsingular. Then $\boldsymbol{L}_{[m-1]}$ and
lyche-numerical-linear-algebra-F01155	92	■ 0.999	$\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{L}_{[m-1]}\boldsymbol{U}_{[m-1]}$, and that $\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{[m-1]}$ are nonsingular. Then $\boldsymbol{L}_{[m-1]}$ and
lyche-numerical-linear-algebra-F01156	92	■ 0.999	$\boldsymbol{L}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}}$	$\boldsymbol{L}_{[m-1]}\boldsymbol{U}_{[m-1]}$, and that $\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{[m-1]}$ are nonsingular. Then $\boldsymbol{L}_{[m-1]}$ and
lyche-numerical-linear-algebra-F01157	92	■ 0.523	$\boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{U}_{[m-1]}$ are nonsingular and
lyche-numerical-linear-algebra-F01158	93	■ 0.934	$\boldsymbol{A}_{\{k\}}=\boldsymbol{L}_{\{k\}}\boldsymbol{U}_{\{k\}}$	$\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}}\boldsymbol{U}_{\{k\}}$	that $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]}\boldsymbol{U}_{[k]}$ is the unique block LU factorization of $\boldsymbol{A}_{[k]}$ for $k = 1, \dots, m$.
lyche-numerical-linear-algebra-F01159	93	■ 0.934	$k=1,\ldots,m$	$k = 1, \dots, m$	that $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]}\boldsymbol{U}_{[k]}$ is the unique block LU factorization of $\boldsymbol{A}_{[k]}$ for $\boxed{k = 1, \dots, m}$.
lyche-numerical-linear-algebra-F01160	93	■ 1.000	$\boldsymbol{U}_{i\ i}=\tilde{\boldsymbol{L}}_{ii}\tilde{\boldsymbol{U}}_{ii}$	$\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii}\tilde{\boldsymbol{U}}_{ii}$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii}\tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}}\hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L} \operatorname{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra-F01161	93	■ 1.000	$\boldsymbol{A}=\hat{\boldsymbol{L}}\hat{\boldsymbol{U}}$	$\boldsymbol{A} = \hat{\boldsymbol{L}}\hat{\boldsymbol{U}}$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii}\tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}}\hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L} \operatorname{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra-F01162	93	■ 1.000	$\hat{\boldsymbol{L}}:=\boldsymbol{L}\operatorname{diag}\left(\tilde{\boldsymbol{L}}_{ii}\right)$	$\hat{\boldsymbol{L}} := \boldsymbol{L} \operatorname{diag}\left(\tilde{\boldsymbol{L}}_{ii}\right)$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii}\tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}}\hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L} \operatorname{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra-F01163	93	■ 0.987	$\hat{\boldsymbol{U}}:=\operatorname{diag}\left(\tilde{\boldsymbol{L}}_{ii}^{-1}\right)\boldsymbol{U}$	$\hat{\boldsymbol{U}} := \operatorname{diag}\left(\tilde{\boldsymbol{L}}_{ii}^{-1}\right)\boldsymbol{U}$	$\hat{\boldsymbol{U}} := \operatorname{diag}(\tilde{\boldsymbol{L}}_{ii}^{-1})\boldsymbol{U}$, and this is an ordinary LU factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F01164	93	■ 0.997	$\left[\boldsymbol{b}_1,\ldots,\boldsymbol{b}_n\right]$	$[\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$	$\boxed{[\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]}$ is also lower triangular. The k th column $\boldsymbol{b}_k$ of $\boldsymbol{B}$ is the solution of the
lyche-numerical-linear-algebra-F01165	93	■ 0.997	$\boldsymbol{b}_{k\ }$	$\boldsymbol{b}_k$	$[\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$ is also lower triangular. The k th column $\boldsymbol{b}_k$ of $\boldsymbol{B}$ is the solution of the
lyche-numerical-linear-algebra-F01166	93	■ 1.000	$\boldsymbol{A}\boldsymbol{b}_{k\ }=\boldsymbol{e}_{k\ }$	$\boldsymbol{A}\boldsymbol{b}_k = \boldsymbol{e}_k$	linear systems $\boldsymbol{A}\boldsymbol{b}_k = \boldsymbol{e}_k$. Show that $b_k(k) = 1/a(k, k)$ for $k = 1, \dots, n$, and
lyche-numerical-linear-algebra-F01167	93	■ 1.000	$b_{k\ }(k)=1\ /\ a(k,\ k)$	$b_k(k) = 1/a(k, k)$	linear systems $\boldsymbol{A}\boldsymbol{b}_k = \boldsymbol{e}_k$. Show that $b_k(k) = 1/a(k, k)$ for $k = 1, \dots, n$, and
lyche-numerical-linear-algebra-F01168	94	■ 1.000	$k=n,\ n-1,\ldots,\ 1$	$k = n, n - 1, \dots, 1$	for $\boxed{k = n, n - 1, \dots, 1}$.
lyche-numerical-linear-algebra-F01169	94	■ 1.000	$\frac{1}{3}n\left(2n^2+1\right)\approx G_n$	$\frac{1}{3}n\left(2n^2+1\right)\approx G_n$	cation $\boldsymbol{A}\boldsymbol{B}$ can be done in $\frac{1}{3}n(2n^2+1) \approx G_n$ arithmetic operations when $\boldsymbol{A} \in \mathbb{R}^{n \times n}$

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lyche-numerical-linear-algebra_F01170	94	■ 1.000	$\boldsymbol{B} \in \mathbb{R}^{n \times n}$	$\boldsymbol{B} \in \mathbb{R}^{n \times n}$	is lower triangular and $\boldsymbol{B} \in \mathbb{R}^{n \times n}$ is upper triangular. What about $\boldsymbol{BA}$?
lyche-numerical-linear-algebra_F01171	94	■ 1.000	$\boldsymbol{B} \boldsymbol{A}$	$\boldsymbol{BA}$	is lower triangular and $\boldsymbol{B} \in \mathbb{R}^{n \times n}$ is upper triangular. What about $\boldsymbol{BA}$?
lyche-numerical-linear-algebra_F01172	94	■ 1.000	$\boldsymbol{P}, \boldsymbol{L}, \boldsymbol{U}$	$\boldsymbol{P}, \boldsymbol{L}, \boldsymbol{U}$	Exercise 3.5 (Using PLU for A^*) Suppose we know the PLU factors $\boldsymbol{P}, \boldsymbol{L}, \boldsymbol{U}$ in a
lyche-numerical-linear-algebra_F01173	94	■ 1.000	$\boldsymbol{A}^* \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{A}^* \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{A}^* \boldsymbol{x} = \boldsymbol{b}$ economically.
lyche-numerical-linear-algebra_F01174	94	■ 1.000	$2G_n$	$2G_n$	$2G_n$ arithmetic operations to compute $\boldsymbol{A}^{-1} = \boldsymbol{U}^{-1} \boldsymbol{L}^{-1} \boldsymbol{P}^T$. Here we have not
lyche-numerical-linear-algebra_F01175	94	■ 1.000	$\boldsymbol{A}^{-1} = \boldsymbol{U}^{-1} \boldsymbol{L}^{-1} \boldsymbol{P}^T$	$\boldsymbol{A}^{-1} = \boldsymbol{U}^{-1} \boldsymbol{L}^{-1} \boldsymbol{P}^T$	$2G_n$ arithmetic operations to compute $\boldsymbol{A}^{-1} = \boldsymbol{U}^{-1} \boldsymbol{L}^{-1} \boldsymbol{P}^T$. Here we have not
lyche-numerical-linear-algebra_F01176	94	■ 1.000	$\boldsymbol{P}^T$	$\boldsymbol{P}^T$	counted the final multiplication with $\boldsymbol{P}^T$ which amounts to n row interchanges.
lyche-numerical-linear-algebra_F01177	95	■ 0.993	$k=1,2, \ldots, n-1$	$k = 1, 2, \dots, n - 1$	1. for $k = 1, 2, \dots, n - 1$
lyche-numerical-linear-algebra_F01178	95	■ 1.000	$a_{k,j}$	$a_{k,j}$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k + 1, \dots, n$
lyche-numerical-linear-algebra_F01179	95	■ 1.000	$a_{k+1,j}$	$a_{k+1,j}$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k + 1, \dots, n$
lyche-numerical-linear-algebra_F01180	95	■ 1.000	$j=k, k+1, \ldots, n$	$j = k, k + 1, \dots, n$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k + 1, \dots, n$
lyche-numerical-linear-algebra_F01181	95	■ 1.000	$i=k+1, k+2, \ldots, n$	$i = k + 1, k + 2, \dots, n$	(b) for $i = k + 1, k + 2, \dots, n$
lyche-numerical-linear-algebra_F01182	95	■ 1.000	$a_{i,k} = m_{i,k} = a_{i,k} / a_{k,k}$	$a_{i,k} = m_{i,k} = a_{i,k} / a_{k,k}$	i. $a_{i,k} = m_{i,k} = a_{i,k} / a_{k,k}$
lyche-numerical-linear-algebra_F01183	95	■ 1.000	$a_{i,j} = a_{i,j} - m_{i,k} a_{k,j}$	$a_{i,j} = a_{i,j} - m_{i,k} a_{k,j}$	ii. $a_{i,j} = a_{i,j} - m_{i,k} a_{k,j}$ for $j = k + 1, k + 2, \dots, n$
lyche-numerical-linear-algebra_F01184	95	■ 1.000	$j=k+1, k+2, \ldots, n$	$j = k + 1, k + 2, \dots, n$	ii. $a_{i,j} = a_{i,j} - m_{i,k} a_{k,j}$ for $j = k + 1, k + 2, \dots, n$
lyche-numerical-linear-algebra_F01185	95	■ 1.000	b_k	b_k	(a) exchange b_k and b_{k+1}
lyche-numerical-linear-algebra_F01186	95	■ 1.000	b_{k+1}	b_{k+1}	(a) exchange b_k and b_{k+1}
lyche-numerical-linear-algebra_F01187	95	■ 1.000	$b_i = b_i - a_{i,k} b_k$	$b_i = b_i - a_{i,k} b_k$	(b) $b_i = b_i - a_{i,k} b_k$ for $i = k + 1, k + 2, \dots, n$
lyche-numerical-linear-algebra_F01188	95	■ 1.000	$x_n = b_n / a_{n,n}$	$x_n = b_n / a_{n,n}$	2. $x_n = b_n / a_{n,n}$
lyche-numerical-linear-algebra_F01189	95	■ 0.981	$=0$	$= 0$	(a) $sum = 0$
lyche-numerical-linear-algebra_F01190	95	■ 0.958	$+a_{k,j} x_j$	$+a_{k,j} x_j$	(b) $sum = sum + a_{k,j} x_j$ for $j = k + 1, k + 2, \dots, n$
lyche-numerical-linear-algebra_F01191	95	■ 1.000	$x_k = (b_k -$	$x_k = (b_k -$	(c) $x_k = (b_k - sum) / a_{k,k}$
lyche-numerical-linear-algebra_F01192	95	■ 1.000	$) / a_{k,k}$	$) / a_{k,k}$	(c) $x_k = (b_k - sum) / a_{k,k}$
lyche-numerical-linear-algebra_F01193	95	■ 1.000	$\boldsymbol{H} \in \mathbb{R}^{n \times n}$	$\boldsymbol{H} \in \mathbb{R}^{n \times n}$	We say that $\boldsymbol{H} \in \mathbb{R}^{n \times n}$ is unreduced upper Hessenberg if it is upper Hessenberg
lyche-numerical-linear-algebra_F01194	95	■ 1.000	$h_{i,i-1} \neq 0$	$h_{i,i-1} \neq 0$	and the subdiagonal elements $h_{i,i-1} \neq 0$ for $i = 2, \dots, n$.
lyche-numerical-linear-algebra_F01195	95	■ 0.983	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$	solving the linear system $\boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$ using suitable specializations of Algorithms 1
lyche-numerical-linear-algebra_F01196	95	■ 1.000	$\boldsymbol{U} \in \mathbb{R}^{n \times n}$	$\boldsymbol{U} \in \mathbb{R}^{n \times n}$	c) Let $\boldsymbol{U} \in \mathbb{R}^{n \times n}$ be upper triangular and nonsingular. We define

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lyche-numerical-linear-algebra_F01197	95	1.000	$\boldsymbol{v} \in \mathbb{R}^n$	$\boldsymbol{v} \in \mathbb{R}^n$	where $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{e}_1$ is the first unit vector in $\mathbb{R}^n$. We also let
lyche-numerical-linear-algebra_F01198	95	1.000	$\boldsymbol{e}_1$	$\boldsymbol{e}_1$	where $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{e}_1$ is the first unit vector in $\mathbb{R}^n$. We also let
lyche-numerical-linear-algebra_F01199	95	0.999	$\boldsymbol{I}_{i,j}$	$\boldsymbol{I}_{i,j}$	where the $\boldsymbol{I}_{i,j}$ are obtained from the identity matrix by interchanging rows i and
lyche-numerical-linear-algebra_F01200	95	0.822	$\boldsymbol{E} := \boldsymbol{C} \boldsymbol{P}$	$\boldsymbol{E} := \boldsymbol{C} \boldsymbol{P}$	j . Explain why the matrix $\boldsymbol{E} := \boldsymbol{C} \boldsymbol{P}$ is unreduced upper Hessenberg.
lyche-numerical-linear-algebra_F01201	96	1.000	$\boldsymbol{W} \in \mathbb{R}^n$	$\boldsymbol{W} \in \mathbb{R}^n$	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{U}$. To a given $\boldsymbol{W} \in \mathbb{R}^n$ we define a rank one modification of $\boldsymbol{A}$ by
lyche-numerical-linear-algebra_F01202	96	1.000	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{H} \boldsymbol{P}^T$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{H} \boldsymbol{P}^T$	Show that $\boldsymbol{B}$ has the factorization $\boldsymbol{B} = \boldsymbol{L} \boldsymbol{H} \boldsymbol{P}^T$, where $\boldsymbol{L}$ is unit lower triangular,
lyche-numerical-linear-algebra_F01203	96	1.000	$\boldsymbol{H}$	$\boldsymbol{H}$	$\boldsymbol{P}$ is given by (3.29) and $\boldsymbol{H}$ is unreduced upper Hessenberg.
lyche-numerical-linear-algebra_F01204	96	1.000	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{b}$, where $\boldsymbol{B}$ is given by (3.30). We assume that the matrices $\boldsymbol{L}$ and $\boldsymbol{U}$ in
lyche-numerical-linear-algebra_F01205	96	0.999	$1 \leq$	$1 \leq$	Exercise 3.9 (# Operations for Banded Triangular Systems) Show that for $1 \leq$
lyche-numerical-linear-algebra_F01206	96	1.000	$d \leq n$	$d \leq n$	$d \leq n$ Algorithm 3.4, with $A(k, k) = 1$ for $k = 1, \dots, n$ in Algorithm 3.1, requires
lyche-numerical-linear-algebra_F01207	96	1.000	$A(k, k) = 1$	$A(k, k) = 1$	$d \leq n$ Algorithm 3.4, with $A(k, k) =$ for $k = 1, \dots, n$ in Algorithm 3.1, requires
lyche-numerical-linear-algebra_F01208	96	1.000	$N_{LU}(n, d) := (2d^2 + d)n - (d^2 + d)(8d + 1)/6 = O(d^2n)$	$N_{LU}(n, d) := (2d^2 + d)n - (d^2 + d)(8d + 1)/6 = O(d^2n)$	exactly $N_{LU}(n, d) := (2d^2 + d)n - (d^2 + d)(8d + 1)/6 = O(d^2n)$ operations. ² In
lyche-numerical-linear-algebra_F01209	96	1.000	$d = n - 1$	$d = n - 1$	particular, for a full matrix $d = n - 1$ and we find $N_{LU}(n, n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$
lyche-numerical-linear-algebra_F01210	96	1.000	$N_{LU}(n, n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$	$N_{LU}(n, n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$	particular, for a full matrix $d = n - 1$ and we find $N_{LU}(n, n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$
lyche-numerical-linear-algebra_F01211	96	1.000	$N_{LU}(n, 1) = 3n - 3 = O(n)$	$N_{LU}(n, 1) = 3n - 3 = O(n)$	tridiagonal matrix $N_{LU}(n, 1) = 3n - 3 = O(n)$.
lyche-numerical-linear-algebra_F01212	96	0.985	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	Exercise 3.12 (Row Interchange) Show that $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has a unique L1U
lyche-numerical-linear-algebra_F01213	96	0.999	$2 \leq k \leq d$	$2 \leq k \leq d$	² Hint: Consider the cases $2 \leq k \leq d$ and $d + 1 \leq k \leq n$ separately.
lyche-numerical-linear-algebra_F01214	96	0.999	$d + 1 \leq k \leq n$	$d + 1 \leq k \leq n$	² Hint: Consider the cases $2 \leq k \leq d$ and $d + 1 \leq k \leq n$ separately.
lyche-numerical-linear-algebra_F01215	97	1.000	u_{kk}	u_{kk}	u_{kk} in the L1U factorization are
lyche-numerical-linear-algebra_F01216	97	1.000	$\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$	$\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.
lyche-numerical-linear-algebra_F01217	97	1.000	$\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}$	$\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.
lyche-numerical-linear-algebra_F01218	97	1.000	$\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$	$\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.
lyche-numerical-linear-algebra_F01219	97	1.000	$l_{i,j}, i > j$	$l_{i,j}, i > j$	b) Suppose that the elements $l_{i,j}, i > j$ in $\boldsymbol{L}$ and $u_{i,j}, j \geq i$ in $\boldsymbol{U}$ are stored in
lyche-numerical-linear-algebra_F01220	97	1.000	$u_{i,j}, j \geq i$	$u_{i,j}, j \geq i$	b) Suppose that the elements $l_{i,j}, i > j$ in $\boldsymbol{L}$ and $u_{i,j}, j \geq i$ in $\boldsymbol{U}$ are stored in
lyche-numerical-linear-algebra_F01221	97	1.000	$a_{i,j}$	$a_{i,j}$	$\boldsymbol{A}$ with elements $a_{i,j}$. Write an algorithm that overwrites the elements in $\boldsymbol{A}$ with
lyche-numerical-linear-algebra_F01222	97	1.000	$\boldsymbol{s} \in \mathbb{R}^n$	$\boldsymbol{s} \in \mathbb{R}^n$	ones in $\boldsymbol{A}^{-1}$. Only one extra vector $\boldsymbol{s} \in \mathbb{R}^n$ should be used.

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lyche-numerical-linear-algebra-F01223	97	1.000	$\boldsymbol{T} \boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{THx} = \boldsymbol{b}$	Exercise 3.18 (Solving $\boldsymbol{THx} = \boldsymbol{b}$ (Exam Exercise 1981-3)) In this exercise we
lyche-numerical-linear-algebra-F01224	97	1.000	$\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$	$\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$	consider nonsingular matrices $\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$ with $\boldsymbol{T} = (t_{ij})$ upper triangular,
lyche-numerical-linear-algebra-F01225	97	1.000	$\boldsymbol{T} = \left(t_{ij}\right)$	$\boldsymbol{T} = (t_{ij})$	consider nonsingular matrices $\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$ with $\boldsymbol{T} = (t_{ij})$ upper triangular,
lyche-numerical-linear-algebra-F01226	97	1.000	$\boldsymbol{H} = \left(h_{ij}\right)$	$\boldsymbol{H} = (h_{ij})$	$\boldsymbol{H} = (h_{ij})$ upper Hessenberg and $\boldsymbol{S} := \boldsymbol{TH}$. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra-F01227	97	1.000	$\boldsymbol{S} := \boldsymbol{T} \boldsymbol{H}$	$\boldsymbol{S} := \boldsymbol{TH}$	$\boldsymbol{H} = (h_{ij})$ upper Hessenberg and $\boldsymbol{S} := \boldsymbol{TH}$. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra-F01228	97	1.000	$\boldsymbol{H} = \boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{H} = \boldsymbol{LU}$	LU1 factorization $\boldsymbol{H} = \boldsymbol{LU}$ with $\ \boldsymbol{L}\ _{\infty} \ \boldsymbol{U}\ _{\infty} \leq K \ \boldsymbol{H}\ _{\infty}$ for a constant K not
lyche-numerical-linear-algebra-F01229	97	1.000	$\ \boldsymbol{L}\ _{\infty} \ \boldsymbol{U}\ _{\infty} \leq K \ \boldsymbol{H}\ _{\infty}$	$\ \boldsymbol{L}\ _{\infty} \ \boldsymbol{U}\ _{\infty} \leq K \ \boldsymbol{H}\ _{\infty}$	LU1 factorization $\boldsymbol{H} = \boldsymbol{LU}$ with $\ \boldsymbol{L}\ _{\infty} \ \boldsymbol{U}\ _{\infty} \leq K \ \boldsymbol{H}\ _{\infty}$ for a constant K not
lyche-numerical-linear-algebra-F01230	97	1.000	$\boldsymbol{S}$	$\boldsymbol{S}$	a) Give an algorithm which computes $\boldsymbol{S}$ from $\boldsymbol{T}$ and $\boldsymbol{H}$ without using the lower parts
lyche-numerical-linear-algebra-F01231	97	0.999	$\left(t_{ij}, i > j\right)$	$(t_{ij}, i > j)$	$(t_{ij}, i > j)$ of $\boldsymbol{T}$ and $(h_{ij}, i > j + 1)$ of $\boldsymbol{H}$. In what order should the elements
lyche-numerical-linear-algebra-F01232	97	0.999	$\left(h_{ij}, i > j + 1\right)$	$(h_{ij}, i > j + 1)$	$(t_{ij}, i > j)$ of $\boldsymbol{T}$ and $(h_{ij}, i > j + 1)$ of $\boldsymbol{H}$. In what order should the elements
lyche-numerical-linear-algebra-F01233	98	0.987	$\boldsymbol{T}, \boldsymbol{H}$	$\boldsymbol{T}, \boldsymbol{H}$	d) Given $\boldsymbol{b} \in \mathbb{R}^n$ and $\boldsymbol{T}, \boldsymbol{H}$ as before. Suppose $\boldsymbol{S}$ and the LU1-factorization are not
lyche-numerical-linear-algebra-F01234	98	1.000	$\boldsymbol{S} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{Sx} = \boldsymbol{b}$	computed. We want to find $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Sx} = \boldsymbol{b}$. We have the 2 following
lyche-numerical-linear-algebra-F01235	98	0.779	$\boldsymbol{S} = \boldsymbol{T} \boldsymbol{H}$	$\boldsymbol{S} = \boldsymbol{TH}$	1. $\boldsymbol{S} = \boldsymbol{TH}$
lyche-numerical-linear-algebra-F01236	98	0.991	$\boldsymbol{T} \boldsymbol{z} = \boldsymbol{b}$	$\boldsymbol{Tz} = \boldsymbol{b}$	1. Solve $\boldsymbol{Tz} = \boldsymbol{b}$
lyche-numerical-linear-algebra-F01237	98	0.894	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{z}$	$\boldsymbol{Hx} = \boldsymbol{z}$	2. Solve $\boldsymbol{Hx} = \boldsymbol{z}$
lyche-numerical-linear-algebra-F01238	98	1.000	$\boldsymbol{a}_1, \boldsymbol{a}_2, \dots, \boldsymbol{a}_n$	$\boldsymbol{a}_1, \boldsymbol{a}_2, \dots, \boldsymbol{a}_n$	$\mathbb{R}^{n \times n}$ be nonsingular with columns $\boldsymbol{a}_1, \boldsymbol{a}_2, \dots, \boldsymbol{a}_n$. We assume that $\boldsymbol{A}$ has a unique
lyche-numerical-linear-algebra-F01239	98	1.000	$p \leq n$	$p \leq n$	For a positive integer $p \leq n$ and $\boldsymbol{b} \in \mathbb{R}^n$ we define
lyche-numerical-linear-algebra-F01240	98	1.000	$\boldsymbol{H} := \boldsymbol{L}^{-1} \boldsymbol{B}$	$\boldsymbol{H} := \boldsymbol{L}^{-1} \boldsymbol{B}$	a) Show that $\boldsymbol{H} := \boldsymbol{L}^{-1} \boldsymbol{B}$ is upper Hessenberg. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra-F01241	98	1.000	$\boldsymbol{H} = \boldsymbol{L}_H \boldsymbol{U}_H$	$\boldsymbol{H} = \boldsymbol{L}_H \boldsymbol{U}_H$	L1U factorization. $\boldsymbol{H} = \boldsymbol{L}_H \boldsymbol{U}_H$.
lyche-numerical-linear-algebra-F01242	98	1.000	$\boldsymbol{H} := \boldsymbol{L}_H \boldsymbol{U}_H$	$\boldsymbol{H} := \boldsymbol{L}_H \boldsymbol{U}_H$	c) Suppose we have found the L1U factorization $\boldsymbol{H} := \boldsymbol{L}_H \boldsymbol{U}_H$ of $\boldsymbol{H}$. Explain how
lyche-numerical-linear-algebra-F01243	98	1.000	$\boldsymbol{L}_H$	$\boldsymbol{L}_H$	we can find the L1U factorization of $\boldsymbol{B}$ from $\boldsymbol{L}_H$ and $\boldsymbol{U}_H$.
lyche-numerical-linear-algebra-F01244	98	1.000	$\boldsymbol{U}_H$	$\boldsymbol{U}_H$	we can find the L1U factorization of $\boldsymbol{B}$ from $\boldsymbol{L}_H$ and $\boldsymbol{U}_H$.
lyche-numerical-linear-algebra-F01245	98	0.999	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{L}$	$\boldsymbol{A} = \boldsymbol{UL}$	$\mathbb{R}^{n \times n}$ has a UIL factorization if $\boldsymbol{A} = \boldsymbol{UL}$ for an upper triangular matrix $\boldsymbol{U} \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01246	98	1.000	$\boldsymbol{P} \in \mathbb{R}^{n \times n}$	$\boldsymbol{P} \in \mathbb{R}^{n \times n}$	b) Let the columns of $\boldsymbol{P} \in \mathbb{R}^{n \times n}$ be the unit vectors in reverse order, i.e.,
lyche-numerical-linear-algebra-F01247	99	1.000	$\boldsymbol{P}^T = \boldsymbol{P}$	$\boldsymbol{P}^T = \boldsymbol{P}$	Show that $\boldsymbol{P}^T = \boldsymbol{P}$ and $\boldsymbol{P}^2 = \boldsymbol{I}$. What is the connection between the elements
lyche-numerical-linear-algebra-F01248	99	1.000	$\boldsymbol{P}^2 = \boldsymbol{I}$	$\boldsymbol{P}^2 = \boldsymbol{I}$	Show that $\boldsymbol{P}^T = \boldsymbol{P}$ and $\boldsymbol{P}^2 = \boldsymbol{I}$. What is the connection between the elements
lyche-numerical-linear-algebra-F01249	99	1.000	$\boldsymbol{P} \boldsymbol{A}$	$\boldsymbol{PA}$	in $\boldsymbol{A}$ and $\boldsymbol{PA}$?
lyche-numerical-linear-algebra-F01250	99	1.000	$\boldsymbol{B} := \boldsymbol{P} \boldsymbol{A} \boldsymbol{P}$	$\boldsymbol{B} := \boldsymbol{PAP}$	c) Let $\boldsymbol{B} := \boldsymbol{PAP}$. Find integers r, s , depending on i, j, n , such that $b_{i,j} = a_{r,s}$.

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lyche-numerical-linear-algebra-F01251	99	1.000	r, s	r, s	c) Let $\mathbf{B} := \mathbf{PAP}$. Find integers r, s , depending on i, j, n , such that $b_{i,j} = a_{r,s}$.
lyche-numerical-linear-algebra-F01252	99	1.000	i, j, n	i, j, n	c) Let $\mathbf{B} := \mathbf{PAP}$. Find integers r, s , depending on i, j, n , such that $b_{i,j} = a_{r,s}$.
lyche-numerical-linear-algebra-F01253	99	1.000	$b_{\{i, j\}} = a_{\{r, s\}}$	$b_{i,j} = a_{r,s}$	c) Let $\mathbf{B} := \mathbf{PAP}$. Find integers r, s , depending on i, j, n , such that $b_{i,j} = a_{r,s}$.
lyche-numerical-linear-algebra-F01254	99	1.000	$b_{\{i, j\}}$	$b_{i,j}$	The elements $b_{i,j}$ in $\mathbf{B}$ should be stored in position i, j in $\mathbf{A}$. You should not use
lyche-numerical-linear-algebra-F01255	99	1.000	$w \in \mathbb{R}$	$w \in \mathbb{R}$	other matrices than $\mathbf{A}$ and a scalar $w \in \mathbb{R}$.
lyche-numerical-linear-algebra-F01256	99	1.000	$\text{\textbf{\textit{P}}}\text{\textbf{\textit{A}}}\text{\textbf{\textit{P}}} = \text{\textbf{\textit{M}}}\text{\textbf{\textit{R}}}$	$\mathbf{PAP} = \mathbf{MR}$	e) Let $\mathbf{PAP} = \mathbf{MR}$ be an LIU factorization of $\mathbf{PAP}$, i.e., $\mathbf{M}$ is lower triangular
lyche-numerical-linear-algebra-F01257	99	1.000	$\text{\textbf{\textit{P}}}\text{\textbf{\textit{A}}}\text{\textbf{\textit{P}}}$	$\mathbf{PAP}$	e) Let $\mathbf{PAP} = \mathbf{MR}$ be an LIU factorization of $\mathbf{PAP}$, i.e., $\mathbf{M}$ is lower triangular
lyche-numerical-linear-algebra-F01258	99	1.000	$\text{\textbf{\textit{M}}}$	$\mathbf{M}$	e) Let $\mathbf{PAP} = \mathbf{MR}$ be an LIU factorization of $\mathbf{PAP}$, i.e., $\mathbf{M}$ is lower triangular
lyche-numerical-linear-algebra-F01259	99	1.000	$\text{\textbf{\textit{M}}}, \text{\textbf{\textit{R}}}$	$\mathbf{M}, \mathbf{R}$	$\mathbf{L}$ in a UIL factorization of $\mathbf{A}$ in terms of $\mathbf{M}, \mathbf{R}$ and $\mathbf{P}$.
lyche-numerical-linear-algebra-F01260	99	0.504	$\hat{\text{\textbf{\textit{L}}}}$	$\hat{\mathbf{L}}$	Exercise 3.21 (Making Block LU into LU) Show that $\hat{\mathbf{L}}$ is unit lower triangular
lyche-numerical-linear-algebra-F01261	99	1.000	$\hat{\text{\textbf{\textit{U}}}}$	$\hat{\mathbf{U}}$	and $\hat{\mathbf{U}}$ is upper triangular.
lyche-numerical-linear-algebra-F01262	100	0.997	$a_{\{j\ i\}} = \bar{a}_{\{i\ j\}}$	$a_{ji} = \bar{a}_{ij}$	matrices. Recall that a matrix $\mathbf{A} \in \mathbb{C}^{n \times n}$ is Hermitian if $\mathbf{A}^* = \mathbf{A}$, i.e., $a_{ji} = \bar{a}_{ij}$
lyche-numerical-linear-algebra-F01263	100	1.000	$a_{\{i\ i\}} = \bar{a}_{\{i\ i\}}$	$a_{ii} = \bar{a}_{ii}$	for all i, j . A real Hermitian matrix is symmetric. Since $a_{ii} = \bar{a}_{ii}$ the diagonal
lyche-numerical-linear-algebra-F01264	100	1.000	$\text{\textbf{\textit{U}}} = \text{\textbf{\textit{L}}}^*$	$\mathbf{U} = \mathbf{L}^*$	1. the LDL* factorization which is an LDU factorization with $\mathbf{U} = \mathbf{L}^*$ and $\mathbf{D}$ a
lyche-numerical-linear-algebra-F01265	100	1.000	$l_{\{i\ i\}} > 0$	$l_{ii} > 0$	2. the LL* factorization which is an LU factorization with $\mathbf{U} = \mathbf{L}^*$ and $l_{ii} > 0$
lyche-numerical-linear-algebra-F01266	100	0.999	$\text{\textbf{\textit{A}}}^* = \left(\text{\textbf{\textit{L}}} \text{\textbf{\textit{D}}} \text{\textbf{\textit{L}}}^* \right)^* = \text{\textbf{\textit{L}}} \text{\textbf{\textit{D}}} \text{\textbf{\textit{L}}}^* = \text{\textbf{\textit{A}}}$	$\mathbf{A}^* = (\mathbf{LDL}^*)^* = \mathbf{LD}^* \mathbf{L}^* = \mathbf{A}$	that $\mathbf{A}^* = (\mathbf{LDL}^*)^* = \mathbf{LD}^* \mathbf{L}^* = \mathbf{A}$. The LL* factorization is called a Cholesky
lyche-numerical-linear-algebra-F01267	100	0.966	$b \in \mathbb{C}$	$b \in \mathbb{C}$	<i>Example 4.1 (LDL* of 2×2 Hermitian Matrix)</i> Let $a, d \in \mathbb{R}$ and $b \in \mathbb{C}$. An LDL*
lyche-numerical-linear-algebra-F01268	101	1.000	$d_{\{1\}}, d_{\{2\}}$	d_1, d_2	for the unknowns l_1 in $\mathbf{L}$ and d_1, d_2 in $\mathbf{D}$. They are determined from
lyche-numerical-linear-algebra-F01269	101	1.000	$d_{\{1\}}$	d_1	1. $a \neq 0$: The matrix has a unique LDL* factorization. Note that d_1 and d_2 are real.
lyche-numerical-linear-algebra-F01270	101	1.000	$d_{\{2\}}$	d_2	1. $a \neq 0$: The matrix has a unique LDL* factorization. Note that d_1 and d_2 are real.
lyche-numerical-linear-algebra-F01271	101	1.000	$a=b=0$	$a = b = 0$	2. $a = b = 0$: The LDL* factorization exists, but it is not unique. Any value for l_1
lyche-numerical-linear-algebra-F01272	101	1.000	$a=0, b \neq 0$	$a = 0, b \neq 0$	3. $a = 0, b \neq 0$: No LDL* factorization exists.
lyche-numerical-linear-algebra-F01273	101	0.830	$\text{\textbf{\textit{A}}} = \text{\textbf{\textit{L}}} \text{\textbf{\textit{D}}} \text{\textbf{\textit{L}}}^*$	$\mathbf{A} = \mathbf{LDL}^*$	Lemma 4.1 (LDL* of Leading Principal Sub Matrices) Suppose $\mathbf{A} = \mathbf{LDL}^*$
lyche-numerical-linear-algebra-F01274	101	0.954	$\text{\textbf{\textit{L}}} \text{\textbf{\textit{D}}} \text{\textbf{\textit{L}}}^*$	$\mathbf{LDL}^*$	is an LDL* factorization of $\mathbf{A} \in \mathbb{C}^{n \times n}$. For $k = 1, \dots, n$ let $\mathbf{A}_{[k]}, \mathbf{L}_{[k]}$ and $\mathbf{D}_{[k]}$
lyche-numerical-linear-algebra-F01275	101	0.954	$\text{\textbf{\textit{A}}}_{\{[k]\}}, \text{\textbf{\textit{L}}}_{\{[k]\}}$	$\mathbf{A}_{[k]}, \mathbf{L}_{[k]}$	is an LDL* factorization of $\mathbf{A} \in \mathbb{C}^{n \times n}$. For $k = 1, \dots, n$ let $\mathbf{A}_{[k]}, \mathbf{L}_{[k]}$ and $\mathbf{D}_{[k]}$
lyche-numerical-linear-algebra-F01276	101	0.954	$\text{\textbf{\textit{D}}}_{\{[k]\}}$	$\mathbf{D}_{[k]}$	is an LDL* factorization of $\mathbf{A} \in \mathbb{C}^{n \times n}$. For $k = 1, \dots, n$ let $\mathbf{A}_{[k]}, \mathbf{L}_{[k]}$ and $\mathbf{D}_{[k]}$
lyche-numerical-linear-algebra-F01277	101	1.000	$\text{\textbf{\textit{A}}}, \text{\textbf{\textit{L}}}$	$\mathbf{A}, \mathbf{L}$	be the leading principal submatrices of $\mathbf{A}, \mathbf{L}$ and $\mathbf{D}$, respectively. Then $\mathbf{A}_{[k]} =$

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lyche-numerical-linear-algebra- F01278	101	1.000	$\boldsymbol{A}_{[k]}=$	$\boldsymbol{A}_{[k]} =$	<i>be the leading principal submatrices of $\boldsymbol{A}, \boldsymbol{L}$ and $\boldsymbol{D}$, respectively. Then $\boldsymbol{A}_{[k]} =$</i>
lyche-numerical-linear-algebra- F01279	101	1.000	$\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$ $\boldsymbol{L}_{[k]}^*$	$\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$	$\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$ is an LDL* factorization of $\boldsymbol{A}_{[k]}$ for $k = 1, \dots, n$.
lyche-numerical-linear-algebra- F01280	101	1.000	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D}$ $\boldsymbol{L}^*$	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^*$	Proof For $k = 1, \dots, n - 1$ we partition $\boldsymbol{A} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^*$ as follows:
lyche-numerical-linear-algebra- F01281	101	1.000	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$	where $\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$. Block multiplication gives $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$.
lyche-numerical-linear-algebra- F01282	101	1.000	$\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$	$\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$	where $\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$. Block multiplication gives $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$.
lyche-numerical-linear-algebra- F01283	101	1.000	$\boldsymbol{A}=\boldsymbol{A}^*$	$\boldsymbol{A} = \boldsymbol{A}^*$	factorization if and only if $\boldsymbol{A} = \boldsymbol{A}^*$ and $\boldsymbol{A}_{[k]}$ is nonsingular for $k = 1, \dots, n - 1$.
lyche-numerical-linear-algebra- F01284	101	1.000	$\boldsymbol{A}_{[k]}^*=\boldsymbol{A}_{[k]}$	$\boldsymbol{A}_{[k]}^* = \boldsymbol{A}_{[k]}$	Note that $\boldsymbol{A}_{[k]}^* = \boldsymbol{A}_{[k]}$ for $k = 1, \dots, n$. We use induction on n to show that $\boldsymbol{A}$
lyche-numerical-linear-algebra- F01285	101	0.983	$\left[\boldsymbol{a}_{11} \right] = \left[\boldsymbol{a}_{11} \right] \left[\boldsymbol{a}_{11} \right]$	$[a_{11}] = [1] [a_{11}] [1]$	unique LDL* factorization of a 1-by-1 matrix is $[a_{11}] = [1][a_{11}][1]$ and a_{11} is real
lyche-numerical-linear-algebra- F01286	101	0.997	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$ $\boldsymbol{L}_{n-1}^*$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$, and that $\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By definition $\boldsymbol{D}_{n-1}$ is
lyche-numerical-linear-algebra- F01287	101	0.997	$\boldsymbol{D}_{n-1}$	$\boldsymbol{D}_{n-1}$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$, and that $\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By definition $\boldsymbol{D}_{n-1}$ is
lyche-numerical-linear-algebra- F01288	102	1.000	$\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$	$\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$, and
lyche-numerical-linear-algebra- F01289	102	1.000	d_{nn}	d_{nn}	nonsingular. Also d_{nn} is real since a_{nn} and $\boldsymbol{D}_{n-1}$ are real.
lyche-numerical-linear-algebra- F01290	102	1.000	a_{nn}	a_{nn}	nonsingular. Also d_{nn} is real since a_{nn} and $\boldsymbol{D}_{n-1}$ are real.
lyche-numerical-linear-algebra- F01291	102	1.000	$\frac{1}{2} G_n$	$\frac{1}{2} G_n$	$\frac{1}{2} G_n$, half the number of operations needed for the LU factorization. Indeed, in the
lyche-numerical-linear-algebra- F01292	102	1.000	$\boldsymbol{m}$	$\boldsymbol{m}$	and $\boldsymbol{m}$, while only one such system is needed to find $\boldsymbol{m}$ in the Hermitian case (4.3).
lyche-numerical-linear-algebra- F01293	102	1.000	$f: \mathbb{C}^n \rightarrow \mathbb{R}$	$f: \mathbb{C}^n \rightarrow \mathbb{R}$	Given $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. The function $f: \mathbb{C}^n \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra- F01294	103	0.998	$\overline{f(\boldsymbol{x})}=\overline{f(\boldsymbol{x})^*}$ $\boldsymbol{A} \boldsymbol{x}=\left(\boldsymbol{x}^* \boldsymbol{A}\right)^*$ $\boldsymbol{A} \boldsymbol{x}\right)^*=\boldsymbol{x}^* \boldsymbol{A}^*$ $\boldsymbol{A} \boldsymbol{x}=\boldsymbol{x}^* \boldsymbol{A}^*$ $\boldsymbol{A} \boldsymbol{x}=\boldsymbol{x}^* \boldsymbol{A}^*$ $\boldsymbol{A} \boldsymbol{x}=\boldsymbol{x}^* \boldsymbol{A}^*$	$\overline{f(\boldsymbol{x})} = \overline{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}} = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x} = f(\boldsymbol{x})$	$\overline{f(\boldsymbol{x})} = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x} = f(\boldsymbol{x})$.
lyche-numerical-linear-algebra- F01295	103	1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} > 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} > 0$	(i) positive definite if $\boldsymbol{A}^* = \boldsymbol{A}$ and $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{C}^n$;
lyche-numerical-linear-algebra- F01296	103	1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0$	(ii) positive semidefinite if $\boldsymbol{A}^* = \boldsymbol{A}$ and $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \in \mathbb{C}^n$;
lyche-numerical-linear-algebra- F01297	103	1.000	$-\boldsymbol{A}$	$-\boldsymbol{A}$	(iii) negative (semi)definite if $-\boldsymbol{A}$ is positive (semi)definite.
lyche-numerical-linear-algebra- F01298	103	1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0 \implies \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0 \implies \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0 \implies \boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra- F01299	103	0.999	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0$	3. A positive definite matrix $\boldsymbol{A}$ is nonsingular. For if $\boldsymbol{A} \boldsymbol{x} = \mathbf{0}$ then $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0$ and
lyche-numerical-linear-algebra- F01300	103	1.000	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$ for
lyche-numerical-linear-algebra- F01301	103	1.000	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$ for
lyche-numerical-linear-algebra- F01302	103	1.000	$\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$	$\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$ for

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lyche-numerical-linear-algebra- F01303	103	■ 0.999	$\boldsymbol{z} \in \mathbb{C}^n$	$z \in \mathbb{C}^n$	all nonzero $z \in \mathbb{C}^n$. For if $z = x + iy \neq \mathbf{0}$ with $x, y \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra- F01304	103	■ 0.999	$\boldsymbol{z} = \boldsymbol{x} + i\boldsymbol{y} \neq \mathbf{0}$	$z = x + iy \neq \mathbf{0}$	all nonzero $z \in \mathbb{C}^n$. For if $z = x + iy \neq \mathbf{0}$ with $x, y \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra- F01305	103	■ 0.991	$\boldsymbol{x}, \boldsymbol{y}$	x, y	and this is positive since at least one of the real vectors x, y is nonzero.
lyche-numerical-linear-algebra- F01306	103	■ 1.000	∇f	∇f	important in nonlinear optimization. Consider (cf. (16.1)) the gradient ∇f and
lyche-numerical-linear-algebra- F01307	103	■ 0.658	$H f$	Hf	hessian Hf of a function $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra- F01308	103	■ 0.658	$f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$	$f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$	hessian Hf of a function $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra- F01309	103	■ 1.000	Ω	Ω	We assume that f has continuous first and second order partial derivatives on Ω .
lyche-numerical-linear-algebra- F01310	103	■ 1.000	$\nabla f(\boldsymbol{x}) = \mathbf{0}$	$\nabla f(x) = \mathbf{0}$	that if $\nabla f(x) = \mathbf{0}$ and $Hf(x)$ is positive definite then x is a local minimum for f .
lyche-numerical-linear-algebra- F01311	103	■ 1.000	$H f(\boldsymbol{x})$	$Hf(x)$	that if $\nabla f(x) = \mathbf{0}$ and $Hf(x)$ is positive definite then x is a local minimum for f .
lyche-numerical-linear-algebra- F01312	103	■ 0.933	$\boldsymbol{A}^* \boldsymbol{A}$	$A^* A$	Lemma 4.2 (The Matrix $A^* A$) The matrix $A^* A$ is positive semidefinite for any
lyche-numerical-linear-algebra- F01313	103	■ 1.000	$m, n \in \mathbb{N}$	$m, n \in \mathbb{N}$	$m, n \in \mathbb{N}$ and $A \in \mathbb{C}^{m \times n}$. It is positive definite if and only if A has linearly
lyche-numerical-linear-algebra- F01314	104	■ 0.986	$\boldsymbol{z} := \boldsymbol{A} \boldsymbol{x}$	$z := Ax$	Proof Clearly $A^* A$ is Hermitian. Let $x \in \mathbb{C}^n$ and set $z := Ax$. By the definition
lyche-numerical-linear-algebra- F01315	104	■ 1.000	$\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{z} = \ \boldsymbol{z}\ _2^2 = \ \boldsymbol{A} \boldsymbol{x}\ _2^2 \geq 0$	$x^* A^* A x = z^* z = \ z\ _2^2 = \ Ax\ _2^2 \geq 0$	(1.11) of the Euclidean norm we have $x^* A^* A x = z^* z = \ z\ _2^2 = \ Ax\ _2^2 \geq 0$
lyche-numerical-linear-algebra- F01316	104	■ 0.515	$\boldsymbol{T} =$	$T =$	Lemma 4.3 (T Is Positive Definite) The second derivative matrix $T =$
lyche-numerical-linear-algebra- F01317	104	■ 0.983	$\text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n \times n}$	$\text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n \times n}$	$\text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n \times n}$ is positive definite.
lyche-numerical-linear-algebra- F01318	104	■ 0.994	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$	$x^T T x \geq 0$	Thus $x^T T x \geq 0$ and if $x^T T x = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra- F01319	104	■ 0.994	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$	$x^T T x = 0$	Thus $x^T T x \geq 0$ and if $x^T T x = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra- F01320	104	■ 0.994	$x_1 = x_n = 0$	$x_1 = x_n = 0$	Thus $x^T T x \geq 0$ and if $x^T T x = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra- F01321	104	■ 0.994	$x_i = x_{i+1}$	$x_i = x_{i+1}$	Thus $x^T T x \geq 0$ and if $x^T T x = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra- F01322	104	■ 0.999	$\boldsymbol{x} = \mathbf{0}$	$x = 0$	$1, \dots, n-1$ which implies that $x = 0$. Hence T is positive definite.
lyche-numerical-linear-algebra- F01323	104	■ 1.000	$\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) \in \mathbb{C}^{k \times k}$	$B = A(r, r) \in \mathbb{C}^{k \times k}$	Recall that a principal submatrix $B = A(r, r) \in \mathbb{C}^{k \times k}$ of a matrix $A \in \mathbb{C}^{n \times n}$ has
lyche-numerical-linear-algebra- F01324	104	■ 1.000	$b_{i,j} = a_{r_i, r_j}$	$b_{i,j} = a_{r_i, r_j}$	elements $b_{i,j} = a_{r_i, r_j}$ for $i, j = 1, \dots, k$, where $1 \leq r_1 < \dots < r_k \leq n$. It is a
lyche-numerical-linear-algebra- F01325	104	■ 1.000	$i, j = 1, \dots, k$	$i, j = 1, \dots, k$	elements $b_{i,j} = a_{r_i, r_j}$ for $i, j = 1, \dots, k$, where $1 \leq r_1 < \dots < r_k \leq n$. It is a
lyche-numerical-linear-algebra- F01326	104	■ 1.000	$\boldsymbol{r} = [1, 2, \dots, k]^T$	$r = [1, 2, \dots, k]^T$	leading principal submatrix , denoted $A_{[k]}$ if $r = [1, 2, \dots, k]^T$. We have
lyche-numerical-linear-algebra- F01327	104	■ 1.000	$\boldsymbol{B} := \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$	$B := A(r, r)$	Proof Let X and $B := A(r, r)$ be given by (4.5). If A is positive semidefinite then
lyche-numerical-linear-algebra- F01328	104	■ 1.000	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = 0$	$y^* B y = 0$	Suppose A is positive definite and $y^* B y = 0$. By (4.6) we have $x = \mathbf{0}$ and since

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lyche-numerical-linear-algebra- F01329	105	■ 1.000	$\boldsymbol{A}^{-*} := \left(\boldsymbol{A}^{-1} \right)^* = \left(\boldsymbol{A}^* \right)^{-1}$	$\boldsymbol{A}^{-*} := (\boldsymbol{A}^{-1})^* = (\boldsymbol{A}^*)^{-1}$	<i>Proof</i> Recall that $\boldsymbol{A}^{-*} := (\boldsymbol{A}^{-1})^* = (\boldsymbol{A}^*)^{-1}$.
lyche-numerical-linear-algebra- F01330	105	■ 1.000	$1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$	$1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$	We show that $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$.
lyche-numerical-linear-algebra- F01331	105	■ 0.999	$1 \Rightarrow$	$1 \Rightarrow$	$1 \Rightarrow 2$: Suppose $\boldsymbol{A}$ is positive definite. By Lemma 4.4 the leading principal
lyche-numerical-linear-algebra- F01332	105	■ 1.000	$\boldsymbol{x}_i := \boldsymbol{L}^{-*} \boldsymbol{e}_i$	$\boldsymbol{x}_i := \boldsymbol{L}^{-*} \boldsymbol{e}_i$	is positive we note that $\boldsymbol{x}_i := \boldsymbol{L}^{-*} \boldsymbol{e}_i$ is nonzero since $\boldsymbol{L}^{-*}$ is nonsingular. But
lyche-numerical-linear-algebra- F01333	105	■ 1.000	$\boldsymbol{L}^{-*}$	$\boldsymbol{L}^{-*}$	is positive we note that $\boldsymbol{x}_i := \boldsymbol{L}^{-*} \boldsymbol{e}_i$ is nonzero since $\boldsymbol{L}^{-*}$ is nonsingular. But
lyche-numerical-linear-algebra- F01334	105	■ 1.000	$d_{ii} = \boldsymbol{e}_i^* \boldsymbol{D} \boldsymbol{e}_i = \boldsymbol{e}_i^* \boldsymbol{L}^{-1} \boldsymbol{A} \boldsymbol{L}^{-*} \boldsymbol{e}_i = \boldsymbol{x}_i^* \boldsymbol{A} \boldsymbol{x}_i > 0$	$d_{ii} = \boldsymbol{e}_i^* \boldsymbol{D} \boldsymbol{e}_i = \boldsymbol{e}_i^* \boldsymbol{L}^{-1} \boldsymbol{A} \boldsymbol{L}^{-*} \boldsymbol{e}_i = \boldsymbol{x}_i^* \boldsymbol{A} \boldsymbol{x}_i > 0$	then $d_{ii} = \boldsymbol{e}_i^* \boldsymbol{D} \boldsymbol{e}_i = \boldsymbol{e}_i^* \boldsymbol{L}^{-1} \boldsymbol{A} \boldsymbol{L}^{-*} \boldsymbol{e}_i = \boldsymbol{x}_i^* \boldsymbol{A} \boldsymbol{x}_i > 0$ since $\boldsymbol{A}$ is positive definite.
lyche-numerical-linear-algebra- F01335	105	■ 0.913	$2 \Rightarrow$	$2 \Rightarrow$	$2 \Rightarrow 3$: Suppose $\boldsymbol{A}$ has an LDL* factorization $\boldsymbol{A} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^*$ with positive
lyche-numerical-linear-algebra- F01336	105	■ 0.913	LDL^*	LDL*	$2 \Rightarrow 3$: Suppose $\boldsymbol{A}$ has an LDL* factorization $\boldsymbol{A} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^*$ with positive
lyche-numerical-linear-algebra- F01337	105	■ 1.000	d_{ii}	d_{ii}	diagonal elements d_{ii} in $\boldsymbol{D}$. Then $\boldsymbol{A} = \boldsymbol{S} \boldsymbol{S}^*$, where $\boldsymbol{S} := \boldsymbol{L} \boldsymbol{D}^{1/2}$ and $\boldsymbol{D}^{1/2} :=$
lyche-numerical-linear-algebra- F01338	105	■ 1.000	$\boldsymbol{A} = \boldsymbol{S} \boldsymbol{S}^*$	$\boldsymbol{A} = \boldsymbol{S} \boldsymbol{S}^*$	diagonal elements d_{ii} in $\boldsymbol{D}$. Then $\boldsymbol{A} = \boldsymbol{S} \boldsymbol{S}^*$, where $\boldsymbol{S} := \boldsymbol{L} \boldsymbol{D}^{1/2}$ and $\boldsymbol{D}^{1/2} :=$
lyche-numerical-linear-algebra- F01339	105	■ 1.000	$\boldsymbol{S} := \boldsymbol{L} \boldsymbol{D}^{1/2}$	$\boldsymbol{S} := \boldsymbol{L} \boldsymbol{D}^{1/2}$	diagonal elements d_{ii} in $\boldsymbol{D}$. Then $\boldsymbol{A} = \boldsymbol{S} \boldsymbol{S}^*$, where $\boldsymbol{S} := \boldsymbol{L} \boldsymbol{D}^{1/2}$ and $\boldsymbol{D}^{1/2} :=$
lyche-numerical-linear-algebra- F01340	105	■ 1.000	$\boldsymbol{D}^{1/2} :=$	$\boldsymbol{D}^{1/2} :=$	diagonal elements d_{ii} in $\boldsymbol{D}$. Then $\boldsymbol{A} = \boldsymbol{S} \boldsymbol{S}^*$, where $\boldsymbol{S} := \boldsymbol{L} \boldsymbol{D}^{1/2}$ and $\boldsymbol{D}^{1/2} :=$
lyche-numerical-linear-algebra- F01341	105	■ 1.000	$\operatorname{diag}(\sqrt{d_{11}}, \dots, \sqrt{d_{nn}})$	$\operatorname{diag}(\sqrt{d_{11}}, \dots, \sqrt{d_{nn}})$	$\operatorname{diag}(\sqrt{d_{11}}, \dots, \sqrt{d_{nn}})$, and this is a Cholesky factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra- F01342	105	■ 1.000	$3 \Rightarrow$	$3 \Rightarrow$	$3 \Rightarrow 1$: Suppose $\boldsymbol{A}$ has a Cholesky factorization $\boldsymbol{A} = \boldsymbol{L} \boldsymbol{L}^*$. Clearly $\boldsymbol{A}^* = \boldsymbol{A}$.
lyche-numerical-linear-algebra- F01343	105	■ 1.000	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{L}^*$	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{L}^*$	$3 \Rightarrow 1$: Suppose $\boldsymbol{A}$ has a Cholesky factorization $\boldsymbol{A} = \boldsymbol{L} \boldsymbol{L}^*$. Clearly $\boldsymbol{A}^* = \boldsymbol{A}$.
lyche-numerical-linear-algebra- F01344	105	■ 1.000	$\boldsymbol{L} \boldsymbol{L}^* = \boldsymbol{S} \boldsymbol{S}^*$	$\boldsymbol{L} \boldsymbol{L}^* = \boldsymbol{S} \boldsymbol{S}^*$	For uniqueness suppose $\boldsymbol{L} \boldsymbol{L}^* = \boldsymbol{S} \boldsymbol{S}^*$ are two Cholesky factorizations of the
lyche-numerical-linear-algebra- F01345	105	■ 1.000	$\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$	$\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$	Then $\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$, where by Lemma 2.5 $\boldsymbol{S}^{-1} \boldsymbol{L}$ is lower triangular and $\boldsymbol{S}^* \boldsymbol{L}^{-*}$
lyche-numerical-linear-algebra- F01346	105	■ 1.000	$2.5 \boldsymbol{S}^{-1} \boldsymbol{L}$	$2.5 \boldsymbol{S}^{-1} \boldsymbol{L}$	Then $\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$, where by Lemma 2.5 $\boldsymbol{S}^{-1} \boldsymbol{L}$ is lower triangular and $\boldsymbol{S}^* \boldsymbol{L}^{-*}$
lyche-numerical-linear-algebra- F01347	105	■ 1.000	$\boldsymbol{S}^* \boldsymbol{L}^{-*}$	$\boldsymbol{S}^* \boldsymbol{L}^{-*}$	Then $\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$, where by Lemma 2.5 $\boldsymbol{S}^{-1} \boldsymbol{L}$ is lower triangular and $\boldsymbol{S}^* \boldsymbol{L}^{-*}$
lyche-numerical-linear-algebra- F01348	105	■ 1.000	ℓ_{ii} / s_{ii}	ℓ_{ii} / s_{ii}	is upper triangular, with diagonal elements ℓ_{ii} / s_{ii} and s_{ii} / ℓ_{ii} , respectively. But then
lyche-numerical-linear-algebra- F01349	105	■ 1.000	s_{ii} / ℓ_{ii}	s_{ii} / ℓ_{ii}	is upper triangular, with diagonal elements ℓ_{ii} / s_{ii} and s_{ii} / ℓ_{ii} , respectively. But then
lyche-numerical-linear-algebra- F01350	105	■ 1.000	$\ell_{ii}^2 = s_{ii}^2$	$\ell_{ii}^2 = s_{ii}^2$	both matrices must be equal to the same diagonal matrix and $\ell_{ii}^2 = s_{ii}^2$. By positivity
lyche-numerical-linear-algebra- F01351	105	■ 1.000	$\ell_{ii} = s_{ii}$	$\ell_{ii} = s_{ii}$	$\ell_{ii} = s_{ii}$ and we conclude that $\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$ which means that $\boldsymbol{L} = \boldsymbol{S}$.
lyche-numerical-linear-algebra- F01352	105	■ 1.000	$\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$	$\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$	$\ell_{ii} = s_{ii}$ and we conclude that $\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$ which means that $\boldsymbol{L} = \boldsymbol{S}$.
lyche-numerical-linear-algebra- F01353	105	■ 1.000	$\boldsymbol{L} = \boldsymbol{S}$	$\boldsymbol{L} = \boldsymbol{S}$	$\ell_{ii} = s_{ii}$ and we conclude that $\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$ which means that $\boldsymbol{L} = \boldsymbol{S}$.

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lyche-numerical-linear-algebra-F01354	105	■ 0.993	$\boldsymbol{A}=\boldsymbol{R}^*\boldsymbol{R}$	$\boldsymbol{A} = \boldsymbol{R}^* \boldsymbol{R}$	A Cholesky factorization can also be written in the equivalent form $\boldsymbol{A} = \boldsymbol{R}^* \boldsymbol{R}$,
lyche-numerical-linear-algebra-F01355	105	■ 1.000	$\boldsymbol{R}=\boldsymbol{L}^*$	$\boldsymbol{R} = \boldsymbol{L}^*$	where $\boldsymbol{R} = \boldsymbol{L}^*$ is upper triangular with positive diagonal elements.
lyche-numerical-linear-algebra-F01356	105	■ 0.993	4.4(2 \times 2)	4.4(2 × 2)	<i>Example 4.4</i> (2 × 2) The matrix $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ has an LDL* and a Cholesky-
lyche-numerical-linear-algebra-F01357	105	■ 0.993	$\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$	<i>Example 4.4</i> (2 × 2) The matrix $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ has an LDL* and a Cholesky-
lyche-numerical-linear-algebra-F01358	106	■ 1.000	$\frac{1}{2} G_n=n^3 / 3$	$\frac{1}{2} G_n = n^3 / 3$	count $\frac{1}{5} G_n = n^3 / 3$ for a full matrix.
lyche-numerical-linear-algebra-F01359	106	■ 1.000	$a_{i i}>0, \left(a_{i i} \geq 0\right)$	$a_{ii} > 0, (a_{ii} \geq 0)$	1. $a_{ij} > 0, (a_{ij} > 0)$,
lyche-numerical-linear-algebra-F01360	106	■ 1.000	$\left \operatorname{Re}\left(a_{i j}\right)\right <\left(a_{i i}+a_{j j}\right) / 2, \left \operatorname{Re}\left(a_{i j}\right)\right \leq\left(a_{i i}+a_{j j}\right) / 2$	$ \operatorname{Re}\left(a_{ij}\right) < \left(a_{ii}+a_{jj}\right) / 2, (\operatorname{Re}\left(a_{ij}\right) \leq\left(a_{ii}+a_{jj}\right) / 2)$	2. $ \operatorname{Re}\left(a_{ij}\right) < \left(a_{ii}+a_{jj}\right) / 2, (\operatorname{Re}\left(a_{ij}\right) \leq\left(a_{ii}+a_{jj}\right) / 2)$,
lyche-numerical-linear-algebra-F01361	106	■ 1.000	$\left a_{i j}\right <\sqrt{a_{i i} a_{j j}},\left a_{i j}\right \leq \sqrt{a_{i i} a_{j j}}$	$\left a_{ij}\right < \sqrt{a_{ii} a_{jj}}, (a_{ij} \leq \sqrt{a_{ii} a_{jj}})$	3. $ a_{ij} < \sqrt{a_{ii} a_{jj}}, (a_{ij} \leq \sqrt{a_{ii} a_{jj}})$,
lyche-numerical-linear-algebra-F01362	106	■ 1.000	$a_{i i}=0$	$a_{ii} = 0$	4. If $\boldsymbol{A}$ is positive semidefinite and $a_{ii} = 0$ for some i then $a_{ij} = a_{ji} = 0$ for
lyche-numerical-linear-algebra-F01363	106	■ 1.000	$a_{i j}=a_{j i}=0$	$a_{ij} = a_{ji} = 0$	4. If $\boldsymbol{A}$ is positive semidefinite and $a_{ii} = 0$ for some i then $a_{ij} = a_{ji} = 0$ for
lyche-numerical-linear-algebra-F01364	106	■ 1.000	$a_{i i}=\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i>(\geq) 0$	$a_{ii} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > (\geq) 0$	Proof Clearly $a_{ii} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra-F01365	106	■ 1.000	$\alpha, \beta \in \mathbb{C}$	$\alpha, \beta \in \mathbb{C}$	Proof Clearly $a_{ii} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra-F01366	106	■ 1.000	$\alpha \boldsymbol{e}_i+\beta \boldsymbol{e}_j \neq$	$\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j \neq$	Proof Clearly $a_{ii} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra-F01367	106	■ 0.999	$\alpha=1, \beta=\pm 1$	$\alpha = 1, \beta = \pm 1$	Taking $\alpha = 1, \beta = \pm 1$ we obtain $a_{ii} + a_{jj} \pm 2 \operatorname{Re} a_{ij} > 0$ and this implies Part 2.
lyche-numerical-linear-algebra-F01368	106	■ 0.999	$a_{i i}+a_{j j} \pm 2 \operatorname{Re} a_{i j}>0$	$a_{ii} + a_{jj} \pm 2 \operatorname{Re} a_{ij} > 0$	Taking $\alpha = 1, \beta = \pm 1$ we obtain $a_{ii} + a_{jj} \pm 2 \operatorname{Re} a_{ij} > 0$ and this implies Part 2.
lyche-numerical-linear-algebra-F01369	106	■ 0.834	$\alpha=-a_{i j}, \beta=a_{i i}$	$\alpha = -a_{ij}, \beta = a_{ii}$	We first show 3. when $\boldsymbol{A}$ is positive definite. Taking $\alpha = -a_{ij}, \beta = a_{ii}$ in (4.7) we
lyche-numerical-linear-algebra-F01370	106	■ 1.000	$a_{i i}>0$	$a_{ii} > 0$	Since $a_{ii} > 0$ Part 3 follows in the positive definite case.
lyche-numerical-linear-algebra-F01371	106	■ 1.000	$\varepsilon>0$	$\varepsilon > 0$	Suppose now $\boldsymbol{A}$ is positive semidefinite. For $\varepsilon > 0$ we define $\boldsymbol{B}:=\boldsymbol{A}+\varepsilon \boldsymbol{I}$. The
lyche-numerical-linear-algebra-F01372	106	■ 1.000	$\boldsymbol{B}:=\boldsymbol{A}+\varepsilon \boldsymbol{I}$	$\boldsymbol{B}:=\boldsymbol{A} + \varepsilon \boldsymbol{I}$	Suppose now $\boldsymbol{A}$ is positive semidefinite. For $\varepsilon > 0$ we define $\boldsymbol{B}:=\boldsymbol{A} + \varepsilon \boldsymbol{I}$. The
lyche-numerical-linear-algebra-F01373	106	■ 1.000	$\boldsymbol{x}^* \boldsymbol{B} \boldsymbol{x} \geq \varepsilon\ \boldsymbol{x}\ _2^2>0$	$\boldsymbol{x}^* \boldsymbol{B} \boldsymbol{x} \geq \varepsilon\ \boldsymbol{x}\ _2^2 > 0$	matrix $\boldsymbol{B}$ is positive definite since it is Hermitian and $\boldsymbol{x}^* \boldsymbol{B} \boldsymbol{x} \geq \varepsilon\ \boldsymbol{x}\ _2^2 > 0$ for any
lyche-numerical-linear-algebra-F01374	107	■ 1.000	$\left[\begin{array}{ll}2 & 1 \\ 1 & 2\end{array}\right]$	$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	The matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ enjoys all the necessary conditions in Theorem 4.3. But to
lyche-numerical-linear-algebra-F01375	107	■ 1.000	$\boldsymbol{A} \boldsymbol{x}=\lambda \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$	if $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ and $\boldsymbol{x}$ is nonzero, then $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$. This implies that $\lambda > 0 (\geq 0)$
lyche-numerical-linear-algebra-F01376	107	■ 1.000	$\lambda>0(\geq 0)$	$\lambda > 0 (\geq 0)$	if $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ and $\boldsymbol{x}$ is nonzero, then $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$. This implies that $\lambda > 0 (\geq 0)$
lyche-numerical-linear-algebra-F01377	107	■ 1.000	$\boldsymbol{x}^* \boldsymbol{x}=\ \boldsymbol{x}\ _2^2>0$	$\boldsymbol{x}^* \boldsymbol{x} = \ \boldsymbol{x}\ _2^2 > 0$	since $\boldsymbol{A}$ is positive (semi)definite and $\boldsymbol{x}^* \boldsymbol{x} = \ \boldsymbol{x}\ _2^2 > 0$. Conversely, suppose
lyche-numerical-linear-algebra-F01378	107	■ 1.000	$\boldsymbol{U}^* \boldsymbol{U} =$	$\boldsymbol{U}^* \boldsymbol{U} =$	Theorem 6.9 (the spectral theorem) there is a matrix $\boldsymbol{U} \in \mathbb{C}^{n \times n}$ with $\boldsymbol{U}^* \boldsymbol{U} =$

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lyche-numerical-linear-algebra-F01379	107	1.000	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$	$\boldsymbol{U}\boldsymbol{U}^* = \boldsymbol{I}$	$\boldsymbol{U}\boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra-F01380	107	1.000	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$	$\boldsymbol{U}\boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra-F01381	107	1.000	$\boldsymbol{z} :=$	$\boldsymbol{z} :=$	$\boldsymbol{U}\boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra-F01382	107	1.000	$\boldsymbol{U}^* \boldsymbol{x} = [z_1, \dots, z_n]^T \in \mathbb{C}^n$	$\boldsymbol{U}^*\boldsymbol{x} = [z_1, \dots, z_n]^T \in \mathbb{C}^n$	$\boldsymbol{U}^*\boldsymbol{x} = [z_1, \dots, z_n]^T \in \mathbb{C}^n$. Then $\boldsymbol{x} = \boldsymbol{U}\boldsymbol{U}^*\boldsymbol{x} = \boldsymbol{U}\boldsymbol{z}$ and by the spectral theorem
lyche-numerical-linear-algebra-F01383	107	1.000	$\boldsymbol{x} = \boldsymbol{U}\boldsymbol{U}^* \boldsymbol{x} = \boldsymbol{U}\boldsymbol{z}$	$\boldsymbol{x} = \boldsymbol{U}\boldsymbol{U}^*\boldsymbol{x} = \boldsymbol{U}\boldsymbol{z}$	$\boldsymbol{U}^*\boldsymbol{x} = [z_1, \dots, z_n]^T \in \mathbb{C}^n$. Then $\boldsymbol{x} = \boldsymbol{U}\boldsymbol{U}^*\boldsymbol{x} = \boldsymbol{U}\boldsymbol{z}$ and by the spectral theorem
lyche-numerical-linear-algebra-F01384	107	0.982	$\boldsymbol{U}^*$	$\boldsymbol{U}^*$	It follows that $\boldsymbol{A}$ is positive semidefinite. Since $\boldsymbol{U}^*$ is nonsingular we see that $\boldsymbol{z} =$
lyche-numerical-linear-algebra-F01385	107	0.982	$\boldsymbol{z} =$	$\boldsymbol{z} =$	It follows that $\boldsymbol{A}$ is positive semidefinite. Since $\boldsymbol{U}^*$ is nonsingular we see that $\boldsymbol{z} =$
lyche-numerical-linear-algebra-F01386	107	1.000	$\boldsymbol{U}^* \boldsymbol{x}$	$\boldsymbol{U}^*\boldsymbol{x}$	$\boldsymbol{U}^*\boldsymbol{x}$ is nonzero if $\boldsymbol{x}$ is nonzero, and therefore $\boldsymbol{A}$ is positive definite.
lyche-numerical-linear-algebra-F01387	108	1.000	$\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^*$	$\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^*$	4. $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^*$ for a nonsingular $\boldsymbol{B} \in \mathbb{C}^{n \times n}$.
lyche-numerical-linear-algebra-F01388	108	0.793	$\boldsymbol{B} = \boldsymbol{L}$	$\boldsymbol{B} = \boldsymbol{L}$	and by Theorem 4.2 a unique Cholesky factorization $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^*$ with $\boldsymbol{B} = \boldsymbol{L}$.
lyche-numerical-linear-algebra-F01389	108	0.973	4 $\Longrightarrow$	$4 \Longrightarrow$	$4 \Longrightarrow$ 1: This follows from Lemma 4.2.
lyche-numerical-linear-algebra-F01390	108	1.000	$\boldsymbol{A} := \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$	$\boldsymbol{A} := \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$	$\boldsymbol{A} := \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.
lyche-numerical-linear-algebra-F01391	108	0.991	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + 2x_2^2 + (x_1 + x_2)^2 > 0$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + 2x_2^2 + (x_1 + x_2)^2 > 0$	1. We have $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + 2x_2^2 + (x_1 + x_2)^2 > 0$ for all nonzero $\boldsymbol{x}$ showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra-F01392	108	1.000	$\lambda_1 = 2$	$\lambda_1 = 2$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 4$. They are positive showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra-F01393	108	1.000	$\lambda_2 = 4$	$\lambda_2 = 4$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 4$. They are positive showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra-F01394	108	0.998	$\det(\boldsymbol{A}_{[1]}) = 3$	$\det(\boldsymbol{A}_{[1]}) = 3$	3. We find $\det(\boldsymbol{A}_{[1]}) = 3$ and $\det(\boldsymbol{A}_{[2]}) = 8$ showing again that $\boldsymbol{A}$ is positive definite
lyche-numerical-linear-algebra-F01395	108	0.998	$\det(\boldsymbol{A}_{[2]}) = 8$	$\det(\boldsymbol{A}_{[2]}) = 8$	3. We find $\det(\boldsymbol{A}_{[1]}) = 3$ and $\det(\boldsymbol{A}_{[2]}) = 8$ showing again that $\boldsymbol{A}$ is positive definite
lyche-numerical-linear-algebra-F01396	109	1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \ \boldsymbol{L}^* \boldsymbol{x}\ _2^2 \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \ \boldsymbol{L}^* \boldsymbol{x}\ _2^2 \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \ \boldsymbol{L}^* \boldsymbol{x}\ _2^2 \geq 0$ and $\boldsymbol{A}$ is positive semidefinite. For the converse we use
lyche-numerical-linear-algebra-F01397	109	1.000	$\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$	$\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$	There are two cases. Suppose first $\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$. We claim that $\boldsymbol{C} := \boldsymbol{B} -$
lyche-numerical-linear-algebra-F01398	109	1.000	$\boldsymbol{C} := \boldsymbol{B} -$	$\boldsymbol{C} := \boldsymbol{B} -$	There are two cases. Suppose first $\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$. We claim that $\boldsymbol{C} := \boldsymbol{B} -$
lyche-numerical-linear-algebra-F01399	109	1.000	$\boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\boldsymbol{v} \boldsymbol{v}^* / \alpha$ is positive semidefinite. $\boldsymbol{C}$ is Hermitian since $\boldsymbol{B}$ is. To show that $\boldsymbol{C}$ is positive
lyche-numerical-linear-algebra-F01400	109	1.000	$\boldsymbol{y} \in \mathbb{C}^{n-1}$	$\boldsymbol{y} \in \mathbb{C}^{n-1}$	semidefinite we consider any $\boldsymbol{y} \in \mathbb{C}^{n-1}$ and define $\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$. Then
lyche-numerical-linear-algebra-F01401	109	1.000	$\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$	$\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$	semidefinite we consider any $\boldsymbol{y} \in \mathbb{C}^{n-1}$ and define $\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$. Then
lyche-numerical-linear-algebra-F01402	109	1.000	$\boldsymbol{C} \in \mathbb{C}^{(n-1) \times (n-1)}$	$\boldsymbol{C} \in \mathbb{C}^{(n-1) \times (n-1)}$	So $\boldsymbol{C} \in \mathbb{C}^{(n-1) \times (n-1)}$ is positive semidefinite and by the induction hypothesis it has
lyche-numerical-linear-algebra-F01403	109	1.000	$\boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$	$\boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$	a semi-Cholesky factorization $\boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$. The matrix

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lyche-numerical-linear-algebra-F01404	109	1.000	$\alpha=0$	$\alpha = 0$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\mathbf{v} = \mathbf{0}$. Moreover, $\mathbf{B} \in$
lyche-numerical-linear-algebra-F01405	109	1.000	$\boldsymbol{v}=\mathbf{0}$	$\mathbf{v} = \mathbf{0}$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\mathbf{v} = \mathbf{0}$. Moreover, $\mathbf{B} \in$
lyche-numerical-linear-algebra-F01406	109	1.000	$\boldsymbol{B} \in$	$\mathbf{B} \in$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\mathbf{v} = \mathbf{0}$. Moreover, $\mathbf{B} \in$
lyche-numerical-linear-algebra-F01407	109	1.000	$\mathbb{C}^{(n-1) \times (n-1)}$	$\mathbb{C}^{(n-1) \times (n-1)}$	$\mathbb{C}^{(n-1) \times (n-1)}$ in (4.8) is positive semidefinite and therefore has a semi-Cholesky
lyche-numerical-linear-algebra-F01408	110	0.554	$\boldsymbol{B}=\boldsymbol{L}_1 \boldsymbol{L}_1^*$	$\mathbf{B} = \mathbf{L}_1 \mathbf{L}_1^*$	factorization $\mathbf{B} = \mathbf{L}_1 \mathbf{L}_1^*$. But then $\mathbf{L} \mathbf{L}^*$, where $\mathbf{L} = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & \mathbf{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra-F01409	110	0.554	$\boldsymbol{L}^*$	$\mathbf{L}^*$	factorization $\mathbf{B} = \mathbf{L}_1 \mathbf{L}_1^*$. But then $\mathbf{L} \mathbf{L}^*$, where $\mathbf{L} = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & \mathbf{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra-F01410	110	0.554	$\boldsymbol{L}=\left[\begin{array}{cc} 0 & \mathbf{0}^* \\ \mathbf{0} & \mathbf{L}_1 \end{array}\right]$	$\mathbf{L} = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & \mathbf{L}_1 \end{bmatrix}$	factorization $\mathbf{B} = \mathbf{L}_1 \mathbf{L}_1^*$. But then $\mathbf{L} \mathbf{L}^*$, where $\mathbf{L} = \begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & \mathbf{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra-F01411	110	1.000	$\boldsymbol{v}^*=\left[\boldsymbol{u}^*, \mathbf{0}^*\right]$	$\mathbf{v}^* = [\mathbf{u}^*, \mathbf{0}^*]$	Proof Suppose $\mathbf{A} \in \mathbb{C}^{n \times n}$ is d -banded. Then $\mathbf{v}^* = [\mathbf{u}^*, \mathbf{0}^*]$ in (4.8), where $\mathbf{u} \in \mathbb{C}^d$,
lyche-numerical-linear-algebra-F01412	110	1.000	$\boldsymbol{u} \in \mathbb{C}^d$	$\mathbf{u} \in \mathbb{C}^d$	Proof Suppose $\mathbf{A} \in \mathbb{C}^{n \times n}$ is d -banded. Then $\mathbf{v}^* = [\mathbf{u}^*, \mathbf{0}^*]$ in (4.8), where $\mathbf{u} \in \mathbb{C}^d$,
lyche-numerical-linear-algebra-F01413	110	1.000	$\boldsymbol{C}:=\boldsymbol{B}-\boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\mathbf{C} := \mathbf{B} - \mathbf{v} \mathbf{v}^* / \alpha$	and therefore $\mathbf{C} := \mathbf{B} - \mathbf{v} \mathbf{v}^* / \alpha$ differs from $\mathbf{B}$ only in the upper left $d \times d$ corner. It
lyche-numerical-linear-algebra-F01414	110	1.000	$d \times d$	$d \times d$	and therefore $\mathbf{C} := \mathbf{B} - \mathbf{v} \mathbf{v}^* / \alpha$ differs from $\mathbf{B}$ only in the upper left $d \times d$ corner. It
lyche-numerical-linear-algebra-F01415	110	1.000	$n, \boldsymbol{C}=\boldsymbol{L}_1 \boldsymbol{L}_1^*$	$n, \mathbf{C} = \mathbf{L}_1 \mathbf{L}_1^*$	follows that $\mathbf{C}$ has the same bandwidth as $\mathbf{B}$ and $\mathbf{A}$. By induction on n , $\mathbf{C} = \mathbf{L}_1 \mathbf{L}_1^*$,
lyche-numerical-linear-algebra-F01416	110	0.999	$\boldsymbol{L}_1^*$	$\mathbf{L}_1^*$	where $\mathbf{L}_1^*$ has the same bandwidth as $\mathbf{C}$. But then $\mathbf{L}$ in (4.10) has the same bandwidth
lyche-numerical-linear-algebra-F01417	110	1.000	$\left[\beta, \boldsymbol{v}^* / \beta\right]^*$	$[\beta, \mathbf{v}^* / \beta]^*$	of $\mathbf{L}$ is $[\beta, \mathbf{v}^* / \beta]^*$ if $\alpha > 0$. If $\alpha = 0$ then by 4 in Theorem 4.3 the first column of
lyche-numerical-linear-algebra-F01418	110	1.000	$\alpha>0$	$\alpha > 0$	of $\mathbf{L}$ is $[\beta, \mathbf{v}^* / \beta]^*$ if $\alpha > 0$. If $\alpha = 0$ then by 4 in Theorem 4.3 the first column of
lyche-numerical-linear-algebra-F01419	110	0.947	$\boldsymbol{C}=\boldsymbol{B}-\boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\mathbf{C} = \mathbf{B} - \mathbf{v} \mathbf{v}^* / \alpha$	$\mathbf{C} = \mathbf{B} - \mathbf{v} \mathbf{v}^* / \alpha$ in the lower part of $\mathbf{A}(2:n, 2:n)$.
lyche-numerical-linear-algebra-F01420	110	0.947	$\mathbf{A}(2:n, 2:n)$	$\mathbf{A}(2:n, 2:n)$	$\mathbf{C} = \mathbf{B} - \mathbf{v} \mathbf{v}^* / \alpha$ in the lower part of $\mathbf{A}(2:n, 2:n)$.
lyche-numerical-linear-algebra-F01421	110	1.000	$\min (i+d, n)$	$\min(i + d, n)$	replace all occurrences of n by $\min(i + d, n)$. Continuing the reduction we arrive
lyche-numerical-linear-algebra-F01422	111	1.000	$\ell_{k k}=0$	$\ell_{kk} = 0$	Column k of $\mathbf{L}$ is zero for those k where $\ell_{kk} = 0$. We reduce round-off noise by
lyche-numerical-linear-algebra-F01423	111	0.910	$1 \Longleftrightarrow 2$	$1 \iff 2$	$1 \iff 2$: This follows from Lemma 4.5.
lyche-numerical-linear-algebra-F01424	111	0.858	$1 \Longleftrightarrow$	$1 \iff$	$1 \iff 4$: This follows from Theorem 4.5
lyche-numerical-linear-algebra-F01425	111	0.933	$1 \Longleftrightarrow 3$	$1 \iff 3$	$1 \iff 3$: We refer to page 567 of [15], where it is shown that $4 \implies 3$ (and
lyche-numerical-linear-algebra-F01426	111	0.933	$4 \rightarrow 3$	$4 \implies 3$	$1 \iff 3$: We refer to page 567 of [15], where it is shown that $4 \implies 3$ (and
lyche-numerical-linear-algebra-F01427	111	0.835	$1 \rightarrow 3$	$1 \implies 3$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.
lyche-numerical-linear-algebra-F01428	111	0.835	$1 \Longleftrightarrow 4$	$1 \iff 4$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.
lyche-numerical-linear-algebra-F01429	111	0.835	$3 \rightarrow 1$	$3 \implies 1$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.

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lyche-numerical-linear-algebra-112-F01436	112	0.997	$\boldsymbol{A}:=\left[\begin{array}{ll}1 & 1 \\ 1 & 1\end{array}\right]$	$\boldsymbol{A}:=\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	matrix $\boldsymbol{A}:=\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-112-F01431	112	0.996	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=\boldsymbol{x}_1^2+\boldsymbol{x}_2^2+\boldsymbol{x}_1\boldsymbol{x}_2+\boldsymbol{x}_2\boldsymbol{x}_1=\left(\boldsymbol{x}_1+\boldsymbol{x}_2\right)^2\geq 0$ $\boldsymbol{x}_1=\left(\boldsymbol{x}_1+\boldsymbol{x}_2\right)^2\geq 0$	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=x_1^2+x_2^2+x_1x_2+x_2x_1=(x_1+x_2)^2\geq 0$	1. We have $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=x_1^2+x_2^2+x_1x_2+x_2x_1=(x_1+x_2)^2\geq 0$ for all $\boldsymbol{x}\in\mathbb{R}^2$
lyche-numerical-linear-algebra-112-F01432	112	0.996	$\boldsymbol{x}\in\mathbb{R}^2$	$\boldsymbol{x}\in\mathbb{R}^2$	1. We have $\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=x_1^2+x_2^2+x_1x_2+x_2x_1=(x_1+x_2)^2\geq 0$ for all $\boldsymbol{x}\in\mathbb{R}^2$
lyche-numerical-linear-algebra-112-F01433	112	1.000	$\lambda_2=0$	$\lambda_2=0$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1=2$ and $\lambda_2=0$ and they are nonnegative showing
lyche-numerical-linear-algebra-112-F01434	112	0.998	$\operatorname{det}\left(\left[a_{11}\right]\right)=$	$\det\left([a_{11}]\right)=$	3. There are three principal sub matrices, and they have determinants $\det([a_{11}])=$
lyche-numerical-linear-algebra-112-F01435	112	0.971	$\operatorname{det}\left(\left[a_{22}\right]\right)=1$	$\det([a_{22}])=1$	1, $\det([a_{22}])=1$ and $\det(\boldsymbol{A})=0$ and showing again that $\boldsymbol{A}$ is positive
lyche-numerical-linear-algebra-112-F01436	112	0.575	$\boldsymbol{A}=\boldsymbol{B}\boldsymbol{B}^*$	$\boldsymbol{A}=\boldsymbol{B}\boldsymbol{B}^*$	4. Finally $\boldsymbol{A}$ is positive semidefinite since $\boldsymbol{A}=\boldsymbol{B}\boldsymbol{B}^*$, where $\boldsymbol{B}=\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$.
lyche-numerical-linear-algebra-112-F01437	112	0.575	$\boldsymbol{B}=\left[\begin{array}{ll}1 & 0 \\ 1 & 0\end{array}\right]$	$\boldsymbol{B}=\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	4. Finally $\boldsymbol{A}$ is positive semidefinite since $\boldsymbol{A}=\boldsymbol{B}\boldsymbol{B}^*$, where $\boldsymbol{B}=\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$.
lyche-numerical-linear-algebra-112-F01438	112	0.546	$\boldsymbol{A}:=\left[\begin{array}{rr}0 & 0 \\ 0 & -1\end{array}\right]$	$\boldsymbol{A}:=\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$	is not enough consider the matrix $\boldsymbol{A}:=\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$. The leading principal minors are
lyche-numerical-linear-algebra-112-F01439	112	1.000	$\operatorname{det}(\boldsymbol{A})>0$	$\det(\boldsymbol{A})>0$	4. $\det(\boldsymbol{A})>0$,
lyche-numerical-linear-algebra-112-F01440	112	1.000	$a_{ii}a_{jj}>a_{ij}a_{ji}$	$a_{ii}a_{jj}>a_{ij}a_{ji}$	5. $a_{ii}a_{jj}>a_{ij}a_{ji}$, for $i\neq j$.
lyche-numerical-linear-algebra-113-F01441	113	0.752	$\lambda=\frac{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}}>0$	$\lambda=\frac{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}}>0$	$\lambda=\frac{\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^T\boldsymbol{x}}>0$.
lyche-numerical-linear-algebra-113-F01442	113	0.981	$\left[\begin{array}{cc}a_{ii} & a_{ij} \\ a_{ji} & a_{jj}\end{array}\right]$	$\begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$	5. The principal submatrix $\begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$ has a positive determinant.
lyche-numerical-linear-algebra-113-F01443	113	1.000	$a\in\mathbb{R}$	$a\in\mathbb{R}$	is positive definite for any $a\in\mathbb{R}$. Indeed, for any nonzero $\boldsymbol{x}\in\mathbb{R}^2$
lyche-numerical-linear-algebra-113-F01444	113	1.000	$\boldsymbol{A}[a]$	$\boldsymbol{A}[a]$	The eigenvalues of $\boldsymbol{A}[a]$ are positive for $a\in[1-\frac{\sqrt{5}}{2},1+\frac{\sqrt{5}}{2}]$ and complex for
lyche-numerical-linear-algebra-113-F01445	113	1.000	$a\in\left[1-\frac{\sqrt{5}}{2},1+\frac{\sqrt{5}}{2}\right]$	$a\in\left[1-\frac{\sqrt{5}}{2},1+\frac{\sqrt{5}}{2}\right]$	The eigenvalues of $\boldsymbol{A}[a]$ are positive for $a\in[1-\frac{\sqrt{5}}{2},1+\frac{\sqrt{5}}{2}]$ and complex for
lyche-numerical-linear-algebra-113-F01446	113	1.000	a	a	other values of a .
lyche-numerical-linear-algebra-114-F01447	114	1.000	$\boldsymbol{A}:=\left[\begin{array}{cc}1 & 0 \\ -2 & 1\end{array}\right]$	$\boldsymbol{A}:=\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$	the matrix $\boldsymbol{A}:=\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-114-F01448	114	1.000	$\boldsymbol{E}\in\mathbb{R}^{n\times n}$	$\boldsymbol{E}\in\mathbb{R}^{n\times n}$	a) Let $\boldsymbol{E}\in\mathbb{R}^{n\times n}$ be of the form $\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u}\in\mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra-114-F01449	114	1.000	$\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$	$\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$	a) Let $\boldsymbol{E}\in\mathbb{R}^{n\times n}$ be of the form $\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u}\in\mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra-114-F01450	114	1.000	$\boldsymbol{u}\in\mathbb{R}^n$	$\boldsymbol{u}\in\mathbb{R}^n$	a) Let $\boldsymbol{E}\in\mathbb{R}^{n\times n}$ be of the form $\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u}\in\mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra-114-F01451	114	1.000	$\boldsymbol{E}$	$\boldsymbol{E}$	a) Let $\boldsymbol{E}\in\mathbb{R}^{n\times n}$ be of the form $\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u}\in\mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra-114-F01452	114	0.834	$\boldsymbol{E}^{-1}$	$\boldsymbol{E}^{-1}$	symmetric and positive definite, and find an expression for $\boldsymbol{E}^{-1}$.

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lyche-numerical-linear-algebra- F01453	114	■ 1.000	$\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{u}\boldsymbol{u}^T$ $\boldsymbol{u}^{\{T\}}$	$\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{u}\boldsymbol{u}^T$	b) Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be of the form $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{B} \in \mathbb{R}^{n \times n}$ is symmetric and
lyche-numerical-linear-algebra- F01454	114	■ 1.000	$\boldsymbol{A}=\boldsymbol{L}\boldsymbol{L}^T$	$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^T$	symmetric positive definite matrix with a known Cholesky factorization $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^T$.
lyche-numerical-linear-algebra- F01455	114	■ 0.998	$\boldsymbol{A}_+$	$\boldsymbol{A}_+$	Furthermore, let $\boldsymbol{A}_+$ be a corresponding $(n+1) \times (n+1)$ matrix of the form
lyche-numerical-linear-algebra- F01456	114	■ 0.998	$(n+1) \times (n+1)$	$(n+1) \times (n+1)$	Furthermore, let $\boldsymbol{A}_+$ be a corresponding $(n+1) \times (n+1)$ matrix of the form
lyche-numerical-linear-algebra- F01457	114	■ 1.000	α	α	where $\boldsymbol{a}$ is a vector in $\mathbb{R}^n$, and α is a real number. We assume that the matrix $\boldsymbol{A}_+$ is
lyche-numerical-linear-algebra- F01458	114	■ 1.000	$\boldsymbol{A}_+=\boldsymbol{L}_+\boldsymbol{L}_+^T$ $\boldsymbol{L}_+^{\{T\}}$	$\boldsymbol{A}_+ = \boldsymbol{L}_+\boldsymbol{L}_+^T$	a) Show that if $\boldsymbol{A}_+ = \boldsymbol{L}_+\boldsymbol{L}_+^T$ is the Cholesky factorization of $\boldsymbol{A}_+$, then $\boldsymbol{L}_+$ is of the
lyche-numerical-linear-algebra- F01459	114	■ 1.000	$\boldsymbol{L}_+$	$\boldsymbol{L}_+$	a) Show that if $\boldsymbol{A}_+ = \boldsymbol{L}_+\boldsymbol{L}_+^T$ is the Cholesky factorization of $\boldsymbol{A}_+$, then $\boldsymbol{L}_+$ is of the
lyche-numerical-linear-algebra- F01460	114	■ 1.000	$\alpha>\left \left \boldsymbol{L}^{-1}\boldsymbol{a}\right \right _2$ $\left \right _2^{\{2\}}$	$\alpha > \left\ \boldsymbol{L}^{-1} \boldsymbol{a} \right\ _2^2$	b) Explain why $\alpha > \left\ \boldsymbol{L}^{-1} \boldsymbol{a} \right\ _2^2$.
lyche-numerical-linear-algebra- F01461	114	■ 1.000	$\boldsymbol{E}^{-1}$	$\boldsymbol{E}^{-1}$	¹ Hint: The matrix $\boldsymbol{E}^{-1}$ is of the form $\boldsymbol{E}^{-1} = \boldsymbol{I} + \boldsymbol{a}\boldsymbol{u}\boldsymbol{u}^T$ for some $\boldsymbol{a} \in \mathbb{R}$.
lyche-numerical-linear-algebra- F01462	114	■ 1.000	$\boldsymbol{E}^{-1}=\boldsymbol{I}+a\boldsymbol{u}\boldsymbol{u}^T$ $\boldsymbol{u}^{\{T\}}$	$\boldsymbol{E}^{-1} = \boldsymbol{I} + \boldsymbol{a}\boldsymbol{u}\boldsymbol{u}^T$	¹ Hint: The matrix $\boldsymbol{E}^{-1}$ is of the form $\boldsymbol{E}^{-1} = \boldsymbol{I} + \boldsymbol{a}\boldsymbol{u}\boldsymbol{u}^T$ for some $\boldsymbol{a} \in \mathbb{R}$.
lyche-numerical-linear-algebra- F01463	115	■ 0.493	$\left[\begin{array}{lll}10&4&3\\4&0&2\\3&2&5\end{array}\right]$	$\begin{bmatrix} 10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5 \end{bmatrix}$	4.6.3 Is the matrix $\begin{bmatrix} 10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5 \end{bmatrix}$ positive definite?
lyche-numerical-linear-algebra- F01464	117	■ 1.000	$\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$	$\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$	a function $\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$ satisfying for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$ the following
lyche-numerical-linear-algebra- F01465	117	■ 1.000	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$	a function $\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$ satisfying for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$ the following
lyche-numerical-linear-algebra- F01466	117	■ 1.000	$\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$	$\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$	a function $\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$ satisfying for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$ the following
lyche-numerical-linear-algebra- F01467	117	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$	1. $\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$ with equality if and only if $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra- F01468	117	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}$	2. $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}$
lyche-numerical-linear-algebra- F01469	117	■ 1.000	$\langle \boldsymbol{a}\boldsymbol{x} + \boldsymbol{b}\boldsymbol{y}, \boldsymbol{z} \rangle = \boldsymbol{a}\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \boldsymbol{b}\langle \boldsymbol{y}, \boldsymbol{z} \rangle$	$\langle \boldsymbol{a}\boldsymbol{x} + \boldsymbol{b}\boldsymbol{y}, \boldsymbol{z} \rangle = \boldsymbol{a}\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \boldsymbol{b}\langle \boldsymbol{y}, \boldsymbol{z} \rangle$	3. $\langle \boldsymbol{a}\boldsymbol{x} + \boldsymbol{b}\boldsymbol{y}, \boldsymbol{z} \rangle = \boldsymbol{a}\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \boldsymbol{b}\langle \boldsymbol{y}, \boldsymbol{z} \rangle$.
lyche-numerical-linear-algebra- F01470	117	■ 0.996	$(\mathcal{V}, \langle \cdot, \cdot \rangle)$	$(\mathcal{V}, \langle \cdot, \cdot \rangle)$	The pair $(\mathcal{V}, \langle \cdot, \cdot \rangle)$ is called an inner product space .
lyche-numerical-linear-algebra- F01471	117	■ 0.975	$\ \boldsymbol{x}\ = \sqrt{\boldsymbol{x}^* \boldsymbol{x}}$	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ _2 = \sqrt{\boldsymbol{x}^* \boldsymbol{x}}$	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ _2 = \sqrt{\boldsymbol{x}^* \boldsymbol{x}}$.
lyche-numerical-linear-algebra- F01472	118	■ 1.000	$0\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$	$0\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$	Proof If $\boldsymbol{y} = \mathbf{0}$ then $0\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$ and $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly dependent. Moreover the
lyche-numerical-linear-algebra- F01473	118	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, 0\boldsymbol{y} \rangle = 0\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, 0\boldsymbol{y} \rangle = 0\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$	inequality holds with equality since $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, 0\boldsymbol{y} \rangle = 0\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$ and $\ \boldsymbol{y}\ = 0$.
lyche-numerical-linear-algebra- F01474	118	■ 1.000	$\ \boldsymbol{y}\ = 0$	$\ \boldsymbol{y}\ = 0$	inequality holds with equality since $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, 0\boldsymbol{y} \rangle = 0\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$ and $\ \boldsymbol{y}\ = 0$.
lyche-numerical-linear-algebra- F01475	118	■ 1.000	$\boldsymbol{y} \neq \mathbf{0}$	$\boldsymbol{y} \neq \mathbf{0}$	So assume $\boldsymbol{y} \neq \mathbf{0}$. Define
lyche-numerical-linear-algebra- F01476	118	■ 0.862	$\langle \boldsymbol{z}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle - \boldsymbol{a}\langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0$	$\langle \boldsymbol{z}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle - \boldsymbol{a}\langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0$	By linearity $\langle \boldsymbol{z}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle - \boldsymbol{a}\langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0$ so that by 2. and (5.1)
lyche-numerical-linear-algebra- F01477	118	■ 1.000	$\ \boldsymbol{y}\ ^2$	$\ \boldsymbol{y}\ ^2$	Multiplying by $\ \boldsymbol{y}\ ^2$ gives (5.3). We have equality if and only if $\boldsymbol{z} = \mathbf{0}$, which means

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lyche-numerical-linear-algebra- F01478	118	1.000	$\boldsymbol{z}=\mathbf{0}$	$\boldsymbol{z} = \mathbf{0}$	Multiplying by $\ \boldsymbol{y}\ ^2$ gives (5.3). We have equality if and only if $\boldsymbol{z} = \mathbf{0}$, which means
lyche-numerical-linear-algebra- F01479	118	1.000	$\ \boldsymbol{x}\ \geq 0$	$\ \boldsymbol{x}\ \geq 0$	1. $\ \boldsymbol{x}\ \geq 0$ with equality if and only if $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra- F01480	118	1.000	$ a \ \boldsymbol{x}\ = \ a\boldsymbol{x}\ $	$\ a\boldsymbol{x}\ = a \ \boldsymbol{x}\ $	2. $\ a\boldsymbol{x}\ = a \ \boldsymbol{x}\ $.
lyche-numerical-linear-algebra- F01481	118	1.000	$\ \boldsymbol{x} + \boldsymbol{y}\ \leq \ \boldsymbol{x}\ + \ \boldsymbol{y}\ $	$\ \boldsymbol{x} + \boldsymbol{y}\ \leq \ \boldsymbol{x}\ + \ \boldsymbol{y}\ $	3. $\ \boldsymbol{x} + \boldsymbol{y}\ \leq \ \boldsymbol{x}\ + \ \boldsymbol{y}\ $,
lyche-numerical-linear-algebra- F01482	118	1.000	$\ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	$\ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	where $\ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$
lyche-numerical-linear-algebra- F01483	118	1.000	$\left\ \cdot \right\ : \mathbb{C}^n \rightarrow \mathbb{R}$	$\ \cdot\ : \mathbb{C}^n \rightarrow \mathbb{R}$	In general a function $\ \cdot\ : \mathbb{C}^n \rightarrow \mathbb{R}$ that satisfies these three properties is called a
lyche-numerical-linear-algebra- F01484	119	1.000	$\ \boldsymbol{x} + a\boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + a\boldsymbol{y}, \boldsymbol{x} + a\boldsymbol{y} \rangle$	$\ \boldsymbol{x} + a\boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + a\boldsymbol{y}, \boldsymbol{x} + a\boldsymbol{y} \rangle$	second one follows from (5.1). Expanding $\ \boldsymbol{x} + a\boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + a\boldsymbol{y}, \boldsymbol{x} + a\boldsymbol{y} \rangle$ using
lyche-numerical-linear-algebra- F01485	119	1.000	$a=1$	$a = 1$	Now (5.5) with $a = 1$ and the Cauchy-Schwarz inequality implies
lyche-numerical-linear-algebra- F01486	119	0.999	$-\frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \leq 1$	$-\frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \leq 1$	In the real case the Cauchy-Schwarz inequality implies that $-1 \leq \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \leq 1$
lyche-numerical-linear-algebra- F01487	119	1.000	θ	θ	for nonzero $\boldsymbol{x}$ and $\boldsymbol{y}$, so there is a unique angle θ in $[0, \pi]$ such that
lyche-numerical-linear-algebra- F01488	119	1.000	$[0, \pi]$	$[0, \pi]$	for nonzero $\boldsymbol{x}$ and $\boldsymbol{y}$, so there is a unique angle θ in $[0, \pi]$ such that
lyche-numerical-linear-algebra- F01489	119	1.000	$\boldsymbol{x} \perp \boldsymbol{y}$	$\boldsymbol{x} \perp \boldsymbol{y}$	uct space are orthogonal or perpendicular , denoted as $\boldsymbol{x} \perp \boldsymbol{y}$, if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$. The
lyche-numerical-linear-algebra- F01490	119	1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$	uct space are orthogonal or perpendicular , denoted as $\boldsymbol{x} \perp \boldsymbol{y}$, if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$. The
lyche-numerical-linear-algebra- F01491	119	1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{y}\ = 1$	$\ \boldsymbol{x}\ = \ \boldsymbol{y}\ = 1$	vectors are orthonormal if in addition $\ \boldsymbol{x}\ = \ \boldsymbol{y}\ = 1$.
lyche-numerical-linear-algebra- F01492	119	1.000	$\theta = \pi/2$	$\theta = \pi/2$	or $\mathbb{C}^n$ it follows that $\boldsymbol{x} \perp \boldsymbol{y}$ if and only if $\theta = \pi/2$.
lyche-numerical-linear-algebra- F01493	119	0.997	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$	orthogonal basis for $\mathcal{S}$ if it is a basis for $\mathcal{S}$ and $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$ for $i \neq j$. It is an
lyche-numerical-linear-algebra- F01494	119	0.734	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \delta_{ij}$	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \delta_{ij}$	orthonormal basis for $\mathcal{S}$ if it is a basis for $\mathcal{S}$ and $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \delta_{ij}$ for all i, j .
lyche-numerical-linear-algebra- F01495	120	0.999	$\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_k\}$	$\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_k\}$	Theorem 5.4 (Gram-Schmidt) Let $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_k\}$ be a basis for a real or complex
lyche-numerical-linear-algebra- F01496	120	0.833	$(\mathcal{S}, \langle \cdot, \cdot \rangle)$	$(\mathcal{S}, \langle \cdot, \cdot \rangle)$	inner product space $(\mathcal{S}, \langle \cdot, \cdot \rangle)$. Define
lyche-numerical-linear-algebra- F01497	120	0.999	$\mathcal{S}_j := \text{span}\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_j\}$	$\mathcal{S}_j := \text{span}\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_j\}$	k . Define subspaces $\mathcal{S}_j := \text{span}\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_j\}$ for $j = 1, \dots, k$. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$
lyche-numerical-linear-algebra- F01498	120	0.999	$\boldsymbol{v}_1 = \boldsymbol{s}_1$	$\boldsymbol{v}_1 = \boldsymbol{s}_1$	k . Define subspaces $\mathcal{S}_j := \text{span}\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_j\}$ for $j = 1, \dots, k$. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$
lyche-numerical-linear-algebra- F01499	120	1.000	$\mathcal{S}_{\{1\}}$	$\mathcal{S}_1$	is an orthogonal basis for $\mathcal{S}_1$. Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$ is an
lyche-numerical-linear-algebra- F01500	120	1.000	$j \geq 2$	$j \geq 2$	is an orthogonal basis for $\mathcal{S}_1$. Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$ is an
lyche-numerical-linear-algebra- F01501	120	1.000	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$	is an orthogonal basis for $\mathcal{S}_1$. Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$ is an
lyche-numerical-linear-algebra- F01502	120	1.000	$\mathcal{S}_{\{j-1\}}$	$\mathcal{S}_{j-1}$	orthogonal basis for $\mathcal{S}_{j-1}$ and let $\boldsymbol{v}_j$ be given by (5.8) as a linear combination of
lyche-numerical-linear-algebra- F01503	120	1.000	$\boldsymbol{v}_j$	$\boldsymbol{v}_j$	orthogonal basis for $\mathcal{S}_{j-1}$ and let $\boldsymbol{v}_j$ be given by (5.8) as a linear combination of
lyche-numerical-linear-algebra- F01504	120	1.000	$\boldsymbol{v}_i$	$\boldsymbol{v}_i$	$\boldsymbol{s}_j$ and $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$. Now each of these $\boldsymbol{v}_i$ is a linear combination of $\boldsymbol{s}_1, \dots, \boldsymbol{s}_i$,

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lyche-numerical-linear-algebra- F01505	120	1.000	$\boldsymbol{s}_1, \ldots, \boldsymbol{s}_i$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_i$	$\boldsymbol{s}_j$ and $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$. Now each of these $\boldsymbol{v}_i$ is a linear combination of $\boldsymbol{s}_1, \dots, \boldsymbol{s}_i$,
lyche-numerical-linear-algebra- F01506	120	1.000	$\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$	$\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$ for some $a_0, \dots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra- F01507	120	1.000	$a_{\{0\}}, \ldots, a_{\{j\}}$	$a_0, \dots, a_j$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$ for some $a_0, \dots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra- F01508	120	1.000	$a_{\{j\}}=1$	$a_j = 1$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$ for some $a_0, \dots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra- F01509	120	1.000	$\boldsymbol{s}_1, \ldots, \boldsymbol{s}_{\{j\}}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$ for some $a_0, \dots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra- F01510	120	1.000	$a_{\{j\}} \neq 0$	$a_j \neq 0$	are linearly independent and $a_j \neq 0$ we deduce that $\boldsymbol{v}_j \neq 0$. By the induction
lyche-numerical-linear-algebra- F01511	120	1.000	$\boldsymbol{v}_{\{j\}} \neq 0$	$\boldsymbol{v}_j \neq 0$	are linearly independent and $a_j \neq 0$ we deduce that $\boldsymbol{v}_j \neq 0$. By the induction
lyche-numerical-linear-algebra- F01512	120	1.000	$l=1, \ldots, j-1$	$l = 1, \dots, j - 1$	for $l = 1, \dots, j - 1$. Thus $\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$ is an orthogonal basis for S_j .
lyche-numerical-linear-algebra- F01513	120	1.000	$\boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{v}_{\{j\}}$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$	for $l = 1, \dots, j - 1$. Thus $\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$ is an orthogonal basis for S_j .
lyche-numerical-linear-algebra- F01514	120	1.000	$S_{\{j\}}$	S_j	for $l = 1, \dots, j - 1$. Thus $\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$ is an orthogonal basis for S_j .
lyche-numerical-linear-algebra- F01515	120	1.000	$\left\{ \boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{k\}} \right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\}$	If $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$ is an orthogonal basis for $\mathcal{S}$ then clearly $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\}$ is an
lyche-numerical-linear-algebra- F01516	121	1.000	$\mathcal{S} \subset \mathcal{T}$	$\mathcal{S} \subset \mathcal{T}$	Theorem 5.5 (Orthogonal Extension of Basis) Suppose $\mathcal{S} \subset \mathcal{T}$ are finite dimen-
lyche-numerical-linear-algebra- F01517	121	1.000	$\operatorname{dim} \mathcal{S} := k < \operatorname{dim} \mathcal{T} = n$	$\dim \mathcal{S} := k < \dim \mathcal{T} = n$	Proof Suppose $\dim \mathcal{S} := k < \dim \mathcal{T} = n$. Using Theorem 1.3 we first extend
lyche-numerical-linear-algebra- F01518	121	1.000	$\boldsymbol{s}_1, \ldots, \boldsymbol{s}_{\{k\}}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_k$	an orthogonal basis $\boldsymbol{s}_1, \dots, \boldsymbol{s}_k$ for $\mathcal{S}$ to a basis $\boldsymbol{s}_1, \dots, \boldsymbol{s}_k, \boldsymbol{s}_{k+1}, \dots, \boldsymbol{s}_n$ for $\mathcal{T}$, and
lyche-numerical-linear-algebra- F01519	121	1.000	$\boldsymbol{s}_1, \ldots, \boldsymbol{s}_{\{k\}}, \boldsymbol{s}_{\{k+1\}}, \ldots, \boldsymbol{s}_{\{n\}}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_k, \boldsymbol{s}_{k+1}, \dots, \boldsymbol{s}_n$	an orthogonal basis $\boldsymbol{s}_1, \dots, \boldsymbol{s}_k$ for $\mathcal{S}$ to a basis $\boldsymbol{s}_1, \dots, \boldsymbol{s}_k, \boldsymbol{s}_{k+1}, \dots, \boldsymbol{s}_n$ for $\mathcal{T}$, and
lyche-numerical-linear-algebra- F01520	121	0.955	$\boldsymbol{v}_{\{i\}} = \boldsymbol{s}_{\{i\}}$	$\boldsymbol{v}_i = \boldsymbol{s}_i$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ for $\mathcal{T}$. This is an extension of the basis for $\mathcal{S}$ since $\boldsymbol{v}_i = \boldsymbol{s}_i$ for $i =$
lyche-numerical-linear-algebra- F01521	121	1.000	$1, \ldots, k$	$1, \dots, k$	$1, \dots, k$. We show this by induction. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$. Suppose for some $2 \leq r < k$
lyche-numerical-linear-algebra- F01522	121	1.000	$2 \leq r < k$	$2 \leq r < k$	$1, \dots, k$. We show this by induction. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$. Suppose for some $2 \leq r < k$
lyche-numerical-linear-algebra- F01523	121	1.000	$\boldsymbol{v}_{\{j\}} = \boldsymbol{s}_{\{j\}}$	$\boldsymbol{v}_j = \boldsymbol{s}_j$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \dots, r - 1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra- F01524	121	1.000	$j=1, \ldots, r-1$	$j = 1, \dots, r - 1$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \dots, r - 1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra- F01525	121	1.000	$j=r$	$j = r$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \dots, r - 1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra- F01526	121	1.000	$\left\langle \boldsymbol{s}_r, \boldsymbol{v}_{\{i\}} \right\rangle =$	$\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \dots, r - 1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra- F01527	121	0.999	$\left\langle \boldsymbol{s}_r, \boldsymbol{s}_{\{i\}} \right\rangle = 0$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$ for $i < r$ we obtain $\boldsymbol{v}_r = \boldsymbol{s}_r$.
lyche-numerical-linear-algebra- F01528	121	0.999	$i < r$	$i < r$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$ for $i < r$ we obtain $\boldsymbol{v}_r = \boldsymbol{s}_r$.
lyche-numerical-linear-algebra- F01529	121	0.999	$\boldsymbol{v}_{\{r\}} = \boldsymbol{s}_{\{r\}}$	$\boldsymbol{v}_r = \boldsymbol{s}_r$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$ for $i < r$ we obtain $\boldsymbol{v}_r = \boldsymbol{s}_r$.
lyche-numerical-linear-algebra- F01530	121	0.999	$\mathcal{S} = \operatorname{span}(\boldsymbol{s}_{\{1\}}, \ldots, \boldsymbol{s}_{\{k\}})$	$\mathcal{S} = \operatorname{span}(\boldsymbol{s}_1, \dots, \boldsymbol{s}_k)$	Letting $\mathcal{S} = \operatorname{span}(\boldsymbol{s}_1, \dots, \boldsymbol{s}_k)$ and $\mathcal{T}$ be $\mathbb{R}^n$ or $\mathbb{C}^n$ we obtain
lyche-numerical-linear-algebra- F01531	121	1.000	$1 \leq k < n$	$1 \leq k < n$	Corollary 5.1 (Extending Orthogonal Vectors to a Basis) For $1 \leq k < n$ a

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lyche-numerical-linear-algebra-F01532	121	1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle$	an inner product $\langle \boldsymbol{x}, \boldsymbol{y} \rangle$. We define
lyche-numerical-linear-algebra-F01533	121	1.000	$\mathcal{S} + \mathcal{T} := \{ \boldsymbol{s} + \boldsymbol{t} : \boldsymbol{s} \in \mathcal{S}, \boldsymbol{t} \in \mathcal{T} \}$	$\mathcal{S} + \mathcal{T} := \{ \boldsymbol{s} + \boldsymbol{t} : \boldsymbol{s} \in \mathcal{S}, \boldsymbol{t} \in \mathcal{T} \}$	• Sum: $\mathcal{S} + \mathcal{T} := \{ \boldsymbol{s} + \boldsymbol{t} : \boldsymbol{s} \in \mathcal{S}, \boldsymbol{t} \in \mathcal{T} \}$,
lyche-numerical-linear-algebra-F01534	121	1.000	$\boldsymbol{t} \in \mathcal{T}$	$\boldsymbol{t} \in \mathcal{T}$	• Sum: $\mathcal{S} + \mathcal{T} := \{ \boldsymbol{s} + \boldsymbol{t} : \boldsymbol{s} \in \mathcal{S}, \boldsymbol{t} \in \mathcal{T} \}$,
lyche-numerical-linear-algebra-F01535	121	0.967	$\sum \boldsymbol{S} \oplus \mathcal{T}$	$\sum \boldsymbol{S} \oplus \mathcal{T}$	• direct sum $\boldsymbol{S} \oplus \mathcal{T}$: a sum where $\boldsymbol{S} \cap \mathcal{T} = \{\mathbf{0}\}$,
lyche-numerical-linear-algebra-F01536	121	0.998	$\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	$\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	• orthogonal sum $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$.
lyche-numerical-linear-algebra-F01537	121	0.998	$\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$	$\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$	• orthogonal sum $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$.
lyche-numerical-linear-algebra-F01538	121	0.998	$\boldsymbol{s} \in \mathcal{S}$	$\boldsymbol{s} \in \mathcal{S}$	• orthogonal sum $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$.
lyche-numerical-linear-algebra-F01539	121	0.998	$\boldsymbol{t} \in \mathcal{T}$	$\boldsymbol{t} \in \mathcal{T}$	• orthogonal sum $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$ for all $\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$.
lyche-numerical-linear-algebra-F01540	121	1.000	$\boldsymbol{v} \in \boldsymbol{S} \oplus \mathcal{T}$	$\boldsymbol{v} \in \boldsymbol{S} \oplus \mathcal{T}$	• Every $\boldsymbol{v} \in \boldsymbol{S} \oplus \mathcal{T}$ can be decomposed uniquely in the form $\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}$, where
lyche-numerical-linear-algebra-F01541	121	1.000	$\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}$	$\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}$	• Every $\boldsymbol{v} \in \boldsymbol{S} \oplus \mathcal{T}$ can be decomposed uniquely in the form $\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}$, where
lyche-numerical-linear-algebra-F01542	121	1.000	$\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$	$\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$	$\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$. For if $\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$ for $\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$ and $\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra-F01543	121	1.000	$\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$	$\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$	$\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$. For if $\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$ for $\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$ and $\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra-F01544	121	1.000	$\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$	$\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$	$\boldsymbol{s} \in \mathcal{S}$ and $\boldsymbol{t} \in \mathcal{T}$. For if $\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$ for $\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$ and $\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra-F01545	121	0.969	$\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$	$\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$	then $\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$ or $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$. It follows that $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and $\boldsymbol{t}_2 - \boldsymbol{t}_1$
lyche-numerical-linear-algebra-F01546	121	0.969	$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$	$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$	then $\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$ or $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$. It follows that $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and $\boldsymbol{t}_2 - \boldsymbol{t}_1$
lyche-numerical-linear-algebra-F01547	121	0.969	$\boldsymbol{s}_1 - \boldsymbol{s}_2$	$\boldsymbol{s}_1 - \boldsymbol{s}_2$	then $\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$ or $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$. It follows that $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and $\boldsymbol{t}_2 - \boldsymbol{t}_1$
lyche-numerical-linear-algebra-F01548	121	0.969	$\boldsymbol{t}_2 - \boldsymbol{t}_1$	$\boldsymbol{t}_2 - \boldsymbol{t}_1$	then $\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$ or $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$. It follows that $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and $\boldsymbol{t}_2 - \boldsymbol{t}_1$
lyche-numerical-linear-algebra-F01549	121	1.000	$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1 = \mathbf{0}$	$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1 = \mathbf{0}$	belong to both $\mathcal{S}$ and $\mathcal{T}$ and hence to $\mathcal{S} \cap \mathcal{T}$. But then $\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1 = \mathbf{0}$ so
lyche-numerical-linear-algebra-F01550	121	1.000	$\boldsymbol{s}_1 = \boldsymbol{s}_2$	$\boldsymbol{s}_1 = \boldsymbol{s}_2$	$\boldsymbol{s}_1 = \boldsymbol{s}_2$ and $\boldsymbol{t}_2 = \boldsymbol{t}_1$.
lyche-numerical-linear-algebra-F01551	121	1.000	$\boldsymbol{t}_2 = \boldsymbol{t}_1$	$\boldsymbol{t}_2 = \boldsymbol{t}_1$	$\boldsymbol{s}_1 = \boldsymbol{s}_2$ and $\boldsymbol{t}_2 = \boldsymbol{t}_1$.
lyche-numerical-linear-algebra-F01552	121	1.000	$\boldsymbol{v} \in \mathcal{S} \cap \mathcal{T}$	$\boldsymbol{v} \in \mathcal{S} \cap \mathcal{T}$	• An orthogonal sum is a direct sum. For if $\boldsymbol{v} \in \mathcal{S} \cap \mathcal{T}$ then $\boldsymbol{v}$ is orthogonal to itself,
lyche-numerical-linear-algebra-F01553	121	1.000	$\langle \boldsymbol{v}, \boldsymbol{v} \rangle = 0$	$\langle \boldsymbol{v}, \boldsymbol{v} \rangle = 0$	$\langle \boldsymbol{v}, \boldsymbol{v} \rangle = 0$, which implies that $\boldsymbol{v} = \mathbf{0}$. We often write $\mathcal{T} := \mathcal{S}^\perp$.
lyche-numerical-linear-algebra-F01554	121	1.000	$\boldsymbol{v} = \mathbf{0}$	$\boldsymbol{v} = \mathbf{0}$	$\langle \boldsymbol{v}, \boldsymbol{v} \rangle = 0$, which implies that $\boldsymbol{v} = \mathbf{0}$. We often write $\mathcal{T} := \mathcal{S}^\perp$.
lyche-numerical-linear-algebra-F01555	121	1.000	$\mathcal{T} := \mathcal{S}^\perp$	$\mathcal{T} := \mathcal{S}^\perp$	$\langle \boldsymbol{v}, \boldsymbol{v} \rangle = 0$, which implies that $\boldsymbol{v} = \mathbf{0}$. We often write $\mathcal{T} := \mathcal{S}^\perp$.
lyche-numerical-linear-algebra-F01556	121	1.000	$\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$	$\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$	• Suppose $\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$, where $\boldsymbol{s}_0 \in \mathcal{S}$ and $\boldsymbol{t}_0 \in \mathcal{T}$. The vector $\boldsymbol{s}_0$ is
lyche-numerical-linear-algebra-F01557	121	1.000	$\boldsymbol{s}_0 \in \mathcal{S}$	$\boldsymbol{s}_0 \in \mathcal{S}$	• Suppose $\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$, where $\boldsymbol{s}_0 \in \mathcal{S}$ and $\boldsymbol{t}_0 \in \mathcal{T}$. The vector $\boldsymbol{s}_0$ is
lyche-numerical-linear-algebra-F01558	121	1.000	$\boldsymbol{t}_0 \in \mathcal{T}$	$\boldsymbol{t}_0 \in \mathcal{T}$	• Suppose $\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0 \in \mathcal{S} \oplus \mathcal{T}$, where $\boldsymbol{s}_0 \in \mathcal{S}$ and $\boldsymbol{t}_0 \in \mathcal{T}$. The vector $\boldsymbol{s}_0$ is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F01559	121	■ 1.000	$\boldsymbol{s}_{\{0\}}$	s_0	• Suppose $\boldsymbol{v} = s_0 + \boldsymbol{t}_0 \in S \oplus \mathcal{T}$, where $s_0 \in S$ and $\boldsymbol{t}_0 \in \mathcal{T}$. The vector s_0 is
lyche-numerical-linear-algebra- F01560	121	■ 1.000	$\boldsymbol{t}_{\{0\}}$	$\boldsymbol{t}_0$	called the oblique projection of $\boldsymbol{v}$ into S along $\mathcal{T}$. Similarly, The vector $\boldsymbol{t}_0$ is
lyche-numerical-linear-algebra- F01561	122	■ 1.000	$\mathcal{T} = \mathcal{S}^\perp$	$\mathcal{T} = \mathcal{S}^\perp$	called the orthogonal projection of $\boldsymbol{v}$ in $\mathcal{T} = \mathcal{S}^\perp$. The orthogonal projections
lyche-numerical-linear-algebra- F01562	122	■ 1.000	$\langle \cdot, \cdot \rangle$	$\langle \cdot, \cdot \rangle$	<i>dimensional real or complex vector space $\mathcal{V}$ with an inner product $\langle \cdot, \cdot \rangle$. The</i>
lyche-numerical-linear-algebra- F01563	122	■ 0.984	$\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	$\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	<i>orthogonal projections s_0 of $\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$ into S and $\boldsymbol{t}_0$ of $\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$ into $\mathcal{T}$ satisfy</i>
lyche-numerical-linear-algebra- F01564	122	■ 1.000	$\boldsymbol{v} = \boldsymbol{s}_{\{0\}} + \boldsymbol{t}_{\{0\}}$	$\boldsymbol{v} = s_0 + \boldsymbol{t}_0$	$\boldsymbol{v} = s_0 + \boldsymbol{t}_0$, and
lyche-numerical-linear-algebra- F01565	122	■ 1.000	$\langle \boldsymbol{s}_{\{0\}}, \boldsymbol{s} \rangle = \langle \boldsymbol{v} - \boldsymbol{t}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v}, \boldsymbol{s} \rangle$	$\langle s_0, s \rangle = \langle \boldsymbol{v} - \boldsymbol{t}_0, s \rangle = \langle \boldsymbol{v}, s \rangle$	Proof We have $\langle s_0, s \rangle = \langle \boldsymbol{v} - \boldsymbol{t}_0, s \rangle = \langle \boldsymbol{v}, s \rangle$, since $\langle \boldsymbol{t}_0, s \rangle = 0$ for all $s \in S$ and
lyche-numerical-linear-algebra- F01566	122	■ 1.000	$\langle \boldsymbol{t}_0, \boldsymbol{s} \rangle = 0$	$\langle \boldsymbol{t}_0, s \rangle = 0$	Proof We have $\langle s_0, s \rangle = \langle \boldsymbol{v} - \boldsymbol{t}_0, s \rangle = \langle \boldsymbol{v}, s \rangle$, since $\langle \boldsymbol{t}_0, s \rangle = 0$ for all $s \in S$ and
lyche-numerical-linear-algebra- F01567	123	■ 1.000	$\ \boldsymbol{v}\ := \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$	$\ \boldsymbol{v}\ := \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$	$\ \boldsymbol{v}\ := \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$. If $s_0 \in S$ is the orthogonal projection of $\boldsymbol{v} \in \mathcal{V}$ then
lyche-numerical-linear-algebra- F01568	123	■ 1.000	$\boldsymbol{v} \in \mathcal{V}$	$\boldsymbol{v} \in \mathcal{V}$	$\ \boldsymbol{v}\ := \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$. If $s_0 \in S$ is the orthogonal projection of $\boldsymbol{v} \in \mathcal{V}$ then
lyche-numerical-linear-algebra- F01569	123	■ 0.975	$s_{\{0\}} \neq s \in \mathcal{S}$	$s_0 \neq s \in S$	Proof Let $s_0 \neq s \in S$ and $0 \neq \boldsymbol{u} := s_0 - s \in S$. It follows from (5.9) that
lyche-numerical-linear-algebra- F01570	123	■ 0.975	$0 \neq \boldsymbol{u} := s_{\{0\}} - s \in \mathcal{S}$	$0 \neq \boldsymbol{u} := s_0 - s \in S$	Proof Let $s_0 \neq s \in S$ and $0 \neq \boldsymbol{u} := s_0 - s \in S$. It follows from (5.9) that
lyche-numerical-linear-algebra- F01571	123	■ 1.000	$\langle \boldsymbol{v} - \boldsymbol{s}_{\{0\}}, \boldsymbol{u} \rangle = 0$	$\langle \boldsymbol{v} - s_0, \boldsymbol{u} \rangle = 0$	$\langle \boldsymbol{v} - s_0, \boldsymbol{u} \rangle = 0$. By (5.7) (Pythagoras) we obtain
lyche-numerical-linear-algebra- F01572	123	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \sum_{j=1}^n x_j \overline{y_j}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \sum_{j=1}^n x_j \overline{y_j}$	$\mathbb{C}^n$ given by $\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \sum_{j=1}^n x_j \overline{y_j}$. For symmetric and Hermitian matrices
lyche-numerical-linear-algebra- F01573	123	■ 1.000	$\boldsymbol{A}^T = \boldsymbol{A} \iff \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle$	$\boldsymbol{A}^T = \boldsymbol{A} \iff \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle$	1. $\boldsymbol{A}^T = \boldsymbol{A} \iff \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra- F01574	123	■ 1.000	$\boldsymbol{A}^* = \boldsymbol{A} \iff \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle$	$\boldsymbol{A}^* = \boldsymbol{A} \iff \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle$	2. $\boldsymbol{A}^* = \boldsymbol{A} \iff \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra- F01575	123	■ 1.000	$\boldsymbol{x} = \boldsymbol{e}_j$	$\boldsymbol{x} = \boldsymbol{e}_j$	For the converse we take $\boldsymbol{x} = \boldsymbol{e}_j$ and $\boldsymbol{y} = \boldsymbol{e}_i$ for some i, j and obtain
lyche-numerical-linear-algebra- F01576	123	■ 1.000	$\boldsymbol{y} = \boldsymbol{e}_i$	$\boldsymbol{y} = \boldsymbol{e}_i$	For the converse we take $\boldsymbol{x} = \boldsymbol{e}_j$ and $\boldsymbol{y} = \boldsymbol{e}_i$ for some i, j and obtain
lyche-numerical-linear-algebra- F01577	123	■ 0.815	$\boldsymbol{A} = \boldsymbol{A}^T$	$\boldsymbol{A} = \boldsymbol{A}^T$	Thus, $\boldsymbol{A} = \boldsymbol{A}^T$ since they have the same i, j element for all i, j . The proof of 2. is
lyche-numerical-linear-algebra- F01578	123	■ 1.000	$\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$	$\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$	A square matrix $\boldsymbol{U} \in \mathbb{C}^{n \times n}$ is unitary if $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$. If $\boldsymbol{U}$ is real then $\boldsymbol{U}^T \boldsymbol{U} =$
lyche-numerical-linear-algebra- F01579	123	■ 1.000	$\boldsymbol{U}^T \boldsymbol{U} =$	$\boldsymbol{U}^T \boldsymbol{U} =$	A square matrix $\boldsymbol{U} \in \mathbb{C}^{n \times n}$ is unitary if $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$. If $\boldsymbol{U}$ is real then $\boldsymbol{U}^T \boldsymbol{U} =$
lyche-numerical-linear-algebra- F01580	123	■ 1.000	$\boldsymbol{U}^{-1} = \boldsymbol{U}^*$	$\boldsymbol{U}^{-1} = \boldsymbol{U}^*$	If $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$ the matrix $\boldsymbol{U}$ is nonsingular, $\boldsymbol{U}^{-1} = \boldsymbol{U}^*$ and therefore $\boldsymbol{U} \boldsymbol{U}^* =$
lyche-numerical-linear-algebra- F01581	123	■ 1.000	$\boldsymbol{U} \boldsymbol{U}^* =$	$\boldsymbol{U} \boldsymbol{U}^* =$	If $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$ the matrix $\boldsymbol{U}$ is nonsingular, $\boldsymbol{U}^{-1} = \boldsymbol{U}^*$ and therefore $\boldsymbol{U} \boldsymbol{U}^* =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F01582	123	1.000	$\boldsymbol{U} \boldsymbol{U}^{-1}=\boldsymbol{I}$	$\boldsymbol{U}\boldsymbol{U}^{-1} = \boldsymbol{I}$	$\boldsymbol{U}\boldsymbol{U}^{-1} = \boldsymbol{I}$ as well. Moreover, both the columns and rows of a unitary matrix of
lyche-numerical-linear-algebra- F01583	123	1.000	$\boldsymbol{U}_1^* \boldsymbol{U}_1=\boldsymbol{I}$	$\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1 \boldsymbol{U}_2)^*(\boldsymbol{U}_1 \boldsymbol{U}_2) =$
lyche-numerical-linear-algebra- F01584	123	1.000	$\boldsymbol{U}_2^* \boldsymbol{U}_2=\boldsymbol{I}$	$\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1 \boldsymbol{U}_2)^*(\boldsymbol{U}_1 \boldsymbol{U}_2) =$
lyche-numerical-linear-algebra- F01585	123	1.000	$\left(\boldsymbol{U}_1 \boldsymbol{U}_2\right)^* \left(\boldsymbol{U}_1 \boldsymbol{U}_2\right)=$	$(\boldsymbol{U}_1 \boldsymbol{U}_2)^* (\boldsymbol{U}_1 \boldsymbol{U}_2) =$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1 \boldsymbol{U}_2)^*(\boldsymbol{U}_1 \boldsymbol{U}_2) =$
lyche-numerical-linear-algebra- F01586	123	0.984	$\boldsymbol{U}_2^* \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_2=\boldsymbol{I}$	$\boldsymbol{U}_2^* \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_2 = \boldsymbol{I}$	$\boldsymbol{U}_2^* \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_2 = \boldsymbol{I}$.
lyche-numerical-linear-algebra- F01587	124	0.997	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$. In particular, if $\boldsymbol{U}$ is unitary then $\ \boldsymbol{U} \boldsymbol{x}\ _2 =$
lyche-numerical-linear-algebra- F01588	124	0.997	$\ \boldsymbol{U} \boldsymbol{x}\ _2 =$	$\ \boldsymbol{U} \boldsymbol{x}\ _2 =$	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$. In particular, if $\boldsymbol{U}$ is unitary then $\ \boldsymbol{U} \boldsymbol{x}\ _2 =$
lyche-numerical-linear-algebra- F01589	124	1.000	$\ \boldsymbol{x}\ _2$	$\ \boldsymbol{x}\ _2$	$\ \boldsymbol{x}\ _2$ for all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra- F01590	124	1.000	$i, j=1, \ldots, n$	$i, j = 1, \ldots, n$	$i, j = 1, \ldots, n$
lyche-numerical-linear-algebra- F01591	124	1.000	$\left(\boldsymbol{U}^* \boldsymbol{U}\right)_{i, j}=\delta_{i, j}$	$(\boldsymbol{U}^* \boldsymbol{U})_{i, j} = \delta_{i, j}$	so that $(\boldsymbol{U}^* \boldsymbol{U})_{i, j} = \delta_{i, j}$ for all i, j . The last part of the theorem follows immediately
lyche-numerical-linear-algebra- F01592	124	1.000	$\boldsymbol{y}=\boldsymbol{x}$	$\boldsymbol{y} = \boldsymbol{x}$	by taking $\boldsymbol{y}=\boldsymbol{x}$:
lyche-numerical-linear-algebra- F01593	124	1.000	$\boldsymbol{H} \in \mathbb{C}^{n \times n}$	$\boldsymbol{H} \in \mathbb{C}^{n \times n}$	Definition 5.4 (Householder Transformation) A matrix $\boldsymbol{H} \in \mathbb{C}^{n \times n}$ of the form
lyche-numerical-linear-algebra- F01594	124	1.000	$\boldsymbol{H}^*=(\boldsymbol{I}-$	$\boldsymbol{H}^* = (\boldsymbol{I}-$	A Householder transformation is Hermitian and unitary. Indeed, $\boldsymbol{H}^* = (\boldsymbol{I}-$
lyche-numerical-linear-algebra- F01595	124	1.000	$\left.\boldsymbol{u} \boldsymbol{u}^*\right)^*=\boldsymbol{H}$	$\boldsymbol{u} \boldsymbol{u}^*)^* = \boldsymbol{H}$	$\boldsymbol{u} \boldsymbol{u}^*)^* = \boldsymbol{H}$ and
lyche-numerical-linear-algebra- F01596	124	1.000	$\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^*$	$\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^*$	used $\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u}^* \boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra- F01597	124	1.000	$\boldsymbol{u}^* \boldsymbol{u}=1$	$\boldsymbol{u}^* \boldsymbol{u} = 1$	used $\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u}^* \boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra- F01598	124	1.000	$\boldsymbol{v} \in \mathbb{C}^n$	$\boldsymbol{v} \in \mathbb{C}^n$	used $\boldsymbol{I}-2 \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u}^* \boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra- F01599	125	1.000	$\boldsymbol{H}=\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*$	$\boldsymbol{H} = \boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*$	is a Householder transformation. Indeed, $\boldsymbol{H} = \boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u}:=\sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$ has
lyche-numerical-linear-algebra- F01600	125	1.000	$\boldsymbol{u}:=\sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$	$\boldsymbol{u}:=\sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$	is a Householder transformation. Indeed, $\boldsymbol{H} = \boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u}:=\sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$ has
lyche-numerical-linear-algebra- F01601	125	1.000	$\sqrt{2}$	$\sqrt{2}$	length $\sqrt{2}$.
lyche-numerical-linear-algebra- F01602	125	1.000	$\ \boldsymbol{x}\ _2=\ \boldsymbol{y}\ _2, \boldsymbol{y}^* \boldsymbol{x}$	$\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2, \boldsymbol{y}^* \boldsymbol{x}$	Lemma 5.2 Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ are two vectors such that $\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2$, $\boldsymbol{y}^* \boldsymbol{x}$ is
lyche-numerical-linear-algebra- F01603	125	0.993	$\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq \mathbf{0}$	$\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq \mathbf{0}$	real and $\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq \mathbf{0}$. Then $\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}\right) \boldsymbol{x}=\boldsymbol{y}$.
lyche-numerical-linear-algebra- F01604	125	0.993	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}\right) \boldsymbol{x}=\boldsymbol{y}$	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}\right) \boldsymbol{x} = \boldsymbol{y}$	real and $\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq \mathbf{0}$. Then $\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}\right) \boldsymbol{x} = \boldsymbol{y}$.
lyche-numerical-linear-algebra- F01605	125	1.000	$\boldsymbol{x}^* \boldsymbol{x}=\boldsymbol{y}^* \boldsymbol{y}$	$\boldsymbol{x}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{y}$	Proof Since $\boldsymbol{x}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{y}$ and $\operatorname{Re}\left(\boldsymbol{y}^* \boldsymbol{x}\right)=\boldsymbol{y}^* \boldsymbol{x}$ we have
lyche-numerical-linear-algebra- F01606	125	1.000	$\operatorname{Re}\left(\boldsymbol{y}^* \boldsymbol{x}\right)=\boldsymbol{y}^* \boldsymbol{x}$	$\operatorname{Re}\left(\boldsymbol{y}^* \boldsymbol{x}\right) = \boldsymbol{y}^* \boldsymbol{x}$	Proof Since $\boldsymbol{x}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{y}$ and $\operatorname{Re}\left(\boldsymbol{y}^* \boldsymbol{x}\right)=\boldsymbol{y}^* \boldsymbol{x}$ we have

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F01607	125	■ 0.998	$\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{x}-\frac{2\boldsymbol{v}^*\boldsymbol{x}}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{v}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$	$\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{x}-\frac{2\boldsymbol{v}^*\boldsymbol{x}}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{v}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$	But then $\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{x}-\frac{2\boldsymbol{v}^*\boldsymbol{x}}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{v}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$.
lyche-numerical-linear-algebra- F01608	125	■ 1.000	$\boldsymbol{H}\boldsymbol{x}$	$\boldsymbol{H}\boldsymbol{x}$	If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ it follows that $\boldsymbol{H}\boldsymbol{x}$ is the reflected image of $\boldsymbol{x}$. The “mirror” $\mathcal{M} :=$
lyche-numerical-linear-algebra- F01609	125	■ 1.000	$\mathcal{M} :=$	$\mathcal{M} :=$	If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ it follows that $\boldsymbol{H}\boldsymbol{x}$ is the reflected image of $\boldsymbol{x}$. The “mirror” $\mathcal{M} :=$
lyche-numerical-linear-algebra- F01610	125	■ 1.000	$\left\{\boldsymbol{w} \in \mathbb{R}^n: \boldsymbol{w}^*\boldsymbol{v}=0\right\}$	$\{\boldsymbol{w} \in \mathbb{R}^n: \boldsymbol{w}^*\boldsymbol{v}=0\}$	$\{\boldsymbol{w} \in \mathbb{R}^n: \boldsymbol{w}^*\boldsymbol{v}=0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra- F01611	125	■ 1.000	$(\boldsymbol{x}+\boldsymbol{y}) / 2$	$(\boldsymbol{x}+\boldsymbol{y})/2$	$\{\boldsymbol{w} \in \mathbb{R}^n: \boldsymbol{w}^*\boldsymbol{v}=0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra- F01612	125	■ 1.000	$\boldsymbol{x}-\boldsymbol{y}$	$\boldsymbol{x}-\boldsymbol{y}$	$\{\boldsymbol{w} \in \mathbb{R}^n: \boldsymbol{w}^*\boldsymbol{v}=0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra- F01613	126	■ 1.000	$\boldsymbol{x}:=[1,0,1]^T$	$\boldsymbol{x}:=[1,0,1]^T$	<i>Example 5.1 (Reflector)</i> Suppose $\boldsymbol{x}:=[1,0,1]^T$ and $\boldsymbol{y}:=[-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra- F01614	126	■ 1.000	$\boldsymbol{y}:=[-1,0,1]^T$	$\boldsymbol{y}:=[-1,0,1]^T$	<i>Example 5.1 (Reflector)</i> Suppose $\boldsymbol{x}:=[1,0,1]^T$ and $\boldsymbol{y}:=[-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra- F01615	126	■ 1.000	$\boldsymbol{v} =$	$\boldsymbol{v} =$	<i>Example 5.1 (Reflector)</i> Suppose $\boldsymbol{x}:=[1,0,1]^T$ and $\boldsymbol{y}:=[-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra- F01616	126	■ 1.000	$\boldsymbol{x}-\boldsymbol{y}=[2,0,0]^T$	$\boldsymbol{x}-\boldsymbol{y}=[2,0,0]^T$	$\boldsymbol{x}-\boldsymbol{y}=[2,0,0]^T$ and
lyche-numerical-linear-algebra- F01617	126	■ 0.999	$\boldsymbol{y} \boldsymbol{z}$	$\boldsymbol{y}\boldsymbol{z}$	is the $\boldsymbol{y}\boldsymbol{z}$ plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x}=[0,0,1]^T=$
lyche-numerical-linear-algebra- F01618	126	■ 0.999	$\boldsymbol{H}\boldsymbol{x} \boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$	$\boldsymbol{H}\boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$	is the $\boldsymbol{y}\boldsymbol{z}$ plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x}=[0,0,1]^T=$
lyche-numerical-linear-algebra- F01619	126	■ 0.999	$\boldsymbol{P}\boldsymbol{x} \boldsymbol{x}=[0,0,1]^T=$	$\boldsymbol{P}\boldsymbol{x}=[0,0,1]^T=$	is the $\boldsymbol{y}\boldsymbol{z}$ plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x}=[0,0,1]^T=$
lyche-numerical-linear-algebra- F01620	126	■ 1.000	$(\boldsymbol{x}+\boldsymbol{y}) / 2 \in \mathcal{M}$	$(\boldsymbol{x}+\boldsymbol{y})/2 \in \mathcal{M}$	$(\boldsymbol{x}+\boldsymbol{y})/2 \in \mathcal{M}$.
lyche-numerical-linear-algebra- F01621	126	■ 1.000	$\boldsymbol{H}\boldsymbol{x} \boldsymbol{x}=\mathbf{0}$	$\boldsymbol{H}\boldsymbol{x}=\mathbf{0}$	<i>Proof</i> If $\boldsymbol{x}=\mathbf{0}$ then $\boldsymbol{H}\boldsymbol{x}=\mathbf{0}$ and $a=0$. Any $\boldsymbol{u}$ with $\ \boldsymbol{u}\ _2=\sqrt{2}$ will work, and
lyche-numerical-linear-algebra- F01622	126	■ 1.000	$a=0$	$a=0$	<i>Proof</i> If $\boldsymbol{x}=\mathbf{0}$ then $\boldsymbol{H}\boldsymbol{x}=\mathbf{0}$ and $a=0$. Any $\boldsymbol{u}$ with $\ \boldsymbol{u}\ _2=\sqrt{2}$ will work, and
lyche-numerical-linear-algebra- F01623	126	■ 1.000	$\ \boldsymbol{u}\ _2=\sqrt{2}$	$\ \boldsymbol{u}\ _2=\sqrt{2}$	<i>Proof</i> If $\boldsymbol{x}=\mathbf{0}$ then $\boldsymbol{H}\boldsymbol{x}=\mathbf{0}$ and $a=0$. Any $\boldsymbol{u}$ with $\ \boldsymbol{u}\ _2=\sqrt{2}$ will work, and
lyche-numerical-linear-algebra- F01624	126	■ 1.000	$\boldsymbol{u}:=\sqrt{2} \boldsymbol{e}_1$	$\boldsymbol{u}:=\sqrt{2} \boldsymbol{e}_1$	we choose $\boldsymbol{u}:=\sqrt{2} \boldsymbol{e}_1$ in this case. For $\boldsymbol{x} \neq \mathbf{0}$ we define
lyche-numerical-linear-algebra- F01625	126	■ 1.000	$\boldsymbol{x} \neq \mathbf{0}$	$\boldsymbol{x} \neq \mathbf{0}$	we choose $\boldsymbol{u}:=\sqrt{2} \boldsymbol{e}_1$ in this case. For $\boldsymbol{x} \neq \mathbf{0}$ we define
lyche-numerical-linear-algebra- F01626	126	■ 0.971	$ \rho =1$	$ \rho =1$	Since $ \rho =1$ we have $\rho\ \boldsymbol{x}\ _2 \boldsymbol{z}= \rho ^2 \boldsymbol{x}=\boldsymbol{x}$. Moreover, $\ \boldsymbol{z}\ _2=1$ and $z_1=$
lyche-numerical-linear-algebra- F01627	126	■ 0.971	$\rho\ \boldsymbol{x}\ _2 \boldsymbol{z}= \rho ^2 \boldsymbol{x}=\boldsymbol{x}$	$\rho\ \boldsymbol{x}\ _2 \boldsymbol{z}= \rho ^2 \boldsymbol{x}=\boldsymbol{x}$	Since $ \rho =1$ we have $\rho\ \boldsymbol{x}\ _2 \boldsymbol{z}= \rho ^2 \boldsymbol{x}=\boldsymbol{x}$. Moreover, $\ \boldsymbol{z}\ _2=1$ and $z_1=$
lyche-numerical-linear-algebra- F01628	126	■ 0.971	$\ \boldsymbol{z}\ _2=1$	$\ \boldsymbol{z}\ _2=1$	Since $ \rho =1$ we have $\rho\ \boldsymbol{x}\ _2 \boldsymbol{z}= \rho ^2 \boldsymbol{x}=\boldsymbol{x}$. Moreover, $\ \boldsymbol{z}\ _2=1$ and $z_1=$
lyche-numerical-linear-algebra- F01629	126	■ 0.971	$z_1=$	$z_1=$	Since $ \rho =1$ we have $\rho\ \boldsymbol{x}\ _2 \boldsymbol{z}= \rho ^2 \boldsymbol{x}=\boldsymbol{x}$. Moreover, $\ \boldsymbol{z}\ _2=1$ and $z_1=$
lyche-numerical-linear-algebra- F01630	126	■ 0.969	$\left x_1\right /\ \boldsymbol{x}\ _2$	$\left x_1\right /\ \boldsymbol{x}\ _2$	$\left x_1\right /\ \boldsymbol{x}\ _2$ is real so that $\boldsymbol{u}^* \boldsymbol{u}=\frac{\left(\boldsymbol{z}+\boldsymbol{e}_1\right)^*(\boldsymbol{z}+\boldsymbol{e}_1)}{1+z_1}=\frac{2+2 z_1}{1+z_1}=2$. Finally,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F01631	126	0.969	$\boldsymbol{u}^* \boldsymbol{u} = \frac{(z + \boldsymbol{e}_1)^* (z + \boldsymbol{e}_1)}{1 + z_1} = \frac{2 + 2z_1}{1 + z_1} = 2$	$\boldsymbol{u}^* \boldsymbol{u} = \frac{(z + \boldsymbol{e}_1)^* (z + \boldsymbol{e}_1)}{1 + z_1} = \frac{2 + 2z_1}{1 + z_1} = 2$	$ x_1 /\ \boldsymbol{x}\ _2$ is real so that $\boldsymbol{u}^* \boldsymbol{u} = \frac{(z + \boldsymbol{e}_1)^* (z + \boldsymbol{e}_1)}{1 + z_1} = \frac{2 + 2z_1}{1 + z_1} = 2$. Finally,
lyche-numerical-linear-algebra F01632	127	1.000	$\boldsymbol{u}^* \boldsymbol{u} = 2$	$\boldsymbol{u}^* \boldsymbol{u} = 2$	adapted from [17]. To any given $\boldsymbol{x} \in \mathbb{C}^n$, a number a and a vector $\boldsymbol{u}$ with $\boldsymbol{u}^* \boldsymbol{u} = 2$
lyche-numerical-linear-algebra F01633	127	0.986	$\left(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^* \right) \boldsymbol{x} = a \boldsymbol{e}_1$	$(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) \boldsymbol{x} = a \boldsymbol{e}_1$	is computed so that $(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) \boldsymbol{x} = a \boldsymbol{e}_1$:
lyche-numerical-linear-algebra F01634	127	0.999	$z_1 = \left x_1 \right / \ \boldsymbol{x}\ _2 \geq 0$	$z_1 = x_1 / \ \boldsymbol{x}\ _2 \geq 0$	• In Theorem 5.8 the first component of $\boldsymbol{z}$ is $z_1 = x_1 / \ \boldsymbol{x}\ _2 \geq 0$. Since $\ \boldsymbol{z}\ _2 =$
lyche-numerical-linear-algebra F01635	127	0.999	$\ \boldsymbol{z}\ _2 =$	$\ \boldsymbol{z}\ _2 =$	• In Theorem 5.8 the first component of $\boldsymbol{z}$ is $z_1 = x_1 / \ \boldsymbol{x}\ _2 \geq 0$. Since $\ \boldsymbol{z}\ _2 =$
lyche-numerical-linear-algebra F01636	127	1.000	$1 \leq 1 + z_1 \leq 2$	$1 \leq 1 + z_1 \leq 2$	1 we have $1 \leq 1 + z_1 \leq 2$. It follows that we avoid cancelation error when
lyche-numerical-linear-algebra F01637	127	1.000	$1 + z_1$	$1 + z_1$	computing $1 + z_1$ and $\boldsymbol{u}$ and a are computed in a numerically stable way.
lyche-numerical-linear-algebra F01638	127	0.887	$\boldsymbol{H} \boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) \boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u}$	$\boldsymbol{H} \boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) \boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u}$	Indeed, $\boldsymbol{H} \boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) \boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$
lyche-numerical-linear-algebra F01639	127	0.887	$\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$	$\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$	Indeed, $\boldsymbol{H} \boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) \boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$
lyche-numerical-linear-algebra F01640	127	0.887	$2m$	$2m$	Indeed, $\boldsymbol{H} \boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) \boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$
lyche-numerical-linear-algebra F01641	127	0.994	$\boldsymbol{u}^T \boldsymbol{x}, m$	$\boldsymbol{u}^T \boldsymbol{x}, m$	operations to find $\boldsymbol{u}^T \boldsymbol{x}$, m operations for $(\boldsymbol{u}^T \boldsymbol{x}) \boldsymbol{u}$ and m operations for the final
lyche-numerical-linear-algebra F01642	127	0.994	$(\boldsymbol{u}^T \boldsymbol{x}) \boldsymbol{u}$	$(\boldsymbol{u}^T \boldsymbol{x}) \boldsymbol{u}$	operations to find $\boldsymbol{u}^T \boldsymbol{x}$, m operations for $(\boldsymbol{u}^T \boldsymbol{x}) \boldsymbol{u}$ and m operations for the final
lyche-numerical-linear-algebra F01643	127	1.000	$4m$	$4m$	subtraction of the two vectors, a total of $4m$ arithmetic operations. If $\boldsymbol{A} \in \mathbb{R}^{m \times n}$
lyche-numerical-linear-algebra F01644	127	1.000	$4mn$	$4mn$	then $4mn$ operations are required for $\boldsymbol{H} \boldsymbol{A} = \boldsymbol{A} - (\boldsymbol{u}^T \boldsymbol{A}) \boldsymbol{u}$, i.e., $4m$ operations
lyche-numerical-linear-algebra F01645	127	1.000	$\boldsymbol{H} \boldsymbol{A} = \boldsymbol{A} - (\boldsymbol{u}^T \boldsymbol{A}) \boldsymbol{u}$	$\boldsymbol{H} \boldsymbol{A} = \boldsymbol{A} - (\boldsymbol{u}^T \boldsymbol{A}) \boldsymbol{u}$	then $4mn$ operations are required for $\boldsymbol{H} \boldsymbol{A} = \boldsymbol{A} - (\boldsymbol{u}^T \boldsymbol{A}) \boldsymbol{u}$, i.e., $4m$ operations
lyche-numerical-linear-algebra F01646	127	1.000	$\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$	$\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$	a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k$, $\boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \leq k <$
lyche-numerical-linear-algebra F01647	127	1.000	$\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$	$\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$	a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \leq k <$
lyche-numerical-linear-algebra F01648	127	1.000	$1 \leq k <$	$1 \leq k <$	a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \leq k <$
lyche-numerical-linear-algebra F01649	127	0.915	$[\hat{\boldsymbol{u}}, a] := \text{housegen}(\boldsymbol{z})$	$[\hat{\boldsymbol{u}}, a] := \text{housegen}(\boldsymbol{z})$	n . The command $[\hat{\boldsymbol{u}}, a] := \text{housegen}(\boldsymbol{z})$ defines a Householder transformation
lyche-numerical-linear-algebra F01650	127	0.384	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}} \hat{\boldsymbol{u}}^*$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}} \hat{\boldsymbol{u}}^*$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}} \hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}} \boldsymbol{z} = a \boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$ we see that
lyche-numerical-linear-algebra F01651	127	0.384	$\hat{\boldsymbol{H}} \boldsymbol{z} = a \boldsymbol{e}_1 \in \mathbb{C}^{n-k}$	$\hat{\boldsymbol{H}} \boldsymbol{z} = a \boldsymbol{e}_1 \in \mathbb{C}^{n-k}$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}} \hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}} \boldsymbol{z} = a \boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$ we see that
lyche-numerical-linear-algebra F01652	127	0.384	$\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$	$\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}} \hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}} \boldsymbol{z} = a \boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$ we see that
lyche-numerical-linear-algebra F01653	127	1.000	$\boldsymbol{u}^* \boldsymbol{u} = \hat{\boldsymbol{u}}^* \hat{\boldsymbol{u}} = 2$	$\boldsymbol{u}^* \boldsymbol{u} = \hat{\boldsymbol{u}}^* \hat{\boldsymbol{u}} = 2$	$\boldsymbol{u}^* \boldsymbol{u} = \hat{\boldsymbol{u}}^* \hat{\boldsymbol{u}} = 2$, and
lyche-numerical-linear-algebra F01654	128	1.000	$\boldsymbol{R} \in \mathbb{C}^{m \times n}$	$\boldsymbol{R} \in \mathbb{C}^{m \times n}$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m \times n}$ is upper trapezoidal , if $r_{i,j} = 0$ for $j < i$ and
lyche-numerical-linear-algebra F01655	128	1.000	$r_{i,j} = 0$	$r_{i,j} = 0$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m \times n}$ is upper trapezoidal , if $r_{i,j} = 0$ for $j < i$ and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01656	128	■ 1.000	$j < i$	$j < i$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m \times n}$ is upper trapezoidal , if $r_{i,j} = 0$ for $j < i$ and
lyche-numerical-linear-algebra-F01657	128	■ 1.000	$i=2,3 \ \text{\ldots}, \ m$	$i = 2, 3 \dots, m$	$\equiv 2, 3 \dots m$. Upper trapezoidal matrices corresponding to $m < n$, $m = n$, and
lyche-numerical-linear-algebra-F01658	128	■ 1.000	$m \leq n$	$m \leq n$	We treat the cases $m > n$ and $m \leq n$ separately and consider first $m > n$. We
lyche-numerical-linear-algebra-F01659	128	■ 0.996	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_n$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_n$	describe how to find a sequence $\boldsymbol{H}_1, \dots, \boldsymbol{H}_n$ of Householder transformations such
lyche-numerical-linear-algebra-F01660	128	■ 1.000	$\boldsymbol{R}_1$	$\boldsymbol{R}_1$	and where $\boldsymbol{R}_1$ is square and upper triangular. We define
lyche-numerical-linear-algebra-F01661	129	■ 1.000	$\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k \hat{\boldsymbol{u}}_k^*$	$\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k \hat{\boldsymbol{u}}_k^*$	Let $\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k \hat{\boldsymbol{u}}_k^*$ be a Householder transformation that maps the first column
lyche-numerical-linear-algebra-F01662	129	■ 1.000	$\left[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)} \right]^T$	$\left[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)} \right]^T$	$[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra-F01663	129	■ 1.000	$\boldsymbol{D}_k$	$\boldsymbol{D}_k$	$[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra-F01664	129	■ 1.000	$\boldsymbol{e}_1, \hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$	$\boldsymbol{e}_1, \hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$	$[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra-F01665	129	■ 0.865	$[\hat{\boldsymbol{u}}_k, a_k] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$	$[\hat{\boldsymbol{u}}_k, a_k] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$	we have $[\hat{\boldsymbol{u}}_k, a_k] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$. Then $\boldsymbol{H}_k := \begin{bmatrix} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{bmatrix}$ is a Householder
lyche-numerical-linear-algebra-F01666	129	■ 0.865	$\boldsymbol{H}_k := \begin{bmatrix} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{bmatrix}$	$\boldsymbol{H}_k := \begin{bmatrix} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{bmatrix}$	we have $[\hat{\boldsymbol{u}}_k, a_k] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$. Then $\boldsymbol{H}_k := \begin{bmatrix} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{bmatrix}$ is a Householder
lyche-numerical-linear-algebra-F01667	129	■ 1.000	$\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$	$\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra-F01668	129	■ 1.000	$\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$	$\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra-F01669	129	■ 1.000	$\boldsymbol{A}_{k+1}$	$\boldsymbol{A}_{k+1}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra-F01670	129	■ 1.000	$\boldsymbol{R} := \boldsymbol{A}_{n+1} = \begin{bmatrix} \boldsymbol{R}_1 \\ \mathbf{0} \end{bmatrix}$	$\boldsymbol{R} := \boldsymbol{A}_{n+1} = \begin{bmatrix} \boldsymbol{R}_1 \\ \mathbf{0} \end{bmatrix}$	further. At the end $\boldsymbol{R} := \boldsymbol{A}_{n+1} = \begin{bmatrix} \boldsymbol{R}_1 \\ \mathbf{0} \end{bmatrix}$, where $\boldsymbol{R}_1$ is upper triangular.
lyche-numerical-linear-algebra-F01671	129	■ 1.000	$m > 1$	$m > 1$	already in upper trapezoidal form. Suppose $m > 1$. In this case $m - 1$ Householder
lyche-numerical-linear-algebra-F01672	129	■ 1.000	$m - 1$	$m - 1$	already in upper trapezoidal form. Suppose $m > 1$. In this case $m - 1$ Householder
lyche-numerical-linear-algebra-F01673	129	■ 0.999	$\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$	transformations will suffice and $\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$ is upper trapezoidal.
lyche-numerical-linear-algebra-F01674	129	■ 1.000	$\hat{\boldsymbol{u}}_k = [u_{kk}, \dots, u_{mk}]^T$	$\hat{\boldsymbol{u}}_k = [u_{kk}, \dots, u_{mk}]^T$	In an algorithm we can store most of the vector $\hat{\boldsymbol{u}}_k = [u_{kk}, \dots, u_{mk}]^T$ and the
lyche-numerical-linear-algebra-F01675	129	■ 1.000	$u_{k,k}$	$u_{k,k}$	matrix and $\boldsymbol{A}_k$ in $\boldsymbol{A}$. However, the elements $u_{k,k}$ and $a_k = r_{k,k}$ have to compete for
lyche-numerical-linear-algebra-F01676	129	■ 1.000	$a_k = r_{k,k}$	$a_k = r_{k,k}$	matrix and $\boldsymbol{A}_k$ in $\boldsymbol{A}$. However, the elements $u_{k,k}$ and $a_k = r_{k,k}$ have to compete for
lyche-numerical-linear-algebra-F01677	129	■ 1.000	$m = 4$	$m = 4$	the diagonal in $\boldsymbol{A}$. For $m = 4$ and $n = 3$ the two possibilities look as follows:
lyche-numerical-linear-algebra-F01678	129	■ 1.000	$\boldsymbol{B} \in \mathbb{C}^{m \times r}$	$\boldsymbol{B} \in \mathbb{C}^{m \times r}$	$\boldsymbol{B} \in \mathbb{C}^{m \times r}$ as input, and uses housegen to compute Householder transformations
lyche-numerical-linear-algebra-F01679	129	■ 1.000	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_s$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_s$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_s$ so that $\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{A}$ is upper trapezoidal, and $\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{B}$.

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lyche-numerical-linear-algebra-F01680	129	1.000	$\boldsymbol{R}=\boldsymbol{H}_{\{s\}} \cdots \boldsymbol{H}_{\{1\}} \boldsymbol{A}$	$\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_s$ so that $\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{A}$ is upper trapezoidal, and $\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{B}$.
lyche-numerical-linear-algebra-F01681	129	1.000	$\boldsymbol{C}=\boldsymbol{H}_{\{s\}} \cdots \boldsymbol{H}_{\{1\}} \boldsymbol{B}$	$\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{B}$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_s$ so that $\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{A}$ is upper trapezoidal, and $\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{B}$.
lyche-numerical-linear-algebra-F01682	129	1.000	$r_{\{k, k\}}$	$r_{k,k}$	matrix with m rows and 0 columns. $r_{k,k}$ is stored in $\boldsymbol{A}$, and $u_{k,k}$ is stored in a
lyche-numerical-linear-algebra-F01683	129	0.961	$\operatorname{side}(\mathrm{s}) \boldsymbol{B}$	side(s) $\boldsymbol{B}$	and least squares problems with right hand side(s) $\boldsymbol{B}$, and to compute the product of
lyche-numerical-linear-algebra-F01684	129	0.999	$\boldsymbol{B}=\boldsymbol{I}$	$\boldsymbol{B} = \boldsymbol{I}$	the Householder transformations by choosing $\boldsymbol{B} = \boldsymbol{I}$.
lyche-numerical-linear-algebra-F01685	130	0.998	$v=\hat{\boldsymbol{u}}_{\{k\}}$	$v = \hat{\boldsymbol{u}}_k$	Here $v = \hat{\boldsymbol{u}}_k$ and the update is computed as $\hat{\boldsymbol{H}}_k \boldsymbol{C} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^*) \boldsymbol{C} = \boldsymbol{C} - \boldsymbol{v}(\boldsymbol{v}^* \boldsymbol{C})$.
lyche-numerical-linear-algebra-F01686	130	0.998	$\hat{\boldsymbol{H}}_{\{k\}} \boldsymbol{C}=\left(\boldsymbol{I}-\boldsymbol{v} \boldsymbol{v}^*\right) \boldsymbol{C}=\boldsymbol{C}-\boldsymbol{v}\left(\boldsymbol{v}^* \boldsymbol{C}\right)$	$\hat{\boldsymbol{H}}_k \boldsymbol{C} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^*) \boldsymbol{C} = \boldsymbol{C} - \boldsymbol{v}(\boldsymbol{v}^* \boldsymbol{C})$	Here $v = \hat{\boldsymbol{u}}_k$ and the update is computed as $\hat{\boldsymbol{H}}_k \boldsymbol{C} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^*) \boldsymbol{C} = \boldsymbol{C} - \boldsymbol{v}(\boldsymbol{v}^* \boldsymbol{C})$.
lyche-numerical-linear-algebra-F01687	130	1.000	$n+1, \ldots, m$	$n+1, \dots, m$	zeros in rows $n+1, \dots, m$.
lyche-numerical-linear-algebra-F01688	130	1.000	$\boldsymbol{C}-\boldsymbol{v} *\left(\boldsymbol{v}^* *\boldsymbol{C}\right)$	$\boldsymbol{C} - \boldsymbol{v} * (\boldsymbol{v}^* * \boldsymbol{C})$	The bulk of the work in Algorithm 5.2 is the computation of $\boldsymbol{C} - \boldsymbol{v} * (\boldsymbol{v}^* * \boldsymbol{C})$ for each
lyche-numerical-linear-algebra-F01689	130	1.000	$\boldsymbol{C} \in \mathbb{C}^{(m-k+1) \times(n+r-k)}$	$\boldsymbol{C} \in \mathbb{C}^{(m-k+1) \times (n+r-k)}$	k . Since in Algorithm 5.2, $\boldsymbol{C} \in \mathbb{C}^{(m-k+1) \times (n+r-k)}$ and $m \geq n$ the cost of computing
lyche-numerical-linear-algebra-F01690	130	1.000	$m \geq n$	$m \geq n$	k . Since in Algorithm 5.2, $\boldsymbol{C} \in \mathbb{C}^{(m-k+1) \times (n+r-k)}$ and $m \geq n$ the cost of computing
lyche-numerical-linear-algebra-F01691	130	1.000	$\boldsymbol{C}-\boldsymbol{v} *\left(\boldsymbol{v}^T *\boldsymbol{C}\right)$	$\boldsymbol{C} - \boldsymbol{v} * (\boldsymbol{v}^T * \boldsymbol{C})$	the update $\boldsymbol{C} - \boldsymbol{v} * (\boldsymbol{v}^T * \boldsymbol{C})$ in the real case is $4(m-k+1)(n+r-k)$ arithmetic
lyche-numerical-linear-algebra-F01692	130	1.000	$4(m-k+1)(n+r-k)$	$4(m-k+1)(n+r-k)$	the update $\boldsymbol{C} - \boldsymbol{v} * (\boldsymbol{v}^T * \boldsymbol{C})$ in the real case is $4(m-k+1)(n+r-k)$ arithmetic
lyche-numerical-linear-algebra-F01693	130	1.000	$r=0$	$r = 0$	For $m = n$ and $r = 0$ this gives $4n^3/3 = 2G_n$ for the number of arithmetic
lyche-numerical-linear-algebra-F01694	130	1.000	$4 n^3 / 3=2 G_{\{n\}}$	$4n^3/3 = 2G_n$	For $m = n$ and $r = 0$ this gives $4n^3/3 = 2G_n$ for the number of arithmetic
lyche-numerical-linear-algebra-F01695	130	1.000	$\boldsymbol{R} \boldsymbol{x}=\boldsymbol{c}$	$\boldsymbol{R} \boldsymbol{x} = \boldsymbol{c}$	obtain an upper triangular system $\boldsymbol{R} \boldsymbol{x} = \boldsymbol{c}$ that is upper triangular and nonsingular if
lyche-numerical-linear-algebra-F01696	131	1.000	$4 n^3 / 3=$	$4n^3/3 =$	reason is that the number of arithmetic operations in (5.16) when $m = n$ is $4n^3/3 =$
lyche-numerical-linear-algebra-F01697	131	0.998	$\boldsymbol{R}=\boldsymbol{H}_{\{n-1\}} \cdots \boldsymbol{H}_{\{1\}} \boldsymbol{A}$	$\boldsymbol{R} = \boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$	Algorithm 5.2 gives $\boldsymbol{R} = \boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$ implying the factorization $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$,
lyche-numerical-linear-algebra-F01698	131	1.000	$\boldsymbol{Q}=\boldsymbol{H}_{\{1\}} \cdots \boldsymbol{H}_{\{n-1\}}$	$\boldsymbol{Q} = \boldsymbol{H}_1 \cdots \boldsymbol{H}_{n-1}$	where $\boldsymbol{Q} = \boldsymbol{H}_1 \cdots \boldsymbol{H}_{n-1}$ is unitary and $\boldsymbol{R}$ is upper triangular. This is known as a
lyche-numerical-linear-algebra-F01699	131	0.982	$\mathbf{Q} \boldsymbol{R}$	$\boldsymbol{Q} \boldsymbol{R}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$ is a QR decomposition of $\boldsymbol{A}$ if $\boldsymbol{Q} \in \mathbb{C}^{m,m}$ is square and unitary and
lyche-numerical-linear-algebra-F01700	131	0.982	$\boldsymbol{Q} \in \mathbb{C}^{m, m}$	$\boldsymbol{Q} \in \mathbb{C}^{m,m}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$ is a QR decomposition of $\boldsymbol{A}$ if $\boldsymbol{Q} \in \mathbb{C}^{m,m}$ is square and unitary and
lyche-numerical-linear-algebra-F01701	131	1.000	$\boldsymbol{R}_{\{1\}} \in \mathbb{C}^{n \times n}$	$\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$	where $\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$ is upper triangular and $\mathbf{0}_{m-n, n}$ is the zero matrix with $m - n$
lyche-numerical-linear-algebra-F01702	131	1.000	$\mathbf{0}_{\{m-n, n\}}$	$\mathbf{0}_{m-n, n}$	where $\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$ is upper triangular and $\mathbf{0}_{m-n, n}$ is the zero matrix with $m - n$
lyche-numerical-linear-algebra-F01703	131	1.000	$m-n$	$m - n$	where $\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$ is upper triangular and $\mathbf{0}_{m-n, n}$ is the zero matrix with $m - n$
lyche-numerical-linear-algebra-F01704	131	0.961	$\boldsymbol{A}=\boldsymbol{Q}_{\{1\}} \boldsymbol{R}_{\{1\}}$	$\boldsymbol{A} = \boldsymbol{Q}_1 \boldsymbol{R}_1$	rows and n columns. For $m \geq n$ we call $\boldsymbol{A} = \boldsymbol{Q}_1 \boldsymbol{R}_1$ a QR factorization of $\boldsymbol{A}$ if
lyche-numerical-linear-algebra-F01705	131	1.000	$\boldsymbol{Q}_{\{1\}} \in \mathbb{C}^{m \times n}$	$\boldsymbol{Q}_1 \in \mathbb{C}^{m \times n}$	$\boldsymbol{Q}_1 \in \mathbb{C}^{m \times n}$ has orthonormal columns and $\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$ is upper triangular.

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lyche-numerical-linear-algebra-F01706	131	1.000	$\boldsymbol{Q}$ $\boldsymbol{R}$	QR	QR by simply using the first n columns of Q and the first n rows of R . Indeed, if
lyche-numerical-linear-algebra-F01707	131	1.000	$\left[\boldsymbol{Q}_1, \boldsymbol{Q}_2\right]$	$[Q_1, Q_2]$	we partition Q as $[Q_1, Q_2]$ and $R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$, where $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01708	131	1.000	$\boldsymbol{R}=\left[\begin{array}{c}\boldsymbol{R}_1 \\ \mathbf{0}\end{array}\right]$	$R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$	we partition Q as $[Q_1, Q_2]$ and $R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$, where $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01709	131	1.000	$\boldsymbol{Q}_1 \in \mathbb{R}^{m \times n}$	$Q_1 \in \mathbb{R}^{m \times n}$	we partition Q as $[Q_1, Q_2]$ and $R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$, where $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01710	131	1.000	$\boldsymbol{R}_1 \in \mathbb{R}^{n \times n}$	$R_1 \in \mathbb{R}^{n \times n}$	we partition Q as $[Q_1, Q_2]$ and $R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$, where $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01711	131	1.000	$\left\{\boldsymbol{q}_1, \dots, \boldsymbol{q}_n\right\}$	$\{q_1, \dots, q_n\}$	columns $\{q_1, \dots, q_n\}$ of Q_1 into an orthonormal basis $\{q_1, \dots, q_n, q_{n+1}, \dots, q_m\}$
lyche-numerical-linear-algebra-F01712	131	1.000	$\boldsymbol{Q}_1$	Q_1	columns $\{q_1, \dots, q_n\}$ of Q_1 into an orthonormal basis $\{q_1, \dots, q_n, q_{n+1}, \dots, q_m\}$
lyche-numerical-linear-algebra-F01713	131	1.000	$\left\{\boldsymbol{q}_1, \dots, \boldsymbol{q}_n, \boldsymbol{q}_{n+1}, \dots, \boldsymbol{q}_m\right\}$	$\{q_1, \dots, q_n, q_{n+1}, \dots, q_m\}$	columns $\{q_1, \dots, q_n\}$ of Q_1 into an orthonormal basis $\{q_1, \dots, q_n, q_{n+1}, \dots, q_m\}$
lyche-numerical-linear-algebra-F01714	131	1.000	$\boldsymbol{Q}=\left[\boldsymbol{q}_1, \dots, \boldsymbol{q}_m\right]$	$Q = [q_1, \dots, q_m]$	$A = QR$, where $Q = [q_1, \dots, q_m]$ and $R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$.
lyche-numerical-linear-algebra-F01715	132	0.743	$\boldsymbol{Q}^T \boldsymbol{Q}=\boldsymbol{I}$	$Q^T Q = I$	Since $Q^T Q = I$ and R is upper trapezoidal, this is a QR decomposition of A . A
lyche-numerical-linear-algebra-F01716	132	1.000	Q R	QR	$m, n \in \mathbb{N}$ has a QR decomposition.
lyche-numerical-linear-algebra-F01717	132	1.000	$\boldsymbol{A}=[1] \boldsymbol{A}$	$A = [1]A$	Proof If $m = 1$ then A is already in upper trapezoidal form and $A = [1]A$ is a
lyche-numerical-linear-algebra-F01718	132	1.000	$s:=\min (m-1, n)$	$s := \min(m - 1, n)$	QR decomposition of A . Suppose $m > 1$ and set $s := \min(m - 1, n)$. Note that
lyche-numerical-linear-algebra-F01719	132	0.999	$\boldsymbol{R}=\boldsymbol{C} \boldsymbol{A}$	$R = CA$	for any vector x . With $B = I$ in Algorithm 5.2 we obtain $R = CA$ and $C =$
lyche-numerical-linear-algebra-F01720	132	0.999	$\boldsymbol{C} =$	$C =$	for any vector x . With $B = I$ in Algorithm 5.2 we obtain $R = CA$ and $C =$
lyche-numerical-linear-algebra-F01721	132	0.615	$\boldsymbol{H}_s \cdots \boldsymbol{H}_2 \boldsymbol{H}_1$	$H_s \cdots H_2 H_1$	$H_s \cdots H_2 H_1$. Thus $A = QR$ is a QR decomposition of A since $Q := C^* =$
lyche-numerical-linear-algebra-F01722	132	0.615	$\boldsymbol{Q}:=\boldsymbol{C}^* =$	$Q := C^* =$	$H_s \cdots H_2 H_1$. Thus $A = QR$ is a QR decomposition of A since $Q := C^* =$
lyche-numerical-linear-algebra-F01723	132	1.000	$\boldsymbol{H}_1 \cdots \boldsymbol{H}_s$	$H_1 \cdots H_s$	$H_1 \cdots H_s$ is a product of unitary matrices and therefore unitary.
lyche-numerical-linear-algebra-F01724	132	1.000	$\boldsymbol{A}^* \boldsymbol{A} =$	$A^* A =$	Proof Let $A = Q_1 R_1$ be a QR factorization of $A \in \mathbb{C}^{m \times n}$. Now $A^* A =$
lyche-numerical-linear-algebra-F01725	132	1.000	$\boldsymbol{R}_1^* \boldsymbol{Q}_1^* \boldsymbol{Q}_1 \boldsymbol{R}_1 = \boldsymbol{R}_1^* \boldsymbol{R}_1$	$R_1^* Q_1^* Q_1 R_1 = R_1^* R_1$	$R_1^* Q_1^* Q_1 R_1 = R_1^* R_1$. By Lemma 4.2 the matrix $A^* A$ is positive definite, the
lyche-numerical-linear-algebra-F01726	132	1.000	$\boldsymbol{R}_1^* \boldsymbol{R}_1$	$R_1^* R_1$	matrix R_1 is nonsingular, and if its diagonal elements are positive then $R_1^* R_1$ is the
lyche-numerical-linear-algebra-F01727	132	1.000	$\boldsymbol{Q}_1 = \boldsymbol{A} \boldsymbol{R}_1^{-1}$	$Q_1 = AR_1^{-1}$	that R_1 is unique and since necessarily $Q_1 = AR_1^{-1}$, it must also be unique.
lyche-numerical-linear-algebra-F01728	132	1.000	$\boldsymbol{B}:=\boldsymbol{A}^T \boldsymbol{A}=\left[\begin{array}{cc} 5 & -4 \\ -4 & 5 \end{array}\right]$	$B := A^T A = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$	uniqueness proof of Theorem 5.10. We find $B := A^T A = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$. The Cholesky
lyche-numerical-linear-algebra-F01729	132	1.000	$\boldsymbol{B}=\boldsymbol{R}^T \boldsymbol{R}$	$B = R^T R$	factorization of $B = R^T R$ is given by $R = \frac{1}{\sqrt{5}} \begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$. Now $R^{-1} = \frac{1}{3\sqrt{5}} \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$
lyche-numerical-linear-algebra-F01730	132	1.000	$\boldsymbol{R}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5 & -4 \\ 0 & 3 \end{array}\right]$	$R = \frac{1}{\sqrt{5}} \begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$	factorization of $B = R^T R$ is given by $R = \frac{1}{\sqrt{5}} \begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$. Now $R^{-1} = \frac{1}{3\sqrt{5}} \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$

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lyche-numerical-linear-algebra- F01731	132	1.000	$\boldsymbol{R}^{-1}=\frac{1}{3\sqrt{5}}\left[\begin{array}{ll}3 & 4 \\ 0 & 5\end{array}\right]$	$\boldsymbol{R}^{-1} = \frac{1}{3\sqrt{5}} \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$	factorization of $\boldsymbol{B} = \boldsymbol{R}^T \boldsymbol{R}$ is given by $\boldsymbol{R} = \frac{1}{\sqrt{5}} \begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$. Now $\boldsymbol{R}^{-1} = \frac{1}{3\sqrt{5}} \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$
lyche-numerical-linear-algebra- F01732	132	0.889	$\boldsymbol{Q}=\boldsymbol{A} \boldsymbol{R}^{-1}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc}2 & 1 \\ -1 & 2\end{array}\right]$	$\boldsymbol{Q} = \boldsymbol{A} \boldsymbol{R}^{-1} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$	so $\boldsymbol{Q} = \boldsymbol{A} \boldsymbol{R}^{-1} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$. Since $\boldsymbol{A}$ is square $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$ is both the QR
lyche-numerical-linear-algebra- F01733	133	0.992	$\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$	$\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ be the result of applying Gram Schmidt to the columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ of $\boldsymbol{A}$,
lyche-numerical-linear-algebra- F01734	133	1.000	$\mathcal{R}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A})$	has orthonormal columns, since $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\}$ is an orthonormal basis for $\mathcal{R}(\boldsymbol{A})$ by
lyche-numerical-linear-algebra- F01735	133	1.000	$\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]=\left[\boldsymbol{a}_1, \boldsymbol{a}_2\right]$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = [\boldsymbol{a}_1, \boldsymbol{a}_2]$	and factorization of the matrix $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = [\boldsymbol{a}_1, \boldsymbol{a}_2]$ using Gram-Schmidt.
lyche-numerical-linear-algebra- F01736	133	0.999	$\boldsymbol{v}_1=\boldsymbol{a}_1$	$\boldsymbol{v}_1 = \boldsymbol{a}_1$	Using (5.17) we find $\boldsymbol{v}_1 = \boldsymbol{a}_1$ and $\boldsymbol{v}_2 = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1 = \frac{3}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Thus $\boldsymbol{Q} = [\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-algebra- F01737	133	0.999	$\boldsymbol{v}_2=\boldsymbol{a}_2-\frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1=\frac{3}{5}\left[\begin{array}{c}1 \\ 2\end{array}\right]$	$\boldsymbol{v}_2 = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1 = \frac{3}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$	Using (5.17) we find $\boldsymbol{v}_1 = \boldsymbol{a}_1$ and $\boldsymbol{v}_2 = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1 = \frac{3}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Thus $\boldsymbol{Q} = [\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-algebra- F01738	133	0.999	$\boldsymbol{Q}=\left[\boldsymbol{q}_1, \boldsymbol{q}_2\right]$	$\boldsymbol{Q} = [\boldsymbol{q}_1, \boldsymbol{q}_2]$	Using (5.17) we find $\boldsymbol{v}_1 = \boldsymbol{a}_1$ and $\boldsymbol{v}_2 = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1 = \frac{3}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Thus $\boldsymbol{Q} = [\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-algebra- F01739	133	1.000	$\boldsymbol{q}_1=\frac{1}{\sqrt{5}}\left[\begin{array}{c}2 \\ -1\end{array}\right]$	$\boldsymbol{q}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$	where $\boldsymbol{q}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\boldsymbol{q}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. By (5.18) we find
lyche-numerical-linear-algebra- F01740	133	1.000	$\boldsymbol{q}_2=\frac{1}{\sqrt{5}}\left[\begin{array}{c}1 \\ 2\end{array}\right]$	$\boldsymbol{q}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$	where $\boldsymbol{q}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\boldsymbol{q}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. By (5.18) we find
lyche-numerical-linear-algebra- F01741	134	1.000	$n=4$	$n = 4$	by a Wilkinson diagram for $n=4$
lyche-numerical-linear-algebra- F01742	134	1.000	$\boldsymbol{P} \in \mathbb{R}^{2,2}$	$\boldsymbol{P} \in \mathbb{R}^{2,2}$	a Given's rotation) is a matrix $\boldsymbol{P} \in \mathbb{R}^{2,2}$ of the form
lyche-numerical-linear-algebra- F01743	134	1.000	$\theta \in[0,2 \pi)$	$\theta \in [0, 2\pi)$	A plane rotation is an orthogonal matrix and there is a unique angle $\theta \in [0, 2\pi)$
lyche-numerical-linear-algebra- F01744	134	1.000	$c=\cos \theta$	$c = \cos \theta$	such that $c = \cos \theta$ and $s = \sin \theta$. Moreover, the identity matrix is a plane rotation
lyche-numerical-linear-algebra- F01745	134	1.000	$s=\sin \theta$	$s = \sin \theta$	such that $c = \cos \theta$ and $s = \sin \theta$. Moreover, the identity matrix is a plane rotation
lyche-numerical-linear-algebra- F01746	134	1.000	$\theta=0$	$\theta = 0$	corresponding to $\theta = 0$. A vector $\boldsymbol{x}$ in the plane is rotated an angle θ clockwise by
lyche-numerical-linear-algebra- F01747	134	1.000	$\boldsymbol{P}=\boldsymbol{R}$	$\boldsymbol{P} = \boldsymbol{R}$	$\boldsymbol{P} = \boldsymbol{R}$. See Exercise 5.16 and Fig. 5.4.
lyche-numerical-linear-algebra- F01748	135	1.000	$\boldsymbol{P}=\boldsymbol{I}$	$\boldsymbol{P} = \boldsymbol{I}$	and we have introduced a zero in $\boldsymbol{x}$. We can take $\boldsymbol{P} = \boldsymbol{I}$ when $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra- F01749	135	1.000	$1 \leq i < j \leq n$	$1 \leq i < j \leq n$	For an n -vector $\boldsymbol{x} \in \mathbb{R}^n$ and $1 \leq i < j \leq n$ we define a rotation in the i, j-plane
lyche-numerical-linear-algebra- F01750	135	0.998	$\boldsymbol{P}_{i j}=\left(p_{k l}\right) \in \mathbb{R}^{n \times n}$	$\boldsymbol{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n \times n}$	as a matrix $\boldsymbol{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n \times n}$ by $p_{kl} = \delta_{kl}$ except for positions ii, jj, ij, ji ,
lyche-numerical-linear-algebra- F01751	135	0.998	$p_{k l}=\delta_{k l}$	$p_{kl} = \delta_{kl}$	as a matrix $\boldsymbol{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n \times n}$ by $p_{kl} = \delta_{kl}$ except for positions ii, jj, ij, ji ,
lyche-numerical-linear-algebra- F01752	135	0.998	$i i, j j, i j, j i$	ii, jj, ij, ji	as a matrix $\boldsymbol{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n \times n}$ by $p_{kl} = \delta_{kl}$ except for positions ii, jj, ij, ji ,
lyche-numerical-linear-algebra- F01753	135	1.000	$\boldsymbol{B}=\boldsymbol{P}_{i j} \boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{P}_{ij} \boldsymbol{A}$	column i and j . In particular, if $\boldsymbol{B} = \boldsymbol{P}_{ij} \boldsymbol{A}$ and $\boldsymbol{C} = \boldsymbol{A} \boldsymbol{P}_{ij}$ then $\boldsymbol{B}(k, :)=\boldsymbol{A}(k, :)$,

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lyche-numerical-linear-algebra- F01754	135	■ 1.000	$\boldsymbol{C}=\boldsymbol{A} \boldsymbol{P}_{ij}$	$\boldsymbol{C} = \boldsymbol{A}\boldsymbol{P}_{ij}$	column i and j . In particular, if $\boldsymbol{B} = \boldsymbol{P}_{ij}\boldsymbol{A}$ and $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{P}_{ij}$ then $\boldsymbol{B}(k, :) = \boldsymbol{A}(k, :)$,
lyche-numerical-linear-algebra- F01755	135	■ 1.000	$\boldsymbol{B}(k, :)=\boldsymbol{A}(k, :)$	$\boldsymbol{B}(k, :) = \boldsymbol{A}(k, :)$	column i and j . In particular, if $\boldsymbol{B} = \boldsymbol{P}_{ij}\boldsymbol{A}$ and $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{P}_{ij}$ then $\boldsymbol{B}(k, :) = \boldsymbol{A}(k, :)$,
lyche-numerical-linear-algebra- F01756	135	■ 1.000	$\boldsymbol{C}(:, k)=\boldsymbol{A}(:, k)$	$\boldsymbol{C}(:, k) = \boldsymbol{A}(:, k)$	$\boldsymbol{C}(:, k) = \boldsymbol{A}(:, k)$ for all $k \neq i, j$ and
lyche-numerical-linear-algebra- F01757	135	■ 1.000	$k \neq i, j$	$k \neq i, j$	$\boldsymbol{C}(:, k) = \boldsymbol{A}(:, k)$ for all $k \neq i, j$ and
lyche-numerical-linear-algebra- F01758	135	■ 0.999	$2n^3$	$2n^3$	number of arithmetic operations is asymptotically $2n^3$, corresponding to the work
lyche-numerical-linear-algebra- F01759	135	■ 1.000	$\boldsymbol{P}_{i, i+1}$	$\boldsymbol{P}_{i, i+1}$	using rotations $\boldsymbol{P}_{i, i+1}$ for $i = 1, \dots, n-1$. For $n = 4$ the process can be illustrated
lyche-numerical-linear-algebra- F01760	136	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}$	dent columns. Show that $\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}$ defines an inner product on $\mathbb{C}^n$.
lyche-numerical-linear-algebra- F01761	136	■ 0.998	$[0, \pi / 2]$	$[0, \pi / 2]$	plex case there is a unique angle θ in $[0, \pi / 2]$ such that
lyche-numerical-linear-algebra- F01762	136	■ 1.000	$\boldsymbol{x}=\boldsymbol{e}_1$	$\boldsymbol{x} = \boldsymbol{e}_1$	Exercise 5.4 (What Does Algorithm Housegen Do When $\boldsymbol{x} = \boldsymbol{e}_1$?) Determine
lyche-numerical-linear-algebra- F01763	136	■ 1.000	$\ \boldsymbol{x}\ _2=\ \boldsymbol{y}\ _2$	$\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2$	$\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2$ and $\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$ then it follows from Example 5.1 that
lyche-numerical-linear-algebra- F01764	136	■ 0.998	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}\right) \boldsymbol{x}=\boldsymbol{y}$	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}\right) \boldsymbol{x} = \boldsymbol{y}$	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}\right) \boldsymbol{x} = \boldsymbol{y}$. Use this to construct a Householder transformation $\boldsymbol{H}$ such
lyche-numerical-linear-algebra- F01765	136	■ 1.000	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{y}$	that $\boldsymbol{H} \boldsymbol{x} = \boldsymbol{y}$ in the following cases.
lyche-numerical-linear-algebra- F01766	136	■ 0.995	$\boldsymbol{x}=\left[\begin{array}{l} 3 \\ 4 \end{array}\right], \quad \boldsymbol{y}=\left[\begin{array}{l} 5 \\ 0 \end{array}\right]$	$\boldsymbol{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$	a) $\boldsymbol{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$.
lyche-numerical-linear-algebra- F01767	136	■ 1.000	$\boldsymbol{x}=\left[\begin{array}{l} 2 \\ 2 \\ 1 \end{array}\right], \quad \boldsymbol{y}=\left[\begin{array}{l} 0 \\ 3 \\ 0 \end{array}\right]$	$\boldsymbol{x} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$	b) $\boldsymbol{x} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$.
lyche-numerical-linear-algebra- F01768	137	■ 1.000	$\boldsymbol{x}=[\cos \phi, \sin \phi]^T$	$\boldsymbol{x} = [\cos \phi, \sin \phi]^T$	Find $\boldsymbol{H} \boldsymbol{x}$ if $\boldsymbol{x} = [\cos \phi, \sin \phi]^T$.
lyche-numerical-linear-algebra- F01769	137	■ 0.997	$\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq \mathbf{0}$	$\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$	a) Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ with $\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2$ and $\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$. Show that
lyche-numerical-linear-algebra- F01770	137	■ 1.000	$\boldsymbol{B} \in \mathbb{R}^{4,4}$	$\boldsymbol{B} \in \mathbb{R}^{4,4}$	b) Let $\boldsymbol{B} \in \mathbb{R}^{4,4}$ be given by
lyche-numerical-linear-algebra- F01771	137	■ 1.000	$0<\epsilon<1$	$0 < \epsilon < 1$	where $0 < \epsilon < 1$. Compute a Householder transformation $\boldsymbol{H}$ and a matrix $\boldsymbol{B}_1$
lyche-numerical-linear-algebra- F01772	137	■ 1.000	$\boldsymbol{B}_1$	$\boldsymbol{B}_1$	where $0 < \epsilon < 1$. Compute a Householder transformation $\boldsymbol{H}$ and a matrix $\boldsymbol{B}_1$
lyche-numerical-linear-algebra- F01773	137	■ 0.857	$\boldsymbol{B}_1:=\boldsymbol{H} \boldsymbol{B} \boldsymbol{H}$	$\boldsymbol{B}_1 := \boldsymbol{H} \boldsymbol{B} \boldsymbol{H}$	such that the first column of $\boldsymbol{B}_1 := \boldsymbol{H} \boldsymbol{B} \boldsymbol{H}$ has a zero in the last two positions.
lyche-numerical-linear-algebra- F01774	138	■ 1.000	$\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3 \times 3}$	$\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3 \times 3}$	Find Householder transformations $\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3 \times 3}$ such that $\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$ is
lyche-numerical-linear-algebra- F01775	138	■ 1.000	$\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$	Find Householder transformations $\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3 \times 3}$ such that $\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$ is
lyche-numerical-linear-algebra- F01776	138	■ 0.999	$\boldsymbol{A}=\left[\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n\right] \in \mathbb{C}^{n \times n}$	$\boldsymbol{A} = [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n] \in \mathbb{C}^{n \times n}$	tion to prove a classical determinant inequality. For any $\boldsymbol{A} = [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n] \in \mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra- F01777	138	■ 1.000	$ \operatorname{det}(\boldsymbol{Q}) =1$	$ \det(\boldsymbol{Q}) = 1$	a) Show that if $\boldsymbol{Q}$ is unitary then $ \det(\boldsymbol{Q}) = 1$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F01778	138	1.000	$\boldsymbol{R}=\left[\boldsymbol{r}_{\{1\}}, \ldots, \boldsymbol{r}_{\{n\}}\right]$	$\boldsymbol{R}=\left[\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n\right]$	b) Let $\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$ be a QR factorization of $\boldsymbol{A}$ and let $\boldsymbol{R}=\left[\boldsymbol{r}_1, \ldots, \boldsymbol{r}_n\right]$. Show that
lyche-numerical-linear-algebra- F01779	138	0.986	$\left(\boldsymbol{A}^* \boldsymbol{A}\right)_{j j}=\left\ \boldsymbol{a}_{\{j\}}\right\ _2^2=\left(\boldsymbol{R}^* \boldsymbol{R}\right)_{j j}=\left\ \boldsymbol{r}_{\{j\}}\right\ _2^2$	$\left(\boldsymbol{A}^* \boldsymbol{A}\right)_{j j}=\left\ \boldsymbol{a}_j\right\ _2^2=\left(\boldsymbol{R}^* \boldsymbol{R}\right)_{j j}=\left\ \boldsymbol{r}_j\right\ _2^2$	$\left(\boldsymbol{A}^* \boldsymbol{A}\right)_{j j}=\left\ \boldsymbol{a}_j\right\ _2^2=\left(\boldsymbol{R}^* \boldsymbol{R}\right)_{j j}=\left\ \boldsymbol{r}_j\right\ _2^2$.
lyche-numerical-linear-algebra- F01780	138	1.000	$\left \operatorname{det}(\boldsymbol{A})\right =\prod_{j=1}^n\left \boldsymbol{r}_{\{j\}}\right \leq \prod_{j=1}^n\left\ \boldsymbol{a}_{\{j\}}\right\ _2$	$ \det(\boldsymbol{A}) =\prod_{j=1}^n\left r_{j j}\right \leq \prod_{j=1}^n\left\ \boldsymbol{a}_j\right\ _2$	c) Show that $ \det(\boldsymbol{A}) =\prod_{i=1}^n\left r_{i i}\right \leq \prod_{j=1}^n\left\ \boldsymbol{a}_j\right\ _2$.
lyche-numerical-linear-algebra- F01781	138	0.949	$\boldsymbol{B}=\boldsymbol{L}^T \boldsymbol{L}$	$\boldsymbol{B}=\boldsymbol{L}^T \boldsymbol{L}$	form $\boldsymbol{B}=\boldsymbol{L}^T \boldsymbol{L}$, where $\boldsymbol{L}$ is lower triangular with positive diagonal elements (you
lyche-numerical-linear-algebra- F01782	138	0.995	$\boldsymbol{B}=\boldsymbol{L} \boldsymbol{L}^T$	$\boldsymbol{B}=\boldsymbol{L} \boldsymbol{L}^T$	$\boldsymbol{B}=\boldsymbol{L} \boldsymbol{L}^T$.
lyche-numerical-linear-algebra- F01783	138	1.000	$l_{\{i, j\}}$	$l_{i, j}$	a) Suppose $\boldsymbol{B}=\boldsymbol{L}^T \boldsymbol{L}$. Write down the equations to determine the elements $l_{i, j}$ of
lyche-numerical-linear-algebra- F01784	138	1.000	$i=n, n-1, \ldots, 1$	$i=n, n-1, \ldots, 1$	$\boldsymbol{L}$, in the order $i=n, n-1, \ldots, 1$ and $j=i, 1, 2, \ldots, i-1$.
lyche-numerical-linear-algebra- F01785	138	1.000	$j=i, 1, 2, \ldots, i-1$	$j=i, 1, 2, \ldots, i-1$	$\boldsymbol{L}$, in the order $i=n, n-1, \ldots, 1$ and $j=i, 1, 2, \ldots, i-1$.
lyche-numerical-linear-algebra- F01786	138	1.000	$\boldsymbol{L}^T \boldsymbol{L}$	$\boldsymbol{L}^T \boldsymbol{L}$	b) Explain (without making a detailed algorithm) how the $\boldsymbol{L}^T \boldsymbol{L}$ factorization can be
lyche-numerical-linear-algebra- F01787	138	1.000	$\boldsymbol{B} \boldsymbol{x}=\boldsymbol{c}$	$\boldsymbol{B} \boldsymbol{x}=\boldsymbol{c}$	used to solve the linear system $\boldsymbol{B} \boldsymbol{x}=\boldsymbol{c}$. Compute $\ \boldsymbol{L}\ _F$. Is the algorithm stable?
lyche-numerical-linear-algebra- F01788	138	1.000	$\ \boldsymbol{L}\ _F$	$\ \boldsymbol{L}\ _F$	used to solve the linear system $\boldsymbol{B} \boldsymbol{x}=\boldsymbol{c}$. Compute $\ \boldsymbol{L}\ _F$. Is the algorithm stable?
lyche-numerical-linear-algebra- F01789	138	1.000	$\boldsymbol{Q} \boldsymbol{L}$	$\boldsymbol{Q} \boldsymbol{L}$	$\boldsymbol{Q} \boldsymbol{L}$, where $\boldsymbol{Q} \in \mathbb{R}^{n \times n}$ is orthogonal and $\boldsymbol{L} \in \mathbb{R}^{n \times n}$ is lower triangular with
lyche-numerical-linear-algebra- F01790	138	1.000	$\boldsymbol{Q} \in \mathbb{R}^{n \times n}$	$\boldsymbol{Q} \in \mathbb{R}^{n \times n}$	$\boldsymbol{Q} \boldsymbol{L}$, where $\boldsymbol{Q} \in \mathbb{R}^{n \times n}$ is orthogonal and $\boldsymbol{L} \in \mathbb{R}^{n \times n}$ is lower triangular with
lyche-numerical-linear-algebra- F01791	138	1.000	$\boldsymbol{a}:=\left[\boldsymbol{a}_{\{1\}}, \ldots, \boldsymbol{a}_{\{n\}}\right]^T \in \mathbb{R}^n$	$\boldsymbol{a}:=\left[a_1, \ldots, a_n\right]^T \in \mathbb{R}^n$	a) Given vectors $\boldsymbol{a}:=\left[a_1, \ldots, a_n\right]^T \in \mathbb{R}^n$ and $\boldsymbol{e}_n:=\left[0, \ldots, 0, 1\right]^T$. Find $\boldsymbol{v} \in \mathbb{R}^n$
lyche-numerical-linear-algebra- F01792	138	1.000	$\boldsymbol{e}_{\{n\}}:=\left[0, \ldots, 0, 1\right]^T$	$\boldsymbol{e}_n:=\left[0, \ldots, 0, 1\right]^T$	a) Given vectors $\boldsymbol{a}:=\left[a_1, \ldots, a_n\right]^T \in \mathbb{R}^n$ and $\boldsymbol{e}_n:=\left[0, \ldots, 0, 1\right]^T$. Find $\boldsymbol{v} \in \mathbb{R}^n$
lyche-numerical-linear-algebra- F01793	138	0.724	$\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$	$\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$	such that the Householder transformation $\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$ satisfies $\boldsymbol{H} \boldsymbol{a}=-s \boldsymbol{e}_n$,
lyche-numerical-linear-algebra- F01794	138	0.724	$\boldsymbol{H} \boldsymbol{a}=-s \boldsymbol{e}_{\{n\}}$	$\boldsymbol{H} \boldsymbol{a}=-s \boldsymbol{e}_n$	such that the Householder transformation $\boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$ satisfies $\boldsymbol{H} \boldsymbol{a}=-s \boldsymbol{e}_n$,
lyche-numerical-linear-algebra- F01795	138	1.000	$ s =\left\ \boldsymbol{a}\right\ _2$	$ s =\left\ \boldsymbol{a}\right\ _2$	where $ s =\left\ \boldsymbol{a}\right\ _2$. How should we choose the sign of s ?
lyche-numerical-linear-algebra- F01796	138	1.000	s	s	where $ s =\left\ \boldsymbol{a}\right\ _2$. How should we choose the sign of s ?
lyche-numerical-linear-algebra- F01797	138	0.630	$\Longleftrightarrow \boldsymbol{R}$	$\Longleftrightarrow \boldsymbol{R}$	¹ Show that we have equality $\Longleftrightarrow \boldsymbol{R}$ is diagonal $\Longleftrightarrow \boldsymbol{A}^* \boldsymbol{A}$ is diagonal.
lyche-numerical-linear-algebra- F01798	138	0.630	$\Longleftrightarrow \boldsymbol{A}^* \boldsymbol{A}$	$\Longleftrightarrow \boldsymbol{A}^* \boldsymbol{A}$	¹ Show that we have equality $\Longleftrightarrow \boldsymbol{R}$ is diagonal $\Longleftrightarrow \boldsymbol{A}^* \boldsymbol{A}$ is diagonal.
lyche-numerical-linear-algebra- F01799	139	1.000	$1 \leq r \leq n, \boldsymbol{v}_{\{r\}} \in \mathbb{R}^r, \boldsymbol{v}_{\{r\}} \neq \mathbf{0}$	$1 \leq r \leq n, \boldsymbol{v}_r \in \mathbb{R}^r, \boldsymbol{v}_r \neq \mathbf{0}$	b) Let $1 \leq r \leq n, \boldsymbol{v}_r \in \mathbb{R}^r, \boldsymbol{v}_r \neq \mathbf{0}$, and
lyche-numerical-linear-algebra- F01800	139	1.000	$\boldsymbol{H}:=\left[\begin{array}{cc} \boldsymbol{V}_{\{r\}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r} \end{array}\right]$	$\boldsymbol{H}:=\left[\begin{array}{cc} \boldsymbol{V}_r & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r} \end{array}\right]$	Show that $\boldsymbol{H}:=\left[\begin{array}{cc} \boldsymbol{V}_r & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r} \end{array}\right]$ is a Householder transformation. Show also that if
lyche-numerical-linear-algebra- F01801	139	1.000	$\boldsymbol{a}_{\{i, j\}}=0$	$\boldsymbol{a}_{i, j}=0$	$\boldsymbol{a}_{i, j}=0$ for $i=1, \ldots, r$ and $j=r+1, \ldots, n$ then the last r columns of $\boldsymbol{A}$ and
lyche-numerical-linear-algebra- F01802	139	1.000	$i=1, \ldots, r$	$i=1, \ldots, r$	$\boldsymbol{a}_{i, j}=0$ for $i=1, \ldots, r$ and $j=r+1, \ldots, n$ then the last r columns of $\boldsymbol{A}$ and

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lyche-numerical-linear-algebra-F01803	139	1.000	$j=r+1, \ldots, n$	$j = r + 1, \dots, n$	$a_{i,j} = 0$ for $i = 1, \dots, r$ and $i = r + 1, \dots, n$ then the last r columns of A and
lyche-numerical-linear-algebra-F01804	139	1.000	$\text{\boldsymbol{H}} \text{\boldsymbol{A}}$	$\boldsymbol{H} \boldsymbol{A}$	$\boldsymbol{H} \boldsymbol{A}$ are the same.
lyche-numerical-linear-algebra-F01805	139	1.000	$\text{\boldsymbol{H}}_1, \ldots, \text{\boldsymbol{H}}_{n-1}$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_{n-1}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$ can find Householder transformations $\boldsymbol{H}_1, \dots, \boldsymbol{H}_{n-1}$ such that
lyche-numerical-linear-algebra-F01806	139	1.000	$\text{\boldsymbol{H}}_{n-1}, \ldots, \text{\boldsymbol{H}}_1 \text{\boldsymbol{A}}$	$\boldsymbol{H}_{n-1}, \dots, \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_{n-1}, \dots, \boldsymbol{H}_1 \boldsymbol{A}$ is lower triangular. Give a $\boldsymbol{QL}$ factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F01807	139	1.000	$d \leq n-1$	$d \leq n - 1$	$\mathbb{R}^{n \times n}$ be a nonsingular symmetric band matrix with bandwidth $d \leq n - 1$, so that
lyche-numerical-linear-algebra-F01808	139	1.000	$\text{\boldsymbol{B}} := \text{\boldsymbol{A}}^T \text{\boldsymbol{A}}$	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A}$	$a_{ij} = 0$ for all i, j with $ i - j > d$. We define $\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A}$ and let $\boldsymbol{A} = \boldsymbol{QR}$ be the
lyche-numerical-linear-algebra-F01809	139	1.000	$\leq 2d$	$\leq 2d$	b) Show that $\boldsymbol{B}$ has bandwidth $\leq 2d$.
lyche-numerical-linear-algebra-F01810	139	0.426	$\text{\mathrm{B}} =$	$\text{B} =$	c) Write a MATLAB function <code>B=ata(A,d)</code> which computes $\boldsymbol{B}$. You shall
lyche-numerical-linear-algebra-F01811	139	0.426	$(\text{\mathrm{A}}, \text{\mathrm{d}})$	(A, d)	c) Write a MATLAB function <code>B=ata(A,d)</code> which computes $\boldsymbol{B}$. You shall
lyche-numerical-linear-algebra-F01812	139	1.000	$\mathcal{O}(cn^2)$	$\mathcal{O}(cn^2)$	exploit the symmetry and the function should only use $\mathcal{O}(cn^2)$ flops, where c
lyche-numerical-linear-algebra-F01813	139	1.000	$\text{\boldsymbol{A}}^T \text{\boldsymbol{A}} = \text{\boldsymbol{R}}^T \text{\boldsymbol{R}}$	$\boldsymbol{A}^T \boldsymbol{A} = \boldsymbol{R}^T \boldsymbol{R}$	e) Show that $\boldsymbol{A}^T \boldsymbol{A} = \boldsymbol{R}^T \boldsymbol{R}$.
lyche-numerical-linear-algebra-F01814	139	1.000	$2d$	$2d$	f) Explain why $\boldsymbol{R}$ has upper bandwidth $2d$.
lyche-numerical-linear-algebra-F01815	140	0.928	$\text{\boldsymbol{x}} = \begin{bmatrix} r \cos \alpha \\ r \sin \alpha \end{bmatrix}$	$\boldsymbol{x} = \begin{bmatrix} r \cos \alpha \\ r \sin \alpha \end{bmatrix}$	Exercise 5.16 (Plane Rotation) Show that if $\boldsymbol{x} = \begin{bmatrix} r \cos \alpha \\ r \sin \alpha \end{bmatrix}$ then $\boldsymbol{Px} =$
lyche-numerical-linear-algebra-F01816	140	0.928	$\text{\boldsymbol{P}} \text{\boldsymbol{x}} =$	$\boldsymbol{Px} =$	Exercise 5.16 (Plane Rotation) Show that if $\boldsymbol{x} = \begin{bmatrix} r \cos \alpha \\ r \sin \alpha \end{bmatrix}$ then $\boldsymbol{Px} =$
lyche-numerical-linear-algebra-F01817	140	0.711	$\begin{bmatrix} r \cos(\alpha - \theta) \\ r \sin(\alpha - \theta) \end{bmatrix}$	$\begin{bmatrix} r \cos(\alpha - \theta) \\ r \sin(\alpha - \theta) \end{bmatrix}$	$\begin{bmatrix} r \cos(\alpha - \theta) \\ r \sin(\alpha - \theta) \end{bmatrix}$.
lyche-numerical-linear-algebra-F01818	140	0.988	$\text{\boldsymbol{H}} \in \mathbb{R}^{4,4}$	$\boldsymbol{H} \in \mathbb{R}^{4,4}$	Exercise 5.17 (Updating the QR Decomposition) Let $\boldsymbol{H} \in \mathbb{R}^{4,4}$ be upper
lyche-numerical-linear-algebra-F01819	140	1.000	$\text{\boldsymbol{G}}_1, \text{\boldsymbol{G}}_2, \text{\boldsymbol{G}}_3$	$\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3$	Hessenberg. Find Givens rotation matrices $\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3$ such that
lyche-numerical-linear-algebra-F01820	140	1.000	$\text{\boldsymbol{G}}_k = \text{\boldsymbol{P}}_{i,j}$	$\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$	is upper triangular. (Here each $\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$ for suitable i, j, c og s , and for each k
lyche-numerical-linear-algebra-F01821	140	1.000	i, j, c	i, j, c	is upper triangular. (Here each $\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$ for suitable i, j, c og s , and for each k
lyche-numerical-linear-algebra-F01822	140	0.999	$\text{\boldsymbol{P}}_{k,k+1}$	$\boldsymbol{P}_{k,k+1}$	solves the linear system $\boldsymbol{Ax} = \boldsymbol{b}$ using rotations $\boldsymbol{P}_{k,k+1}$ for $k = 1, \dots, n - 1$. It
lyche-numerical-linear-algebra-F01823	140	0.838	$\text{\boldsymbol{G}} := \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{G} := \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \in \mathbb{R}^{2 \times 2}$	rotation of order 2 has the form $\boldsymbol{G} := \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, where $s^2 + c^2 = 1$.
lyche-numerical-linear-algebra-F01824	140	0.838	$s^2 + c^2 = 1$	$s^2 + c^2 = 1$	rotation of order 2 has the form $\boldsymbol{G} := \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, where $s^2 + c^2 = 1$.
lyche-numerical-linear-algebra-F01825	141	1.000	$\text{\boldsymbol{G}}$	$\boldsymbol{G}$	a) Is $\boldsymbol{G}$ symmetric and unitary?
lyche-numerical-linear-algebra-F01826	141	1.000	$x_1, x_2 \in \mathbb{R}$	$x_1, x_2 \in \mathbb{R}$	b) Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2 + x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra-F01827	141	1.000	$r := \sqrt{x_1^2 + x_2^2}$	$r := \sqrt{x_1^2 + x_2^2}$	b) Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2 + x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,

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lyche-numerical-linear-algebra-F01828	141	1.000	$y_{\{1\}}, y_{\{2\}}$	y_1, y_2	b) Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2 + x_2^2}$. Find G and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra-F01829	141	1.000	$y_{\{1\}}=y_{\{2\}}$	$y_1 = y_2$	b) Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2 + x_2^2}$. Find G and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra-F01830	141	1.000	$\left[\begin{array}{l} y_{\{1\}} \\ y_{\{2\}} \end{array}\right] = G \left[\begin{array}{l} x_{\{1\}} \\ x_{\{2\}} \end{array}\right]$	$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = G \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$	where $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = G \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.
lyche-numerical-linear-algebra-F01831	141	1.000	$i \ j$	ij	rotation in the ij -plane is an $m \times m$ -matrix, denoted $P_{i,j}$, which differs from the
lyche-numerical-linear-algebra-F01832	141	1.000	$\boldsymbol{P}_{\{i, j\}}$	$P_{i,j}$	rotation in the ij -plane is an $m \times m$ -matrix, denoted $P_{i,j}$, which differs from the
lyche-numerical-linear-algebra-F01833	141	0.938	$i \ i, i \ j, j \ i, j \ j$	ii, ij, ji, jj	identity matrix only in the entries ii, ij, ji, jj , which equal
lyche-numerical-linear-algebra-F01834	141	1.000	$\theta \in \mathbb{R}$	$\theta \in \mathbb{R}$	a) For $\theta \in \mathbb{R}$, let P be a Givens rotation of the form
lyche-numerical-linear-algebra-F01835	141	1.000	$\theta \in (-\pi/2, \pi/2]$	$\theta \in (-\pi/2, \pi/2]$	and let x be a fixed vector in $\mathbb{R}^2$. Show that there exists a unique $\theta \in (-\pi/2, \pi/2]$
lyche-numerical-linear-algebra-F01836	141	0.910	$\boldsymbol{P} \boldsymbol{x} = \pm \ \boldsymbol{x}\ _2 \boldsymbol{e}_1$	$Px = \pm \ x\ _2 e_1$	so that $Px = \pm \ x\ _2 e_1$, where $e_1 = (1, 0)^T$.
lyche-numerical-linear-algebra-F01837	141	0.910	$\boldsymbol{e}_1 = (1, 0)^T$	$e_1 = (1, 0)^T$	so that $Px = \pm \ x\ _2 e_1$, where $e_1 = (1, 0)^T$.
lyche-numerical-linear-algebra-F01838	141	1.000	$\boldsymbol{w} \in \mathbb{R}^m$	$w \in \mathbb{R}^m$	b) Show that, for any vector $w \in \mathbb{R}^m$, one can find rotations in the 12-plane, 23-
lyche-numerical-linear-algebra-F01839	141	0.481	$(m-1) \ m$	$(m-1)m$	plane, ..., $(m-1)m$ -plane, so that
lyche-numerical-linear-algebra-F01840	141	0.989	$\alpha = \pm \ \boldsymbol{w}\ _2$	$\alpha = \pm \ w\ _2$	where $\alpha = \pm \ w\ _2$.
lyche-numerical-linear-algebra-F01841	142	1.000	$\boldsymbol{A}_-$	A_-	d) Let again A be an $m \times n$ -matrix with $m \geq n$, and let A_- be the matrix obtained
lyche-numerical-linear-algebra-F01842	142	0.916	$\boldsymbol{A}_-^2$	A_-^2	A_- , when we already have a QR decomposition $A = QR$ of A . ²
lyche-numerical-linear-algebra-F01843	142	0.936	z	z	Assume also that z is a given column vector of length n .
lyche-numerical-linear-algebra-F01844	142	0.999	$\boldsymbol{A} + \boldsymbol{z} \boldsymbol{z}^*$	$A + zz^*$	a) Explain why $A + zz^*$ has a unique Cholesky factorization.
lyche-numerical-linear-algebra-F01845	142	1.000	$\boldsymbol{R}^* \boldsymbol{R}$	$R^* R$	factorization $R^* R$.
lyche-numerical-linear-algebra-F01846	142	0.987	$P_{\{i_{\{1\}}, n+1\}}, P_{\{i_{\{2\}}, n+1\}}, \ldots, P_{\{i_{\{n\}}, n+1\}}$	$P_{i_1, n+1}, P_{i_2, n+1}, \dots, P_{i_n, n+1}$	c) Explain how one can find plane rotations $P_{i_1, n+1}, P_{i_2, n+1}, \dots, P_{i_n, n+1}$ so that
lyche-numerical-linear-algebra-F01847	142	1.000	$\boldsymbol{R}'$	R'	with R' upper triangular, and explain how to obtain a QR decomposition of
lyche-numerical-linear-algebra-F01848	142	0.996	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array}\right]$	$\begin{bmatrix} L^* \\ z^* \end{bmatrix}$	$\begin{bmatrix} L^* \\ z^* \end{bmatrix}$ from this. In particular you should write down the numbers $i_1, \dots, i_n$.
lyche-numerical-linear-algebra-F01849	142	0.996	$i_{\{1\}}, \ldots, i_{\{n\}}$	$i_1, \dots, i_n$	$\begin{bmatrix} L^* \\ z^* \end{bmatrix}$ from this. In particular you should write down the numbers $i_1, \dots, i_n$.
lyche-numerical-linear-algebra-F01850	142	1.000	$\boldsymbol{Q}^T \boldsymbol{A}_-$	$Q^T A_-$	² Consider the matrix $Q^T A_-$.
lyche-numerical-linear-algebra-F01851	146	0.999	$(\lambda_k, \boldsymbol{x}_k)$, $k =$	$(\lambda_k, x_k), k =$	Proof The proof is by contradiction. Suppose A has eigenpairs (λ_k, x_k) , $k =$
lyche-numerical-linear-algebra-F01852	146	0.999	$\{\boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{m\}}\}$	$\{x_1, \dots, x_m\}$	that $\{x_1, \dots, x_m\}$ is linearly dependent. Thus $\sum_{j=1}^m c_j x_j = \mathbf{0}$, where at least one

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01853	146	■ 0.999	$\sum_{j=1}^m c_{\{j\}} \boldsymbol{x}_{\{j\}} = \mathbf{0}$	$\sum_{j=1}^m c_j \mathbf{x}_j = \mathbf{0}$	that $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ is linearly dependent. Thus $\sum_{j=1}^m c_j \mathbf{x}_j = \mathbf{0}$, where at least one
lyche-numerical-linear-algebra-F01854	146	■ 1.000	$c_{\{j\}}$	c_j	c_j is nonzero. We must have $m \geq 2$ since eigenvectors are nonzero. We find
lyche-numerical-linear-algebra-F01855	146	■ 1.000	$m \geq 2$	$m \geq 2$	c_j is nonzero. We must have $m \geq 2$ since eigenvectors are nonzero. We find
lyche-numerical-linear-algebra-F01856	146	■ 1.000	$\sum_{j=1}^m c_{\{j\}} \lambda_{\{m\}} \boldsymbol{x}_{\{j\}} = \mathbf{0}$	$\sum_{j=1}^m c_j \lambda_m \mathbf{x}_j = \mathbf{0}$	From the last relation we subtract $\sum_{j=1}^m c_j \lambda_m \mathbf{x}_j = \mathbf{0}$ and find $\sum_{j=1}^{m-1} c_j (\lambda_j -$
lyche-numerical-linear-algebra-F01857	146	■ 1.000	$\sum_{j=1}^{m-1} c_{\{j\}} \left(\lambda_{\{j\}} - \right.$	$\sum_{j=1}^{m-1} c_j (\lambda_j -$	From the last relation we subtract $\sum_{j=1}^m c_j \lambda_m \mathbf{x}_j = \mathbf{0}$ and find $\sum_{j=1}^{m-1} c_j (\lambda_j -$
lyche-numerical-linear-algebra-F01858	146	■ 1.000	$\left. \lambda_{\{m\}} \right) \boldsymbol{x}_{\{j\}} = \mathbf{0}$	$\lambda_m) \mathbf{x}_j = \mathbf{0}$	$\lambda_m) \mathbf{x}_j = \mathbf{0}$. But since $\lambda_j - \lambda_m \neq 0$ for $j = 1, \dots, m-1$ and at least one
lyche-numerical-linear-algebra-F01859	146	■ 1.000	$\lambda_{\{j\}} - \lambda_{\{m\}} \neq 0$	$\lambda_j - \lambda_m \neq 0$	$\lambda_m) \mathbf{x}_j = \mathbf{0}$. But since $\lambda_j - \lambda_m \neq 0$ for $j = 1, \dots, m-1$ and at least one
lyche-numerical-linear-algebra-F01860	146	■ 1.000	$j=1, \dots, m-1$	$j = 1, \dots, m-1$	$\lambda_m) \mathbf{x}_j = \mathbf{0}$. But since $\lambda_j - \lambda_m \neq 0$ for $j = 1, \dots, m-1$ and at least one
lyche-numerical-linear-algebra-F01861	146	■ 1.000	$c_{\{j\}} \neq 0$	$c_j \neq 0$	$c_j \neq 0$ for $j < m$ we see that $\{\mathbf{x}_1, \dots, \mathbf{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra-F01862	146	■ 1.000	$j < m$	$j < m$	$c_j \neq 0$ for $j < m$ we see that $\{\mathbf{x}_1, \dots, \mathbf{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra-F01863	146	■ 1.000	$\left\{ \boldsymbol{x}_{\{1\}}, \dots, \boldsymbol{x}_{\{m-1\}} \right\}$	$\{\mathbf{x}_1, \dots, \mathbf{x}_{m-1}\}$	$c_j \neq 0$ for $j < m$ we see that $\{\mathbf{x}_1, \dots, \mathbf{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra-F01864	146	■ 1.000	$\boldsymbol{I} \boldsymbol{x} = \boldsymbol{x}$	$\boldsymbol{I} \mathbf{x} = \mathbf{x}$	Since $\boldsymbol{I} \mathbf{x} = \mathbf{x}$ and $\lambda_1 = \lambda_2 = 1$ any vector $\mathbf{x} \in \mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra-F01865	146	■ 1.000	$\lambda_{\{1\}} = \lambda_{\{2\}} = 1$	$\lambda_1 = \lambda_2 = 1$	Since $\boldsymbol{I} \mathbf{x} = \mathbf{x}$ and $\lambda_1 = \lambda_2 = 1$ any vector $\mathbf{x} \in \mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra-F01866	146	■ 1.000	$\boldsymbol{x} \in \mathbb{C}^2$	$\mathbf{x} \in \mathbb{C}^2$	Since $\boldsymbol{I} \mathbf{x} = \mathbf{x}$ and $\lambda_1 = \lambda_2 = 1$ any vector $\mathbf{x} \in \mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra-F01867	146	■ 1.000	$\boldsymbol{e}_{\{2\}}$	$\mathbf{e}_2$	particular the two unit vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ are eigenvectors and form an orthonormal
lyche-numerical-linear-algebra-F01868	146	■ 1.000	$\mathbb{C}^2$	$\mathbb{C}^2$	basis for $\mathbb{C}^2$. We conclude that the identity matrix is nondefective. The matrix $\boldsymbol{J}$
lyche-numerical-linear-algebra-F01869	146	■ 1.000	$\boldsymbol{J}$	$\boldsymbol{J}$	basis for $\mathbb{C}^2$. We conclude that the identity matrix is nondefective. The matrix $\boldsymbol{J}$
lyche-numerical-linear-algebra-F01870	146	■ 1.000	$\boldsymbol{J} \boldsymbol{x} = \boldsymbol{x}$	$\boldsymbol{J} \mathbf{x} = \mathbf{x}$	also has the eigenvalue one with multiplicity two, but since $\boldsymbol{J} \mathbf{x} = \mathbf{x}$ if and only if
lyche-numerical-linear-algebra-F01871	146	■ 1.000	$x_{\{2\}} = 0$	$x_2 = 0$	$x_2 = 0$, any eigenvector must be a multiple of $\mathbf{e}_1$. Thus $\boldsymbol{J}$ is defective.
lyche-numerical-linear-algebra-F01872	146	■ 1.000	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$	<i>Example 6.2 (Eigenvector Expansion Example)</i> Eigenpairs of $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ are
lyche-numerical-linear-algebra-F01873	146	■ 1.000	$(1, [1, 1]^T)$	$(1, [1, 1]^T)$	$(1, [1, 1]^T)$ and $(3, [1, -1]^T)$. Any $\mathbf{x} = [x_1, x_2]^T \in \mathbb{C}^2$ has the eigenvector
lyche-numerical-linear-algebra-F01874	146	■ 1.000	$(3, [1, -1]^T)$	$(3, [1, -1]^T)$	$(1, [1, 1]^T)$ and $(3, [1, -1]^T)$. Any $\mathbf{x} = [x_1, x_2]^T \in \mathbb{C}^2$ has the eigenvector
lyche-numerical-linear-algebra-F01875	146	■ 1.000	$\boldsymbol{x} = \left[x_{\{1\}}, x_{\{2\}} \right]^T \in \mathbb{C}^2$	$\mathbf{x} = [x_1, x_2]^T \in \mathbb{C}^2$	$(1, [1, 1]^T)$ and $(3, [1, -1]^T)$. Any $\mathbf{x} = [x_1, x_2]^T \in \mathbb{C}^2$ has the eigenvector
lyche-numerical-linear-algebra-F01876	147	■ 1.000	$\boldsymbol{S} \in \mathbb{C}^{n \times n}$	$\boldsymbol{S} \in \mathbb{C}^{n \times n}$	similar if there is a nonsingular matrix $\boldsymbol{S} \in \mathbb{C}^{n \times n}$ such that $\boldsymbol{B} = \boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$. The
lyche-numerical-linear-algebra-F01877	147	■ 1.000	$\boldsymbol{B} = \boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	$\boldsymbol{B} = \boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	similar if there is a nonsingular matrix $\boldsymbol{S} \in \mathbb{C}^{n \times n}$ such that $\boldsymbol{B} = \boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$. The

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lyche-numerical-linear-algebra-F01878	147	■ 1.000	$\boldsymbol{A} \rightarrow \boldsymbol{B}$	$\boldsymbol{A} \rightarrow \boldsymbol{B}$	transformation $\boldsymbol{A} \rightarrow \boldsymbol{B}$ is called a similarity transformation . The columns of $\boldsymbol{S}$
lyche-numerical-linear-algebra-F01879	147	■ 1.000	$\boldsymbol{s}_1, \boldsymbol{s}_2, \dots, \boldsymbol{s}_n$	$\boldsymbol{s}_1, \boldsymbol{s}_2, \dots, \boldsymbol{s}_n$	are denoted by $\boldsymbol{s}_1, \boldsymbol{s}_2, \dots, \boldsymbol{s}_n$.
lyche-numerical-linear-algebra-F01880	147	■ 1.000	$\operatorname{det}(\boldsymbol{A} \boldsymbol{C}) =$	$\det(\boldsymbol{AC}) =$	teristic polynomial. Indeed, by the product rule for determinants $\det(\boldsymbol{AC}) =$
lyche-numerical-linear-algebra-F01881	147	■ 1.000	$\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{C})$	$\det(\boldsymbol{A}) \det(\boldsymbol{C})$	$\det(\boldsymbol{A}) \det(\boldsymbol{C})$ so that
lyche-numerical-linear-algebra-F01882	147	■ 0.866	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	$\boldsymbol{S}^{-1} \boldsymbol{AS}$	2. $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{S}^{-1} \boldsymbol{AS}$ if and only if $(\lambda, \boldsymbol{Sx})$ is an eigenpair for $\boldsymbol{A}$. In
lyche-numerical-linear-algebra-F01883	147	■ 0.866	$\lambda, \boldsymbol{Sx}$	$\lambda, \boldsymbol{Sx}$	2. $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{S}^{-1} \boldsymbol{AS}$ if and only if $(\lambda, \boldsymbol{Sx})$ is an eigenpair for $\boldsymbol{A}$. In
lyche-numerical-linear-algebra-F01884	147	■ 1.000	$(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}) \boldsymbol{x} = \lambda \boldsymbol{x}$	$(\boldsymbol{S}^{-1} \boldsymbol{AS}) \boldsymbol{x} = \lambda \boldsymbol{x}$	fact $(\boldsymbol{S}^{-1} \boldsymbol{AS}) \boldsymbol{x} = \lambda \boldsymbol{x}$ if and only if $\boldsymbol{A}(\boldsymbol{Sx}) = \lambda(\boldsymbol{Sx})$.
lyche-numerical-linear-algebra-F01885	147	■ 1.000	$\boldsymbol{A}(\boldsymbol{Sx}) = \lambda(\boldsymbol{Sx})$	$\boldsymbol{A}(\boldsymbol{Sx}) = \lambda(\boldsymbol{Sx})$	fact $(\boldsymbol{S}^{-1} \boldsymbol{AS}) \boldsymbol{x} = \lambda \boldsymbol{x}$ if and only if $\boldsymbol{A}(\boldsymbol{Sx}) = \lambda(\boldsymbol{Sx})$.
lyche-numerical-linear-algebra-F01886	147	■ 1.000	$\boldsymbol{S}^{-1} \boldsymbol{AS} = \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$	$\boldsymbol{S}^{-1} \boldsymbol{AS} = \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$	3. If $\boldsymbol{S}^{-1} \boldsymbol{AS} = \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ we can partition $\boldsymbol{AS} = \boldsymbol{SD}$ by
lyche-numerical-linear-algebra-F01887	147	■ 1.000	$\boldsymbol{AS} = \boldsymbol{SD}$	$\boldsymbol{AS} = \boldsymbol{SD}$	3. If $\boldsymbol{S}^{-1} \boldsymbol{AS} = \boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ we can partition $\boldsymbol{AS} = \boldsymbol{SD}$ by
lyche-numerical-linear-algebra-F01888	147	■ 1.000	$[\boldsymbol{A} \boldsymbol{s}_1, \dots, \boldsymbol{A} \boldsymbol{s}_n] = [\lambda_1 \boldsymbol{s}_1, \dots, \lambda_n \boldsymbol{s}_n]$	$[\boldsymbol{As}_1, \dots, \boldsymbol{As}_n] = [\lambda_1 \boldsymbol{s}_1, \dots, \lambda_n \boldsymbol{s}_n]$	columns to obtain $[\boldsymbol{As}_1, \dots, \boldsymbol{As}_n] = [\lambda_1 \boldsymbol{s}_1, \dots, \lambda_n \boldsymbol{s}_n]$ Thus the columns of
lyche-numerical-linear-algebra-F01889	147	■ 1.000	$\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	4. For any square matrices $\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$ the two products $\boldsymbol{AC}$ and $\boldsymbol{CA}$ have the
lyche-numerical-linear-algebra-F01890	147	■ 1.000	$\boldsymbol{AC}$	$\boldsymbol{AC}$	4. For any square matrices $\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$ the two products $\boldsymbol{AC}$ and $\boldsymbol{CA}$ have the
lyche-numerical-linear-algebra-F01891	147	■ 1.000	$\boldsymbol{CA}$	$\boldsymbol{CA}$	4. For any square matrices $\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$ the two products $\boldsymbol{AC}$ and $\boldsymbol{CA}$ have the
lyche-numerical-linear-algebra-F01892	147	■ 1.000	$\boldsymbol{C} \in \mathbb{C}^{n \times m}$	$\boldsymbol{C} \in \mathbb{C}^{n \times m}$	$\mathbb{C}^{m \times n}$ and $\boldsymbol{C} \in \mathbb{C}^{n \times m}$, with say $m > n$, the bigger matrix has $m - n$ extra zero
lyche-numerical-linear-algebra-F01893	147	■ 1.000	$n + m$	$n + m$	To show this define for any $m, n \in \mathbb{N}$ block triangular matrices of order $n + m$ by
lyche-numerical-linear-algebra-F01894	147	■ 0.997	$\boldsymbol{S}^{-1} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A} \\ \mathbf{0} & \boldsymbol{I} \end{bmatrix}$	$\boldsymbol{S}^{-1} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A} \\ \mathbf{0} & \boldsymbol{I} \end{bmatrix}$	The matrix $\boldsymbol{S}$ is nonsingular with $\boldsymbol{S}^{-1} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A} \\ \mathbf{0} & \boldsymbol{I} \end{bmatrix}$. Moreover, $\boldsymbol{ES} = \boldsymbol{SF}$ so
lyche-numerical-linear-algebra-F01895	147	■ 0.997	$\boldsymbol{ES} = \boldsymbol{SF}$	$\boldsymbol{ES} = \boldsymbol{SF}$	The matrix $\boldsymbol{S}$ is nonsingular with $\boldsymbol{S}^{-1} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{A} \\ \mathbf{0} & \boldsymbol{I} \end{bmatrix}$. Moreover, $\boldsymbol{ES} = \boldsymbol{SF}$ so
lyche-numerical-linear-algebra-F01896	147	■ 1.000	$\boldsymbol{F}$	$\boldsymbol{F}$	$\boldsymbol{E}$ and $\boldsymbol{F}$ are similar and therefore have the same characteristic polynomials.
lyche-numerical-linear-algebra-F01897	147	■ 1.000	$\pi_{\boldsymbol{E}}(\lambda) = \lambda^n \pi_{\boldsymbol{AC}}(\lambda) = \pi_{\boldsymbol{F}}(\lambda) = \lambda^m \pi_{\boldsymbol{CA}}(\lambda)$	$\pi_{\boldsymbol{E}}(\lambda) = \lambda^n \pi_{\boldsymbol{AC}}(\lambda) = \pi_{\boldsymbol{F}}(\lambda) = \lambda^m \pi_{\boldsymbol{CA}}(\lambda)$	diagonal blocks. But then $\pi_{\boldsymbol{E}}(\lambda) = \lambda^n \pi_{\boldsymbol{AC}}(\lambda) = \pi_{\boldsymbol{F}}(\lambda) = \lambda^m \pi_{\boldsymbol{CA}}(\lambda)$ This
lyche-numerical-linear-algebra-F01898	148	■ 1.000	$\lambda_1, \dots, \lambda_k$	$\lambda_1, \dots, \lambda_k$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has k distinct eigenvalues $\lambda_1, \dots, \lambda_k$ with multiplicities
lyche-numerical-linear-algebra-F01899	148	■ 1.000	$a_1, \dots, a_k$	$a_1, \dots, a_k$	$a_1, \dots, a_k$ so that
lyche-numerical-linear-algebra-F01900	148	■ 0.890	$a_i = a(\lambda_i) = a_{\boldsymbol{A}}(\lambda_i)$	$a_i = a(\lambda_i) = a_{\boldsymbol{A}}(\lambda_i)$	The positive integer $a_i = a(\lambda_i) = a_{\boldsymbol{A}}(\lambda_i)$ is called the multiplicity , or more

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lyche-numerical-linear-algebra-F01901	148	1.000	$\lambda_{\mathbf{i}}$	λ_i	precisely the algebraic multiplicity of the eigenvalue λ_i . The multiplicity of an
lyche-numerical-linear-algebra-F01902	148	1.000	$a_{\mathbf{i}}$	a_i	eigenvalue is simple (double, triple) if a_i is equal to one (two, three).
lyche-numerical-linear-algebra-F01903	148	1.000	$\lambda \in \sigma(\mathbf{A})$	$\lambda \in \sigma(\mathbf{A})$	To define a second kind of multiplicity we consider for each $\lambda \in \sigma(\mathbf{A})$ the
lyche-numerical-linear-algebra-F01904	148	1.000	$g(\lambda) =$	$g = g(\lambda) =$	Definition 6.2 (Geometric Multiplicity) The geometric multiplicity $g = g(\lambda) =$
lyche-numerical-linear-algebra-F01905	148	1.000	$g_{\mathbf{A}}(\lambda)$	$g_{\mathbf{A}}(\lambda)$	$g_{\mathbf{A}}(\lambda)$ of an eigenvalue λ of $\mathbf{A}$ is the dimension of the nullspace $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$.
lyche-numerical-linear-algebra-F01906	148	1.000	$\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$	$\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$	$g_{\mathbf{A}}(\lambda)$ of an eigenvalue λ of $\mathbf{A}$ is the dimension of the nullspace $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$.
lyche-numerical-linear-algebra-F01907	148	1.000	$\lambda = 1$	$\lambda = 1$	value $\lambda = 1$ with $\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$. Since $\mathbf{I} - \lambda \mathbf{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra-F01908	148	1.000	$\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$	$\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$	value $\lambda = 1$ with $\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$. Since $\mathbf{I} - \lambda \mathbf{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra-F01909	148	1.000	$\mathbf{I} - \lambda \mathbf{I}$	$\mathbf{I} - \lambda \mathbf{I}$	value $\lambda = 1$ with $\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$. Since $\mathbf{I} - \lambda \mathbf{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra-F01910	148	1.000	$a = g = n$	$a = g = n$	the nullspace of $\mathbf{I} - \lambda \mathbf{I}$ is all of n -space and it follows that $a = g = n$. On the other
lyche-numerical-linear-algebra-F01911	148	0.985	$\mathbf{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\mathbf{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	hand we saw in Example 6.1 that the matrix $\mathbf{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has the eigenvalue $\lambda = 1$
lyche-numerical-linear-algebra-F01912	148	1.000	$a = 2$	$a = 2$	with $a = 2$ and any eigenvector is a multiple of $\mathbf{e}_1$. Thus $g = 1$.
lyche-numerical-linear-algebra-F01913	148	1.000	$g = 1$	$g = 1$	with $a = 2$ and any eigenvector is a multiple of $\mathbf{e}_1$. Thus $g = 1$.
lyche-numerical-linear-algebra-F01914	148	0.991	$\dim \mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I}) = k$	$\dim \mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I}) = k$	$\dim \mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I}) = k$, and $\dim \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}) = \ell$. We need to show that
lyche-numerical-linear-algebra-F01915	148	0.991	$\dim \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}) = \ell$	$\dim \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}) = \ell$	$\dim \mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I}) = k$, and $\dim \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}) = \ell$. We need to show that
lyche-numerical-linear-algebra-F01916	148	1.000	$k = \ell$	$k = \ell$	$k = \ell$. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis for $\mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$. Then $\mathbf{S}^{-1} \mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{v}_i$
lyche-numerical-linear-algebra-F01917	148	1.000	$\mathbf{v}_1, \dots, \mathbf{v}_k$	$\mathbf{v}_1, \dots, \mathbf{v}_k$	$k = \ell$. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis for $\mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$. Then $\mathbf{S}^{-1} \mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{v}_i$
lyche-numerical-linear-algebra-F01918	148	1.000	$\mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$	$\mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$	$k = \ell$. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis for $\mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$. Then $\mathbf{S}^{-1} \mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{v}_i$
lyche-numerical-linear-algebra-F01919	148	1.000	$\mathbf{S}^{-1} \mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{v}_i$	$\mathbf{S}^{-1} \mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{v}_i$	$k = \ell$. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis for $\mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$. Then $\mathbf{S}^{-1} \mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{v}_i$
lyche-numerical-linear-algebra-F01920	148	1.000	$\mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{S} \mathbf{v}_i, i = 1, \dots, k$	$\mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{S} \mathbf{v}_i, i = 1, \dots, k$	or $\mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{S} \mathbf{v}_i, i = 1, \dots, k$. But then $\{\mathbf{S} \mathbf{v}_1, \dots, \mathbf{S} \mathbf{v}_k\} \subset \mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$, which
lyche-numerical-linear-algebra-F01921	148	1.000	$\{\mathbf{S} \mathbf{v}_1, \dots, \mathbf{S} \mathbf{v}_k\} \subset \mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$	$\{\mathbf{S} \mathbf{v}_1, \dots, \mathbf{S} \mathbf{v}_k\} \subset \mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$	or $\mathbf{A} \mathbf{S} \mathbf{v}_i = \lambda \mathbf{S} \mathbf{v}_i, i = 1, \dots, k$. But then $\{\mathbf{S} \mathbf{v}_1, \dots, \mathbf{S} \mathbf{v}_k\} \subset \mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$, which
lyche-numerical-linear-algebra-F01922	148	1.000	$k \leq \ell$	$k \leq \ell$	implies that $k \leq \ell$. Similarly, if $\mathbf{w}_1, \dots, \mathbf{w}_{\ell}$ is a basis for $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$ then
lyche-numerical-linear-algebra-F01923	148	1.000	$\mathbf{w}_1, \dots, \mathbf{w}_{\ell}$	$\mathbf{w}_1, \dots, \mathbf{w}_{\ell}$	implies that $k \leq \ell$. Similarly, if $\mathbf{w}_1, \dots, \mathbf{w}_{\ell}$ is a basis for $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$ then
lyche-numerical-linear-algebra-F01924	148	1.000	$\{\mathbf{S}^{-1} \mathbf{w}_1, \dots, \mathbf{S}^{-1} \mathbf{w}_{\ell}\} \subset \mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$	$\{\mathbf{S}^{-1} \mathbf{w}_1, \dots, \mathbf{S}^{-1} \mathbf{w}_{\ell}\} \subset \mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$	$\{\mathbf{S}^{-1} \mathbf{w}_1, \dots, \mathbf{S}^{-1} \mathbf{w}_{\ell}\} \subset \mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$, which implies that $k \geq \ell$. We conclude
lyche-numerical-linear-algebra-F01925	148	1.000	$k \geq \ell$	$k \geq \ell$	$\{\mathbf{S}^{-1} \mathbf{w}_1, \dots, \mathbf{S}^{-1} \mathbf{w}_{\ell}\} \subset \mathcal{N}(\mathbf{S}^{-1} \mathbf{A} \mathbf{S} - \lambda \mathbf{I})$, which implies that $k \geq \ell$. We conclude

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01926	149	1.000	$\boldsymbol{J}_m(\lambda)$	$\boldsymbol{J}_m(\lambda)$	A Jordan block of order m , denoted $\boldsymbol{J}_m(\lambda)$ is an $m \times m$ matrix of the form
lyche-numerical-linear-algebra-F01927	149	0.764	$\boldsymbol{J}_3(\lambda)=\left[\begin{array}{ccc}\lambda&1&0\\0&\lambda&1\\0&0&\lambda\end{array}\right]$	$\boldsymbol{J}_3(\lambda)=\left[\begin{array}{ccc}\lambda&1&0\\0&\lambda&1\\0&0&\lambda\end{array}\right]$	A 3×3 Jordan block has the form $\boldsymbol{J}_3(\lambda)=\left[\begin{array}{ccc}\lambda&1&0\\0&\lambda&1\\0&0&\lambda\end{array}\right]$. Since a Jordan block is upper
lyche-numerical-linear-algebra-F01928	149	1.000	$\boldsymbol{J}_m(\lambda)\boldsymbol{v}=\lambda\boldsymbol{v}$	$\boldsymbol{J}_m(\lambda)\boldsymbol{v}=\lambda\boldsymbol{v}$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda)\boldsymbol{v}=\lambda\boldsymbol{v}$ for some $\boldsymbol{v}=[v_1,\dots,v_m]$ then $\lambda v_{i-1}+v_i=\lambda v_{i-1}$,
lyche-numerical-linear-algebra-F01929	149	1.000	$\boldsymbol{v}=[v_1,\dots,v_m]$	$\boldsymbol{v}=[v_1,\dots,v_m]$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda)\boldsymbol{v}=\lambda\boldsymbol{v}$ for some $\boldsymbol{v}=[v_1,\dots,v_m]$ then $\lambda v_{i-1}+v_i=\lambda v_{i-1}$,
lyche-numerical-linear-algebra-F01930	149	1.000	$\lambda v_{i-1}+v_i=\lambda v_{i-1}$	$\lambda v_{i-1}+v_i=\lambda v_{i-1}$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda)\boldsymbol{v}=\lambda\boldsymbol{v}$ for some $\boldsymbol{v}=[v_1,\dots,v_m]$ then $\lambda v_{i-1}+v_i=\lambda v_{i-1}$,
lyche-numerical-linear-algebra-F01931	149	1.000	$i=2,\dots,m$	$i=2,\dots,m$	$i=2,\dots,m$ which shows that $v_2=\dots=v_m=0$. Thus, the eigenvalue λ of
lyche-numerical-linear-algebra-F01932	149	1.000	$v_2=\dots=v_m=0$	$v_2=\dots=v_m=0$	$i=2,\dots,m$ which shows that $v_2=\dots=v_m=0$. Thus, the eigenvalue λ of
lyche-numerical-linear-algebra-F01933	149	0.997	$a=m$	$a=m$	$\boldsymbol{J}_m(\lambda)$ has algebraic multiplicity $a=m$ and geometric multiplicity $g=1$.
lyche-numerical-linear-algebra-F01934	149	1.000	$g_1,\dots,g_k$	$g_1,\dots,g_k$	$a_1,\dots,a_k$ and geometric multiplicities $g_1,\dots,g_k$. There is a nonsingular matrix
lyche-numerical-linear-algebra-F01935	150	1.000	$\boldsymbol{U}_i$	$\boldsymbol{U}_i$	where each $\boldsymbol{U}_i$ is block diagonal having g_i Jordan blocks along the diagonal
lyche-numerical-linear-algebra-F01936	150	1.000	g_i	g_i	where each $\boldsymbol{U}_i$ is block diagonal having g_i Jordan blocks along the diagonal
lyche-numerical-linear-algebra-F01937	150	1.000	$m_{i,1},\dots,m_{i,g_i}$	$m_{i,1},\dots,m_{i,g_i}$	Here $m_{i,1},\dots,m_{i,g_i}$ are positive integers and they are unique if they are ordered so
lyche-numerical-linear-algebra-F01938	150	1.000	$m_{i,1}\geq m_{i,2}\geq\dots\geq m_{i,g_i}$	$m_{i,1}\geq m_{i,2}\geq\dots\geq m_{i,g_i}$	that $m_{i,1}\geq m_{i,2}\geq\dots\geq m_{i,g_i}$. Moreover, $a_i=\sum_{j=1}^{g_i}m_{i,j}$ for all i .
lyche-numerical-linear-algebra-F01939	150	1.000	$a_i=\sum_{j=1}^{g_i}m_{i,j}$	$a_i=\sum_{j=1}^{g_i}m_{i,j}$	that $m_{i,1}\geq m_{i,2}\geq\dots\geq m_{i,g_i}$. Moreover, $a_i=\sum_{j=1}^{g_i}m_{i,j}$ for all i .
lyche-numerical-linear-algebra-F01940	150	1.000	$\boldsymbol{A}\boldsymbol{S}=\boldsymbol{S}\boldsymbol{J}$	$\boldsymbol{A}\boldsymbol{S}=\boldsymbol{S}\boldsymbol{J}$	They satisfy the matrix equation $\boldsymbol{A}\boldsymbol{S}=\boldsymbol{S}\boldsymbol{J}$.
lyche-numerical-linear-algebra-F01941	150	1.000	$\boldsymbol{U}_1=\text{diag}(\boldsymbol{J}_3(2),\boldsymbol{J}_2(2),\boldsymbol{J}_1(2))$	$\boldsymbol{U}_1=\text{diag}(\boldsymbol{J}_3(2),\boldsymbol{J}_2(2),\boldsymbol{J}_1(2))$	• $\boldsymbol{U}_1=\text{diag}(\boldsymbol{J}_3(2),\boldsymbol{J}_2(2),\boldsymbol{J}_1(2))$ and $\boldsymbol{U}_2=\boldsymbol{J}_2(3)$.
lyche-numerical-linear-algebra-F01942	150	1.000	$\boldsymbol{U}_2=\boldsymbol{J}_2(3)$	$\boldsymbol{U}_2=\boldsymbol{J}_2(3)$	• $\boldsymbol{U}_1=\text{diag}(\boldsymbol{J}_3(2),\boldsymbol{J}_2(2),\boldsymbol{J}_1(2))$ and $\boldsymbol{U}_2=\boldsymbol{J}_2(3)$.
lyche-numerical-linear-algebra-F01943	150	1.000	$\lambda=2$	$\lambda=2$	number of Jordan blocks corresponding to $\lambda=2$.
lyche-numerical-linear-algebra-F01944	150	1.000	$\boldsymbol{S}=[\boldsymbol{s}_1,\dots,\boldsymbol{s}_8]$	$\boldsymbol{S}=[\boldsymbol{s}_1,\dots,\boldsymbol{s}_8]$	The columns of $\boldsymbol{S}=[\boldsymbol{s}_1,\dots,\boldsymbol{s}_8]$ are determined from the columns of $\boldsymbol{J}$ as
lyche-numerical-linear-algebra-F01945	151	1.000	$\boldsymbol{J}_2(3),\boldsymbol{J}_2(2),\boldsymbol{J}_1(2),\boldsymbol{J}_3(2)$	$\boldsymbol{J}_2(3),\boldsymbol{J}_2(2),\boldsymbol{J}_1(2),\boldsymbol{J}_3(2)$	the 4 Jordan blocks $\boldsymbol{J}_2(3),\boldsymbol{J}_2(2),\boldsymbol{J}_1(2),\boldsymbol{J}_3(2)$ are uniquely given.
lyche-numerical-linear-algebra-F01946	151	1.000	$m_{i,j}\geq 1$	$m_{i,j}\geq 1$	corresponding $\boldsymbol{U}_i$. Moreover each $\boldsymbol{U}_i$ contains g_i Jordan blocks of size $m_{i,j}\geq 1$.
lyche-numerical-linear-algebra-F01947	151	1.000	$g_i\leq a_i$	$g_i\leq a_i$	Thus $g_i\leq a_i$.
lyche-numerical-linear-algebra-F01948	151	1.000	$\boldsymbol{e}_1,\boldsymbol{e}_4,\boldsymbol{e}_6,\boldsymbol{e}_7$	$\boldsymbol{e}_1,\boldsymbol{e}_4,\boldsymbol{e}_6,\boldsymbol{e}_7$	should then be clear. There are only 4 eigenvectors of $\boldsymbol{J}$, namely $\boldsymbol{e}_1,\boldsymbol{e}_4,\boldsymbol{e}_6,\boldsymbol{e}_7$
lyche-numerical-linear-algebra-F01949	151	1.000	$k=2$	$k=2$	independent. Moreover there are $k=2$ distinct eigenvalues and $g_1+g_2=$
lyche-numerical-linear-algebra-F01950	151	1.000	$g_1+g_2=$	$g_1+g_2=$	independent. Moreover there are $k=2$ distinct eigenvalues and $g_1+g_2=$
lyche-numerical-linear-algebra-F01951	151	1.000	$3+1=4$	$3+1=4$	$3+1=4$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01952	151	1.000	$\sum_{i} a_i = n$	$\sum_i a_i = n$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-algebra-F01953	151	1.000	$\sum_{i} g_i = n$	$\sum_i g_i = n$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-algebra-F01954	151	1.000	$a_i = g_i$	$a_i = g_i$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-algebra-F01955	151	1.000	$i=1, \ldots, k$	$i = 1, \dots, k$	for $i = 1, \dots, k$.
lyche-numerical-linear-algebra-F01956	151	1.000	$\boldsymbol{S} = \boldsymbol{U}$	$\boldsymbol{S} = \boldsymbol{U}$	We turn now to unitary similarity transformations $\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$, where $\boldsymbol{S} = \boldsymbol{U}$ is
lyche-numerical-linear-algebra-F01957	151	1.000	$\boldsymbol{S}^{-1} = \boldsymbol{U}^*$	$\boldsymbol{S}^{-1} = \boldsymbol{U}^*$	unitary. Thus $\boldsymbol{S}^{-1} = \boldsymbol{U}^*$ and a unitary similarity transformation takes the form
lyche-numerical-linear-algebra-F01958	151	1.000	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$.
lyche-numerical-linear-algebra-F01959	152	1.000	$\boldsymbol{R} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{R} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	matrix $\boldsymbol{U} \in \mathbb{C}^{n \times n}$ such that $\boldsymbol{R} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$ is upper triangular.
lyche-numerical-linear-algebra-F01960	152	1.000	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^*$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^*$	call $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^*$ the Schur factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F01961	152	1.000	$k \times k$	$k \times k$	Assume that the theorem is true for all $k \times k$ matrices, and suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$, where
lyche-numerical-linear-algebra-F01962	152	0.993	$n := k + 1$	$n := k + 1$	$n := k + 1$. Let $(\lambda_1, \boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\ \boldsymbol{v}_1\ _2 = 1$. By Theorem 5.5
lyche-numerical-linear-algebra-F01963	152	0.993	$(\lambda_1, \boldsymbol{v}_1)$	$(\lambda_1, \boldsymbol{v}_1)$	$n := k + 1$. Let $(\lambda_1, \boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\ \boldsymbol{v}_1\ _2 = 1$. By Theorem 5.5
lyche-numerical-linear-algebra-F01964	152	0.993	$\ \boldsymbol{v}_1\ _2 = 1$	$\ \boldsymbol{v}_1\ _2 = 1$	$n := k + 1$. Let $(\lambda_1, \boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\ \boldsymbol{v}_1\ _2 = 1$. By Theorem 5.5
lyche-numerical-linear-algebra-F01965	152	1.000	$\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$	$\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$	we can extend $\boldsymbol{v}_1$ to an orthonormal basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ for $\mathbb{C}^n$. The matrix $\boldsymbol{V} :=$
lyche-numerical-linear-algebra-F01966	152	1.000	$\boldsymbol{V} :=$	$\boldsymbol{V} :=$	we can extend $\boldsymbol{v}_1$ to an orthonormal basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ for $\mathbb{C}^n$. The matrix $\boldsymbol{V} :=$
lyche-numerical-linear-algebra-F01967	152	1.000	$[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	$[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	$[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$ is unitary, and
lyche-numerical-linear-algebra-F01968	152	1.000	$\boldsymbol{W}_1 \in \mathbb{C}^{(n-1) \times (n-1)}$	$\boldsymbol{W}_1 \in \mathbb{C}^{(n-1) \times (n-1)}$	By the induction hypothesis there is a unitary matrix $\boldsymbol{W}_1 \in \mathbb{C}^{(n-1) \times (n-1)}$ such that
lyche-numerical-linear-algebra-F01969	152	1.000	$\boldsymbol{W}_1^* \boldsymbol{M} \boldsymbol{W}_1$	$\boldsymbol{W}_1^* \boldsymbol{M} \boldsymbol{W}_1$	$\boldsymbol{W}_1^* \boldsymbol{M} \boldsymbol{W}_1$ is upper triangular. Define
lyche-numerical-linear-algebra-F01970	152	1.000	$\boldsymbol{W}$	$\boldsymbol{W}$	Then $\boldsymbol{W}$ and $\boldsymbol{U}$ are unitary and
lyche-numerical-linear-algebra-F01971	152	0.570	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}$	eigenvalues there exists an orthogonal matrix $\boldsymbol{U} \in \mathbb{R}^{n \times n}$ such that $\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}$ is upper
lyche-numerical-linear-algebra-F01972	152	1.000	λ_1	λ_1	Proof Consider the proof of Theorem 6.5. Since $\boldsymbol{A}$ and λ_1 are real the eigenvector
lyche-numerical-linear-algebra-F01973	152	1.000	$\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}$	$\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}$	$\boldsymbol{v}_1$ is real and the matrix $\boldsymbol{W}$ is real and $\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}$. By the induction hypothesis $\boldsymbol{V}$
lyche-numerical-linear-algebra-F01974	152	1.000	$\boldsymbol{V}$	$\boldsymbol{V}$	$\boldsymbol{v}_1$ is real and the matrix $\boldsymbol{W}$ is real and $\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}$. By the induction hypothesis $\boldsymbol{V}$
lyche-numerical-linear-algebra-F01975	152	1.000	$\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}$	$\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}$	is real and $\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}$. But then also $\boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}$ is real and $\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$.
lyche-numerical-linear-algebra-F01976	152	1.000	$\boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}$	$\boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}$	is real and $\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}$. But then also $\boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}$ is real and $\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$.
lyche-numerical-linear-algebra-F01977	152	1.000	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$	is real and $\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}$. But then also $\boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}$ is real and $\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$.
lyche-numerical-linear-algebra-F01978	153	1.000	$n - 1$	$n - 1$	$n \times n$ matrix $\boldsymbol{A}$, we obtain a matrix $\boldsymbol{M}$ of order $n - 1$. $\boldsymbol{M}$ has the same eigenvalues

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lyche-numerical-linear-algebra- F01979	153	1.000	$\boldsymbol{T}:=\left(\begin{array}{ccc}2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2\end{array}\right)$	$\boldsymbol{T}:=\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$	numerical work. The second derivative matrix $\boldsymbol{T}:=\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ has an eigenpair
lyche-numerical-linear-algebra- F01980	153	0.675	$\left(2, \boldsymbol{x}_1\right)$	$(2, \boldsymbol{x}_1)$	$(2, \boldsymbol{x}_1)$, where $\boldsymbol{x}_1 = [-1, 0, 1]^T$. Find the remaining eigenvalues using deflation.
lyche-numerical-linear-algebra- F01981	153	0.675	$\boldsymbol{x}_1=[-1,0,1]^T$	$\boldsymbol{x}_1 = [-1, 0, 1]^T$	$(2, \boldsymbol{x}_1)$, where $\boldsymbol{x}_1 = [-1, 0, 1]^T$. Find the remaining eigenvalues using deflation.
lyche-numerical-linear-algebra- F01982	153	1.000	$\boldsymbol{x}_1$	$\boldsymbol{x}_1$	For this we extend $\boldsymbol{x}_1$ to a basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$ for $\mathbb{R}^3$ by defining $\boldsymbol{x}_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra- F01983	153	1.000	$\left\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\right\}$	$\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$	For this we extend $\boldsymbol{x}_1$ to a basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$ for $\mathbb{R}^3$ by defining $\boldsymbol{x}_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra- F01984	153	1.000	$\boldsymbol{x}_2=[0,1,0]^T$	$\boldsymbol{x}_2 = [0, 1, 0]^T$	For this we extend $\boldsymbol{x}_1$ to a basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$ for $\mathbb{R}^3$ by defining $\boldsymbol{x}_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra- F01985	153	1.000	$\boldsymbol{x}_3=[1,0,1]^T$	$\boldsymbol{x}_3 = [1, 0, 1]^T$	$\boldsymbol{x}_3 = [1, 0, 1]^T$. This is already an orthogonal basis and normalizing we obtain the
lyche-numerical-linear-algebra- F01986	153	1.000	$\boldsymbol{A}^* \boldsymbol{A}=\boldsymbol{A} \boldsymbol{A}^*$	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^*$	A matrix $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is normal if $\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^*$. In this section we show that a matrix
lyche-numerical-linear-algebra- F01987	153	1.000	$\boldsymbol{A}^*=-\boldsymbol{A}$	$\boldsymbol{A}^* = -\boldsymbol{A}$	2. $\boldsymbol{A}^* = -\boldsymbol{A}$,
lyche-numerical-linear-algebra- F01988	153	1.000	$\boldsymbol{A}^*=\boldsymbol{A}^{-1}$	$\boldsymbol{A}^* = \boldsymbol{A}^{-1}$	3. $\boldsymbol{A}^* = \boldsymbol{A}^{-1}$,
lyche-numerical-linear-algebra- F01989	153	0.999	$\boldsymbol{A}=\operatorname{diag}\left(d_1, \ldots, d_n\right)$	$\boldsymbol{A} = \operatorname{diag}\left(d_1, \ldots, d_n\right)$	4. $\boldsymbol{A} = \operatorname{diag}\left(d_1, \ldots, d_n\right)$.
lyche-numerical-linear-algebra- F01990	154	0.989	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}=\boldsymbol{D}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$	normal if and only if there exists a unitary matrix $\boldsymbol{U} \in \mathbb{C}^{n \times n}$ such that $\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$
lyche-numerical-linear-algebra- F01991	154	1.000	$\boldsymbol{D}=\operatorname{diag}\left(\lambda_1, \ldots, \lambda_n\right)$	$\boldsymbol{D} = \operatorname{diag}\left(\lambda_1, \ldots, \lambda_n\right)$	is diagonal. If $\boldsymbol{D} = \operatorname{diag}\left(\lambda_1, \ldots, \lambda_n\right)$ and $\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n]$ then $(\lambda_j, \boldsymbol{u}_j)$, $j =$
lyche-numerical-linear-algebra- F01992	154	1.000	$\boldsymbol{U}=\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\right]$	$\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n]$	is diagonal. If $\boldsymbol{D} = \operatorname{diag}\left(\lambda_1, \ldots, \lambda_n\right)$ and $\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n]$ then $(\lambda_j, \boldsymbol{u}_j)$, $j =$
lyche-numerical-linear-algebra- F01993	154	1.000	$\left(\lambda_j, \boldsymbol{u}_j\right), j=$	$(\lambda_j, \boldsymbol{u}_j), j =$	is diagonal. If $\boldsymbol{D} = \operatorname{diag}\left(\lambda_1, \ldots, \lambda_n\right)$ and $\boldsymbol{U} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n]$ then $(\lambda_j, \boldsymbol{u}_j)$, $j =$
lyche-numerical-linear-algebra- F01994	154	0.996	$\boldsymbol{B}=\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{B} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	Proof If $\boldsymbol{B} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$, with $\boldsymbol{B}$ diagonal, and $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$, then $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*$ and
lyche-numerical-linear-algebra- F01995	154	0.996	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*$	Proof If $\boldsymbol{B} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$, with $\boldsymbol{B}$ diagonal, and $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$, then $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*$ and
lyche-numerical-linear-algebra- F01996	154	1.000	$\boldsymbol{B} \boldsymbol{B}^*=\boldsymbol{B}^* \boldsymbol{B}$	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{B}^* \boldsymbol{B}$	Now $\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{B}^* \boldsymbol{B}$ since $\boldsymbol{B}$ is diagonal, and $\boldsymbol{A}$ is normal.
lyche-numerical-linear-algebra- F01997	154	1.000	$\boldsymbol{B}:=\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{B} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{I}$ such that $\boldsymbol{B} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$ is upper triangular. Since $\boldsymbol{A}$ is normal $\boldsymbol{B}$ is normal. Indeed,
lyche-numerical-linear-algebra- F01998	154	0.458	e_{ii}	e_{ii}	must be diagonal. The diagonal elements e_{ii} in $\boldsymbol{E} := \boldsymbol{B}^* \boldsymbol{B}$ and f_{ii} in $\boldsymbol{F} := \boldsymbol{B} \boldsymbol{B}^*$
lyche-numerical-linear-algebra- F01999	154	0.458	$\boldsymbol{E}:=\boldsymbol{B}^* \boldsymbol{B}$	$\boldsymbol{E} := \boldsymbol{B}^* \boldsymbol{B}$	must be diagonal. The diagonal elements e_{ii} in $\boldsymbol{E} := \boldsymbol{B}^* \boldsymbol{B}$ and f_{ii} in $\boldsymbol{F} := \boldsymbol{B} \boldsymbol{B}^*$
lyche-numerical-linear-algebra- F02000	154	0.458	f_{ii}	f_{ii}	must be diagonal. The diagonal elements e_{ii} in $\boldsymbol{E} := \boldsymbol{B}^* \boldsymbol{B}$ and f_{ii} in $\boldsymbol{F} := \boldsymbol{B} \boldsymbol{B}^*$
lyche-numerical-linear-algebra- F02001	154	0.458	$\boldsymbol{F}:=\boldsymbol{B} \boldsymbol{B}^*$	$\boldsymbol{F} := \boldsymbol{B} \boldsymbol{B}^*$	must be diagonal. The diagonal elements e_{ii} in $\boldsymbol{E} := \boldsymbol{B}^* \boldsymbol{B}$ and f_{ii} in $\boldsymbol{F} := \boldsymbol{B} \boldsymbol{B}^*$
lyche-numerical-linear-algebra- F02002	154	1.000	$\left b_{11}\right ^2=\left b_{11}\right ^2+\left b_{12}\right ^2+\cdots+\left b_{1 n}\right ^2$	$\left b_{11}\right ^2=\left b_{11}\right ^2+\left b_{12}\right ^2+\cdots+\left b_{1 n}\right ^2$	$i = 1$ we have $\left b_{11}\right ^2=\left b_{11}\right ^2+\left b_{12}\right ^2+\cdots+\left b_{1 n}\right ^2$, so $b_{1 k}=0$ for $k=2,3, \ldots, n$.
lyche-numerical-linear-algebra- F02003	154	1.000	$b_{1 k}=0$	$b_{1 k} = 0$	$i = 1$ we have $\left b_{11}\right ^2=\left b_{11}\right ^2+\left b_{12}\right ^2+\cdots+\left b_{1 n}\right ^2$, so $b_{1 k}=0$ for $k=2,3, \ldots, n$.
lyche-numerical-linear-algebra- F02004	154	1.000	$k=2,3, \ldots, n$	$k=2,3, \ldots, n$	$i = 1$ we have $\left b_{11}\right ^2=\left b_{11}\right ^2+\left b_{12}\right ^2+\cdots+\left b_{1 n}\right ^2$, so $b_{1 k}=0$ for $k=2,3, \ldots, n$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F02005	154	■ 1.000	i-1	$i - 1$	Suppose $\mathbf{B}$ is diagonal in its first $i - 1$ rows so that $b_{jk} = 0$ for $j = 1, \dots, i - 1$,
lyche-numerical-linear-algebra- F02006	154	■ 1.000	b_{j k}=0	$b_{jk} = 0$	Suppose $\mathbf{B}$ is diagonal in its first $i - 1$ rows so that $b_{jk} = 0$ for $j = 1, \dots, i - 1$,
lyche-numerical-linear-algebra- F02007	154	■ 1.000	j=1, \ldots, i-1	$j = 1, \dots, i - 1$	Suppose $\mathbf{B}$ is diagonal in its first $i - 1$ rows so that $b_{jk} = 0$ for $j = 1, \dots, i - 1$
lyche-numerical-linear-algebra- F02008	154	■ 0.999	k=j+1, \ldots, n	$k = j + 1, \dots, n$	$k = j + 1, \dots, n$. Then
lyche-numerical-linear-algebra- F02009	154	■ 1.000	b_{i k}=0, k=i+1, \ldots, n	$b_{ik} = 0, k = i + 1, \dots, n$	and it follows that $b_{ik} = 0, k = i + 1, \dots, n$. By induction on the rows we see that
lyche-numerical-linear-algebra- F02010	154	■ 0.877	\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}=	$\mathbf{U}^T \mathbf{A} \mathbf{U} =$	<i>Example 6.6</i> The orthogonal diagonalization of $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ is $\mathbf{U}^T \mathbf{A} \mathbf{U} =$
lyche-numerical-linear-algebra- F02011	154	■ 0.987	\operatorname{diag}(1,3)	$\operatorname{diag}(1, 3)$	$\operatorname{diag}(1, 3)$, where $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.
lyche-numerical-linear-algebra- F02012	154	■ 0.987	\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc}1&1\\1&-1\end{array}\right]	$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	$\operatorname{diag}(1, 3)$, where $\mathbf{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.
lyche-numerical-linear-algebra- F02013	155	■ 0.835	R(\boldsymbol{x})=\frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{\lambda}	$R(\mathbf{x}) = \frac{\mathbf{x}^* \mathbf{A} \mathbf{x}}{\mathbf{x}^* \mathbf{x}} = \lambda$	If $(\lambda, \mathbf{x})$ is an eigenpair for $\mathbf{A}$ then $R(\mathbf{x}) = \frac{\mathbf{x}^* \mathbf{A} \mathbf{x}}{\mathbf{x}^* \mathbf{x}} = \lambda$.
lyche-numerical-linear-algebra- F02014	155	■ 0.761	\lambda_{j}, \boldsymbol{u}_{j}	$\lambda_j, \mathbf{u}_j$	is normal with orthonormal eigenpairs $(\lambda_j, \mathbf{u}_j)$, for $j = 1, 2, \dots, n$. Then the
lyche-numerical-linear-algebra- F02015	155	■ 0.761	j=1,2, \ldots, n	$j = 1, 2, \dots, n$	is normal with orthonormal eigenpairs $(\lambda_j, \mathbf{u}_j)$, for $j = 1, 2, \dots, n$. Then the
lyche-numerical-linear-algebra- F02016	155	■ 1.000	\boldsymbol{x}^* \boldsymbol{x}=\sum_{i=1}^n \bar{c}_{i} \overline{\boldsymbol{u}_{i}} \boldsymbol{c}_{j} \boldsymbol{u}_{j}	$\mathbf{x}^* \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\mathbf{u}_i} c_j \mathbf{u}_j =$	Proof By orthonormality of the eigenvectors $\mathbf{x}^* \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\mathbf{u}_i} c_j \mathbf{u}_j =$
lyche-numerical-linear-algebra- F02017	155	■ 1.000	\sum_{j=1}^n\left c_{j}\right ^2	$\sum_{j=1}^n c_j ^2$	$\sum_{j=1}^n c_j ^2$. Similarly, $\mathbf{x}^* \mathbf{A} \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\mathbf{u}_i} c_j \lambda_j \mathbf{u}_j = \sum_{i=1}^n \lambda_i c_i ^2$. and
lyche-numerical-linear-algebra- F02018	155	■ 1.000	\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}=\sum_{i=1}^n \sum_{j=1}^n \bar{c}_{i} \overline{\boldsymbol{u}_{i}} c_{j} \boldsymbol{u}_{j}=\sum_{i=1}^n \lambda_{i}\left c_{i}\right ^2	$\mathbf{x}^* \mathbf{A} \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\mathbf{u}_i} c_j \lambda_j \mathbf{u}_j = \sum_{i=1}^n \lambda_i c_i ^2$	$\sum_{j=1}^n c_j ^2$. Similarly, $\mathbf{x}^* \mathbf{A} \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\mathbf{u}_i} c_j \lambda_j \mathbf{u}_j = \sum_{i=1}^n \lambda_i c_i ^2$. and
lyche-numerical-linear-algebra- F02019	155	■ 1.000	\sum_{i=1}^n\left c_{i}\right ^2 / \sum_{j=1}^n\left c_{j}\right ^2=1	$\sum_{i=1}^n c_i ^2 / \sum_{j=1}^n c_j ^2 = 1$	combination since $\sum_{i=1}^n c_i ^2 / \sum_{j=1}^n c_j ^2 = 1$.
lyche-numerical-linear-algebra- F02020	155	■ 0.998	\lambda=\mu+i \nu, \bar{\lambda}=\mu-i \nu	$\lambda = \mu + i\nu, \bar{\lambda} = \mu - i\nu$	eigenvalues of a real matrix occur in conjugate pairs, $\lambda = \mu + i\nu, \bar{\lambda} = \mu - i\nu$,
lyche-numerical-linear-algebra- F02021	155	■ 0.758	\mu, \nu	μ, ν	where μ, ν are real. The real 2×2 matrix
lyche-numerical-linear-algebra- F02022	155	■ 0.999	\lambda=\mu+i \nu	$\lambda = \mu + i\nu$	has eigenvalues $\lambda = \mu + i\nu$ and $\bar{\lambda} = \mu - i\nu$.
lyche-numerical-linear-algebra- F02023	155	■ 0.999	\bar{\lambda}=\mu-i \nu	$\bar{\lambda} = \mu - i\nu$	has eigenvalues $\lambda = \mu + i\nu$ and $\bar{\lambda} = \mu - i\nu$.
lyche-numerical-linear-algebra- F02024	156	■ 1.000	\pi_{\boldsymbol{R}}=	$\pi_{\mathbf{R}} =$	Since $\mathbf{R}$ is block triangular the characteristic polynomial of $\mathbf{R}$ is given by $\pi_{\mathbf{R}} =$
lyche-numerical-linear-algebra- F02025	156	■ 0.831	\pi_{\boldsymbol{D}_1} \pi_{\boldsymbol{D}_2} \pi_{\boldsymbol{D}_3}	$\pi_{\mathbf{D}_1} \pi_{\mathbf{D}_2} \pi_{\mathbf{D}_3}$	$\pi_{\mathbf{D}_1} \pi_{\mathbf{D}_2} \pi_{\mathbf{D}_3}$. We find
lyche-numerical-linear-algebra- F02026	156	■ 1.000	\boldsymbol{D}_1	$\mathbf{D}_1$	and the eigenvalues $\mathbf{D}_1$ and $\mathbf{D}_3$ are $\lambda_1 = 2 + i, \lambda_2 = 2 - i$, while $\mathbf{D}_2$ obviously has
lyche-numerical-linear-algebra- F02027	156	■ 1.000	\boldsymbol{D}_3	$\mathbf{D}_3$	and the eigenvalues $\mathbf{D}_1$ and $\mathbf{D}_3$ are $\lambda_1 = 2 + i, \lambda_2 = 2 - i$, while $\mathbf{D}_2$ obviously has
lyche-numerical-linear-algebra- F02028	156	■ 1.000	\lambda_1=2+i, \lambda_2=2-i	$\lambda_1 = 2 + i, \lambda_2 = 2 - i$	and the eigenvalues $\mathbf{D}_1$ and $\mathbf{D}_3$ are $\lambda_1 = 2 + i, \lambda_2 = 2 - i$, while $\mathbf{D}_2$ obviously has

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lyche-numerical-linear-algebra- F02029	156	■ 1.000	$\boldsymbol{D}_{\{2\}}$	D_2	and the eigenvalues D_1 and D_3 are $\lambda_1 = 2+i, \lambda_2 = 2-i$, while D_2 obviously has
lyche-numerical-linear-algebra- F02030	156	■ 1.000	$\left\{\boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{n\}}\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_n\}$	For any such $\boldsymbol{U}$ the columns $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_n\}$ of $\boldsymbol{U}$ are orthonormal eigenvectors of $\boldsymbol{A}$
lyche-numerical-linear-algebra- F02031	156	■ 1.000	$\boldsymbol{A} \boldsymbol{u}_{\{j\}}=\lambda_{\{j\}} \boldsymbol{u}_{\{j\}}$	$\boldsymbol{A} \boldsymbol{u}_j = \lambda_j \boldsymbol{u}_j$	and $\boldsymbol{A} \boldsymbol{u}_j = \lambda_j \boldsymbol{u}_j$ for $j = 1, \dots, n$.
lyche-numerical-linear-algebra- F02032	156	■ 0.995	$\lambda_{\{1\}}, \lambda_{\{2\}}, \ldots, \lambda_{\{n\}}$	$\lambda_1, \lambda_2, \dots, \lambda_n$	ric. Then $\boldsymbol{A}$ has real eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Moreover, there is an orthogonal
lyche-numerical-linear-algebra- F02033	157	■ 0.999	$R(\boldsymbol{x})=\frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$	$R(\boldsymbol{x}) = R_{\boldsymbol{A}}(\boldsymbol{x}) := \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$	terms of the Rayleigh quotient $R(\boldsymbol{x}) = R_{\boldsymbol{A}}(\boldsymbol{x}) := \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$. First we show
lyche-numerical-linear-algebra- F02034	157	■ 0.924	$\left(\lambda_{\{j\}}, \boldsymbol{u}_{\{j\}}\right), 1 \leq j \leq n$	$(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq n$	eigenpairs $(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq n$, ordered so that $\lambda_1 \geq \dots \geq \lambda_n$. Let $1 \leq k \leq n$. For
lyche-numerical-linear-algebra- F02035	157	■ 0.924	$\lambda_{\{1\}} \geq \cdots \geq \lambda_{\{n\}}$	$\lambda_1 \geq \dots \geq \lambda_n$	eigenpairs $(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq n$, ordered so that $\lambda_1 \geq \dots \geq \lambda_n$. Let $1 \leq k \leq n$. For
lyche-numerical-linear-algebra- F02036	157	■ 0.924	$1 \leq k \leq n$	$1 \leq k \leq n$	eigenpairs $(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq n$, ordered so that $\lambda_1 \geq \dots \geq \lambda_n$. Let $1 \leq k \leq n$. For
lyche-numerical-linear-algebra- F02037	157	■ 1.000	$\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span}\left(\boldsymbol{u}_{\{k\}}, \ldots, \boldsymbol{u}_{\{n\}}\right)$	$\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$	with equality for $\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$ and $\boldsymbol{x} = \boldsymbol{u}_k$.
lyche-numerical-linear-algebra- F02038	157	■ 1.000	$\boldsymbol{x}=\boldsymbol{u}_{\{k\}}$	$\boldsymbol{x} = \boldsymbol{u}_k$	with equality for $\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$ and $\boldsymbol{x} = \boldsymbol{u}_k$.
lyche-numerical-linear-algebra- F02039	157	■ 1.000	$\mathcal{S}^{\prime}:=$	$\mathcal{S}' :=$	Proof Let $\mathcal{S}$ be any subspace of $\mathbb{C}^n$ of dimension $n - k + 1$ and define $\mathcal{S}' :=$
lyche-numerical-linear-algebra- F02040	157	■ 1.000	$\operatorname{span}\left(\boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{k\}}\right)$	$\operatorname{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$	$\operatorname{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S} + \mathcal{S}' :=$
lyche-numerical-linear-algebra- F02041	157	■ 1.000	$\boldsymbol{y} \in \mathcal{S}$	$\boldsymbol{y} \in \mathcal{S}$	$\operatorname{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S} + \mathcal{S}' :=$
lyche-numerical-linear-algebra- F02042	157	■ 1.000	$R(\boldsymbol{y}) \geq \lambda_{\{k\}}$	$R(\boldsymbol{y}) \geq \lambda_k$	$\operatorname{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S} + \mathcal{S}' :=$
lyche-numerical-linear-algebra- F02043	157	■ 1.000	$\mathcal{S}+\mathcal{S}^{\prime}:=$	$\mathcal{S} + \mathcal{S}' :=$	$\operatorname{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S} + \mathcal{S}' :=$
lyche-numerical-linear-algebra- F02044	157	■ 1.000	$\left\{\boldsymbol{s}+\boldsymbol{s}^{\prime}: \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}^{\prime} \in \mathcal{S}^{\prime}\right\}$	$\{\boldsymbol{s} + \boldsymbol{s}' : \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}' \in \mathcal{S}'\}$	$\{\boldsymbol{s} + \boldsymbol{s}' : \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}' \in \mathcal{S}'\}$ is a subspace of $\mathbb{C}^n$ and by (1.7)
lyche-numerical-linear-algebra- F02045	157	■ 0.996	$\mathcal{S} \cap \mathcal{S}^{\prime}$	$\mathcal{S} \cap \mathcal{S}'$	It follows that $\mathcal{S} \cap \mathcal{S}'$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^k c_j \boldsymbol{u}_j$ with $\sum_{j=1}^k c_j ^2 =$
lyche-numerical-linear-algebra- F02046	157	■ 0.996	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}=\sum_{j=1}^k c_{\{j\}} \boldsymbol{u}_{\{j\}}$	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^k c_j \boldsymbol{u}_j$	It follows that $\mathcal{S} \cap \mathcal{S}'$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^k c_j \boldsymbol{u}_j$ with $\sum_{j=1}^k c_j ^2 =$
lyche-numerical-linear-algebra- F02047	157	■ 0.996	$\sum_{j=1}^k\left c_{\{j\}}\right ^2=$	$\sum_{j=1}^k c_j ^2 =$	It follows that $\mathcal{S} \cap \mathcal{S}'$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^k c_j \boldsymbol{u}_j$ with $\sum_{j=1}^k c_j ^2 =$
lyche-numerical-linear-algebra- F02048	157	■ 0.997	$c_{\{j\}}=0$	$c_j = 0$	1. Defining $c_j = 0$ for $k + 1 \leq j \leq n$, we obtain by Theorem 6.8
lyche-numerical-linear-algebra- F02049	157	■ 0.997	$k+1 \leq j \leq n$	$k + 1 \leq j \leq n$	1. Defining $c_j = 0$ for $k + 1 \leq j \leq n$, we obtain by Theorem 6.8
lyche-numerical-linear-algebra- F02050	157	■ 0.877	$\boldsymbol{z} \in \mathcal{S}=\tilde{\mathcal{S}}$	$\boldsymbol{z} \in \mathcal{S} = \tilde{\mathcal{S}}$	and (6.11) follows. To show equality suppose $\boldsymbol{z} \in \mathcal{S} = \tilde{\mathcal{S}}$. Now $\boldsymbol{z} = \sum_{j=k}^n d_j \boldsymbol{u}_j$ for
lyche-numerical-linear-algebra- F02051	157	■ 0.877	$\boldsymbol{z}=\sum_{j=k}^n d_{\{j\}} \boldsymbol{u}_{\{j\}}$	$\boldsymbol{z} = \sum_{j=k}^n d_j \boldsymbol{u}_j$	and (6.11) follows. To show equality suppose $\boldsymbol{z} \in \mathcal{S} = \tilde{\mathcal{S}}$. Now $\boldsymbol{z} = \sum_{j=k}^n d_j \boldsymbol{u}_j$ for
lyche-numerical-linear-algebra- F02052	157	■ 0.999	$d_{\{k\}}, \ldots, d_{\{n\}}$	$d_k, \dots, d_n$	some $d_k, \dots, d_n$ with $\sum_{j=k}^n d_j ^2 = 1$ and by Lemma 6.8 $R(\boldsymbol{z}) = \sum_{j=k}^n \lambda_j d_j ^2 \leq$

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lyche-numerical-linear-algebra-F02053	157	0.999	$\sum_{j=k}^n \left d_j \right ^2 = 1$	$\sum_{j=k}^n d_j ^2 = 1$	some $d_k, \dots, d_n$ with $\sum_{j=k}^n d_j ^2 = 1$ and by Lemma 6.8 $R(z) = \sum_{j=k}^n \lambda_j d_j ^2 \leq$
lyche-numerical-linear-algebra-F02054	157	0.999	6.8 $R(z) = \sum_{j=k}^n \lambda_j \left d_j \right ^2 \leq$	$6.8 R(z) = \sum_{j=k}^n \lambda_j d_j ^2 \leq$	some $d_k, \dots, d_n$ with $\sum_{j=k}^n d_j ^2 = 1$ and by Lemma 6.8 $R(z) = \sum_{j=k}^n \lambda_j d_j ^2 \leq$
lyche-numerical-linear-algebra-F02055	157	0.959	$\lambda_{\{k\}}$	λ_k	λ_k . Since $z \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$ and equality in (6.11)
lyche-numerical-linear-algebra-F02056	157	0.959	$\boldsymbol{z} \in \tilde{\mathcal{S}}$	$z \in \tilde{\mathcal{S}}$	λ_k . Since $z \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$ and equality in (6.11)
lyche-numerical-linear-algebra-F02057	157	0.959	$\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$	$\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$	λ_k . Since $z \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$ and equality in (6.11)
lyche-numerical-linear-algebra-F02058	157	1.000	$\mathcal{S} = \tilde{\mathcal{S}}$	$\mathcal{S} = \tilde{\mathcal{S}}$	follows for $\mathcal{S} = \tilde{\mathcal{S}}$. Moreover, $R(\boldsymbol{u}_k) = \lambda_k$.
lyche-numerical-linear-algebra-F02059	157	1.000	$R(\boldsymbol{u}_k) = \lambda_k$	$R(\boldsymbol{u}_k) = \lambda_k$	follows for $\mathcal{S} = \tilde{\mathcal{S}}$. Moreover, $R(\boldsymbol{u}_k) = \lambda_k$.
lyche-numerical-linear-algebra-F02060	158	1.000	$\mathcal{S} = \tilde{\mathcal{S}} := \text{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$	$\mathcal{S} = \tilde{\mathcal{S}} := \text{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$	with equality for $\mathcal{S} = \tilde{\mathcal{S}} := \text{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$ and $x = \boldsymbol{u}_k$. Here $(\lambda_j, \boldsymbol{u}_j)$, $1 \leq j \leq$
lyche-numerical-linear-algebra-F02061	158	1.000	$(\lambda_j, \boldsymbol{u}_j)$, $1 \leq j \leq$	$(\lambda_j, \boldsymbol{u}_j)$, $1 \leq j \leq$	with equality for $\mathcal{S} = \tilde{\mathcal{S}} := \text{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$ and $x = \boldsymbol{u}_k$. Here $(\lambda_j, \boldsymbol{u}_j)$, $1 \leq j \leq$
lyche-numerical-linear-algebra-F02062	158	1.000	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$ and show that $R(\boldsymbol{y}) \leq \lambda_k$ for some $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-algebra-F02063	158	1.000	$R(\boldsymbol{y}) \leq \lambda_k$	$R(\boldsymbol{y}) \leq \lambda_k$	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$ and show that $R(\boldsymbol{y}) \leq \lambda_k$ for some $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-algebra-F02064	158	1.000	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$ and show that $R(\boldsymbol{y}) \leq \lambda_k$ for some $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-algebra-F02065	158	1.000	$\boldsymbol{y} \in \tilde{\mathcal{S}}$	$\boldsymbol{y} \in \tilde{\mathcal{S}}$	that $R(\boldsymbol{y}) \geq \lambda_k$ for any $\boldsymbol{y} \in \tilde{\mathcal{S}}$.
lyche-numerical-linear-algebra-F02066	158	0.995	$\boldsymbol{A}, \boldsymbol{B} \in$	$\boldsymbol{A}, \boldsymbol{B} \in$	Theorem 6.13 (Eigenvalue Perturbation for Hermitian Matrices) Let $\boldsymbol{A}, \boldsymbol{B} \in$
lyche-numerical-linear-algebra-F02067	158	0.998	$\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_n$	$\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_n$	$\mathbb{C}^{n \times n}$ be Hermitian with eigenvalues $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_n$ and $\beta_1 \geq \beta_2 \geq \dots \geq \beta_n$.
lyche-numerical-linear-algebra-F02068	158	0.998	$\beta_1 \geq \beta_2 \geq \dots \geq \beta_n$	$\beta_1 \geq \beta_2 \geq \dots \geq \beta_n$	$\mathbb{C}^{n \times n}$ be Hermitian with eigenvalues $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_n$ and $\beta_1 \geq \beta_2 \geq \dots \geq \beta_n$.
lyche-numerical-linear-algebra-F02069	158	1.000	$\varepsilon_1 \geq \varepsilon_2 \geq \dots \geq \varepsilon_n$	$\varepsilon_1 \geq \varepsilon_2 \geq \dots \geq \varepsilon_n$	where $\varepsilon_1 \geq \varepsilon_2 \geq \dots \geq \varepsilon_n$ are the eigenvalues of $\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$.
lyche-numerical-linear-algebra-F02070	158	1.000	$\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$	$\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$	where $\varepsilon_1 \geq \varepsilon_2 \geq \dots \geq \varepsilon_n$ are the eigenvalues of $\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$.
lyche-numerical-linear-algebra-F02071	158	0.997	$(\alpha_j, \boldsymbol{u}_j)$, $j = 1, \dots, n$	$(\alpha_j, \boldsymbol{u}_j)$, $j = 1, \dots, n$	eigenvalues are real. Let $(\alpha_j, \boldsymbol{u}_j)$, $j = 1, \dots, n$ be orthonormal eigenpairs for $\boldsymbol{A}$
lyche-numerical-linear-algebra-F02072	158	1.000	$\mathcal{S} := \text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$	$\mathcal{S} := \text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$	and let $\mathcal{S} := \text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$. By Theorem 6.11 we obtain
lyche-numerical-linear-algebra-F02073	158	1.000	$\boldsymbol{D} := -\boldsymbol{E}$	$\boldsymbol{D} := -\boldsymbol{E}$	and this proves the upper inequality. For the lower one we define $\boldsymbol{D} := -\boldsymbol{E}$ and
lyche-numerical-linear-algebra-F02074	158	1.000	$-\varepsilon_n$	$-\varepsilon_n$	observe that $-\varepsilon_n$ is the largest eigenvalue of $\boldsymbol{D}$. Since $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$ it follows from the
lyche-numerical-linear-algebra-F02075	158	1.000	$\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$	$\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$	observe that $-\varepsilon_n$ is the largest eigenvalue of $\boldsymbol{D}$. Since $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$ it follows from the
lyche-numerical-linear-algebra-F02076	158	1.000	$\alpha_k \leq \beta_k - \varepsilon_n$	$\alpha_k \leq \beta_k - \varepsilon_n$	result just proved that $\alpha_k \leq \beta_k - \varepsilon_n$, which is the same as the lower inequality.

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lyche-numerical-linear-algebra-F02077	159	■ 1.000	$\mu_1, \ldots, \mu_n$	$\mu_1, \dots, \mu_n$	<i>normal matrices with eigenvalues $\lambda_1, \dots, \lambda_n$ and $\mu_1, \dots, \mu_n$, respectively. Then</i>
lyche-numerical-linear-algebra-F02078	159	■ 0.655	$\boldsymbol{y} \in \mathbb{C}^n$	$\boldsymbol{y} \in \mathbb{C}^n$	$\lambda \in \mathbb{C}$ and $\boldsymbol{y} \in \mathbb{C}^n$ is nonzero. We say that $(\lambda, \boldsymbol{y})$ is a left eigenpair for $\boldsymbol{A}$ if
lyche-numerical-linear-algebra-F02079	159	■ 0.655	$\lambda, \boldsymbol{y}$	$\lambda, \boldsymbol{y}$	$\lambda \in \mathbb{C}$ and $\boldsymbol{y} \in \mathbb{C}^n$ is nonzero. We say that $(\lambda, \boldsymbol{y})$ is a left eigenpair for $\boldsymbol{A}$ if
lyche-numerical-linear-algebra-F02080	159	■ 0.979	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$ or equivalently $\boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{y}$. We say that $(\lambda, \boldsymbol{y})$ is a right eigenpair for
lyche-numerical-linear-algebra-F02081	159	■ 0.979	$\boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{y}$	$\boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{y}$	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$ or equivalently $\boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{y}$. We say that $(\lambda, \boldsymbol{y})$ is a right eigenpair for
lyche-numerical-linear-algebra-F02082	159	■ 0.988	$\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$	$\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$	$\boldsymbol{A}$ if $\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$. If $(\lambda, \boldsymbol{y})$ is a left eigenpair then λ is called a left eigenvalue and
lyche-numerical-linear-algebra-F02083	159	■ 0.988	$(\lambda, \boldsymbol{y})$	$(\lambda, \boldsymbol{y})$	$\boldsymbol{A}$ if $\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$. If $(\lambda, \boldsymbol{y})$ is a left eigenpair then λ is called a left eigenvalue and
lyche-numerical-linear-algebra-F02084	159	■ 0.779	$(\lambda, \boldsymbol{y})$	$(\lambda, \boldsymbol{y})$	$\boldsymbol{y}$ a left eigenvector . Similarly if $(\lambda, \boldsymbol{y})$ is a right eigenpair then λ is called a right
lyche-numerical-linear-algebra-F02085	159	■ 0.999	$\bar{\lambda}$	$\bar{\lambda}$	is an eigenvector of $\boldsymbol{A}^*$. If λ is a left eigenvalue of $\boldsymbol{A}$ then $\bar{\lambda}$ is an eigenvalue of $\boldsymbol{A}^*$
lyche-numerical-linear-algebra-F02086	159	■ 1.000	$\{\boldsymbol{y}_1, \dots, \boldsymbol{y}_n\}$	$\{\boldsymbol{y}_1, \dots, \boldsymbol{y}_n\}$	<i>eigenvectors $\{\boldsymbol{y}_1, \dots, \boldsymbol{y}_n\}$ with $\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$. Conversely, if $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has</i>
lyche-numerical-linear-algebra-F02087	159	■ 1.000	$\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$	$\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$	<i>eigenvectors $\{\boldsymbol{y}_1, \dots, \boldsymbol{y}_n\}$ with $\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$. Conversely, if $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has</i>
lyche-numerical-linear-algebra-F02088	160	■ 1.000	$(\lambda_1, \boldsymbol{x}_1), \dots, (\lambda_n, \boldsymbol{x}_n)$	$(\lambda_1, \boldsymbol{x}_1), \dots, (\lambda_n, \boldsymbol{x}_n)$	Proof For any right eigenpairs $(\lambda_1, \boldsymbol{x}_1), \dots, (\lambda_n, \boldsymbol{x}_n)$ and left eigenpairs
lyche-numerical-linear-algebra-F02089	160	■ 0.998	$(\lambda_1, \boldsymbol{y}_1), \dots, (\lambda_n, \boldsymbol{y}_n)$	$(\lambda_1, \boldsymbol{y}_1), \dots, (\lambda_n, \boldsymbol{y}_n)$	$(\lambda_1, \boldsymbol{y}_1), \dots, (\lambda_n, \boldsymbol{y}_n)$ of $\boldsymbol{A}$ we have $\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D}$, $\boldsymbol{Y}^* \boldsymbol{A} = \boldsymbol{D} \boldsymbol{Y}^*$, where
lyche-numerical-linear-algebra-F02090	160	■ 0.998	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^* \boldsymbol{A} = \boldsymbol{D} \boldsymbol{Y}^*$	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^* \boldsymbol{A} = \boldsymbol{D} \boldsymbol{Y}^*$	$(\lambda_1, \boldsymbol{y}_1), \dots, (\lambda_n, \boldsymbol{y}_n)$ of $\boldsymbol{A}$ we have $\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^* \boldsymbol{A} = \boldsymbol{D} \boldsymbol{Y}^*$, where
lyche-numerical-linear-algebra-F02091	160	■ 1.000	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D} \implies \boldsymbol{A} = \boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \implies \boldsymbol{X}^{-1} \boldsymbol{A} =$	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D} \implies \boldsymbol{A} = \boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \implies \boldsymbol{X}^{-1} \boldsymbol{A} =$	Suppose $\boldsymbol{X}$ is nonsingular. Then $\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D} \implies \boldsymbol{A} = \boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \implies \boldsymbol{X}^{-1} \boldsymbol{A} =$
lyche-numerical-linear-algebra-F02092	160	■ 1.000	$\boldsymbol{D} \boldsymbol{X}^{-1}$	$\boldsymbol{D} \boldsymbol{X}^{-1}$	$\boldsymbol{D} \boldsymbol{X}^{-1}$ and it follows that $\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$ contains a collection of left eigenvectors such
lyche-numerical-linear-algebra-F02093	160	■ 1.000	$\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$	$\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$	$\boldsymbol{D} \boldsymbol{X}^{-1}$ and it follows that $\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$ contains a collection of left eigenvectors such
lyche-numerical-linear-algebra-F02094	160	■ 1.000	$\boldsymbol{Y}^* \boldsymbol{X} = \boldsymbol{I}$	$\boldsymbol{Y}^* \boldsymbol{X} = \boldsymbol{I}$	that $\boldsymbol{Y}^* \boldsymbol{X} = \boldsymbol{I}$. Thus the columns of $\boldsymbol{Y}$ are linearly independent and $\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$.
lyche-numerical-linear-algebra-F02095	160	■ 1.000	$\boldsymbol{Y}$	$\boldsymbol{Y}$	that $\boldsymbol{Y}^* \boldsymbol{X} = \boldsymbol{I}$. Thus the columns of $\boldsymbol{Y}$ are linearly independent and $\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$.
lyche-numerical-linear-algebra-F02096	160	■ 1.000	$\boldsymbol{A} \boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*} \boldsymbol{D}$	$\boldsymbol{A} \boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*} \boldsymbol{D}$	Similarly, if $\boldsymbol{Y}$ is nonsingular then $\boldsymbol{A} \boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*} \boldsymbol{D}$ and it follows that $\boldsymbol{X} := \boldsymbol{Y}^{-*}$
lyche-numerical-linear-algebra-F02097	160	■ 1.000	$\boldsymbol{X} := \boldsymbol{Y}^{-*}$	$\boldsymbol{X} := \boldsymbol{Y}^{-*}$	Similarly, if $\boldsymbol{Y}$ is nonsingular then $\boldsymbol{A} \boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*} \boldsymbol{D}$ and it follows that $\boldsymbol{X} := \boldsymbol{Y}^{-*}$
lyche-numerical-linear-algebra-F02098	160	■ 0.996	$\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$	$\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$	If $\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$ then $\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$ for
lyche-numerical-linear-algebra-F02099	160	■ 0.996	$\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$	$\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$	If $\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$ then $\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$ for
lyche-numerical-linear-algebra-F02100	160	■ 0.996	$c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$	$c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$	If $\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$ then $\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$ for
lyche-numerical-linear-algebra-F02101	160	■ 0.995	$\mu, \boldsymbol{y}$	$\mu, \boldsymbol{y}$	Theorem 6.16 (Biorthogonality) Suppose $(\mu, \boldsymbol{y})$ and $(\lambda, \boldsymbol{x})$ are left and right

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lyche-numerical-linear-algebra-F02102	160	1.000	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{y}^* \boldsymbol{x} = \mu \boldsymbol{y}^* \boldsymbol{x}$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{y}^* \boldsymbol{x} = \mu \boldsymbol{y}^* \boldsymbol{x}$	Proof Using the eigenpair relation in two ways we obtain $\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{y}^* \boldsymbol{x} = \mu \boldsymbol{y}^* \boldsymbol{x}$
lyche-numerical-linear-algebra-F02103	160	1.000	$\boldsymbol{y}^* \boldsymbol{x} \neq 0$	$\boldsymbol{y}^* \boldsymbol{x} \neq 0$	eigenpairs of $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. If λ has algebraic multiplicity one then $\boldsymbol{y}^* \boldsymbol{x} \neq 0$.
lyche-numerical-linear-algebra-F02104	160	1.000	$\ \boldsymbol{x}\ _2 = 1$	$\ \boldsymbol{x}\ _2 = 1$	Proof Assume that $\ \boldsymbol{x}\ _2 = 1$. We have (cf. (6.8))
lyche-numerical-linear-algebra-F02105	161	1.000	$\boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{x}$	$\boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{x}$	where $\boldsymbol{V}$ is unitary and $\boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{x}$. We show that if $\boldsymbol{y}^* \boldsymbol{x} = 0$ then λ is also an
lyche-numerical-linear-algebra-F02106	161	1.000	$\boldsymbol{u} := \boldsymbol{V}^* \boldsymbol{y}$	$\boldsymbol{u} := \boldsymbol{V}^* \boldsymbol{y}$	eigenvalue of $\boldsymbol{M}$ contradicting the multiplicity assumption of λ . Let $\boldsymbol{u} := \boldsymbol{V}^* \boldsymbol{y}$.
lyche-numerical-linear-algebra-F02107	161	1.000	$(\bar{\lambda}, \boldsymbol{u})$	$(\bar{\lambda}, \boldsymbol{u})$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$. But then $\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra-F02108	161	1.000	$\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$	$\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$. But then $\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra-F02109	161	1.000	$\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$	$\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$. But then $\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra-F02110	161	0.596	$\boldsymbol{u}^* \boldsymbol{e}_1 = 0$	$\boldsymbol{u}^* \boldsymbol{e}_1 = 0$	that $\boldsymbol{u}^* \boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \begin{bmatrix} 0 \\ v \end{bmatrix}$ for some nonzero $v \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra-F02111	161	0.596	$\boldsymbol{u} = \begin{bmatrix} 0 \\ v \end{bmatrix}$	$\boldsymbol{u} = \begin{bmatrix} 0 \\ v \end{bmatrix}$	that $\boldsymbol{u}^* \boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \begin{bmatrix} 0 \\ v \end{bmatrix}$ for some nonzero $v \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra-F02112	161	0.596	$\boldsymbol{v} \in \mathbb{C}^{n-1}$	$\boldsymbol{v} \in \mathbb{C}^{n-1}$	that $\boldsymbol{u}^* \boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \begin{bmatrix} 0 \\ v \end{bmatrix}$ for some nonzero $v \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra-F02113	161	1.000	$\boldsymbol{A} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\boldsymbol{A} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\boldsymbol{A} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has one eigenvalue $\lambda = 1$ of algebraic multiplicity two, one right
lyche-numerical-linear-algebra-F02114	161	1.000	$\boldsymbol{y} = \boldsymbol{e}_2$	$\boldsymbol{y} = \boldsymbol{e}_2$	eigenvector $\boldsymbol{x} = \boldsymbol{e}_1$ and one left eigenvector $\boldsymbol{y} = \boldsymbol{e}_2$. Thus $\boldsymbol{x}$ and $\boldsymbol{y}$ are orthogonal.
lyche-numerical-linear-algebra-F02115	161	1.000	$\det(\boldsymbol{B}^T) =$	$\det(\boldsymbol{B}^T) =$	Exercise 6.2 (Characteristic Polynomial of Transpose) We have $\det(\boldsymbol{B}^T) =$
lyche-numerical-linear-algebra-F02116	161	1.000	$\det(\boldsymbol{B})$	$\det(\boldsymbol{B})$	$\det(\boldsymbol{B})$ and $\det(\bar{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$ for any square matrix $\boldsymbol{B}$. Use this to show that
lyche-numerical-linear-algebra-F02117	161	1.000	$\det(\bar{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$	$\det(\bar{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$	$\det(\boldsymbol{B})$ and $\det(\bar{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$ for any square matrix $\boldsymbol{B}$. Use this to show that
lyche-numerical-linear-algebra-F02118	161	0.991	$\pi_{\boldsymbol{A}^T} = \pi_{\boldsymbol{A}}$	$\pi_{\boldsymbol{A}^T} = \pi_{\boldsymbol{A}}$	a) $\pi_{\boldsymbol{A}^T} = \pi_{\boldsymbol{A}}$,
lyche-numerical-linear-algebra-F02119	161	1.000	$\pi_{\boldsymbol{A}^*}(\bar{\lambda}) = \overline{\pi_{\boldsymbol{A}}(\lambda)}$	$\pi_{\boldsymbol{A}^*}(\bar{\lambda}) = \overline{\pi_{\boldsymbol{A}}(\lambda)}$	b) $\pi_{\boldsymbol{A}^*}(\bar{\lambda}) = \overline{\pi_{\boldsymbol{A}}(\lambda)}$.
lyche-numerical-linear-algebra-F02120	161	1.000	$(\lambda^{-1}, \boldsymbol{x})$	$(\lambda^{-1}, \boldsymbol{x})$	a) If $\boldsymbol{A}$ is nonsingular then $(\lambda^{-1}, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra-F02121	161	0.562	$(\lambda^k, \boldsymbol{x})$	$(\lambda^k, \boldsymbol{x})$	b) $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.
lyche-numerical-linear-algebra-F02122	161	0.562	$\boldsymbol{A}^k$	$\boldsymbol{A}^k$	b) $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.
lyche-numerical-linear-algebra-F02123	161	0.562	$k \in \mathbb{Z}$	$k \in \mathbb{Z}$	b) $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.
lyche-numerical-linear-algebra-F02124	162	1.000	$(\lambda_j, \boldsymbol{x}_j), j = 1, \dots, n$	$(\lambda_j, \boldsymbol{x}_j), j = 1, \dots, n$	is nondefective with eigenpairs $(\lambda_j, \boldsymbol{x}_j), j = 1, \dots, n$ then for any $\boldsymbol{x} \in \mathbb{C}^n$ and
lyche-numerical-linear-algebra-F02125	162	1.000	$k \in \mathbb{N}$	$k \in \mathbb{N}$	$k \in \mathbb{N}$
lyche-numerical-linear-algebra-F02126	162	1.000	$\boldsymbol{A}^2 =$	$\boldsymbol{A}^2 =$	Exercise 6.5 (Eigenvalues of an Idempotent Matrix) Let $\lambda \in \sigma(\boldsymbol{A})$ where $\boldsymbol{A}^2 =$

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lyche-numerical-linear-algebra-162-F02127	162	1.000	$\backslash\lambda=0$	$\lambda = 0$	$A \in \mathbb{C}^{n \times n}$. Show that $\lambda = 0$ or $\lambda = 1$. (A matrix is called idempotent if $A^2 = A$).
lyche-numerical-linear-algebra-162-F02128	162	1.000	$\backslash\boldsymbol{A}^{\wedge\{2\}}=\boldsymbol{A}$	$A^2 = A$	$A \in \mathbb{C}^{n \times n}$. Show that $\lambda = 0$ or $\lambda = 1$. (A matrix is called idempotent if $A^2 = A$).
lyche-numerical-linear-algebra-162-F02129	162	1.000	$\backslash\boldsymbol{A}^{\wedge\{k\}}=0$	$A^k = 0$	Exercise 6.6 (Eigenvalues of a Nilpotent Matrix) Let $\lambda \in \sigma(A)$ where $A^k = 0$
lyche-numerical-linear-algebra-162-F02130	162	1.000	$\backslash\boldsymbol{A}^{\wedge\{*\}} \ \backslash\boldsymbol{A}=\boldsymbol{I}$	$A^* A = I$	Exercise 6.7 (Eigenvalues of a Unitary Matrix) Let $\lambda \in \sigma(A)$, where $A^* A = I$.
lyche-numerical-linear-algebra-162-F02131	162	1.000	$ \backslash\lambda =1$	$ \lambda = 1$	Show that $ \lambda = 1$.
lyche-numerical-linear-algebra-162-F02132	162	1.000	$\backslash\epsilon_{\{0\}}>0$	$\epsilon_0 > 0$	$\mathbb{C}^{n \times n}$ is singular. Then we can find $\epsilon_0 > 0$ such that $A + \epsilon I$ is nonsingular for all
lyche-numerical-linear-algebra-162-F02133	162	1.000	$\backslash\boldsymbol{A}+\backslash\epsilon \ \backslash\boldsymbol{I}$	$A + \epsilon I$	$\mathbb{C}^{n \times n}$ is singular. Then we can find $\epsilon_0 > 0$ such that $A + \epsilon I$ is nonsingular for all
lyche-numerical-linear-algebra-162-F02134	162	0.997	$\backslash\epsilon \ \backslash\in \ \backslash\mathbb{C}$	$\epsilon \in \mathbb{C}$	$\epsilon \in \mathbb{C}$ with $ \epsilon < \epsilon_0$. Hint: $\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$, where λ_i are the eigenvalues of A .
lyche-numerical-linear-algebra-162-F02135	162	0.997	$ \backslash\epsilon <\backslash\epsilon_{\{0\}}$	$ \epsilon < \epsilon_0$	$\epsilon \in \mathbb{C}$ with $ \epsilon < \epsilon_0$. Hint: $\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$, where λ_i are the eigenvalues of A .
lyche-numerical-linear-algebra-162-F02136	162	0.997	$\backslash\operatorname{det}(\backslash\boldsymbol{A})=\backslash\lambda_{\{1\}} \backslash\lambda_{\{2\}} \ \backslash\cdots \ \backslash\lambda_{\{n\}}$	$\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$	$\epsilon \in \mathbb{C}$ with $ \epsilon < \epsilon_0$. Hint: $\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$, where λ_i are the eigenvalues of A .
lyche-numerical-linear-algebra-162-F02137	162	1.000	$q_{\{0\}}, \ \backslash\ldots, \ q_{\{n-1\}} \ \backslash\in \ \backslash\mathbb{C}$	$q_0, \dots, q_{n-1} \in \mathbb{C}$	Exercise 6.9 (Companion Matrix) For $q_0, \dots, q_{n-1} \in \mathbb{C}$ let $p(\lambda) = \lambda^n +$
lyche-numerical-linear-algebra-162-F02138	162	1.000	$p(\backslash\lambda)=\backslash\lambda^{\wedge\{n\}}+$	$p(\lambda) = \lambda^n +$	Exercise 6.9 (Companion Matrix) For $q_0, \dots, q_{n-1} \in \mathbb{C}$ let $p(\lambda) = \lambda^n +$
lyche-numerical-linear-algebra-162-F02139	162	1.000	$q_{\{n-1\}} \ \backslash\lambda^{\wedge\{n-1\}}+\backslash\cdots+q_{\{0\}}$	$q_{n-1} \lambda^{n-1} + \cdots + q_0$	$q_{n-1} \lambda^{n-1} + \cdots + q_0$ be a polynomial of degree n in λ . We derive two matrices
lyche-numerical-linear-algebra-162-F02140	162	1.000	$(-1)^{\wedge\{n\}} \ p$	$(-1)^n p$	that have $(-1)^n p$ as its characteristic polynomial.
lyche-numerical-linear-algebra-162-F02141	162	0.998	$p=(-1)^{\wedge\{n\}} \ \backslash\pi_{\{\backslash\boldsymbol{A}\}}$	$p = (-1)^n \pi_A$	a) Show that $p = (-1)^n \pi_A$ where
lyche-numerical-linear-algebra-162-F02142	162	1.000	$p=(-1)^{\wedge\{n\}} \ \backslash\pi_{\{\backslash\boldsymbol{B}\}}$	$p = (-1)^n \pi_B$	b) Show that $p = (-1)^n \pi_B$ where
lyche-numerical-linear-algebra-163-F02143	163	0.998	$\backslash\boldsymbol{A}=\backslash\left[\backslash\begin{array}{lll}1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2\end{array}\backslash\right]$	$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$. Is A defective?
lyche-numerical-linear-algebra-163-F02144	163	1.000	$\backslash\boldsymbol{Y} \ \backslash\in \ \backslash\mathbb{R}^{\wedge\{2\}} \ \backslash\times \ 2\}$	$Y \in \mathbb{R}^{2 \times 2}$	where $a \in \mathbb{R}$. There is a nonsingular matrix $Y \in \mathbb{R}^{2 \times 2}$ and a diagonal matrix
lyche-numerical-linear-algebra-163-F02145	163	1.000	$\backslash\boldsymbol{D} \ \backslash\in \ \backslash\mathbb{R}^{\wedge\{2\}} \ \backslash\times \ 2\}$	$D \in \mathbb{R}^{2 \times 2}$	$D \in \mathbb{R}^{2 \times 2}$ such that $A = Y D Y^{-1}$?
lyche-numerical-linear-algebra-163-F02146	163	1.000	$\backslash\boldsymbol{A}=\backslash\boldsymbol{Y} \ \backslash\boldsymbol{D} \ \backslash\boldsymbol{Y}^{\wedge\{-1\}}$	$A = Y D Y^{-1}$	$D \in \mathbb{R}^{2 \times 2}$ such that $A = Y D Y^{-1}$?
lyche-numerical-linear-algebra-163-F02147	163	1.000	$\backslash\boldsymbol{A} \ \backslash\in \ \backslash\mathbb{R}^{\wedge\{n, \ n\}}$	$A \in \mathbb{R}^{n, n}$	$A \in \mathbb{R}^{n, n}$ be tridiagonal (i.e. $a_{ij} = 0$ when $ i - j > 1$) and suppose also that
lyche-numerical-linear-algebra-163-F02148	163	1.000	$a_{\{i+1, \ i\}} \ a_{\{i, \ i+1\}}>0$	$a_{i+1, i} a_{i, i+1} > 0$	$a_{i+1, i} a_{i, i+1} > 0$ for $i = 1, \dots, n - 1$. Show that the eigenvalues of A are real. ²
lyche-numerical-linear-algebra-163-F02149	163	1.000	$\backslash\boldsymbol{A}=\backslash\left[\backslash\begin{array}{rrr}3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1\end{array}\backslash\right]$	$A = \begin{bmatrix} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{bmatrix}$	For the Jordan factorization of the matrix $A = \begin{bmatrix} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{bmatrix}$ we have $J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-163-F02150	163	1.000	$\backslash\boldsymbol{J}=\backslash\left[\backslash\begin{array}{lll}1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{array}\backslash\right]$	$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	For the Jordan factorization of the matrix $A = \begin{bmatrix} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{bmatrix}$ we have $J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

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lyche-numerical-linear-algebra-163 F02151	163	■ 1.000	$\left(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I}\right)^r = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_{m-r} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^r = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_{m-r} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	Exercise 6.14 (A Nilpotent Matrix) Show that $(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^r = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_{m-r} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$ for
lyche-numerical-linear-algebra-163 F02152	163	■ 0.998	$1 \leq r \leq m-1$	$1 \leq r \leq m-1$	$1 \leq r \leq m-1$ and conclude that $(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^m = \mathbf{0}$.
lyche-numerical-linear-algebra-163 F02153	163	■ 0.998	$\left(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I}\right)^m = \mathbf{0}$	$(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^m = \mathbf{0}$	$1 \leq r \leq m-1$ and conclude that $(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^m = \mathbf{0}$.
lyche-numerical-linear-algebra-163 F02154	163	■ 1.000	$r=0,1,2, \ldots, m=2,3, \ldots$	$r = 0, 1, 2, \dots, m = 2, 3, \dots$	Then for $r = 0, 1, 2, \dots, m = 2, 3, \dots$, and any $\lambda \in \mathbb{C}$
lyche-numerical-linear-algebra-163 F02155	163	■ 0.998	$\boldsymbol{A}^r = \boldsymbol{S} \boldsymbol{J}^r \boldsymbol{S}^{-1}$	$\boldsymbol{A}^r = \boldsymbol{S} \boldsymbol{J}^r \boldsymbol{S}^{-1}$	a) $\boldsymbol{A}^r = \boldsymbol{S} \boldsymbol{J}^r \boldsymbol{S}^{-1}$,
lyche-numerical-linear-algebra-163 F02156	163	■ 1.000	$\boldsymbol{J}^r = \text{diag}(\boldsymbol{U}_1^r, \dots, \boldsymbol{U}_k^r)$	$\boldsymbol{J}^r = \text{diag}(\boldsymbol{U}_1^r, \dots, \boldsymbol{U}_k^r)$	b) $\boldsymbol{J}^r = \text{diag}(\boldsymbol{U}_1^r, \dots, \boldsymbol{U}_k^r)$,
lyche-numerical-linear-algebra-163 F02157	163	■ 1.000	$\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$	$\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$	² Hint: show that there is a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$ is symmetric.
lyche-numerical-linear-algebra-164 F02158	164	■ 0.985	$\boldsymbol{U}_i^r = \text{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i)^r, \dots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)^r)$	$\boldsymbol{U}_i^r = \text{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i)^r, \dots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)^r)$	c) $\boldsymbol{U}_i^r = \text{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i)^r, \dots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)^r)$,
lyche-numerical-linear-algebra-164 F02159	164	■ 0.968	$\boldsymbol{J}_m(\lambda)^r = (\boldsymbol{E}_m + \lambda \boldsymbol{I}_m)^r = \sum_{k=0}^{\min\{r, m-1\}} \binom{r}{k} \lambda^{r-k} \boldsymbol{E}_m^k$	$\boldsymbol{J}_m(\lambda)^r = (\boldsymbol{E}_m + \lambda \boldsymbol{I}_m)^r = \sum_{k=0}^{\min\{r, m-1\}} \binom{r}{k} \lambda^{r-k} \boldsymbol{E}_m^k$	d) $\boldsymbol{J}_m(\lambda)^r = (\boldsymbol{E}_m + \lambda \boldsymbol{I}_m)^r = \sum_{k=0}^{\min\{r, m-1\}} \binom{r}{k} \lambda^{r-k} \boldsymbol{E}_m^k$.
lyche-numerical-linear-algebra-164 F02160	164	■ 1.000	$\boldsymbol{J}^{100}$	$\boldsymbol{J}^{100}$	Exercise 6.16 (Powers of a Jordan Block) Find $\boldsymbol{J}^{100}$ and $\boldsymbol{A}^{100}$ for the matrix in
lyche-numerical-linear-algebra-164 F02161	164	■ 1.000	$\boldsymbol{A}^{100}$	$\boldsymbol{A}^{100}$	Exercise 6.16 (Powers of a Jordan Block) Find $\boldsymbol{J}^{100}$ and $\boldsymbol{A}^{100}$ for the matrix in
lyche-numerical-linear-algebra-164 F02162	164	■ 1.000	$\mu_{\boldsymbol{A}}(\boldsymbol{A})$	$\mu_{\boldsymbol{A}}(\boldsymbol{A})$	is called the minimal polynomial of $\boldsymbol{A}$. We define the matrix polynomial $\mu_{\boldsymbol{A}}(\boldsymbol{A})$
lyche-numerical-linear-algebra-164 F02163	164	■ 1.000	$\lambda_i - \lambda$	$\lambda_i - \lambda$	by replacing the factors $\lambda_i - \lambda$ by $\lambda_i \boldsymbol{I} - \boldsymbol{A}$.
lyche-numerical-linear-algebra-164 F02164	164	■ 1.000	$\lambda_i \boldsymbol{I} - \boldsymbol{A}$	$\lambda_i \boldsymbol{I} - \boldsymbol{A}$	by replacing the factors $\lambda_i - \lambda$ by $\lambda_i \boldsymbol{I} - \boldsymbol{A}$.
lyche-numerical-linear-algebra-164 F02165	164	■ 1.000	$\pi_{\boldsymbol{A}}(\lambda) = \prod_{i=1}^k \prod_{j=1}^{g_i} (\lambda_i - \lambda)^{m_{i,j}}$	$\pi_{\boldsymbol{A}}(\lambda) = \prod_{i=1}^k \prod_{j=1}^{g_i} (\lambda_i - \lambda)^{m_{i,j}}$	a) We have $\pi_{\boldsymbol{A}}(\lambda) = \prod_{i=1}^k \prod_{j=1}^{g_i} (\lambda_i - \lambda)^{m_{i,j}}$. Use this to show that the minimal
lyche-numerical-linear-algebra-164 F02166	164	■ 0.629	$\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$	$\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$	polynomial divides the characteristic polynomial, i.e., $\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$ for some
lyche-numerical-linear-algebra-164 F02167	164	■ 0.953	$\nu_{\boldsymbol{A}}$	$\nu_{\boldsymbol{A}}$	polynomial $\nu_{\boldsymbol{A}}$.
lyche-numerical-linear-algebra-164 F02168	164	■ 0.986	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0} \iff \mu_{\boldsymbol{A}}(\boldsymbol{J}) = \mathbf{0}$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0} \iff \mu_{\boldsymbol{A}}(\boldsymbol{J}) = \mathbf{0}$	b) Show that $\mu_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0} \iff \mu_{\boldsymbol{A}}(\boldsymbol{J}) = \mathbf{0}$
lyche-numerical-linear-algebra-164 F02169	164	■ 1.000	m_i	m_i	c) (can be difficult) Use Exercises 6.14, 6.15 and the maximality of m_i to show that
lyche-numerical-linear-algebra-164 F02170	164	■ 1.000	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$. Thus a matrix satisfies its minimal equation. Finally show that the
lyche-numerical-linear-algebra-164 F02171	164	■ 1.000	$p(\boldsymbol{A}) = \mathbf{0}$	$p(\boldsymbol{A}) = \mathbf{0}$	degree of any polynomial p such that $p(\boldsymbol{A}) = \mathbf{0}$ is at least as large as the degree
lyche-numerical-linear-algebra-164 F02172	164	■ 1.000	$\pi_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$	$\pi_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$	its characteristic equation $\pi_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$.

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lyche-numerical-linear-algebra- F02173	164	1.000	$p(t) := \sum_{j=0}^r b_j t^j$	$p(t) := \sum_{j=0}^r b_j t^j$	is a polynomial given by $p(t) := \sum_{j=0}^r b_j t^j$, where $b_j \in \mathbb{C}$ and $A \in \mathbb{C}^{n \times n}$. We
lyche-numerical-linear-algebra- F02174	164	1.000	$b_j \in \mathbb{C}$	$b_j \in \mathbb{C}$	is a polynomial given by $p(t) := \sum_{j=0}^r b_j t^j$, where $b_j \in \mathbb{C}$ and $A \in \mathbb{C}^{n \times n}$. We
lyche-numerical-linear-algebra- F02175	164	1.000	$p(A) \in \mathbb{C}^{n \times n}$	$p(A) \in \mathbb{C}^{n \times n}$	define the matrix $p(A) \in \mathbb{C}^{n \times n}$ by
lyche-numerical-linear-algebra- F02176	164	1.000	$A^0 := I$	$A^0 := I$	where $A^0 := I$. From this it follows that if $p(t) := (t - \alpha_1) \cdots (t - \alpha_r)$ for some
lyche-numerical-linear-algebra- F02177	164	1.000	$p(t) := (t - \alpha_1) \cdots (t - \alpha_r)$	$p(t) := (t - \alpha_1) \cdots (t - \alpha_r)$	where $A^0 := I$. From this it follows that if $p(t) := (t - \alpha_1) \cdots (t - \alpha_r)$ for some
lyche-numerical-linear-algebra- F02178	164	1.000	$\alpha_0, \dots, \alpha_r \in \mathbb{C}$	$\alpha_0, \dots, \alpha_r \in \mathbb{C}$	$\alpha_0, \dots, \alpha_r \in \mathbb{C}$ then $p(A) = (A - \alpha_1) \cdots (A - \alpha_r)$. We accept this without proof.
lyche-numerical-linear-algebra- F02179	164	1.000	$p(A) = (A - \alpha_1) \cdots (A - \alpha_r)$	$p(A) = (A - \alpha_1) \cdots (A - \alpha_r)$	$\alpha_0, \dots, \alpha_r \in \mathbb{C}$ then $p(A) = (A - \alpha_1) \cdots (A - \alpha_r)$. We accept this without proof.
lyche-numerical-linear-algebra- F02180	164	1.000	$U^* A U = T$	$U^* A U = T$	Let $U^* A U = T$, where U is unitary and T upper triangular with the eigenvalues
lyche-numerical-linear-algebra- F02181	164	0.997	$\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$	a) Find the characteristic polynomial π_A to $\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Show that $\pi(A) = 0$.
lyche-numerical-linear-algebra- F02182	164	0.997	$\pi(A) = 0$	$\pi(A) = 0$	a) Find the characteristic polynomial π_A to $\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Show that $\pi(A) = 0$.
lyche-numerical-linear-algebra- F02183	164	1.000	$p(A) =$	$p(A) =$	b) Let now $A \in \mathbb{C}^{n \times n}$ be arbitrary. For any polynomial p show that $p(A) =$
lyche-numerical-linear-algebra- F02184	164	0.877	$U p(T) U^*$	$U p(T) U^*$	$U p(T) U^*$.
lyche-numerical-linear-algebra- F02185	164	1.000	$n, k \in \mathbb{N}$	$n, k \in \mathbb{N}$	c) Let $n, k \in \mathbb{N}$ with $1 \leq k < n$. Let $C, D \in \mathbb{C}^{n \times n}$ be upper triangular. Moreover,
lyche-numerical-linear-algebra- F02186	164	1.000	$C, D \in \mathbb{C}^{n \times n}$	$C, D \in \mathbb{C}^{n \times n}$	c) Let $n, k \in \mathbb{N}$ with $1 \leq k < n$. Let $C, D \in \mathbb{C}^{n \times n}$ be upper triangular. Moreover,
lyche-numerical-linear-algebra- F02187	164	1.000	$c_{i,j} = 0$	$c_{i,j} = 0$	$c_{i,j} = 0$ for $i, j \leq k$ and $d_{k+1,k+1} = 0$. Define $E := CD$ and show that $e_{i,j} = 0$
lyche-numerical-linear-algebra- F02188	164	1.000	$i, j \leq k$	$i, j \leq k$	$c_{i,j} = 0$ for $i, j \leq k$ and $d_{k+1,k+1} = 0$. Define $E := CD$ and show that $e_{i,j} = 0$
lyche-numerical-linear-algebra- F02189	164	1.000	$d_{k+1,k+1} = 0$	$d_{k+1,k+1} = 0$	$c_{i,j} = 0$ for $i, j \leq k$ and $d_{k+1,k+1} = 0$. Define $E := CD$ and show that $e_{i,j} = 0$
lyche-numerical-linear-algebra- F02190	164	1.000	$E := CD$	$E := CD$	$c_{i,j} = 0$ for $i, j \leq k$ and $d_{k+1,k+1} = 0$. Define $E := CD$ and show that $e_{i,j} = 0$
lyche-numerical-linear-algebra- F02191	164	1.000	$e_{i,j} = 0$	$e_{i,j} = 0$	$c_{i,j} = 0$ for $i, j \leq k$ and $d_{k+1,k+1} = 0$. Define $E := CD$ and show that $e_{i,j} = 0$
lyche-numerical-linear-algebra- F02192	164	1.000	$i, j \leq k+1$	$i, j \leq k+1$	for $i, j \leq k+1$.
lyche-numerical-linear-algebra- F02193	165	0.773	$p := \pi_A$	$p := \pi_A$	d) Now let $p := \pi_A$ be the characteristic polynomial of A . Show that $p(T) = 0$. ³
lyche-numerical-linear-algebra- F02194	165	0.773	$p(T) = 0$. ³	$p(T) = 0$. ³	d) Now let $p := \pi_A$ be the characteristic polynomial of A . Show that $p(T) = 0$. ³
lyche-numerical-linear-algebra- F02195	165	1.000	$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ is $U^T A U = \begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$, where $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra- F02196	165	1.000	$U^T A U = \begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$	$U^T A U = \begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ is $U^T A U = \begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$, where $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra- F02197	165	1.000	$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$	$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ is $U^T A U = \begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$, where $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.

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lyche-numerical-linear-algebra- F02198	165	■ 0.998	$\backslash\boldsymbol{C}=\backslash\boldsymbol{A}+i\ \backslash\boldsymbol{B}$	$C = A + iB$	Exercise 6.20 (Skew-Hermitian Matrix) Suppose $C = A + iB$, where $A, B \in$
lyche-numerical-linear-algebra- F02199	165	■ 1.000	$\backslash\boldsymbol{A}^{\wedge T}=-\backslash\boldsymbol{A}$	$A^T = -A$	$\mathbb{R}^{n \times n}$. Show that C is skew-Hermitian if and only if $A^T = -A$ and $B^T = B$.
lyche-numerical-linear-algebra- F02200	165	■ 1.000	$\backslash\boldsymbol{B}^{\wedge T}=\backslash\boldsymbol{B}$	$B^T = B$	$\mathbb{R}^{n \times n}$. Show that C is skew-Hermitian if and only if $A^T = -A$ and $B^T = B$.
lyche-numerical-linear-algebra- F02201	165	■ 1.000	$\backslash\left(\backslash\lambda_{\{1\}},\ \backslash\boldsymbol{u}_{\{1\}}\backslash\right),\ \backslash\ldots,\ \backslash\left(\backslash\lambda_{\{n\}},\ \backslash\boldsymbol{u}_{\{n\}}\backslash\right)$	$(\lambda_1, \boldsymbol{u}_1), \dots, (\lambda_n, \boldsymbol{u}_n)$	that if the eigenpairs $\boxed{\lambda_1, \boldsymbol{u}_1), \dots, (\lambda_n, \boldsymbol{u}_n)}$ of $A \in \mathbb{C}^{n \times n}$ are orthogonal, i.e.,
lyche-numerical-linear-algebra- F02202	165	■ 0.999	$\backslash\boldsymbol{u}_{\{j\}}^{\wedge *} \ \backslash\boldsymbol{u}_{\{k\}}=0$	$\boldsymbol{u}_j^* \boldsymbol{u}_k = 0$	$\boldsymbol{u}_j^* \boldsymbol{u}_k = 0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $A\boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-algebra- F02203	165	■ 0.999	$j\ \backslash\neq\ k$	$j \neq k$	$\boldsymbol{u}_j^* \boldsymbol{u}_k = 0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $A\boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-algebra- F02204	165	■ 0.999	$\backslash\boldsymbol{A}\ \backslash\boldsymbol{x}\ \backslash\mathrm{in}\ \backslash\mathbb{C}^{\wedge n}$	$A\boldsymbol{x} \in \mathbb{C}^n$	$\boldsymbol{u}_j^* \boldsymbol{u}_k = 0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $A\boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-algebra- F02205	165	■ 0.996	$\backslash\lambda_{\{\text{min}\ \}}$	$\lambda_{\min}$	where $\lambda_{\min}$ is the smallest eigenvalue of A , and $R(\boldsymbol{x})$ is the Rayleigh quotient
lyche-numerical-linear-algebra- F02206	165	■ 0.996	$R(\backslash\boldsymbol{x})$	$R(\boldsymbol{x})$	where $\lambda_{\min}$ is the smallest eigenvalue of A , and $R(\boldsymbol{x})$ is the Rayleigh quotient
lyche-numerical-linear-algebra- F02207	166	■ 0.997	$\backslash\boldsymbol{y}\ \backslash\neq\ 0$	$\boldsymbol{y} \neq 0$	b) Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ such that $\ \boldsymbol{x}\ _2 = 1$ and $\boldsymbol{y} \neq 0$. Show that
lyche-numerical-linear-algebra- F02208	166	■ 1.000	$t>0$	$t > 0$	where $t > 0$ is small. ⁴
lyche-numerical-linear-algebra- F02209	166	■ 1.000	$\backslash\boldsymbol{x}^{\wedge 0}\ \backslash\mathrm{in}\ B_{\{1\}}$	$\boldsymbol{x}^0 \in B_1$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $A\boldsymbol{x}^0 - R(\boldsymbol{x}^0)\boldsymbol{x}^0 \neq 0$, i.e., $(R(\boldsymbol{x}^0), \boldsymbol{x}^0)$ is not an
lyche-numerical-linear-algebra- F02210	166	■ 1.000	$\backslash\boldsymbol{A}\ \backslash\boldsymbol{x}^{\wedge 0}-R\backslash\left(\backslash\boldsymbol{x}^{\wedge 0}\backslash\right)\ \backslash\boldsymbol{x}^{\wedge 0}\ \backslash\neq\ 0$	$A\boldsymbol{x}^0 - R(\boldsymbol{x}^0)\boldsymbol{x}^0 \neq 0$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $\boxed{A\boldsymbol{x}^0 - R(\boldsymbol{x}^0)\boldsymbol{x}^0 \neq 0}$, i.e., $(R(\boldsymbol{x}^0), \boldsymbol{x}^0)$ is not an
lyche-numerical-linear-algebra- F02211	166	■ 1.000	$\backslash\left(R\backslash\left(\backslash\boldsymbol{x}^{\wedge 0}\backslash\right),\ \backslash\boldsymbol{x}^{\wedge 0}\backslash\right)$	$(R(\boldsymbol{x}^0), \boldsymbol{x}^0)$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $A\boldsymbol{x}^0 - R(\boldsymbol{x}^0)\boldsymbol{x}^0 \neq 0$, i.e., $(R(\boldsymbol{x}^0), \boldsymbol{x}^0)$ is not an
lyche-numerical-linear-algebra- F02212	166	■ 1.000	$\backslash\boldsymbol{x}^{\wedge 1}\ \backslash\mathrm{in}\ B_{\{1\}}$	$\boldsymbol{x}^1 \in B_1$	eigenpair for A . Explain how we can find a vector $\boldsymbol{x}^1 \in B_1$ such that $R(\boldsymbol{x}^1) <$
lyche-numerical-linear-algebra- F02213	166	■ 1.000	$R\backslash\left(\backslash\boldsymbol{x}^{\wedge 1}\backslash\right)<$	$R(\boldsymbol{x}^1) <$	eigenpair for A . Explain how we can find a vector $\boldsymbol{x}^1 \in B_1$ such that $R(\boldsymbol{x}^1) <$
lyche-numerical-linear-algebra- F02214	166	■ 0.993	$R\backslash\left(\backslash\boldsymbol{x}^{\wedge 0}\backslash\right)$	$R(\boldsymbol{x}^0)$	$R(\boldsymbol{x}^0)$.
lyche-numerical-linear-algebra- F02215	166	■ 1.000	$\backslash\beta_{\{i\}}\ \backslash\geq\ \backslash\alpha_{\{i\}}$	$\beta_i \geq \alpha_i$	Theorem 6.13, if E is symmetric positive semidefinite then $\boxed{\beta_i \geq \alpha_i}$.
lyche-numerical-linear-algebra- F02216	166	■ 1.000	$\backslash\boldsymbol{A}:=\backslashleft[\backslashbegin{array}{ll}0\ \&\ 0\ \backslash\ 0\ \&\ 4\ \backslash\end{array}\backslash\right]$	$A := \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$	ces $A := \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$ and $B := \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$. Why does this not contradict the Hoffman-
lyche-numerical-linear-algebra- F02217	166	■ 1.000	$\backslash\boldsymbol{B}:=\backslashleft[\backslashbegin{array}{cc}-1\ \&\ -1\ \backslash\ 1\ \&\ 1\ \backslash\end{array}\backslash\right]$	$B := \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$	ces $A := \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$ and $B := \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$. Why does this not contradict the Hoffman-
lyche-numerical-linear-algebra- F02218	166	■ 1.000	$\backslash\boldsymbol{A}:=\backslashleft[\backslashbegin{array}{ll}3\ \&\ 1\ \backslash\ 2\ \&\ 2\ \backslash\end{array}\backslash\right]$	$A := \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$	the matrix $A := \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$ and the two expansions in (6.16) for any $\boldsymbol{v} \in \mathbb{R}^2$.
lyche-numerical-linear-algebra- F02219	166	■ 1.000	$\backslash\boldsymbol{v}\ \backslash\mathrm{in}\ \backslash\mathbb{R}^{\wedge 2}$	$\boldsymbol{v} \in \mathbb{R}^2$	the matrix $A := \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$ and the two expansions in (6.16) for any $\boldsymbol{v} \in \mathbb{R}^2$.
lyche-numerical-linear-algebra- F02220	166	■ 1.000	$\backslash\boldsymbol{y},\ \backslash\boldsymbol{x}\ \backslash\mathrm{in}$	$\boldsymbol{y}, \boldsymbol{x} \in$	Exercise 6.27 (Generalized Rayleigh Quotient) For $A \in \mathbb{C}^{n \times n}$ and any $\boldsymbol{y}, \boldsymbol{x} \in$
lyche-numerical-linear-algebra- F02221	166	■ 1.000	$R(\backslash\boldsymbol{y},\ \backslash\boldsymbol{x})=R_{\{\backslash\boldsymbol{A}\}}(\backslash\boldsymbol{y},\ \backslash\boldsymbol{x}):=\frac{\backslash\boldsymbol{y}^{\wedge *} \ \backslash\boldsymbol{A}\ \backslash\boldsymbol{x}}{\backslash\boldsymbol{y}^{\wedge *} \ \backslash\boldsymbol{x}}$	$R(\boldsymbol{y}, \boldsymbol{x}) = R_A(\boldsymbol{y}, \boldsymbol{x}) := \frac{\boldsymbol{y}^* A \boldsymbol{x}}{\boldsymbol{y}^* \boldsymbol{x}}$	$\mathbb{C}^n$ with $\boldsymbol{y}^* \boldsymbol{x} \neq 0$ the quantity $\boxed{R(\boldsymbol{y}, \boldsymbol{x}) = R_A(\boldsymbol{y}, \boldsymbol{x}) := \frac{\boldsymbol{y}^* A \boldsymbol{x}}{\boldsymbol{y}^* \boldsymbol{x}}}$ is called a generalized
lyche-numerical-linear-algebra- F02222	166	■ 1.000	$R(\backslash\boldsymbol{y},\ \backslash\boldsymbol{x})=\backslash\lambda$	$R(\boldsymbol{y}, \boldsymbol{x}) = \lambda$	$\boxed{R(\boldsymbol{y}, \boldsymbol{x}) = \lambda}$ for any $\boldsymbol{y}$ with $\boldsymbol{y}^* \boldsymbol{x} \neq 0$. Also show that if $(\lambda, \boldsymbol{y})$ is a left eigenpair for

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lyche-numerical-linear-algebra- F02223	166	1.000	$\boldsymbol{A}, \boldsymbol{A}^T$	$\boldsymbol{A}, \boldsymbol{A}^T$	6.7.1 Does $\boldsymbol{A}, \boldsymbol{A}^T$ and $\boldsymbol{A}^*$ have the same eigenvalues? What about $\boldsymbol{A}^* \boldsymbol{A}$ and
lyche-numerical-linear-algebra- F02224	166	1.000	$\boldsymbol{A} \boldsymbol{A}^*$	$\boldsymbol{A} \boldsymbol{A}^*$	$\boldsymbol{A} \boldsymbol{A}^*$?
lyche-numerical-linear-algebra- F02225	166	0.993	$f(t)=R(\boldsymbol{x}-t \boldsymbol{y})$	$f(t) = R(\boldsymbol{x} - t\boldsymbol{y})$	⁴ Hint: Use Taylor's theorem for the function $f(t) = R(\boldsymbol{x} - t\boldsymbol{y})$.
lyche-numerical-linear-algebra- F02226	168	1.000	$\boldsymbol{U}:=\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\right] \in \mathbb{C}^{n \times n}$	$\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_n] \in \mathbb{C}^{n \times n}$	eigenpairs $(\lambda_1, \boldsymbol{u}_1), \dots, (\lambda_n, \boldsymbol{u}_n)$. Letting $\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_n] \in \mathbb{C}^{n \times n}$ and $\boldsymbol{D} :=$
lyche-numerical-linear-algebra- F02227	168	1.000	$\boldsymbol{D}:=$	$\boldsymbol{D} :=$	eigenpairs $(\lambda_1, \boldsymbol{u}_1), \dots, (\lambda_n, \boldsymbol{u}_n)$. Letting $\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_n] \in \mathbb{C}^{n \times n}$ and $\boldsymbol{D} :=$
lyche-numerical-linear-algebra- F02228	168	1.000	$\operatorname{diag}\left(\lambda_1, \ldots, \lambda_n\right)$	$\operatorname{diag}(\lambda_1, \dots, \lambda_n)$	$\operatorname{diag}(\lambda_1, \dots, \lambda_n)$ we have the spectral decomposition
lyche-numerical-linear-algebra- F02229	168	1.000	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$, where $\boldsymbol{U}$ and $\boldsymbol{V}$ are unitary, and $\boldsymbol{\Sigma}$ is a nonnegative diagonal matrix,
lyche-numerical-linear-algebra- F02230	168	1.000	$\boldsymbol{\Sigma}$	$\boldsymbol{\Sigma}$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$, where $\boldsymbol{U}$ and $\boldsymbol{V}$ are unitary, and $\boldsymbol{\Sigma}$ is a nonnegative diagonal matrix,
lyche-numerical-linear-algebra- F02231	168	1.000	$\Sigma_{i j}=0$	$\Sigma_{ij} = 0$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i . The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra- F02232	168	1.000	$\Sigma_{i i} \geq 0$	$\Sigma_{ii} \geq 0$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i . The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra- F02233	168	1.000	$\sigma_i:=\Sigma_{i i}$	$\sigma_i := \Sigma_{ii}$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i . The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra- F02234	168	1.000	$\sigma_i \geq \sigma_{i+1}$	$\sigma_i \geq \sigma_{i+1}$	$\sigma_i \geq \sigma_{i+1}$ for all i .
lyche-numerical-linear-algebra- F02235	169	1.000	$\sigma_1=2$	$\sigma_1 = 2$	mal, and $\boldsymbol{\Sigma}$ is a nonnegative diagonal matrix with singular values $\sigma_1 = 2$ and
lyche-numerical-linear-algebra- F02236	169	1.000	$\sigma_2=1$	$\sigma_2 = 1$	$\sigma_2 = 1$.
lyche-numerical-linear-algebra- F02237	169	1.000	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A})$	To start we show that bases for the four fundamental subspaces $\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A})$,
lyche-numerical-linear-algebra- F02238	169	1.000	$\mathcal{R}\left(\boldsymbol{A}^*\right)$	$\mathcal{R}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{A}^*)$ and $\mathcal{N}(\boldsymbol{A}^*)$ of a matrix $\boldsymbol{A}$ can be determined from the eigenpairs of $\boldsymbol{A}^* \boldsymbol{A}$
lyche-numerical-linear-algebra- F02239	169	1.000	$\mathcal{N}\left(\boldsymbol{A}^*\right)$	$\mathcal{N}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{A}^*)$ and $\mathcal{N}(\boldsymbol{A}^*)$ of a matrix $\boldsymbol{A}$ can be determined from the eigenpairs of $\boldsymbol{A}^* \boldsymbol{A}$
lyche-numerical-linear-algebra- F02240	169	0.978	$\left.\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^*\right)$	$\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^*)$	Theorem 7.1 (The Matrices $\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^*$) Suppose $m, n \in \mathbb{N}$ and $\boldsymbol{A} \in \mathbb{C}^{m \times n}$.
lyche-numerical-linear-algebra- F02241	169	1.000	$\boldsymbol{A}^* \boldsymbol{A} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}^* \boldsymbol{A} \in \mathbb{C}^{n \times n}$	1. The matrices $\boldsymbol{A}^* \boldsymbol{A} \in \mathbb{C}^{n \times n}$ and $\boldsymbol{A} \boldsymbol{A}^* \in \mathbb{C}^{m \times m}$ have the same nonzero eigenval-
lyche-numerical-linear-algebra- F02242	169	1.000	$\boldsymbol{A} \boldsymbol{A}^* \in \mathbb{C}^{m \times m}$	$\boldsymbol{A} \boldsymbol{A}^* \in \mathbb{C}^{m \times m}$	1. The matrices $\boldsymbol{A}^* \boldsymbol{A} \in \mathbb{C}^{n \times n}$ and $\boldsymbol{A} \boldsymbol{A}^* \in \mathbb{C}^{m \times m}$ have the same nonzero eigenval-
lyche-numerical-linear-algebra- F02243	169	0.992	$\left(\lambda_j, \boldsymbol{v}_j\right)$	$(\lambda_j, \boldsymbol{v}_j)$	3. Let $(\lambda_j, \boldsymbol{v}_j)$ be orthonormal eigenpairs for $\boldsymbol{A}^* \boldsymbol{A}$ with
lyche-numerical-linear-algebra- F02244	169	1.000	$\left\{\boldsymbol{A} \boldsymbol{v}_1, \ldots, \boldsymbol{A} \boldsymbol{v}_r\right\}$	$\{\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_r\}$	Then $\{\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_r\}$ is an orthogonal basis for the column space $\mathcal{R}(\boldsymbol{A}) :=$
lyche-numerical-linear-algebra- F02245	169	1.000	$\mathcal{R}(\boldsymbol{A}) :=$	$\mathcal{R}(\boldsymbol{A}) :=$	Then $\{\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_r\}$ is an orthogonal basis for the column space $\mathcal{R}(\boldsymbol{A}) :=$
lyche-numerical-linear-algebra- F02246	169	1.000	$\left\{\boldsymbol{A} \boldsymbol{y} \in \mathbb{C}^m: \boldsymbol{y} \in \mathbb{C}^n\right\}$	$\{\boldsymbol{A} \boldsymbol{y} \in \mathbb{C}^m : \boldsymbol{y} \in \mathbb{C}^n\}$	$\{\boldsymbol{A} \boldsymbol{y} \in \mathbb{C}^m : \boldsymbol{y} \in \mathbb{C}^n\}$ and $\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n\}$ is an <u>orthonormal basis for</u> the
lyche-numerical-linear-algebra- F02247	169	1.000	$\left\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\right\}$	$\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n\}$	$\{\boldsymbol{A} \boldsymbol{y} \in \mathbb{C}^m : \boldsymbol{y} \in \mathbb{C}^n\}$ and $\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n\}$ is an orthonormal basis for the
lyche-numerical-linear-algebra- F02248	169	0.899	$\mathcal{N}(\boldsymbol{A}) :=\left\{\boldsymbol{y} \in \mathbb{C}^n: \boldsymbol{A} \boldsymbol{y}=\mathbf{0}\right\}$	$\mathcal{N}(\boldsymbol{A}) := \{\boldsymbol{y} \in \mathbb{C}^n : \boldsymbol{A} \boldsymbol{y} = \mathbf{0}\}$	nullspace <u>$\mathcal{N}(\boldsymbol{A}) := \{\boldsymbol{y} \in \mathbb{C}^n : \boldsymbol{A} \boldsymbol{y} = \mathbf{0}\}$</u> .
lyche-numerical-linear-algebra- F02249	169	0.996	$\left(\lambda_j, \boldsymbol{u}_j\right)$	$(\lambda_j, \boldsymbol{u}_j)$	4. Let $(\lambda_j, \boldsymbol{u}_j)$ be orthonormal eigenpairs for $\boldsymbol{A} \boldsymbol{A}^*$. If $\lambda_j > 0, j = 1, \dots, r$ and

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lyche-numerical-linear-algebra- F02250	169	■ 0.996	$\lambda_j > 0, j=1, \ldots, r$	$\lambda_j > 0, j = 1, \dots, r$	4. Let $(\lambda_j, \mathbf{u}_j)$ be orthonormal eigenpairs for $\mathbf{A}\mathbf{A}^*$. If $\lambda_j > 0, j = 1, \dots, r$ and
lyche-numerical-linear-algebra- F02251	169	■ 1.000	$\lambda_j = 0, j=r+1, \ldots, m$	$\lambda_j = 0, j = r + 1, \dots, m$	$\lambda_j = 0, j = r + 1, \dots, m$ then $\{\mathbf{A}^*\mathbf{u}_1, \dots, \mathbf{A}^*\mathbf{u}_r\}$ is an orthogonal basis for
lyche-numerical-linear-algebra- F02252	169	■ 1.000	$\left\{ \mathbf{A}^* \mathbf{u}_1, \dots, \mathbf{A}^* \mathbf{u}_r \right\}$	$\{\mathbf{A}^*\mathbf{u}_1, \dots, \mathbf{A}^*\mathbf{u}_r\}$	$\lambda_j = 0, j = r + 1, \dots, m$ then $\{\mathbf{A}^*\mathbf{u}_1, \dots, \mathbf{A}^*\mathbf{u}_r\}$ is an orthogonal basis for
lyche-numerical-linear-algebra- F02253	169	■ 1.000	$\left\{ \mathbf{u}_{r+1}, \dots, \mathbf{u}_m \right\}$	$\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_m\}$	the column space $\mathcal{R}(\mathbf{A}^*)$ and $\{\mathbf{u}_{r+1}, \dots, \mathbf{u}_m\}$ is an orthonormal basis for the
lyche-numerical-linear-algebra- F02254	169	■ 1.000	$\pi_{\mathbf{A}^* \mathbf{A}}$	$\pi_{\mathbf{A}^* \mathbf{A}}$	1. Consider the characteristic polynomials $\pi_{\mathbf{A}^* \mathbf{A}}$ and $\pi_{\mathbf{A} \mathbf{A}^*}$. By (6.1) we have
lyche-numerical-linear-algebra- F02255	169	■ 1.000	$\pi_{\mathbf{A} \mathbf{A}^*}$	$\pi_{\mathbf{A} \mathbf{A}^*}$	1. Consider the characteristic polynomials $\pi_{\mathbf{A}^* \mathbf{A}}$ and $\pi_{\mathbf{A} \mathbf{A}^*}$. By (6.1) we have
lyche-numerical-linear-algebra- F02256	170	■ 1.000	$\mathbf{A}^* \mathbf{A} \mathbf{v} = \lambda \mathbf{v}$	$\mathbf{A}^* \mathbf{A} \mathbf{v} = \lambda \mathbf{v}$	$\mathbf{A}^* \mathbf{A} \mathbf{v} = \lambda \mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then
lyche-numerical-linear-algebra- F02257	170	■ 1.000	$\mathbf{v} \neq \mathbf{0}$	$\mathbf{v} \neq \mathbf{0}$	$\mathbf{A}^* \mathbf{A} \mathbf{v} = \lambda \mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then
lyche-numerical-linear-algebra- F02258	170	■ 1.000	$\left(\mathbf{A} \mathbf{v}_j \right)^* \mathbf{A} \mathbf{v}_k = \mathbf{v}_j^* \mathbf{A}^* \mathbf{A} \mathbf{v}_k = \lambda_k \mathbf{v}_j^* \mathbf{v}_k$	$(\mathbf{A} \mathbf{v}_j)^* \mathbf{A} \mathbf{v}_k = \mathbf{v}_j^* \mathbf{A}^* \mathbf{A} \mathbf{v}_k = \lambda_k \mathbf{v}_j^* \mathbf{v}_k$	3. By orthonormality of $\mathbf{v}_1, \dots, \mathbf{v}_n$ we have $(\mathbf{A} \mathbf{v}_j)^* \mathbf{A} \mathbf{v}_k = \mathbf{v}_j^* \mathbf{A}^* \mathbf{A} \mathbf{v}_k = \lambda_k \mathbf{v}_j^* \mathbf{v}_k$
lyche-numerical-linear-algebra- F02259	170	■ 0.578	$\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_n$	$\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_n$	$= 0$ for $j \neq k$, showing that $\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_n$ are orthogonal vectors. Moreover,
lyche-numerical-linear-algebra- F02260	170	■ 1.000	$\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r$	$\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r$	(7.3) implies that $\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r$ are nonzero and $\mathbf{A} \mathbf{v}_j = \mathbf{0}$ for $j = r +$
lyche-numerical-linear-algebra- F02261	170	■ 1.000	$\mathbf{A} \mathbf{v}_j = \mathbf{0}$	$\mathbf{A} \mathbf{v}_j = \mathbf{0}$	(7.3) implies that $\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r$ are nonzero and $\mathbf{A} \mathbf{v}_j = \mathbf{0}$ for $j = r +$
lyche-numerical-linear-algebra- F02262	170	■ 1.000	$j = r +$	$j = r +$	(7.3) implies that $\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r$ are nonzero and $\mathbf{A} \mathbf{v}_j = \mathbf{0}$ for $j = r +$
lyche-numerical-linear-algebra- F02263	170	■ 1.000	$\mathcal{N}(\mathbf{A})$	$\mathcal{N}(\mathbf{A})$	linearly independent vectors in $\mathcal{R}(\mathbf{A})$ and $\mathcal{N}(\mathbf{A})$, respectively. The proof will
lyche-numerical-linear-algebra- F02264	170	■ 1.000	$\mathcal{R}(\mathbf{A}) \subset \text{span}(\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r)$	$\mathcal{R}(\mathbf{A}) \subset \text{span}(\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r)$	be complete once it is shown that $\mathcal{R}(\mathbf{A}) \subset \text{span}(\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r)$ and $\mathcal{N}(\mathbf{A}) \subset$
lyche-numerical-linear-algebra- F02265	170	■ 1.000	$\mathcal{N}(\mathbf{A}) \subset$	$\mathcal{N}(\mathbf{A}) \subset$	be complete once it is shown that $\mathcal{R}(\mathbf{A}) \subset \text{span}(\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r)$ and $\mathcal{N}(\mathbf{A}) \subset$
lyche-numerical-linear-algebra- F02266	170	■ 0.999	$\text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$	$\text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$	$\text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$. Suppose $\mathbf{x} \in \mathcal{R}(\mathbf{A})$. Then $\mathbf{x} = \mathbf{A} \mathbf{y}$ for some $\mathbf{y} \in \mathbb{C}^n$,
lyche-numerical-linear-algebra- F02267	170	■ 0.999	$\mathbf{x} \in \mathcal{R}(\mathbf{A})$	$\mathbf{x} \in \mathcal{R}(\mathbf{A})$	$\text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$. Suppose $\mathbf{x} \in \mathcal{R}(\mathbf{A})$. Then $\mathbf{x} = \mathbf{A} \mathbf{y}$ for some $\mathbf{y} \in \mathbb{C}^n$,
lyche-numerical-linear-algebra- F02268	170	■ 0.999	$\mathbf{x} = \mathbf{A} \mathbf{y}$	$\mathbf{x} = \mathbf{A} \mathbf{y}$	$\text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_n)$. Suppose $\mathbf{x} \in \mathcal{R}(\mathbf{A})$. Then $\mathbf{x} = \mathbf{A} \mathbf{y}$ for some $\mathbf{y} \in \mathbb{C}^n$,
lyche-numerical-linear-algebra- F02269	170	■ 1.000	$\mathbf{y} = \sum_{j=1}^n c_j \mathbf{v}_j$	$\mathbf{y} = \sum_{j=1}^n c_j \mathbf{v}_j$	Let $\mathbf{y} = \sum_{j=1}^n c_j \mathbf{v}_j$ be an eigenvector expansion of $\mathbf{y}$. Since $\mathbf{A} \mathbf{v}_j = \mathbf{0}$ for
lyche-numerical-linear-algebra- F02270	170	■ 1.000	$\mathbf{x} = \mathbf{A} \mathbf{y} = \sum_{j=1}^n c_j \mathbf{A} \mathbf{v}_j = \sum_{j=1}^r c_j \mathbf{A} \mathbf{v}_j \in$	$\mathbf{x} = \mathbf{A} \mathbf{y} = \sum_{j=1}^n c_j \mathbf{A} \mathbf{v}_j = \sum_{j=1}^r c_j \mathbf{A} \mathbf{v}_j \in$	$j = r + 1, \dots, n$ we obtain $\mathbf{x} = \mathbf{A} \mathbf{y} = \sum_{j=1}^n c_j \mathbf{A} \mathbf{v}_j = \sum_{j=1}^r c_j \mathbf{A} \mathbf{v}_j \in$
lyche-numerical-linear-algebra- F02271	170	■ 1.000	$\text{span}(\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r)$	$\text{span}(\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r)$	$\text{span}(\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r)$. Finally, if $\mathbf{y} = \sum_{j=1}^n c_j \mathbf{v}_j \in \mathcal{N}(\mathbf{A})$, then we have
lyche-numerical-linear-algebra- F02272	170	■ 1.000	$\mathbf{y} = \sum_{j=1}^n c_j \mathbf{v}_j \in \mathcal{N}(\mathbf{A})$	$\mathbf{y} = \sum_{j=1}^n c_j \mathbf{v}_j \in \mathcal{N}(\mathbf{A})$	$\text{span}(\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r)$. Finally, if $\mathbf{y} = \sum_{j=1}^n c_j \mathbf{v}_j \in \mathcal{N}(\mathbf{A})$, then we have

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lyche-numerical-linear-algebra- F02273	170	1.000	$\boldsymbol{A} \boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{v}_j = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A} \boldsymbol{v}_j = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A} \boldsymbol{v}_j = \mathbf{0}$, and $c_1 = \dots = c_r = 0$ since $\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_r$ are
lyche-numerical-linear-algebra- F02274	170	1.000	$c_1 = \dots = c_r = 0$	$c_1 = \dots = c_r = 0$	$\boldsymbol{A} \boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A} \boldsymbol{v}_j = \mathbf{0}$, and $c_1 = \dots = c_r = 0$ since $\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_r$ are
lyche-numerical-linear-algebra- F02275	170	1.000	$\boldsymbol{y} = \sum_{j=r+1}^n c_j \boldsymbol{v}_j \in \text{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$	$\boldsymbol{y} = \sum_{j=r+1}^n c_j \boldsymbol{v}_j \in \text{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$	linearly independent. But then $\boldsymbol{y} = \sum_{j=r+1}^n c_j \boldsymbol{v}_j \in \text{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$.
lyche-numerical-linear-algebra- F02276	170	1.000	$\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{B}^* \boldsymbol{B}$	$\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{B}^* \boldsymbol{B}$	4. Since $\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{B}^* \boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra- F02277	170	1.000	$\boldsymbol{B} := \boldsymbol{A}^*$	$\boldsymbol{B} := \boldsymbol{A}^*$	4. Since $\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{B}^* \boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra- F02278	170	1.000	$\boldsymbol{A} = \boldsymbol{B}$	$\boldsymbol{A} = \boldsymbol{B}$	4. Since $\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{B}^* \boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra- F02279	170	1.000	$2 \boldsymbol{A}^* \boldsymbol{A}$	$2 \boldsymbol{A}^* \boldsymbol{A}$	5. By part 1 and 2 $\boldsymbol{A}^* \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^*$ have the same number r of positive eigenvalues
lyche-numerical-linear-algebra- F02280	170	1.000	$4r$	$4r$	and by part 3 and $4r$ is the rank of $\boldsymbol{A}$.
lyche-numerical-linear-algebra- F02281	170	1.000	$m, n, r \in \mathbb{N}$	$m, n, r \in \mathbb{N}$	Theorem 7.2 (Existence of SVD) Suppose for $m, n, r \in \mathbb{N}$ that $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ has
lyche-numerical-linear-algebra- F02282	170	1.000	$\lambda_1 \geq \dots \geq$	$\lambda_1 \geq \dots \geq$	rank r , and that $(\lambda_j, \boldsymbol{v}_j)$ are orthonormal eigenpairs for $\boldsymbol{A}^* \boldsymbol{A}$ with $\lambda_1 \geq \dots \geq$
lyche-numerical-linear-algebra- F02283	170	1.000	$\lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$	$\lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$	$\lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$. Define
lyche-numerical-linear-algebra- F02284	170	1.000	$\boldsymbol{V} := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	$\boldsymbol{V} := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	1. $\boldsymbol{V} := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra- F02285	170	1.000	$\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$	$\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$	2. $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$
lyche-numerical-linear-algebra- F02286	170	1.000	$\sigma_j := \sqrt{\lambda_j}$	$\sigma_j := \sqrt{\lambda_j}$	2. $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$
lyche-numerical-linear-algebra- F02287	170	1.000	$j =$	$j =$	2. $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$
lyche-numerical-linear-algebra- F02288	170	1.000	$1, \dots, \min(m, n)$	$1, \dots, \min(m, n)$	$1, \dots, \min(m, n)$.
lyche-numerical-linear-algebra- F02289	170	0.999	$\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$	$\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$	3. $\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$, where $\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A} \boldsymbol{v}_j$ for $j = 1, \dots, r$ and
lyche-numerical-linear-algebra- F02290	170	0.999	$\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A} \boldsymbol{v}_j$	$\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A} \boldsymbol{v}_j$	3. $\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$, where $\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A} \boldsymbol{v}_j$ for $j = 1, \dots, r$ and
lyche-numerical-linear-algebra- F02291	170	0.999	$j=1, \dots, r$	$j = 1, \dots, r$	3. $\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$, where $\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A} \boldsymbol{v}_j$ for $j = 1, \dots, r$ and
lyche-numerical-linear-algebra- F02292	170	1.000	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$ is any extension of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ to an orthonormal basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra- F02293	170	1.000	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$ is any extension of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ to an orthonormal basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra- F02294	170	1.000	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$ is any extension of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ to an orthonormal basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra- F02295	170	1.000	$\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$	$\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$	Proof Let $\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$ be as in the theorem. The vectors $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ are orthonormal
lyche-numerical-linear-algebra- F02296	170	1.000	$\sigma_j = \ \boldsymbol{A} \boldsymbol{v}_j\ _2 > 0, j=1, \dots, r$	$\sigma_j = \ \boldsymbol{A} \boldsymbol{v}_j\ _2 > 0, j = 1, \dots, r$	since $\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_r$ are orthogonal and $\sigma_j = \ \boldsymbol{A} \boldsymbol{v}_j\ _2 > 0, j = 1, \dots, r$ by (7.3).
lyche-numerical-linear-algebra- F02297	171	1.000	$\boldsymbol{U} \boldsymbol{\Sigma} = \boldsymbol{A} \boldsymbol{V}$	$\boldsymbol{U} \boldsymbol{\Sigma} = \boldsymbol{A} \boldsymbol{V}$	Thus $\boldsymbol{U} \boldsymbol{\Sigma} = \boldsymbol{A} \boldsymbol{V}$ and since $\boldsymbol{V}$ is square and unitary we find $\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = \boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^* = \boldsymbol{A}$

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lyche-numerical-linear-algebra-F02298	171	1.000	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = \boldsymbol{A}$ $\boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^* = \boldsymbol{A}$	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = \boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^* = \boldsymbol{A}$	Thus $\boldsymbol{U} \boldsymbol{\Sigma} = \boldsymbol{A} \boldsymbol{V}$ and since $\boldsymbol{V}$ is square and unitary we find $\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = \boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^* = \boldsymbol{A}$
lyche-numerical-linear-algebra-F02299	171	1.000	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$	and we have an SVD of $\boldsymbol{A}$ with $\sigma_1 > \sigma_2 > \cdots > \sigma_r$.
lyche-numerical-linear-algebra-F02300	171	1.000	$\boldsymbol{A} = \frac{1}{15} \begin{bmatrix} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{bmatrix}$	$\boldsymbol{A} = \frac{1}{15} \begin{bmatrix} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{bmatrix}$	<i>Example 7.2 (Find SVD)</i> To derive the SVD in (7.2) where $\boldsymbol{A} = \frac{1}{15} \begin{bmatrix} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{bmatrix}$, we
lyche-numerical-linear-algebra-F02301	171	0.990	$\sigma_1=2, \sigma_2=1$	$\sigma_1 = 2, \sigma_2 = 1$	Thus $\sigma_1 = 2, \sigma_2 = 1$, and $\boldsymbol{V} = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$. Now $\boldsymbol{u}_1 = \boldsymbol{A} \boldsymbol{v}_1 / \sigma_1 = [1, 2, 2]^T / 3$,
lyche-numerical-linear-algebra-F02302	171	0.990	$\boldsymbol{V} = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	$\boldsymbol{V} = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	Thus $\sigma_1 = 2, \sigma_2 = 1$, and $\boldsymbol{V} = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$. Now $\boldsymbol{u}_1 = \boldsymbol{A} \boldsymbol{v}_1 / \sigma_1 = [1, 2, 2]^T / 3$,
lyche-numerical-linear-algebra-F02303	171	0.990	$\boldsymbol{u}_1 = \boldsymbol{A} \boldsymbol{v}_1 / \sigma_1 = [1, 2, 2]^T / 3$	$\boldsymbol{u}_1 = \boldsymbol{A} \boldsymbol{v}_1 / \sigma_1 = [1, 2, 2]^T / 3$	Thus $\sigma_1 = 2, \sigma_2 = 1$, and $\boldsymbol{V} = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$. Now $\boldsymbol{u}_1 = \boldsymbol{A} \boldsymbol{v}_1 / \sigma_1 = [1, 2, 2]^T / 3$,
lyche-numerical-linear-algebra-F02304	171	0.998	$\boldsymbol{u}_2 = \boldsymbol{A} \boldsymbol{v}_2 / \sigma_2 = [2, -2, 1]^T / 3$	$\boldsymbol{u}_2 = \boldsymbol{A} \boldsymbol{v}_2 / \sigma_2 = [2, -2, 1]^T / 3$	$\boldsymbol{u}_2 = \boldsymbol{A} \boldsymbol{v}_2 / \sigma_2 = [2, -2, 1]^T / 3$. For an SVD we also need $\boldsymbol{u}_3$ which is any vector
lyche-numerical-linear-algebra-F02305	171	0.998	$\boldsymbol{u}_3$	$\boldsymbol{u}_3$	$\boldsymbol{u}_2 = \boldsymbol{A} \boldsymbol{v}_2 / \sigma_2 = [2, -2, 1]^T / 3$. For an SVD we also need $\boldsymbol{u}_3$ which is any vector
lyche-numerical-linear-algebra-F02306	171	1.000	$\boldsymbol{u}_1$	$\boldsymbol{u}_1$	of length one orthogonal to $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$. $\boldsymbol{u}_3 = [2, 1, -2]^T / 3$ is such a vector and we
lyche-numerical-linear-algebra-F02307	171	1.000	$\boldsymbol{u}_2 \cdot \boldsymbol{u}_3 = [2, 1, -2]^T / 3$	$\boldsymbol{u}_2 \cdot \boldsymbol{u}_3 = [2, 1, -2]^T / 3$	of length one orthogonal to $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$. $\boldsymbol{u}_3 = [2, 1, -2]^T / 3$ is such a vector and we
lyche-numerical-linear-algebra-F02308	171	1.000	$\boldsymbol{\Sigma}_1$	$\boldsymbol{\Sigma}_1$	Thus $\boldsymbol{\Sigma}_1$ contains the r positive singular values on the diagonal and for $k, l \geq 0$ the
lyche-numerical-linear-algebra-F02309	171	1.000	$k, l \geq 0$	$k, l \geq 0$	Thus $\boldsymbol{\Sigma}_1$ contains the r positive singular values on the diagonal and for $k, l \geq 0$ the
lyche-numerical-linear-algebra-F02310	171	1.000	$\mathbf{0}_{k,l} = []$	$\mathbf{0}_{k,l} = []$	symbol $\mathbf{0}_{k,l} = []$ denotes the empty matrix if $k = 0$ or $l = 0$, and the zero matrix
lyche-numerical-linear-algebra-F02311	171	1.000	$k=0$	$k = 0$	symbol $\mathbf{0}_{k,l} = []$ denotes the empty matrix if $k = 0$ or $l = 0$, and the zero matrix
lyche-numerical-linear-algebra-F02312	171	1.000	$l=0$	$l = 0$	symbol $\mathbf{0}_{k,l} = []$ denotes the empty matrix if $k = 0$ or $l = 0$, and the zero matrix
lyche-numerical-linear-algebra-F02313	172	1.000	l	l	with k rows and l columns otherwise. We obtain by block multiplication a reduced
lyche-numerical-linear-algebra-F02314	172	1.000	$\boldsymbol{A} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$	$\boldsymbol{A} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$	factorization of $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ of the form $\boldsymbol{A} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$, where $\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$ and
lyche-numerical-linear-algebra-F02315	172	1.000	$\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$	$\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$	factorization of $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ of the form $\boldsymbol{A} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$, where $\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$ and
lyche-numerical-linear-algebra-F02316	172	1.000	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$ have orthonormal columns, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$ is a diagonal matrix with
lyche-numerical-linear-algebra-F02317	172	1.000	$\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$	$\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$ have orthonormal columns, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$ is a diagonal matrix with
lyche-numerical-linear-algebra-F02318	172	1.000	$\sigma_1 \geq \cdots \geq \sigma_r > 0$	$\sigma_1 \geq \cdots \geq \sigma_r > 0$	$\sigma_1 > \cdots > \sigma_r > 0$.
lyche-numerical-linear-algebra-F02319	172	1.000	$\boldsymbol{U}_1, \boldsymbol{V}_1$	$\boldsymbol{U}_1, \boldsymbol{V}_1$	$\boldsymbol{U}_1, \boldsymbol{V}_1$ contain the first r columns of $\boldsymbol{U}, \boldsymbol{V}$ respectively, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$ is a
lyche-numerical-linear-algebra-F02320	172	1.000	$\boldsymbol{U}, \boldsymbol{V}$	$\boldsymbol{U}, \boldsymbol{V}$	$\boldsymbol{U}_1, \boldsymbol{V}_1$ contain the first r columns of $\boldsymbol{U}, \boldsymbol{V}$ respectively, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$ is a
lyche-numerical-linear-algebra-F02321	172	1.000	$\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$	$\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$	$\boldsymbol{U}_1, \boldsymbol{V}_1$ contain the first r columns of $\boldsymbol{U}, \boldsymbol{V}$ respectively, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$ is a

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lyche-numerical-linear-algebra- F02322	172	1.000	$\boldsymbol{U}_1$	U_1	U_1 and V_1 in any way to unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$, and let Σ
lyche-numerical-linear-algebra- F02323	172	1.000	$\boldsymbol{V}_1$	V_1	U_1 and V_1 in any way to unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$, and let Σ
lyche-numerical-linear-algebra- F02324	172	1.000	$\boldsymbol{U} \in \mathbb{C}^{m \times m}$	$U \in \mathbb{C}^{m \times m}$	U_1 and V_1 in any way to unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$, and let Σ
lyche-numerical-linear-algebra- F02325	172	1.000	$\boldsymbol{V} \in \mathbb{C}^{n \times n}$	$V \in \mathbb{C}^{n \times n}$	U_1 and V_1 in any way to unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$, and let Σ
lyche-numerical-linear-algebra- F02326	172	1.000	$\boldsymbol{A} = \left[\boldsymbol{u}_1, \dots, \boldsymbol{u}_r \right] \operatorname{diag} \left(\sigma_1, \dots, \sigma_r \right) \left[\boldsymbol{v}_1, \dots, \boldsymbol{v}_r \right]^*$	$A = [u_1, \dots, u_r] \operatorname{diag}(\sigma_1, \dots, \sigma_r) [v_1, \dots, v_r]^*$	3. If $A = [u_1, \dots, u_r] \operatorname{diag}(\sigma_1, \dots, \sigma_r) [v_1, \dots, v_r]^*$ is a singular value factoriza-
lyche-numerical-linear-algebra- F02327	172	1.000	7.3($r < n < m$)	7.3($r < n < m$)	Example 7.3($r < n < m$) To find the SVF and SVD of
lyche-numerical-linear-algebra- F02328	173	1.000	$\sigma_1=2, \sigma_2=0$	$\sigma_1 = 2, \sigma_2 = 0$	and we find $\sigma_1 = 2, \sigma_2 = 0$, Thus $r = 1, m = 3, n = 2$ and
lyche-numerical-linear-algebra- F02329	173	1.000	$r=1, m=3, n=2$	$r = 1, m = 3, n = 2$	and we find $\sigma_1 = 2, \sigma_2 = 0$, Thus $r = 1, m = 3, n = 2$ and
lyche-numerical-linear-algebra- F02330	173	1.000	$\boldsymbol{u}_1 = \boldsymbol{A} \boldsymbol{v}_1 / \sigma_1 = \boldsymbol{s}_1 / \sqrt{2}$	$u_1 = A v_1 / \sigma_1 = s_1 / \sqrt{2}$	We find $u_1 = A v_1 / \sigma_1 = s_1 / \sqrt{2}$, where $s_1 = [1, 1, 0]^T$, and the SVF of A is given
lyche-numerical-linear-algebra- F02331	173	1.000	$\boldsymbol{s}_1 = [1, 1, 0]^T$	$s_1 = [1, 1, 0]^T$	We find $u_1 = A v_1 / \sigma_1 = s_1 / \sqrt{2}$, where $s_1 = [1, 1, 0]^T$, and the SVF of A is given
lyche-numerical-linear-algebra- F02332	173	1.000	$\boldsymbol{s}_1$	s_1	s_1 to a basis $\{s_1, s_2, s_3\}$ for $\mathbb{R}^3$, apply the Gram-Schmidt orthogonalization process
lyche-numerical-linear-algebra- F02333	173	1.000	$\{\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3\}$	$\{s_1, s_2, s_3\}$	s_1 to a basis $\{s_1, s_2, s_3\}$ for $\mathbb{R}^3$, apply the Gram-Schmidt orthogonalization process
lyche-numerical-linear-algebra- F02334	173	0.979	$\boldsymbol{w}_1 = \boldsymbol{s}_1, \boldsymbol{w}_2 = \boldsymbol{s}_2 - \frac{\boldsymbol{s}_2^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 = \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \end{bmatrix}, \boldsymbol{w}_3 = \boldsymbol{s}_3 -$	$w_1 = s_1, w_2 = s_2 - \frac{s_2^T w_1}{w_1^T w_1} w_1 = \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \end{bmatrix}, w_3 = s_3 -$	we find from (5.8) $w_1 = s_1, w_2 = s_2 - \frac{s_2^T w_1}{w_1^T w_1} w_1 = \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \end{bmatrix}, w_3 = s_3 -$
lyche-numerical-linear-algebra- F02335	173	0.763	$\frac{\boldsymbol{s}_3^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 - \frac{\boldsymbol{s}_3^T \boldsymbol{w}_2}{\boldsymbol{w}_2^T \boldsymbol{w}_2} \boldsymbol{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	$\frac{s_3^T w_1}{w_1^T w_1} w_1 - \frac{s_3^T w_2}{w_2^T w_2} w_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	$\frac{s_3^T w_1}{w_1^T w_1} w_1 - \frac{s_3^T w_2}{w_2^T w_2} w_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Normalizing the w_i 's we obtain $u_1 = w_1 / \ w_1\ _2 =$
lyche-numerical-linear-algebra- F02336	173	0.763	$\boldsymbol{w}_i$	w_i	$\frac{s_3^T w_1}{w_1^T w_1} w_1 - \frac{s_3^T w_2}{w_2^T w_2} w_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Normalizing the w_i 's we obtain $u_1 = w_1 / \ w_1\ _2 =$
lyche-numerical-linear-algebra- F02337	173	0.763	$\boldsymbol{u}_1 = \boldsymbol{w}_1 / \ \boldsymbol{w}_1\ _2 =$	$u_1 = w_1 / \ w_1\ _2 =$	$\frac{s_3^T w_1}{w_1^T w_1} w_1 - \frac{s_3^T w_2}{w_2^T w_2} w_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Normalizing the w_i 's we obtain $u_1 = w_1 / \ w_1\ _2 =$
lyche-numerical-linear-algebra- F02338	173	0.999	$[1 / \sqrt{2}, 1 / \sqrt{2}, 0]^T, \boldsymbol{u}_2 = \boldsymbol{w}_2 / \ \boldsymbol{w}_2\ _2 = [-1 / \sqrt{2}, 1 / \sqrt{2}, 0]^T$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T, u_2 = w_2 / \ w_2\ _2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T, u_2 = w_2 / \ w_2\ _2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$, and $u_3 = s_3 / \ s_3\ _2 =$
lyche-numerical-linear-algebra- F02339	173	0.999	$\boldsymbol{u}_3 = \boldsymbol{s}_3 / \ \boldsymbol{s}_3\ _2 =$	$u_3 = s_3 / \ s_3\ _2 =$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T, u_2 = w_2 / \ w_2\ _2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$, and $u_3 = s_3 / \ s_3\ _2 =$
lyche-numerical-linear-algebra- F02340	173	1.000	$[0, 0, 1]^T$	$[0, 0, 1]^T$	$[0, 0, 1]^T$. Therefore, $A = U \Sigma V^T$, is an SVD, where
lyche-numerical-linear-algebra- F02341	173	1.000	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^T$	$A = U \Sigma V^T$	$[0, 0, 1]^T$. Therefore, $A = U \Sigma V^T$, is an SVD, where

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lyche-numerical-linear-algebra- F02342	174	1.000	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^*), \mathcal{N}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^*)$, and $\mathcal{N}(\boldsymbol{A}^*)$.
lyche-numerical-linear-algebra- F02343	174	1.000	m, n	m, n	m, n let $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ have rank r and a singular value decomposition $\boldsymbol{A} =$
lyche-numerical-linear-algebra- F02344	174	0.985	$\left[\boldsymbol{u}_1, \dots, \boldsymbol{u}_m \right] \boldsymbol{\Sigma} \left[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n \right]^* = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$	$[\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \boldsymbol{\Sigma} [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n]^* = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$	$[\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \boldsymbol{\Sigma} [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n]^* = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$. Then the singular vectors satisfy
lyche-numerical-linear-algebra- F02345	174	1.000	$\left\{ \boldsymbol{u}_1, \dots, \boldsymbol{u}_r \right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_r\}$	1. $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_r\}$ is an orthonormal basis for $\mathcal{R}(\boldsymbol{A})$,
lyche-numerical-linear-algebra- F02346	174	1.000	$\left\{ \boldsymbol{v}_1, \dots, \boldsymbol{v}_r \right\}$	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_r\}$	3. $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_r\}$ is an orthonormal basis for $\mathcal{R}(\boldsymbol{A}^*)$,
lyche-numerical-linear-algebra- F02347	174	1.000	$\boldsymbol{A} \boldsymbol{V} = \boldsymbol{U} \boldsymbol{\Sigma}$	$\boldsymbol{A} \boldsymbol{V} = \boldsymbol{U} \boldsymbol{\Sigma}$	Proof If $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$ then $\boldsymbol{A} \boldsymbol{V} = \boldsymbol{U} \boldsymbol{\Sigma}$, or in terms of the block partition (7.4)
lyche-numerical-linear-algebra- F02348	174	0.980	$\boldsymbol{A} \left[\boldsymbol{V}_1, \boldsymbol{V}_2 \right] = \left[\boldsymbol{U}_1, \boldsymbol{U}_2 \right] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$\boldsymbol{A} [\boldsymbol{V}_1, \boldsymbol{V}_2] = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$\boldsymbol{A} [\boldsymbol{V}_1, \boldsymbol{V}_2] = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$. But then $\boldsymbol{A} \boldsymbol{V}_1 = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1$, $\boldsymbol{A} \boldsymbol{V}_2 = \mathbf{0}$, and this
lyche-numerical-linear-algebra- F02349	174	0.980	$\boldsymbol{A} \boldsymbol{V}_1 = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1, \boldsymbol{A} \boldsymbol{V}_2 = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{V}_1 = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1, \boldsymbol{A} \boldsymbol{V}_2 = \mathbf{0}$	$\boldsymbol{A} [\boldsymbol{V}_1, \boldsymbol{V}_2] = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$. But then $\boldsymbol{A} \boldsymbol{V}_1 = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1$, $\boldsymbol{A} \boldsymbol{V}_2 = \mathbf{0}$, and this
lyche-numerical-linear-algebra- F02350	174	1.000	$\boldsymbol{A}^* \boldsymbol{V} = \boldsymbol{U} \boldsymbol{\Sigma}^*$	$\boldsymbol{A}^* = \boldsymbol{V} \boldsymbol{\Sigma}^* \boldsymbol{U}^*$	$\boldsymbol{A}^* = \boldsymbol{V} \boldsymbol{\Sigma}^* \boldsymbol{U}^*$ or $\boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{V} \boldsymbol{\Sigma}^*$. Using the block partition as before we obtain the
lyche-numerical-linear-algebra- F02351	174	1.000	$\boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{V} \boldsymbol{\Sigma}^*$	$\boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{V} \boldsymbol{\Sigma}^*$	$\boldsymbol{A}^* = \boldsymbol{V} \boldsymbol{\Sigma}^* \boldsymbol{U}^*$ or $\boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{V} \boldsymbol{\Sigma}^*$. Using the block partition as before we obtain the
lyche-numerical-linear-algebra- F02352	174	1.000	$\mathcal{R}(\boldsymbol{A}), \left\{ \boldsymbol{A}^* \boldsymbol{u}_1, \dots, \boldsymbol{A}^* \boldsymbol{u}_r \right\}$	$\mathcal{R}(\boldsymbol{A}), \{\boldsymbol{A}^* \boldsymbol{u}_1, \dots, \boldsymbol{A}^* \boldsymbol{u}_r\}$	$\mathcal{R}(\boldsymbol{A}), \{\boldsymbol{A}^* \boldsymbol{u}_1, \dots, \boldsymbol{A}^* \boldsymbol{u}_r\}$ is an orthogonal basis for $\mathcal{R}(\boldsymbol{A}^*)$, $\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_m\}$ is an
lyche-numerical-linear-algebra- F02353	174	1.000	$\mathcal{R}(\boldsymbol{A}^*), \left\{ \boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_m \right\}$	$\mathcal{R}(\boldsymbol{A}^*), \{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_m\}$	$\mathcal{R}(\boldsymbol{A}), \{\boldsymbol{A}^* \boldsymbol{u}_1, \dots, \boldsymbol{A}^* \boldsymbol{u}_r\}$ is an orthogonal basis for $\mathcal{R}(\boldsymbol{A}^*)$, $\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_m\}$ is an
lyche-numerical-linear-algebra- F02354	174	1.000	$\boldsymbol{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$\boldsymbol{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$	a linear transformation. Consider the linear transformation $\boldsymbol{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by
lyche-numerical-linear-algebra- F02355	174	0.825	$\boldsymbol{T} \boldsymbol{z} := \boldsymbol{A} \boldsymbol{z}$	$\boldsymbol{T} \boldsymbol{z} := \boldsymbol{A} \boldsymbol{z}$	$\boldsymbol{T} \boldsymbol{z} := \boldsymbol{A} \boldsymbol{z}$ where $\boldsymbol{A} \in \mathbb{R}^{m \times n}$. Assume that $\text{rank}(\boldsymbol{A}) = n$. In the following theorem
lyche-numerical-linear-algebra- F02356	174	0.825	$(\boldsymbol{A}) = n$	$(\boldsymbol{A}) = n$	$\boldsymbol{T} \boldsymbol{z} := \boldsymbol{A} \boldsymbol{z}$ where $\boldsymbol{A} \in \mathbb{R}^{m \times n}$. Assume that $\text{rank}(\boldsymbol{A}) = n$. In the following theorem
lyche-numerical-linear-algebra- F02357	174	0.672	$\mathcal{S} := \{ \boldsymbol{z} \in \mathbb{R}^n \}$	$\mathcal{S} := \{ \boldsymbol{z} \in \mathbb{R}^n$	we show that the function $\boldsymbol{T}$ maps the unit sphere in $\mathbb{R}^n$ given by $\mathcal{S} := \{ \boldsymbol{z} \in \mathbb{R}^n :$
lyche-numerical-linear-algebra- F02358	174	0.943	$\ \boldsymbol{z} \ _2 = 1$	$\ \boldsymbol{z} \ _2 = 1$	$\ \boldsymbol{z} \ _2 = 1$ onto an ellipsoid $\mathcal{E} := \boldsymbol{A} \mathcal{S} = \{ \boldsymbol{A} \boldsymbol{z} : \boldsymbol{z} \in \mathcal{S} \}$ in $\mathbb{R}^m$.
lyche-numerical-linear-algebra- F02359	174	0.943	$\mathcal{E} := \boldsymbol{A} \mathcal{S} = \{ \boldsymbol{A} \boldsymbol{z} : \boldsymbol{z} \in \mathcal{S} \}$	$\mathcal{E} := \boldsymbol{A} \mathcal{S} = \{ \boldsymbol{A} \boldsymbol{z} : \boldsymbol{z} \in \mathcal{S} \}$	$\ \boldsymbol{z} \ _2 = 1$ onto an ellipsoid $\mathcal{E} := \boldsymbol{A} \mathcal{S} = \{ \boldsymbol{A} \boldsymbol{z} : \boldsymbol{z} \in \mathcal{S} \}$ in $\mathbb{R}^m$.
lyche-numerical-linear-algebra- F02360	174	1.000	$r = n$	$r = n$	Theorem 7.4 (SVF Ellipse) Suppose $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ has rank $r = n$, and let $\boldsymbol{A} =$
lyche-numerical-linear-algebra- F02361	174	0.977	$\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T$	$\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T$	$\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T$ be a singular value factorization of $\boldsymbol{A}$. Then
lyche-numerical-linear-algebra- F02362	175	0.407	$\boldsymbol{z} \in \mathcal{S}$	$\boldsymbol{z} \in \mathcal{S}$	Proof Suppose $\boldsymbol{z} \in \mathcal{S}$. Now $\boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y} := \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$.
lyche-numerical-linear-algebra- F02363	175	0.407	$\boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$	$\boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$	Proof Suppose $\boldsymbol{z} \in \mathcal{S}$. Now $\boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y} := \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$.
lyche-numerical-linear-algebra- F02364	175	0.407	$\boldsymbol{y} := \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$	$\boldsymbol{y} := \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$	Proof Suppose $\boldsymbol{z} \in \mathcal{S}$. Now $\boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y} := \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$.

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lyche-numerical-linear-algebra- F02365	175	■ 0.997	$\operatorname{rank}(\boldsymbol{A})=n$	$\operatorname{rank}(\boldsymbol{A}) = n$	Since $\operatorname{rank}(\boldsymbol{A}) = n$ it follows that $\boldsymbol{V}_1 = \boldsymbol{V}$ is square so that $\boldsymbol{V}_1 \boldsymbol{V}_1^T = \boldsymbol{I}$. But then
lyche-numerical-linear-algebra- F02366	175	■ 0.997	$\boldsymbol{V}_{\{1\}}=\boldsymbol{V}$	$\boldsymbol{V}_1 = \boldsymbol{V}$	Since $\operatorname{rank}(\boldsymbol{A}) = n$ it follows that $\boldsymbol{V}_1 = \boldsymbol{V}$ is square so that $\boldsymbol{V}_1 \boldsymbol{V}_1^T = \boldsymbol{I}$. But then
lyche-numerical-linear-algebra- F02367	175	■ 0.997	$\boldsymbol{V}_{\{1\}} \boldsymbol{V}_{\{1\}}^T=\boldsymbol{I}$	$\boldsymbol{V}_1 \boldsymbol{V}_1^T = \boldsymbol{I}$	Since $\operatorname{rank}(\boldsymbol{A}) = n$ it follows that $\boldsymbol{V}_1 = \boldsymbol{V}$ is square so that $\boldsymbol{V}_1 \boldsymbol{V}_1^T = \boldsymbol{I}$. But then
lyche-numerical-linear-algebra- F02368	175	■ 0.984	$\boldsymbol{V}_{\{1\}} \boldsymbol{\Sigma}_{\{1\}}^{-1} \boldsymbol{y}=\boldsymbol{z}$	$\boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y} = \boldsymbol{z}$	$\boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y} = \boldsymbol{z}$ and we obtain
lyche-numerical-linear-algebra- F02369	175	■ 1.000	$\boldsymbol{y} \in \tilde{\mathcal{E}}$	$\boldsymbol{y} \in \tilde{\mathcal{E}}$	This implies that $\boldsymbol{y} \in \tilde{\mathcal{E}}$. Finally, $\boldsymbol{x} = \boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y} \in \tilde{\mathcal{E}}$
lyche-numerical-linear-algebra- F02370	175	■ 1.000	$\boldsymbol{x}=\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$	$\boldsymbol{x} = \boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$	This implies that $\boldsymbol{y} \in \tilde{\mathcal{E}}$. Finally, $\boldsymbol{x} = \boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y} \in \tilde{\mathcal{E}}$
lyche-numerical-linear-algebra- F02371	175	■ 1.000	$\mathcal{E}=\boldsymbol{U}_{\{1\}} \tilde{\mathcal{E}}$	$\mathcal{E} = \boldsymbol{U}_1 \tilde{\mathcal{E}}$	implies that $\mathcal{E} = \boldsymbol{U}_1 \tilde{\mathcal{E}}$.
lyche-numerical-linear-algebra- F02372	175	■ 1.000	$1=\frac{y_{\{1\}}^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}$	$1 = \frac{y_1^2}{\sigma_1^2} + \cdots + \frac{y_n^2}{\sigma_n^2}$	The equation $\frac{y_1^2}{\sigma_1^2} + \cdots + \frac{y_n^2}{\sigma_n^2}$ describes an ellipsoid in $\mathbb{R}^n$ with semiaxes
lyche-numerical-linear-algebra- F02373	175	■ 1.000	$\sigma_{\{j\}}$	σ_j	of length σ_j along the unit vectors $\boldsymbol{e}_j$ for $j = 1, \dots, n$. Since the orthonormal
lyche-numerical-linear-algebra- F02374	175	■ 1.000	$\boldsymbol{e}_{\{j\}}$	$\boldsymbol{e}_j$	of length σ_j along the unit vectors $\boldsymbol{e}_j$ for $j = 1, \dots, n$. Since the orthonormal
lyche-numerical-linear-algebra- F02375	175	■ 0.992	$\boldsymbol{U}_{\{1\}} \boldsymbol{y} \rightarrow \boldsymbol{x}$	$\boldsymbol{U}_1 \boldsymbol{y} \rightarrow \boldsymbol{x}$	transformation $\boldsymbol{U}_1 \boldsymbol{y} \rightarrow \boldsymbol{x}$ preserves length, the image $\mathcal{E} = \boldsymbol{A} \mathcal{S}$ is a rotated ellipsoid
lyche-numerical-linear-algebra- F02376	175	■ 0.992	$\mathcal{E}=\boldsymbol{A} \mathcal{S}$	$\mathcal{E} = \boldsymbol{A} \mathcal{S}$	transformation $\boldsymbol{U}_1 \boldsymbol{y} \rightarrow \boldsymbol{x}$ preserves length, the image $\mathcal{E} = \boldsymbol{A} \mathcal{S}$ is a rotated ellipsoid
lyche-numerical-linear-algebra- F02377	175	■ 1.000	$\boldsymbol{u}_{\{j\}}=\boldsymbol{U} \boldsymbol{e}_{\{j\}}$	$\boldsymbol{u}_j = \boldsymbol{U} \boldsymbol{e}_j$	with semiaxes along the left singular vectors $\boldsymbol{u}_j = \boldsymbol{U} \boldsymbol{e}_j$, of length σ_j , $j = 1, \dots, n$.
lyche-numerical-linear-algebra- F02378	175	■ 1.000	$\sigma_{\{j\}}, j=1, \ldots, n$	$\sigma_j, j = 1, \dots, n$	with semiaxes along the left singular vectors $\boldsymbol{u}_j = \boldsymbol{U} \boldsymbol{e}_j$, of length $\sigma_j, j = 1, \dots, n$.
lyche-numerical-linear-algebra- F02379	175	■ 1.000	$\boldsymbol{A} \boldsymbol{v}_{\{j\}}=\sigma_{\{j\}} \boldsymbol{u}_{\{j\}}$	$\boldsymbol{A} \boldsymbol{v}_j = \sigma_j \boldsymbol{u}_j$	Since $\boldsymbol{A} \boldsymbol{v}_j = \sigma_j \boldsymbol{u}_j$, for $j = 1, \dots, n$ the right singular vectors defines points in $\mathcal{S}$
lyche-numerical-linear-algebra- F02380	175	■ 1.000	$\mathcal{E}$	$\mathcal{E}$	that are mapped onto the semiaxes of $\mathcal{E}$.
lyche-numerical-linear-algebra- F02381	175	■ 1.000	$\boldsymbol{A}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$	$\boldsymbol{A}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$	<i>Example 7.4 (Ellipse)</i> Consider the transformation $\boldsymbol{A}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by the
lyche-numerical-linear-algebra- F02382	175	■ 1.000	$\sigma_1=3, \sigma_2=1, \boldsymbol{u}_{\{1\}}=[3,4]^T / 5$	$\sigma_1 = 3, \sigma_2 = 1, \boldsymbol{u}_1 = [3, 4]^T / 5$	in Example 7.8. Recall that $\sigma_1 = 3, \sigma_2 = 1, \boldsymbol{u}_1 = [3, 4]^T / 5$ and $\boldsymbol{u}_2 = [-4, 3]^T / 5$.
lyche-numerical-linear-algebra- F02383	175	■ 1.000	$\boldsymbol{u}_{\{2\}}=[-4,3]^T / 5$	$\boldsymbol{u}_2 = [-4, 3]^T / 5$	in Example 7.8. Recall that $\sigma_1 = 3, \sigma_2 = 1, \boldsymbol{u}_1 = [3, 4]^T / 5$ and $\boldsymbol{u}_2 = [-4, 3]^T / 5$.
lyche-numerical-linear-algebra- F02384	175	■ 1.000	$y_{\{1\}}^2 / \sigma_1^2+y_{\{2\}}^2 / \sigma_2^2=1$	$y_1^2 / \sigma_1^2 + y_2^2 / \sigma_2^2 = 1$	The ellipses $y_1^2 / \sigma_1^2 + y_2^2 / \sigma_2^2 = 1$ and $\mathcal{E} = \boldsymbol{A} \mathcal{S} = \boldsymbol{U}_1 \tilde{\mathcal{E}}$ are shown in Fig. 7.1. Since
lyche-numerical-linear-algebra- F02385	175	■ 1.000	$\mathcal{E}=\boldsymbol{A} \mathcal{S}=\boldsymbol{U}_{\{1\}} \tilde{\mathcal{E}}$	$\mathcal{E} = \boldsymbol{A} \mathcal{S} = \boldsymbol{U}_1 \tilde{\mathcal{E}}$	The ellipses $y_1^2 / \sigma_1^2 + y_2^2 / \sigma_2^2 = 1$ and $\mathcal{E} = \boldsymbol{A} \mathcal{S} = \boldsymbol{U}_1 \tilde{\mathcal{E}}$ are shown in Fig. 7.1. Since

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lyche-numerical-linear-algebra-F02386	175	—	$\boldsymbol{A} \in \mathcal{S}$	$\mathcal{AS}$	
lyche-numerical-linear-algebra-F02387	176	■ 0.997	$\boldsymbol{y} = \boldsymbol{U}_1^T \boldsymbol{x} = \left[3/5 x_1 + 4/5 x_2, -4/5 x_1 + 3/5 x_2 \right]^T$	$\boldsymbol{y} = \boldsymbol{U}_1^T \boldsymbol{x} = [3/5 x_1 + 4/5 x_2, -4/5 x_1 + 3/5 x_2]^T$	$\boldsymbol{y} = \boldsymbol{U}_1^T \boldsymbol{x} = [3/5 x_1 + 4/5 x_2, -4/5 x_1 + 3/5 x_2]^T$, the equation for the ellipse on the
lyche-numerical-linear-algebra-F02388	176	■ 1.000	$\ \boldsymbol{A}^*\ _F = \ \boldsymbol{A}\ _F$	$\ \boldsymbol{A}^*\ _F = \ \boldsymbol{A}\ _F$	1. $\ \boldsymbol{A}^*\ _F = \ \boldsymbol{A}\ _F$,
lyche-numerical-linear-algebra-F02389	176	■ 1.000	$\ \boldsymbol{A}\ _F^2 = \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2$	$\ \boldsymbol{A}\ _F^2 = \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2$	2. $\ \boldsymbol{A}\ _F^2 = \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2$,
lyche-numerical-linear-algebra-F02390	177	■ 1.000	$\ \boldsymbol{U}\boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F$ for any unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m \times m}$ and $\boldsymbol{V} \in \mathbb{C}^{n \times n}$	$\ \boldsymbol{U}\boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F$	3. $\ \boldsymbol{U}\boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F$ for any unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m \times m}$ and $\boldsymbol{V} \in \mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra-F02391	177	■ 1.000	$\boldsymbol{V} \in \mathbb{C}^{n \times n}$	$\boldsymbol{V} \in \mathbb{C}^{n \times n}$	3. $\ \boldsymbol{U}\boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F$ for any unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m \times m}$ and $\boldsymbol{V} \in \mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra-F02392	177	■ 0.999	$\ \boldsymbol{A}\boldsymbol{B}\ _F \leq \ \boldsymbol{A}\ _F \ \boldsymbol{B}\ _F$	$\ \boldsymbol{A}\boldsymbol{B}\ _F \leq \ \boldsymbol{A}\ _F \ \boldsymbol{B}\ _F$	4. $\ \boldsymbol{A}\boldsymbol{B}\ _F \leq \ \boldsymbol{A}\ _F \ \boldsymbol{B}\ _F$ for any $\boldsymbol{B} \in \mathbb{C}^{n \times k}$, $k \in \mathbb{N}$,
lyche-numerical-linear-algebra-F02393	177	■ 0.999	$\boldsymbol{B} \in \mathbb{C}^{n \times k}$, $k \in \mathbb{N}$	$\boldsymbol{B} \in \mathbb{C}^{n \times k}$, $k \in \mathbb{N}$	4. $\ \boldsymbol{A}\boldsymbol{B}\ _F \leq \ \boldsymbol{A}\ _F \ \boldsymbol{B}\ _F$ for any $\boldsymbol{B} \in \mathbb{C}^{n \times k}$, $k \in \mathbb{N}$,
lyche-numerical-linear-algebra-F02394	177	■ 0.980	$\ \boldsymbol{A}\boldsymbol{x}\ _2 \leq \ \boldsymbol{A}\ _F \ \boldsymbol{x}\ _2$	$\ \boldsymbol{A}\boldsymbol{x}\ _2 \leq \ \boldsymbol{A}\ _F \ \boldsymbol{x}\ _2$	5. $\ \boldsymbol{A}\boldsymbol{x}\ _2 \leq \ \boldsymbol{A}\ _F \ \boldsymbol{x}\ _2$, for all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra-F02395	177	■ 1.000	$\ \boldsymbol{A}^*\ _F^2 = \sum_{j=1}^n \sum_{i=1}^m \bar{a}_{ij} ^2 = \sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2 = \ \boldsymbol{A}\ _F^2$	$\ \boldsymbol{A}^*\ _F^2 = \sum_{j=1}^n \sum_{i=1}^m \bar{a}_{ij} ^2 = \sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2 = \ \boldsymbol{A}\ _F^2$	1. $\ \boldsymbol{A}^*\ _F^2 = \sum_{j=1}^n \sum_{i=1}^m \bar{a}_{ij} ^2 = \sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2 = \ \boldsymbol{A}\ _F^2$.
lyche-numerical-linear-algebra-F02396	177	■ 1.000	$\ \boldsymbol{A}\ _F := \ \text{vec}(\boldsymbol{A})\ _2$	$\ \boldsymbol{A}\ _F := \ \text{vec}(\boldsymbol{A})\ _2$	$\ \boldsymbol{A}\ _F := \ \text{vec}(\boldsymbol{A})\ _2$, where $\text{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$ is the vector obtained by stacking
lyche-numerical-linear-algebra-F02397	177	■ 1.000	$\text{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$	$\text{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$	$\ \boldsymbol{A}\ _F := \ \text{vec}(\boldsymbol{A})\ _2$, where $\text{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$ is the vector obtained by stacking
lyche-numerical-linear-algebra-F02398	177	■ 1.000	$\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$	$\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$	3. Recall that if $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$ then $\ \boldsymbol{U}\boldsymbol{x}\ _2 = \ \boldsymbol{x}\ _2$ for all $\boldsymbol{x} \in \mathbb{C}^n$. Applying this to each
lyche-numerical-linear-algebra-F02399	177	■ 1.000	$\ \boldsymbol{U}\boldsymbol{x}\ _2 = \ \boldsymbol{x}\ _2$	$\ \boldsymbol{U}\boldsymbol{x}\ _2 = \ \boldsymbol{x}\ _2$	3. Recall that if $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$ then $\ \boldsymbol{U}\boldsymbol{x}\ _2 = \ \boldsymbol{x}\ _2$ for all $\boldsymbol{x} \in \mathbb{C}^n$. Applying this to each
lyche-numerical-linear-algebra-F02400	177	■ 0.997	$\ \boldsymbol{U}\boldsymbol{A}\ _F^2 = \sum_{j=1}^n \ \boldsymbol{U}\boldsymbol{a}_{:j}\ _2^2 = \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2 = \ \boldsymbol{A}\ _F^2$	$\ \boldsymbol{U}\boldsymbol{A}\ _F^2 = \sum_{j=1}^n \ \boldsymbol{U}\boldsymbol{a}_{:j}\ _2^2 = \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2 = \ \boldsymbol{A}\ _F^2$	column $\boldsymbol{a}_{:j}$ of $\boldsymbol{A}$ we find $\ \boldsymbol{U}\boldsymbol{A}\ _F^2 = \sum_{j=1}^n \ \boldsymbol{U}\boldsymbol{a}_{:j}\ _2^2 = \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2 = \ \boldsymbol{A}\ _F^2$.
lyche-numerical-linear-algebra-F02401	177	■ 0.933	$\boldsymbol{V}\boldsymbol{V}^* = \boldsymbol{I}$	$\boldsymbol{V}\boldsymbol{V}^* = \boldsymbol{I}$	Similarly, since $\boldsymbol{V}\boldsymbol{V}^* = \boldsymbol{I}$ we find $\ \boldsymbol{A}\boldsymbol{V}\ _F = \ \boldsymbol{A}\ _F$.
lyche-numerical-linear-algebra-F02402	177	■ 0.933	$\ \boldsymbol{A}\boldsymbol{V}\ _F = \ \boldsymbol{A}\ _F$	$\ \boldsymbol{A}\boldsymbol{V}\ _F = \ \boldsymbol{A}\ _F$	Similarly, since $\boldsymbol{V}\boldsymbol{V}^* = \boldsymbol{I}$ we find $\ \boldsymbol{A}\boldsymbol{V}\ _F = \ \boldsymbol{A}\ _F$.

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lyche-numerical-linear-algebra-F02403	177	■ 0.986	$\ \boldsymbol{v}\ _F = \ \boldsymbol{v}\ _2$	$\ \boldsymbol{v}\ _F = \ \boldsymbol{v}\ _2$	5. Since $\ \boldsymbol{v}\ _F = \ \boldsymbol{v}\ _2$ for a vector this follows by taking $k = 1$ and $\boldsymbol{B} = \boldsymbol{x}$ in 4.
lyche-numerical-linear-algebra-F02404	177	■ 0.986	$\boldsymbol{B} = \boldsymbol{x}$	$\boldsymbol{B} = \boldsymbol{x}$	5. Since $\ \boldsymbol{v}\ _F = \ \boldsymbol{v}\ _2$ for a vector this follows by taking $k = 1$ and $\boldsymbol{B} = \boldsymbol{x}$ in 4.
lyche-numerical-linear-algebra-F02405	177	■ 0.996	$\ \boldsymbol{A}\ _F =$	$\ \boldsymbol{A}\ _F =$	Theorem 7.5 (Frobenius Norm and Singular Values) We have $\ \boldsymbol{A}\ _F =$
lyche-numerical-linear-algebra-F02406	177	■ 0.999	$\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$	$\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$	$\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$, where $\sigma_1, \dots, \sigma_n$ are the singular values of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F02407	177	■ 0.999	$\sigma_1, \ldots, \sigma_n$	$\sigma_1, \dots, \sigma_n$	$\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$, where $\sigma_1, \dots, \sigma_n$ are the singular values of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F02408	177	■ 1.000	$m \geq n \geq 1$	$m \geq n \geq 1$	Suppose $m \geq n \geq 1$ and $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ has a singular value decomposition $\boldsymbol{A} =$
lyche-numerical-linear-algebra-F02409	177	■ 1.000	$\boldsymbol{U} \begin{bmatrix} \boldsymbol{D} \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$	$\boldsymbol{U} \begin{bmatrix} \boldsymbol{D} \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$	$\boldsymbol{U} \begin{bmatrix} \boldsymbol{D} \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$, where $\boldsymbol{D} = \text{diag}(\sigma_1, \dots, \sigma_n)$. We choose $\epsilon > 0$ and let $1 \leq r \leq n$ be
lyche-numerical-linear-algebra-F02410	177	■ 1.000	$\boldsymbol{D} = \text{diag}(\sigma_1, \ldots, \sigma_n)$	$\boldsymbol{D} = \text{diag}(\sigma_1, \dots, \sigma_n)$	$\boldsymbol{U} \begin{bmatrix} \boldsymbol{D} \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$, where $\boldsymbol{D} = \text{diag}(\sigma_1, \dots, \sigma_n)$. We choose $\epsilon > 0$ and let $1 \leq r \leq n$ be
lyche-numerical-linear-algebra-F02411	177	■ 1.000	$\epsilon > 0$	$\epsilon > 0$	$\boldsymbol{U} \begin{bmatrix} \boldsymbol{D} \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$, where $\boldsymbol{D} = \text{diag}(\sigma_1, \dots, \sigma_n)$. We choose $\epsilon > 0$ and let $1 \leq r \leq n$ be
lyche-numerical-linear-algebra-F02412	177	■ 1.000	$1 \leq r \leq n$	$1 \leq r \leq n$	$\boldsymbol{U} \begin{bmatrix} \boldsymbol{D} \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$, where $\boldsymbol{D} = \text{diag}(\sigma_1, \dots, \sigma_n)$. We choose $\epsilon > 0$ and let $1 \leq r \leq n$ be
lyche-numerical-linear-algebra-F02413	177	■ 1.000	$\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$	$\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$	the smallest integer such that $\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$. Define $\boldsymbol{A}' := \boldsymbol{U} \begin{bmatrix} \boldsymbol{D}' \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$, where
lyche-numerical-linear-algebra-F02414	177	■ 1.000	$\boldsymbol{A}' := \boldsymbol{U} \begin{bmatrix} \boldsymbol{D}' \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$	$\boldsymbol{A}' := \boldsymbol{U} \begin{bmatrix} \boldsymbol{D}' \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$	the smallest integer such that $\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$. Define $\boldsymbol{A}' := \boldsymbol{U} \begin{bmatrix} \boldsymbol{D}' \\ \mathbf{0} \end{bmatrix} \boldsymbol{V}^*$, where
lyche-numerical-linear-algebra-F02415	177	■ 1.000	$\boldsymbol{D}' := \text{diag}(\sigma_1, \ldots, \sigma_r, 0, \ldots, 0) \in \mathbb{R}^{n \times n}$	$\boldsymbol{D}' := \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0) \in \mathbb{R}^{n \times n}$	$\boldsymbol{D}' := \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0) \in \mathbb{R}^{n \times n}$. By Lemma 7.1
lyche-numerical-linear-algebra-F02416	178	■ 1.000	$\boldsymbol{A}'$	$\boldsymbol{A}'$	Thus, if ϵ is small then $\boldsymbol{A}$ is near a matrix $\boldsymbol{A}'$ of rank r . This can be used to determine
lyche-numerical-linear-algebra-F02417	178	■ 1.000	$\sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2}$	$\sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2}$	rank numerically. We choose an r such that $\sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2}$ is “small”. Then we
lyche-numerical-linear-algebra-F02418	178	■ 0.972	$\text{rank}(\boldsymbol{A}) = r$	$\text{rank}(\boldsymbol{A}) = r$	postulate that $\text{rank}(\boldsymbol{A}) = r$ since $\boldsymbol{A}$ is close to a matrix of rank r .
lyche-numerical-linear-algebra-F02419	178	■ 1.000	$r, \boldsymbol{A}'$	$r, \boldsymbol{A}'$	The following theorem shows that of all $m \times n$ matrices of rank r , $\boldsymbol{A}'$ is closest
lyche-numerical-linear-algebra-F02420	178	■ 1.000	$\sigma_1 \geq \cdots \geq \sigma_n \geq 0$	$\sigma_1 \geq \cdots \geq \sigma_n \geq 0$	values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. For any $r \leq \text{rank}(\boldsymbol{A})$ we have
lyche-numerical-linear-algebra-F02421	178	■ 1.000	$r \leq \text{rank}(\boldsymbol{A})$	$r \leq \text{rank}(\boldsymbol{A})$	values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. For any $r \leq \text{rank}(\boldsymbol{A})$ we have
lyche-numerical-linear-algebra-F02422	178	■ 1.000	$\boldsymbol{A} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$	a) $\boldsymbol{A} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
lyche-numerical-linear-algebra-F02423	178	■ 1.000	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$	b) $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$.
lyche-numerical-linear-algebra-F02424	179	■ 1.000	$\boldsymbol{A} = \boldsymbol{e}_1$	$\boldsymbol{A} = \boldsymbol{e}_1$	a) $\boldsymbol{A} = \boldsymbol{e}_1$ the first unit vector in $\mathbb{R}^m$.
lyche-numerical-linear-algebra-F02425	179	■ 1.000	$\boldsymbol{A} = \boldsymbol{e}_n^T$	$\boldsymbol{A} = \boldsymbol{e}_n^T$	b) $\boldsymbol{A} = \boldsymbol{e}_n^T$ the last unit vector in $\mathbb{R}^n$.

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lyche-numerical-linear-algebra-F02426	179	1.000	$\boldsymbol{A}=\left[\begin{array}{rr}-1 & 0 \\ 0 & 3\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$	c) $\boldsymbol{A} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$.
lyche-numerical-linear-algebra-F02427	179	1.000	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}$	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}$	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}$ is a singular value decomposition of $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ then
lyche-numerical-linear-algebra-F02428	179	1.000	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{V} \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2) \boldsymbol{V}^*$	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{V} \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2) \boldsymbol{V}^*$	a) $\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{V} \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2) \boldsymbol{V}^*$ is a spectral decomposition of $\boldsymbol{A}^* \boldsymbol{A}$.
lyche-numerical-linear-algebra-F02429	179	1.000	$\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{U} \operatorname{diag}(\sigma_1^2, \dots, \sigma_m^2) \boldsymbol{U}^*$	$\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{U} \operatorname{diag}(\sigma_1^2, \dots, \sigma_m^2) \boldsymbol{U}^*$	b) $\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{U} \operatorname{diag}(\sigma_1^2, \dots, \sigma_m^2) \boldsymbol{U}^*$ is a spectral decomposition of $\boldsymbol{A} \boldsymbol{A}^*$.
lyche-numerical-linear-algebra-F02430	179	0.803	$(\sigma_1, \dots, \sigma_n)$	$(\sigma_1, \dots, \sigma_n)$	“If $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ has singular values $\sigma_1, \dots, \sigma_n$ then $\boldsymbol{A}^2$ has singular values
lyche-numerical-linear-algebra-F02431	179	0.803	$\boldsymbol{A}^2$	$\boldsymbol{A}^2$	“If $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ has singular values $(\sigma_1, \dots, \sigma_n)$ then $\boldsymbol{A}^2$ has singular values
lyche-numerical-linear-algebra-F02432	179	1.000	$(\sigma_1^2, \dots, \sigma_n^2)$	$(\sigma_1^2, \dots, \sigma_n^2)$	$(\sigma_1^2, \dots, \sigma_n^2)$ ”. Find a class of matrices for which the statement is true. Show that
lyche-numerical-linear-algebra-F02433	179	0.884	$\boldsymbol{A}=\frac{1}{25}\left[\begin{array}{ll}11 & 48 \\ 48 & 39\end{array}\right]$	$\boldsymbol{A} = \frac{1}{25} \begin{bmatrix} 11 & 48 \\ 48 & 39 \end{bmatrix}$	$\boldsymbol{A} = \frac{1}{25} \begin{bmatrix} 11 & 48 \\ 48 & 39 \end{bmatrix}$. Also, using possibly a computer, find its spectral decomposition
lyche-numerical-linear-algebra-F02434	179	0.998	$\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^T$	$\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^T$	$\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^T$. The matrix $\boldsymbol{A}$ is normal, but the spectral decomposition is not an SVD.
lyche-numerical-linear-algebra-F02435	179	1.000	$\boldsymbol{A}=\frac{1}{5}\left[\begin{array}{cc}3 & -4 \\ 4 & 3\end{array}\right]\left[\begin{array}{cc}3 & 0 \\ 0 & 1\end{array}\right]\frac{1}{5}\left[\begin{array}{cc}3 & 4 \\ 4 & -3\end{array}\right]$	$\boldsymbol{A} = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	¹ Answer: $\boldsymbol{A} = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$.
lyche-numerical-linear-algebra-F02436	180	1.000	$\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A}^*)$	$\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A}^*)$	a) $\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A}^*)$.
lyche-numerical-linear-algebra-F02437	180	1.000	$\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}) = n$	$\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}) = n$	b) $\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}) = n$,
lyche-numerical-linear-algebra-F02438	180	1.000	$\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}^*) = m$	$\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}^*) = m$	c) $\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}^*) = m$,
lyche-numerical-linear-algebra-F02439	180	1.000	$\operatorname{null}(\boldsymbol{A})$	$\operatorname{null}(\boldsymbol{A})$	where $\operatorname{null}(\boldsymbol{A})$ is defined as the dimension of $\mathcal{N}(\boldsymbol{A})$.
lyche-numerical-linear-algebra-F02440	180	1.000	$\operatorname{rank} \boldsymbol{A} = \operatorname{rank}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A} \boldsymbol{A}^*)$	$\operatorname{rank} \boldsymbol{A} = \operatorname{rank}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A} \boldsymbol{A}^*)$	a) $\operatorname{rank} \boldsymbol{A} = \operatorname{rank}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A} \boldsymbol{A}^*)$,
lyche-numerical-linear-algebra-F02441	180	0.997	$\operatorname{null}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{null} \boldsymbol{A}$	$\operatorname{null}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{null} \boldsymbol{A}$	b) $\operatorname{null}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{null} \boldsymbol{A}$, and $\operatorname{null}(\boldsymbol{A} \boldsymbol{A}^*) = \operatorname{null}(\boldsymbol{A}^*)$.
lyche-numerical-linear-algebra-F02442	180	0.997	$\operatorname{null}(\boldsymbol{A} \boldsymbol{A}^*) = \operatorname{null}(\boldsymbol{A}^*)$	$\operatorname{null}(\boldsymbol{A} \boldsymbol{A}^*) = \operatorname{null}(\boldsymbol{A}^*)$	b) $\operatorname{null}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{null} \boldsymbol{A}$, and $\operatorname{null}(\boldsymbol{A} \boldsymbol{A}^*) = \operatorname{null}(\boldsymbol{A}^*)$.
lyche-numerical-linear-algebra-F02443	180	1.000	$\mathcal{R}(\boldsymbol{B})$	$\mathcal{R}(\boldsymbol{B})$	Give orthonormal bases for $\mathcal{R}(\boldsymbol{B})$ and $\mathcal{N}(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F02444	180	1.000	$\mathcal{N}(\boldsymbol{B})$	$\mathcal{N}(\boldsymbol{B})$	Give orthonormal bases for $\mathcal{R}(\boldsymbol{B})$ and $\mathcal{N}(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F02445	180	1.000	$\mathcal{R}(\boldsymbol{A}^* \boldsymbol{A}) =$	$\mathcal{R}(\boldsymbol{A}^* \boldsymbol{A}) =$	Exercise 7.13 (Some Spanning Sets) Show for any $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ that $\mathcal{R}(\boldsymbol{A}^* \boldsymbol{A}) =$
lyche-numerical-linear-algebra-F02446	180	1.000	$\mathcal{R}(\boldsymbol{V}_1) = \mathcal{R}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{V}_1) = \mathcal{R}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{V}_1) = \mathcal{R}(\boldsymbol{A}^*)$
lyche-numerical-linear-algebra-F02447	180	1.000	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m \in \mathbb{C}^m$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m \in \mathbb{C}^m$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m \in \mathbb{C}^m$, and right singular vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n \in \mathbb{C}^n$. Show that the matrix
lyche-numerical-linear-algebra-F02448	180	1.000	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n \in \mathbb{C}^n$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n \in \mathbb{C}^n$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m \in \mathbb{C}^m$, and right singular vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n \in \mathbb{C}^n$. Show that the matrix

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lyche-numerical-linear-algebra-F02449	181	1.000	$\boldsymbol{B} :=$	$\boldsymbol{B} :=$	c) Use the singular value decomposition of $\boldsymbol{A}$ to give a suitable definition of $\boldsymbol{B} :=$
lyche-numerical-linear-algebra-F02450	181	0.981	$\sqrt{\boldsymbol{A}^T \boldsymbol{A}}$	$\sqrt{\boldsymbol{A}^T \boldsymbol{A}}$	$\sqrt{\boldsymbol{A}^T \boldsymbol{A}}$ so that $\boldsymbol{P} = \boldsymbol{B}$.
lyche-numerical-linear-algebra-F02451	181	0.981	$\boldsymbol{P} = \boldsymbol{B}$	$\boldsymbol{P} = \boldsymbol{B}$	$\sqrt{\boldsymbol{A}^T \boldsymbol{A}}$ so that $\boldsymbol{P} = \boldsymbol{B}$.
lyche-numerical-linear-algebra-F02452	181	1.000	$\boldsymbol{X}_k = \boldsymbol{U} \boldsymbol{\Sigma}_k \boldsymbol{V}^T$	$\boldsymbol{X}_k = \boldsymbol{U} \boldsymbol{\Sigma}_k \boldsymbol{V}^T$	d) Show that the iteration (7.14) is well defined by showing that $\boldsymbol{X}_k = \boldsymbol{U} \boldsymbol{\Sigma}_k \boldsymbol{V}^T$,
lyche-numerical-linear-algebra-F02453	181	0.985	$\boldsymbol{\Sigma}_k$	$\boldsymbol{\Sigma}_k$	for a diagonal matrix $\boldsymbol{\Sigma}_k$ with positive diagonal elements, $k = 0, 1, 2, \dots$
lyche-numerical-linear-algebra-F02454	181	0.985	$k=0,1,2, \ldots$	$k = 0, 1, 2, \dots$	for a diagonal matrix $\boldsymbol{\Sigma}_k$ with positive diagonal elements, $k = 0, 1, 2, \dots$.
lyche-numerical-linear-algebra-F02455	181	0.749	$[\boldsymbol{Q}, \boldsymbol{P}, k] =$	$[\boldsymbol{Q}, \boldsymbol{P}, k] =$	function $[\boldsymbol{Q}, \boldsymbol{P}, k] = \text{polardecomp}(\boldsymbol{A}, \text{tol}, K)$ to carry out the iter-
lyche-numerical-linear-algebra-F02456	181	0.749	$(\boldsymbol{A}, \text{tol}, K)$	$(\boldsymbol{A}, \text{tol}, K)$	function $[\boldsymbol{Q}, \boldsymbol{P}, k] = \text{polardecomp}(\boldsymbol{A}, \text{tol}, K)$ to carry out the iter-
lyche-numerical-linear-algebra-F02457	181	1.000	$\boldsymbol{P} = \boldsymbol{Q}^T \boldsymbol{A}$	$\boldsymbol{P} = \boldsymbol{Q}^T \boldsymbol{A}$	ation in (7.14). The output is approximations $\boldsymbol{Q}$ and $\boldsymbol{P} = \boldsymbol{Q}^T \boldsymbol{A}$ to the polar
lyche-numerical-linear-algebra-F02458	181	0.977	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$	decomposition $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$ of $\boldsymbol{A}$ and the number of iterations k such that
lyche-numerical-linear-algebra-F02459	181	0.919	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F <$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F <$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F < \text{tol} * \ \boldsymbol{X}_{k+1}\ _F$. Set $k = K + 1$ if convergence is not achieved
lyche-numerical-linear-algebra-F02460	181	0.919	$* \ \boldsymbol{X}_{k+1}\ _F$	$* \ \boldsymbol{X}_{k+1}\ _F$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F < \text{tol} * \ \boldsymbol{X}_{k+1}\ _F$. Set $k = K + 1$ if convergence is not achieved
lyche-numerical-linear-algebra-F02461	181	0.919	$k=K+1$	$k = K + 1$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F < \text{tol} * \ \boldsymbol{X}_{k+1}\ _F$. Set $k = K + 1$ if convergence is not achieved
lyche-numerical-linear-algebra-F02462	181	1.000	$\ \boldsymbol{A}\ _1$	$\ \boldsymbol{A}\ _1$	Compute $\ \boldsymbol{A}\ _1$ and $\ \boldsymbol{A}\ _\infty$.
lyche-numerical-linear-algebra-F02463	181	1.000	$\ \boldsymbol{A}\ _\infty$	$\ \boldsymbol{A}\ _\infty$	Compute $\ \boldsymbol{A}\ _1$ and $\ \boldsymbol{A}\ _\infty$.
lyche-numerical-linear-algebra-F02464	181	0.972	$(\boldsymbol{B}^T)$	$(\boldsymbol{B}^T)$	Find the spaces $\text{span}(\boldsymbol{B}^T)$ and $\ker(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F02465	181	0.972	$\ker(\boldsymbol{B})$	$\ker(\boldsymbol{B})$	Find the spaces $\text{span}(\boldsymbol{B}^T)$ and $\ker(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F02466	182	1.000	$\boldsymbol{x} \in \mathbb{R}^3$	$\boldsymbol{x} \in \mathbb{R}^3$	Find the solution $\boldsymbol{x} \in \mathbb{R}^3$ with $\ \boldsymbol{x}\ _2$ as small as possible.
lyche-numerical-linear-algebra-F02467	182	1.000	$\boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{b} \in \mathbb{R}^m$	d) Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a matrix with linearly independent columns, and $\boldsymbol{b} \in \mathbb{R}^m$
lyche-numerical-linear-algebra-F02468	182	1.000	$\boldsymbol{A}^T \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$	$\boldsymbol{A}^T \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$	the normal equations $\boldsymbol{A}^T \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$. Will the method converge? Justify your
lyche-numerical-linear-algebra-F02469	182	1.000	$\mathcal{R}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^T), \mathcal{N}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^T), \mathcal{N}(\boldsymbol{A})$	a) Give orthonormal bases for $\mathcal{R}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^T), \mathcal{N}(\boldsymbol{A})$ and $\mathcal{N}(\boldsymbol{A}^T)$.
lyche-numerical-linear-algebra-F02470	182	1.000	$\mathcal{N}(\boldsymbol{A}^T)$	$\mathcal{N}(\boldsymbol{A}^T)$	a) Give orthonormal bases for $\mathcal{R}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^T), \mathcal{N}(\boldsymbol{A})$ and $\mathcal{N}(\boldsymbol{A}^T)$.
lyche-numerical-linear-algebra-F02471	182	0.999	$\boldsymbol{B} \in \mathbb{R}^{4,3}$	$\boldsymbol{B} \in \mathbb{R}^{4,3}$	b) Explain why for all matrices $\boldsymbol{B} \in \mathbb{R}^{4,3}$ of rank one we have $\ \boldsymbol{A} - \boldsymbol{B}\ _F \geq 6$.
lyche-numerical-linear-algebra-F02472	182	0.999	$\ \boldsymbol{A} - \boldsymbol{B}\ _F \geq 6$	$\ \boldsymbol{A} - \boldsymbol{B}\ _F \geq 6$	b) Explain why for all matrices $\boldsymbol{B} \in \mathbb{R}^{4,3}$ of rank one we have $\ \boldsymbol{A} - \boldsymbol{B}\ _F \geq 6$.
lyche-numerical-linear-algebra-F02473	182	1.000	$\ \boldsymbol{A} - \boldsymbol{A}_1\ _F = 6$	$\ \boldsymbol{A} - \boldsymbol{A}_1\ _F = 6$	c) Give a matrix $\boldsymbol{A}_1$ of rank one such that $\ \boldsymbol{A} - \boldsymbol{A}_1\ _F = 6$.
lyche-numerical-linear-algebra-F02474	182	0.980	$(n, 1)$	$(n, 1)$	diagonal equal to -1 . Let $\boldsymbol{B}$ be the matrix obtained from $\boldsymbol{A}$ by changing the $(n, 1)$
lyche-numerical-linear-algebra-F02475	182	1.000	-2^{2-n}	-2^{2-n}	element from zero to -2^{2-n} .

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lyche-numerical-linear-algebra-182-F02476	182	1.000	$\backslash\boldsymbol{x}:=\left[2^{\{n-2\}}, 2^{\{n-3\}}, \ldots, 2^{\{0\}}, 1\right]^T$	$\boldsymbol{x} := [2^{n-2}, 2^{n-3}, \dots, 2^0, 1]^T$	a) Show that $\boldsymbol{B}\boldsymbol{x} = \mathbf{0}$, where $\boldsymbol{x} := [2^{n-2}, 2^{n-3}, \dots, 2^0, 1]^T$. Conclude that $\boldsymbol{B}$ is
lyche-numerical-linear-algebra-182-F02477	182	1.000	$\operatorname{det}(\boldsymbol{A})=1$	$\det(\boldsymbol{A}) = 1$	singular, $\det(\boldsymbol{A}) = 1$, and $\ \boldsymbol{A} - \boldsymbol{B}\ _F = 2^{2-n}$. Thus even if $\det(\boldsymbol{A})$ is not small
lyche-numerical-linear-algebra-182-F02478	182	1.000	$\ \boldsymbol{A}-\boldsymbol{B}\ _F=2^{2-n}$	$\ \boldsymbol{A} - \boldsymbol{B}\ _F = 2^{2-n}$	singular, $\det(\boldsymbol{A}) = 1$, and $\ \boldsymbol{A} - \boldsymbol{B}\ _F = 2^{2-n}$. Thus even if $\det(\boldsymbol{A})$ is not small
lyche-numerical-linear-algebra-182-F02479	182	1.000	$\boldsymbol{A}-\boldsymbol{B}$	$\boldsymbol{A} - \boldsymbol{B}$	the Frobenius norm of $\boldsymbol{A} - \boldsymbol{B}$ is small for large n , and the matrix $\boldsymbol{A}$ is very close
lyche-numerical-linear-algebra-182-F02480	182	1.000	σ_n	σ_n	b) Use Theorem 7.6 to show that the smallest singular vale σ_n of $\boldsymbol{A}$ is bounded
lyche-numerical-linear-algebra-182-F02481	182	1.000	2^{2-n}	2^{2-n}	above by 2^{2-n} .
lyche-numerical-linear-algebra-183-F02482	183	1.000	$\ \boldsymbol{A}\ _1,\ \boldsymbol{A}\ _\infty$	$\ \boldsymbol{A}\ _1, \ \boldsymbol{A}\ _\infty$	Compute $\ \boldsymbol{A}\ _1, \ \boldsymbol{A}\ _\infty$ and $\ \boldsymbol{A}\ _F$.
lyche-numerical-linear-algebra-183-F02483	183	1.000	$\ \boldsymbol{A}\ _F$	$\ \boldsymbol{A}\ _F$	Compute $\ \boldsymbol{A}\ _1, \ \boldsymbol{A}\ _\infty$ and $\ \boldsymbol{A}\ _F$.
lyche-numerical-linear-algebra-183-F02484	183	1.000	$\boldsymbol{T}=\boldsymbol{L}\boldsymbol{L}^T$	$\boldsymbol{T} = \boldsymbol{L}\boldsymbol{L}^T$	$\boldsymbol{T} = \boldsymbol{L}\boldsymbol{L}^T$ of $\boldsymbol{T}$.
lyche-numerical-linear-algebra-183-F02485	183	1.000	$\boldsymbol{A}^{\prime}=\sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$	$\boldsymbol{A}' = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$	$m \geq n$, and let $\boldsymbol{A}' = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$, where $1 \leq r \leq n$, σ_i are the singular values
lyche-numerical-linear-algebra-183-F02486	183	1.000	$1 \leq r \leq n, \sigma_i$	$1 \leq r \leq n, \sigma_i$	$m \geq n$, and let $\boldsymbol{A}' = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$, where $1 \leq r \leq n$, σ_i are the singular values
lyche-numerical-linear-algebra-183-F02487	183	1.000	$\boldsymbol{u}_i, \boldsymbol{v}_i$	$\boldsymbol{u}_i, \boldsymbol{v}_i$	of $\boldsymbol{A}$, and where $\boldsymbol{u}_i, \boldsymbol{v}_i$ are the columns of $\boldsymbol{U}$ and $\boldsymbol{V}$. Prove that
lyche-numerical-linear-algebra-185-F02488	185	1.000	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}$	space $\mathcal{V}$ is a function $\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}$ that satisfies for all $\boldsymbol{x}, \boldsymbol{y}$ in $\mathcal{V}$ and all a in $\mathbb{R}$
lyche-numerical-linear-algebra-185-F02489	185	0.728	$(\mathcal{V}, \mathbb{R}, \ \cdot\)$	$(\mathcal{V}, \mathbb{R}, \ \cdot\)$	The triple $(\mathcal{V}, \mathbb{R}, \ \cdot\)$ (resp. $(\mathcal{V}, \mathbb{C}, \ \cdot\)$) is called a normed vector space and the
lyche-numerical-linear-algebra-185-F02490	185	0.728	$(\mathcal{V}, \mathbb{C}, \ \cdot\)$	$(\mathcal{V}, \mathbb{C}, \ \cdot\)$	The triple $(\mathcal{V}, \mathbb{R}, \ \cdot\)$ (resp. $(\mathcal{V}, \mathbb{C}, \ \cdot\)$) is called a normed vector space and the
lyche-numerical-linear-algebra-185-F02491	185	1.000	$\mathbb{R}^n, \mathbb{C}^n$	$\mathbb{R}^n, \mathbb{C}^n$	In this book the vector space will be one of $\mathbb{R}^n, \mathbb{C}^n$ or one of the matrix spaces
lyche-numerical-linear-algebra-185-F02492	185	1.000	$\mathcal{V}=\mathbb{C}^n$	$\mathcal{V} = \mathbb{C}^n$	book we will use the following family of vector norms on $\mathcal{V} = \mathbb{C}^n$ and $\mathcal{V} = \mathbb{R}^n$.
lyche-numerical-linear-algebra-185-F02493	185	1.000	$\mathcal{V}=\mathbb{R}^n$	$\mathcal{V} = \mathbb{R}^n$	book we will use the following family of vector norms on $\mathcal{V} = \mathbb{C}^n$ and $\mathcal{V} = \mathbb{R}^n$.
lyche-numerical-linear-algebra-186-F02494	186	0.987	$p \geq 1$	$p \geq 1$	Definition 8.2 (Vector p-Norms) We define for $p \geq 1$ and $\boldsymbol{x} \in \mathbb{R}^n$ or $\boldsymbol{x} \in \mathbb{C}^n$ the
lyche-numerical-linear-algebra-186-F02495	186	1.000	$p=1,2, \infty$	$p = 1, 2, \infty$	The most important cases are $p = 1, 2, \infty$:
lyche-numerical-linear-algebra-186-F02496	186	1.000	$\ \boldsymbol{x}\ _1:=\sum_{j=1}^n\left x_j\right $	$\ \boldsymbol{x}\ _1 := \sum_{j=1}^n x_j $	1. $\ \boldsymbol{x}\ _1 := \sum_{j=1}^n x_j $,
lyche-numerical-linear-algebra-186-F02497	186	1.000	$\ \boldsymbol{x}\ _2:=\sqrt{\sum_{j=1}^n\left x_j\right ^2}$	$\ \boldsymbol{x}\ _2 := \sqrt{\sum_{j=1}^n x_j ^2}$	2. $\ \boldsymbol{x}\ _2 := \sqrt{\sum_{j=1}^n x_j ^2}$,
lyche-numerical-linear-algebra-186-F02498	186	1.000	l_2	l_2	(the two-norm, l_2 -norm, or Euclidian norm)
lyche-numerical-linear-algebra-186-F02499	186	1.000	$\ \boldsymbol{x}\ _\infty:=\max _{1 \leq j \leq n}\left x_j\right $	$\ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j $	3. $\ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j $,
lyche-numerical-linear-algebra-F02500	186	1.000	l_{∞}	l_∞	(the infinity-norm, l_∞ -norm, or max norm)
lyche-numerical-linear-algebra-F02501	186	1.000	$1 \leq p \leq \infty$	$1 \leq p \leq \infty$	1. In Sect. 8.4, we show that the p -norms are vector norms for $1 \leq p \leq \infty$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F02502	186	■ 1.000	$\ \boldsymbol{x} + \boldsymbol{y}\ _p \leq \ \boldsymbol{x}\ _p + \ \boldsymbol{y}\ _p$	$\ \boldsymbol{x} + \boldsymbol{y}\ _p \leq \ \boldsymbol{x}\ _p + \ \boldsymbol{y}\ _p$	2. The triangle inequality $\ \boldsymbol{x} + \boldsymbol{y}\ _p \leq \ \boldsymbol{x}\ _p + \ \boldsymbol{y}\ _p$ is called Minkowski's
lyche-numerical-linear-algebra-F02503	186	■ 1.000	$\frac{1}{p} + \frac{1}{q} = 1$	$\frac{1}{p} + \frac{1}{q} = 1$	The relation $\frac{1}{p} + \frac{1}{q} = 1$ means that if $p = 1$ then $q = \infty$ and vice versa.
lyche-numerical-linear-algebra-F02504	186	■ 1.000	$p=1$	$p = 1$	The relation $\frac{1}{p} + \frac{1}{q} = 1$ means that if $p = 1$ then $q = \infty$ and vice versa.
lyche-numerical-linear-algebra-F02505	186	■ 1.000	$q=\infty$	$q = \infty$	The relation $\frac{1}{p} + \frac{1}{q} = 1$ means that if $p = 1$ then $q = \infty$ and vice versa.
lyche-numerical-linear-algebra-F02506	186	■ 1.000	$p=2$	$p = 2$	Theorem 5.1) for the Euclidian norm $p = 2$.
lyche-numerical-linear-algebra-F02507	186	■ 1.000	$\lim_{p \rightarrow \infty} n^{1/p} = 1$	$\lim_{p \rightarrow \infty} n^{1/p} = 1$	Since $\lim_{p \rightarrow \infty} n^{1/p} = 1$ for any fixed $n \in \mathbb{N}$, we see that (8.4) follows.
lyche-numerical-linear-algebra-F02508	187	■ 0.987	$\ \cdot\ $	$\ \cdot\ $	Definition 8.3 (Equivalent Norms) We say that two norms $\ \cdot\ $ and $\ \cdot\ '$ on $\mathcal{V}$ are
lyche-numerical-linear-algebra-F02509	187	■ 0.987	$\ \cdot\ '$	$\ \cdot\ '$	Definition 8.3 (Equivalent Norms) We say that two norms $\ \cdot\ $ and $\ \cdot\ '$ on $\mathcal{V}$ are
lyche-numerical-linear-algebra-F02510	187	■ 1.000	M	M	equivalent if there are positive constants m and M such that for all vectors $\boldsymbol{x} \in \mathcal{V}$
lyche-numerical-linear-algebra-F02511	187	■ 1.000	$\boldsymbol{x} \in \mathcal{V}$	$\boldsymbol{x} \in \mathcal{V}$	equivalent if there are positive constants m and M such that for all vectors $\boldsymbol{x} \in \mathcal{V}$
lyche-numerical-linear-algebra-F02512	187	■ 1.000	∞	∞	By (8.5) the p - and ∞ -norms are equivalent for any $p \geq 1$. This result is
lyche-numerical-linear-algebra-F02513	187	■ 1.000	$\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $	$\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $	1. $\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $, for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ (inverse triangle inequality).
lyche-numerical-linear-algebra-F02514	187	■ 1.000	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	1. $\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $, for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ (inverse triangle inequality).
lyche-numerical-linear-algebra-F02515	187	■ 1.000	$\mathcal{V} \rightarrow \mathbb{R}$	$\mathcal{V} \rightarrow \mathbb{R}$	2. The vector norm is a continuous function $\mathcal{V} \rightarrow \mathbb{R}$.
lyche-numerical-linear-algebra-F02516	187	■ 1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{x} - \boldsymbol{y} + \boldsymbol{y}\ \leq \ \boldsymbol{x} - \boldsymbol{y}\ + \ \boldsymbol{y}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{x} - \boldsymbol{y} + \boldsymbol{y}\ \leq \ \boldsymbol{x} - \boldsymbol{y}\ + \ \boldsymbol{y}\ $	1. Since $\ \boldsymbol{x}\ = \ \boldsymbol{x} - \boldsymbol{y} + \boldsymbol{y}\ \leq \ \boldsymbol{x} - \boldsymbol{y}\ + \ \boldsymbol{y}\ $ we obtain $\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $. By
lyche-numerical-linear-algebra-F02517	187	■ 1.000	$\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $	$\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $	1. Since $\ \boldsymbol{x}\ = \ \boldsymbol{x} - \boldsymbol{y} + \boldsymbol{y}\ \leq \ \boldsymbol{x} - \boldsymbol{y}\ + \ \boldsymbol{y}\ $ we obtain $\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $. By
lyche-numerical-linear-algebra-F02518	187	■ 1.000	$\ \boldsymbol{x} - \boldsymbol{y}\ = \ \boldsymbol{y} - \boldsymbol{x}\ \geq \ \boldsymbol{y}\ - \ \boldsymbol{x}\ $	$\ \boldsymbol{x} - \boldsymbol{y}\ = \ \boldsymbol{y} - \boldsymbol{x}\ \geq \ \boldsymbol{y}\ - \ \boldsymbol{x}\ $	symmetry $\ \boldsymbol{x} - \boldsymbol{y}\ = \ \boldsymbol{y} - \boldsymbol{x}\ \geq \ \boldsymbol{y}\ - \ \boldsymbol{x}\ $ and we obtain the inverse triangle
lyche-numerical-linear-algebra-F02519	187	■ 1.000	$f: \mathcal{S} \rightarrow \mathbb{R}$	$f: \mathcal{S} \rightarrow \mathbb{R}$	The set $\mathcal{S}$ is a closed and bounded set and the function $f: \mathcal{S} \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra-F02520	187	■ 1.000	$f(\boldsymbol{y}) = \ \boldsymbol{y}\ $	$f(\boldsymbol{y}) = \ \boldsymbol{y}\ $	$f(\boldsymbol{y}) = \ \boldsymbol{y}\ $ is continuous by what we just showed. Therefore f attains its
lyche-numerical-linear-algebra-F02521	187	■ 1.000	$\boldsymbol{y} := \boldsymbol{x} / \ \boldsymbol{x}\ ' \in \mathcal{S}$	$\boldsymbol{y} := \boldsymbol{x} / \ \boldsymbol{x}\ ' \in \mathcal{S}$	For any $\boldsymbol{x} \in \mathcal{V}$ we have $\boldsymbol{y} := \boldsymbol{x} / \ \boldsymbol{x}\ ' \in \mathcal{S}$, and (8.7) follows if we apply (8.8) to
lyche-numerical-linear-algebra-F02522	188	■ 0.997	$(\mathbb{C}^{m \times n}, \mathbb{C})$	$(\mathbb{C}^{m \times n}, \mathbb{C})$	For simplicity we consider only norms on the vector space $(\mathbb{C}^{m \times n}, \mathbb{C})$. All results
lyche-numerical-linear-algebra-F02523	188	■ 1.000	$(\mathbb{R}^{m \times n}, \mathbb{R})$	$(\mathbb{R}^{m \times n}, \mathbb{R})$	also holds for $(\mathbb{R}^{m \times n}, \mathbb{R})$. A matrix norm $\ \cdot\ : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ is simply a vector norm
lyche-numerical-linear-algebra-F02524	188	■ 1.000	$\ \cdot\ : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$	$\ \cdot\ : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$	also holds for $(\mathbb{R}^{m \times n}, \mathbb{R})$. A matrix norm $\ \cdot\ : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ is simply a vector norm
lyche-numerical-linear-algebra-F02525	188	■ 1.000	$m n$	mn	is a matrix norm. Indeed, writing all elements in $\boldsymbol{A}$ in a string of length mn we see
lyche-numerical-linear-algebra-F02526	188	■ 1.000	$\mathbb{C}^{mn}$	$\mathbb{C}^{mn}$	that the Frobenius norm is the Euclidian norm on the space $\mathbb{C}^{mn}$.
lyche-numerical-linear-algebra-F02527	188	■ 1.000	$\ \cdot\ _V$	$\ \cdot\ _V$	Any vector norm $\ \cdot\ _V$ on $\mathbb{C}^{mn}$ defines a matrix norm on $\mathbb{C}^{m \times n}$ given by $\ \boldsymbol{A}\ :=$

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lyche-numerical-linear-algebra-F02528	188	■ 1.000	$\ \boldsymbol{A}\ :=$	$\ \boldsymbol{A}\ :=$	Any vector norm $\ \cdot\ _V$ on $\mathbb{C}^{mn}$ defines a matrix norm on $\mathbb{C}^{m \times n}$ given by $\ \boldsymbol{A}\ :=$
lyche-numerical-linear-algebra-F02529	188	■ 1.000	$\ \operatorname{vec}(\boldsymbol{A})\ _V$	$\ \operatorname{vec}(\boldsymbol{A})\ _V$	$\ \operatorname{vec}(\boldsymbol{A})\ _V$, where $\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$ is the vector obtained by stacking the columns
lyche-numerical-linear-algebra-F02530	189	■ 1.000	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ $	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ $	4. $\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ $
lyche-numerical-linear-algebra-F02531	189	■ 0.998	$\ \cdot\ $	$\ \cdot\ $	Definition 8.5 (Subordinate Matrix Norms) Suppose $m, n \in \mathbb{N}$ are given, let $\ \cdot\ $
lyche-numerical-linear-algebra-F02532	189	■ 1.000	$\ \cdot\ _{\beta}$	$\ \cdot\ _{\beta}$	on $\mathbb{C}^m$ and $\ \cdot\ _{\beta}$ on $\mathbb{C}^n$ be vector norms, and let $\ \cdot\ $ be a matrix norm on $\mathbb{C}^{m \times n}$. We
lyche-numerical-linear-algebra-F02533	189	■ 1.000	$\ \cdot\ $	$\ \cdot\ $	on $\mathbb{C}^m$ and $\ \cdot\ _{\beta}$ on $\mathbb{C}^n$ be vector norms, and let $\ \cdot\ $ be a matrix norm on $\mathbb{C}^{m \times n}$. We
lyche-numerical-linear-algebra-F02534	189	■ 0.999	$\ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ _{\beta}$	$\ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ _{\beta}$	$\ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ _{\beta}$ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra-F02535	189	■ 1.000	$m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}$	$m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}$	Proposition 8.1 For $m, n \in \mathbb{N}$, $\boldsymbol{A} \in \mathbb{C}^{m \times n}$, all $\boldsymbol{x} \in \mathbb{C}^n$ and any consistent matrix
lyche-numerical-linear-algebra-F02536	189	■ 1.000	$\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n \times 1}$	$\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n \times 1}$	consistency implies that (8.11) holds for $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n \times 1}$. The last
lyche-numerical-linear-algebra-F02537	190	■ 0.969	$n \in \mathbb{N}.$	$n \in \mathbb{N}.$	defined on $\mathbb{C}^n$ for all $n \in \mathbb{N}$
lyche-numerical-linear-algebra-F02538	190	■ 1.000	$\ \cdot\ $	$\ \cdot\ $	Definition 8.6 (Operator Norm) Let $\ \cdot\ $ be a vector norm defined on $\mathbb{C}^n$ for all
lyche-numerical-linear-algebra-F02539	190	■ 1.000	$\ \cdot\ : \mathcal{S} \rightarrow \mathbb{R}$	$\ \cdot\ : \mathcal{S} \rightarrow \mathbb{R}$	norm $\ \cdot\ : \mathcal{S} \rightarrow \mathbb{R}$ is a continuous function, it follows that the function
lyche-numerical-linear-algebra-F02540	190	■ 0.995	$f(\boldsymbol{x}) = \ \boldsymbol{A} \boldsymbol{x}\ $	$f(\boldsymbol{x}) = \ \boldsymbol{A} \boldsymbol{x}\ $	$f : \mathcal{S} \rightarrow \mathbb{R}$ given by $f(\boldsymbol{x}) = \ \boldsymbol{A} \boldsymbol{x}\ $ is continuous. But then f attains its max and
lyche-numerical-linear-algebra-F02541	190	■ 1.000	$\ \cdot\ _{\beta}$	$\ \cdot\ _{\beta}$	¹ In the case of one vector norm $\ \cdot\ $ on $\mathbb{C}^m$ and another vector norm $\ \cdot\ _{\beta}$ on $\mathbb{C}^n$ we would define
lyche-numerical-linear-algebra-F02542	190	■ 0.587	$\ \boldsymbol{A}\ := \max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ _{\beta}}$	$\ \boldsymbol{A}\ := \max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ _{\beta}}$	$\ \boldsymbol{A}\ := \max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{A} \boldsymbol{x}\ }{\ \boldsymbol{x}\ _{\beta}}$.
lyche-numerical-linear-algebra-F02543	191	■ 1.000	$\ \boldsymbol{I}\ = 1$	$\ \boldsymbol{I}\ = 1$	consistent matrix norm. Moreover, $\ \boldsymbol{I}\ = 1$.
lyche-numerical-linear-algebra-F02544	191	■ 1.000	$\ \boldsymbol{A}\ = 0$	$\ \boldsymbol{A}\ = 0$	1. Nonnegativity is obvious. If $\ \boldsymbol{A}\ = 0$ then $\ \boldsymbol{A} \boldsymbol{y}\ = 0$ for each $\boldsymbol{y} \in \mathbb{C}^n$. In
lyche-numerical-linear-algebra-F02545	191	■ 1.000	$\ \boldsymbol{A} \boldsymbol{y}\ = 0$	$\ \boldsymbol{A} \boldsymbol{y}\ = 0$	1. Nonnegativity is obvious. If $\ \boldsymbol{A}\ = 0$ then $\ \boldsymbol{A} \boldsymbol{y}\ = 0$ for each $\boldsymbol{y} \in \mathbb{C}^n$. In
lyche-numerical-linear-algebra-F02546	191	■ 1.000	$\boldsymbol{A} = 0$	$\boldsymbol{A} = 0$	particular, each column $\boldsymbol{A} \boldsymbol{e}_j$ in $\boldsymbol{A}$ is zero. Hence $\boldsymbol{A} = 0$.
lyche-numerical-linear-algebra-F02547	191	■ 1.000	$\ c \boldsymbol{A}\ = \max_{\boldsymbol{x}} \ c \boldsymbol{A} \boldsymbol{x}\ = \max_{\boldsymbol{x}} c \ \boldsymbol{A} \boldsymbol{x}\ = c \ \boldsymbol{A}\ $	$\ c \boldsymbol{A}\ = \max_{\boldsymbol{x}} \ c \boldsymbol{A} \boldsymbol{x}\ = \max_{\boldsymbol{x}} c \ \boldsymbol{A} \boldsymbol{x}\ = c \ \boldsymbol{A}\ $	2. $\ c \boldsymbol{A}\ = \max_{\boldsymbol{x}} \ c \boldsymbol{A} \boldsymbol{x}\ = \max_{\boldsymbol{x}} c \ \boldsymbol{A} \boldsymbol{x}\ = c \ \boldsymbol{A}\ $.
lyche-numerical-linear-algebra-F02548	191	■ 1.000	$\ \boldsymbol{A} + \boldsymbol{B}\ = \max_{\boldsymbol{x}} \ (\boldsymbol{A} + \boldsymbol{B}) \boldsymbol{x}\ \leq \max_{\boldsymbol{x}} \ \boldsymbol{A} \boldsymbol{x}\ + \max_{\boldsymbol{x}} \ \boldsymbol{B} \boldsymbol{x}\ = \ \boldsymbol{A}\ + \ \boldsymbol{B}\ $	$\ \boldsymbol{A} + \boldsymbol{B}\ = \max_{\boldsymbol{x}} \ (\boldsymbol{A} + \boldsymbol{B}) \boldsymbol{x}\ \leq \max_{\boldsymbol{x}} \ \boldsymbol{A} \boldsymbol{x}\ + \max_{\boldsymbol{x}} \ \boldsymbol{B} \boldsymbol{x}\ = \ \boldsymbol{A}\ + \ \boldsymbol{B}\ $	3. $\ \boldsymbol{A} + \boldsymbol{B}\ = \max_{\boldsymbol{x}} \ (\boldsymbol{A} + \boldsymbol{B}) \boldsymbol{x}\ \leq \max_{\boldsymbol{x}} \ \boldsymbol{A} \boldsymbol{x}\ + \max_{\boldsymbol{x}} \ \boldsymbol{B} \boldsymbol{x}\ = \ \boldsymbol{A}\ + \ \boldsymbol{B}\ $.
lyche-numerical-linear-algebra-F02549	191	■ 1.000	$\ \boldsymbol{I}\ _F = \sqrt{n}$	$\ \boldsymbol{I}\ _F = \sqrt{n}$	Since $\ \boldsymbol{I}\ _F = \sqrt{n}$, we see that the Frobenius norm is not an operator norm for
lyche-numerical-linear-algebra-F02550	191	■ 1.000	$n > 1$	$n > 1$	$n > 1$.
lyche-numerical-linear-algebra-F02552	191	■ 1.000	ℓ_p	ℓ_p	Recall that the p or ℓ_p vector norms (8.1) are given by
lyche-numerical-linear-algebra-F02553	191	■ 1.000	$\ \cdot\ _p$	$\ \cdot\ _p$	The operator norms $\ \cdot\ _p$ defined from these p -vector norms are used quite

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lyche-numerical-linear-algebra-192-F02554	192	■ 1.000	$\ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}\ _2$	The two-norm $\ \boldsymbol{A}\ _2$ is also called the spectral norm of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-192-F02555	192	■ 0.998	$K_{\{p\}}$	K_p	(a) We derive a constant K_p such that $\ \boldsymbol{A}\boldsymbol{x}\ _p \leq K_p$ for any $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _p = 1$.
lyche-numerical-linear-algebra-192-F02556	192	■ 0.998	$\ \boldsymbol{A}\boldsymbol{x}\ _p \leq K_p$	$\ \boldsymbol{A}\boldsymbol{x}\ _p \leq K_p$	(a) We derive a constant K_p such that $\ \boldsymbol{A}\boldsymbol{x}\ _p \leq K_p$ for any $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _p = 1$.
lyche-numerical-linear-algebra-192-F02557	192	■ 0.998	$\ \boldsymbol{x}\ _p = 1$	$\ \boldsymbol{x}\ _p = 1$	(a) We derive a constant K_p such that $\ \boldsymbol{A}\boldsymbol{x}\ _p \leq K_p$ for any $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _p = 1$.
lyche-numerical-linear-algebra-192-F02558	192	■ 1.000	$\boldsymbol{y}_e \in \mathbb{C}^n$	$\boldsymbol{y}_e \in \mathbb{C}^n$	(b) We give an extremal vector $\boldsymbol{y}_e \in \mathbb{C}^n$ with $\ \boldsymbol{y}_e\ _p = 1$ so that $\ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$.
lyche-numerical-linear-algebra-192-F02559	192	■ 1.000	$\ \boldsymbol{y}_e\ _p = 1$	$\ \boldsymbol{y}_e\ _p = 1$	(b) We give an extremal vector $\boldsymbol{y}_e \in \mathbb{C}^n$ with $\ \boldsymbol{y}_e\ _p = 1$ so that $\ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$.
lyche-numerical-linear-algebra-192-F02560	192	■ 1.000	$\ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$	$\ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$	(b) We give an extremal vector $\boldsymbol{y}_e \in \mathbb{C}^n$ with $\ \boldsymbol{y}_e\ _p = 1$ so that $\ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$.
lyche-numerical-linear-algebra-192-F02561	192	■ 1.000	$\ \boldsymbol{A}\ _p = \ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$	$\ \boldsymbol{A}\ _p = \ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$	It then follows from (8.17) that $\ \boldsymbol{A}\ _p = \ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$.
lyche-numerical-linear-algebra-192-F02562	192	■ 1.000	$K_2 =$	$K_2 =$	2-norm: Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*$ be a singular value decomposition of $\boldsymbol{A}$, define $K_2 =$
lyche-numerical-linear-algebra-192-F02563	192	■ 0.998	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^* \boldsymbol{x}$	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^* \boldsymbol{x}$	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^* \boldsymbol{x}$, and $\boldsymbol{y}_e = \boldsymbol{v}_1$ the singular vector corresponding to σ_1 . Then $\boldsymbol{x} =$
lyche-numerical-linear-algebra-192-F02564	192	■ 0.998	$\boldsymbol{y}_e = \boldsymbol{v}_1$	$\boldsymbol{y}_e = \boldsymbol{v}_1$	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^* \boldsymbol{x}$, and $\boldsymbol{y}_e = \boldsymbol{v}_1$ the singular vector corresponding to σ_1 . Then $\boldsymbol{x} =$
lyche-numerical-linear-algebra-192-F02565	192	■ 0.998	σ_1	σ_1	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^* \boldsymbol{x}$, and $\boldsymbol{y}_e = \boldsymbol{v}_1$ the singular vector corresponding to σ_1 . Then $\boldsymbol{x} =$
lyche-numerical-linear-algebra-192-F02566	192	■ 1.000	$\ \boldsymbol{V}\boldsymbol{c}\ _2 = \ \boldsymbol{x}\ _2 = 1$	$\ \boldsymbol{V}\boldsymbol{c}\ _2 = \ \boldsymbol{x}\ _2 = 1$	$\ \boldsymbol{V}\boldsymbol{c}\ _2 = \ \boldsymbol{x}\ _2 = 1$, and using (7.7) in (b) we find
lyche-numerical-linear-algebra-192-F02567	192	■ 0.998	$\ \boldsymbol{A}\boldsymbol{x}\ _2^2 = \ \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^* \boldsymbol{x}\ _2^2 = \ \boldsymbol{\Sigma}\boldsymbol{c}\ _2^2 = \sum_{j=1}^n \sigma_j^2 c_j ^2 \leq \sigma_1^2 \sum_{j=1}^n c_j ^2 = \sigma_1^2$	$\ \boldsymbol{A}\boldsymbol{x}\ _2^2 = \ \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^* \boldsymbol{x}\ _2^2 = \ \boldsymbol{\Sigma}\boldsymbol{c}\ _2^2 = \sum_{j=1}^n \sigma_j^2 c_j ^2 \leq \sigma_1^2 \sum_{j=1}^n c_j ^2 = \sigma_1^2$	(a) $\ \boldsymbol{A}\boldsymbol{x}\ _2^2 = \ \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^* \boldsymbol{x}\ _2^2 = \ \boldsymbol{\Sigma}\boldsymbol{c}\ _2^2 = \sum_{j=1}^n \sigma_j^2 c_j ^2 \leq \sigma_1^2 \sum_{j=1}^n c_j ^2 = \sigma_1^2$.
lyche-numerical-linear-algebra-192-F02568	192	■ 1.000	$\ \boldsymbol{A}\boldsymbol{v}_1\ _2 = \ \sigma_1 \boldsymbol{u}_1\ _2 = \sigma_1$	$\ \boldsymbol{A}\boldsymbol{v}_1\ _2 = \ \sigma_1 \boldsymbol{u}_1\ _2 = \sigma_1$	(b) $\ \boldsymbol{A}\boldsymbol{v}_1\ _2 = \ \sigma_1 \boldsymbol{u}_1\ _2 = \sigma_1$.
lyche-numerical-linear-algebra-192-F02569	192	■ 1.000	K_1, c	K_1, c	1-norm: Define K_1, c and $\boldsymbol{y}_e$ by $K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$ and $\boldsymbol{y}_e :=$
lyche-numerical-linear-algebra-192-F02570	192	■ 1.000	$\boldsymbol{y}_e$	$\boldsymbol{y}_e$	1-norm: Define K_1, c and $\boldsymbol{y}_e$ by $K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$ and $\boldsymbol{y}_e :=$
lyche-numerical-linear-algebra-192-F02571	192	■ 1.000	$K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$	$K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$	1-norm: Define K_1, c and $\boldsymbol{y}_e$ by $K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$ and $\boldsymbol{y}_e :=$
lyche-numerical-linear-algebra-192-F02572	192	■ 1.000	$\boldsymbol{y}_e :=$	$\boldsymbol{y}_e :=$	1-norm: Define K_1, c and $\boldsymbol{y}_e$ by $K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$ and $\boldsymbol{y}_e :=$
lyche-numerical-linear-algebra-192-F02573	192	■ 1.000	$\boldsymbol{e}_c$	$\boldsymbol{e}_c$	$\boldsymbol{e}_c$, a unit vector. Then $\ \boldsymbol{y}_e\ _1 = 1$ and we obtain
lyche-numerical-linear-algebra-192-F02574	192	■ 1.000	$\ \boldsymbol{y}_e\ _1 = 1$	$\ \boldsymbol{y}_e\ _1 = 1$	$\boldsymbol{e}_c$, a unit vector. Then $\ \boldsymbol{y}_e\ _1 = 1$ and we obtain
lyche-numerical-linear-algebra-192-F02575	192	■ 1.000	$\ \boldsymbol{A}\boldsymbol{y}_e\ _1 = K_1$	$\ \boldsymbol{A}\boldsymbol{y}_e\ _1 = K_1$	(b) $\ \boldsymbol{A}\boldsymbol{y}_e\ _1 = K_1$.
lyche-numerical-linear-algebra-192-F02576	192	■ 0.997	K_∞, r	K_∞, r	∞-norm: Define K_∞, r and $\boldsymbol{y}_e$ by $K_\infty := \ \boldsymbol{e}_r^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1$ and
lyche-numerical-linear-algebra-192-F02577	192	■ 0.997	$K_\infty := \ \boldsymbol{e}_r^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1$	$K_\infty := \ \boldsymbol{e}_r^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1$	∞-norm: Define K_∞, r and $\boldsymbol{y}_e$ by $K_\infty := \ \boldsymbol{e}_r^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1$ and

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lyche-numerical-linear-algebra-F02578	192	0.998	$\boldsymbol{y}_e := \left[e^{-i\theta_1}, \dots, e^{-i\theta_n} \right]^T$	$\boldsymbol{y}_e := [e^{-i\theta_1}, \dots, e^{-i\theta_n}]^T$	$\boldsymbol{y}_e := [e^{-i\theta_1}, \dots, e^{-i\theta_n}]^T$, where $a_{rj} = a_{rj} e^{i\theta_j}$ for $j = 1, \dots, n$.
lyche-numerical-linear-algebra-F02579	192	0.998	$a_{\{r\}j} = a_{rj} e^{i\theta_j}$	$a_{rj} = a_{rj} e^{i\theta_j}$	$\boldsymbol{y}_e := [e^{-i\theta_1}, \dots, e^{-i\theta_n}]^T$, where $a_{rj} = a_{rj} e^{i\theta_j}$ for $j = 1, \dots, n$.
lyche-numerical-linear-algebra-F02580	192	1.000	$\left\ \boldsymbol{A} \boldsymbol{y}^* \right\ _{\infty} = \max_{1 \leq k \leq m} \left \sum_{j=1}^n a_{kj} e^{-i\theta_j} \right = K_{\infty}$	$\left\ \boldsymbol{A} \boldsymbol{y}^* \right\ _{\infty} = \max_{1 \leq k \leq m} \left \sum_{j=1}^n a_{kj} e^{-i\theta_j} \right = K_{\infty}$	(b) $\left\ \boldsymbol{A} \boldsymbol{y}^* \right\ _{\infty} = \max_{1 \leq k \leq m} \left \sum_{j=1}^n a_{kj} e^{-i\theta_j} \right = K_{\infty}$.
lyche-numerical-linear-algebra-F02581	192	1.000	$\left \sum_{j=1}^n a_{\{k\}j} e^{-i\theta_j} \right \leq \sum_{j=1}^n a_{\{k\}j} \leq K_{\infty}$	$\left \sum_{j=1}^n a_{kj} e^{-i\theta_j} \right \leq \sum_{j=1}^n a_{kj} \leq K_{\infty}$	The last equality is correct because $\left \sum_{j=1}^n a_{kj} e^{-i\theta_j} \right \leq \sum_{j=1}^n a_{kj} \leq K_{\infty}$ with
lyche-numerical-linear-algebra-F02582	192	1.000	$k=r$	$k = r$	equality for $k=r$.
lyche-numerical-linear-algebra-F02583	192	0.999	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix}$	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix}$	matrix $\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix}$, is $\sigma_1 = 2$ (cf. Example 7.9). We find
lyche-numerical-linear-algebra-F02584	193	1.000	$\sigma_1 \geq \sigma_2 \geq$	$\sigma_1 \geq \sigma_2 \geq$	Theorem 8.4 (Spectral Norm) Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has singular values $\sigma_1 \geq \sigma_2 \geq$
lyche-numerical-linear-algebra-F02585	193	1.000	$\cdots \geq \sigma_n$	$\cdots \geq \sigma_n$	$\cdots \geq \sigma_n$ and eigenvalues $ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $. Then
lyche-numerical-linear-algebra-F02586	193	1.000	$ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $	$ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $	$\cdots \geq \sigma_n$ and eigenvalues $ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $. Then
lyche-numerical-linear-algebra-F02587	193	1.000	$1/\sigma_n$	$1/\sigma_n$	Proof Since $1/\sigma_n$ is the largest singular value of $\boldsymbol{A}^{-1}$, (8.19) follows. By Exer-
lyche-numerical-linear-algebra-F02588	193	1.000	$\ \boldsymbol{A}\ _2^2 \leq$	$\ \boldsymbol{A}\ _2^2 \leq$	Theorem 8.5 (Spectral Norm Bound) For any $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ we have $\ \boldsymbol{A}\ _2^2 \leq$
lyche-numerical-linear-algebra-F02589	193	0.994	$\ \boldsymbol{A}\ _1 \ \boldsymbol{A}\ _{\infty}$	$\ \boldsymbol{A}\ _1 \ \boldsymbol{A}\ _{\infty}$	$\ \boldsymbol{A}\ _1 \ \boldsymbol{A}\ _{\infty}$.
lyche-numerical-linear-algebra-F02590	193	0.994	$(\sigma^2, \boldsymbol{v})$	$(\sigma^2, \boldsymbol{v})$	Proof Let $(\sigma^2, \boldsymbol{v})$ be an eigenpair for $\boldsymbol{A}^* \boldsymbol{A}$ corresponding to the largest singular
lyche-numerical-linear-algebra-F02591	193	1.000	$\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _{\infty}$	$\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _{\infty}$	Observing that $\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _{\infty}$ by Theorem 8.3 and canceling $\ \boldsymbol{v}\ _1$ proves the
lyche-numerical-linear-algebra-F02592	193	1.000	$\ \boldsymbol{v}\ _1$	$\ \boldsymbol{v}\ _1$	Observing that $\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _{\infty}$ by Theorem 8.3 and canceling $\ \boldsymbol{v}\ _1$ proves the
lyche-numerical-linear-algebra-F02593	193	1.000	$\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\ = \ \boldsymbol{A}\ $	$\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\ = \ \boldsymbol{A}\ $	unitary invariant if $\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\ = \ \boldsymbol{A}\ $ for any $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and any unitary matrices
lyche-numerical-linear-algebra-F02594	193	1.000	$\boldsymbol{U}(\boldsymbol{A} +$	$\boldsymbol{U}(\boldsymbol{A} +$	increased by a unitary transformation. Thus if $\boldsymbol{U}$ and $\boldsymbol{V}$ are unitary then $\boldsymbol{U}(\boldsymbol{A} +$
lyche-numerical-linear-algebra-F02595	193	1.000	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$, where $\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $.
lyche-numerical-linear-algebra-F02596	193	1.000	$\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $	$\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$, where $\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $.
lyche-numerical-linear-algebra-F02597	194	1.000	$\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$	Theorem 8.3 that $\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$. Moreover $\boldsymbol{V}^*$ is unitary. Using these facts we
lyche-numerical-linear-algebra-F02598	194	1.000	$\boldsymbol{V}^*$	$\boldsymbol{V}^*$	Theorem 8.3 that $\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$. Moreover $\boldsymbol{V}^*$ is unitary. Using these facts we
lyche-numerical-linear-algebra-F02599	194	1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ $	A vector norm on $\mathbb{C}^n$ is an absolute norm if $\ \boldsymbol{x}\ = \ \boldsymbol{x}\ $ for all $\boldsymbol{x} \in \mathbb{C}^n$. Here
lyche-numerical-linear-algebra-F02600	194	1.000	$ \boldsymbol{x} := [x_1 , \dots, x_n]^T$	$ \boldsymbol{x} := [x_1 , \dots, x_n]^T$	$ \boldsymbol{x} := [x_1 , \dots, x_n]^T$, the absolute values of the components of $\boldsymbol{x}$. Clearly the
lyche-numerical-linear-algebra-F02601	195	1.000	$x_1 = x_2 = 10$	$x_1 = x_2 = 10$	whose exact solution is $x_1 = x_2 = 10$. If we replace the second equation by
lyche-numerical-linear-algebra-F02602	195	1.000	$x_1 = 30, x_2 = -10$	$x_1 = 30, x_2 = -10$	the exact solution changes to $x_1 = 30, x_2 = -10$. Here a small change in one of
lyche-numerical-linear-algebra-F02603	195	1.000	$1 - 10^{-16}$	$1 - 10^{-16}$	the coefficients, from $1 - 10^{-16}$ to $1 + 10^{-16}$, changed the exact solution by a large

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lyche-numerical-linear-algebra- F02604	195	■ 1.000	$1+10^{-16}$	$1 + 10^{-16}$	the coefficients, from $1 - 10^{-16}$ to $1 + 10^{-16}$, changed the exact solution by a large
lyche-numerical-linear-algebra- F02605	195	■ 0.994	$\boldsymbol{A}, \boldsymbol{b}$	$\boldsymbol{A}, \boldsymbol{b}$	$\boldsymbol{A}, \boldsymbol{b}$ has on the inverse of $\boldsymbol{A}$ and on the solution $\boldsymbol{x}$ of a linear system $\boldsymbol{Ax} = \boldsymbol{b}$. To
lyche-numerical-linear-algebra- F02606	195	■ 1.000	$\boldsymbol{I} \in \mathbb{R}^{n \times n}$	$\boldsymbol{I} \in \mathbb{R}^{n \times n}$	vector norm. This follows from Lemma 7.1. We recall that if $\boldsymbol{I} \in \mathbb{R}^{n \times n}$ then $\ \boldsymbol{I}\ = 1$
lyche-numerical-linear-algebra- F02607	195	■ 0.996	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	Suppose $\boldsymbol{x}, \boldsymbol{y}$ solve $\boldsymbol{Ax} = \boldsymbol{b}$ and $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$, respectively. where $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$
lyche-numerical-linear-algebra- F02608	195	■ 0.996	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$	Suppose $\boldsymbol{x}, \boldsymbol{y}$ solve $\boldsymbol{Ax} = \boldsymbol{b}$ and $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$, respectively. where $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$
lyche-numerical-linear-algebra- F02609	195	■ 1.000	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$	$\mathbb{C}^{n \times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$. How large can $\boldsymbol{y} - \boldsymbol{x}$ be? The difference $\ \boldsymbol{y} - \boldsymbol{x}\ $
lyche-numerical-linear-algebra- F02610	195	■ 1.000	$\boldsymbol{y} - \boldsymbol{x}$	$\boldsymbol{y} - \boldsymbol{x}$	$\mathbb{C}^{n \times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$. How large can $\boldsymbol{y} - \boldsymbol{x}$ be? The difference $\ \boldsymbol{y} - \boldsymbol{x}\ $
lyche-numerical-linear-algebra- F02611	195	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ $	$\mathbb{C}^{n \times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$. How large can $\boldsymbol{y} - \boldsymbol{x}$ be? The difference $\ \boldsymbol{y} - \boldsymbol{x}\ $
lyche-numerical-linear-algebra- F02612	195	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ $	measures the absolute error in $\boldsymbol{y}$ as an approximation to $\boldsymbol{x}$, while $\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ $ and
lyche-numerical-linear-algebra- F02613	195	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{y}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{y}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{y}\ $ are measures for the relative error .
lyche-numerical-linear-algebra- F02614	195	■ 0.546	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$	nonsingular, $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$ and $\boldsymbol{Ax} = \boldsymbol{b}, \boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$. Then
lyche-numerical-linear-algebra- F02615	195	■ 0.546	$\boldsymbol{Ax} = \boldsymbol{b}, \boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$	$\boldsymbol{Ax} = \boldsymbol{b}, \boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$	nonsingular, $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$ and $\boldsymbol{Ax} = \boldsymbol{b}, \boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$ Then
lyche-numerical-linear-algebra- F02616	195	■ 1.000	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	where $K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $ is the condition number of $\boldsymbol{A}$.
lyche-numerical-linear-algebra- F02617	195	■ 1.000	$\boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$	$\boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$	Proof Subtracting $\boldsymbol{Ax} = \boldsymbol{b}$ from $\boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$ or $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra- F02618	195	■ 1.000	$\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$	$\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$	Proof Subtracting $\boldsymbol{Ax} = \boldsymbol{b}$ from $\boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$ or $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra- F02619	195	■ 1.000	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$	Proof Subtracting $\boldsymbol{Ax} = \boldsymbol{b}$ from $\boldsymbol{Ay} = \boldsymbol{b} + \boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$ or $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra- F02620	195	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $	Combining $\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $ and $\ \boldsymbol{b}\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $ we
lyche-numerical-linear-algebra- F02621	195	■ 1.000	$\ \boldsymbol{b}\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{b}\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	Combining $\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $ and $\ \boldsymbol{b}\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $ we
lyche-numerical-linear-algebra- F02622	195	■ 1.000	$\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $	$\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $	obtain the upper bound in (8.22). Combining $\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $ and $\ \boldsymbol{x}\ \leq$
lyche-numerical-linear-algebra- F02623	195	■ 1.000	$\ \boldsymbol{x}\ \leq$	$\ \boldsymbol{x}\ \leq$	obtain the upper bound in (8.22). Combining $\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $ and $\ \boldsymbol{x}\ \leq$
lyche-numerical-linear-algebra- F02624	195	■ 1.000	$\ \boldsymbol{A}^{-1}\ \ \boldsymbol{b}\ $	$\ \boldsymbol{A}^{-1}\ \ \boldsymbol{b}\ $	$\ \boldsymbol{A}^{-1}\ \ \boldsymbol{b}\ $ we obtain the lower bound.
lyche-numerical-linear-algebra- F02625	196	■ 1.000	$\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $	$\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $	Consider (8.22). $\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $ is a measure of the size of the perturbation $\boldsymbol{e}$ relative
lyche-numerical-linear-algebra- F02626	196	■ 1.000	$\boldsymbol{e}$	$\boldsymbol{e}$	Consider (8.22). $\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $ is a measure of the size of the perturbation $\boldsymbol{e}$ relative
lyche-numerical-linear-algebra- F02627	196	■ 0.999	$K(\boldsymbol{A})$	$K(\boldsymbol{A})$	$K(\boldsymbol{A})$ times as large as $\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $.
lyche-numerical-linear-algebra- F02628	196	■ 1.000	$K(\boldsymbol{A}) \geq 1$	$K(\boldsymbol{A}) \geq 1$	well-conditioned (with respect to inversion). We always have $K(\boldsymbol{A}) \geq 1$. For since
lyche-numerical-linear-algebra- F02629	196	■ 1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{Ix}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{Ix}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{Ix}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $ for any $\boldsymbol{x}$ we have $\ \boldsymbol{I}\ \geq 1$ and therefore $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$

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lyche-numerical-linear-algebra- F02630	196	1.000	$\ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $ for any $\boldsymbol{x}$ we have $\ \boldsymbol{I}\ \geq 1$ and therefore $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$
lyche-numerical-linear-algebra- F02631	196	1.000	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $ for any $\boldsymbol{x}$ we have $\ \boldsymbol{I}\ \geq 1$ and therefore $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ >$
lyche-numerical-linear-algebra- F02632	196	1.000	$\ \boldsymbol{A}\boldsymbol{A}^{-1}\ = \ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{A}\boldsymbol{A}^{-1}\ = \ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{A}\boldsymbol{A}^{-1}\ = \ \boldsymbol{I}\ >$.
lyche-numerical-linear-algebra- F02633	196	1.000	ℓ_1, ℓ_∞	ℓ_1, ℓ_∞	when discussing properties of the condition number, and the ℓ_1, ℓ_∞ , or Frobenius
lyche-numerical-linear-algebra- F02634	196	0.874	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$	Suppose we have computed an approximate solution $\boldsymbol{y}$ to $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. The vector
lyche-numerical-linear-algebra- F02635	196	1.000	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$ is called the residual vector , or just the residual. We can bound
lyche-numerical-linear-algebra- F02636	196	1.000	$\boldsymbol{r}$	$\boldsymbol{r}$	$\boldsymbol{x} - \boldsymbol{y}$ in terms of $\boldsymbol{r}$.
lyche-numerical-linear-algebra- F02637	196	0.987	$\boldsymbol{b} \neq \mathbf{0}$	$\boldsymbol{b} \neq \mathbf{0}$	nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. Let $\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$ for $\boldsymbol{y} \in \mathbb{C}^n$. If $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ then
lyche-numerical-linear-algebra- F02638	196	0.987	$\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$	nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. Let $\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}$ for $\boldsymbol{y} \in \mathbb{C}^n$. If $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ then
lyche-numerical-linear-algebra- F02639	196	1.000	$\boldsymbol{e} = \boldsymbol{r}(\boldsymbol{y})$	$\boldsymbol{e} = \boldsymbol{r}(\boldsymbol{y})$	Proof We simply take $\boldsymbol{e} = \boldsymbol{r}(\boldsymbol{y})$ in Theorem 8.7.
lyche-numerical-linear-algebra- F02640	196	0.999	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$ with $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$ nonsingular. We like to compare the solution $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra- F02641	196	0.999	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$ with $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$ nonsingular. We like to compare the solution $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra- F02642	196	0.999	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$	of the systems $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$.
lyche-numerical-linear-algebra- F02643	196	0.615	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$	Theorem 8.9 (Perturbation in Matrix) Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$ with $\boldsymbol{A}$
lyche-numerical-linear-algebra- F02644	196	0.999	$r := \ \boldsymbol{A}^{-1}\boldsymbol{E}\ < 1$	$r := \ \boldsymbol{A}^{-1}\boldsymbol{E}\ < 1$	nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. If $r := \ \boldsymbol{A}^{-1}\boldsymbol{E}\ < 1$ then $\boldsymbol{A} + \boldsymbol{E}$ is nonsingular. If $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$
lyche-numerical-linear-algebra- F02645	196	0.999	$\boldsymbol{A} + \boldsymbol{E}$	$\boldsymbol{A} + \boldsymbol{E}$	nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. If $r := \ \boldsymbol{A}^{-1}\boldsymbol{E}\ < 1$ then $\boldsymbol{A} + \boldsymbol{E}$ is nonsingular. If $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$
lyche-numerical-linear-algebra- F02646	196	1.000	$r \geq 1$	$r \geq 1$	Proof We show $\boldsymbol{A} + \boldsymbol{E}$ singular implies $r \geq 1$. Suppose $\boldsymbol{A} + \boldsymbol{E}$ is singular. Then
lyche-numerical-linear-algebra- F02647	196	1.000	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ for some nonzero $\boldsymbol{x} \in \mathbb{C}^n$. Multiplying by $\boldsymbol{A}^{-1}$ it follows that
lyche-numerical-linear-algebra- F02648	196	1.000	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ and this implies that $\ \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\ \leq r\ \boldsymbol{x}\ $. But then $r \geq 1$.
lyche-numerical-linear-algebra- F02649	196	1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\ \leq r\ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\ \leq r\ \boldsymbol{x}\ $	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ and this implies that $\ \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\ < r\ \boldsymbol{x}\ $. But then $r \geq 1$.
lyche-numerical-linear-algebra- F02650	196	0.967	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{E}\boldsymbol{y}$	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{E}\boldsymbol{y}$	Subtracting $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$ from $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ gives $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{E}\boldsymbol{y}$ or
lyche-numerical-linear-algebra- F02651	196	1.000	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$. Taking norms and dividing by $\ \boldsymbol{y}\ $ proves (8.24). Solving
lyche-numerical-linear-algebra- F02652	196	1.000	$\ \boldsymbol{y}\ $	$\ \boldsymbol{y}\ $	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$. Taking norms and dividing by $\ \boldsymbol{y}\ $ proves (8.24). Solving
lyche-numerical-linear-algebra- F02653	197	1.000	$\boldsymbol{y} = (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\boldsymbol{x}$	$\boldsymbol{y} = (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\boldsymbol{x}$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$ for $\boldsymbol{y}$ we obtain $\boldsymbol{y} = (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\boldsymbol{x}$. By Theorem 12.14 we
lyche-numerical-linear-algebra- F02654	197	0.995	$\ \boldsymbol{y}\ \leq \left\ (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1} \right\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$	$\ \boldsymbol{y}\ \leq \left\ (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1} \right\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$	have $\ \boldsymbol{y}\ \leq \left\ (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1} \right\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$ and (8.24) implies $\ \boldsymbol{y} - \boldsymbol{x}\ \leq r\ \boldsymbol{y}\ \leq$
lyche-numerical-linear-algebra- F02655	197	0.995	$\ \boldsymbol{y} - \boldsymbol{x}\ \leq r\ \boldsymbol{y}\ \leq$	$\ \boldsymbol{y} - \boldsymbol{x}\ \leq r\ \boldsymbol{y}\ \leq$	have $\ \boldsymbol{y}\ \leq \left\ (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1} \right\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$ and (8.24) implies $\ \boldsymbol{y} - \boldsymbol{x}\ < r\ \boldsymbol{y}\ <$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F02656	197	0.669	$\frac{r}{1-r} \ \mathbf{x}\ \leq \frac{K(\mathbf{A})}{1-r} \ \mathbf{E}\ \ \mathbf{A}\ $	$\frac{r}{1-r} \ \mathbf{x}\ \leq \frac{K(\mathbf{A})}{1-r} \ \mathbf{E}\ \ \mathbf{A}\ $	Dividing by $\ \mathbf{x}\ $ gives (8.25).
lyche-numerical-linear-algebra-F02657	197	0.669	$\ \mathbf{x}\ $	$\ \mathbf{x}\ $	Dividing by $\ \mathbf{x}\ $ gives (8.25).
lyche-numerical-linear-algebra-F02658	197	0.986	$\ \mathbf{x}\ \cdot \ \mathbf{E}\ / \ \mathbf{A}\ $	$\ \mathbf{x}\ \ \mathbf{E}\ / \ \mathbf{A}\ $	to $\mathbf{y}$ and the relative error in $\mathbf{y}$ as an approximation to $\mathbf{E} \cdot \ \mathbf{E}\ / \ \mathbf{A}\ $ is a measure for
lyche-numerical-linear-algebra-F02659	197	1.000	$\ \mathbf{E}\ / \ \mathbf{A}\ $	$\ \mathbf{E}\ / \ \mathbf{A}\ $	again plays a crucial role. $\ \mathbf{y} - \mathbf{x}\ / \ \mathbf{y}\ $ can be as large as $K(\mathbf{A})$ times $\ \mathbf{E}\ / \ \mathbf{A}\ $. It
lyche-numerical-linear-algebra-F02660	197	1.000	$\ \mathbf{A}^{-1} \mathbf{E} \mathbf{y}\ \leq \ \mathbf{A}^{-1}\ \ \mathbf{E}\ \ \mathbf{y}\ $	$\ \mathbf{A}^{-1} \mathbf{E} \mathbf{y}\ \leq \ \mathbf{A}^{-1}\ \ \mathbf{E}\ \ \mathbf{y}\ $	the upper bound we used the inequality $\ \mathbf{A}^{-1} \mathbf{E} \mathbf{y}\ < \ \mathbf{A}^{-1}\ \ \mathbf{E}\ \ \mathbf{y}\ $. For a more
lyche-numerical-linear-algebra-F02661	197	1.000	$\ \mathbf{A}^{-1} \mathbf{E} \mathbf{y}\ $	$\ \mathbf{A}^{-1} \mathbf{E} \mathbf{y}\ $	or less random perturbation $\mathbf{E}$ this is not a severe overestimate for $\ \mathbf{A}^{-1} \mathbf{E} \mathbf{y}\ $. In the
lyche-numerical-linear-algebra-F02662	197	1.000	$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$	$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$	with singular values $\sigma_1 > \sigma_2 > \dots > \sigma_n > 0$ and eigenvalues $ \lambda_1 \geq \lambda_2 \geq \dots \geq$
lyche-numerical-linear-algebra-F02663	197	1.000	$ \lambda_1 \geq \lambda_2 \geq \dots \geq$	$ \lambda_1 \geq \lambda_2 \geq \dots \geq$	with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$ and eigenvalues $ \lambda_1 > \lambda_2 > \dots >$
lyche-numerical-linear-algebra-F02664	197	1.000	$ \lambda_n > 0$	$ \lambda_n > 0$	$ \lambda_n > 0$. Then
lyche-numerical-linear-algebra-F02665	197	0.999	σ_1 / σ_n	σ_1 / σ_n	It follows that $\mathbf{A}$ is ill-conditioned with respect to inversion if and only if σ_1 / σ_n
lyche-numerical-linear-algebra-F02666	197	1.000	λ_1 / λ_n	λ_1 / λ_n	is large, or λ_1 / λ_n is large when $\mathbf{A}$ is positive definite.
lyche-numerical-linear-algebra-F02667	197	1.000	$\ \mathbf{y} - \mathbf{x}\ / \ \mathbf{x}\ \approx \ \mathbf{r}(\mathbf{y})\ / \ \mathbf{b}\ $	$\ \mathbf{y} - \mathbf{x}\ / \ \mathbf{x}\ \approx \ \mathbf{r}(\mathbf{y})\ / \ \mathbf{b}\ $	If $\mathbf{A}$ is well-conditioned, (8.23) says that $\ \mathbf{y} - \mathbf{x}\ / \ \mathbf{x}\ \approx \ \mathbf{r}(\mathbf{y})\ / \ \mathbf{b}\ $. In other
lyche-numerical-linear-algebra-F02668	197	1.000	$\ \mathbf{b}\ \approx 1$	$\ \mathbf{b}\ \approx 1$	long as $\ \mathbf{b}\ \approx 1$. If $\mathbf{A}$ is ill-conditioned, anything can happen. We can for example
lyche-numerical-linear-algebra-F02669	197	1.000	$\mathbf{B} := \mathbf{A} + \mathbf{E}$	$\mathbf{B} := \mathbf{A} + \mathbf{E}$	Suppose $\mathbf{A}$ is nonsingular and $\mathbf{E}$ a perturbation of $\mathbf{A}$. We expect $\mathbf{B} := \mathbf{A} + \mathbf{E}$ to be
lyche-numerical-linear-algebra-F02670	197	1.000	$\ \mathbf{B}^{-1}\ $	$\ \mathbf{B}^{-1}\ $	to have bounds on $\ \mathbf{B}^{-1}\ $ in terms of $\ \mathbf{A}^{-1}\ $ and the difference $\ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ $.
lyche-numerical-linear-algebra-F02671	197	1.000	$\ \mathbf{A}^{-1}\ $	$\ \mathbf{A}^{-1}\ $	to have bounds on $\ \mathbf{B}^{-1}\ $ in terms of $\ \mathbf{A}^{-1}\ $ and the difference $\ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ $.
lyche-numerical-linear-algebra-F02672	197	1.000	$\ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ $	$\ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ $	to have bounds on $\ \mathbf{B}^{-1}\ $ in terms of $\ \mathbf{A}^{-1}\ $ and the difference $\ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ $.
lyche-numerical-linear-algebra-F02673	197	1.000	$\ \mathbf{B}^{-1}\ / \ \mathbf{A}^{-1}\ , \ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ / \ \mathbf{B}^{-1}\ $	$\ \mathbf{B}^{-1}\ / \ \mathbf{A}^{-1}\ , \ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ / \ \mathbf{B}^{-1}\ $	We consider the relative errors $\ \mathbf{B}^{-1}\ / \ \mathbf{A}^{-1}\ , \ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ / \ \mathbf{B}^{-1}\ $ and $\ \mathbf{B}^{-1} -$
lyche-numerical-linear-algebra-F02674	197	1.000	$\ \mathbf{B}^{-1} -$	$\ \mathbf{B}^{-1} -$	We consider the relative errors $\ \mathbf{B}^{-1}\ / \ \mathbf{A}^{-1}\ , \ \mathbf{B}^{-1} - \mathbf{A}^{-1}\ / \ \mathbf{B}^{-1}\ $ and $\ \mathbf{B}^{-1} -$
lyche-numerical-linear-algebra-F02675	197	1.000	$\mathbf{A}^{-1} / \ \mathbf{A}^{-1}\ $	$\mathbf{A}^{-1} / \ \mathbf{A}^{-1}\ $	.
lyche-numerical-linear-algebra-F02676	198	1.000	$\mathbf{B} := \mathbf{A} + \mathbf{E} \in \mathbb{C}^{n \times n}$	$\mathbf{B} := \mathbf{A} + \mathbf{E} \in \mathbb{C}^{n \times n}$	gular and let $\mathbf{B} := \mathbf{A} + \mathbf{E} \in \mathbb{C}^{n \times n}$ be nonsingular. For any consistent matrix norm
lyche-numerical-linear-algebra-F02677	198	0.998	$K(\mathbf{A}) := \ \mathbf{A}\ \ \mathbf{A}^{-1}\ $	$K(\mathbf{A}) := \ \mathbf{A}\ \ \mathbf{A}^{-1}\ $	where $K(\mathbf{A}) := \ \mathbf{A}\ \ \mathbf{A}^{-1}\ $. If $r := \ \mathbf{A}^{-1} \mathbf{E}\ < 1$ then $\mathbf{B}$ is nonsingular and
lyche-numerical-linear-algebra-F02678	198	0.922	$\ \mathbf{A}^{-1} \mathbf{E}\ $	$\ \mathbf{A}^{-1} \mathbf{E}\ $	We can replace $\ \mathbf{A}^{-1} \mathbf{E}\ $ by $\ \mathbf{E} \mathbf{A}^{-1}\ $ everywhere.
lyche-numerical-linear-algebra-F02679	198	0.922	$\ \mathbf{E} \mathbf{A}^{-1}\ $	$\ \mathbf{E} \mathbf{A}^{-1}\ $	We can replace $\ \mathbf{A}^{-1} \mathbf{E}\ $ by $\ \mathbf{E} \mathbf{A}^{-1}\ $ everywhere.

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lyche-numerical-linear-algebra-F02680	198	■ 0.999	$r < 1$	$r < 1$	Proof That B is nonsingular if $r < 1$ follows from Theorem 8.9. We have $-E =$
lyche-numerical-linear-algebra-F02681	198	■ 0.999	$-\boldsymbol{E} =$	$-\boldsymbol{E} =$	Proof That B is nonsingular if $r < 1$ follows from Theorem 8.9. We have $-E =$
lyche-numerical-linear-algebra-F02682	198	■ 1.000	$\boldsymbol{A} - \boldsymbol{B} = \boldsymbol{A}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{B} = \boldsymbol{B}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{A}$	$\boldsymbol{A} - \boldsymbol{B} = \boldsymbol{A}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{B} = \boldsymbol{B}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{A}$	$\boldsymbol{A} - \boldsymbol{B} = \boldsymbol{A}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{B} = \boldsymbol{B}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{A}$ so that
lyche-numerical-linear-algebra-F02683	198	■ 1.000	$\ \boldsymbol{A}^{-1}\ \leq \ \boldsymbol{B}^{-1}\ + r\ \boldsymbol{B}^{-1}\ $	$\ \boldsymbol{A}^{-1}\ \leq \ \boldsymbol{B}^{-1}\ + r\ \boldsymbol{B}^{-1}\ $	Similarly we obtain the lower bound in (8.28) from $\ \boldsymbol{A}^{-1}\ \leq \ \boldsymbol{B}^{-1}\ + r\ \boldsymbol{B}^{-1}\ $.
lyche-numerical-linear-algebra-F02684	198	■ 1.000	$\ \boldsymbol{B}^{-1}\ / \ \boldsymbol{A}^{-1}\ $	$\ \boldsymbol{B}^{-1}\ / \ \boldsymbol{A}^{-1}\ $	The bound in (8.29) follows by multiplying (8.27) by $\ \boldsymbol{B}^{-1}\ / \ \boldsymbol{A}^{-1}\ $ and using
lyche-numerical-linear-algebra-F02685	199	■ 1.000	$1 \leq p \leq \infty, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	$1 \leq p \leq \infty, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	Then for all $1 < p < \infty, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ and all $a \in \mathbb{C}$
lyche-numerical-linear-algebra-F02686	199	■ 1.000	$\ \boldsymbol{x}\ _p \geq 0$	$\ \boldsymbol{x}\ _p \geq 0$	1. $\ \boldsymbol{x}\ _p \geq 0$ with equality if and only if $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra-F02687	199	■ 1.000	$\ a\boldsymbol{x}\ _p = a \ \boldsymbol{x}\ _p$	$\ a\boldsymbol{x}\ _p = a \ \boldsymbol{x}\ _p$	2. $\ a\boldsymbol{x}\ _p = a \ \boldsymbol{x}\ _p$.
lyche-numerical-linear-algebra-F02688	199	■ 1.000	$I \subset \mathbb{R}$	$I \subset \mathbb{R}$	Definition 8.8 (Convex Function) Let $I \subset \mathbb{R}$ be an interval. A function $f : I \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra-F02689	199	■ 1.000	$f : I \rightarrow \mathbb{R}$	$f : I \rightarrow \mathbb{R}$	Definition 8.8 (Convex Function) Let $I \subset \mathbb{R}$ be an interval. A function $f : I \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra-F02690	199	■ 1.000	$x_1, x_2 \in I$	$x_1, x_2 \in I$	for all $x_1, x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in [0, 1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a
lyche-numerical-linear-algebra-F02691	199	■ 1.000	$x_1 < x_2$	$x_1 < x_2$	for all $x_1, x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in [0, 1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a
lyche-numerical-linear-algebra-F02692	199	■ 1.000	$\lambda \in [0, 1]$	$\lambda \in [0, 1]$	for all $x_1, x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in [0, 1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a
lyche-numerical-linear-algebra-F02693	199	■ 1.000	$\sum_{j=1}^n \lambda_j x_j$	$\sum_{j=1}^n \lambda_j x_j$	for all $x_1, x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in [0, 1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a
lyche-numerical-linear-algebra-F02694	199	■ 1.000	$x_1, \dots, x_n$	$x_1, \dots, x_n$	convex combination of $x_1, \dots, x_n$ if $\lambda_j \geq 0$ for $j = 1, \dots, n$ and $\sum_{j=1}^n \lambda_j = 1$.
lyche-numerical-linear-algebra-F02695	199	■ 1.000	$\lambda_j \geq 0$	$\lambda_j \geq 0$	convex combination of $x_1, \dots, x_n$ if $\lambda_j \geq 0$ for $j = 1, \dots, n$ and $\sum_{j=1}^n \lambda_j = 1$.
lyche-numerical-linear-algebra-F02696	199	■ 1.000	$\sum_{j=1}^n \lambda_j = 1$	$\sum_{j=1}^n \lambda_j = 1$	convex combination of $x_1, \dots, x_n$ if $\lambda_j \geq 0$ for $j = 1, \dots, n$ and $\sum_{j=1}^n \lambda_j = 1$.
lyche-numerical-linear-algebra-F02697	200	■ 1.000	$f \in C^2[a, b]$	$f \in C^2[a, b]$	Lemma 8.2 (A Sufficient Condition for Convexity) If $f \in C^2[a, b]$ and $f''(x) \geq$
lyche-numerical-linear-algebra-F02698	200	■ 1.000	$f''(x) \geq$	$f''(x) \geq$	Lemma 8.2 (A Sufficient Condition for Convexity) If $f \in C^2[a, b]$ and $f''(x) \geq$
lyche-numerical-linear-algebra-F02699	200	■ 1.000	$a \leq x_1 \leq x \leq x_2 \leq b$	$a \leq x_1 \leq x \leq x_2 \leq b$	numerical methods) For any $a \leq x_1 \leq x \leq x_2 \leq b$ there is a $c \in [x_1, x_2]$ such that
lyche-numerical-linear-algebra-F02700	200	■ 1.000	$c \in [x_1, x_2]$	$c \in [x_1, x_2]$	numerical methods) For any $a \leq x_1 \leq x \leq x_2 \leq b$ there is a $c \in [x_1, x_2]$ such that
lyche-numerical-linear-algebra-F02701	200	■ 1.000	$f(x) \leq (1 - \lambda)f(x_1) + \lambda f(x_2)$	$f(x) \leq (1 - \lambda)f(x_1) + \lambda f(x_2)$	Since $\lambda \in [0, 1]$ we have $f(x) \leq (1 - \lambda)f(x_1) + \lambda f(x_2)$. Moreover,
lyche-numerical-linear-algebra-F02702	200	■ 0.999	$I \in \mathbb{R}$	$I \in \mathbb{R}$	Theorem 8.13 (Jensen's Inequality) Suppose $I \in \mathbb{R}$ is an interval and $f : I \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra-F02703	200	■ 1.000	$z_1, \dots, z_n \in I$	$z_1, \dots, z_n \in I$	$\sum_{j=1}^n \lambda_j = 1$, and all $z_1, \dots, z_n \in I$
lyche-numerical-linear-algebra-F02704	200	■ 1.000	$n \geq 2$	$n \geq 2$	Proof We use induction on n . The result is trivial for $n = 1$. Let $n \geq 2$, assume

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lyche-numerical-linear-algebra-F02705	200	■ 0.999	λ_j, z_j	λ_j, z_j	the inequality holds for $n - 1$, and let λ_j, z_j for $j = 1, \dots, n$ be given as in the
lyche-numerical-linear-algebra-F02706	200	■ 1.000	$\lambda_i < 1$	$\lambda_i < 1$	theorem. Since $n \geq 2$ we have $\lambda_i < 1$ for at least one i so assume without loss of
lyche-numerical-linear-algebra-F02707	200	■ 1.000	$\lambda_1 < 1$	$\lambda_1 < 1$	generality that $\lambda_1 < 1$, and define $u := \sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} z_j$. Since $\sum_{j=2}^n \lambda_j = 1 - \lambda_1$
lyche-numerical-linear-algebra-F02708	200	■ 1.000	$u := \sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} z_j$	$u := \sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} z_j$	generality that $\lambda_1 < 1$, and define $u := \sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} z_j$. Since $\sum_{j=2}^n \lambda_j = 1 - \lambda_1$
lyche-numerical-linear-algebra-F02709	200	■ 1.000	$\sum_{j=2}^n \lambda_j = 1 - \lambda_1$	$\sum_{j=2}^n \lambda_j = 1 - \lambda_1$	generality that $\lambda_1 < 1$, and define $u := \sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} z_j$. Since $\sum_{j=2}^n \lambda_j = 1 - \lambda_1$
lyche-numerical-linear-algebra-F02710	200	■ 1.000	$f(u) \leq \sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} f(z_j)$	$f(u) \leq \sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} f(z_j)$	that $f(u) \leq \sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} f(z_j)$. But then by the convexity of f
lyche-numerical-linear-algebra-F02711	201	■ 1.000	$\sum_{j=1}^n \lambda_j a_j$	$\sum_{j=1}^n \lambda_j a_j$	$\sum_{j=1}^n \lambda_j a_j$ is a convex combination of nonnegative numbers $a_1, \dots, a_n$. Then
lyche-numerical-linear-algebra-F02712	201	■ 1.000	$a_1, \dots, a_n$	$a_1, \dots, a_n$	$\sum_{j=1}^n \lambda_j a_j$ is a convex combination of nonnegative numbers $a_1, \dots, a_n$. Then
lyche-numerical-linear-algebra-F02713	201	■ 1.000	$0^0 := 0$	$0^0 := 0$	where $0^0 := 0$.
lyche-numerical-linear-algebra-F02714	201	■ 1.000	a_j	a_j	Proof The result is trivial if one or more of the a_j 's are zero so assume $a_j > 0$
lyche-numerical-linear-algebra-F02715	201	■ 1.000	$a_j > 0$	$a_j > 0$	Proof The result is trivial if one or more of the a_j 's are zero so assume $a_j > 0$
lyche-numerical-linear-algebra-F02716	201	■ 1.000	$f: (0, \infty) \rightarrow \mathbb{R}$	$f: (0, \infty) \rightarrow \mathbb{R}$	for all j . Consider the function $f: (0, \infty) \rightarrow \mathbb{R}$ given by $f(x) = -\log x$. Since
lyche-numerical-linear-algebra-F02717	201	■ 1.000	$f(x) = -\log x$	$f(x) = -\log x$	for all j . Consider the function $f: (0, \infty) \rightarrow \mathbb{R}$ given by $f(x) = -\log x$. Since
lyche-numerical-linear-algebra-F02718	201	■ 1.000	$f'(x) = 1/x^2 > 0$	$f''(x) = 1/x^2 > 0$	$f''(x) = 1/x^2 > 0$ for $x \in (0, \infty)$, it follows from Lemma 8.2 that this function is
lyche-numerical-linear-algebra-F02719	201	■ 1.000	$x \in (0, \infty)$	$x \in (0, \infty)$	$f''(x) = 1/x^2 > 0$ for $x \in (0, \infty)$, it follows from Lemma 8.2 that this function is
lyche-numerical-linear-algebra-F02720	201	■ 1.000	$\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n}) \leq \log(\sum_{j=1}^n \lambda_j a_j)$	$\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n}) \leq \log(\sum_{j=1}^n \lambda_j a_j)$	or $\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n}) \leq \log(\sum_{j=1}^n \lambda_j a_j)$. The inequality follows since $\exp(\log x)$
lyche-numerical-linear-algebra-F02721	201	■ 1.000	$\exp(\log x)$	$\exp(\log x)$	or $\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n}) \leq \log(\sum_{j=1}^n \lambda_j a_j)$. The inequality follows since $\exp(\log x)$
lyche-numerical-linear-algebra-F02722	201	■ 1.000	$= x$	$= x$	$= x$ for $x > 0$ and the exponential function is monotone increasing.
lyche-numerical-linear-algebra-F02723	201	■ 1.000	$x > 0$	$x > 0$	$= x$ for $x > 0$ and the exponential function is monotone increasing.
lyche-numerical-linear-algebra-F02724	201	■ 1.000	$\lambda_j = \frac{1}{n}$	$\lambda_j = \frac{1}{n}$	Taking $\lambda_j = \frac{1}{n}$ for all j in (8.32) we obtain the classical geometric/arithmetic
lyche-numerical-linear-algebra-F02725	201	■ 1.000	$p = \infty$	$p = \infty$	Proof We leave the proof for $p = 1$ and $p = \infty$ as an exercise so assume $1 < p <$
lyche-numerical-linear-algebra-F02726	201	■ 1.000	$1 < p <$	$1 < p <$	Proof We leave the proof for $p = 1$ and $p = \infty$ as an exercise so assume $1 < p <$
lyche-numerical-linear-algebra-F02727	201	■ 1.000	$a, b \geq 0$	$a, b \geq 0$	∞ . For any $a, b \geq 0$ the weighted arithmetic/geometric mean inequality implies
lyche-numerical-linear-algebra-F02728	202	■ 1.000	$(p-1)q = p$	$(p-1)q = p$	relation $(p-1)q = p$ the result is
lyche-numerical-linear-algebra-F02729	202	■ 1.000	$a = \pm 1$	$a = \pm 1$	Proof We set $a = \pm 1$ in (5.5) and add the two equations.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-2022-F02730	202	1.000	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ ^2$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ ^2$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ ^2$ if and only if the parallelogram identity (8.35) holds for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$.
lyche-numerical-linear-algebra-2022-F02731	202	1.000	$\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ ^2$ if and only if the parallelogram identity (8.35) holds for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$.
lyche-numerical-linear-algebra-2022-F02732	203	0.670	$\langle \boldsymbol{y}, \boldsymbol{z} \rangle = 0$	$\langle \boldsymbol{y}, \boldsymbol{z} \rangle = 0$	In particular, since $\boldsymbol{y} = \mathbf{0}$ implies $\langle \boldsymbol{y}, \boldsymbol{z} \rangle = 0$ we obtain $\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \rangle$ for all
lyche-numerical-linear-algebra-2022-F02733	203	0.670	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \rangle$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \rangle$	In particular, since $\boldsymbol{y} = \mathbf{0}$ implies $\langle \boldsymbol{y}, \boldsymbol{z} \rangle = 0$ we obtain $\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \rangle$ for all
lyche-numerical-linear-algebra-2022-F02734	203	1.000	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$. This means that $2\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and (8.37)
lyche-numerical-linear-algebra-2022-F02735	203	1.000	$2\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$	$2\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$. This means that $2\langle \frac{\boldsymbol{x}+\boldsymbol{y}}{2}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and (8.37)
lyche-numerical-linear-algebra-2022-F02736	203	1.000	$a = n$	$a = n$	We first show (8.38) when $a = n$ is a positive integer. By induction
lyche-numerical-linear-algebra-2022-F02737	203	1.000	$a > 0$	$a > 0$	Now if $a > 0$ there is a sequence $\{a_n\}$ of positive rational numbers converging to a .
lyche-numerical-linear-algebra-2022-F02738	203	1.000	$\{a_n\}$	$\{a_n\}$	Now if $a > 0$ there is a sequence $\{a_n\}$ of positive rational numbers converging to a .
lyche-numerical-linear-algebra-2022-F02739	204	1.000	$a\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle a\boldsymbol{x}, \boldsymbol{y} \rangle$	$a\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle a\boldsymbol{x}, \boldsymbol{y} \rangle$	Taking limits and using continuity of norms we obtain $a\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle a\boldsymbol{x}, \boldsymbol{y} \rangle$. This also
lyche-numerical-linear-algebra-2022-F02740	204	0.651	$a < 0$	$a < 0$	holds for $a = 0$. Finally, if $a < 0$ then $(-a) > 0$ and from what we just showed
lyche-numerical-linear-algebra-2022-F02741	204	0.651	$(-a) > 0$	$(-a) > 0$	holds for $a = 0$. Finally, if $a < 0$ then $(-a) > 0$ and from what we just showed
lyche-numerical-linear-algebra-2022-F02742	204	1.000	$\mathcal{V} = \mathbb{C}^n, 1 \leq p \leq \infty, n \geq 2$	$\mathcal{V} = \mathbb{C}^n, 1 \leq p \leq \infty, n \geq 2$	on $\mathcal{V} = \mathbb{R}^n$ or $\mathcal{V} = \mathbb{C}^n, 1 < p < \infty, n \geq 2$, there is an inner product on $\mathcal{V}$ such that
lyche-numerical-linear-algebra-2022-F02743	204	1.000	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _p^2$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _p^2$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ ^2$ for all $\boldsymbol{x} \in \mathcal{V}$ if and only if $p = 2$.
lyche-numerical-linear-algebra-2022-F02744	204	1.000	$p \neq 2$	$p \neq 2$	standard inner product. If $p \neq 2$ then the parallelogram identity (8.35) does not
lyche-numerical-linear-algebra-2022-F02745	204	1.000	$\boldsymbol{x} := \boldsymbol{e}_1$	$\boldsymbol{x} := \boldsymbol{e}_1$	hold for say $\boldsymbol{x} := \boldsymbol{e}_1$ and $\boldsymbol{y} := \boldsymbol{e}_2$.
lyche-numerical-linear-algebra-2022-F02746	204	1.000	$\boldsymbol{y} := \boldsymbol{e}_2$	$\boldsymbol{y} := \boldsymbol{e}_2$	hold for say $\boldsymbol{x} := \boldsymbol{e}_1$ and $\boldsymbol{y} := \boldsymbol{e}_2$.
lyche-numerical-linear-algebra-2022-F02747	204	1.000	$0 < \lambda_n \leq \dots \leq \lambda_1$	$0 < \lambda_n \leq \dots \leq \lambda_1$	metric positive definite matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ with eigenvalues $0 < \lambda_n < \dots < \lambda_1$.
lyche-numerical-linear-algebra-2022-F02748	204	1.000	$\boldsymbol{b}_1, \dots, \boldsymbol{b}_n$	$\boldsymbol{b}_1, \dots, \boldsymbol{b}_n$	symmetric and positive definite matrix and assume $\boldsymbol{b}_1, \dots, \boldsymbol{b}_n$ is a basis for $\mathbb{R}^n$. We
lyche-numerical-linear-algebra-2022-F02749	204	1.000	$\boldsymbol{B}_k := [\boldsymbol{b}_1, \dots, \boldsymbol{b}_k] \in \mathbb{R}^{n \times k}$	$\boldsymbol{B}_k := [\boldsymbol{b}_1, \dots, \boldsymbol{b}_k] \in \mathbb{R}^{n \times k}$	define $\boldsymbol{B}_k := [\boldsymbol{b}_1, \dots, \boldsymbol{b}_k] \in \mathbb{R}^{n \times k}$ for $k = 1, \dots, n$. We consider in this exercise the
lyche-numerical-linear-algebra-2022-F02750	204	0.973	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$	inner product $\langle \cdot, \cdot \rangle$ defined by $\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and the corresponding
lyche-numerical-linear-algebra-2022-F02751	204	1.000	$\ \boldsymbol{x}\ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$	$\ \boldsymbol{x}\ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$	norm $\ \boldsymbol{x}\ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$. We define $\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$ and
lyche-numerical-linear-algebra-2022-F02752	204	1.000	$\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$	$\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$	norm $\ \boldsymbol{x}\ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$. We define $\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$ and
lyche-numerical-linear-algebra-2022-F02753	204	1.000	$\boldsymbol{B}_k^T \boldsymbol{A} \boldsymbol{B}_k$	$\boldsymbol{B}_k^T \boldsymbol{A} \boldsymbol{B}_k$	a) Show that $\boldsymbol{B}_k^T \boldsymbol{A} \boldsymbol{B}_k$ is positive definite for $k = 1, \dots, n$.
lyche-numerical-linear-algebra-2022-F02754	204	0.999	$\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$	$\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$	b) Show that for $k = 2, \dots, n$ we have (i) $\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$ for $j = 1, \dots, k-1$ and
lyche-numerical-linear-algebra-2022-F02755	204	0.999	$j=1, \dots, k-1$	$j = 1, \dots, k-1$	b) Show that for $k = 2, \dots, n$ we have (i) $\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$ for $j = 1, \dots, k-1$ and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-204-F02756	204	■ 1.000	$\tilde{\boldsymbol{b}}_k - \boldsymbol{b}_k \in \text{span}(\boldsymbol{b}_1, \dots, \boldsymbol{b}_{k-1})$	$\tilde{\boldsymbol{b}}_k - \boldsymbol{b}_k \in \text{span}(\boldsymbol{b}_1, \dots, \boldsymbol{b}_{k-1})$	(ii) $\tilde{\boldsymbol{b}}_k - \boldsymbol{b}_k \in \text{span}(\boldsymbol{b}_1, \dots, \boldsymbol{b}_{k-1})$.
lyche-numerical-linear-algebra-205-F02757	205	■ 0.985	$\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n$	$\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n$	c) Explain why $\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n$ is a basis for $\mathbb{R}^n$ which in addition is $\boldsymbol{A}$ -orthogonal, i.e.,
lyche-numerical-linear-algebra-205-F02758	205	■ 0.984	$\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \rangle = 0$	$\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \rangle = 0$	$\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \rangle = 0$ for all $i, j \leq n, i \neq j$.
lyche-numerical-linear-algebra-205-F02759	205	■ 0.984	$i, j \leq n, i \neq j$	$i, j \leq n, i \neq j$	$\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \rangle = 0$ for all $i, j \leq n, i \neq j$.
lyche-numerical-linear-algebra-205-F02760	205	■ 1.000	$\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n]$	$\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n]$	d) Define $\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n]$. Show that there is an upper triangular matrix $\boldsymbol{T} \in$
lyche-numerical-linear-algebra-205-F02761	205	■ 1.000	$\boldsymbol{T} \in$	$\boldsymbol{T} \in$	d) Define $\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n]$. Show that there is an upper triangular matrix $\boldsymbol{T} \in$
lyche-numerical-linear-algebra-205-F02762	205	■ 1.000	$\boldsymbol{B}_n = \tilde{\boldsymbol{B}}_n \boldsymbol{T}$	$\boldsymbol{B}_n = \tilde{\boldsymbol{B}}_n \boldsymbol{T}$	$\mathbb{R}^{n \times n}$ with ones on the diagonal and satisfies $\boldsymbol{B}_n = \tilde{\boldsymbol{B}}_n \boldsymbol{T}$.
lyche-numerical-linear-algebra-205-F02763	205	■ 1.000	$ t_{ij} \leq \frac{1}{2}$	$ t_{ij} \leq \frac{1}{2}$	e) Assume that the matrix $\boldsymbol{T}$ in d) is such that $ t_{ij} \leq \frac{1}{2}$ for all $i, j \leq n, i \neq j$.
lyche-numerical-linear-algebra-205-F02764	205	■ 0.947	$\ \tilde{\boldsymbol{b}}_k\ _{\boldsymbol{A}}^2 \leq 2 \ \tilde{\boldsymbol{b}}_{k+1}\ _{\boldsymbol{A}}^2$	$\ \tilde{\boldsymbol{b}}_k\ _{\boldsymbol{A}}^2 \leq 2 \ \tilde{\boldsymbol{b}}_{k+1}\ _{\boldsymbol{A}}^2$	Assume also that $\ \tilde{\boldsymbol{b}}_k\ _{\boldsymbol{A}}^2 \leq 2 \ \tilde{\boldsymbol{b}}_{k+1}\ _{\boldsymbol{A}}^2$ for $k = 1, \dots, n-1$ and that $\det(\boldsymbol{B}_n) = 1$.
lyche-numerical-linear-algebra-205-F02765	205	■ 0.947	$\det(\boldsymbol{B}_n) = 1$	$\det(\boldsymbol{B}_n) = 1$	Assume also that $\ \tilde{\boldsymbol{b}}_k\ _{\boldsymbol{A}}^2 \leq 2 \ \tilde{\boldsymbol{b}}_{k+1}\ _{\boldsymbol{A}}^2$ for $k = 1, \dots, n-1$ and that $\det(\boldsymbol{B}_n) = 1$.
lyche-numerical-linear-algebra-205-F02766	205	■ 0.948	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	consistent by considering $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-205-F02767	205	■ 1.000	$\sqrt{mn}$	$\sqrt{mn}$	b) Show that the constant $\sqrt{mn}$ can be replaced by m and by n .
lyche-numerical-linear-algebra-205-F02768	205	■ 0.985	$\ \boldsymbol{Ax}\ _{\infty} \leq$	$\ \boldsymbol{Ax}\ _{\infty} \leq$	a) Show that the max norm is subordinate to the ∞ and 1 norm, i.e., $\ \boldsymbol{Ax}\ _{\infty} \leq$
lyche-numerical-linear-algebra-205-F02769	205	■ 1.000	$\ \boldsymbol{A}\ _M \ \boldsymbol{x}\ _1$	$\ \boldsymbol{A}\ _M \ \boldsymbol{x}\ _1$	$\ \boldsymbol{A}\ _M \ \boldsymbol{x}\ _1$ holds for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra-205-F02770	205	■ 1.000	$\ \boldsymbol{A}\ _M = a_{kl} $	$\ \boldsymbol{A}\ _M = a_{kl} $	b) Show that if $\ \boldsymbol{A}\ _M = a_{kl} $, then $\ \boldsymbol{A} \boldsymbol{e}_l\ _{\infty} = \ \boldsymbol{A}\ _M \ \boldsymbol{e}_l\ _1$.
lyche-numerical-linear-algebra-205-F02771	205	■ 1.000	$\ \boldsymbol{A} \boldsymbol{e}_l\ _{\infty} = \ \boldsymbol{A}\ _M \ \boldsymbol{e}_l\ _1$	$\ \boldsymbol{A} \boldsymbol{e}_l\ _{\infty} = \ \boldsymbol{A}\ _M \ \boldsymbol{e}_l\ _1$	b) Show that if $\ \boldsymbol{A}\ _M = a_{kl} $, then $\ \boldsymbol{A} \boldsymbol{e}_l\ _{\infty} = \ \boldsymbol{A}\ _M \ \boldsymbol{e}_l\ _1$.
lyche-numerical-linear-algebra-205-F02772	205	■ 1.000	$\ \boldsymbol{A}\ _M = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{Ax}\ _{\infty}}{\ \boldsymbol{x}\ _1}$	$\ \boldsymbol{A}\ _M = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{Ax}\ _{\infty}}{\ \boldsymbol{x}\ _1}$	c) Show that $\ \boldsymbol{A}\ _M = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{Ax}\ _{\infty}}{\ \boldsymbol{x}\ _1}$.
lyche-numerical-linear-algebra-205-F02773	205	■ 0.480	$\ \tilde{\boldsymbol{b}}_1\ _{\boldsymbol{A}}^2 \cdots \ \tilde{\boldsymbol{b}}_n\ _{\boldsymbol{A}}^2 = \det(\boldsymbol{A})$	$\ \tilde{\boldsymbol{b}}_1\ _{\boldsymbol{A}}^2 \cdots \ \tilde{\boldsymbol{b}}_n\ _{\boldsymbol{A}}^2 = \det(\boldsymbol{A})$	² Hint: Show that $\ \tilde{\boldsymbol{b}}_1\ _{\boldsymbol{A}}^2 \cdots \ \tilde{\boldsymbol{b}}_n\ _{\boldsymbol{A}}^2 = \det(\boldsymbol{A})$.
lyche-numerical-linear-algebra-206-F02774	206	■ 0.991	$\ \boldsymbol{Ax}\ _2 \geq \sigma_n$	$\ \boldsymbol{Ax}\ _2 \geq \sigma_n$	Show that $\ \boldsymbol{Ax}\ _2 \geq \sigma_n$ for all $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _2 = 1$. Show that
lyche-numerical-linear-algebra-206-F02775	206	■ 1.000	$\ \boldsymbol{A}\ _p$	$\ \boldsymbol{A}\ _p$	Compute $\ \boldsymbol{A}\ _p$ and $\ \boldsymbol{A}^{-1}\ _p$ for $p = 1, 2, \infty$.
lyche-numerical-linear-algebra-206-F02776	206	■ 1.000	$\ \boldsymbol{A}^{-1}\ _p$	$\ \boldsymbol{A}^{-1}\ _p$	Compute $\ \boldsymbol{A}\ _p$ and $\ \boldsymbol{A}^{-1}\ _p$ for $p = 1, 2, \infty$.
lyche-numerical-linear-algebra-206-F02777	206	■ 1.000	$\ \boldsymbol{VA}\ _2 =$	$\ \boldsymbol{VA}\ _2 =$	Exercise 8.11 (Unitary Invariance of the Spectral Norm) Show that $\ \boldsymbol{VA}\ _2 =$
lyche-numerical-linear-algebra-206-F02778	206	■ 1.000	$\boldsymbol{V}^* \boldsymbol{V} = \boldsymbol{I}$	$\boldsymbol{V}^* \boldsymbol{V} = \boldsymbol{I}$	$\ \boldsymbol{A}\ _2$ holds even for a rectangular $\boldsymbol{V}$ as long as $\boldsymbol{V}^* \boldsymbol{V} = \boldsymbol{I}$.

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lyche-numerical-linear-algebra-206-F02779	206	1.000	$8.12 \backslash \left(\backslash \boldsymbol{A} \right) \boldsymbol{U} \backslash \right. \left. \right) \right. \left. \right)$	$8.12 (\ \boldsymbol{AU} \ _2$	Exercise 8.12 ($\ \boldsymbol{AU} \ _2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$ with
lyche-numerical-linear-algebra-206-F02780	206	1.000	$\backslash \left. \right) \boldsymbol{A} \backslash \right)$	$\boldsymbol{A})$	Exercise 8.12 ($\ \boldsymbol{AU} \ _2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$ with
lyche-numerical-linear-algebra-206-F02781	206	1.000	$\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$	Exercise 8.12 ($\ \boldsymbol{AU} \ _2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$ with
lyche-numerical-linear-algebra-206-F02782	206	1.000	$\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$	$\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$	Exercise 8.12 ($\ \boldsymbol{AU} \ _2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$ with
lyche-numerical-linear-algebra-206-F02783	206	1.000	$\backslash \backslash \boldsymbol{A} \backslash \boldsymbol{U} \backslash \right) \backslash \boldsymbol{A} \backslash \right)$	$\ \boldsymbol{AU} \ _2 < \ \boldsymbol{A} \ _2$	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$ such that $\ \boldsymbol{AU} \ _2 < \ \boldsymbol{A} \ _2$. Thus, in general, $\ \boldsymbol{AU} \ _2 = \ \boldsymbol{A} \ _2$ does
lyche-numerical-linear-algebra-206-F02784	206	1.000	$\backslash \backslash \boldsymbol{A} \backslash \boldsymbol{U} \backslash \right) \backslash \boldsymbol{A} \backslash \right)$	$\ \boldsymbol{AU} \ _2 = \ \boldsymbol{A} \ _2$	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$ such that $\ \boldsymbol{AU} \ _2 < \ \boldsymbol{A} \ _2$. Thus, in general, $\ \boldsymbol{AU} \ _2 = \ \boldsymbol{A} \ _2$ does
lyche-numerical-linear-algebra-206-F02785	206	0.999	$\backslash \backslash \boldsymbol{A} \backslash \right) \backslash \boldsymbol{A} \backslash \right)$	$\ \boldsymbol{A} \ _p = \rho(\boldsymbol{A}) :=$	Exercise 8.13 (p -Norm of Diagonal Matrix) Show that $\ \boldsymbol{A} \ _p = \rho(\boldsymbol{A}) :=$
lyche-numerical-linear-algebra-206-F02786	206	0.972	$\max \left \lambda_i \right $	$\max \lambda_i $	$\max \lambda_i $ (the largest eigenvalue of $\boldsymbol{A}$), $1 \leq p \leq \infty$, when $\boldsymbol{A}$ is a diagonal matrix.
lyche-numerical-linear-algebra-206-F02787	206	1.000	$\boldsymbol{a} \in \mathbb{C}^m$	$\boldsymbol{a} \in \mathbb{C}^m$	Exercise 8.14 (Spectral Norm of a Column Vector) A vector $\boldsymbol{a} \in \mathbb{C}^m$ can also be
lyche-numerical-linear-algebra-206-F02788	206	1.000	$\boldsymbol{A} \in \mathbb{C}^{m, 1}$	$\boldsymbol{A} \in \mathbb{C}^{m, 1}$	considered as a matrix $\boldsymbol{A} \in \mathbb{C}^{m, 1}$.
lyche-numerical-linear-algebra-206-F02789	206	1.000	$\backslash \backslash \boldsymbol{A} \backslash \right) \backslash \boldsymbol{a} \backslash \right)$	$\ \boldsymbol{A} \ _p = \ \boldsymbol{a} \ _p$	b) Show that $\ \boldsymbol{A} \ _p = \ \boldsymbol{a} \ _p$ for $1 \leq p \leq \infty$.
lyche-numerical-linear-algebra-206-F02790	206	1.000	$\boldsymbol{A} \in \mathbb{R}^{m \times n}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}$	let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be the matrix with elements $ a_{ij} $.
lyche-numerical-linear-algebra-206-F02791	206	1.000	$\left a_{ij} \right $	$ a_{ij} $	let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be the matrix with elements $ a_{ij} $.
lyche-numerical-linear-algebra-206-F02792	206	0.999	$\boldsymbol{A}$	$\boldsymbol{A}$	a) Compute $\boldsymbol{A}$ if $\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}$, $i = \sqrt{-1}$.
lyche-numerical-linear-algebra-206-F02793	206	0.999	$\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}$, $i = \sqrt{-1}$	$\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}$, $i = \sqrt{-1}$	a) Compute $\boldsymbol{A}$ if $\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}$, $i = \sqrt{-1}$.
lyche-numerical-linear-algebra-206-F02794	206	1.000	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A} \ _F = \ \boldsymbol{A} \ _F, \ \boldsymbol{A} \ _p = \ \boldsymbol{A} \ _p$	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A} \ _F = \ \boldsymbol{A} \ _F, \ \boldsymbol{A} \ _p = \ \boldsymbol{A} \ _p$	b) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A} \ _F = \ \boldsymbol{A} \ _F, \ \boldsymbol{A} \ _p = \ \boldsymbol{A} \ _p$ for $p = 1, \infty$.
lyche-numerical-linear-algebra-206-F02795	206	1.000	$p=1, \infty$	$p = 1, \infty$	b) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A} \ _F = \ \boldsymbol{A} \ _F, \ \boldsymbol{A} \ _p = \ \boldsymbol{A} \ _p$ for $p = 1, \infty$.
lyche-numerical-linear-algebra-206-F02796	206	1.000	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A} \ _2 \leq \ \boldsymbol{A} \ _2$	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A} \ _2 \leq \ \boldsymbol{A} \ _2$	c) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A} \ _2 \leq \ \boldsymbol{A} \ _2$.
lyche-numerical-linear-algebra-206-F02797	206	0.806	$\backslash \backslash \boldsymbol{A} \backslash \right) \backslash \boldsymbol{A} \backslash \right)$	$\ \boldsymbol{A} \ _2 < \ \boldsymbol{A} \ _2$	d) Find a real symmetric 2×2 matrix $\boldsymbol{A}$ such that $\ \boldsymbol{A} \ _2 < \ \boldsymbol{A} \ _2$.
lyche-numerical-linear-algebra-206-F02798	206	1.000	$\left\{ \lambda_1, \lambda_2, \dots, \lambda_m \right\}$	$\{ \lambda_1, \lambda_2, \dots, \lambda_m \}$	$\{ \lambda_1, \lambda_2, \dots, \lambda_m \}$ in no particular order. We have that $m \leq n$, since some of the
lyche-numerical-linear-algebra-206-F02799	206	1.000	$\boldsymbol{x}_0 \in \mathbb{C}^n$	$\boldsymbol{x}_0 \in \mathbb{C}^n$	eigenvalues may have multiplicity larger than one. Given $\boldsymbol{x}_0 \in \mathbb{C}^n$, and $k \geq 0$, we
lyche-numerical-linear-algebra-207-F02800	207	1.000	$\left\{ \boldsymbol{x}_k \right\}_{k=0}^{m-1}$	$\{ \boldsymbol{x}_k \}_{k=0}^{m-1}$	define the sequence $\{ \boldsymbol{x}_k \}_{k=0}^{m-1}$ by
lyche-numerical-linear-algebra-207-F02801	207	1.000	c_{ik}	c_{ik}	a) Let the coefficients c_{ik} be defined by
lyche-numerical-linear-algebra-207-F02802	207	1.000	$\left\{ (\sigma_i, \boldsymbol{u}_i) \right\}_{i=1}^n$	$\{ (\sigma_i, \boldsymbol{u}_i) \}_{i=1}^n$	where $\{ (\sigma_i, \boldsymbol{u}_i) \}_{i=1}^n$ are the eigenpairs of $\boldsymbol{A}$. Show that
lyche-numerical-linear-algebra-207-F02803	207	1.000	$l \leq m$	$l \leq m$	b) Show that for some $l \leq m$, we have that $\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \dots = \boldsymbol{x}_m = \boldsymbol{x}$, where

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lyche-numerical-linear-algebra- F02804	207	■ 1.000	$\boldsymbol{x}_{\{l\}}=\boldsymbol{x}_{\{l+1\}}=\cdots=\boldsymbol{x}_{\{m\}}=\boldsymbol{x}$	$\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \cdots = \boldsymbol{x}_m = \boldsymbol{x}$	b) Show that for some $l \leq m$, we have that $\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \cdots = \boldsymbol{x}_m = \boldsymbol{x}$, where
lyche-numerical-linear-algebra- F02805	207	■ 1.000	$\boldsymbol{T}=\operatorname{tridiag}(c, d, c)$	$\boldsymbol{T} = \operatorname{tridiag}(c, d, c)$	c) Consider this iteration for the $n \times n$ matrix $\boldsymbol{T} = \operatorname{tridiag}(c, d, c)$, where d and c
lyche-numerical-linear-algebra- F02806	207	■ 1.000	$d>2 \ c$	$d > 2c$	are positive real numbers and $d > 2c$. The eigenvalues of $\boldsymbol{T}$ are
lyche-numerical-linear-algebra- F02807	207	■ 1.000	$\boldsymbol{T} \ \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$	What is the operation count for solving $\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$ using the iterative algorithm
lyche-numerical-linear-algebra- F02808	207	■ 1.000	$b_{\{i \ j\}}=0$	$b_{ij} = 0$	$b_{ij} = 0$ if $ i - j \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.
lyche-numerical-linear-algebra- F02809	207	■ 1.000	$ i-j \ \leq 1$	$ i - j \leq 1$	$b_{ij} = 0$ if $ i - j \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.
lyche-numerical-linear-algebra- F02810	207	■ 1.000	$\boldsymbol{A}=\boldsymbol{T}+\boldsymbol{B}$	$\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$	$b_{ij} = 0$ if $ i - j \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.
lyche-numerical-linear-algebra- F02811	207	■ 0.991	$\rho(\boldsymbol{E}):=\max_{\{i\}} \lambda_{\{i\}} $	$\rho(\boldsymbol{E}) := \max_i \lambda_i $	Recall that if $\boldsymbol{E} \in \mathbb{R}^{n \times n}$ has eigenvalues $\lambda_1, \dots, \lambda_n$ then $\rho(\boldsymbol{E}) := \max_i \lambda_i $ is
lyche-numerical-linear-algebra- F02812	207	■ 1.000	$\rho(\boldsymbol{T}^{-1} \boldsymbol{B}) \leq \rho(\boldsymbol{T}^{-1}) \rho(\boldsymbol{B})$	$\rho(\boldsymbol{T}^{-1} \boldsymbol{B}) \leq \rho(\boldsymbol{T}^{-1}) \rho(\boldsymbol{B})$	the spectral radius of $\boldsymbol{E}$. Show that $\rho(\boldsymbol{T}^{-1} \boldsymbol{B}) < \rho(\boldsymbol{T}^{-1}) \rho(\boldsymbol{B})$.
lyche-numerical-linear-algebra- F02813	208	■ 1.000	$\delta:=\ \boldsymbol{x}-\boldsymbol{y}\ _2/\ \boldsymbol{x}\ _2$	$\delta := \ \boldsymbol{x} - \boldsymbol{y}\ _2 / \ \boldsymbol{x}\ _2$	We define $\delta := \ \boldsymbol{x} - \boldsymbol{y}\ _2 / \ \boldsymbol{x}\ _2$.
lyche-numerical-linear-algebra- F02814	208	■ 1.000	$K_2(\boldsymbol{A})=\ \boldsymbol{A}\ _2\ \boldsymbol{A}^{-1}\ _2$	$K_2(\boldsymbol{A}) = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2$	a) Determine $K_2(\boldsymbol{A}) = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2$. Give an upper bound and a positive lower
lyche-numerical-linear-algebra- F02815	208	■ 1.000	δ	δ	bound for δ without computing $\boldsymbol{x}$ and $\boldsymbol{y}$.
lyche-numerical-linear-algebra- F02816	208	■ 1.000	$b_{\{2\}}=2.0$	$b_2 = 2.0$	b) Suppose as before that $b_2 = 2.0$ and $\ \boldsymbol{e}\ _2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which
lyche-numerical-linear-algebra- F02817	208	■ 1.000	$\ \boldsymbol{e}\ _2=0.1$	$\ \boldsymbol{e}\ _2 = 0.1$	b) Suppose as before that $b_2 = 2.0$ and $\ \boldsymbol{e}\ _2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which
lyche-numerical-linear-algebra- F02818	208	■ 1.000	$b_{\{1\}}$	b_1	b) Suppose as before that $b_2 = 2.0$ and $\ \boldsymbol{e}\ _2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which
lyche-numerical-linear-algebra- F02819	208	■ 1.000	$\boldsymbol{y}_{\{\boldsymbol{A}\}}$	$\boldsymbol{y}_{\boldsymbol{A}}$	special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_{\boldsymbol{A}}$ and $\boldsymbol{y}_{\boldsymbol{A}^{-1}}$ are vectors with $\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^{-1}}\ = 1$
lyche-numerical-linear-algebra- F02820	208	■ 1.000	$\boldsymbol{y}_{\{\boldsymbol{A}^{-1}\}}$	$\boldsymbol{y}_{\boldsymbol{A}^{-1}}$	special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_{\boldsymbol{A}}$ and $\boldsymbol{y}_{\boldsymbol{A}^{-1}}$ are vectors with $\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^{-1}}\ = 1$
lyche-numerical-linear-algebra- F02821	208	■ 1.000	$\ \boldsymbol{y}_{\{\boldsymbol{A}\}}\ =\ \boldsymbol{y}_{\{\boldsymbol{A}^{-1}\}}\ =1$	$\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^{-1}}\ = 1$	special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_{\boldsymbol{A}}$ and $\boldsymbol{y}_{\boldsymbol{A}^{-1}}$ are vectors with $\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^{-1}}\ = 1$
lyche-numerical-linear-algebra- F02822	208	■ 0.968	$\ \boldsymbol{A}\ =\ \boldsymbol{A} \boldsymbol{y}_{\{\boldsymbol{A}\}}\ $	$\ \boldsymbol{A}\ = \ \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}\ $	and $\ \boldsymbol{A}\ = \ \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}\ $ and $\ \boldsymbol{A}^{-1}\ = \ \boldsymbol{A}^{-1} \boldsymbol{y}_{\boldsymbol{A}^{-1}}\ $.
lyche-numerical-linear-algebra- F02823	208	■ 0.968	$\ \boldsymbol{A}^{-1}\ =\ \boldsymbol{A}^{-1} \boldsymbol{y}_{\{\boldsymbol{A}^{-1}\}}\ $	$\ \boldsymbol{A}^{-1}\ = \ \boldsymbol{A}^{-1} \boldsymbol{y}_{\boldsymbol{A}^{-1}}\ $	and $\ \boldsymbol{A}\ = \ \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}\ $ and $\ \boldsymbol{A}^{-1}\ = \ \boldsymbol{A}^{-1} \boldsymbol{y}_{\boldsymbol{A}^{-1}}\ $.
lyche-numerical-linear-algebra- F02824	208	■ 1.000	$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{y}_{\{\boldsymbol{A}\}}$	$\boldsymbol{b} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$	a) Show that the upper bound in (8.22) is attained if $\boldsymbol{b} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$ and $\boldsymbol{e} = \boldsymbol{y}_{\boldsymbol{A}^{-1}}$.
lyche-numerical-linear-algebra- F02825	208	■ 1.000	$\boldsymbol{e}=\boldsymbol{y}_{\{\boldsymbol{A}^{-1}\}}$	$\boldsymbol{e} = \boldsymbol{y}_{\boldsymbol{A}^{-1}}$	a) Show that the upper bound in (8.22) is attained if $\boldsymbol{b} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$ and $\boldsymbol{e} = \boldsymbol{y}_{\boldsymbol{A}^{-1}}$.
lyche-numerical-linear-algebra- F02826	208	■ 0.980	$\boldsymbol{b}=\boldsymbol{y}_{\{\boldsymbol{A}^{-1}\}}$	$\boldsymbol{b} = \boldsymbol{y}_{\boldsymbol{A}^{-1}}$	b) Show that the lower bound is attained if $\boldsymbol{b} = \boldsymbol{y}_{\boldsymbol{A}^{-1}}$ and $\boldsymbol{e} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$.
lyche-numerical-linear-algebra- F02827	208	■ 0.980	$\boldsymbol{e}=\boldsymbol{A} \boldsymbol{y}_{\{\boldsymbol{A}\}}$	$\boldsymbol{e} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$	b) Show that the lower bound is attained if $\boldsymbol{b} = \boldsymbol{y}_{\boldsymbol{A}^{-1}}$ and $\boldsymbol{e} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$.
lyche-numerical-linear-algebra- F02828	208	■ 1.000	$m \geq 1$	$m \geq 1$	will show that for $m \geq 1$
lyche-numerical-linear-algebra- F02829	208	■ 1.000	$\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1) \ \text{in} \ \mathbb{R}^{m \times m}$	$\boldsymbol{T} := \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$	where $\boldsymbol{T} := \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ and $\operatorname{cond}_p(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \ \boldsymbol{T}^{-1}\ _p$ is the p -

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lyche-numerical-linear-algebra-F02830	208	1.000	$\operatorname{cond}_{\{p\}}(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \ \boldsymbol{T}^{-1}\ _p$	$\operatorname{cond}_p(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \ \boldsymbol{T}^{-1}\ _p$	where $\boldsymbol{T} := \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ and $\operatorname{cond}_p(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \ \boldsymbol{T}^{-1}\ _p$ is the p -
lyche-numerical-linear-algebra-F02831	208	1.000	$m \geq 3$	$m \geq 3$	a) Show that for $m \geq 3$
lyche-numerical-linear-algebra-F02832	208	1.000	$\operatorname{cond}_{\{1\}}(\boldsymbol{T}) = \operatorname{cond}_{\{\infty\}}(\boldsymbol{T}) = 3$	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = 3$	and that $\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = 3$ for $m = 2$.
lyche-numerical-linear-algebra-F02833	208	1.000	$m=2$	$m = 2$	and that $\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = 3$ for $m = 2$.
lyche-numerical-linear-algebra-F02834	209	1.000	$\tan x > x$	$\tan x > x$	Hint: For the upper bound use the inequality $\tan x > x$ valid for $0 < x < \pi/2$.
lyche-numerical-linear-algebra-F02835	209	1.000	$0 < x < \pi/2$	$0 < x < \pi/2$	Hint: For the upper bound use the inequality $\tan x > x$ valid for $0 < x < \pi/2$.
lyche-numerical-linear-algebra-F02836	209	1.000	$\cot^2 x > \frac{1}{x^2} - \frac{2}{3}$	$\cot^2 x > \frac{1}{x^2} - \frac{2}{3}$	For the lower bound we use (without proof) the inequality $\cot^2 x > \frac{1}{x^2} - \frac{2}{3}$ for
lyche-numerical-linear-algebra-F02837	209	1.000	$\boldsymbol{I} - \boldsymbol{E}$	$\boldsymbol{I} - \boldsymbol{E}$	a) Show that if $\boldsymbol{I} - \boldsymbol{E}$ is nonsingular then
lyche-numerical-linear-algebra-F02838	209	1.000	$\ \boldsymbol{E}\ < 1$	$\ \boldsymbol{E}\ < 1$	b) If $\ \boldsymbol{E}\ < 1$ then $(\boldsymbol{I} - \boldsymbol{E})^{-1}$ is nonsingular by exists and
lyche-numerical-linear-algebra-F02839	209	1.000	$(\boldsymbol{I} - \boldsymbol{E})^{-1}$	$(\boldsymbol{I} - \boldsymbol{E})^{-1}$	b) If $\ \boldsymbol{E}\ < 1$ then $(\boldsymbol{I} - \boldsymbol{E})^{-1}$ is nonsingular by exists and
lyche-numerical-linear-algebra-F02840	209	1.000	$\boldsymbol{x} = [x_0, \dots, x_n]^T \in \mathbb{R}^{n+1}$	$\boldsymbol{x} = [x_0, \dots, x_n]^T \in \mathbb{R}^{n+1}$	components of $\boldsymbol{x} = [x_0, \dots, x_n]^T \in \mathbb{R}^{n+1}$ define a partition of the interval $[a, b]$,
lyche-numerical-linear-algebra-F02841	210	0.999	$\boldsymbol{y} := [y_0, \dots, y_n]^T \in \mathbb{R}^{n+1}$	$\boldsymbol{y} := [y_0, \dots, y_n]^T \in \mathbb{R}^{n+1}$	and given a dataset $\boldsymbol{y} := [y_0, \dots, y_n]^T \in \mathbb{R}^{n+1}$, where we assume $y_0 = y_n$. The
lyche-numerical-linear-algebra-F02842	210	0.999	$y_0 = y_n$	$y_0 = y_n$	and given a dataset $\boldsymbol{y} := [y_0, \dots, y_n]^T \in \mathbb{R}^{n+1}$, where we assume $y_0 = y_n$. The
lyche-numerical-linear-algebra-F02843	210	0.902	$\left(x_{i-1}, x_i\right)$	(x_{i-1}, x_i)	(Recall that g is a cubic polynomial on each interval (x_{i-1}, x_i) , for $i = 1, \dots, n$
lyche-numerical-linear-algebra-F02844	210	1.000	$C^2[a, b]$	$C^2[a, b]$	with smoothness $C^2[a, b]$.)
lyche-numerical-linear-algebra-F02845	210	0.664	$s_i := g'(\prime)\left(x_i\right), i=0, \ldots, n$	$s_i := g'(x_i), i = 0, \dots, n$	We define $s_i := g'(x_i), i = 0, \dots, n$. It can be shown that the vector $\boldsymbol{s} :=$
lyche-numerical-linear-algebra-F02846	210	1.000	$\left[s_1, \ldots, s_n\right]^T$	$[s_1, \dots, s_n]^T$	$[s_1, \dots, s_n]^T$ is determined from a linear system
lyche-numerical-linear-algebra-F02847	211	1.000	$\left\ \boldsymbol{A}^{-1}\right\ _{\infty} \leq 1$	$\ \boldsymbol{A}^{-1}\ _\infty \leq 1$	b) Show that $\ \boldsymbol{A}^{-1}\ _\infty < 1$.
lyche-numerical-linear-algebra-F02848	211	1.000	$\left\ \boldsymbol{A}^{-1}\right\ _1 \leq \frac{3}{2}$	$\ \boldsymbol{A}^{-1}\ _1 \leq \frac{3}{2}$	c) Show that $\ \boldsymbol{A}^{-1}\ _1 \leq \frac{3}{2}$.
lyche-numerical-linear-algebra-F02849	211	1.000	$\boldsymbol{e} \in \mathbb{R}^n$	$\boldsymbol{e} \in \mathbb{R}^n$	d) Let $\boldsymbol{s}$ and $\boldsymbol{b}$ be as in (8.44), where we assume $\boldsymbol{b} \neq \mathbf{0}$. Let $\boldsymbol{e} \in \mathbb{R}^n$ be such that
lyche-numerical-linear-algebra-F02850	211	1.000	$\left\ \boldsymbol{e}\right\ _p /\left\ \boldsymbol{b}\right\ _p \leq 0.01$	$\ \boldsymbol{e}\ _p / \ \boldsymbol{b}\ _p \leq 0.01$	$\ \boldsymbol{e}\ _p / \ \boldsymbol{b}\ _p \leq 0.01$. Suppose $\hat{\boldsymbol{s}}$ satisfies
lyche-numerical-linear-algebra-F02851	211	1.000	$\hat{\boldsymbol{s}}$	$\hat{\boldsymbol{s}}$	$\ \boldsymbol{e}\ _p / \ \boldsymbol{b}\ _p \leq 0.01$. Suppose $\hat{\boldsymbol{s}}$ satisfies
lyche-numerical-linear-algebra-F02852	211	1.000	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^m$, where $\boldsymbol{A}$ has rank n and let $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^T$ be a singular
lyche-numerical-linear-algebra-F02853	211	1.000	$\boldsymbol{U} \in \mathbb{R}^{m \times n}$	$\boldsymbol{U} \in \mathbb{R}^{m \times n}$	value factorization of $\boldsymbol{A}$. Thus $\boldsymbol{U} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$. Write a MATLAB
lyche-numerical-linear-algebra-F02854	211	1.000	$\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$	$\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$	value factorization of $\boldsymbol{A}$. Thus $\boldsymbol{U} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$. Write a MATLAB
lyche-numerical-linear-algebra-F02855	211	0.932	$[\mathrm{x}, \mathrm{K}]=\mathrm{lsq}(\mathrm{A}, \mathrm{b})$	$[x, K] = \operatorname{lsq}(A, b)$	function <code>[x, K]=lsq(A, b)</code> that uses the singular value factorization of $\boldsymbol{A}$
lyche-numerical-linear-algebra-F02856	211	1.000	$\boldsymbol{x}=\boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$	$\boldsymbol{x} = \boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$	to calculate a least squares solution $\boldsymbol{x} = \boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$ to the system $\boldsymbol{A} \boldsymbol{x} =$

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lyche-numerical-linear-algebra-2021-F02857	211	1.000	$\boldsymbol{A} \boldsymbol{x} =$	$\boldsymbol{A} \boldsymbol{x} =$	to calculate a least squares solution $\boldsymbol{x} = \boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$ to the system $\boldsymbol{A} \boldsymbol{x} =$
lyche-numerical-linear-algebra-2021-F02858	211	0.909	$[\mathrm{U}, \mathrm{Sigma}, \mathrm{V}] = \mathrm{svd}(\mathrm{A}, 0)$	$[U, \text{Sigma}, V] = \text{svd}(A, 0)$	$[U, \text{Sigma}, V] = \text{svd}(A, 0)$ computes the singular value factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-2021-F02859	211	0.998	$\ \cdot\ _p$	$\ \cdot\ _p$	Exercise 8.25 (p Norm for $p = 1$ and $p = \infty$) Show that $\ \cdot\ _p$ is a vector norm
lyche-numerical-linear-algebra-2021-F02860	211	1.000	$p=1, p=\infty$	$p = 1, p = \infty$	in $\mathbb{R}^n$ for $p = 1, p = \infty$.
lyche-numerical-linear-algebra-2021-F02861	211	0.992	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = s(\boldsymbol{x}, \boldsymbol{y}) + is(\boldsymbol{x}, i\boldsymbol{y})$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = s(\boldsymbol{x}, \boldsymbol{y}) + is(\boldsymbol{x}, i\boldsymbol{y})$	⁴ Hint: We have $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = s(\boldsymbol{x}, \boldsymbol{y}) + is(\boldsymbol{x}, i\boldsymbol{y})$, where $s(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{4}(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2)$.
lyche-numerical-linear-algebra-2021-F02862	211	0.992	$s(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{4}(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2)$	$s(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{4}(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2)$	⁴ Hint: We have $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = s(\boldsymbol{x}, \boldsymbol{y}) + is(\boldsymbol{x}, i\boldsymbol{y})$, where $s(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{4}(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2)$.
lyche-numerical-linear-algebra-2021-F02863	212	1.000	S_p	S_p	is called the unit sphere in $\mathbb{R}^n$ with respect to p . Draw S_p for $p = 1, 2, \infty$ for $n = 2$.
lyche-numerical-linear-algebra-2021-F02864	212	1.000	$\ \boldsymbol{x}\ _\infty \leq \ \boldsymbol{x}\ _p \leq n^{1/p} \ \boldsymbol{x}\ _\infty$	$\ \boldsymbol{x}\ _\infty \leq \ \boldsymbol{x}\ _p \leq n^{1/p} \ \boldsymbol{x}\ _\infty$	have $\ \boldsymbol{x}\ _\infty \leq \ \boldsymbol{x}\ _p \leq n^{1/p} \ \boldsymbol{x}\ _\infty$ (cf. (8.5)).
lyche-numerical-linear-algebra-2021-F02865	212	1.000	$\boldsymbol{x}_l$	$\boldsymbol{x}_l$	Produce a vector $\boldsymbol{x}_l$ such that $\ \boldsymbol{x}_l\ _\infty = \ \boldsymbol{x}_l\ _p$ and another vector $\boldsymbol{x}_u$ such that
lyche-numerical-linear-algebra-2021-F02866	212	1.000	$\ \boldsymbol{x}_l\ _\infty = \ \boldsymbol{x}_l\ _p$	$\ \boldsymbol{x}_l\ _\infty = \ \boldsymbol{x}_l\ _p$	Produce a vector $\boldsymbol{x}_l$ such that $\ \boldsymbol{x}_l\ _\infty = \ \boldsymbol{x}_l\ _p$ and another vector $\boldsymbol{x}_u$ such that
lyche-numerical-linear-algebra-2021-F02867	212	1.000	$\boldsymbol{x}_u$	$\boldsymbol{x}_u$	Produce a vector $\boldsymbol{x}_l$ such that $\ \boldsymbol{x}_l\ _\infty = \ \boldsymbol{x}_l\ _p$ and another vector $\boldsymbol{x}_u$ such that
lyche-numerical-linear-algebra-2021-F02868	212	1.000	$\ \boldsymbol{x}_u\ _p = n^{1/p} \ \boldsymbol{x}_u\ _\infty$	$\ \boldsymbol{x}_u\ _p = n^{1/p} \ \boldsymbol{x}_u\ _\infty$	$\ \boldsymbol{x}_u\ _p = n^{1/p} \ \boldsymbol{x}_u\ _\infty$. Thus, these inequalities are sharp.
lyche-numerical-linear-algebra-2021-F02869	212	1.000	$1 \leq q \leq p \leq \infty$	$1 \leq q \leq p \leq \infty$	Exercise 8.28 (p-Norm Inequalities for Arbitrary p) If $1 \leq q \leq p \leq \infty$ then
lyche-numerical-linear-algebra-2021-F02870	212	1.000	$f(z) = z^{p/q}$	$f(z) = z^{p/q}$	$f(z) = z^{p/q}$ and $z_i = x_i ^q$. For the left inequality consider first $y_i = x_i / \ \boldsymbol{x}\ _\infty$,
lyche-numerical-linear-algebra-2021-F02871	212	1.000	$z_i = x_i ^q$	$z_i = x_i ^q$	$f(z) = z^{p/q}$ and $z_i = x_i ^q$. For the left inequality consider first $y_i = x_i / \ \boldsymbol{x}\ _\infty$,
lyche-numerical-linear-algebra-2021-F02872	212	1.000	$y_i = x_i / \ \boldsymbol{x}\ _\infty$	$y_i = x_i / \ \boldsymbol{x}\ _\infty$	$f(z) = z^{p/q}$ and $z_i = x_i ^q$. For the left inequality consider first $y_i = x_i / \ \boldsymbol{x}\ _\infty$,
lyche-numerical-linear-algebra-2021-F02873	212	0.551	$\ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ _F \ \boldsymbol{x}\ $	$\ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ _F \ \boldsymbol{x}\ $	8.6.2 Does there exist a vector norm $\ \cdot\ $ such that $\ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ _F \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in$
lyche-numerical-linear-algebra-2021-F02874	212	0.980	$\mathbb{C}^{n \times n}, \boldsymbol{x} \in \mathbb{C}^n, m, n \in \mathbb{N}$	$\mathbb{C}^{n \times n}, \boldsymbol{x} \in \mathbb{C}^n, m, n \in \mathbb{N}$	$\mathbb{C}^{n \times n}, \boldsymbol{x} \in \mathbb{C}^n, m, n \in \mathbb{N}$?
lyche-numerical-linear-algebra-2021-F02875	212	1.000	$\ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _F$	$\ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _F$	8.6.3 Why is $\ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _F$ for any matrix $\boldsymbol{A}$?
lyche-numerical-linear-algebra-2021-F02876	213	1.000	$\boldsymbol{b} \notin \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{b} \notin \mathcal{R}(\boldsymbol{A})$	either case the system can only be solved approximately if $\boldsymbol{b} \notin \mathcal{R}(\boldsymbol{A})$, the column
lyche-numerical-linear-algebra-2021-F02877	213	0.999	$\ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ $	$\ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ $	$\ \cdot\ $, say a p -norm, and look for $\boldsymbol{x} \in \mathbb{C}^n$ which minimizes $\ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ $. The use of
lyche-numerical-linear-algebra-2021-F02878	213	1.000	$\boldsymbol{b} \in \mathbb{C}^m$	$\boldsymbol{b} \in \mathbb{C}^m$	and $\boldsymbol{b} \in \mathbb{C}^m$. To find $\boldsymbol{x} \in \mathbb{C}^n$ that minimizes $E : \mathbb{C}^n \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra-2021-F02879	213	1.000	$E : \mathbb{C}^n \rightarrow \mathbb{R}$	$E : \mathbb{C}^n \rightarrow \mathbb{R}$	and $\boldsymbol{b} \in \mathbb{C}^m$. To find $\boldsymbol{x} \in \mathbb{C}^n$ that minimizes $E : \mathbb{C}^n \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra-2021-F02880	213	1.000	$E(\boldsymbol{x})$	$E(\boldsymbol{x})$	Since the square root function is monotone, minimizing $E(\boldsymbol{x})$ or $\sqrt{E(\boldsymbol{x})}$ is
lyche-numerical-linear-algebra-2021-F02881	213	1.000	$\sqrt{E(\boldsymbol{x})}$	$\sqrt{E(\boldsymbol{x})}$	Since the square root function is monotone, minimizing $E(\boldsymbol{x})$ or $\sqrt{E(\boldsymbol{x})}$ is
lyche-numerical-linear-algebra-2021-F02882	214	1.000	$6x_1 - 8 = 0$	$6x_1 - 8 = 0$	Setting the first derivative with respect to x_1 equal to zero we obtain $6x_1 - 8 = 0$ or

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lyche-numerical-linear-algebra- F02883	214	1.000	$x_{\{1\}}=4 / 3$	$x_1 = 4/3$	$x_1 = 4/3$, the average of b_1, b_2, b_3 . The second derivative is positive and $x_1 = 4/3$
lyche-numerical-linear-algebra- F02884	214	1.000	$b_{\{1\}}, b_{\{2\}}, b_{\{3\}}$	b_1, b_2, b_3	$x_1 = 4/3$, the average of b_1, b_2, b_3 . The second derivative is positive and $x_1 = 4/3$
lyche-numerical-linear-algebra- F02885	214	1.000	$\boldsymbol{A}^* \boldsymbol{A}^* \boldsymbol{x} = \boldsymbol{A}^* \boldsymbol{b}$	$\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^* \boldsymbol{b}$	if and only if $\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^* \boldsymbol{b}$.
lyche-numerical-linear-algebra- F02886	214	1.000	$p(t)=x_{\{1\}}+x_{\{2\}} t$	$p(t) = x_1 + x_2 t$	Example 9.2 (Linear Regression) We want to fit a straight line $p(t) = x_1 + x_2 t$ to
lyche-numerical-linear-algebra- F02887	214	1.000	$\left(t_{\{k\}}, y_{\{k\}}\right) \in \mathbb{R}^2, k=1, \ldots, m$	$(t_k, y_k) \in \mathbb{R}^2, k = 1, \dots, m$	$m \geq 2$ given data $(t_k, y_k) \in \mathbb{R}^2, k = 1, \dots, m$. This is part of the linear regression
lyche-numerical-linear-algebra- F02888	214	1.000	$m>2$	$m > 2$	This is square for $m = 2$ and overdetermined for $m > 2$. The matrix $\boldsymbol{A}$ has linearly
lyche-numerical-linear-algebra- F02889	214	1.000	$\left\{t_{\{1\}}, \ldots, t_{\{m\}}\right\}$	$\{t_1, \dots, t_m\}$	independent columns if and only if the set $\{t_1, \dots, t_m\}$ of sites contains at least two
lyche-numerical-linear-algebra- F02890	214	1.000	$t_{\{i\}} \neq t_{\{j\}}$	$t_i \neq t_j$	distinct elements. For if say $t_i \neq t_j$ then
lyche-numerical-linear-algebra- F02891	215	1.000	$t_{\{1\}}=\cdots=t_{\{m\}}$	$t_1 = \dots = t_m$	Conversely, if $t_1 = \dots = t_m$ then the columns of $\boldsymbol{A}$ are linearly dependent. The
lyche-numerical-linear-algebra- F02892	215	0.970	$t_{\{1\}} \neq t_{\{2\}}$	$t_1 \neq t_2$	$m = 2$ and $t_1 \neq t_2$ then both systems $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ and $\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^* \boldsymbol{b}$ are square, and p
lyche-numerical-linear-algebra- F02893	215	1.000	$p\left(t_{\{k\}}\right)=y_{\{k\}}, k=1,2$	$p(t_k) = y_k, k = 1, 2$	is the linear interpolant to the data. Indeed, p is linear and $p(t_k) = y_k, k = 1, 2$.
lyche-numerical-linear-algebra- F02894	215	1.000	$\left[\begin{array}{cc} 4 & 10 \\ 10 & 30 \end{array}\right] \left[\begin{array}{c} x_{\{1\}} \\ x_{\{2\}} \end{array}\right] = \left[\begin{array}{c} 6 \\ 10.1 \end{array}\right]$	$\left[\begin{array}{cc} 4 & 10 \\ 10 & 30 \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right] = \left[\begin{array}{c} 6 \\ 10.1 \end{array}\right]$	the normal equations become $\left[\begin{array}{cc} 4 & 10 \\ 10 & 30 \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right] = \left[\begin{array}{c} 6 \\ 10.1 \end{array}\right]$. The data and the least
lyche-numerical-linear-algebra- F02895	215	1.000	$p(t)=x_{\{1\}}+x_{\{2\}} t=3.95-0.98 t$	$p(t) = x_1 + x_2 t = 3.95 - 0.98 t$	squares polynomial $p(t) = x_1 + x_2 t = 3.95 - 0.98 t$ are shown in Fig. 9.1.
lyche-numerical-linear-algebra- F02896	215	1.000	$y \in \mathbb{R}$	$y \in \mathbb{R}$	To every input $\boldsymbol{u} \in \mathbb{R}^n$ we obtain an output $y \in \mathbb{R}$. Assuming we have a linear
lyche-numerical-linear-algebra- F02897	216	0.994	$\min \ \boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\ _2^2$	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$	We can estimate $\boldsymbol{x}$ by solving the least squares problem $\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$.
lyche-numerical-linear-algebra- F02898	216	1.000	$1 \leq n \leq m$	$1 \leq n \leq m$	• size: $1 \leq n \leq m$,
lyche-numerical-linear-algebra- F02899	216	1.000	$\mathcal{S}:=\left\{t_{\{1\}}, t_{\{2\}}, \ldots, t_{\{m\}}\right\} \subset[a, b]$	$\mathcal{S} := \{t_1, t_2, \dots, t_m\} \subset [a, b]$	• sites: $\mathcal{S} := \{t_1, t_2, \dots, t_m\} \subset [a, b]$,
lyche-numerical-linear-algebra- F02900	216	1.000	$\boldsymbol{y}=\left[y_{\{1\}}, y_{\{2\}}, \ldots, y_{\{m\}}\right]^* \in \mathbb{R}^m$	$\boldsymbol{y} = [y_1, y_2, \dots, y_m]^* \in \mathbb{R}^m$	• y-values: $\boldsymbol{y} = [y_1, y_2, \dots, y_m]^* \in \mathbb{R}^m$
lyche-numerical-linear-algebra- F02901	216	1.000	$\phi_{\{j\}}:[a, b] \rightarrow \mathbb{R}, j=1, \ldots, n$	$\phi_j : [a, b] \rightarrow \mathbb{R}, j = 1, \dots, n$	• functions: $\phi_j : [a, b] \rightarrow \mathbb{R}, j = 1, \dots, n$.
lyche-numerical-linear-algebra- F02902	216	0.999	$p:[a, b] \rightarrow \mathbb{R}$	$p : [a, b] \rightarrow \mathbb{R}$	Find a function (curve fit) $p : [a, b] \rightarrow \mathbb{R}$ given by $p := \sum_{j=1}^n x_j \phi_j$ such that
lyche-numerical-linear-algebra- F02903	216	0.999	$p:=\sum_{j=1}^n x_{\{j\}} \phi_{\{j\}}$	$p := \sum_{j=1}^n x_j \phi_j$	Find a function (curve fit) $p : [a, b] \rightarrow \mathbb{R}$ given by $p := \sum_{j=1}^n x_j \phi_j$ such that
lyche-numerical-linear-algebra- F02904	216	1.000	$p\left(t_{\{k\}}\right) \approx y_{\{k\}}$	$p(t_k) \approx y_k$	$p(t_k) \approx y_k$ for $k = 1, \dots, m$.
lyche-numerical-linear-algebra- F02905	217	1.000	$\phi_{\{j\}}$	ϕ_j	Typical examples of functions ϕ_j are polynomials, trigonometric functions, expo-
lyche-numerical-linear-algebra- F02906	217	1.000	$w_{\{k\}}>0$	$w_k > 0$	In (9.2) one can also include weights $w_k > 0$ for $k = 1, \dots, m$ and minimize
lyche-numerical-linear-algebra- F02907	217	1.000	$y_{\{k\}}$	y_k	If y_k is an accurate observation, we can choose a large weight w_k . This will force
lyche-numerical-linear-algebra- F02908	217	1.000	$w_{\{k\}}$	w_k	If y_k is an accurate observation, we can choose a large weight w_k . This will force

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lyche-numerical-linear-algebra-2020-06-10-F02935	218	1.000	$\boldsymbol{b}_1$	$\boldsymbol{b}_1$	uniquely as $\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2$, where $\boldsymbol{b}_1$ is the orthogonal projection of $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$
lyche-numerical-linear-algebra-2020-06-10-F02936	218	1.000	$\boldsymbol{b}_2$	$\boldsymbol{b}_2$	and $\boldsymbol{b}_2$ is the orthogonal projection of $\boldsymbol{b}$ into $\mathcal{N}(\boldsymbol{A}^*)$. Suppose $\boldsymbol{x} \in \mathbb{C}^n$. Clearly
lyche-numerical-linear-algebra-2020-06-10-F02937	218	1.000	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$
lyche-numerical-linear-algebra-2020-06-10-F02938	218	1.000	$\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$
lyche-numerical-linear-algebra-2020-06-10-F02939	218	1.000	$\langle \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$	$\langle \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$
lyche-numerical-linear-algebra-2020-06-10-F02940	219	1.000	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}_1$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}_1$	with equality if and only if $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}_1$. It follows that the set of all least squares
lyche-numerical-linear-algebra-2020-06-10-F02941	219	1.000	$\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	This set is nonempty since $\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$.
lyche-numerical-linear-algebra-2020-06-10-F02942	219	1.000	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 = \mathbf{0}$	Proof of Theorem 9.3 If $\boldsymbol{x}$ solves the least squares problem then $\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 = \mathbf{0}$ and
lyche-numerical-linear-algebra-2020-06-10-F02943	219	0.699	$\boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1) = \mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1) = \mathbf{0}$	it follows that $\boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1) = \mathbf{0}$ since $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. This shows that
lyche-numerical-linear-algebra-2020-06-10-F02944	219	1.000	$\boldsymbol{A}^* \boldsymbol{b}_2 = \mathbf{0}$	$\boldsymbol{A}^* \boldsymbol{b}_2 = \mathbf{0}$	the normal equations hold. Conversely, if $\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^* \boldsymbol{b}$ then $\boldsymbol{A}^* \boldsymbol{b}_2 = \mathbf{0}$ implies that
lyche-numerical-linear-algebra-2020-06-10-F02945	219	0.995	$\boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1) = \mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1) = \mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1) = \mathbf{0}$. But then $\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$ showing that $\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 = \mathbf{0}$,
lyche-numerical-linear-algebra-2020-06-10-F02946	219	0.995	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{A}^*(\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1) = \mathbf{0}$. But then $\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$ showing that $\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 = \mathbf{0}$,
lyche-numerical-linear-algebra-2020-06-10-F02947	219	1.000	$m \geq n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^m$	$m \geq n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^m$	We assume that $m > n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^m$. We consider numerical methods based
lyche-numerical-linear-algebra-2020-06-10-F02948	219	1.000	$\boldsymbol{B} := \boldsymbol{A}^* \boldsymbol{A}$	$\boldsymbol{B} := \boldsymbol{A}^* \boldsymbol{A}$	coefficient matrix $\boldsymbol{B} := \boldsymbol{A}^* \boldsymbol{A}$ in the normal equations is positive definite, and we
lyche-numerical-linear-algebra-2020-06-10-F02949	219	0.998	$(\boldsymbol{A}^* \boldsymbol{A})_{i,j} = \sum_{k=1}^m \bar{a}_{k,i} a_{k,j}, i, j = 1, \dots, n$	$(\boldsymbol{A}^* \boldsymbol{A})_{i,j} = \sum_{k=1}^m \bar{a}_{k,i} a_{k,j}, i, j = 1, \dots, n$	1. inner product: $(\boldsymbol{A}^* \boldsymbol{A})_{i,j} = \sum_{k=1}^m \bar{a}_{k,i} a_{k,j}, i, j = 1, \dots, n$,
lyche-numerical-linear-algebra-2020-06-10-F02950	219	1.000	$\boldsymbol{A}^* \boldsymbol{A} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} [a_{k1} \cdots a_{kn}], \boldsymbol{A}^* \boldsymbol{b} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} b_k$	$\boldsymbol{A}^* \boldsymbol{A} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} [a_{k1} \cdots a_{kn}], \boldsymbol{A}^* \boldsymbol{b} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} b_k$	2. outer product: $\boldsymbol{A}^* \boldsymbol{A} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} [a_{k1} \cdots a_{kn}], \boldsymbol{A}^* \boldsymbol{b} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} b_k$.
lyche-numerical-linear-algebra-2020-06-10-F02951	219	1.000	$n(n+1)/2$	$n(n+1)/2$	we only need to compute $n(n+1)/2$ such inner products. It follows that $\boldsymbol{B}$ can be
lyche-numerical-linear-algebra-2020-06-10-F02952	219	1.000	mn^2	mn^2	computed in approximately mn^2 arithmetic operations. In conclusion the number
lyche-numerical-linear-algebra-2020-06-10-F02953	220	1.000	$\boldsymbol{B}, 2mn$	$\boldsymbol{B}, 2mn$	of operations are mn^2 to find $\boldsymbol{B}$, $2mn$ to find $\boldsymbol{c} := \boldsymbol{A}^* \boldsymbol{b}$, $n^3/3$ to find $\boldsymbol{L}$ such that
lyche-numerical-linear-algebra-2020-06-10-F02954	220	1.000	$\boldsymbol{c} := \boldsymbol{A}^* \boldsymbol{b}, n^3/3$	$\boldsymbol{c} := \boldsymbol{A}^* \boldsymbol{b}, n^3/3$	of operations are mn^2 to find $\boldsymbol{B}$, $2mn$ to find $\boldsymbol{c} := \boldsymbol{A}^* \boldsymbol{b}$, $n^3/3$ to find $\boldsymbol{L}$ such that
lyche-numerical-linear-algebra-2020-06-10-F02955	220	1.000	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^*, n^2$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^*, n^2$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^*, n^2$ to solve $\boldsymbol{L} \boldsymbol{y} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^* \boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra-2020-06-10-F02956	220	1.000	$\boldsymbol{L} \boldsymbol{y} = \boldsymbol{c}$	$\boldsymbol{L} \boldsymbol{y} = \boldsymbol{c}$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^*, n^2$ to solve $\boldsymbol{L} \boldsymbol{y} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^* \boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra-2020-06-10-F02957	220	1.000	$\boldsymbol{L}^* \boldsymbol{x} = \boldsymbol{y}$	$\boldsymbol{L}^* \boldsymbol{x} = \boldsymbol{y}$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^*, n^2$ to solve $\boldsymbol{L} \boldsymbol{y} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^* \boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$

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lyche-numerical-linear-algebra- F02958	220	1.000	$m \approx n$	$m \approx n$	$\mathbf{B} = \mathbf{L}\mathbf{L}^*, n^2$ to solve $\mathbf{L}\mathbf{y} = \mathbf{c}$ and n^2 to solve $\mathbf{L}^*\mathbf{x} = \mathbf{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra- F02959	220	1.000	$\frac{4}{3}n^3 =$	$\frac{4}{3}n^3 =$	$\mathbf{B} = \mathbf{L}\mathbf{L}^*, n^2$ to solve $\mathbf{L}\mathbf{y} = \mathbf{c}$ and n^2 to solve $\mathbf{L}^*\mathbf{x} = \mathbf{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra- F02960	220	1.000	$\lambda_1 \geq \cdots \geq \lambda_n > 0$	$\lambda_1 \geq \cdots \geq \lambda_n > 0$	where $\lambda_1 \geq \cdots \geq \lambda_n > 0$ are the eigenvalues of $\mathbf{A}^*\mathbf{A}$, and $\sigma_1 \geq \cdots \geq \sigma_n > 0$ are
lyche-numerical-linear-algebra- F02961	220	1.000	$\sigma_1 \geq \cdots \geq \sigma_n > 0$	$\sigma_1 \geq \cdots \geq \sigma_n > 0$	where $\lambda_1 \geq \cdots \geq \lambda_n > 0$ are the eigenvalues of $\mathbf{A}^*\mathbf{A}$, and $\sigma_1 \geq \cdots \geq \sigma_n > 0$ are
lyche-numerical-linear-algebra- F02962	220	1.000	$K_2(\mathbf{A}) = \frac{\sigma_1}{\sigma_n}$	$K_2(\mathbf{A}) = \frac{\sigma_1}{\sigma_n}$	Proof Since $\mathbf{A}^*\mathbf{A}$ is Hermitian it follows from Theorem 8.10 that $K_2(\mathbf{A}) = \frac{\sigma_1}{\sigma_n}$ and
lyche-numerical-linear-algebra- F02963	220	0.999	$K_2(\mathbf{A}^*\mathbf{A}) = \frac{\lambda_1}{\lambda_n}$	$K_2(\mathbf{A}^*\mathbf{A}) = \frac{\lambda_1}{\lambda_n}$	$K_2(\mathbf{A}^*\mathbf{A}) = \frac{\lambda_1}{\lambda_n}$. But $\lambda_i = \sigma_i^2$ by Theorem 7.2 and the proof is complete.
lyche-numerical-linear-algebra- F02964	220	0.999	$\lambda_i = \sigma_i^2$	$\lambda_i = \sigma_i^2$	$K_2(\mathbf{A}^*\mathbf{A}) = \frac{\lambda_1}{\lambda_n}$. But $\lambda_i = \sigma_i^2$ by Theorem 7.2 and the proof is complete.
lyche-numerical-linear-algebra- F02965	220	1.000	$\mathbf{R}_1^*$	$\mathbf{R}_1^*$	Since $\mathbf{A}$ has rank n the matrix $\mathbf{R}_1^*$ is nonsingular and can be canceled. Thus
lyche-numerical-linear-algebra- F02966	220	1.000	$\mathbf{c}_1$	$\mathbf{c}_1$	We can use Householder transformations or Givens rotations to find $\mathbf{R}_1$ and $\mathbf{c}_1$.
lyche-numerical-linear-algebra- F02967	220	0.997	$\mathbf{R} = \mathbf{Q}^*\mathbf{A}$	$\mathbf{R} = \mathbf{Q}^*\mathbf{A}$	$\mathbf{R} = \mathbf{Q}^*\mathbf{A}$ and $\mathbf{c} = \mathbf{Q}^*\mathbf{b}$, where $\mathbf{A} = \mathbf{Q}\mathbf{R}$ is the QR decomposition of $\mathbf{A}$.
lyche-numerical-linear-algebra- F02968	220	0.997	$\mathbf{c} = \mathbf{Q}^*\mathbf{b}$	$\mathbf{c} = \mathbf{Q}^*\mathbf{b}$	$\mathbf{R} = \mathbf{Q}^*\mathbf{A}$ and $\mathbf{c} = \mathbf{Q}^*\mathbf{b}$, where $\mathbf{A} = \mathbf{Q}\mathbf{R}$ is the QR decomposition of $\mathbf{A}$.
lyche-numerical-linear-algebra- F02969	220	1.000	$\mathbf{c}$	$\mathbf{c}$	The matrices $\mathbf{R}_1$ and $\mathbf{c}_1$ are located in the first n rows of $\mathbf{R}$ and $\mathbf{c}$. Using also
lyche-numerical-linear-algebra- F02970	220	0.769	$[\mathbf{R}, \mathbf{c}] =$	$[\mathbf{R}, \mathbf{c}] =$	1. $[\mathbf{R}, \mathbf{c}] = \text{housetriang}(\mathbf{A}, \mathbf{b})$.
lyche-numerical-linear-algebra- F02971	221	0.958	$\mathbf{x} = [1, 0, 0]^*$	$\mathbf{x} = [1, 0, 0]^*$	and we find $\mathbf{x} = [1, 0, 0]^*$.
lyche-numerical-linear-algebra- F02972	221	1.000	$\mathbf{R}_1\mathbf{x} = \mathbf{c}_1$	$\mathbf{R}_1\mathbf{x} = \mathbf{c}_1$	problems, see [2]. The 2 norm condition number for the system $\mathbf{R}_1\mathbf{x} = \mathbf{c}_1$ is
lyche-numerical-linear-algebra- F02973	221	1.000	$K_2(\mathbf{R}_1) = K_2(\mathbf{Q}_1\mathbf{R}_1) = K_2(\mathbf{A})$	$K_2(\mathbf{R}_1) = K_2(\mathbf{Q}_1\mathbf{R}_1) = K_2(\mathbf{A})$	$K_2(\mathbf{R}_1) = K_2(\mathbf{Q}_1\mathbf{R}_1) = K_2(\mathbf{A})$, and as discussed in the previous section this
lyche-numerical-linear-algebra- F02974	221	1.000	$K_2(\mathbf{A}^*\mathbf{A})$	$K_2(\mathbf{A}^*\mathbf{A})$	is the square root of $K_2(\mathbf{A}^*\mathbf{A})$, the condition number for the normal equations.
lyche-numerical-linear-algebra- F02975	221	1.000	$2mn^2 -$	$2mn^2 -$	the number of arithmetic operations in Algorithm 5.2 is approximately $2mn^2 -$
lyche-numerical-linear-algebra- F02976	221	1.000	$\mathbf{x} = \mathbf{A} \backslash \mathbf{b}$	$\mathbf{x} = \mathbf{A} \backslash \mathbf{b}$	Using MATLAB a least squares solution can be found using $\mathbf{x} = \mathbf{A} \backslash \mathbf{b}$ if $\mathbf{A}$ has
lyche-numerical-linear-algebra- F02977	221	0.996	$\mathbf{x} = \text{lscov}(\mathbf{A}, \mathbf{b})$	$\mathbf{x} = \text{lscov}(\mathbf{A}, \mathbf{b})$	full rank. For rank deficient problems the function $\mathbf{x} = \text{lscov}(\mathbf{A}, \mathbf{b})$ finds a least
lyche-numerical-linear-algebra- F02978	222	1.000	$\mathbf{U}_2$	$\mathbf{U}_2$	• the set of columns of $\mathbf{U}_2$ is an orthonormal basis for $\mathcal{N}(\mathbf{A}^*)$,
lyche-numerical-linear-algebra- F02979	222	1.000	$\mathbf{V}_2$	$\mathbf{V}_2$	• the set of columns of $\mathbf{V}_2$ is an orthonormal basis for $\mathcal{N}(\mathbf{A})$.
lyche-numerical-linear-algebra- F02980	222	1.000	$\mathbf{A}^\dagger \in \mathbb{C}^{n \times m}$	$\mathbf{A}^\dagger \in \mathbb{C}^{n \times m}$	there is a unique matrix $\mathbf{A}^\dagger \in \mathbb{C}^{n \times m}$ such that
lyche-numerical-linear-algebra- F02981	222	1.000	$\mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^*$	$\mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^*$	If $\mathbf{U}_1\mathbf{\Sigma}_1\mathbf{V}_1^*$ is a singular value factorization of $\mathbf{A}$ then
lyche-numerical-linear-algebra- F02982	222	1.000	$(\mathbf{A}^\dagger\mathbf{A})^* = \mathbf{V}_1\mathbf{V}_1^*$	$(\mathbf{A}^\dagger\mathbf{A})^* = \mathbf{V}_1\mathbf{V}_1^*$	A similar calculation shows that $(\mathbf{A}^\dagger\mathbf{A})^* = \mathbf{V}_1\mathbf{V}_1^*$ and $(\mathbf{A}\mathbf{A}^\dagger)^* = \mathbf{U}_1\mathbf{U}_1^*$ showing
lyche-numerical-linear-algebra- F02983	222	1.000	$(\mathbf{A}\mathbf{A}^\dagger)^* = \mathbf{U}_1\mathbf{U}_1^*$	$(\mathbf{A}\mathbf{A}^\dagger)^* = \mathbf{U}_1\mathbf{U}_1^*$	A similar calculation shows that $(\mathbf{A}^\dagger\mathbf{A})^* = \mathbf{V}_1\mathbf{V}_1^*$ and $(\mathbf{A}\mathbf{A}^\dagger)^* = \mathbf{U}_1\mathbf{U}_1^*$ showing

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lyche-numerical-linear-algebra-2022-F02984	222	1.000	$\boldsymbol{A}^{\dagger} \boldsymbol{A}$	$\boldsymbol{A}^{\dagger} \boldsymbol{A}$	that $\boldsymbol{A}^{\dagger} \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^{\dagger}$ are Hermitian. Moreover, by (9.10)
lyche-numerical-linear-algebra-2022-F02985	222	1.000	$\boldsymbol{A} \boldsymbol{A}^{\dagger}$	$\boldsymbol{A} \boldsymbol{A}^{\dagger}$	that $\boldsymbol{A}^{\dagger} \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^{\dagger}$ are Hermitian. Moreover, by (9.10)
lyche-numerical-linear-algebra-2022-F02986	223	1.000	$\boldsymbol{A}^{\dagger}$	$\boldsymbol{A}^{\dagger}$	The matrix $\boldsymbol{A}^{\dagger}$ is called the generalized inverse of $\boldsymbol{A}$. We note that
lyche-numerical-linear-algebra-2022-F02987	223	0.999	$\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$	$\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$	1. If $\boldsymbol{A}$ is square and nonsingular then $\boldsymbol{A}^{-1}$ satisfies (9.8) so that $\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$. Indeed,
lyche-numerical-linear-algebra-2022-F02988	223	1.000	$\boldsymbol{A}^{-1} \boldsymbol{A}$	$\boldsymbol{A}^{-1} \boldsymbol{A}$	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$ implies that $\boldsymbol{A}^{-1} \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^{-1}$ are Hermitian. Moreover,
lyche-numerical-linear-algebra-2022-F02989	223	1.000	$\boldsymbol{A} \boldsymbol{A}^{-1}$	$\boldsymbol{A} \boldsymbol{A}^{-1}$	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$ implies that $\boldsymbol{A}^{-1} \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^{-1}$ are Hermitian. Moreover,
lyche-numerical-linear-algebra-2022-F02990	223	1.000	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}$	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}$	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}$. By uniqueness $\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$.
lyche-numerical-linear-algebra-2022-F02991	223	1.000	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}^*)$	and let $\mathcal{S}$ be one of the subspaces $\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}^*)$. The orthogonal projection of
lyche-numerical-linear-algebra-2022-F02992	223	1.000	$\boldsymbol{v} \in \mathbb{C}^m$	$\boldsymbol{v} \in \mathbb{C}^m$	$\boldsymbol{v} \in \mathbb{C}^m$ into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$,
lyche-numerical-linear-algebra-2022-F02993	223	1.000	$\boldsymbol{P}_{\mathcal{S}}$	$\boldsymbol{P}_{\mathcal{S}}$	$\boldsymbol{v} \in \mathbb{C}^m$ into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$,
lyche-numerical-linear-algebra-2022-F02994	223	1.000	$\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$	$\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$	$\boldsymbol{v} \in \mathbb{C}^m$ into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$,
lyche-numerical-linear-algebra-2022-F02995	223	0.998	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* \in \mathbb{C}^{m \times n}$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* \in \mathbb{C}^{m \times n}$	where $\boldsymbol{A}^{\dagger}$ is the generalized inverse of $\boldsymbol{A}$, and $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* \in \mathbb{C}^{m \times n}$ is a singular
lyche-numerical-linear-algebra-2022-F02996	223	1.000	$\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$	where $\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$ and $\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$. By uniqueness and (9.10)
lyche-numerical-linear-algebra-2022-F02997	223	1.000	$\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$	where $\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$ and $\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$. By uniqueness and (9.10)
lyche-numerical-linear-algebra-2022-F02998	223	1.000	$\boldsymbol{v} = (\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^*) \boldsymbol{v}$	$\boldsymbol{v} = (\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^*) \boldsymbol{v}$	we obtain the first equation in (9.12). Since $\boldsymbol{v} = (\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^*) \boldsymbol{v}$ for any $\boldsymbol{v} \in \mathbb{C}^m$
lyche-numerical-linear-algebra-2022-F02999	223	0.999	$\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$	$\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$	we have $\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$, and hence $\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger}$ and the
lyche-numerical-linear-algebra-2022-F03000	223	0.999	$\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger}$	$\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger}$	we have $\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$, and hence $\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger}$ and the
lyche-numerical-linear-algebra-2022-F03001	223	0.848	$\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$	$\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$	solves the least squares problem $\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$ if and only if $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$,
lyche-numerical-linear-algebra-2022-F03002	223	0.848	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$	solves the least squares problem $\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$ if and only if $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$,
lyche-numerical-linear-algebra-2022-F03003	223	0.893	$\boldsymbol{z} \in \mathcal{N}(\boldsymbol{A})$	$\boldsymbol{z} \in \mathcal{N}(\boldsymbol{A})$	where $\boldsymbol{A}^{\dagger}$ is the generalized inverse of $\boldsymbol{A}$ and $\boldsymbol{z} \in \mathcal{N}(\boldsymbol{A})$.
lyche-numerical-linear-algebra-2022-F03004	224	1.000	$\boldsymbol{b}_1 := \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}$	$\boldsymbol{b}_1 := \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}$	Proof It follows from Theorem 9.6 that $\boldsymbol{b}_1 := \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}$ is the orthogonal projection
lyche-numerical-linear-algebra-2022-F03005	224	0.998	$\boldsymbol{z} := \boldsymbol{x} - \boldsymbol{A}^{\dagger} \boldsymbol{b}$	$\boldsymbol{z} := \boldsymbol{x} - \boldsymbol{A}^{\dagger} \boldsymbol{b}$	Let $\boldsymbol{x}$ be a least squares solution, i.e., $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}_1$. If $\boldsymbol{z} := \boldsymbol{x} - \boldsymbol{A}^{\dagger} \boldsymbol{b}$ then $\boldsymbol{A} \boldsymbol{z} =$
lyche-numerical-linear-algebra-2022-F03006	224	0.998	$\boldsymbol{A} \boldsymbol{z} =$	$\boldsymbol{A} \boldsymbol{z} =$	Let $\boldsymbol{x}$ be a least squares solution, i.e., $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}_1$. If $\boldsymbol{z} := \boldsymbol{x} - \boldsymbol{A}^{\dagger} \boldsymbol{b}$ then $\boldsymbol{A} \boldsymbol{z} =$
lyche-numerical-linear-algebra-2022-F03007	224	1.000	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{b}_1 - \boldsymbol{b}_1 = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{b}_1 - \boldsymbol{b}_1 = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{b}_1 - \boldsymbol{b}_1 = \mathbf{0}$ and $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$.

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lyche-numerical-linear-algebra-224 F03008	224	■ 0.997	$\boldsymbol{A} \boldsymbol{z} = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{z} = \mathbf{0}$	Conversely, if $\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}$ with $\boldsymbol{A} \boldsymbol{z} = \mathbf{0}$ then $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$ and $\boldsymbol{x}$ is
lyche-numerical-linear-algebra-224 F03009	224	■ 0.997	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$	Conversely, if $\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}$ with $\boldsymbol{A} \boldsymbol{z} = \mathbf{0}$ then $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$ and $\boldsymbol{x}$ is
lyche-numerical-linear-algebra-224 F03010	224	■ 1.000	$\boldsymbol{A}^\dagger \boldsymbol{b}$	$\boldsymbol{A}^\dagger \boldsymbol{b}$	The least squares solution $\boldsymbol{A}^\dagger \boldsymbol{b}$ has an interesting property.
lyche-numerical-linear-algebra-224 F03011	224	■ 1.000	$\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}$	$\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}$	Euclidian norm is $\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}$ corresponding to $\boldsymbol{z} = \mathbf{0}$.
lyche-numerical-linear-algebra-224 F03012	224	■ 0.999	$\boldsymbol{z} = \boldsymbol{V}_2 \boldsymbol{y}$	$\boldsymbol{z} = \boldsymbol{V}_2 \boldsymbol{y}$	$\mathcal{N}(\boldsymbol{A})$ we have $\boldsymbol{z} = \boldsymbol{V}_2 \boldsymbol{y}$ for some $\boldsymbol{y}$. Moreover, $\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$ since $\boldsymbol{V}$ has orthonormal
lyche-numerical-linear-algebra-224 F03013	224	■ 0.999	$\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$	$\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$	$\mathcal{N}(\boldsymbol{A})$ we have $\boldsymbol{z} = \boldsymbol{V}_2 \boldsymbol{y}$ for some $\boldsymbol{y}$. Moreover, $\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$ since $\boldsymbol{V}$ has orthonormal
lyche-numerical-linear-algebra-224 F03014	224	■ 1.000	$\boldsymbol{z}^* \boldsymbol{A}^\dagger \boldsymbol{b} = \boldsymbol{y}^* \boldsymbol{V}_2^* \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* \boldsymbol{b} = \mathbf{0}$	$\boldsymbol{z}^* \boldsymbol{A}^\dagger \boldsymbol{b} = \boldsymbol{y}^* \boldsymbol{V}_2^* \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* \boldsymbol{b} = \mathbf{0}$	columns. But then $\boldsymbol{z}^* \boldsymbol{A}^\dagger \boldsymbol{b} = \boldsymbol{y}^* \boldsymbol{V}_2^* \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* \boldsymbol{b} = \mathbf{0}$. Thus $\boldsymbol{z}$ and $\boldsymbol{A}^\dagger \boldsymbol{b}$ are orthogonal
lyche-numerical-linear-algebra-224 F03015	224	■ 1.000	$\ \boldsymbol{x}\ _2^2 = \ \boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}\ _2^2 = \ \boldsymbol{A}^\dagger \boldsymbol{b}\ _2^2 + \ \boldsymbol{z}\ _2^2 \geq \ \boldsymbol{A}^\dagger \boldsymbol{b}\ _2^2$	$\ \boldsymbol{x}\ _2^2 = \ \boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}\ _2^2 = \ \boldsymbol{A}^\dagger \boldsymbol{b}\ _2^2 + \ \boldsymbol{z}\ _2^2 \geq \ \boldsymbol{A}^\dagger \boldsymbol{b}\ _2^2$	so that by Pythagoras $\ \boldsymbol{x}\ _2^2 = \ \boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z}\ _2^2 = \ \boldsymbol{A}^\dagger \boldsymbol{b}\ _2^2 + \ \boldsymbol{z}\ _2^2 \geq \ \boldsymbol{A}^\dagger \boldsymbol{b}\ _2^2$ with
lyche-numerical-linear-algebra-224 F03016	224	■ 0.997	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	problem with $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\boldsymbol{b} = [1, 1]^*$. The singular value factorization, $\boldsymbol{A}^\dagger$ and
lyche-numerical-linear-algebra-224 F03017	224	■ 0.997	$\boldsymbol{b} = [1, 1]^*$	$\boldsymbol{b} = [1, 1]^*$	problem with $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\boldsymbol{b} = [1, 1]^*$. The singular value factorization, $\boldsymbol{A}^\dagger$ and
lyche-numerical-linear-algebra-224 F03018	224	■ 1.000	$[1/2, 1/2] + [a, -a]$	$[1/2, 1/2] + [a, -a]$	Using Corollary 9.1 we find the general solution $[1/2, 1/2] + [a, -a]$ for any $a \in \mathbb{C}$.
lyche-numerical-linear-algebra-224 F03019	224	■ 0.653	$a = 1/2$	$a = 1/2$	The MATLAB function <code>lsqcov</code> gives the solution $[1, 0]^*$ corresponding to $a = 1/2$,
lyche-numerical-linear-algebra-224 F03020	224	■ 0.761	$[1/2, 1/2]$	$[1/2, 1/2]$	while the minimal norm solution is $[1/2, 1/2]$ obtained for $a = 0$.
lyche-numerical-linear-algebra-224 F03021	224	■ 0.933	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$	solution $\boldsymbol{x}$ of the least squares problem $\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$.
lyche-numerical-linear-algebra-225 F03022	225	■ 1.000	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$	independent columns, and let $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$. Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ be the solutions of
lyche-numerical-linear-algebra-225 F03023	225	■ 0.994	$\min \ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\ _2$	$\min \ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\ _2$	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$ and $\min \ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\ _2$. Finally, let $\boldsymbol{b}_1, \boldsymbol{e}_1$ be the orthogonal
lyche-numerical-linear-algebra-225 F03024	225	■ 0.994	$\boldsymbol{b}_1, \boldsymbol{e}_1$	$\boldsymbol{b}_1, \boldsymbol{e}_1$	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$ and $\min \ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\ _2$. Finally, let $\boldsymbol{b}_1, \boldsymbol{e}_1$ be the orthogonal
lyche-numerical-linear-algebra-225 F03025	225	■ 1.000	$\boldsymbol{b}_1 \neq \mathbf{0}$	$\boldsymbol{b}_1 \neq \mathbf{0}$	projections of $\boldsymbol{b}$ and $\boldsymbol{e}$ into $\mathcal{R}(\boldsymbol{A})$. If $\boldsymbol{b}_1 \neq \mathbf{0}$, we have for any operator norm
lyche-numerical-linear-algebra-225 F03026	225	■ 1.000	$\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}_1$	$\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}_1$	Proof Subtracting $\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^\dagger \boldsymbol{b}_1 + \boldsymbol{A}^\dagger \boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{e}_1$.
lyche-numerical-linear-algebra-225 F03027	225	■ 1.000	$\boldsymbol{y} = \boldsymbol{A}^\dagger \boldsymbol{b}_1 + \boldsymbol{A}^\dagger \boldsymbol{e}_1$	$\boldsymbol{y} = \boldsymbol{A}^\dagger \boldsymbol{b}_1 + \boldsymbol{A}^\dagger \boldsymbol{e}_1$	Proof Subtracting $\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^\dagger \boldsymbol{b}_1 + \boldsymbol{A}^\dagger \boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{e}_1$.
lyche-numerical-linear-algebra-225 F03028	225	■ 1.000	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{e}_1$	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{e}_1$	Proof Subtracting $\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^\dagger \boldsymbol{b}_1 + \boldsymbol{A}^\dagger \boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{e}_1$.
lyche-numerical-linear-algebra-225 F03029	225	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^\dagger \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^\dagger\ \ \boldsymbol{e}_1\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^\dagger \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^\dagger\ \ \boldsymbol{e}_1\ $	Thus $\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^\dagger \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^\dagger\ \ \boldsymbol{e}_1\ $. Moreover, $\ \boldsymbol{b}_1\ = \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $.

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lyche-numerical-linear-algebra-225 F03030	225	1.000	$\ \boldsymbol{b}_1\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{b}_1\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	Thus $\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^\dagger \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^\dagger\ \ \boldsymbol{e}_1\ $. Moreover, $\ \boldsymbol{b}_1\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $.
lyche-numerical-linear-algebra-225 F03031	225	1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ \ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ \ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $	Therefore $\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ \ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $ proving the rightmost inequality.
lyche-numerical-linear-algebra-225 F03032	225	1.000	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{e}_1$	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{e}_1$	From $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{e}_1$ and $\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b}_1$ we obtain the leftmost inequality.
lyche-numerical-linear-algebra-225 F03033	225	1.000	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ $	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ $	the number $K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ $ generalizes the condition number $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $ for
lyche-numerical-linear-algebra-225 F03034	225	1.000	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	the number $K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ $ generalizes the condition number $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $ for
lyche-numerical-linear-algebra-225 F03035	225	0.997	$\ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $	$\ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $	$\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $ in (8.22) has been replaced by $\ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $, the orthogonal projections of
lyche-numerical-linear-algebra-225 F03036	225	1.000	$\ \boldsymbol{b}\ / \ \boldsymbol{b}_1\ $	$\ \boldsymbol{b}\ / \ \boldsymbol{b}_1\ $	$\boldsymbol{e}$ and $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$. If $\boldsymbol{b}$ lies almost entirely in $\mathcal{N}(\boldsymbol{A}^*)$, i.e. $\ \boldsymbol{b}\ / \ \boldsymbol{b}_1\ $ is large, then
lyche-numerical-linear-algebra-226 F03037	226	1.000	$K_\infty(\boldsymbol{A}) = \ \boldsymbol{A}\ _\infty \ \boldsymbol{A}^\dagger\ _\infty = 2 \cdot 2 = 4$	$K_\infty(\boldsymbol{A}) = \ \boldsymbol{A}\ _\infty \ \boldsymbol{A}^\dagger\ _\infty = 2 \cdot 2 = 4$	Thus $K_\infty(\boldsymbol{A}) = \ \boldsymbol{A}\ _\infty \ \boldsymbol{A}^\dagger\ _\infty = 2 \cdot 2 = 4$ is quite small.
lyche-numerical-linear-algebra-226 F03038	226	0.999	$\boldsymbol{A} \boldsymbol{A}^\dagger = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	$\boldsymbol{A} \boldsymbol{A}^\dagger = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	Consider now the projections $\boldsymbol{b}_1$ and $\boldsymbol{e}_1$. We find $\boldsymbol{A} \boldsymbol{A}^\dagger = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Hence
lyche-numerical-linear-algebra-226 F03039	226	1.000	$\ \boldsymbol{e}_1\ _\infty / \ \boldsymbol{b}_1\ _\infty = 10^{-2}$	$\ \boldsymbol{e}_1\ _\infty / \ \boldsymbol{b}_1\ _\infty = 10^{-2}$	Thus $\ \boldsymbol{e}_1\ _\infty / \ \boldsymbol{b}_1\ _\infty = 10^{-2}$ and (9.13) takes the form
lyche-numerical-linear-algebra-226 F03040	226	1.000	$\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b} = [10^{-4}, 0]^*$	$\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b} = [10^{-4}, 0]^*$	To verify the bounds we compute the solutions as $\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b} = [10^{-4}, 0]^*$ and
lyche-numerical-linear-algebra-226 F03041	226	0.999	$\boldsymbol{y} = \boldsymbol{A}^\dagger(\boldsymbol{b} + \boldsymbol{e}) = [10^{-4} + 10^{-6}, 0]^*$	$\boldsymbol{y} = \boldsymbol{A}^\dagger(\boldsymbol{b} + \boldsymbol{e}) = [10^{-4} + 10^{-6}, 0]^*$	$\boldsymbol{y} = \boldsymbol{A}^\dagger(\boldsymbol{b} + \boldsymbol{e}) = [10^{-4} + 10^{-6}, 0]^*$. Hence
lyche-numerical-linear-algebra-226 F03042	226	1.000	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m \times n}, m > n$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m \times n}, m > n$	Theorem 9.9 (Perturbing the Matrix) Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m \times n}, m > n$, where $\boldsymbol{A}$
lyche-numerical-linear-algebra-226 F03043	226	1.000	$\alpha := 1 - \ \boldsymbol{E}\ _2 \ \boldsymbol{A}^\dagger\ _2 > 0$	$\alpha := 1 - \ \boldsymbol{E}\ _2 \ \boldsymbol{A}^\dagger\ _2 > 0$	has linearly independent columns and $\alpha := 1 - \ \boldsymbol{E}\ _2 \ \boldsymbol{A}^\dagger\ _2 > 0$. Then $\boldsymbol{A} + \boldsymbol{E}$
lyche-numerical-linear-algebra-226 F03044	226	1.000	$\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2 \in \mathbb{C}^m$	$\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2 \in \mathbb{C}^m$	has linearly independent columns. Let $\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2 \in \mathbb{C}^m$ where $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are the
lyche-numerical-linear-algebra-226 F03045	226	0.755	$\min \ (\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} - \boldsymbol{b}\ _2$	$\min \ (\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} - \boldsymbol{b}\ _2$	and $\boldsymbol{y}$ be the solutions of $\min \ \boldsymbol{Ax} - \boldsymbol{b}\ _2$ and $\min \ (\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} - \boldsymbol{b}\ _2$. Then
lyche-numerical-linear-algebra-226 F03046	226	1.000	$K(1 + \beta K) / \alpha$	$K(1 + \beta K) / \alpha$	most $K(1 + \beta K) / \alpha$ times as large as the size $\ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$ of the relative perturbation
lyche-numerical-linear-algebra-226 F03047	226	1.000	$\ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	$\ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	most $K(1 + \beta K) / \alpha$ times as large as the size $\ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$ of the relative perturbation
lyche-numerical-linear-algebra-227 F03048	227	0.993	$\boldsymbol{A} \cdot \beta$	$\boldsymbol{A} \cdot \beta$	in $\boldsymbol{A}$. β will be small if $\boldsymbol{b}$ lies almost entirely in $\mathcal{R}(\boldsymbol{A})$, and we have approximately
lyche-numerical-linear-algebra-227 F03049	227	1.000	$\rho \leq \frac{1}{\alpha} K \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	$\rho \leq \frac{1}{\alpha} K \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	$\rho \leq \frac{1}{\alpha} K \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$. This corresponds to the estimate (8.25) for linear systems.
lyche-numerical-linear-algebra-227 F03050	227	1.000	β	β	If β is not small, the term $\frac{1}{\alpha} K^2 \beta \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$ will dominate. In other words, the
lyche-numerical-linear-algebra-227 F03051	227	1.000	$\frac{1}{\alpha} K^2 \beta \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	$\frac{1}{\alpha} K^2 \beta \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	If β is not small, the term $\frac{1}{\alpha} K^2 \beta \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$ will dominate. In other words, the
lyche-numerical-linear-algebra-227 F03052	227	1.000	$K(\boldsymbol{A})^2 \beta$	$K(\boldsymbol{A})^2 \beta$	condition number is roughly $K(\boldsymbol{A})$ if β is small and $K(\boldsymbol{A})^2 \beta$ if β is not small. Note
lyche-numerical-linear-algebra-227 F03053	227	0.999	$\boldsymbol{b}_2 = \boldsymbol{b} - \boldsymbol{Ax}$	$\boldsymbol{b}_2 = \boldsymbol{b} - \boldsymbol{Ax}$	that β is large if $\boldsymbol{b}$ is almost orthogonal to $\mathcal{R}(\boldsymbol{A})$ and that $\boldsymbol{b}_2 = \boldsymbol{b} - \boldsymbol{Ax}$ is the residual

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lyche-numerical-linear-algebra-227-F03054	227	■ 0.999	$\sigma_1, \sigma_2, \dots, \sigma_n$	$\sigma_1, \sigma_2, \dots, \sigma_n$	$A \in \mathbb{C}^{m \times n}$ has singular values $\sigma_1, \sigma_2, \dots, \sigma_n$ ordered so that $\sigma_1 \geq \dots \geq \sigma_n$. Then
lyche-numerical-linear-algebra-227-F03055	227	■ 0.999	$\sigma_1 \geq \dots \geq \sigma_n$	$\sigma_1 \geq \dots \geq \sigma_n$	$A \in \mathbb{C}^{m \times n}$ has singular values $\sigma_1, \sigma_2, \dots, \sigma_n$ ordered so that $\sigma_1 \geq \dots \geq \sigma_n$. Then
lyche-numerical-linear-algebra-228-F03056	228	■ 1.000	$A, B \in \mathbb{R}^{m \times n}$	$A, B \in \mathbb{R}^{m \times n}$	Theorem 9.11 (Perturbation of Singular Values) Let $A, B \in \mathbb{R}^{m \times n}$ be rectangu-
lyche-numerical-linear-algebra-228-F03057	228	■ 1.000	$n - j + 1$	$n - j + 1$	Proof Fix j and let S be the $n - j + 1$ dimensional subspace for which the minimum
lyche-numerical-linear-algebra-228-F03058	228	■ 1.000	$\beta_j \leq \alpha_j + \ A - B\ _2$	$\beta_j \leq \alpha_j + \ A - B\ _2$	By symmetry we obtain $\beta_j \leq \alpha_j + \ A - B\ _2$ and the proof is complete.
lyche-numerical-linear-algebra-228-F03059	228	■ 1.000	$A, E \in$	$A, E \in$	Theorem 9.12 (Generalized Inverse When Perturbing the Matrix) Let $A, E \in$
lyche-numerical-linear-algebra-228-F03060	228	■ 1.000	$\alpha_1 \geq \dots \geq \alpha_n$	$\alpha_1 \geq \dots \geq \alpha_n$	$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \dots \geq \alpha_n$ and $\epsilon_1 \geq \dots \geq \epsilon_n$. If $\text{rank}(A + E) \leq$
lyche-numerical-linear-algebra-228-F03061	228	■ 1.000	$\epsilon_1 \geq \dots \geq \epsilon_n$	$\epsilon_1 \geq \dots \geq \epsilon_n$	$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \dots \geq \alpha_n$ and $\epsilon_1 \geq \dots \geq \epsilon_n$. If $\text{rank}(A + E) \leq$
lyche-numerical-linear-algebra-228-F03062	228	■ 1.000	$\text{rank}(A + E) \leq$	$\text{rank}(A + E) \leq$	$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \dots \geq \alpha_n$ and $\epsilon_1 \geq \dots \geq \epsilon_n$. If $\text{rank}(A + E) \leq$
lyche-numerical-linear-algebra-228-F03063	228	■ 0.893	$\ A^\dagger\ _2 \ E\ _2 < 1$	$\ A^\dagger\ _2 \ E\ _2 < 1$	$\text{rank}(A) = r$ and $\ A^\dagger\ _2 \ E\ _2 < 1$ then
lyche-numerical-linear-algebra-228-F03064	228	■ 1.000	$\text{rank}(A + E) = \text{rank}(A)$	$\text{rank}(A + E) = \text{rank}(A)$	1. $\text{rank}(A + E) = \text{rank}(A)$,
lyche-numerical-linear-algebra-228-F03065	228	■ 1.000	$\ (A + E)^\dagger\ _2 \leq \frac{\ A^\dagger\ _2}{1 - \ A^\dagger\ _2 \ E\ _2} = \frac{1}{\alpha_r - \epsilon_1}$	$\ (A + E)^\dagger\ _2 \leq \frac{\ A^\dagger\ _2}{1 - \ A^\dagger\ _2 \ E\ _2} = \frac{1}{\alpha_r - \epsilon_1}$	2. $\ (A + E)^\dagger\ _2 \leq \frac{\ A^\dagger\ _2}{1 - \ A^\dagger\ _2 \ E\ _2} = \frac{1}{\alpha_r - \epsilon_1}$.
lyche-numerical-linear-algebra-228-F03066	228	■ 1.000	$\beta_1 \geq$	$\beta_1 \geq$	Proof Suppose A has rank r and let $B := A + E$ have singular values $\beta_1 \geq$
lyche-numerical-linear-algebra-228-F03067	228	■ 1.000	$\dots \geq \beta_n$	$\dots \geq \beta_n$	$\dots \geq \beta_n$. In terms of singular values the inequality $\ A^\dagger\ _2 \ E\ _2 < 1$ can be written
lyche-numerical-linear-algebra-228-F03068	228	■ 1.000	$\epsilon_1 / \alpha_r < 1$	$\epsilon_1 / \alpha_r < 1$	$\epsilon_1 / \alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \ E\ _2 = \epsilon_1$, which
lyche-numerical-linear-algebra-228-F03069	228	■ 1.000	$\alpha_r > \epsilon_1$	$\alpha_r > \epsilon_1$	$\epsilon_1 / \alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \ E\ _2 = \epsilon_1$, which
lyche-numerical-linear-algebra-228-F03070	228	■ 1.000	$\alpha_r - \beta_r \leq \ E\ _2 = \epsilon_1$	$\alpha_r - \beta_r \leq \ E\ _2 = \epsilon_1$	$\epsilon_1 / \alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \ E\ _2 = \epsilon_1$, which
lyche-numerical-linear-algebra-228-F03071	228	■ 0.703	$\beta_r \geq \alpha_r - \epsilon_1 > 0$	$\beta_r \geq \alpha_r - \epsilon_1 > 0$	implies $\beta_r \geq \alpha_r - \epsilon_1 > 0$, and this shows that $\text{rank}(A + E) \geq r$. Thus 1. follows.
lyche-numerical-linear-algebra-228-F03072	228	■ 0.703	$\text{rank}(A + E) \geq r$	$\text{rank}(A + E) \geq r$	implies $\beta_r \geq \alpha_r - \epsilon_1 > 0$, and this shows that $\text{rank}(A + E) \geq r$. Thus 1. follows.
lyche-numerical-linear-algebra-228-F03073	228	■ 0.960	$\beta_r \geq \alpha_r - \epsilon_1$	$\beta_r \geq \alpha_r - \epsilon_1$	To prove 2., the inequality $\beta_r \geq \alpha_r - \epsilon_1$ implies that
lyche-numerical-linear-algebra-228-F03074	228	■ 1.000	$\mu_1 \geq \dots \geq \mu_n$	$\mu_1 \geq \dots \geq \mu_n$	and $\mu_1 \geq \dots \geq \mu_n$, respectively.
lyche-numerical-linear-algebra-229-F03075	229	■ 1.000	$A, B \in \mathbb{C}^{m \times n}$	$A, B \in \mathbb{C}^{m \times n}$	$m, n \in \mathbb{N}$ and $A, B \in \mathbb{C}^{m \times n}$ we have
lyche-numerical-linear-algebra-229-F03076	229	■ 1.000	$\beta_1 \geq \dots \geq \beta_n$	$\beta_1 \geq \dots \geq \beta_n$	where $\alpha_1 \geq \dots \geq \alpha_n$ and $\beta_1 \geq \dots \geq \beta_n$ are the singular values of A and B ,
lyche-numerical-linear-algebra-229-F03077	229	■ 1.000	$\lambda_1 \geq \dots \geq \lambda_{m+n}$	$\lambda_1 \geq \dots \geq \lambda_{m+n}$	If C and D have eigenvalues $\lambda_1 \geq \dots \geq \lambda_{m+n}$ and $\mu_1 \geq \dots \geq \mu_{m+n}$, respectively
lyche-numerical-linear-algebra-229-F03078	229	■ 1.000	$\mu_1 \geq \dots \geq \mu_{m+n}$	$\mu_1 \geq \dots \geq \mu_{m+n}$	If C and D have eigenvalues $\lambda_1 \geq \dots \geq \lambda_{m+n}$ and $\mu_1 \geq \dots \geq \mu_{m+n}$, respectively
lyche-numerical-linear-algebra-229-F03079	229	■ 0.651	$\text{SVD}[u_1, \dots, u_m] \Sigma [v_1, \dots, v_n]^*$	$\text{SVD}[u_1, \dots, u_m] \Sigma [v_1, \dots, v_n]^*$	Suppose A has rank r and $\text{SVD}[u_1, \dots, u_m] \Sigma [v_1, \dots, v_n]^*$. We use (7.7) and

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lyche-numerical-linear-algebra-F03080	229	1.000	$2 \ r$	$2r$	Thus C has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$ and $m + n - 2r$ additional
lyche-numerical-linear-algebra-F03081	229	1.000	$\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$	$\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$	Thus C has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$ and $m + n - 2r$ additional
lyche-numerical-linear-algebra-F03082	229	1.000	$m+n-2 \ r$	$m + n - 2r$	Thus C has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$ and $m + n - 2r$ additional
lyche-numerical-linear-algebra-F03083	229	1.000	$2 \ s$	$2s$	zero eigenvalues. Similarly, if B has rank s then D has the $2s$ eigenvalues
lyche-numerical-linear-algebra-F03084	229	0.999	$\beta_1, -\beta_1, \dots, \beta_s, -\beta_s$	$\beta_1, -\beta_1, \dots, \beta_s, -\beta_s$	$\beta_1, -\beta_1, \dots, \beta_s, -\beta_s$ and $m + n - 2s$ additional zero eigenvalues. Let
lyche-numerical-linear-algebra-F03085	229	0.999	$m+n-2 \ s$	$m + n - 2s$	$\beta_1, -\beta_1, \dots, \beta_s, -\beta_s$ and $m + n - 2s$ additional zero eigenvalues. Let
lyche-numerical-linear-algebra-F03086	230	1.000	$\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ B - A\ _F^2$	$\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ B - A\ _F^2$	But then (9.21) implies $\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ B - A\ _F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra-F03087	230	1.000	$t \leq n$	$t \leq n$	But then (9.21) implies $\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ B - A\ _F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra-F03088	230	1.000	$\alpha_i = \beta_i = 0$	$\alpha_i = \beta_i = 0$	But then (9.21) implies $\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ B - A\ _F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra-F03089	230	1.000	$i=t+1, \dots, n$	$i = t + 1, \dots, n$	for $i = t + 1, \dots, n$ we obtain (9.20).
lyche-numerical-linear-algebra-F03090	230	1.000	$(t - c_1)^2 + (y - c_2)^2 = r^2$	$(t - c_1)^2 + (y - c_2)^2 = r^2$	to fit a circle $(t - c_1)^2 + (y - c_2)^2 = r^2$ to $m \geq 3$ given points $(t_i, y_i)_{i=1}^m$ in the
lyche-numerical-linear-algebra-F03091	230	1.000	$(t_i, y_i)_{i=1}^m$	$(t_i, y_i)_{i=1}^m$	to fit a circle $(t - c_1)^2 + (y - c_2)^2 = r^2$ to $m \geq 3$ given points $(t_i, y_i)_{i=1}^m$ in the
lyche-numerical-linear-algebra-F03092	230	0.999	(t, y)	(t, y)	(t, y) -plane. We obtain the overdetermined system
lyche-numerical-linear-algebra-F03093	230	1.000	c_1, c_2	c_1, c_2	of m equations in the three unknowns c_1, c_2 and r . This system is nonlinear, but it
lyche-numerical-linear-algebra-F03094	230	1.000	$c_1 = x_1/2, c_2 = x_2/2$	$c_1 = x_1/2, c_2 = x_2/2$	and then setting $c_1 = x_1/2, c_2 = x_2/2$ and $r^2 = c_1^2 + c_2^2 + x_3$,
lyche-numerical-linear-algebra-F03095	230	1.000	$r^2 = c_1^2 + c_2^2 + x_3$	$r^2 = c_1^2 + c_2^2 + x_3$	and then setting $c_1 = x_1/2, c_2 = x_2/2$ and $r^2 = c_1^2 + c_2^2 + x_3$.
lyche-numerical-linear-algebra-F03096	230	1.000	c_1, c_2, r	c_1, c_2, r	a) Derive (9.23) from (9.22). Explain how we can find c_1, c_2, r once $[x_1, x_2, x_3]$ is
lyche-numerical-linear-algebra-F03097	230	1.000	$[x_1, x_2, x_3]$	$[x_1, x_2, x_3]$	a) Derive (9.23) from (9.22). Explain how we can find c_1, c_2, r once $[x_1, x_2, x_3]$ is
lyche-numerical-linear-algebra-F03098	230	0.593	$(1, 4), (3, 2), (1, 0)$	$(1, 4), (3, 2), (1, 0)$	d) Use (9.23) to find the circle passing through the three points $(1, 4), (3, 2), (1, 0)$.
lyche-numerical-linear-algebra-F03099	231	1.000	$\begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}$	$\begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}$	a) Let A be the matrix $\begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}$. Find the singular values of A , and compute
lyche-numerical-linear-algebra-F03100	231	0.993	$A = \begin{bmatrix} 3 & \alpha \\ \alpha & 1 \end{bmatrix}$	$A = \begin{bmatrix} 3 & \alpha \\ \alpha & 1 \end{bmatrix}$	b) Consider the matrix $A = \begin{bmatrix} 3 & \alpha \\ \alpha & 1 \end{bmatrix}$, where α is a real number. For which values of
lyche-numerical-linear-algebra-F03101	231	1.000	$p_1 = (0, 1), p_2 = (1, 0), p_3 = (2, 1)$	$p_1 = (0, 1), p_2 = (1, 0), p_3 = (2, 1)$	c) We would like to fit the points $p_1 = (0, 1), p_2 = (1, 0), p_3 = (2, 1)$ to a straight
lyche-numerical-linear-algebra-F03102	231	1.000	$p(x) = mx + b$	$p(x) = mx + b$	line in the plane. Find a line $p(x) = mx + b$ which minimizes
lyche-numerical-linear-algebra-F03103	231	1.000	$p_i = (x_i, y_i)$	$p_i = (x_i, y_i)$	where $p_i = (x_i, y_i)$. Is this solution unique?
lyche-numerical-linear-algebra-F03104	231	1.000	$F: \mathbb{R}^n \rightarrow \mathbb{R}$	$F: \mathbb{R}^n \rightarrow \mathbb{R}$	$\mathbb{R}^{m \times n}$ and let $I \in \mathbb{R}^{n \times n}$ be the identity matrix. We define $F: \mathbb{R}^n \rightarrow \mathbb{R}$ by

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lyche-numerical-linear-algebra- F03105	231	1.000	$\boldsymbol{B}:=\boldsymbol{I}+\boldsymbol{A}^T\boldsymbol{A}$ $\boldsymbol{A}$	$\boldsymbol{B} := \boldsymbol{I} + \boldsymbol{A}^T \boldsymbol{A}$	a) Show that the matrix $\boldsymbol{B} := \boldsymbol{I} + \boldsymbol{A}^T \boldsymbol{A}$ is symmetric and positive definite.
lyche-numerical-linear-algebra- F03106	231	1.000	$\left(\boldsymbol{I}+\boldsymbol{A}^T\boldsymbol{A}\right)\boldsymbol{x}=\boldsymbol{A}^T\boldsymbol{b}$ $\boldsymbol{x}=\boldsymbol{A}^T\boldsymbol{b}$	$(\boldsymbol{I} + \boldsymbol{A}^T \boldsymbol{A}) \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$	is the unique solution of the linear system $(\boldsymbol{I} + \boldsymbol{A}^T \boldsymbol{A}) \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$.
lyche-numerical-linear-algebra- F03107	231	1.000	$\operatorname{diag}(d_1, d_2, \ldots, d_m)$ $\in \mathbb{R}^{m \times m}$	$\operatorname{diag}(d_1, d_2, \ldots, d_m) \in \mathbb{R}^{m \times m}$	$\operatorname{diag}(d_1, d_2, \ldots, d_m) \in \mathbb{R}^{m \times m}$, where $d_i > 0, i = 1, 2, \ldots, m$. We want to minimize
lyche-numerical-linear-algebra- F03108	231	1.000	$d_i > 0, i = 1, 2, \ldots, m$	$d_i > 0, i = 1, 2, \ldots, m$	$\operatorname{diag}(d_1, d_2, \ldots, d_m) \in \mathbb{R}^{m \times m}$, where $d_i > 0, i = 1, 2, \ldots, m$. We want to minimize
lyche-numerical-linear-algebra- F03109	232	1.000	$r_i = r_i(\boldsymbol{x}), i = 1, 2, \ldots, m$	$r_i = r_i(\boldsymbol{x}), i = 1, 2, \ldots, m$	where $r_i = r_i(\boldsymbol{x}), i = 1, 2, \ldots, m$ are the components of the vector
lyche-numerical-linear-algebra- F03110	232	1.000	$\ \boldsymbol{r}(\boldsymbol{x})\ _D^2$	$\ \boldsymbol{r}(\boldsymbol{x})\ _D^2$	a) Show that $\ \boldsymbol{r}(\boldsymbol{x})\ _D^2$ in (9.24) obtains a unique minimum when $\boldsymbol{x} = \boldsymbol{x}_{\min}$ is the
lyche-numerical-linear-algebra- F03111	232	1.000	$\boldsymbol{x} = \boldsymbol{x}_{\min}$	$\boldsymbol{x} = \boldsymbol{x}_{\min}$	a) Show that $\ \boldsymbol{r}(\boldsymbol{x})\ _D^2$ in (9.24) obtains a unique minimum when $\boldsymbol{x} = \boldsymbol{x}_{\min}$ is the
lyche-numerical-linear-algebra- F03112	232	1.000	$K_2(\boldsymbol{B}) := \ \boldsymbol{B}\ _2 \ \boldsymbol{B}^{-1}\ _2$	$K_2(\boldsymbol{B}) := \ \boldsymbol{B}\ _2 \ \boldsymbol{B}^{-1}\ _2$	where for any nonsingular matrix $K_2(\boldsymbol{B}) := \ \boldsymbol{B}\ _2 \ \boldsymbol{B}^{-1}\ _2$.
lyche-numerical-linear-algebra- F03113	232	0.999	$\boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times m}$	$\boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times m}$	pose $\boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times m}$ satisfy
lyche-numerical-linear-algebra- F03114	232	1.000	$\boldsymbol{B} = \boldsymbol{C}$	$\boldsymbol{B} = \boldsymbol{C}$	Verify the following proof that $\boldsymbol{B} = \boldsymbol{C}$.
lyche-numerical-linear-algebra- F03115	232	1.000	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$	generalized inverse of $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ is $\boldsymbol{A}^\dagger = \frac{1}{4} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ without using the singular
lyche-numerical-linear-algebra- F03116	232	1.000	$\boldsymbol{A}^\dagger = \frac{1}{4} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$	$\boldsymbol{A}^\dagger = \frac{1}{4} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$	generalized inverse of $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ is $\boldsymbol{A}^\dagger = \frac{1}{4} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ without using the singular
lyche-numerical-linear-algebra- F03117	232	1.000	$\boldsymbol{A}^\dagger = (\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^*$	$\boldsymbol{A}^\dagger = (\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^*$	and $\boldsymbol{A}^\dagger = (\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^*$. If $\boldsymbol{A}$ has linearly independent rows, then show that $\boldsymbol{A} \boldsymbol{A}^*$ is
lyche-numerical-linear-algebra- F03118	232	1.000	$\boldsymbol{A}^\dagger = \boldsymbol{A}^* (\boldsymbol{A} \boldsymbol{A}^*)^{-1}$	$\boldsymbol{A}^\dagger = \boldsymbol{A}^* (\boldsymbol{A} \boldsymbol{A}^*)^{-1}$	nonsingular and $\boldsymbol{A}^\dagger = \boldsymbol{A}^* (\boldsymbol{A} \boldsymbol{A}^*)^{-1}$.
lyche-numerical-linear-algebra- F03119	233	1.000	$\mathcal{R}(\boldsymbol{A}^*), \mathcal{N}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A}^*), \mathcal{N}(\boldsymbol{A})$	rank r , and let $\mathcal{S}$ be one of the subspaces $\mathcal{R}(\boldsymbol{A}^*), \mathcal{N}(\boldsymbol{A})$. Show that the orthogonal
lyche-numerical-linear-algebra- F03120	233	0.996	$\mathbb{C}^n = \mathcal{R}(\boldsymbol{A}^*) \oplus \mathcal{N}(\boldsymbol{A})$	$\mathbb{C}^n = \mathcal{R}(\boldsymbol{A}^*) \oplus \mathcal{N}(\boldsymbol{A})$	orthogonal sum $\mathbb{C}^n = \mathcal{R}(\boldsymbol{A}^*) \oplus \mathcal{N}(\boldsymbol{A})$
lyche-numerical-linear-algebra- F03121	233	1.000	$\boldsymbol{u}^\dagger = (\boldsymbol{u}^* \boldsymbol{u})^{-1} \boldsymbol{u}^*$	$\boldsymbol{u}^\dagger = (\boldsymbol{u}^* \boldsymbol{u})^{-1} \boldsymbol{u}^*$	Exercise 9.9 (The Generalized Inverse of a Vector) Show that $\boldsymbol{u}^\dagger = (\boldsymbol{u}^* \boldsymbol{u})^{-1} \boldsymbol{u}^*$
lyche-numerical-linear-algebra- F03122	233	1.000	$\boldsymbol{u} \in \mathbb{C}^{n,1}$	$\boldsymbol{u} \in \mathbb{C}^{n,1}$	if $\boldsymbol{u} \in \mathbb{C}^{n,1}$ is nonzero.
lyche-numerical-linear-algebra- F03123	233	1.000	$\boldsymbol{A} = \boldsymbol{u} \boldsymbol{v}^*$	$\boldsymbol{A} = \boldsymbol{u} \boldsymbol{v}^*$	Exercise 9.10 (The Generalized Inverse of an Outer Product) If $\boldsymbol{A} = \boldsymbol{u} \boldsymbol{v}^*$ where
lyche-numerical-linear-algebra- F03124	233	1.000	$\boldsymbol{u} \in \mathbb{C}^m, \boldsymbol{v} \in \mathbb{C}^n$	$\boldsymbol{u} \in \mathbb{C}^m, \boldsymbol{v} \in \mathbb{C}^n$	$\boldsymbol{u} \in \mathbb{C}^m, \boldsymbol{v} \in \mathbb{C}^n$ are nonzero, show that
lyche-numerical-linear-algebra- F03125	233	0.990	$\operatorname{diag}(\lambda_1, \ldots, \lambda_n)^\dagger = \operatorname{diag}(\lambda_1^\dagger, \ldots, \lambda_n^\dagger)$	$\operatorname{diag}(\lambda_1, \ldots, \lambda_n)^\dagger = \operatorname{diag}(\lambda_1^\dagger, \ldots, \lambda_n^\dagger)$	$\operatorname{diag}(\lambda_1, \ldots, \lambda_n)^\dagger = \operatorname{diag}(\lambda_1^\dagger, \ldots, \lambda_n^\dagger)$ where
lyche-numerical-linear-algebra- F03126	233	1.000	$(\boldsymbol{A}^*)^\dagger = (\boldsymbol{A}^\dagger)^*$	$(\boldsymbol{A}^*)^\dagger = (\boldsymbol{A}^\dagger)^*$	a) $(\boldsymbol{A}^*)^\dagger = (\boldsymbol{A}^\dagger)^*$.
lyche-numerical-linear-algebra- F03127	233	0.999	$(\boldsymbol{A}^\dagger)^\dagger = \boldsymbol{A}$	$(\boldsymbol{A}^\dagger)^\dagger = \boldsymbol{A}$	b) $(\boldsymbol{A}^\dagger)^\dagger = \boldsymbol{A}$.
lyche-numerical-linear-algebra- F03128	233	0.995	$(\alpha \boldsymbol{A})^\dagger = \frac{1}{\alpha} \boldsymbol{A}^\dagger, \alpha \neq 0$	$(\alpha \boldsymbol{A})^\dagger = \frac{1}{\alpha} \boldsymbol{A}^\dagger, \alpha \neq 0$	c) $(\alpha \boldsymbol{A})^\dagger = \frac{1}{\alpha} \boldsymbol{A}^\dagger, \alpha \neq 0$.

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lyche-numerical-linear-algebra-233-F03129	233	1.000	$k, m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{B} \in \mathbb{C}^{n \times k}$	$k, m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{B} \in \mathbb{C}^{n \times k}$	Exercise 9.13 (The Generalized Inverse of a Product) Suppose $\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{B} \in \mathbb{C}^{n \times k}$. Suppose $\boldsymbol{A}$ has linearly independent columns and $\boldsymbol{B}$ has linearly
lyche-numerical-linear-algebra-233-F03130	233	0.623	$(\boldsymbol{AB})^\dagger = \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$	$(\boldsymbol{AB})^\dagger = \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$	a) Show that $(\boldsymbol{AB})^\dagger = \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$.
lyche-numerical-linear-algebra-233-F03131	233	0.993	$\boldsymbol{E} = \boldsymbol{AB}, \boldsymbol{F} = \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$	$\boldsymbol{E} = \boldsymbol{AB}, \boldsymbol{F} = \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$	Hint: Let $\boldsymbol{E} = \boldsymbol{AB}, \boldsymbol{F} = \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$. Show by using $\boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{BB}^\dagger = \boldsymbol{I}$ that $\boldsymbol{F}$ is the
lyche-numerical-linear-algebra-233-F03132	233	1.000	$\boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{BB}^\dagger = \boldsymbol{I}$	$\boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{BB}^\dagger = \boldsymbol{I}$	Hint: Let $\boldsymbol{E} = \boldsymbol{AB}, \boldsymbol{F} = \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$. Show by using $\boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{BB}^\dagger = \boldsymbol{I}$ that $\boldsymbol{F}$ is the
lyche-numerical-linear-algebra-233-F03133	233	1.000	$\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}$	$\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}$	b) Find $\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}$ such that $(\boldsymbol{AB})^\dagger \neq \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$.
lyche-numerical-linear-algebra-233-F03134	233	0.998	$(\boldsymbol{AB})^\dagger \neq \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$	$(\boldsymbol{AB})^\dagger \neq \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$	b) Find $\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}$ such that $(\boldsymbol{AB})^\dagger \neq \boldsymbol{B}^\dagger \boldsymbol{A}^\dagger$.
lyche-numerical-linear-algebra-233-F03135	233	0.998	$\boldsymbol{A}^* = \boldsymbol{A}^\dagger$	$\boldsymbol{A}^* = \boldsymbol{A}^\dagger$	$\boldsymbol{A}^* = \boldsymbol{A}^\dagger$ if and only if all singular values of $\boldsymbol{A}$ are either zero or one.
lyche-numerical-linear-algebra-234-F03136	234	1.000	$\boldsymbol{A}(\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^* \boldsymbol{b}$	$\boldsymbol{A}(\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^* \boldsymbol{b}$	$\boldsymbol{A}(\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^* \boldsymbol{b}$ is the projection of $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$. (Cf. Exercise 9.8.)
lyche-numerical-linear-algebra-234-F03137	234	0.998	$r > 0$	$r > 0$	system $\boldsymbol{Ax} = \boldsymbol{b}$ where $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has rank $r > 0$ and $\boldsymbol{b} \in \mathbb{C}^n$. Let
lyche-numerical-linear-algebra-234-F03138	234	1.000	$\boldsymbol{c} = [c_1, \dots, c_n]^* = \boldsymbol{U}^* \boldsymbol{b}$	$\boldsymbol{c} = [c_1, \dots, c_n]^* = \boldsymbol{U}^* \boldsymbol{b}$	a) Let $\boldsymbol{c} = [c_1, \dots, c_n]^* = \boldsymbol{U}^* \boldsymbol{b}$ and $\boldsymbol{y} = [y_1, \dots, y_n]^* = \boldsymbol{V}^* \boldsymbol{x}$. Show that $\boldsymbol{Ax} = \boldsymbol{b}$
lyche-numerical-linear-algebra-234-F03139	234	1.000	$\boldsymbol{y} = [y_1, \dots, y_n]^* = \boldsymbol{V}^* \boldsymbol{x}$	$\boldsymbol{y} = [y_1, \dots, y_n]^* = \boldsymbol{V}^* \boldsymbol{x}$	a) Let $\boldsymbol{c} = [c_1, \dots, c_n]^* = \boldsymbol{U}^* \boldsymbol{b}$ and $\boldsymbol{y} = [y_1, \dots, y_n]^* = \boldsymbol{V}^* \boldsymbol{x}$. Show that $\boldsymbol{Ax} = \boldsymbol{b}$
lyche-numerical-linear-algebra-234-F03140	234	1.000	$\boldsymbol{c}_{r+1} = \dots = c_n = 0$	$c_{r+1} = \dots = c_n = 0$	b) Show that $\boldsymbol{Ax} = \boldsymbol{b}$ has a solution $\boldsymbol{x}$ if and only if $c_{r+1} = \dots = c_n = 0$.
lyche-numerical-linear-algebra-234-F03141	234	0.999	$\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^n$	Exercise 9.17 (Fredholm's Alternative) For any $\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^n$ show that
lyche-numerical-linear-algebra-234-F03142	234	1.000	$\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$	In other words either $\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$, or we can find $\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$ such that $\boldsymbol{y}^* \boldsymbol{b} \neq 0$. This
lyche-numerical-linear-algebra-234-F03143	234	1.000	$\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$	In other words either $\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$, or we can find $\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$ such that $\boldsymbol{y}^* \boldsymbol{b} \neq 0$. This
lyche-numerical-linear-algebra-234-F03144	234	1.000	$\boldsymbol{y}^* \boldsymbol{b} \neq 0$	$\boldsymbol{y}^* \boldsymbol{b} \neq 0$	In other words either $\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$, or we can find $\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$ such that $\boldsymbol{y}^* \boldsymbol{b} \neq 0$. This
lyche-numerical-linear-algebra-234-F03145	234	1.000	$\mathbb{C}^{(m-n) \times n}$	$\mathbb{C}^{(m-n) \times n}$	where $\boldsymbol{B}$ is a non-singular $n \times n$ matrix and $\boldsymbol{C}$ is in $\mathbb{C}^{(m-n) \times n}$. Let $\boldsymbol{A}^\dagger$ denote the
lyche-numerical-linear-algebra-234-F03146	234	1.000	$\ \boldsymbol{A}^\dagger\ _2 \leq \ \boldsymbol{B}^{-1}\ _2$	$\ \boldsymbol{A}^\dagger\ _2 \leq \ \boldsymbol{B}^{-1}\ _2$	pseudoinverse of $\boldsymbol{A}$. Show that $\ \boldsymbol{A}^\dagger\ _2 \leq \ \boldsymbol{B}^{-1}\ _2$.
lyche-numerical-linear-algebra-234-F03147	234	1.000	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^\dagger\ _2$	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^\dagger\ _2$	b) Compute $K(\boldsymbol{A}) = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^\dagger\ _2$.
lyche-numerical-linear-algebra-235-F03148	235	1.000	$\boldsymbol{y}_{\boldsymbol{A}^\dagger}$	$\boldsymbol{y}_{\boldsymbol{A}^\dagger}$	and $\boldsymbol{y}_{\boldsymbol{A}^\dagger}$ are vectors with $\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^\dagger}\ = 1$ and $\ \boldsymbol{A}\ = \ \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}\ $ and $\ \boldsymbol{A}^\dagger\ =$
lyche-numerical-linear-algebra-235-F03149	235	0.999	$\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^\dagger}\ = 1$	$\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^\dagger}\ = 1$	and $\boldsymbol{y}_{\boldsymbol{A}^\dagger}$ are vectors with $\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^\dagger}\ = 1$ and $\ \boldsymbol{A}\ = \ \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}\ $ and $\ \boldsymbol{A}^\dagger\ =$
lyche-numerical-linear-algebra-235-F03150	235	0.999	$\ \boldsymbol{A}^\dagger\ =$	$\ \boldsymbol{A}^\dagger\ =$	and $\boldsymbol{y}_{\boldsymbol{A}^\dagger}$ are vectors with $\ \boldsymbol{y}_{\boldsymbol{A}}\ = \ \boldsymbol{y}_{\boldsymbol{A}^\dagger}\ = 1$ and $\ \boldsymbol{A}\ = \ \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}\ $ and $\ \boldsymbol{A}^\dagger\ =$
lyche-numerical-linear-algebra-235-F03151	235	0.999	$\ \boldsymbol{A}^\dagger \boldsymbol{y}_{\boldsymbol{A}^\dagger}\ $	$\ \boldsymbol{A}^\dagger \boldsymbol{y}_{\boldsymbol{A}^\dagger}\ $	$\ \boldsymbol{A}^\dagger \boldsymbol{y}_{\boldsymbol{A}^\dagger}\ $.
lyche-numerical-linear-algebra-235-F03152	235	0.999	$\boldsymbol{b} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}, \boldsymbol{e}_1 = \boldsymbol{y}_{\boldsymbol{A}^\dagger}$	$\boldsymbol{b} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}, \boldsymbol{e}_1 = \boldsymbol{y}_{\boldsymbol{A}^\dagger}$	a) Show that we have equality to the right in (9.13) if $\boldsymbol{b} = \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}, \boldsymbol{e}_1 = \boldsymbol{y}_{\boldsymbol{A}^\dagger}$.
lyche-numerical-linear-algebra-235-F03153	235	1.000	$\boldsymbol{A}^\dagger = \boldsymbol{A}^{-1}$	$\boldsymbol{A}^\dagger = \boldsymbol{A}^{-1}$	nonsingular then $\boldsymbol{A}^\dagger = \boldsymbol{A}^{-1}$ and $\boldsymbol{e}_1 = \boldsymbol{e}$.
lyche-numerical-linear-algebra-235-F03154	235	1.000			

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lyche-numerical-linear-algebra-235-2023-01-15-155	235	1.000	$\boldsymbol{e}_1=\boldsymbol{e}$	$\boldsymbol{e}_1 = \boldsymbol{e}$	nonsingular then $\boldsymbol{A}^\dagger = \boldsymbol{A}^{-1}$ and $\boldsymbol{e}_1 = \boldsymbol{e}$.
lyche-numerical-linear-algebra-235-2023-01-15-156	235	1.000	(2,2)	(2,2)	b) Suppose ϵ is small and we replace the (2, 2) entry $3+2\epsilon+\epsilon^2$ in $\boldsymbol{A}^*\boldsymbol{A}$ by $3+2\epsilon$.
lyche-numerical-linear-algebra-235-2023-01-15-157	235	1.000	$3+2\ \epsilonpsilon+\epsilonpsilon^{\{2\}}$	$3 + 2\epsilon + \epsilon^2$	b) Suppose ϵ is small and we replace the (2, 2) entry $3+2\epsilon+\epsilon^2$ in $\boldsymbol{A}^*\boldsymbol{A}$ by $3+2\epsilon$.
lyche-numerical-linear-algebra-235-2023-01-15-158	235	1.000	$3+2\ \epsilonpsilon$	$3 + 2\epsilon$	b) Suppose ϵ is small and we replace the (2, 2) entry $3+2\epsilon+\epsilon^2$ in $\boldsymbol{A}^*\boldsymbol{A}$ by $3+2\epsilon$.
lyche-numerical-linear-algebra-235-2023-01-15-159	235	1.000	$\epsilonpsilon<\sqrt{u},\ u$	$\epsilon < \sqrt{u}, u$	(This will be done in a computer if $\epsilon < \sqrt{u}$, u being the round-off unit). For
lyche-numerical-linear-algebra-235-2023-01-15-160	235	1.000	$u{=}10^{\{-16\}}$	$u = 10^{-16}$	example, if $u = 10^{-16}$ then $\sqrt{u} = 10^{-8}$. Solve $\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^*\boldsymbol{b}$ for $\boldsymbol{x}$ and compare
lyche-numerical-linear-algebra-235-2023-01-15-161	235	1.000	$\sqrt{u}{=}10^{\{-8\}}$	$\sqrt{u} = 10^{-8}$	example, if $u = 10^{-16}$ then $\sqrt{u} = 10^{-8}$. Solve $\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^*\boldsymbol{b}$ for $\boldsymbol{x}$ and compare
lyche-numerical-linear-algebra-235-2023-01-15-162	235	1.000	$\boldsymbol{A}(\epsilonpsilon)\ \in\ \mathbb{R}^{\{n\ \times\ n\}}$	$\boldsymbol{A}(\epsilon) \in \mathbb{R}^{n \times n}$	$\boldsymbol{A}(\epsilon) \in \mathbb{R}^{n \times n}$ be bidiagonal with $a_{i,j} = 0$ for $i, j = 1, \dots, n$ and $j \neq i, i+1$.
lyche-numerical-linear-algebra-235-2023-01-15-163	235	1.000	$j\ \neq\ i,\ i+1$	$j \neq i, i+1$	$\boldsymbol{A}(\epsilon) \in \mathbb{R}^{n \times n}$ be bidiagonal with $a_{i,j} = 0$ for $i, j = 1, \dots, n$ and $j \neq i, i+1$.
lyche-numerical-linear-algebra-235-2023-01-15-164	235	1.000	$a_{\{k,\ k+1\}}=\epsilonpsilon\ \in\ \mathbb{R}$	$a_{k,k+1} = \epsilon \in \mathbb{R}$	Moreover, for some $1 \leq k \leq n-1$ we have $a_{k,k+1} = \epsilon \in \mathbb{R}$. Show that
lyche-numerical-linear-algebra-235-2023-01-15-165	235	1.000	$\sigma_{i}(\epsilonpsilon),\ i{=}1,\ \ldots,\ n$	$\sigma_i(\epsilon), i = 1, \dots, n$	where $\sigma_i(\epsilon), i = 1, \dots, n$ are the singular values of $\boldsymbol{A}(\epsilon)$.
lyche-numerical-linear-algebra-235-2023-01-15-166	235	1.000	$\boldsymbol{A}(\epsilonpsilon)$	$\boldsymbol{A}(\epsilon)$	where $\sigma_i(\epsilon), i = 1, \dots, n$ are the singular values of $\boldsymbol{A}(\epsilon)$.
lyche-numerical-linear-algebra-238-2023-01-15-167	238	1.000	$\Omega{:=}(0,1)^{\{2\}}{=}\{(x,\ y)\colon\ 0<x,\ y<1\}$	$\Omega := (0,1)^2 = \{(x,y) : 0 < x, y < 1\}$	Let $\Omega := (0,1)^2 = \{(x,y) : 0 < x, y < 1\}$ be the open unit square with boundary
lyche-numerical-linear-algebra-238-2023-01-15-168	238	1.000	$\partial\Omega$	$\partial\Omega$	$\partial\Omega$. Consider the problem
lyche-numerical-linear-algebra-238-2023-01-15-169	238	1.000	$u{=}$	$u =$	Here the function f is given and continuous on Ω , and we seek a function $u =$
lyche-numerical-linear-algebra-238-2023-01-15-170	238	1.000	$u(x,\ y)$	$u(x,y)$	$u(x,y)$ such that (10.1) holds and which is zero on $\partial\Omega$.
lyche-numerical-linear-algebra-238-2023-01-15-171	238	1.000	$v_{\{j,\ k\}}\ \approx\ u(j\ h,\ k\ h)$	$v_{j,k} \approx u(jh, kh)$	approximations $v_{j,k} \approx u(jh, kh)$ on a grid of points given by
lyche-numerical-linear-algebra-238-2023-01-15-172	238	1.000	$\Omega_h{:=}\{(j\ h,\ k\ h)\colon\ j,\ k{=}1,\ \ldots,\ m\}$	$\Omega_h := \{(jh, kh) : j, k = 1, \dots, m\}$	The points $\Omega_h := \{(jh, kh) : j, k = 1, \dots, m\}$ are called the interior points, while
lyche-numerical-linear-algebra-238-2023-01-15-173	238	1.000	$\bar{\Omega}_h\backslash\Omega_h$	$\bar{\Omega}_h \setminus \Omega_h$	$\bar{\Omega}_h \setminus \Omega_h$ are the boundary points. The solution is zero at the boundary points. Using
lyche-numerical-linear-algebra-239-2023-01-15-174	239	1.000	$f_{\{j,\ k\}}{:=}f(j\ h,\ k\ h)$	$f_{j,k} := f(jh, kh)$	where $f_{j,k} := f(jh, kh)$ and
lyche-numerical-linear-algebra-239-2023-01-15-175	239	1.000	$n{:=}m^{\{2\}}$	$n := m^2$	Let us take a closer look at (10.2). It consists of $n := m^2$ linear equations. Since the
lyche-numerical-linear-algebra-239-2023-01-15-176	239	1.000	$v_{\{j,\ k\}}$	$v_{j,k}$	values at the boundary points are known, the unknowns are the n numbers $v_{j,k}$ at
lyche-numerical-linear-algebra-239-2023-01-15-177	239	1.000	$\boldsymbol{T}{=}\operatorname{tridiag}(-1,2,-1)\ \in\ \mathbb{R}^{\{m\ \times\ m\}}$	$\boldsymbol{T} = \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$	where $\boldsymbol{T} = \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ is the second derivative matrix given by
lyche-numerical-linear-algebra-239-2023-01-15-178	239	0.913	$j,\ k$	j, k	Indeed, the (j, k) element in $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T}$ is given by
lyche-numerical-linear-algebra-239-2023-01-15-179	239	0.913	$\boldsymbol{T}\ \boldsymbol{V}+\boldsymbol{V}\ \boldsymbol{T}$	$\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T}$	Indeed, the (j, k) element in $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T}$ is given by
lyche-numerical-linear-algebra-239-2023-01-15-180	239	1.000	$\boldsymbol{B}\ \in\ \mathbb{R}^{\{m\ \times\ n\}}$	$\boldsymbol{B} \in \mathbb{R}^{m \times n}$	Definition 10.1 (vec Operation) For any $\boldsymbol{B} \in \mathbb{R}^{m \times n}$ we define the vector
lyche-numerical-linear-algebra-240-2023-01-15-181	240	1.000	$\boldsymbol{x}{:=}\operatorname{vec}(\boldsymbol{V})\ \in\ \mathbb{R}^{\{n\}}$	$\boldsymbol{x} := \operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^n$	Let $\boldsymbol{x} := \operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^n$, where $n = m^2$. Note that forming $\boldsymbol{x}$ by stacking the

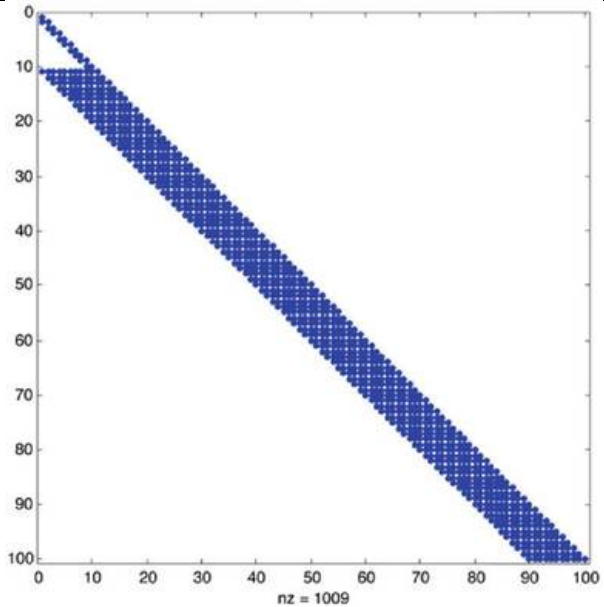
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lyche-numerical-linear-algebra- F03182	240	■ 1.000	$n=m^2$	$n = m^2$	Let $\mathbf{x} := \text{vec}(\mathbf{V}) \in \mathbb{R}^n$, where $n = m^2$. Note that forming $\mathbf{x}$ by stacking the
lyche-numerical-linear-algebra- F03183	240	■ 1.000	$x_{\{i\}}=v_{\{j, k\}}$	$x_i = v_{j,k}$	where $x_i = v_{j,k}$ and $b_i = h^2 f_{j,k}$. This must be modified close to the boundary. We
lyche-numerical-linear-algebra- F03184	240	■ 1.000	$b_{\{i\}}=h^2 f_{\{j, k\}}$	$b_i = h^2 f_{j,k}$	where $x_i = v_{j,k}$ and $b_i = h^2 f_{j,k}$. This must be modified close to the boundary. We
lyche-numerical-linear-algebra- F03185	240	■ 1.000	$\boldsymbol{x}=\operatorname{vec}(\boldsymbol{V}),$ $\boldsymbol{b}=h^2 \operatorname{vec}(\boldsymbol{F})$	$\mathbf{x} = \text{vec}(\mathbf{V}), \mathbf{b} = h^2 \text{vec}(\mathbf{F})$	where $\mathbf{x} = \text{vec}(\mathbf{V}), \mathbf{b} = h^2 \text{vec}(\mathbf{F})$ and $\mathbf{A}$ is the Poisson matrix given by
lyche-numerical-linear-algebra- F03186	241	■ 0.905	$\boldsymbol{T}_1 \in \mathbb{R}^{m \times m}$	$\mathbf{T}_1 \in \mathbb{R}^{m \times m}$	In Sect. 2.4 we encountered the 1-dimensional test matrix $\mathbf{T}_1 \in \mathbb{R}^{m \times m}$ defined for
lyche-numerical-linear-algebra- F03187	241	■ 1.000	a, d	a, d	any real numbers a, d by
lyche-numerical-linear-algebra- F03188	241	■ 1.000	$\boldsymbol{T}_2=\left[a_{i j}\right] \in$	$\mathbf{T}_2 = [a_{ij}] \in$	The (2-dimensional) Poisson matrix is a special case of the matrix $\mathbf{T}_2 = [a_{ij}] \in$
lyche-numerical-linear-algebra- F03189	242	■ 0.956	$\boldsymbol{T}_2$	$\mathbf{T}_2$	The partition into 3×3 sub matrices shows that $\mathbf{T}_2$ is block tridiagonal.
lyche-numerical-linear-algebra- F03190	242	■ 1.000	p, q, r, s	p, q, r, s	Definition 10.2 (Kronecker Product) For any positive integers p, q, r, s we
lyche-numerical-linear-algebra- F03191	242	■ 1.000	$\boldsymbol{A} \in \mathbb{R}^{p \times q}$	$\mathbf{A} \in \mathbb{R}^{p \times q}$	define the Kronecker product of two matrices $\mathbf{A} \in \mathbb{R}^{p \times q}$ and $\mathbf{B} \in \mathbb{R}^{r \times s}$ as a
lyche-numerical-linear-algebra- F03192	242	■ 1.000	$\boldsymbol{B} \in \mathbb{R}^{r \times s}$	$\mathbf{B} \in \mathbb{R}^{r \times s}$	define the Kronecker product of two matrices $\mathbf{A} \in \mathbb{R}^{p \times q}$ and $\mathbf{B} \in \mathbb{R}^{r \times s}$ as a
lyche-numerical-linear-algebra- F03193	242	■ 1.000	$\boldsymbol{C} \in \mathbb{R}^{p r \times q s}$	$\mathbf{C} \in \mathbb{R}^{pr \times qs}$	matrix $\mathbf{C} \in \mathbb{R}^{pr \times qs}$ given in block form as
lyche-numerical-linear-algebra- F03194	242	■ 1.000	$\boldsymbol{C}=\boldsymbol{A} \otimes \boldsymbol{B}$	$\mathbf{C} = \mathbf{A} \otimes \mathbf{B}$	We denote the Kronecker product of $\mathbf{A}$ and $\mathbf{B}$ by $\mathbf{C} = \mathbf{A} \otimes \mathbf{B}$.
lyche-numerical-linear-algebra- F03195	242	■ 1.000	$\boldsymbol{B} \otimes \boldsymbol{A}$	$\mathbf{B} \otimes \mathbf{A}$	which in our notation is given by $\mathbf{B} \otimes \mathbf{A}$.
lyche-numerical-linear-algebra- F03196	242	■ 1.000	$\boldsymbol{u} \otimes \boldsymbol{v}=\left[\boldsymbol{u}^T v_1, \ldots, \boldsymbol{u}^T v_r\right]^T$	$\mathbf{u} \otimes \mathbf{v} = [\mathbf{u}^T v_1, \dots, \mathbf{u}^T v_r]^T$	The Kronecker product $\mathbf{u} \otimes \mathbf{v} = [\mathbf{u}^T v_1, \dots, \mathbf{u}^T v_r]^T$ of two column vectors
lyche-numerical-linear-algebra- F03197	242	■ 1.000	$\boldsymbol{u} \in \mathbb{R}^p$	$\mathbf{u} \in \mathbb{R}^p$	$\mathbf{u} \in \mathbb{R}^p$ and $\mathbf{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra- F03198	242	■ 1.000	$\boldsymbol{v} \in \mathbb{R}^r$	$\mathbf{v} \in \mathbb{R}^r$	$\mathbf{u} \in \mathbb{R}^p$ and $\mathbf{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra- F03199	242	■ 1.000	$p \cdot r$	$p \cdot r$	$\mathbf{u} \in \mathbb{R}^p$ and $\mathbf{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra- F03200	243	■ 1.000	r, s, k	r, s, k	Definition 10.3 (Kronecker Sum) For positive integers r, s, k , let $\mathbf{A} \in \mathbb{R}^{r \times r}, \mathbf{B} \in$
lyche-numerical-linear-algebra- F03201	243	■ 1.000	$\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in$	$\mathbf{A} \in \mathbb{R}^{r \times r}, \mathbf{B} \in$	Definition 10.3 (Kronecker Sum) For positive integers r, s, k , let $\mathbf{A} \in \mathbb{R}^{r \times r}, \mathbf{B} \in$
lyche-numerical-linear-algebra- F03202	243	■ 1.000	$\mathbb{R}^{s \times s}$	$\mathbb{R}^{s \times s}$	$\mathbb{R}^{s \times s}$, and $\mathbf{I}_k$ be the identity matrix of order k . The sum $\mathbf{A} \otimes \mathbf{I}_s + \mathbf{I}_r \otimes \mathbf{B}$ is known
lyche-numerical-linear-algebra- F03203	243	■ 1.000	$\boldsymbol{I}_k$	$\mathbf{I}_k$	$\mathbb{R}^{s \times s}$, and $\mathbf{I}_k$ be the identity matrix of order k . The sum $\mathbf{A} \otimes \mathbf{I}_s + \mathbf{I}_r \otimes \mathbf{B}$ is known
lyche-numerical-linear-algebra- F03204	243	■ 1.000	$\boldsymbol{A} \otimes \boldsymbol{I}_s+\boldsymbol{I}_r \otimes \boldsymbol{B}$	$\mathbf{A} \otimes \mathbf{I}_s + \mathbf{I}_r \otimes \mathbf{B}$	$\mathbb{R}^{s \times s}$, and $\mathbf{I}_k$ be the identity matrix of order k . The sum $\mathbf{A} \otimes \mathbf{I}_s + \mathbf{I}_r \otimes \mathbf{B}$ is known
lyche-numerical-linear-algebra- F03205	243	■ 1.000	λ, μ	λ, μ	λ, μ and matrices $\mathbf{A}, \mathbf{A}_1, \mathbf{A}_2, \mathbf{B}, \mathbf{B}_1, \mathbf{B}_2, \mathbf{C}$ of dimensions such that the operations
lyche-numerical-linear-algebra- F03206	243	■ 1.000	$\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$	$\mathbf{A}, \mathbf{A}_1, \mathbf{A}_2, \mathbf{B}, \mathbf{B}_1, \mathbf{B}_2, \mathbf{C}$	λ, μ and matrices $\mathbf{A}, \mathbf{A}_1, \mathbf{A}_2, \mathbf{B}, \mathbf{B}_1, \mathbf{B}_2, \mathbf{C}$ of dimensions such that the operations
lyche-numerical-linear-algebra- F03207	243	■ 1.000	$\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}$	$\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}$	Note however that in general we have $\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}$, but it can be shown that

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lyche-numerical-linear-algebra- F03208	243	■ 1.000	$\boldsymbol{P}, \boldsymbol{Q}$	$\boldsymbol{P}, \boldsymbol{Q}$	there are permutation matrices $\boldsymbol{P}, \boldsymbol{Q}$ such that $\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$, see [9].
lyche-numerical-linear-algebra- F03209	243	■ 1.000	$\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$	$\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$	there are permutation matrices $\boldsymbol{P}, \boldsymbol{Q}$ such that $\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$, see [9].
lyche-numerical-linear-algebra- F03210	243	■ 0.989	$\boldsymbol{B} \boldsymbol{D}$	$\boldsymbol{B} \boldsymbol{D}$	ces with dimensions so that the products $\boldsymbol{A} \boldsymbol{C}$ and $\boldsymbol{B} \boldsymbol{D}$ are well defined. Then the
lyche-numerical-linear-algebra- F03211	243	■ 1.000	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$	product $\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$ is defined and
lyche-numerical-linear-algebra- F03212	243	■ 1.000	$\boldsymbol{B} \in \mathbb{R}^{r, t}$	$\boldsymbol{B} \in \mathbb{R}^{r, t}$	Proof If $\boldsymbol{B} \in \mathbb{R}^{r, t}$ and $\boldsymbol{D} \in \mathbb{R}^{t, s}$ for some integers r, s, t , then
lyche-numerical-linear-algebra- F03213	243	■ 1.000	$\boldsymbol{D} \in \mathbb{R}^{t, s}$	$\boldsymbol{D} \in \mathbb{R}^{t, s}$	Proof If $\boldsymbol{B} \in \mathbb{R}^{r, t}$ and $\boldsymbol{D} \in \mathbb{R}^{t, s}$ for some integers r, s, t , then
lyche-numerical-linear-algebra- F03214	243	■ 1.000	r, s, t	r, s, t	Proof If $\boldsymbol{B} \in \mathbb{R}^{r, t}$ and $\boldsymbol{D} \in \mathbb{R}^{t, s}$ for some integers r, s, t , then
lyche-numerical-linear-algebra- F03215	244	■ 1.000	$r, s \in \mathbb{N}$	$r, s \in \mathbb{N}$	Theorem 10.1 (Properties of Kronecker Products) Suppose for $r, s \in \mathbb{N}$ that
lyche-numerical-linear-algebra- F03216	244	■ 1.000	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$ and $\boldsymbol{B} \in \mathbb{R}^{s \times s}$ are square matrices with eigenpairs $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$
lyche-numerical-linear-algebra- F03217	244	■ 1.000	$\boldsymbol{B} \in \mathbb{R}^{s \times s}$	$\boldsymbol{B} \in \mathbb{R}^{s \times s}$	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$ and $\boldsymbol{B} \in \mathbb{R}^{s \times s}$ are square matrices with eigenpairs $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$
lyche-numerical-linear-algebra- F03218	244	■ 1.000	$(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$	$(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$ and $\boldsymbol{B} \in \mathbb{R}^{s \times s}$ are square matrices with eigenpairs $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$
lyche-numerical-linear-algebra- F03219	244	■ 1.000	$(\mu_j, \boldsymbol{v}_j)$ $j = 1, \dots, s$	$(\mu_j, \boldsymbol{v}_j)$ $j = 1, \dots, s$	and $(\mu_j, \boldsymbol{v}_j)$ $j = 1, \dots, s$. Moreover, let $\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r \times s}$. Then
lyche-numerical-linear-algebra- F03220	244	■ 1.000	$\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r \times s}$	$\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r \times s}$	and $(\mu_j, \boldsymbol{v}_j)$ $j = 1, \dots, s$. Moreover, let $\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r \times s}$. Then
lyche-numerical-linear-algebra- F03221	244	■ 0.999	$(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T$	$(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T$	1. $\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T$, (this also holds for rectangular matrices).
lyche-numerical-linear-algebra- F03222	244	■ 0.999	$\boldsymbol{A} \otimes \boldsymbol{B}$	$\boldsymbol{A} \otimes \boldsymbol{B}$	2. If $\boldsymbol{A}$ and $\boldsymbol{B}$ are nonsingular then $\boldsymbol{A} \otimes \boldsymbol{B}$ is nonsingular. with $(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$
lyche-numerical-linear-algebra- F03223	244	■ 0.999	$(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$	$(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$	2. If $\boldsymbol{A}$ and $\boldsymbol{B}$ are nonsingular then $\boldsymbol{A} \otimes \boldsymbol{B}$ is nonsingular. with $(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$
lyche-numerical-linear-algebra- F03224	244	■ 1.000	$\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}$	$\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}$	$\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}$.
lyche-numerical-linear-algebra- F03225	244	■ 1.000	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$	4. $(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$,
lyche-numerical-linear-algebra- F03226	244	■ 0.999	$(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\lambda_i + \mu_j)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\lambda_i + \mu_j)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$	5. $(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\lambda_i + \mu_j)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$,
lyche-numerical-linear-algebra- F03227	244	■ 1.000	$\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$	$\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$	$\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$ is positive definite.
lyche-numerical-linear-algebra- F03228	244	■ 0.689	$\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$	$\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$	7. $\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F03229	244	■ 1.000	$\boldsymbol{A} \boldsymbol{V} = \boldsymbol{B}^T \boldsymbol{F}$ $\boldsymbol{B}^T = \boldsymbol{F} \quad \Leftrightarrow$ $\left(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B} \right) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$	$\boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F} \quad \Leftrightarrow \quad (\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$	8. $\boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F} \quad \Leftrightarrow \quad (\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03230	244	■ 1.000	$\boldsymbol{A} = \boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}$	$\boldsymbol{A} = \boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}$	Poisson matrix $\boldsymbol{A} = \boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}$ is also positive definite.
lyche-numerical-linear-algebra- F03231	244	■ 1.000	$\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}$	$\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}$	4. The system $\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}$ in part 7 can be solved by first finding $\boldsymbol{W}$ from $\boldsymbol{A} \boldsymbol{W} =$
lyche-numerical-linear-algebra- F03232	244	■ 1.000	$\boldsymbol{A} \boldsymbol{W} =$	$\boldsymbol{A} \boldsymbol{W} =$	4. The system $\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}$ in part 7 can be solved by first finding $\boldsymbol{W}$ from $\boldsymbol{A} \boldsymbol{W} =$
lyche-numerical-linear-algebra- F03233	244	■ 1.000	$\boldsymbol{B} \boldsymbol{V}^T = \boldsymbol{W}^T$	$\boldsymbol{B} \boldsymbol{V}^T = \boldsymbol{W}^T$	$\boldsymbol{F}$, and then finding $\boldsymbol{V}$ from $\boldsymbol{B} \boldsymbol{V}^T = \boldsymbol{W}^T$. This is preferable to solving the much
lyche-numerical-linear-algebra- F03234	244	■ 0.544	$(\boldsymbol{A} \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$	$(\boldsymbol{A} \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$	larger linear system $(\boldsymbol{A} \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03235	244	■ 0.965	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{F}$	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{F}$	5. A fast way to solve the 2D Poisson problem in the form $\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{F}$ will be
lyche-numerical-linear-algebra- F03236	245	■ 0.998	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}) = (\boldsymbol{A} \boldsymbol{A}^{-1}) \otimes (\boldsymbol{B} \boldsymbol{B}^{-1}) =$ $\left(\boldsymbol{A}^{-1} \otimes \boldsymbol{B} \right) \left(\boldsymbol{A} \otimes \boldsymbol{B}^{-1} \right) =$ $\left(\boldsymbol{A} \otimes \boldsymbol{B}^{-1} \right) \left(\boldsymbol{A}^{-1} \otimes \boldsymbol{B} \right) =$ $\left(\boldsymbol{A} \otimes \boldsymbol{B} \right) (\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}) = (\boldsymbol{A} \boldsymbol{A}^{-1}) \otimes (\boldsymbol{B} \boldsymbol{B}^{-1}) =$	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}) = (\boldsymbol{A} \boldsymbol{A}^{-1}) \otimes (\boldsymbol{B} \boldsymbol{B}^{-1}) =$	2. By the mixed product rule $(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}) = (\boldsymbol{A} \boldsymbol{A}^{-1}) \otimes (\boldsymbol{B} \boldsymbol{B}^{-1}) =$
lyche-numerical-linear-algebra- F03237	245	■ 0.997	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$. Thus $(\boldsymbol{A} \otimes \boldsymbol{B})$ is nonsingular with the indicated inverse.
lyche-numerical-linear-algebra- F03238	245	■ 0.997	$(\boldsymbol{A} \otimes \boldsymbol{B})$	$(\boldsymbol{A} \otimes \boldsymbol{B})$	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$. Thus $(\boldsymbol{A} \otimes \boldsymbol{B})$ is nonsingular with the indicated inverse.
lyche-numerical-linear-algebra- F03239	245	■ 1.000	$1, (\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$	$1, (\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra- F03240	245	■ 1.000	$\boldsymbol{A} \otimes \boldsymbol{I}$	$\boldsymbol{A} \otimes \boldsymbol{I}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra- F03241	245	■ 1.000	$\boldsymbol{I} \otimes \boldsymbol{B}$	$\boldsymbol{I} \otimes \boldsymbol{B}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra- F03242	245	■ 1.000	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\boldsymbol{A} \boldsymbol{u}_i) \otimes (\boldsymbol{B} \boldsymbol{v}_j) = (\lambda_i \boldsymbol{u}_i) \otimes (\mu_j \boldsymbol{v}_j) = (\lambda_i \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\boldsymbol{A} \boldsymbol{u}_i) \otimes (\boldsymbol{B} \boldsymbol{v}_j) = (\lambda_i \boldsymbol{u}_i) \otimes (\mu_j \boldsymbol{v}_j) = (\lambda_i \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	4. $(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\boldsymbol{A} \boldsymbol{u}_i) \otimes (\boldsymbol{B} \boldsymbol{v}_j) = (\lambda_i \boldsymbol{u}_i) \otimes (\mu_j \boldsymbol{v}_j) = (\lambda_i \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, for
lyche-numerical-linear-algebra- F03243	245	■ 0.475	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_s \right) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$ $\left(\boldsymbol{I}_r \otimes \boldsymbol{B} \right) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j).$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$	5. $(\boldsymbol{A} \otimes \boldsymbol{I}_s) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j).$
lyche-numerical-linear-algebra- F03244	245	■ 0.475	$\left(\boldsymbol{I}_r \otimes \boldsymbol{B} \right) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	$(\boldsymbol{I}_r \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	5. $(\boldsymbol{A} \otimes \boldsymbol{I}_s) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j).$
lyche-numerical-linear-algebra- F03245	245	■ 0.892	$1, \boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$	$1, \boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$	6. By $\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$ is symmetric. Moreover, the eigenvalues $\lambda_i + \mu_j$ are positive
lyche-numerical-linear-algebra- F03246	245	■ 0.892	$\lambda_i + \mu_j$	$\lambda_i + \mu_j$	6. By 1, $\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$ is symmetric. Moreover, the eigenvalues $\lambda_i + \mu_j$ are positive
lyche-numerical-linear-algebra- F03247	245	■ 0.500	μ_j	μ_j	since for all i, j , both λ_i and μ_j are nonnegative and one of them is positive. It

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lyche-numerical-linear-algebra-F03248	245	1.000	$\boldsymbol{V}, \boldsymbol{F}$	$\boldsymbol{V}, \boldsymbol{F}$	7. We partition $\boldsymbol{V}, \boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra-F03249	245	1.000	$\boldsymbol{B}^T$	$\boldsymbol{B}^T$	7. We partition $\boldsymbol{V}, \boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra-F03250	245	1.000	$\boldsymbol{V} = \left[\boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{v}_{\{s\}} \right], \boldsymbol{F} = \left[\boldsymbol{f}_{\{1\}}, \ldots, \boldsymbol{f}_{\{s\}} \right]$	$\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$	7. We partition $\boldsymbol{V}, \boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra-F03251	245	1.000	$\boldsymbol{B}^T = \left[\boldsymbol{b}_{\{1\}}, \ldots, \boldsymbol{b}_{\{s\}} \right]$	$\boldsymbol{B}^T = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_s]$	and $\boldsymbol{B}^T = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_s]$. Then we have
lyche-numerical-linear-algebra-F03252	245	0.890	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_s \right) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$	8. From the proof of 7. we have $(\boldsymbol{A} \otimes \boldsymbol{I}_s) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) =$
lyche-numerical-linear-algebra-F03253	245	0.890	$\left(\boldsymbol{I}_r \otimes \boldsymbol{B} \right) \operatorname{vec}(\boldsymbol{V}) =$	$(\boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) =$	8. From the proof of 7. we have $(\boldsymbol{A} \otimes \boldsymbol{I}_s) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) =$
lyche-numerical-linear-algebra-F03254	245	1.000	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T$	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T$	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T$. But then
lyche-numerical-linear-algebra-F03255	246	0.527	$(\lambda_j, \boldsymbol{s}_j)$	$(\lambda_j, \boldsymbol{s}_j)$	where $(\lambda_j, \boldsymbol{s}_j)$ are the eigenpairs of $\boldsymbol{T}_1$ given by (10.15) and (10.16). The
lyche-numerical-linear-algebra-F03256	246	0.975	$\boldsymbol{s}_j \otimes \boldsymbol{s}_k$	$\boldsymbol{s}_j \otimes \boldsymbol{s}_k$	eigenvectors $\boldsymbol{s}_j \otimes \boldsymbol{s}_k$ are orthogonal
lyche-numerical-linear-algebra-F03257	246	1.000	$d > 0$	$d > 0$	and $\boldsymbol{T}_2$ is positive definite if $d > 0$ and $d \geq 2 a $.
lyche-numerical-linear-algebra-F03258	246	1.000	$d \geq 2 a $	$d \geq 2 a $	and $\boldsymbol{T}_2$ is positive definite if $d > 0$ and $d \geq 2 a $.
lyche-numerical-linear-algebra-F03259	246	1.000	$j, k, p, q = 1, \dots, m$	$j, k, p, q = 1, \dots, m$	rule, the mixed product rule and (2.32) we find for $j, k, p, q = 1, \dots, m$
lyche-numerical-linear-algebra-F03260	246	1.000	λ_j	λ_j	and (10.19) follows. Since $\boldsymbol{T}_2$ is symmetric, $\boldsymbol{T}_2$ is positive definite if the λ_j given
lyche-numerical-linear-algebra-F03261	246	0.990	$\mathbb{R}^{m^2 \times m^2}$	$\mathbb{R}^{m^2 \times m^2}$	$\mathbb{R}^{m^2 \times m^2}$ given by (10.8) is given by
lyche-numerical-linear-algebra-F03262	246	1.000	$d=2, a=-1$	$d = 2, a = -1$	Proof Recall that by (10.15) with $d = 2, a = -1$, and (10.18), the eigenvalues $\lambda_{j,k}$
lyche-numerical-linear-algebra-F03263	246	1.000	$\lambda_{j,k}$	$\lambda_{j,k}$	Proof Recall that by (10.15) with $d = 2, a = -1$, and (10.18), the eigenvalues $\lambda_{j,k}$
lyche-numerical-linear-algebra-F03264	247	0.967	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\lambda_{\max}}{\lambda_{\min}}$	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\lambda_{\max}}{\lambda_{\min}}$	$\ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2 = \frac{\lambda_{\max}}{\lambda_{\min}}$ and (10.20) follows.
lyche-numerical-linear-algebra-F03265	247	1.000	$\lambda = 2a + 2d$	$\lambda = 2a + 2d$	Show that $\lambda = 2a + 2d$ is an eigenvalue corresponding to the eigenvector $\boldsymbol{x} =$
lyche-numerical-linear-algebra-F03266	247	1.000	$[1, 1, 1, 1]^T$	$[1, 1, 1, 1]^T$	$[1, 1, 1, 1]^T$. Verify that apart from a scaling of the eigenvector this agrees with
lyche-numerical-linear-algebra-F03267	247	1.000	$j=k=1$	$j = k = 1$	(10.15) and (10.16) for $j = k =$ and $m = 2$.
lyche-numerical-linear-algebra-F03268	247	1.000	$-\Delta u = f, u = 0$	$-\Delta u = f, u = 0$	9 point difference approximation to the Poisson problem $-\Delta u = f, u = 0$ on the
lyche-numerical-linear-algebra-F03269	248	1.000	$-(\square_h v)_{j,k} = (\mu f)_{j,k}$	$-(\square_h v)_{j,k} = (\mu f)_{j,k}$	(a) $-(\square_h v)_{j,k} = (\mu f)_{j,k}$
lyche-numerical-linear-algebra-F03270	248	1.000	$j, k = 1, \dots, m$	$j, k = 1, \dots, m$	$j, k = 1, \dots, m$
lyche-numerical-linear-algebra-F03271	248	1.000	$0 = v_{0,k} = v_{m+1,k} = v_{j,0} = v_{j,m+1}, j, k = 0, 1, \dots, m+1$	$0 = v_{0,k} = v_{m+1,k} = v_{j,0} = v_{j,m+1}, j, k = 0, 1, \dots, m+1$	$0 = v_{0,k} = v_{m+1,k} = v_{j,0} = v_{j,m+1}, j, k = 0, 1, \dots, m+1$,
lyche-numerical-linear-algebra-F03272	248	1.000	$-(\square_h v)_{j,k} = [20v_{j,k} - 4v_{j-1,k} - 4v_{j,k-1} - 4v_{j+1,k} - 4v_{j,k+1}]$	$-(\square_h v)_{j,k} = [20v_{j,k} - 4v_{j-1,k} - 4v_{j,k-1} - 4v_{j+1,k} - 4v_{j,k+1}]$	$-(\square_h v)_{j,k} = [20v_{j,k} - 4v_{j-1,k} - 4v_{j,k-1} - 4v_{j+1,k} - 4v_{j,k+1}]$

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lyche-numerical-linear-algebra- F03273	248	1.000	$\left.-v_{j-1, k-1}-v_{j+1, k-1}-v_{j-1, k+1}-v_{j+1, k+1}\right] /\left(6 h^2\right)$	$-v_{j-1, k-1}-v_{j+1, k-1}-v_{j-1, k+1}-v_{j+1, k+1}] / \left(6 h^2\right)$	$-v_{i-1, k-1}-v_{i+1, k-1}-v_{i-1, k+1}-v_{i+1, k+1}] / \left(6 h^2\right)$.
lyche-numerical-linear-algebra- F03274	248	0.877	$\left(\mu f\right)_{j, k}=\left[8 f_{j, k}+f_{j-1, k}+f_{j, k-1}+f_{j+1, k}+f_{j, k+1}\right] / 12$	$(\mu f)_{j, k}=\left[8 f_{j, k}+f_{j-1, k}+f_{j, k-1}+f_{j+1, k}+f_{j, k+1}\right] / 12$	$(\mu f)_{i, k}=\left[8 f_{i, k}+f_{i-1, k}+f_{i, k-1}+f_{i+1, k}+f_{i, k+1}\right] / 12$.
lyche-numerical-linear-algebra- F03275	248	1.000	$j, k=1,2$	$j, k=1,2$	b) Find $v_{j, k}$ for $j, k=1,2$, if $f(x, y)=2 \pi^2 \sin (\pi x) \sin (\pi y)$ and $m=2$.
lyche-numerical-linear-algebra- F03276	248	1.000	$f(x, y)=2 \pi^2 \sin (\pi x) \sin (\pi y)$	$f(x, y)=2 \pi^2 \sin (\pi x) \sin (\pi y)$	b) Find $v_{j, k}$ for $j, k=1,2$, if $f(x, y)=2 \pi^2 \sin (\pi x) \sin (\pi y)$ and $m=2$.
lyche-numerical-linear-algebra- F03277	248	1.000	$v_{j, k}=5 \pi^2 / 66$	$v_{j, k}=5 \pi^2 / 66$	Answer: $v_{i, k}=5 \pi^2 / 66$.
lyche-numerical-linear-algebra- F03278	248	1.000	$O\left(h^4\right)$	$O\left(h^4\right)$	It can be shown that (10.21) defines an $O\left(h^4\right)$ approximation to (10.1).
lyche-numerical-linear-algebra- F03279	248	0.997	$\mu \boldsymbol{F}$	$\mu \boldsymbol{F}$	Here $\mu \boldsymbol{F}$ has elements $(\mu f)_{j, k}$ given by (10.21d) and $\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1)$.
lyche-numerical-linear-algebra- F03280	248	0.997	$(\mu f)_{j, k}$	$(\mu f)_{j, k}$	Here $\mu \boldsymbol{F}$ has elements $(\mu f)_{j, k}$ given by (10.21d) and $\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1)$.
lyche-numerical-linear-algebra- F03281	248	1.000	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{V})$	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{V})$	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{V})$, and $\boldsymbol{b}=h^2 \operatorname{vec}(\mu \boldsymbol{F})$.
lyche-numerical-linear-algebra- F03282	248	1.000	$\boldsymbol{b}=h^2 \operatorname{vec}(\mu \boldsymbol{F})$	$\boldsymbol{b}=h^2 \operatorname{vec}(\mu \boldsymbol{F})$	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{V})$, and $\boldsymbol{b}=h^2 \operatorname{vec}(\mu \boldsymbol{F})$.
lyche-numerical-linear-algebra- F03283	248	1.000	$\Delta u=0$	$\Delta u=0$	Here Ω is the open unit square. The condition $\Delta u=0$ is called the <i>Navier boundary</i>
lyche-numerical-linear-algebra- F03284	248	0.994	$\Delta^2 u=u_{x x x x}+2 u_{x x y y}+u_{y y y y}$	$\Delta^2 u=u_{x x x x}+2 u_{x x y y}+u_{y y y y}$	condition. Moreover, $\Delta^2 u=u_{x x x x}+2 u_{x x y y}+u_{y y y y}$.
lyche-numerical-linear-algebra- F03285	248	1.000	$v=-\Delta u$	$v=-\Delta u$	a) Let $v=-\Delta u$. Show that (10.23) can be written as a system
lyche-numerical-linear-algebra- F03286	248	1.000	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+1)$	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+1)$	b) Discretizing, using (10.4), with $\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+1)$
lyche-numerical-linear-algebra- F03287	248	0.584	$\boldsymbol{F}=\left(f(j h, k h)\right)_{j, k=1}^m$	$\boldsymbol{F}=\left(f(j h, k h)\right)_{j, k=1}^m$	1), and $\boldsymbol{F}=\left(f(j h, k h)\right)_{j, k=1}^m$ we get two matrix equations
lyche-numerical-linear-algebra- F03288	249	1.000	$\boldsymbol{A}=\left(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}\right)^2$	$\boldsymbol{A}=\left(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}\right)^2$	and hence $\boldsymbol{A}=\left(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}\right)^2$ is the matrix for the standard form of the
lyche-numerical-linear-algebra- F03289	249	0.999	$\boldsymbol{A}=\boldsymbol{T}^2 \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{U})$	$\boldsymbol{A}=\boldsymbol{T}^2 \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{U})$	where $\boldsymbol{A}=\boldsymbol{T}^2 \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{U})$, and $\boldsymbol{b}=h^4 \operatorname{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03290	249	0.999	$\boldsymbol{b}=h^4 \operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{b}=h^4 \operatorname{vec}(\boldsymbol{F})$	where $\boldsymbol{A}=\boldsymbol{T}^2 \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{U})$, and $\boldsymbol{b}=h^4 \operatorname{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03291	249	0.831	$\mathbf{b}$	$\mathbf{b}$	representations for the matrix $\boldsymbol{A}$, the product one in b) and the one in c) . Which
lyche-numerical-linear-algebra- F03292	249	0.831	$\mathbf{c}$	$\mathbf{c}$	representations for the matrix $\boldsymbol{A}$, the product one in b) and the one in c) . Which
lyche-numerical-linear-algebra- F03293	249	1.000	$\boldsymbol{T}_1:=\operatorname{tridiagonal}(a, d, a)$	$\boldsymbol{T}_1:=\operatorname{tridiagonal}(a, d, a)$	10.5.2 What are the eigenpairs of $\boldsymbol{T}_1:=\operatorname{tridiagonal}(a, d, a)$?
lyche-numerical-linear-algebra- F03294	249	1.000	$(\boldsymbol{A} \otimes$	$(\boldsymbol{A} \otimes$	• give an economical general way to solve the linear system $(\boldsymbol{A} \otimes$
lyche-numerical-linear-algebra- F03295	249	1.000	$\boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$?

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lyche-numerical-linear-algebra- F03296	249	■ 0.999	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$	Same for $\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$ $\operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03297	250	■ 1.000	$n=9$	$n = 9$	matrix (10.8). Thus, for $n=9$
lyche-numerical-linear-algebra- F03298	251	■ 1.000	$d=\sqrt{n}$	$d = \sqrt{n}$	lower triangular, to solve $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. Since $\boldsymbol{A}$ and $\boldsymbol{L}$ has the same bandwidth $d = \sqrt{n}$
lyche-numerical-linear-algebra- F03299	251	■ 1.000	$0\left(n\ d^2\right)=0\left(n^2\right)$	$O\left(nd^2\right)=O\left(n^2\right)$	the complexity of this factorization is $O(nd^2) = O(n^2)$, cf. Algorithm 4.2. We need
lyche-numerical-linear-algebra- F03300	251	—	(n=100)	(n = 100)	
lyche-numerical-linear-algebra- F03301	252	■ 1.000	$\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$	$\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$	Here $\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1,$
lyche-numerical-linear-algebra- F03302	252	■ 1.000	$\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$	$\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$	Here $\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1,$
lyche-numerical-linear-algebra- F03303	252	■ 1.000	$\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1$	$\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1$	Here $\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1,$
lyche-numerical-linear-algebra- F03304	252	■ 1.000	$\dots, \boldsymbol{L}_{m-1}$	$\dots, \boldsymbol{L}_{m-1}$	$\dots, \boldsymbol{L}_{m-1}$ and $\boldsymbol{C}_1, \dots, \boldsymbol{C}_{m-1}$ can be rectangular.
lyche-numerical-linear-algebra- F03305	252	■ 1.000	$\boldsymbol{C}_1, \dots, \boldsymbol{C}_{m-1}$	$\boldsymbol{C}_1, \dots, \boldsymbol{C}_{m-1}$	$\dots, \boldsymbol{L}_{m-1}$ and $\boldsymbol{C}_1, \dots, \boldsymbol{C}_{m-1}$ can be rectangular.
lyche-numerical-linear-algebra- F03306	252	■ 0.995	$\boldsymbol{b}^T =$	$\boldsymbol{b}^T =$	To solve the system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ we partition $\boldsymbol{b}$ conformally with $\boldsymbol{A}$ in the form $\boldsymbol{b}^T =$
lyche-numerical-linear-algebra- F03307	252	■ 1.000	$\left[\boldsymbol{b}_1^T, \dots, \boldsymbol{b}_m^T\right]$	$[\boldsymbol{b}_1^T, \dots, \boldsymbol{b}_m^T]$	$[\boldsymbol{b}_1^T, \dots, \boldsymbol{b}_m^T]$. The formulas for solving $\boldsymbol{L}\boldsymbol{y} = \boldsymbol{b}$ and $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{y}$ are as follows:
lyche-numerical-linear-algebra- F03308	252	■ 1.000	$\boldsymbol{x}^T = \left[\boldsymbol{x}_1^T, \dots, \boldsymbol{x}_m^T\right]$	$\boldsymbol{x}^T = [\boldsymbol{x}_1^T, \dots, \boldsymbol{x}_m^T]$	The solution is then $\boldsymbol{x}^T = [\boldsymbol{x}_1^T, \dots, \boldsymbol{x}_m^T]$. To find $\boldsymbol{L}_k$ in (11.2) we solve the linear
lyche-numerical-linear-algebra- F03309	252	■ 1.000	$\boldsymbol{L}_k$	$\boldsymbol{L}_k$	The solution is then $\boldsymbol{x}^T = [\boldsymbol{x}_1^T, \dots, \boldsymbol{x}_m^T]$. To find $\boldsymbol{L}_k$ in (11.2) we solve the linear
lyche-numerical-linear-algebra- F03310	252	■ 1.000	$\boldsymbol{L}_k \boldsymbol{U}_k = \boldsymbol{A}_k$	$\boldsymbol{L}_k \boldsymbol{U}_k = \boldsymbol{A}_k$	systems $\boldsymbol{L}_k \boldsymbol{U}_k = \boldsymbol{A}_k$. Similarly we need to solve a linear system to find $\boldsymbol{x}_k$ in (11.3).
lyche-numerical-linear-algebra- F03311	252	■ 1.000	$\boldsymbol{x}_k$	$\boldsymbol{x}_k$	systems $\boldsymbol{L}_k \boldsymbol{U}_k = \boldsymbol{A}_k$. Similarly we need to solve a linear system to find $\boldsymbol{x}_k$ in (11.3).
lyche-numerical-linear-algebra- F03312	252	■ 1.000	$0\left(n^3 / 2\right)$	$O\left(n^{3 / 2}\right)$	The algorithm we now derive will only require $O(n^{3/2})$ arithmetic operations and
lyche-numerical-linear-algebra- F03313	252	■ 1.000	$0(n \log n)$	$O(n \log n)$	number of arithmetic operations can be reduced further to $O(n \log n)$.

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lyche-numerical-linear-algebra-2024-03-14 F03314	252	1.000	$\boldsymbol{V} =$	$V =$	where $T = \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ is the second derivative matrix, $V =$
lyche-numerical-linear-algebra-2024-03-15 F03315	252	1.000	$\left(v_{\{j, k\}}\right) \in \mathbb{R}^{m \times m}$	$(v_{j,k}) \in \mathbb{R}^{m \times m}$	$(v_{j,k}) \in \mathbb{R}^{m \times m}$ are the unknowns, and $F = (f_{j,k}) = (f(jh, kh)) \in \mathbb{R}^{m \times m}$ contains
lyche-numerical-linear-algebra-2024-03-16 F03316	252	1.000	$\boldsymbol{F} = \left(f_{\{j, k\}}\right) = (f(jh, kh)) \in \mathbb{R}^{m \times m}$	$F = (f_{j,k}) = (f(jh, kh)) \in \mathbb{R}^{m \times m}$	$(v_{j,k}) \in \mathbb{R}^{m \times m}$ are the unknowns, and $F = (f_{j,k}) = (f(jh, kh)) \in \mathbb{R}^{m \times m}$ contains
lyche-numerical-linear-algebra-2024-03-17 F03317	253	0.999	$\boldsymbol{X} \in \mathbb{R}^{m \times m}$	$X \in \mathbb{R}^{m \times m}$	Define $X \in \mathbb{R}^{m \times m}$ by $V = SXS$, where V is the solution of $TV + VT = h^2 F$.
lyche-numerical-linear-algebra-2024-03-18 F03318	253	0.999	$\boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}$	$V = SXS$	Define $X \in \mathbb{R}^{m \times m}$ by $V = SXS$, where V is the solution of $TV + VT = h^2 F$.
lyche-numerical-linear-algebra-2024-03-19 F03319	253	0.999	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F}$	$TV + VT = h^2 F$	Define $X \in \mathbb{R}^{m \times m}$ by $V = SXS$, where V is the solution of $TV + VT = h^2 F$.
lyche-numerical-linear-algebra-2024-03-20 F03320	253	1.000	$\boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4 h^4 \boldsymbol{G}$	$DX + XD = 4h^4 G$	Since D is diagonal, the equation $DX + XD = 4h^4 G$, is easy to solve. For the j, k
lyche-numerical-linear-algebra-2024-03-21 F03321	254	1.000	$\boldsymbol{G} = \boldsymbol{S} \boldsymbol{F} \boldsymbol{S}$	$G = SFS$	1. $G = SES$,
lyche-numerical-linear-algebra-2024-03-22 F03322	254	1.000	$x_{\{j, k\}} = h^4 g_{\{j, k\}} / (\sigma_j + \sigma_k)$	$x_{j,k} = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad j, k = 1, \dots, m$	2. $x_{i,k} = h^4 g_{i,k} / (\sigma_i + \sigma_k), \quad j, k = 1, \dots, n$,
lyche-numerical-linear-algebra-2024-03-23 F03323	254	1.000	$\boldsymbol{X}, \boldsymbol{S}$	X, S	We can compute X, S and the σ 's without using loops. Using outer products,
lyche-numerical-linear-algebra-2024-03-24 F03324	254	0.999	$-\Delta u = f$	$-\Delta u = f$	$-\Delta u = f$ on $\Omega = (0, 1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$
lyche-numerical-linear-algebra-2024-03-25 F03325	254	0.999	$\Omega = (0, 1)^2$	$\Omega = (0, 1)^2$	$-\Delta u = f$ on $\Omega = (0, 1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$
lyche-numerical-linear-algebra-2024-03-26 F03326	254	0.999	$m \in$	$m \in$	$-\Delta u = f$ on $\Omega = (0, 1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$
lyche-numerical-linear-algebra-2024-03-27 F03327	254	1.000	$\mathbb{N}, h = 1/(m+1)$	$\mathbb{N}, h = 1/(m+1)$	$\mathbb{N}, h = 1/(m+1)$, and $F = (f(jh, kh)) \in \mathbb{R}^{m \times m}$. We compute $V \in \mathbb{R}^{(m+2) \times (m+2)}$
lyche-numerical-linear-algebra-2024-03-28 F03328	254	1.000	$\boldsymbol{F} = (f(jh, kh)) \in \mathbb{R}^{m \times m}$	$F = (f(jh, kh)) \in \mathbb{R}^{m \times m}$	$\mathbb{N}, h = 1/(m+1)$, and $F = (f(jh, kh)) \in \mathbb{R}^{m \times m}$. We compute $V \in \mathbb{R}^{(m+2) \times (m+2)}$
lyche-numerical-linear-algebra-2024-03-29 F03329	254	1.000	$\boldsymbol{V} \in \mathbb{R}^{(m+2) \times (m+2)}$	$V \in \mathbb{R}^{(m+2) \times (m+2)}$	$\mathbb{N}, h = 1/(m+1)$, and $F = (f(jh, kh)) \in \mathbb{R}^{m \times m}$. We compute $V \in \mathbb{R}^{(m+2) \times (m+2)}$
lyche-numerical-linear-algebra-2024-03-30 F03330	254	1.000	$O(m^3)$	$O(m^3)$	requires $O(m^3)$ arithmetic operations, the complexity of this algorithm is for large
lyche-numerical-linear-algebra-2024-03-31 F03331	254	1.000	$O(4 \times 2m^3) =$	$O(4 \times 2m^3) =$	m determined by 4 m -by- m matrix multiplications and is given by $O(4 \times 2m^3) =$
lyche-numerical-linear-algebra-2024-03-32 F03332	254	0.943	$O(8n^{3/2})$	$O(8n^{3/2})$	$O(8n^{3/2})$. ¹ The method is very fast and will be used as a preconditioner for a more
lyche-numerical-linear-algebra-2024-03-33 F03333	254	1.000	10^6	10^6	find the 10^6 unknowns $v_{j,k}$ on a 1000×1000 grid.
lyche-numerical-linear-algebra-2024-03-34 F03334	254	1.000	1000×1000	1000×1000	find the 10^6 unknowns $v_{j,k}$ on a 1000×1000 grid.
lyche-numerical-linear-algebra-2024-03-35 F03335	254	1.000	$O(4n^{3/2})$	$O(4n^{3/2})$	to $O(4n^{3/2})$. This is detailed in Problem 11.4.
lyche-numerical-linear-algebra-2024-03-36 F03336	255	1.000	$\boldsymbol{A} \in \mathbb{R}^{m \times m}$	$A \in \mathbb{R}^{m \times m}$	given by (11.4) and a matrix $A \in \mathbb{R}^{m \times m}$. Since the matrices are m -by- m this will
lyche-numerical-linear-algebra-2024-03-37 F03337	255	1.000	$\boldsymbol{S} \boldsymbol{A}$	SA	calculate the products SA and AS in $O(m^2 \log_2 m)$ operations.
lyche-numerical-linear-algebra-2024-03-38 F03338	255	1.000	$\boldsymbol{A} \boldsymbol{S}$	AS	calculate the products SA and AS in $O(m^2 \log_2 m)$ operations.
lyche-numerical-linear-algebra-2024-03-39 F03339	255	1.000	$O(m^2 \log_2 m)$	$O(m^2 \log_2 m)$	calculate the products SA and AS in $O(m^2 \log_2 m)$ operations.

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lyche-numerical-linear-algebra- F03340	255	1.000	$\boldsymbol{v}=\left[v_1,\ldots,v_m\right]^T\in\mathbb{R}^m$	$\boldsymbol{v} = [v_1, \dots, v_m]^T \in \mathbb{R}^m$	Given $\boldsymbol{v} = [v_1, \dots, v_m]^T \in \mathbb{R}^m$ we say that the vector $\boldsymbol{w} = [w_1, \dots, w_m]^T$ given
lyche-numerical-linear-algebra- F03341	255	1.000	$\boldsymbol{w}=\left[w_1,\ldots,w_m\right]^T$	$\boldsymbol{w} = [w_1, \dots, w_m]^T$	Given $\boldsymbol{v} = [v_1, \dots, v_m]^T \in \mathbb{R}^m$ we say that the vector $\boldsymbol{w} = [w_1, \dots, w_m]^T$ given
lyche-numerical-linear-algebra- F03342	255	1.000	$\boldsymbol{w}=\boldsymbol{S}\boldsymbol{v}$	$\boldsymbol{w} = \boldsymbol{S}\boldsymbol{v}$	as the matrix times vector $\boldsymbol{w} = \boldsymbol{S}\boldsymbol{v}$, where $\boldsymbol{S}$ is the sine matrix given by (11.4). We
lyche-numerical-linear-algebra- F03343	255	1.000	$\boldsymbol{B}=\boldsymbol{S}\boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{S}\boldsymbol{A}$	can then identify the matrix $\boldsymbol{B} = \boldsymbol{S}\boldsymbol{A}$ as the DST of $\boldsymbol{A} \in \mathbb{R}^{m,n}$, i.e. as the DST of
lyche-numerical-linear-algebra- F03344	255	1.000	$\boldsymbol{A}\in\mathbb{R}^{m,n}$	$\boldsymbol{A} \in \mathbb{R}^{m,n}$	can then identify the matrix $\boldsymbol{B} = \boldsymbol{S}\boldsymbol{A}$ as the DST of $\boldsymbol{A} \in \mathbb{R}^{m,n}$, i.e. as the DST of
lyche-numerical-linear-algebra- F03345	255	1.000	$\boldsymbol{B}=\boldsymbol{A}\boldsymbol{S}$	$\boldsymbol{B} = \boldsymbol{A}\boldsymbol{S}$	the columns of $\boldsymbol{A}$. The product $\boldsymbol{B} = \boldsymbol{A}\boldsymbol{S}$ can also be interpreted as a DST. Indeed,
lyche-numerical-linear-algebra- F03346	255	1.000	$\boldsymbol{B}=\left(\boldsymbol{S}\boldsymbol{A}\right)^T$	$\boldsymbol{B} = (\boldsymbol{S}\boldsymbol{A})^T$	since $\boldsymbol{S}$ is symmetric we have $\boldsymbol{B} = (\boldsymbol{S}\boldsymbol{A})^T$ which means that $\boldsymbol{B}$ is the transpose
lyche-numerical-linear-algebra- F03347	255	1.000	$N\in\mathbb{N}$	$N \in \mathbb{N}$	as the fast Fourier transform (FFT). To define the DFT let for $N \in \mathbb{N}$
lyche-numerical-linear-algebra- F03348	256	1.000	$\boldsymbol{y}=\left[y_1,\ldots,y_N\right]^T\in\mathbb{R}^N$	$\boldsymbol{y} = [y_1, \dots, y_N]^T \in \mathbb{R}^N$	where $i = \sqrt{-1}$ is the imaginary unit. Given $\boldsymbol{y} = [y_1, \dots, y_N]^T \in \mathbb{R}^N$ we say that
lyche-numerical-linear-algebra- F03349	256	1.000	$\boldsymbol{z}=\left[z_1,\ldots,z_N\right]^T$	$\boldsymbol{z} = [z_1, \dots, z_N]^T$	$\boldsymbol{z} = [z_1, \dots, z_N]^T$ given by
lyche-numerical-linear-algebra- F03350	256	0.991	$\boldsymbol{z}=\boldsymbol{F}_N\boldsymbol{y}$	$\boldsymbol{z} = \boldsymbol{F}_N\boldsymbol{y}$	vector product $\boldsymbol{z} = \boldsymbol{F}_N\boldsymbol{y}$, where the Fourier matrix $\boldsymbol{F}_N \in \mathbb{C}^{N \times N}$ has elements
lyche-numerical-linear-algebra- F03351	256	0.991	$\boldsymbol{F}_N\in\mathbb{C}^{N\times N}$	$\boldsymbol{F}_N \in \mathbb{C}^{N \times N}$	vector product $\boldsymbol{z} = \boldsymbol{F}_N\boldsymbol{y}$, where the Fourier matrix $\boldsymbol{F}_N \in \mathbb{C}^{N \times N}$ has elements
lyche-numerical-linear-algebra- F03352	256	1.000	$\omega_N^{jk},j,k=0,1,\ldots,N-1$	$\omega_N^{jk}, j, k = 0, 1, \dots, N - 1$	$\omega_N^{jk}, j, k = 0, 1, \dots, N - 1$. For a matrix we say that $\boldsymbol{B} = \boldsymbol{F}_N\boldsymbol{A}$ is the DFT of $\boldsymbol{A}$.
lyche-numerical-linear-algebra- F03353	256	1.000	$\boldsymbol{B}=\boldsymbol{F}_N\boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{F}_N\boldsymbol{A}$	$\omega_N^{jk}, j, k = 0, 1, \dots, N - 1$. For a matrix we say that $\boldsymbol{B} = \boldsymbol{F}_N\boldsymbol{A}$ is the DFT of $\boldsymbol{A}$.
lyche-numerical-linear-algebra- F03354	256	1.000	$\omega_4^2=(-i)^2=-1,\omega_4^3=(-i)(-1)=i,\omega_4^4=(-1)^2=1,\omega_4^6=i^2=-1$	$\omega_4^2 = (-i)^2 = -1, \omega_4^3 = (-i)(-1) = i, \omega_4^4 = (-1)^2 = 1, \omega_4^6 = i^2 = -1$	we find $\omega_4^2 = (-i)^2 = -1, \omega_4^3 = (-i)(-1) = i, \omega_4^4 = (-1)^2 = 1, \omega_4^6 = i^2 = -1$,
lyche-numerical-linear-algebra- F03355	256	1.000	$\omega_4^9=i^3=-i$	$\omega_4^9 = i^3 = -i$	$\omega_4^9 = i^3 = -i$, and so
lyche-numerical-linear-algebra- F03356	256	1.000	$2m+2$	$2m + 2$	computed from the discrete Fourier transform of order $2m + 2$. We recall that for
lyche-numerical-linear-algebra- F03357	256	1.000	w	w	any complex number w
lyche-numerical-linear-algebra- F03358	256	1.000	$\boldsymbol{S}\boldsymbol{x}$	$\boldsymbol{S}\boldsymbol{x}$	and a vector $\boldsymbol{x} \in \mathbb{R}^m$. Component k of $\boldsymbol{S}\boldsymbol{x}$ is equal to $i/2$ times component $k + 1$ of
lyche-numerical-linear-algebra- F03359	256	1.000	$i/2$	$i/2$	and a vector $\boldsymbol{x} \in \mathbb{R}^m$. Component k of $\boldsymbol{S}\boldsymbol{x}$ is equal to $i/2$ times component $k + 1$ of
lyche-numerical-linear-algebra- F03360	256	1.000	$\boldsymbol{F}_{2m+2}\boldsymbol{z}$	$\boldsymbol{F}_{2m+2}\boldsymbol{z}$	$\boldsymbol{F}_{2m+2}\boldsymbol{z}$ where
lyche-numerical-linear-algebra- F03361	256	1.000	$\omega=\omega_{2m+2}=e^{-2\pi i/(2m+2)}=e^{-\pi i/(m+1)}$	$\omega = \omega_{2m+2} = e^{-2\pi i/(2m+2)} = e^{-\pi i/(m+1)}$	Proof Let $\omega = \omega_{2m+2} = e^{-2\pi i/(2m+2)} = e^{-\pi i/(m+1)}$. We note that
lyche-numerical-linear-algebra- F03362	257	1.000	$-2i$	$-2i$	Dividing both sides by $-2i$ and noting $-1/(2i) = -i/(2i^2) = i/2$, proves the
lyche-numerical-linear-algebra- F03363	257	1.000	$-1/(2i)=-i/(2i^2)=i/2$	$-1/(2i) = -i/(2i^2) = i/2$	Dividing both sides by $-2i$ and noting $-1/(2i) = -i/(2i^2) = i/2$, proves the
lyche-numerical-linear-algebra- F03364	257	1.000	$N=2m+2$	$N = 2m + 2$	from the DFT of length $N = 2m + 2$.
lyche-numerical-linear-algebra- F03365	257	1.000	$\boldsymbol{F}_N\boldsymbol{y}$	$\boldsymbol{F}_N\boldsymbol{y}$	the matrix- vector product $\boldsymbol{F}_N\boldsymbol{y}$. Suppose N is even. The key to the FFT is a

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lyche-numerical-linear-algebra-F03366	257	1.000	$\boldsymbol{F}_N$	$\boldsymbol{F}_N$	connection between $\boldsymbol{F}_N$ and $\boldsymbol{F}_{N/2}$ which makes it possible to compute the FFT
lyche-numerical-linear-algebra-F03367	257	1.000	$\boldsymbol{F}_{N/2}$	$\boldsymbol{F}_{N/2}$	connection between $\boldsymbol{F}_N$ and $\boldsymbol{F}_{N/2}$ which makes it possible to compute the FFT
lyche-numerical-linear-algebra-F03368	257	1.000	$N/2$	$N/2$	of order N as two FFT's of order $N/2$. By repeating this process we can reduce the
lyche-numerical-linear-algebra-F03369	257	1.000	$O(N^2)$	$O(N^2)$	number of arithmetic operations to compute a DFT from $O(N^2)$ to $O(N \log_2 N)$.
lyche-numerical-linear-algebra-F03370	257	1.000	$O(N \log_2 N)$	$O(N \log_2 N)$	number of arithmetic operations to compute a DFT from $O(N^2)$ to $O(N \log_2 N)$.
lyche-numerical-linear-algebra-F03371	257	1.000	$\boldsymbol{P}_N \in \mathbb{R}^{N \times N}$	$\boldsymbol{P}_N \in \mathbb{R}^{N \times N}$	tion matrix $\boldsymbol{P}_N \in \mathbb{R}^{N \times N}$ given by
lyche-numerical-linear-algebra-F03372	257	0.998	$\boldsymbol{e}_k = (\delta_{j,k})$	$\boldsymbol{e}_k = (\delta_{j,k})$	where the $\boldsymbol{e}_k = (\delta_{j,k})$ are unit vectors. If $\boldsymbol{A}$ is a matrix with N columns $[\boldsymbol{a}_1, \dots, \boldsymbol{a}_N]$
lyche-numerical-linear-algebra-F03373	257	0.998	$[\boldsymbol{a}_1, \dots, \boldsymbol{a}_N]$	$[\boldsymbol{a}_1, \dots, \boldsymbol{a}_N]$	where the $\boldsymbol{e}_k = (\delta_{j,k})$ are unit vectors. If $\boldsymbol{A}$ is a matrix with N columns $[\boldsymbol{a}_1, \dots, \boldsymbol{a}_N]$
lyche-numerical-linear-algebra-F03374	257	0.965	$\boldsymbol{P}_N$	$\boldsymbol{P}_N$	i.e. post multiplying $\boldsymbol{A}$ by $\boldsymbol{P}_N$ permutes the columns of $\boldsymbol{A}$ so that all the odd-indexed
lyche-numerical-linear-algebra-F03375	258	1.000	$\boldsymbol{F}_4 \boldsymbol{P}_4$	$\boldsymbol{F}_4 \boldsymbol{P}_4$	where we have indicated a certain block structure of $\boldsymbol{F}_4 \boldsymbol{P}_4$. These blocks can be
lyche-numerical-linear-algebra-F03376	258	1.000	$\boldsymbol{F}_2$	$\boldsymbol{F}_2$	related to the 2-by-2 matrix $\boldsymbol{F}_2$. We define the diagonal scaling matrix $\boldsymbol{D}_2$ by
lyche-numerical-linear-algebra-F03377	258	1.000	$\omega_2 = \exp^{-2\pi i/2} = -1$	$\omega_2 = \exp^{-2\pi i/2} = -1$	Since $\omega_2 = \exp^{-2\pi i/2} = -1$ we find
lyche-numerical-linear-algebra-F03378	258	1.000	$N = 2m$	$N = 2m$	Theorem 11.1 (Fast Fourier Transform) <i>If $N = 2m$ is even then</i>
lyche-numerical-linear-algebra-F03379	258	1.000	p, q	p, q	Proof Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p - 1$ and $k := q - 1$.
lyche-numerical-linear-algebra-F03380	258	1.000	$1 \leq p, q \leq m$	$1 \leq p, q \leq m$	Proof Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p - 1$ and $k := q - 1$.
lyche-numerical-linear-algebra-F03381	258	1.000	$j := p - 1$	$j := p - 1$	Proof Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p - 1$ and $k := q - 1$.
lyche-numerical-linear-algebra-F03382	258	1.000	$k := q - 1$	$k := q - 1$	Proof Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p - 1$ and $k := q - 1$.
lyche-numerical-linear-algebra-F03383	258	1.000	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}$	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}$	It follows that the four m -by- m blocks of $\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}$ have the required structure.
lyche-numerical-linear-algebra-F03384	258	1.000	$\boldsymbol{y} \in \mathbb{R}^{2m}$	$\boldsymbol{y} \in \mathbb{R}^{2m}$	$\boldsymbol{y} \in \mathbb{R}^{2m}$ and set $\boldsymbol{w} = \boldsymbol{P}_{2m}^T \boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$, where
lyche-numerical-linear-algebra-F03385	258	1.000	$\boldsymbol{w} = \boldsymbol{P}_{2m}^T \boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$	$\boldsymbol{w} = \boldsymbol{P}_{2m}^T \boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$	$\boldsymbol{y} \in \mathbb{R}^{2m}$ and set $\boldsymbol{w} = \boldsymbol{P}_{2m}^T \boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$, where
lyche-numerical-linear-algebra-F03386	259	1.000	$\boldsymbol{F}_{2m} \boldsymbol{y}$	$\boldsymbol{F}_{2m} \boldsymbol{y}$	In order to compute $\boldsymbol{F}_{2m} \boldsymbol{y}$ we need to compute $\boldsymbol{F}_m \boldsymbol{w}_1$ and $\boldsymbol{F}_m \boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-algebra-F03387	259	1.000	$\boldsymbol{F}_m \boldsymbol{w}_1$	$\boldsymbol{F}_m \boldsymbol{w}_1$	In order to compute $\boldsymbol{F}_{2m} \boldsymbol{y}$ we need to compute $\boldsymbol{F}_m \boldsymbol{w}_1$ and $\boldsymbol{F}_m \boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-algebra-F03388	259	1.000	$\boldsymbol{F}_m \boldsymbol{w}_2$	$\boldsymbol{F}_m \boldsymbol{w}_2$	In order to compute $\boldsymbol{F}_{2m} \boldsymbol{y}$ we need to compute $\boldsymbol{F}_m \boldsymbol{w}_1$ and $\boldsymbol{F}_m \boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-algebra-F03389	259	1.000	$n = 2^k$	$n = 2^k$	combining two FFT's of order m we obtain an FFT of order $2m$. If $n = 2^k$ then
lyche-numerical-linear-algebra-F03390	259	1.000	$\boldsymbol{y} \in \mathbb{R}^n$	$\boldsymbol{y} \in \mathbb{R}^n$	Statement 3 is included so that the input $\boldsymbol{y} \in \mathbb{R}^n$ can be either a row or column
lyche-numerical-linear-algebra-F03391	259	1.000	$\gamma N \log_2 N$	$\gamma N \log_2 N$	The complexity of the FFT is given by $\gamma N \log_2 N$ for some constant γ

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lyche-numerical-linear-algebra-259 F03392	259	■ 1.000	γ	γ	The complexity of the FFT is given by $\gamma N \log_2 N$ for some constant γ
lyche-numerical-linear-algebra-259 F03393	259	■ 1.000	$N=2^{\{k\}}$	$N = 2^k$	let x_k be the complexity (the number of arithmetic operations) when $N = 2^k$.
lyche-numerical-linear-algebra-259 F03394	259	■ 1.000	$N / 2=2^{\{k-1\}}$	$N/2 = 2^{k-1}$	Since we need two FFT's of order $N/2 = 2^{k-1}$ and a multiplication with the
lyche-numerical-linear-algebra-259 F03395	259	■ 1.000	$\boldsymbol{D}_{N / 2}$	$\boldsymbol{D}_{N/2}$	diagonal matrix $\boldsymbol{D}_{N/2}$, it is reasonable to assume that $x_k = 2x_{k-1} + \gamma 2^k$ for
lyche-numerical-linear-algebra-259 F03396	259	■ 1.000	$x_{\{k\}}=2 x_{\{k-1\}}+\gamma 2^{\{k\}}$	$x_k = 2x_{k-1} + \gamma 2^k$	diagonal matrix $\boldsymbol{D}_{N/2}$, it is reasonable to assume that $x_k = 2x_{k-1} + \gamma 2^k$ for
lyche-numerical-linear-algebra-259 F03397	259	■ 1.000	$x_{\{0\}}=0$	$x_0 = 0$	some constant γ independent of k . Since $x_0 = 0$ we obtain by induction on k
lyche-numerical-linear-algebra-259 F03398	259	■ 1.000	$x_{\{k\}}=\gamma k 2^{\{k\}}$	$x_k = \gamma k 2^k$	that $x_k = \gamma k 2^k$. Indeed, this holds for $k = 0$ and if $x_{k-1} = \gamma(k-1)2^{k-1}$ then
lyche-numerical-linear-algebra-259 F03399	259	■ 1.000	$x_{\{k-1\}}=\gamma(k-1) 2^{\{k-1\}}$	$x_{k-1} = \gamma(k-1)2^{k-1}$	that $x_k = \gamma k 2^k$. Indeed, this holds for $k = 0$ and if $x_{k-1} = \gamma(k-1)2^{k-1}$ then
lyche-numerical-linear-algebra-259 F03400	259	■ 1.000	$x_{\{k\}}=2 x_{\{k-1\}}+\gamma 2^{\{k\}}=2 \gamma(k-1) 2^{\{k-1\}}+\gamma 2^{\{k\}}$ $2^{\{k\}}=\gamma k 2^{\{k\}}$	$x_k = 2x_{k-1} + \gamma 2^k = 2\gamma(k-1)2^{k-1} + \gamma 2^k = \gamma k 2^k$	$x_k = 2x_{k-1} + \gamma 2^k = 2\gamma(k-1)2^{k-1} + \gamma 2^k = \gamma k 2^k$. Reasonable implementations
lyche-numerical-linear-algebra-259 F03401	259	■ 1.000	$\gamma \approx 5$	$\gamma \approx 5$	of FFT typically have $\gamma \approx 5$, see [14].
lyche-numerical-linear-algebra-259 F03402	259	■ 1.000	$O\left(8 n^{\{2\}}\right)$	$O(8n^2)$	large N . The direct multiplication $\boldsymbol{F}_N \boldsymbol{y}$ requires $O(8n^2)$ arithmetic operations since
lyche-numerical-linear-algebra-260 F03403	260	■ 1.000	$5 N \log _{\{2\}} N$	$5N \log_2 N$	complex arithmetic is involved. Assuming that the FFT uses $5N \log_2 N$ arithmetic
lyche-numerical-linear-algebra-260 F03404	260	■ 1.000	$N=2^{\{20\}} \approx 10^{\{6\}}$	$N = 2^{20} \approx 10^6$	operations we find for $N = 2^{20} \approx 10^6$ the ratio
lyche-numerical-linear-algebra-260 F03405	260	■ 1.000	$\boldsymbol{H}=\boldsymbol{S} \boldsymbol{F}$	$\boldsymbol{H} = \boldsymbol{S}\boldsymbol{F}$	Algorithm 11.1. We first compute $\boldsymbol{H} = \boldsymbol{S}\boldsymbol{F}$ using Lemma 11.1 and m FFT's, one
lyche-numerical-linear-algebra-260 F03406	260	■ 0.998	$\boldsymbol{G}=\boldsymbol{H} \boldsymbol{S}$	$\boldsymbol{G} = \boldsymbol{H}\boldsymbol{S}$	for each of the m columns of $\boldsymbol{F}$. We then compute $\boldsymbol{G} = \boldsymbol{H}\boldsymbol{S}$ by m FFT's, one for
lyche-numerical-linear-algebra-260 F03407	260	■ 1.000	$\boldsymbol{Z}=\boldsymbol{S} \boldsymbol{X}$	$\boldsymbol{Z} = \boldsymbol{S}\boldsymbol{X}$	each of the rows of $\boldsymbol{H}$. After $\boldsymbol{X}$ is determined we compute $\boldsymbol{Z} = \boldsymbol{S}\boldsymbol{X}$ and $\boldsymbol{V} = \boldsymbol{Z}\boldsymbol{S}$
lyche-numerical-linear-algebra-260 F03408	260	■ 1.000	$\boldsymbol{V}=\boldsymbol{Z} \boldsymbol{S}$	$\boldsymbol{V} = \boldsymbol{Z}\boldsymbol{S}$	each of the rows of $\boldsymbol{H}$. After $\boldsymbol{X}$ is determined we compute $\boldsymbol{Z} = \boldsymbol{S}\boldsymbol{X}$ and $\boldsymbol{V} = \boldsymbol{Z}\boldsymbol{S}$
lyche-numerical-linear-algebra-260 F03409	260	■ 1.000	$O\left(\gamma(2 m+2) \log _{\{2\}}(2 m+2)\right)$	$O(\gamma(2m+2) \log_2(2m+2))$	one FFT requires $O(\gamma(2m+2) \log_2(2m+2))$ arithmetic operations the $4m$ FFT's
lyche-numerical-linear-algebra-260 F03410	260	■ 1.000	$O\left(8 n^{\{3 / 2\}}\right)$	$O(8n^{3/2})$	if Cholesky factorization was used. This should be compared to the $O(8n^{3/2})$
lyche-numerical-linear-algebra-260 F03411	260	■ 1.000	$\boldsymbol{F}_{\{4\}}$	$\boldsymbol{F}_4$	Exercise 11.1 (Fourier Matrix) Show that the Fourier matrix $\boldsymbol{F}_4$ is symmetric,
lyche-numerical-linear-algebra-261 F03412	261	■ 1.000	$\left(v_{\{i, j\}}\right)_{\{i, j=1\}}^{\{m\}}$	$(v_{i,j})_{i,j=1}^m$	$(v_{i,j})_{i,j=1}^m$, where
lyche-numerical-linear-algebra-261 F03413	261	■ 0.608	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^{\{2\}} \boldsymbol{F}$	$\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$	in $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_m)$, where $\lambda_j = 4 \sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra-261 F03414	261	■ 0.608	$\boldsymbol{D}=\operatorname{diag}\left(\lambda_{\{1\}}, \ldots, \lambda_{\{m\}}\right)$	$\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_m)$	in $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_m)$, where $\lambda_j = 4 \sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra-261 F03415	261	■ 0.608	$\lambda_{\{j\}}=4 \sin ^{\{2\}}(j \pi h / 2)$	$\lambda_j = 4 \sin^2(j\pi h/2)$	in $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_m)$, where $\lambda_j = 4 \sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra-261 F03416	261	■ 1.000	$\boldsymbol{X}=\left[\boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{m\}}\right]$	$\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_m]$	where $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_m]$ and $\boldsymbol{C} = [\boldsymbol{c}_1, \dots, \boldsymbol{c}_m]$. Thus we can find $\boldsymbol{X}$ by solving
lyche-numerical-linear-algebra-261 F03417	261	■ 1.000	$\boldsymbol{C}=\left[\boldsymbol{c}_{\{1\}}, \ldots, \boldsymbol{c}_{\{m\}}\right]$	$\boldsymbol{C} = [\boldsymbol{c}_1, \dots, \boldsymbol{c}_m]$	where $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_m]$ and $\boldsymbol{C} = [\boldsymbol{c}_1, \dots, \boldsymbol{c}_m]$. Thus we can find $\boldsymbol{X}$ by solving

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lyche-numerical-linear-algebra- F03418	261	1.000	$8m - 7$	$8m - 7$	$m \times m$ system can be solved by Algorithms 2.1 and 2.2 in $8m - 7$ arithmetic
lyche-numerical-linear-algebra- F03419	261	1.000	$O(\delta m^2)$	$O(\delta m^2)$	operations. Give an algorithm to find X which only requires $O(\delta m^2)$ arithmetic
lyche-numerical-linear-algebra- F03420	261	1.000	$O(4m^3) = O(4n^{3/2})$	$O(4m^3) = O(4n^{3/2})$	c) Describe a method to compute V which only requires $O(4m^3) = O(4n^{3/2})$
lyche-numerical-linear-algebra- F03421	261	1.000	$O(2\gamma n \log_2 n)$	$O(2\gamma n \log_2 n)$	$O(2\gamma n \log_2 n)$ where γ is the same constant as mentioned at the end of the
lyche-numerical-linear-algebra- F03422	261	1.000	$V = SX S = (x_{j,k})$	$V = SX S = (x_{j,k})$	$V = SX S = (x_{j,k})$ where V is the solution of (10.22). Show that
lyche-numerical-linear-algebra- F03423	262	1.000	$\lambda_j = 4 \sin^2(j\pi h/2), j = 1, \dots, m$	$\lambda_j = 4 \sin^2(j\pi h/2), j = 1, \dots, m$	where $D = \text{diag}(\lambda_1, \dots, \lambda_m)$, with $\lambda_j = 4 \sin^2(j\pi h/2), j = 1, \dots, m$, and that
lyche-numerical-linear-algebra- F03424	262	1.000	$\sigma_j + \sigma_k - \frac{2}{3}\sigma_j\sigma_k > 0$	$\sigma_j + \sigma_k - \frac{2}{3}\sigma_j\sigma_k > 0$	Show that $\sigma_j + \sigma_k - \frac{2}{3}\sigma_j\sigma_k > 0$ for $j, k = 1, 2, \dots, m$. Conclude that the matrix
lyche-numerical-linear-algebra- F03425	262	1.000	$j, k = 1, 2, \dots, m$	$j, k = 1, 2, \dots, m$	Show that $\sigma_j + \sigma_k - \frac{2}{3}\sigma_j\sigma_k > 0$ for $j, k = 1, 2, \dots, m$. Conclude that the matrix
lyche-numerical-linear-algebra- F03426	262	1.000	$X = (x_{j,k})$	$X = (x_{j,k})$	Define the matrix $X = (x_{j,k})$ by $U = SX S$ where U is the solution of (10.25).
lyche-numerical-linear-algebra- F03427	262	1.000	$U = SX S$	$U = SX S$	Define the matrix $X = (x_{j,k})$ by $U = SX S$ where U is the solution of (10.25).
lyche-numerical-linear-algebra- F03428	262	1.000	$O(\delta n^{3/2})$	$O(\delta n^{3/2})$	which requires only $O(\delta n^{3/2})$ operations to find U in Problem 10.9. Here δ is some
lyche-numerical-linear-algebra- F03429	262	0.713	$F =$	$F =$	Exercise 11.8 compute the solution U corresponding to $F = \text{ones}(m, m)$. For
lyche-numerical-linear-algebra- F03430	262	1.000	$x = A \backslash b$	$x = A \backslash b$	standard form $Ax = b$ in (10.25). You can use $x = A \backslash b$ for solving $Ax = b$.
lyche-numerical-linear-algebra- F03431	262	0.581	$x \in \mathbb{R}^{m^2}$	$x \in \mathbb{R}^{m^2}$	Use $F(:)$ to vectorize a matrix and $\text{reshape}(x, m, m)$ to turn a vector $x \in \mathbb{R}^{m^2}$
lyche-numerical-linear-algebra- F03432	262	1.000	U	U	into an $m \times m$ matrix. Use the MATLAB command <code>surf(U)</code> for plotting U for, say,
lyche-numerical-linear-algebra- F03433	262	1.000	$m=50$	$m = 50$	$m = 50$. Compare the result with Exercise 11.8 by plotting the difference between
lyche-numerical-linear-algebra- F03434	264	1.000	x_0	x_0	approximation x_0 to the exact solution x and then compute a sequence $\{x_k\}$ such that
lyche-numerical-linear-algebra- F03435	264	1.000	$\{x_k\}$	$\{x_k\}$	approximation x_0 to the exact solution x and then compute a sequence $\{x_k\}$ such that
lyche-numerical-linear-algebra- F03436	264	1.000	$x_k \rightarrow x$	$x_k \rightarrow x$	hopefully $x_k \rightarrow x$. Iterative methods are mainly used for large sparse systems, i.e.,
lyche-numerical-linear-algebra- F03437	265	1.000	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	onal elements, the linear system $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ can be written in component
lyche-numerical-linear-algebra- F03438	265	1.000	$y = (z+1) / 2$	$y = (z+1)/2$	form as $y = (z+1)/2$ and $z = (y+1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-algebra- F03439	265	1.000	$z = (y+1) / 2$	$z = (y+1)/2$	form as $y = (z+1)/2$ and $z = (y+1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-algebra- F03440	265	1.000	y_0, z_0	y_0, z_0	form as $y = (z+1)/2$ and $z = (y+1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-algebra- F03441	265	0.996	$\{y_k\}$	$\{y_k\}$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k + 1)/2$ and
lyche-numerical-linear-algebra- F03442	265	0.996	$\{z_k\}$	$\{z_k\}$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k + 1)/2$ and
lyche-numerical-linear-algebra- F03443	265	0.996	$y_{k+1} = (z_k + 1) / 2$	$y_{k+1} = (z_k + 1) / 2$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k + 1)/2$ and

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lyche-numerical-linear-algebra- F03444	265	■ 1.000	$z_{k+1}=\left(y_{k+1}\right) / 2$	$z_{k+1}=\left(y_k+1\right) / 2$	$z_{k+1}=\left(y_k+1\right) / 2$. This is an example of Jacobi's method. If $y_0=z_0=0$ then
lyche-numerical-linear-algebra- F03445	265	■ 1.000	$y_{\{0\}}=z_{\{0\}}=0$	$y_0=z_0=0$	$z_{k+1}=\left(y_k+1\right) / 2$. This is an example of Jacobi's method. If $y_0=z_0=0$ then
lyche-numerical-linear-algebra- F03446	265	■ 1.000	$y_{\{1\}}=z_{\{1\}}=1 / 2$	$y_1=z_1=1 / 2$	we find $y_1=z_1=1 / 2$ and in general $y_k=z_k=1-2^{-k}$ for $k=0,1,2,3, \ldots$
lyche-numerical-linear-algebra- F03447	265	■ 1.000	$y_{\{k\}}=z_{\{k\}}=1-2^{\{-k\}}$	$y_k=z_k=1-2^{-k}$	we find $y_1=z_1=1 / 2$ and in general $y_k=z_k=1-2^{-k}$ for $k=0,1,2,3, \ldots$
lyche-numerical-linear-algebra- F03448	265	■ 1.000	$k=0,1,2,3, \backslash \mathrm{ldots}$	$k=0,1,2,3, \ldots$	we find $y_1=z_1=1 / 2$ and in general $y_k=z_k=1-2^{-k}$ for $k=0,1,2,3, \ldots$.
lyche-numerical-linear-algebra- F03449	265	■ 1.000	$[1,1]^{\wedge\{T\}}$	$[1,1]^T$	The iteration converges to the exact solution $[1,1]^T$, and the error is halved in each
lyche-numerical-linear-algebra- F03450	266	■ 1.000	$z_{k+1}=\left(y_{k+1}+1\right) / 2$	$z_{k+1}=\left(y_{k+1}+1\right) / 2$	$z_{k+1}=\left(y_{k+1}+1\right) / 2$. If $y_0=z_0=0$ then we find $y_1=1 / 2, z_1=3 / 4, y_2=7 / 8$,
lyche-numerical-linear-algebra- F03451	266	■ 1.000	$y_{\{1\}}=1 / 2, z_{\{1\}}=3 / 4, y_{\{2\}}=7 / 8$	$y_1=1 / 2, z_1=3 / 4, y_2=7 / 8$	$z_{k+1}=\left(y_{k+1}+1\right) / 2$. If $y_0=z_0=0$ then we find $y_1=1 / 2, z_1=3 / 4, y_2=7 / 8$,
lyche-numerical-linear-algebra- F03452	266	■ 1.000	$z_{\{2\}}=15 / 16$	$z_2=15 / 16$	$z_2=15 / 16$, and in general $y_k=1-2 \cdot 4^{-k}$ and $z_k=1-4^{-k}$ for $k=1,2,3, \ldots$
lyche-numerical-linear-algebra- F03453	266	■ 1.000	$y_{\{k\}}=1-2 \cdot 4^{\{-k\}}$	$y_k=1-2 \cdot 4^{-k}$	$z_2=15 / 16$, and in general $y_k=1-2 \cdot 4^{-k}$ and $z_k=1-4^{-k}$ for $k=1,2,3, \ldots$
lyche-numerical-linear-algebra- F03454	266	■ 1.000	$z_{\{k\}}=1-4^{\{-k\}}$	$z_k=1-4^{-k}$	$z_2=15 / 16$, and in general $y_k=1-2 \cdot 4^{-k}$ and $z_k=1-4^{-k}$ for $k=1,2,3, \ldots$
lyche-numerical-linear-algebra- F03455	266	■ 1.000	$k=1,2,3, \backslash \mathrm{ldots}$	$k=1,2,3, \ldots$	$z_2=15 / 16$, and in general $y_k=1-2 \cdot 4^{-k}$ and $z_k=1-4^{-k}$ for $k=1,2,3, \ldots$.
lyche-numerical-linear-algebra- F03456	266	■ 1.000	$\boldsymbol{x}_{\{k\}}=\left[\boldsymbol{x}_{\{k\}}(1), \backslash \mathrm{ldots}, \boldsymbol{x}_{\{k\}}(n)\right]^{\wedge\{T\}}$	$\boldsymbol{x}_k=\left[\boldsymbol{x}_k(1), \ldots, \boldsymbol{x}_k(n)\right]^T$	Suppose we know an approximation $\boldsymbol{x}_\ell=\left[\boldsymbol{x}_\ell(1), \ldots, \boldsymbol{x}_\ell(n)\right]^T$ to the exact solution
lyche-numerical-linear-algebra- F03457	266	■ 1.000	$\boldsymbol{x}(i)$	$\boldsymbol{x}(i)$	diagonal elements. Solving the i th equation of $\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}$ for $\boldsymbol{x}(i)$, we obtain a fixed-
lyche-numerical-linear-algebra- F03458	266	■ 1.000	$\boldsymbol{x}_{k+1}(i)$	$\boldsymbol{x}_{k+1}(i)$	where we use the new $\boldsymbol{x}_{k+1}(i)$ immediately after it has been computed.
lyche-numerical-linear-algebra- F03459	266	■ 1.000	$0<\omega<2$	$0<\omega<2$	ducing an acceleration parameter $0<\omega<2$ in the GS method. We write
lyche-numerical-linear-algebra- F03460	266	■ 1.000	$\boldsymbol{x}(i)=\omega \boldsymbol{x}(i)+(1-\omega) \boldsymbol{x}(i)$	$\boldsymbol{x}(i)=\omega \boldsymbol{x}(i)+(1-\omega) \boldsymbol{x}(i)$	$\boldsymbol{x}(i)=\omega \boldsymbol{x}(i)+(1-\omega) \boldsymbol{x}(i)$ and this leads to the method
lyche-numerical-linear-algebra- F03461	266	■ 1.000	$\omega=1$	$\omega=1$	The SOR method reduces to the Gauss-Seidel method for $\omega=1$. Denoting the
lyche-numerical-linear-algebra- F03462	266	■ 1.000	$\boldsymbol{x}_{k+1}^{\{g s\}}$	$\boldsymbol{x}_{k+1}^{g s}$	right hand side of (12.3) by $\boldsymbol{x}_{k+1}^{g s}$ we can write (12.4) as $\boldsymbol{x}_{k+1}=\omega \boldsymbol{x}_{k+1}^{g s}+(1-$
lyche-numerical-linear-algebra- F03463	266	■ 1.000	$\boldsymbol{x}_{k+1}=\omega \boldsymbol{x}_{k+1}^{\{g s\}}+(1-$	$\boldsymbol{x}_{k+1}=\omega \boldsymbol{x}_{k+1}^{g s}+(1-$	right hand side of (12.3) by $\boldsymbol{x}_{k+1}^{g s}$ we can write (12.4) as $\boldsymbol{x}_{k+1}=\omega \boldsymbol{x}_{k+1}^{g s}+(1-$
lyche-numerical-linear-algebra- F03464	266	■ 1.000	$\omega) \boldsymbol{x}_{\{k\}}$	$\omega) \boldsymbol{x}_k$	$\omega) \boldsymbol{x}_k$, and we see that $\boldsymbol{x}_{k+1}$ is located on the straight line passing through the
lyche-numerical-linear-algebra- F03465	266	■ 1.000	$\boldsymbol{x}_{k+1}$	$\boldsymbol{x}_{k+1}$	$\omega) \boldsymbol{x}_k$, and we see that $\boldsymbol{x}_{k+1}$ is located on the straight line passing through the
lyche-numerical-linear-algebra- F03466	267	■ 1.000	$1 \leq \omega<2$	$1 \leq \omega<2$	parameter ω in the range $1 \leq \omega<2$ and then $\boldsymbol{x}_{k+1}$ is computed by linear
lyche-numerical-linear-algebra- F03467	267	■ 1.000	$\boldsymbol{x}_{k+1 / 2}$	$\boldsymbol{x}_{k+1 / 2}$	(12.4), computing an approximation denoted $\boldsymbol{x}_{k+1 / 2}$ instead of $\boldsymbol{x}_{k+1}$, is followed
lyche-numerical-linear-algebra- F03468	267	■ 1.000	$i=n, n-1, \backslash \mathrm{ldots} 1$	$i=n, n-1, \ldots 1$	in the order $i=n, n-1, \ldots 1$. The method is slower and more complicated
lyche-numerical-linear-algebra- F03469	267	■ 0.999	$\boldsymbol{v}(i, j)$	$\boldsymbol{v}(i, j)$	with homogenous boundary conditions also given in (10.4). Solving for $\boldsymbol{v}(i, j)$ we
lyche-numerical-linear-algebra- F03470	267	■ 0.570	$e_{\{i, j\}}:=f_{\{i, j\}} /\left(m+1\right)^2$	$e_{i, j}:=f_{i, j} /\left(m+1\right)^2$	where $e_{i, j}:=f_{i, j} /\left(m+1\right)^2$. The J, GS, and SOR methods take the form

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lyche-numerical-linear-algebra-268-F03476	268	■ 0.995	$\backslash\mathrm{left}(i_{\{1\}}, j_{\{1\}}\backslash\mathrm{right})<$	$(i_1, j_1) <$	We note that for GS and SOR we have used the natural ordering , i.e., $(i_1, j_1) <$
lyche-numerical-linear-algebra-268-F03477	268	■ 0.960	$i_{\{2\}}, j_{\{2\}}$	i_2, j_2	(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering
lyche-numerical-linear-algebra-268-F03478	268	■ 0.960	$j_{\{1\}} \backslash\mathrm{leq} j_{\{2\}}$	$j_1 \leq j_2$	(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering
lyche-numerical-linear-algebra-268-F03479	268	■ 0.960	$i_{\{1\}}<i_{\{2\}}$	$i_1 < i_2$	(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering
lyche-numerical-linear-algebra-268-F03480	268	■ 0.960	$j_{\{1\}}=j_{\{2\}}$	$j_1 = j_2$	(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering
lyche-numerical-linear-algebra-268-F03481	268	■ 1.000	$\backslash\mathrm{boldsymbol{F}}=\backslash\mathrm{left}(f_{\{i, j\}}\backslash\mathrm{right}) \backslash\mathrm{in} \backslash\mathrm{mathbb{R}}^{\{m \times m\}}$	$\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m \times m}$	on the linear system (12.6) with $\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m \times m}$, starting with $\boldsymbol{V}_0 = \mathbf{0} \in$
lyche-numerical-linear-algebra-268-F03482	268	■ 1.000	$\backslash\mathrm{boldsymbol{V}}_{\{0\}}=\backslash\mathrm{mathbf{0}} \backslash\mathrm{in}$	$\boldsymbol{V}_0 = \mathbf{0} \in$	on the linear system (12.6) with $\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m \times m}$, starting with $\boldsymbol{V}_0 = \mathbf{0} \in$
lyche-numerical-linear-algebra-268-F03483	268	■ 1.000	$\backslash\mathrm{mathbb{R}}^{\{(m+2) \times (m+2)\}}$	$\mathbb{R}^{(m+2) \times (m+2)}$	$\mathbb{R}^{(m+2) \times (m+2)}$. The output is the number of iterations k , to obtain $\ \boldsymbol{V}^{(k)} - \boldsymbol{U}\ _M :=$
lyche-numerical-linear-algebra-268-F03484	268	■ 1.000	$\backslash\mathrm{left} \backslash\mathrm{boldsymbol{V}}^{\{(k)\}}-\backslash\mathrm{boldsymbol{U}}\backslash\mathrm{right} \backslash\mathrm{in} \backslash\mathrm{mathbb{R}}^{\{(m+2) \times (m+2)\}}:=$	$\ \boldsymbol{V}^{(k)} - \boldsymbol{U}\ _M :=$	$\mathbb{R}^{(m+2) \times (m+2)}$. The output is the number of iterations k , to obtain $\ \boldsymbol{V}^{(k)} - \boldsymbol{U}\ _M :=$
lyche-numerical-linear-algebra-268-F03485	268	■ 0.911	$\backslash\mathrm{max}_{\{i, j\}}\backslash\mathrm{left} v_{\{i j\}}-u_{\{i j\}}\backslash\mathrm{right} <$	$\max_{i,j} v_{ij} - u_{ij} <$	$\max_{i,j} v_{ij} - u_{ij} < tol$. Here $[u_{ij}] \in \mathbb{R}^{(m+2) \times (m+2)}$ is the "exact" solution of (12.6)
lyche-numerical-linear-algebra-268-F03486	268	■ 0.911	$\backslash\mathrm{left}[u_{\{i j\}}\backslash\mathrm{right}] \backslash\mathrm{in} \backslash\mathrm{mathbb{R}}^{\{(m+2) \times (m+2)\}}$	$[u_{ij}] \in \mathbb{R}^{(m+2) \times (m+2)}$	$\max_{i,j} v_{ij} - u_{ij} < tol$. Here $[u_{ij}] \in \mathbb{R}^{(m+2) \times (m+2)}$ is the "exact" solution of (12.6)
lyche-numerical-linear-algebra-268-F03487	268	■ 0.843	$k=k_{\{n\}}$	$k = k_n$	$k = k_n$ from this algorithm using $\boldsymbol{F} = \mathrm{ones}(m, m)$ for $m = 10, 50, K = 10^4$, and
lyche-numerical-linear-algebra-268-F03488	268	■ 0.843	$\backslash\mathrm{boldsymbol{F}}=\backslash\mathrm{operatorname{ones}}(m, m)$	$\boldsymbol{F} = \mathrm{ones}(m, m)$	$k = k_n$ from this algorithm using $\boldsymbol{F} = \mathrm{ones}(m, m)$ for $m = 10, 50, K = 10^4$, and
lyche-numerical-linear-algebra-268-F03489	268	■ 0.843	$m=10,50, K=10^{\{4\}}$	$m = 10, 50, K = 10^4$	$k = k_n$ from this algorithm using $\boldsymbol{F} = \mathrm{ones}(m, m)$ for $m = 10, 50, K = 10^4$, and
lyche-numerical-linear-algebra-268-F03490	268	■ 0.825	$=10^{\{-8\}}$	$= 10^{-8}$	$tol = 10^{-8}$. We also show the number of iterations for Gauss-Seidel and SOR with
lyche-numerical-linear-algebra-268-F03491	268	■ 1.000	$\backslash\mathrm{omega}^{\{*\}}:=2 \backslash\mathrm{left}(1+\backslash\sin \backslash\mathrm{left}(\backslash\mathrm{frac}\{\pi\}\{m+1\}\backslash\mathrm{right}) \backslash\mathrm{right})$	$\omega^* := 2 / \left(1 + \sin \left(\frac{\pi}{m+1} \right) \right)$	a value of ω known as the optimal acceleration parameter $\omega^* := 2 / (1 + \sin(\frac{\pi}{m+1}))$.
lyche-numerical-linear-algebra-269-F03492	269	■ 1.000	$0<w<2$	$0 < w < 2$	is an acceleration parameter with $0 < w < 2$. For $w = 1$ we obtain Gauss-Seidel's
lyche-numerical-linear-algebra-269-F03493	269	■ 1.000	$w=1$	$w = 1$	is an acceleration parameter with $0 < w < 2$. For $w = 1$ we obtain Gauss-Seidel's
lyche-numerical-linear-algebra-269-F03494	269	■ 1.000	$k_{\{n\}}=O(n)$	$k_n = O(n)$	$k_n = O(n)$ for the J and GS method and $k_n = O(\sqrt{n})$ for SOR with optimal ω .
lyche-numerical-linear-algebra-269-F03495	269	■ 1.000	$k_{\{n\}}=O(\backslash\mathrm{sqrt}\{n\})$	$k_n = O(\sqrt{n})$	$k_n = O(n)$ for the J and GS method and $k_n = O(\sqrt{n})$ for SOR with optimal ω .
lyche-numerical-linear-algebra-269-F03496	269	■ 1.000	$\backslash\mathrm{sqrt}\{n\}$	$\sqrt{n}$	The choice of tol will only influence the constants multiplying n or $\sqrt{n}$.
lyche-numerical-linear-algebra-269-F03497	269	■ 1.000	$O\backslash\mathrm{left}(k_{\{n\}} \backslash\mathrm{times} n\backslash\mathrm{right})$	$O(k_n \times n)$	$O(k_n \times n)$. Therefore the number of arithmetic operations for the J and GS
lyche-numerical-linear-algebra-270-F03498	270	■ 1.000	$\backslash\mathrm{boldsymbol{G}} \backslash\mathrm{in} \backslash\mathrm{mathbb{C}}^{\{n \times n\}}$	$\boldsymbol{G} \in \mathbb{C}^{n \times n}$	$\{\boldsymbol{x}_k\}$ of vectors in $\mathbb{C}^n$. For a general $\boldsymbol{G} \in \mathbb{C}^{n \times n}$ and $\boldsymbol{c} \in \mathbb{C}^n$ a solution of $\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$
lyche-numerical-linear-algebra-270-F03499	270	■ 1.000	$\backslash\mathrm{boldsymbol{x}}=\backslash\mathrm{boldsymbol{G}} \backslash\mathrm{boldsymbol{x}}+\backslash\mathrm{boldsymbol{c}}$	$\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$	$\{\boldsymbol{x}_k\}$ of vectors in $\mathbb{C}^n$. For a general $\boldsymbol{G} \in \mathbb{C}^{n \times n}$ and $\boldsymbol{c} \in \mathbb{C}^n$ a solution of $\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$
lyche-numerical-linear-algebra-270-F03500	270	■ 1.000	$\backslash\mathrm{boldsymbol{I}}-\backslash\mathrm{boldsymbol{G}}$	$\boldsymbol{I} - \boldsymbol{G}$	is called a fixed-point . The fixed-point is unique if $\boldsymbol{I} - \boldsymbol{G}$ is nonsingular.
lyche-numerical-linear-algebra-270-F03501	270	■ 1.000	$\backslash\mathrm{lim}_{\{k \rightarrow\infty\}} \backslash\mathrm{boldsymbol{x}}_{\{k\}}=\backslash\mathrm{boldsymbol{x}}$	$\lim_{k \rightarrow \infty} \boldsymbol{x}_k = \boldsymbol{x}$	If $\lim_{k \rightarrow \infty} \boldsymbol{x}_k = \boldsymbol{x}$ for some $\boldsymbol{x} \in \mathbb{C}^n$ then $\boldsymbol{x}$ is a fixed point since
lyche-numerical-linear-algebra-270-F03502	270	■ 1.000	$\backslash\mathrm{boldsymbol{M}}^{\{-1\}} \backslash\mathrm{boldsymbol{A}} \backslash\mathrm{boldsymbol{x}}=\backslash\mathrm{boldsymbol{M}}^{\{-1\}} \backslash\mathrm{boldsymbol{b}}$	$\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ in the equivalent form $\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{M}^{-1} \boldsymbol{b}$ or $\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} + \boldsymbol{M}^{-1} \boldsymbol{b}$.

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lyche-numerical-linear-algebra- F03503	270	1.000	$\boldsymbol{x}=\boldsymbol{x}-\boldsymbol{M}^{-1}\boldsymbol{A}\boldsymbol{x}+\boldsymbol{M}^{-1}\boldsymbol{b}$ $\boldsymbol{A} \ \boldsymbol{x}+\boldsymbol{M}^{-1}$ $\boldsymbol{b}$	$\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{M}^{-1}\boldsymbol{b}$	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ in the equivalent form $\boldsymbol{M}^{-1}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{M}^{-1}\boldsymbol{b}$ or $\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{M}^{-1}\boldsymbol{b}$.
lyche-numerical-linear-algebra- F03504	270	1.000	$\boldsymbol{A}=\boldsymbol{D}-\boldsymbol{A}_L-\boldsymbol{A}_R$ $\boldsymbol{L}}-\boldsymbol{A}_\boldsymbol{R}$	$\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}_R$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}_R$, where $-\boldsymbol{A}_L$, $\boldsymbol{D}$, and $-\boldsymbol{A}_R$ are the lower,
lyche-numerical-linear-algebra- F03505	270	1.000	$-\boldsymbol{A}_L,\boldsymbol{D}$	$-\boldsymbol{A}_L, \boldsymbol{D}$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}_R$, where $-\boldsymbol{A}_L$, $\boldsymbol{D}$, and $-\boldsymbol{A}_R$ are the lower,
lyche-numerical-linear-algebra- F03506	270	1.000	$-\boldsymbol{A}_R$	$-\boldsymbol{A}_R$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}_R$, where $-\boldsymbol{A}_L$, $\boldsymbol{D}$, and $-\boldsymbol{A}_R$ are the lower,
lyche-numerical-linear-algebra- F03507	270	1.000	$\boldsymbol{D}:=\operatorname{diag}(\boldsymbol{a}_{11}, \ldots, \boldsymbol{a}_{nn})$ $\boldsymbol{a}_{nn})\right)$	$\boldsymbol{D} := \operatorname{diag}(a_{11}, \dots, a_{nn})$	diagonal, and upper part of $\boldsymbol{A}$, respectively. Thus $\boldsymbol{D} := \operatorname{diag}(a_{11}, \dots, a_{nn})$,
lyche-numerical-linear-algebra- F03508	271	0.753	$\boldsymbol{M}_J,\boldsymbol{M}_1$	$\boldsymbol{M}_J, \boldsymbol{M}_1$	$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J , GS and SOR methods are given by
lyche-numerical-linear-algebra- F03509	271	0.753	$\boldsymbol{M}_\omega$	$\boldsymbol{M}_\omega$	$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J , GS and SOR methods are given by
lyche-numerical-linear-algebra- F03510	271	0.753	J, G S	J, GS	$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J , GS and SOR methods are given by
lyche-numerical-linear-algebra- F03511	271	0.955	$\boldsymbol{D} \ \boldsymbol{x}_{k+1}=(\boldsymbol{D}-\boldsymbol{A}) \boldsymbol{x}_k+\boldsymbol{b}$ $\boldsymbol{A}) \ \boldsymbol{x}_{k+1}+\boldsymbol{b}$	$\boldsymbol{D}\boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$	obtain the matrix form $\boldsymbol{D}\boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$ showing that $\boldsymbol{M}_J = \boldsymbol{D}$.
lyche-numerical-linear-algebra- F03512	271	0.955	$\boldsymbol{M}_J=\boldsymbol{D}$	$\boldsymbol{M}_J = \boldsymbol{D}$	obtain the matrix form $\boldsymbol{D}\boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$ showing that $\boldsymbol{M}_J = \boldsymbol{D}$.
lyche-numerical-linear-algebra- F03513	271	1.000	$\boldsymbol{A}_L \boldsymbol{x}_{k+1}$	$\boldsymbol{A}_L\boldsymbol{x}_{k+1}$	Dividing both sides by ω and moving $\boldsymbol{A}_L\boldsymbol{x}_{k+1}$ to the left hand side this takes the
lyche-numerical-linear-algebra- F03514	271	1.000	$(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_L) \boldsymbol{x}_{k+1}=\boldsymbol{A}_R \boldsymbol{x}_k+\boldsymbol{b}+(\omega^{-1}-1) \boldsymbol{D} \boldsymbol{x}_k$ $\right) \ \boldsymbol{x}_{k+1}=\boldsymbol{A}_\boldsymbol{R}$ $\boldsymbol{x}_{k+1}+\boldsymbol{b}+\left(\omega^{-1}-1\right. \boldsymbol{D} \boldsymbol{x}_k$ $\right) \ \boldsymbol{D} \ \boldsymbol{x}_{k+1}$	$(\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{x}_{k+1} = \boldsymbol{A}_R\boldsymbol{x}_k + \boldsymbol{b} + (\omega^{-1} - 1)\boldsymbol{D}\boldsymbol{x}_k$	form $(\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{x}_{k+1} = \boldsymbol{A}_R\boldsymbol{x}_k + \boldsymbol{b} + (\omega^{-1} - 1)\boldsymbol{D}\boldsymbol{x}_k$ showing that $\boldsymbol{M}_\omega =$
lyche-numerical-linear-algebra- F03515	271	1.000	$\boldsymbol{M}_\omega=$	$\boldsymbol{M}_\omega =$	form $(\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{x}_{k+1} = \boldsymbol{A}_R\boldsymbol{x}_k + \boldsymbol{b} + (\omega^{-1} - 1)\boldsymbol{D}\boldsymbol{x}_k$ showing that $\boldsymbol{M}_\omega =$
lyche-numerical-linear-algebra- F03516	271	1.000	$\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_L$	$\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L$	$\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L$. We obtain $\boldsymbol{M}_1$ by letting $\omega = 1$ in $\boldsymbol{M}_\omega$.
lyche-numerical-linear-algebra- F03517	271	1.000	$\boldsymbol{M}_1$	$\boldsymbol{M}_1$	$\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L$. We obtain $\boldsymbol{M}_1$ by letting $\omega = 1$ in $\boldsymbol{M}_\omega$.
lyche-numerical-linear-algebra- F03518	271	1.000	$\boldsymbol{G}_\omega=\boldsymbol{I}-\boldsymbol{M}_\omega^{-1} \boldsymbol{A}$ $\omega)^{-1} \ \boldsymbol{A}$	$\boldsymbol{G}_\omega = \boldsymbol{I} - \boldsymbol{M}_\omega^{-1}\boldsymbol{A}$	The iteration matrix $\boldsymbol{G}_\omega = \boldsymbol{I} - \boldsymbol{M}_\omega^{-1}\boldsymbol{A}$ is given by
lyche-numerical-linear-algebra- F03519	272	1.000	$\boldsymbol{x}_{k+1}(1)$	$\boldsymbol{x}_{k+1}(1)$	Substituting the value of $\boldsymbol{x}_{k+1}(1)$ from the first equation into the second equation
lyche-numerical-linear-algebra- F03520	272	1.000	$\boldsymbol{x}_{k+1}:=\boldsymbol{G} \boldsymbol{x}_k+\boldsymbol{c}$ $\boldsymbol{c}$	$\boldsymbol{x}_{k+1} := \boldsymbol{G}\boldsymbol{x}_k + \boldsymbol{c}$	for methods of the form $\boldsymbol{x}_{k+1} := \boldsymbol{G}\boldsymbol{x}_k + \boldsymbol{c}$.
lyche-numerical-linear-algebra- F03521	272	1.000	$\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^n$	$\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^n$	In the following we assume that $\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^n$ and $\boldsymbol{I} - \boldsymbol{G}$ is nonsingular.
lyche-numerical-linear-algebra- F03522	272	1.000	$\lim _k \rightarrow \infty \boldsymbol{G}^k=\mathbf{0}$	$\lim_{k \rightarrow \infty} \boldsymbol{G}^k = \mathbf{0}$	$\boldsymbol{x}_{k+1} := \boldsymbol{G}\boldsymbol{x}_k + \boldsymbol{c}$ converges if and only if $\lim_{k \rightarrow \infty} \boldsymbol{G}^k = \mathbf{0}$.
lyche-numerical-linear-algebra- F03523	272	0.991	$\boldsymbol{x}_{k+1}=\boldsymbol{G} \boldsymbol{x}_k+\boldsymbol{c}$ $\boldsymbol{c}$	$\boldsymbol{x}_{k+1} = \boldsymbol{G}\boldsymbol{x}_k + \boldsymbol{c}$	$\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$ from $\boldsymbol{x}_{k+1} = \boldsymbol{G}\boldsymbol{x}_k + \boldsymbol{c}$ The vector $\boldsymbol{c}$ cancels and we obtain $\boldsymbol{x}_{k+1} - \boldsymbol{x} =$
lyche-numerical-linear-algebra- F03524	272	0.991	$\boldsymbol{x}_{k+1}-\boldsymbol{x}=$	$\boldsymbol{x}_{k+1} - \boldsymbol{x} =$	$\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$ from $\boldsymbol{x}_{k+1} = \boldsymbol{G}\boldsymbol{x}_k + \boldsymbol{c}$. The vector $\boldsymbol{c}$ cancels and we obtain $\boldsymbol{x}_{k+1} - \boldsymbol{x} =$
lyche-numerical-linear-algebra- F03525	272	1.000	$\boldsymbol{G}\left(\boldsymbol{x}_k-\boldsymbol{x}\right)$	$\boldsymbol{G}(\boldsymbol{x}_k - \boldsymbol{x})$	$\boldsymbol{G}(\boldsymbol{x}_k - \boldsymbol{x})$. By induction on k
lyche-numerical-linear-algebra- F03526	272	0.999	$\boldsymbol{x}_k-\boldsymbol{x} \rightarrow \mathbf{0}$	$\boldsymbol{x}_k - \boldsymbol{x} \rightarrow \mathbf{0}$	Clearly $\boldsymbol{x}_k - \boldsymbol{x} \rightarrow \mathbf{0}$ if $\boldsymbol{G}^k \rightarrow \mathbf{0}$ and the method converges. For the converse we let
lyche-numerical-linear-algebra- F03527	272	0.999	$\boldsymbol{G}^k \rightarrow \mathbf{0}$	$\boldsymbol{G}^k \rightarrow \mathbf{0}$	Clearly $\boldsymbol{x}_k - \boldsymbol{x} \rightarrow \mathbf{0}$ if $\boldsymbol{G}^k \rightarrow \mathbf{0}$ and the method converges. For the converse we let

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lyche-numerical-linear-algebra-272 F03528	272	1.000	$\backslash\boldsymbol{x}_{\{0\}}-\backslash\boldsymbol{x}=\backslash\boldsymbol{e}_{\{j\}}$	$\boldsymbol{x}_0 - \boldsymbol{x} = \boldsymbol{e}_j$	$\boldsymbol{x}$ be as before and choose $\boldsymbol{x}_0 - \boldsymbol{x} = \boldsymbol{e}_j$, the j th unit vector for $j = 1, \dots, n$. Since
lyche-numerical-linear-algebra-272 F03529	272	1.000	$\backslash\boldsymbol{G}^{\{k\}} \backslash\boldsymbol{e}_{\{j\}} \rightarrow \backslash\mathbf{0}$	$\boldsymbol{G}^k \boldsymbol{e}_j \rightarrow \mathbf{0}$	$\boldsymbol{x}_k - \boldsymbol{x} \rightarrow \mathbf{0}$ for any $\boldsymbol{x}_0$ we have $\boldsymbol{G}^k \boldsymbol{e}_j \rightarrow \mathbf{0}$ for $j = 1, \dots, n$ which implies that
lyche-numerical-linear-algebra-272 F03530	272	1.000	$\ \backslash\boldsymbol{G} \ < 1$	$\ \boldsymbol{G}\ < 1$	Theorem 12.3 (Sufficient Condition for Convergence) <i>If $\ \boldsymbol{G}\ < 1$ then the</i>
lyche-numerical-linear-algebra-273 F03531	273	1.000	$\rho(\backslash\boldsymbol{G}) < 1$	$\rho(\boldsymbol{G}) < 1$	<i>and only if $\rho(\boldsymbol{G}) < 1$.</i>
lyche-numerical-linear-algebra-273 F03532	273	1.000	$\backslash\boldsymbol{r}_{\{k\}}:=\backslash\boldsymbol{b}-\backslash\boldsymbol{A}\boldsymbol{x}_{\{k\}}$	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$	adding a multiple of the residual vector $\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$. Note that we do not need
lyche-numerical-linear-algebra-273 F03533	273	0.992	$\backslash\lambda_{\{1\}} \geq \backslash\lambda_{\{2\}} \geq \backslash\cdots \geq \backslash\lambda_{\{n\}} > 0$	$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$	values $\lambda_1 > \lambda_2 > \dots > \lambda_n > 0$ then the R method given by $\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$
lyche-numerical-linear-algebra-273 F03534	273	0.992	$\backslash\boldsymbol{x}_{\{k+1\}}=(\backslash\boldsymbol{I}-\backslash\alpha \backslash\boldsymbol{A}) \backslash\boldsymbol{x}_{\{k\}}+\backslash\boldsymbol{b}$	$\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$	values $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$ then the R method given by $\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$
lyche-numerical-linear-algebra-273 F03535	273	1.000	$0 < \backslash\alpha < 2 / \backslash\lambda_{\{1\}}$	$0 < \alpha < 2/\lambda_1$	converges if and only if $0 < \alpha < 2/\lambda_1$. Moreover,
lyche-numerical-linear-algebra-274 F03536	274	1.000	$\backslash\boldsymbol{I}-\backslash\alpha \backslash\boldsymbol{A}$	$\boldsymbol{I} - \alpha \boldsymbol{A}$	Proof The eigenvalues of $\boldsymbol{I} - \alpha \boldsymbol{A}$ are $1 - \alpha \lambda_j$, $j = 1, \dots, n$. We have
lyche-numerical-linear-algebra-274 F03537	274	1.000	$1-\backslash\alpha \backslash\lambda_{\{j\}}, \quad \backslash\text{quad } j=1, \backslash\text{ldots}, n$	$1 - \alpha \lambda_j, \quad j = 1, \dots, n$	Proof The eigenvalues of $\boldsymbol{I} - \alpha \boldsymbol{A}$ are $1 - \alpha \lambda_j$, $j = 1, \dots, n$. We have
lyche-numerical-linear-algebra-274 F03538	274	1.000	$1-\backslash\alpha \backslash\lambda_{\{n\}}=\backslash\alpha \backslash\lambda_{\{1\}}-1$	$1 - \alpha \lambda_n = \alpha \lambda_1 - 1$	see Fig. 12.1. Clearly $1 - \alpha \lambda_n = \alpha \lambda_1 - 1$ for $\alpha = \alpha_o$ and
lyche-numerical-linear-algebra-274 F03539	274	1.000	$\backslash\alpha=\backslash\alpha_{\{o\}}$	$\alpha = \alpha_o$	see Fig. 12.1. Clearly $1 - \alpha \lambda_n = \alpha \lambda_1 - 1$ for $\alpha = \alpha_o$ and
lyche-numerical-linear-algebra-274 F03540	274	1.000	$\rho_{\{\alpha\}} < 1$	$\rho_\alpha < 1$	We have $\rho_\alpha < 1$ if and only if $\alpha > 0$ and $\alpha \lambda_1 - 1 < 1$ showing convergence if
lyche-numerical-linear-algebra-274 F03541	274	1.000	$\backslash\alpha \backslash\lambda_{\{1\}}-1 < 1$	$\alpha \lambda_1 - 1 < 1$	We have $\rho_\alpha < 1$ if and only if $\alpha > 0$ and $\alpha \lambda_1 - 1 < 1$ showing convergence if
lyche-numerical-linear-algebra-274 F03542	274	1.000	$\rho_{\{\alpha\}} > \rho_{\{\alpha_o\}}$	$\rho_\alpha > \rho_{\alpha_o}$	and only if $0 < \alpha < 2/\lambda_1$ and $\rho_\alpha > \rho_{\alpha_o}$ for $\alpha \leq 0$ and $\alpha \geq 2/\lambda_1$. Finally, if
lyche-numerical-linear-algebra-274 F03543	274	1.000	$\backslash\alpha \leq 0$	$\alpha \leq 0$	and only if $0 < \alpha < 2/\lambda_1$ and $\rho_\alpha > \rho_{\alpha_o}$ for $\alpha \leq 0$ and $\alpha \geq 2/\lambda_1$. Finally, if
lyche-numerical-linear-algebra-274 F03544	274	1.000	$\backslash\alpha \geq 2 / \backslash\lambda_{\{1\}}$	$\alpha \geq 2/\lambda_1$	and only if $0 < \alpha < 2/\lambda_1$ and $\rho_\alpha > \rho_{\alpha_o}$ for $\alpha \leq 0$ and $\alpha \geq 2/\lambda_1$. Finally, if
lyche-numerical-linear-algebra-274 F03545	274	1.000	$0 < \backslash\alpha < \backslash\alpha_{\{o\}}$	$0 < \alpha < \alpha_o$	$0 < \alpha < \alpha_o$ then $\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$ and if $\alpha_o < \alpha < 2/\lambda_1$ then
lyche-numerical-linear-algebra-274 F03546	274	1.000	$\rho_{\{\alpha\}}=1-\backslash\alpha \backslash\lambda_{\{n\}}>1-\backslash\alpha_o \backslash\lambda_{\{n\}}=\rho_{\{\alpha_o\}}$	$\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$	$0 < \alpha < \alpha_o$ then $\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$ and if $\alpha_o < \alpha < 2/\lambda_1$ then
lyche-numerical-linear-algebra-274 F03547	274	1.000	$\backslash\alpha_{\{o\}} < \backslash\alpha < 2 / \backslash\lambda_{\{1\}}$	$\alpha_o < \alpha < 2/\lambda_1$	$0 < \alpha < \alpha_o$ then $\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$ and if $\alpha_o < \alpha < 2/\lambda_1$ then
lyche-numerical-linear-algebra-274 F03548	274	0.998	$\rho_{\{\alpha\}}=\backslash\alpha \backslash\lambda_{\{1\}}-1>\backslash\alpha_o \backslash\lambda_{\{1\}}-1=\rho_{\{\alpha_o\}}$	$\rho_\alpha = \alpha \lambda_1 - 1 > \alpha_o \lambda_1 - 1 = \rho_{\alpha_o}$	$\rho_\alpha = \alpha \lambda_1 - 1 > \alpha_o \lambda_1 - 1 = \rho_{\alpha_o}$.
lyche-numerical-linear-algebra-274 F03549	274	0.964	$\backslash\mathbf{R}$	$\mathbf{R}$	Corollary 12.1 (Rate of Convergence for the R Method) <i>Suppose $\boldsymbol{A}$ is positive</i>
lyche-numerical-linear-algebra-274 F03550	274	0.991	$\backslash\lambda_{\{\text{max}\}}$	$\lambda_{\max}$	<i>definite with largest and smallest eigenvalue $\lambda_{\max}$ and $\lambda_{\min}$, respectively. Richard-</i>
lyche-numerical-linear-algebra-274 F03551	274	0.999	$0 < \backslash\alpha < 2 / \backslash\lambda_{\{\text{max}\}}$	$0 < \alpha < 2/\lambda_{\max}$	<i>son's method $\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$ converges if and only if $0 < \alpha < 2/\lambda_{\max}$.</i>
lyche-numerical-linear-algebra-274 F03552	274	1.000	$\backslash\alpha=\backslash\alpha_{\{o\}}:=\frac{2}{\backslash\lambda_{\{\text{max}\}}+\backslash\lambda_{\{\text{min}\}}}$	$\alpha = \alpha_o := \frac{2}{\lambda_{\max} + \lambda_{\min}}$	<i>With $\alpha = \alpha_o := \frac{2}{\lambda_{\max} + \lambda_{\min}}$ we have the error estimate</i>
lyche-numerical-linear-algebra-274 F03553	274	0.960	$\backslash\kappa:=\backslash\lambda_{\{\text{max}\}} / \backslash\lambda_{\{\text{min}\}}$	$\kappa := \lambda_{\max} / \lambda_{\min}$	<i>where $\kappa := \lambda_{\max} / \lambda_{\min}$ is the spectral condition number of $\boldsymbol{A}$.</i>
lyche-numerical-linear-algebra-274 F03554	274	0.988	$\backslash\left\ \backslash\right\ _{\{2\}} \backslash\right\ $	$\ \cdot\ _2$	Proof The spectral norm $\ \cdot\ _2$ is consistent and therefore $\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} -$

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lyche-numerical-linear-algebra- F03555	274	■ 0.988	$\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} - \boldsymbol{A}\ _2$	$\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} - \boldsymbol{A}\ _2$	Proof The spectral norm $\ \cdot\ _2$ is consistent and therefore $\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} - \boldsymbol{A}\ _2 \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2$. But for a positive definite matrix the spectral norm is equal to
lyche-numerical-linear-algebra- F03556	274	■ 0.994	$\alpha_o \ \boldsymbol{A}\ _2^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2$	$\alpha_o \ \boldsymbol{A}\ _2^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2$	
lyche-numerical-linear-algebra- F03557	275	■ 1.000	$\omega \in (0, 2)$	$\omega \in (0, 2)$	The condition $\omega \in (0, 2)$ is necessary for convergence of the SOR method.
lyche-numerical-linear-algebra- F03558	275	■ 1.000	$\boldsymbol{D} \boldsymbol{x}_{k+1} = \omega (\boldsymbol{A}_L \boldsymbol{x}_{k+1} + \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b}) + (1 - \omega) \boldsymbol{D} \boldsymbol{x}_k$	$\boldsymbol{D} \boldsymbol{x}_{k+1} = \omega (\boldsymbol{A}_L \boldsymbol{x}_{k+1} + \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b}) + (1 - \omega) \boldsymbol{D} \boldsymbol{x}_k$	Proof We have (cf. (12.13)) $\boldsymbol{D} \boldsymbol{x}_{k+1} = \omega (\boldsymbol{A}_L \boldsymbol{x}_{k+1} + \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b}) + (1 - \omega) \boldsymbol{D} \boldsymbol{x}_k$
lyche-numerical-linear-algebra- F03559	275	■ 1.000	$\boldsymbol{x}_{k+1} = \omega (\boldsymbol{L} \boldsymbol{x}_{k+1} + \boldsymbol{R} \boldsymbol{x}_k + \boldsymbol{D}^{-1} \boldsymbol{b}) + (1 - \omega) \boldsymbol{x}_k$	$\boldsymbol{x}_{k+1} = \omega (\boldsymbol{L} \boldsymbol{x}_{k+1} + \boldsymbol{R} \boldsymbol{x}_k + \boldsymbol{D}^{-1} \boldsymbol{b}) + (1 - \omega) \boldsymbol{x}_k$	or $\boldsymbol{x}_{k+1} = \omega (\boldsymbol{L} \boldsymbol{x}_{k+1} + \boldsymbol{R} \boldsymbol{x}_k + \boldsymbol{D}^{-1} \boldsymbol{b}) + (1 - \omega) \boldsymbol{x}_k$, where $\boldsymbol{L} := \boldsymbol{D}^{-1} \boldsymbol{A}_L$ and
lyche-numerical-linear-algebra- F03560	275	■ 1.000	$\boldsymbol{L} := \boldsymbol{D}^{-1} \boldsymbol{A}_L$	$\boldsymbol{L} := \boldsymbol{D}^{-1} \boldsymbol{A}_L$	or $\boldsymbol{x}_{k+1} = \omega (\boldsymbol{L} \boldsymbol{x}_{k+1} + \boldsymbol{R} \boldsymbol{x}_k + \boldsymbol{D}^{-1} \boldsymbol{b}) + (1 - \omega) \boldsymbol{x}_k$, where $\boldsymbol{L} := \boldsymbol{D}^{-1} \boldsymbol{A}_L$ and
lyche-numerical-linear-algebra- F03561	275	■ 1.000	$\boldsymbol{R} := \boldsymbol{D}^{-1} \boldsymbol{A}_R$	$\boldsymbol{R} := \boldsymbol{D}^{-1} \boldsymbol{A}_R$	$\boldsymbol{R} := \boldsymbol{D}^{-1} \boldsymbol{A}_R$. Thus $(\boldsymbol{I} - \omega \boldsymbol{L}) \boldsymbol{x}_{k+1} = (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}) \boldsymbol{x}_k + \omega \boldsymbol{D}^{-1} \boldsymbol{b}$ so the
lyche-numerical-linear-algebra- F03562	275	■ 1.000	$(\boldsymbol{I} - \omega \boldsymbol{L}) \boldsymbol{x}_{k+1} = (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}) \boldsymbol{x}_k + \omega \boldsymbol{D}^{-1} \boldsymbol{b}$	$(\boldsymbol{I} - \omega \boldsymbol{L}) \boldsymbol{x}_{k+1} = (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}) \boldsymbol{x}_k + \omega \boldsymbol{D}^{-1} \boldsymbol{b}$	$\boldsymbol{R} := \boldsymbol{D}^{-1} \boldsymbol{A}_R$. Thus $(\boldsymbol{I} - \omega \boldsymbol{L}) \boldsymbol{x}_{k+1} = (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}) \boldsymbol{x}_k + \omega \boldsymbol{D}^{-1} \boldsymbol{b}$ so the
lyche-numerical-linear-algebra- F03563	275	■ 1.000	$\boldsymbol{G}_\omega$	$\boldsymbol{G}_\omega$	We next compute the determinant of $\boldsymbol{G}_\omega$. Since $\boldsymbol{I} - \omega \boldsymbol{L}$ is lower triangular with
lyche-numerical-linear-algebra- F03564	275	■ 1.000	$\boldsymbol{I} - \omega \boldsymbol{L}$	$\boldsymbol{I} - \omega \boldsymbol{L}$	We next compute the determinant of $\boldsymbol{G}_\omega$. Since $\boldsymbol{I} - \omega \boldsymbol{L}$ is lower triangular with
lyche-numerical-linear-algebra- F03565	275	■ 1.000	$\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}$	$\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}$	the determinant of this matrix is equal to one. The matrix $\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}$ is upper
lyche-numerical-linear-algebra- F03566	275	■ 1.000	$1 - \omega$	$1 - \omega$	triangular with $1 - \omega$ on the diagonal and therefore its determinant equals $(1 - \omega)^n$.
lyche-numerical-linear-algebra- F03567	275	■ 1.000	$(1 - \omega)^n$	$(1 - \omega)^n$	triangular with $1 - \omega$ on the diagonal and therefore its determinant equals $(1 - \omega)^n$.
lyche-numerical-linear-algebra- F03568	275	■ 0.998	$\det(\boldsymbol{G}_\omega) = (1 - \omega)^n$	$\det(\boldsymbol{G}_\omega) = (1 - \omega)^n$	It follows that $\det(\boldsymbol{G}_\omega) = (1 - \omega)^n$. Since the determinant of a matrix equals the
lyche-numerical-linear-algebra- F03569	275	■ 1.000	$ \lambda \geq 1 - \omega $	$ \lambda \geq 1 - \omega $	product of its eigenvalues we must have $ \lambda \geq 1 - \omega $ for at least one eigenvalue λ
lyche-numerical-linear-algebra- F03570	275	■ 1.000	$\rho(\boldsymbol{G}_\omega) \geq \omega - 1 $	$\rho(\boldsymbol{G}_\omega) \geq \omega - 1 $	of $\boldsymbol{G}_\omega$ and we conclude that $\rho(\boldsymbol{G}_\omega) \geq \omega - 1 $. But then $\rho(\boldsymbol{G}_\omega) \geq 1$ if ω is not in
lyche-numerical-linear-algebra- F03571	275	■ 1.000	$\rho(\boldsymbol{G}_\omega) \geq 1$	$\rho(\boldsymbol{G}_\omega) \geq 1$	of $\boldsymbol{G}_\omega$ and we conclude that $\rho(\boldsymbol{G}_\omega) \geq \omega - 1 $. But then $\rho(\boldsymbol{G}_\omega) \geq 1$ if ω is not in
lyche-numerical-linear-algebra- F03572	275	■ 1.000	$ \lambda < 1$	$ \lambda < 1$	to show that $ \lambda < 1$. The following identity will be shown below:
lyche-numerical-linear-algebra- F03573	275	■ 1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}$	where $\boldsymbol{D} := \text{diag}(a_{11}, \dots, a_{nn})$. Now $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}$ and $\boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}$ are positive for all nonzero
lyche-numerical-linear-algebra- F03574	275	■ 1.000	$\boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}$	where $\boldsymbol{D} := \text{diag}(a_{11}, \dots, a_{nn})$. Now $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}$ and $\boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}$ are positive for all nonzero
lyche-numerical-linear-algebra- F03575	275	■ 1.000	$\boldsymbol{x} \in \mathbb{C}^n$	$\boldsymbol{x} \in \mathbb{C}^n$	$\boldsymbol{x} \in \mathbb{C}^n$ since a positive definite matrix has positive diagonal elements $a_{ii} =$
lyche-numerical-linear-algebra- F03576	275	■ 1.000	$a_{ii} =$	$a_{ii} =$	$\boldsymbol{x} \in \mathbb{C}^n$ since a positive definite matrix has positive diagonal elements $a_{ii} =$
lyche-numerical-linear-algebra- F03577	275	■ 1.000	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > 0$	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > 0$	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > 0$. It follows that the left hand side of (12.22) is nonnegative and then
lyche-numerical-linear-algebra- F03578	275	■ 1.000	$ \lambda \leq 1$	$ \lambda \leq 1$	the right hand side must be nonnegative as well. This implies $ \lambda \leq 1$. If $ \lambda = 1$

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lyche-numerical-linear-algebra F03579	276	■ 1.000	$\boldsymbol{G}_{\omega} \boldsymbol{x} = \lambda \boldsymbol{x}$	$\boldsymbol{G}_{\omega} \boldsymbol{x} = \lambda \boldsymbol{x}$	and the eigenpair equation $\boldsymbol{G}_{\omega} \boldsymbol{x} = \lambda \boldsymbol{x}$ can be written $\boldsymbol{x} - (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$
lyche-numerical-linear-algebra F03580	276	■ 1.000	$\boldsymbol{x} - \left(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L \right)^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$	$\boldsymbol{x} - \left(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L \right)^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$	and the eigenpair equation $\boldsymbol{G}_{\omega} \boldsymbol{x} = \lambda \boldsymbol{x}$ can be written $\boldsymbol{x} - (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$
lyche-numerical-linear-algebra F03581	276	■ 0.976	$\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}$	Now $\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}$ implies that $\lambda \neq 1$.
lyche-numerical-linear-algebra F03582	276	■ 0.976	$\lambda \neq 1$	$\lambda \neq 1$	Now $\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}$ implies that $\lambda \neq 1$.
lyche-numerical-linear-algebra F03583	276	■ 1.000	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^* \boldsymbol{y} =$	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^* \boldsymbol{y} =$	Again using (12.23), (12.24) and adding $(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^* \boldsymbol{y} =$
lyche-numerical-linear-algebra F03584	276	■ 1.000	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$ and $\boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$ we find
lyche-numerical-linear-algebra F03585	276	■ 1.000	$\boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$	$\boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$ and $\boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$ we find
lyche-numerical-linear-algebra F03586	276	■ 1.000	$(\boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$	$(\boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$	Since $(\boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$ and $\boldsymbol{y}^* = (1 - \bar{\lambda}) \boldsymbol{x}^*$ this also equals
lyche-numerical-linear-algebra F03587	276	■ 1.000	$\boldsymbol{y}^* = (1 - \bar{\lambda}) \boldsymbol{x}^*$	$\boldsymbol{y}^* = (1 - \bar{\lambda}) \boldsymbol{x}^*$	Since $(\boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$ and $\boldsymbol{y}^* = (1 - \bar{\lambda}) \boldsymbol{x}^*$ this also equals
lyche-numerical-linear-algebra F03588	276	■ 0.997	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A} / 4$	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A} / 4$	Consider first the J method. The matrix $\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A} / 4$ has eigenvalues
lyche-numerical-linear-algebra F03589	277	■ 1.000	$\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$	$\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$	It follows that $\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$. Since $\boldsymbol{G}_J$ is symmetric it is normal, and the
lyche-numerical-linear-algebra F03590	277	■ 1.000	$\boldsymbol{G}_J$	$\boldsymbol{G}_J$	It follows that $\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$. Since $\boldsymbol{G}_J$ is symmetric it is normal, and the
lyche-numerical-linear-algebra F03591	277	■ 0.932	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$	The R method given by $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$ with $\alpha = 2 / (\lambda_{\max} + \lambda_{\min}) = 1/4$
lyche-numerical-linear-algebra F03592	277	■ 0.932	$\alpha = 2 / (\lambda_{\max} + \lambda_{\min}) = 1/4$	$\alpha = 2 / (\lambda_{\max} + \lambda_{\min}) = 1/4$	The R method given by $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$ with $\alpha = 2 / (\lambda_{\max} + \lambda_{\min}) = 1/4$
lyche-numerical-linear-algebra F03593	277	■ 1.000	κ	κ	from Corollary 12.1 with κ given by (10.20).
lyche-numerical-linear-algebra F03594	277	■ 1.000	$\rho(\boldsymbol{G}_{\omega})$	$\rho(\boldsymbol{G}_{\omega})$	For the SOR method it is possible to explicitly determine $\rho(\boldsymbol{G}_{\omega})$ for any $\omega \in$
lyche-numerical-linear-algebra F03595	277	■ 1.000	$\omega \in$	$\omega \in$	For the SOR method it is possible to explicitly determine $\rho(\boldsymbol{G}_{\omega})$ for any $\omega \in$
lyche-numerical-linear-algebra F03596	277	■ 1.000	$\beta := \rho(\boldsymbol{G}_J) = \cos(\pi h)$	$\beta := \rho(\boldsymbol{G}_J) = \cos(\pi h)$	where $\beta := \rho(\boldsymbol{G}_J) = \cos(\pi h)$ and
lyche-numerical-linear-algebra F03597	277	■ 1.000	ω^*	ω^*	decreases monotonically to the minimum ω^* . Then it increases linearly to the value
lyche-numerical-linear-algebra F03598	277	■ 1.000	$\omega = 2$	$\omega = 2$	one for $\omega = 2$. We call ω^* the optimal relaxation parameter .
lyche-numerical-linear-algebra F03599	277	■ 1.000	$\beta = \cos(\pi h)$	$\beta = \cos(\pi h)$	For the discrete Poisson problem we have $\beta = \cos(\pi h)$ and it follows from
lyche-numerical-linear-algebra F03600	277	■ 0.998	$\rho(\boldsymbol{G}_1) = \beta^2 = \rho(\boldsymbol{G}_J)^2 = \cos^2(\pi h)$	$\rho(\boldsymbol{G}_1) = \beta^2 = \rho(\boldsymbol{G}_J)^2 = \cos^2(\pi h)$	Letting $\omega = 1$ in (12.27) we find $\rho(\boldsymbol{G}_1) = \beta^2 = \rho(\boldsymbol{G}_J)^2 = \cos^2(\pi h)$ for the GS
lyche-numerical-linear-algebra F03601	277	■ 0.999	$\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$	$\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$	The values of $\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$, and $\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$ are shown in Table 12.2 for
lyche-numerical-linear-algebra F03602	277	■ 0.999	$\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$	$\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$	The values of $\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$, and $\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$ are shown in Table 12.2 for

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lyche-numerical-linear-algebra F03603	277	■ 1.000	$\rho(\boldsymbol{G})^{k_n} \leq$	$\rho(\boldsymbol{G})^{k_n} \leq$	$n = 100$ and $n = 2500$. We also show the smallest integer k_n such that $\rho(\boldsymbol{G})^{k_n} \leq$
lyche-numerical-linear-algebra F03606	278	■ 1.000	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x})$	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x})$	know how fast the iterative method converges. Recall that $\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x})$.
lyche-numerical-linear-algebra F03607	278	■ 1.000	$\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} =$	$\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} =$	For the last formula we apply Theorem 12.13 which says that $\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} =$
lyche-numerical-linear-algebra F03608	278	■ 1.000	$\rho(\boldsymbol{G})$	$\rho(\boldsymbol{G})$	$\rho(\boldsymbol{G})$. For Jacobi's method and the spectral norm we have $\ \boldsymbol{G}_J^k\ _2 = \rho(\boldsymbol{G}_J)^k$ (cf.
lyche-numerical-linear-algebra F03609	278	■ 1.000	$\ \boldsymbol{G}_J^k\ _2 = \rho(\boldsymbol{G}_J)^k$	$\ \boldsymbol{G}_J^k\ _2 = \rho(\boldsymbol{G}_J)^k$	$\rho(\boldsymbol{G})$. For Jacobi's method and the spectral norm we have $\ \boldsymbol{G}_J^k\ _2 = \rho(\boldsymbol{G}_J)^k$ (cf.
lyche-numerical-linear-algebra F03610	278	■ 1.000	$\rho(\boldsymbol{G}) = 1 - \eta$	$\rho(\boldsymbol{G}) = 1 - \eta$	Lemma 12.1 (Number of Iterations) Suppose $\rho(\boldsymbol{G}) = 1 - \eta$ for some $0 < \eta < 1$
lyche-numerical-linear-algebra F03611	278	■ 1.000	$0 < \eta < 1$	$0 < \eta < 1$	Lemma 12.1 (Number of Iterations) Suppose $\rho(\boldsymbol{G}) = 1 - \eta$ for some $0 < \eta < 1$
lyche-numerical-linear-algebra F03612	278	■ 1.000	$s \in \mathbb{N}$	$s \in \mathbb{N}$	and let $s \in \mathbb{N}$. Then
lyche-numerical-linear-algebra F03613	278	■ 1.000	$\rho(\boldsymbol{G})^k \leq 10^{-s}$	$\rho(\boldsymbol{G})^k \leq 10^{-s}$	is an estimate for the smallest number of iterations k so that $\rho(\boldsymbol{G})^k \leq 10^{-s}$.
lyche-numerical-linear-algebra F03614	279	■ 1.000	$\tilde{k}$	$\tilde{k}$	Proof The estimate $\tilde{k}$ is an approximate solution of the equation $\rho(\boldsymbol{G})^{\tilde{k}} = 10^{-s}$.
lyche-numerical-linear-algebra F03615	279	■ 1.000	$\rho(\boldsymbol{G})^k = 10^{-s}$	$\rho(\boldsymbol{G})^k = 10^{-s}$	Proof The estimate $\tilde{k}$ is an approximate solution of the equation $\rho(\boldsymbol{G})^{\tilde{k}} = 10^{-s}$.
lyche-numerical-linear-algebra F03616	279	■ 1.000	$-\log(1 - \eta) \approx \eta$	$-\log(1 - \eta) \approx \eta$	Thus, since $-\log(1 - \eta) \approx \eta$ when η is small
lyche-numerical-linear-algebra F03617	279	■ 1.000	η	η	Thus, since $-\log(1 - \eta) \approx \eta$ when η is small
lyche-numerical-linear-algebra F03618	279	■ 0.996	$\rho(\boldsymbol{G}_J) = \cos(\pi h) = 1 - \eta, \eta = 1 - \cos(\pi h) = \frac{1}{2}\pi^2 h^2 + O(h^4) =$	$\rho(\boldsymbol{G}_J) = \cos(\pi h) = 1 - \eta, \eta = 1 - \cos(\pi h) = \frac{1}{2}\pi^2 h^2 + O(h^4) =$	• R and J: $\rho(\boldsymbol{G}_J) = \cos(\pi h) = 1 - \eta, \eta = 1 - \cos(\pi h) = \frac{1}{2}\pi^2 h^2 + O(h^4) =$
lyche-numerical-linear-algebra F03619	279	■ 1.000	$\frac{\pi^2}{2}/n + O(n^{-2})$	$\frac{\pi^2}{2}/n + O(n^{-2})$	$\frac{\pi^2}{2}/n + O(n^{-2})$. Thus,
lyche-numerical-linear-algebra F03620	279	■ 0.512	$\operatorname{GS}: \rho(\boldsymbol{G}_1) = \cos^2(\pi h) = 1 - \eta, \eta = 1 - \cos^2(\pi h) = \sin^2 \pi h = \pi^2 h^2 +$	$\operatorname{GS}: \rho(\boldsymbol{G}_1) = \cos^2(\pi h) = 1 - \eta, \eta = 1 - \cos^2(\pi h) = \sin^2 \pi h = \pi^2 h^2 +$	• $\operatorname{GS}: \rho(\boldsymbol{G}_1) = \cos^2(\pi h) = 1 - \eta, \eta = 1 - \cos^2(\pi h) = \sin^2 \pi h = \pi^2 h^2 +$
lyche-numerical-linear-algebra F03621	279	■ 1.000	$O(h^4) = \pi^2/n + O(n^{-2})$	$O(h^4) = \pi^2/n + O(n^{-2})$	$O(h^4) = \pi^2/n + O(n^{-2})$. Thus,
lyche-numerical-linear-algebra F03622	279	■ 0.906	$\rho(\boldsymbol{G}_{\omega^*}) = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)} = 1 - 2\pi h + O(h^2)$	$\rho(\boldsymbol{G}_{\omega^*}) = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)} = 1 - 2\pi h + O(h^2)$	• $\operatorname{SOR}: \rho(\boldsymbol{G}_{\omega^*}) = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)} = 1 - 2\pi h + O(h^2)$ Thus,
lyche-numerical-linear-algebra F03623	279	■ 1.000	$\boldsymbol{G}^k$	$\boldsymbol{G}^k$	1. The convergence depends on the behavior of the powers $\boldsymbol{G}^k$ as k increases.
lyche-numerical-linear-algebra F03624	279	■ 0.998	$\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} = \rho(\boldsymbol{G})$	$\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} = \rho(\boldsymbol{G})$	2. The convergence $\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} = \rho(\boldsymbol{G})$ can be quite slow (cf. Exer-
lyche-numerical-linear-algebra F03625	279	■ 1.000	$\ \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\ $	$\ \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\ $	One possibility is to choose a vector norm, keep track of $\ \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\ $, and stop
lyche-numerical-linear-algebra F03626	280	■ 1.000	$\ \boldsymbol{x}_k - \boldsymbol{x}\ $	$\ \boldsymbol{x}_k - \boldsymbol{x}\ $	when this number is sufficiently small. The following result indicates that $\ \boldsymbol{x}_k - \boldsymbol{x}\ $
lyche-numerical-linear-algebra F03627	280	■ 1.000	$\ \boldsymbol{G}\ $	$\ \boldsymbol{G}\ $	can be quite large if $\ \boldsymbol{G}\ $ is close to one.
lyche-numerical-linear-algebra F03628	280	■ 1.000	$\boldsymbol{x}_k = \boldsymbol{G}\boldsymbol{x}_{k-1} + \boldsymbol{c}, \boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$	$\boldsymbol{x}_k = \boldsymbol{G}\boldsymbol{x}_{k-1} + \boldsymbol{c}, \boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$	Lemma 12.2 (Be Careful When Stopping) If $\boldsymbol{x}_\ell = \boldsymbol{G}\boldsymbol{x}_{\ell-1} + \boldsymbol{c}, \boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$ and

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lyche-numerical-linear-algebra-F03629	280	■ 0.998	$(1 - \ \boldsymbol{G}\) \ \boldsymbol{x}_k - \boldsymbol{x}\ \leq \ \boldsymbol{G}\ \ \boldsymbol{x}_{k-1} - \boldsymbol{x}_k\ $	$(1 - \ \boldsymbol{G}\) \ \boldsymbol{x}_k - \boldsymbol{x}\ \leq \ \boldsymbol{G}\ \ \boldsymbol{x}_{k-1} - \boldsymbol{x}_k\ $	Thus $\boxed{1 - \ \boldsymbol{G}\ \ \boldsymbol{x}_k - \boldsymbol{x}\ < \ \boldsymbol{G}\ \ \boldsymbol{x}_{k-1} - \boldsymbol{x}_k\ }$ which implies (12.32).
lyche-numerical-linear-algebra-F03630	280	■ 0.997	$\boldsymbol{M} \boldsymbol{x}_{k+1} = \boldsymbol{M} \boldsymbol{x}_k - \boldsymbol{A} \boldsymbol{x}_k + \boldsymbol{b}$	$\boldsymbol{M} \boldsymbol{x}_{k+1} = \boldsymbol{M} \boldsymbol{x}_k - \boldsymbol{A} \boldsymbol{x}_k + \boldsymbol{b}$	splitting matrix of the method then by (12.9) we have $\boxed{\boldsymbol{M} \boldsymbol{x}_{k+1} = \boldsymbol{M} \boldsymbol{x}_k - \boldsymbol{A} \boldsymbol{x}_k + \boldsymbol{b}}$.
lyche-numerical-linear-algebra-F03631	280	■ 1.000	$\{\boldsymbol{A}^k\}$	$\{\boldsymbol{A}^k\}$	sequence $\{\boldsymbol{A}^k\}$ of powers of $\boldsymbol{A}$. We want to know when this sequence converges to
lyche-numerical-linear-algebra-F03632	281	■ 1.000	$\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$	$\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$	Theorem 12.10 (When Is $\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$?) For any $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ we have
lyche-numerical-linear-algebra-F03633	281	■ 1.000	$\rho(\boldsymbol{A})$	$\rho(\boldsymbol{A})$	where $\rho(\boldsymbol{A})$ is the spectral radius of $\boldsymbol{A}$ given by (12.17).
lyche-numerical-linear-algebra-F03634	281	■ 0.913	$\rho(\boldsymbol{A}) < 1$	$\rho(\boldsymbol{A}) < 1$	Clearly $\rho(\boldsymbol{A}) < 1$ is a necessary condition for $\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$. For if $(\lambda, \boldsymbol{x})$ is
lyche-numerical-linear-algebra-F03635	281	■ 1.000	$ \lambda \geq 1$	$ \lambda \geq 1$	an eigenpair of $\boldsymbol{A}$ with $ \lambda \geq 1$ and $\ \boldsymbol{x}\ _2 = 1$ then $\boldsymbol{A}^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$, and this implies
lyche-numerical-linear-algebra-F03636	281	■ 1.000	$\boldsymbol{A}^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$	$\boldsymbol{A}^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$	an eigenpair of $\boldsymbol{A}$ with $ \lambda \geq 1$ and $\ \boldsymbol{x}\ _2 = 1$ then $\boldsymbol{A}^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$, and this implies
lyche-numerical-linear-algebra-F03637	281	■ 1.000	$\ \boldsymbol{A}^k\ _2 \geq \ \boldsymbol{A}^k \boldsymbol{x}\ _2 = \ \lambda^k \boldsymbol{x}\ _2 = \lambda ^k$	$\ \boldsymbol{A}^k\ _2 \geq \ \boldsymbol{A}^k \boldsymbol{x}\ _2 = \ \lambda^k \boldsymbol{x}\ _2 = \lambda ^k$	$\boxed{\ \boldsymbol{A}^k\ _2 \geq \ \boldsymbol{A}^k \boldsymbol{x}\ _2 = \ \lambda^k \boldsymbol{x}\ _2 = \lambda ^k}$, and it follows that $\boldsymbol{A}^k$ does not tend to zero.
lyche-numerical-linear-algebra-F03638	281	■ 1.000	$\ \boldsymbol{A}\ < 1$	$\ \boldsymbol{A}\ < 1$	norm on $\mathbb{C}^{n \times n}$ such that $\ \boldsymbol{A}\ < 1$ and then use Theorems 12.2 and 12.3.
lyche-numerical-linear-algebra-F03639	281	■ 1.000	$\rho(\boldsymbol{A}) \leq$	$\rho(\boldsymbol{A}) \leq$	matrix norm $\ \cdot\ $ that is consistent on $\mathbb{C}^{n \times n}$ and any $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ we have $\rho(\boldsymbol{A}) \leq$
lyche-numerical-linear-algebra-F03640	281	■ 1.000	$\ \boldsymbol{A}\ $	$\ \boldsymbol{A}\ $	$\boxed{\ \boldsymbol{A}\ }$.
lyche-numerical-linear-algebra-F03641	281	■ 0.679	$\boldsymbol{A}, \ \cdot\ $	$\boldsymbol{A}, \ \cdot\ $	Proof Let $(\lambda, \boldsymbol{x})$ be an eigenpair for $\boldsymbol{A}$, $\ \cdot\ $ a consistent matrix norm on $\mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra-F03642	281	■ 1.000	$\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$	$\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$	and define $\boxed{\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}}$. Then $\lambda \boldsymbol{X} = \boldsymbol{A} \boldsymbol{X}$, which implies $ \lambda \ \boldsymbol{X}\ =$
lyche-numerical-linear-algebra-F03643	281	■ 1.000	$\lambda \boldsymbol{X} = \boldsymbol{A} \boldsymbol{X}$	$\lambda \boldsymbol{X} = \boldsymbol{A} \boldsymbol{X}$	and define $\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$. Then $\lambda \boldsymbol{X} = \boldsymbol{A} \boldsymbol{X}$, which implies $ \lambda \ \boldsymbol{X}\ =$
lyche-numerical-linear-algebra-F03644	281	■ 1.000	$ \lambda \ \boldsymbol{X}\ =$	$ \lambda \ \boldsymbol{X}\ =$	and define $\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$. Then $\lambda \boldsymbol{X} = \boldsymbol{A} \boldsymbol{X}$, which implies $ \lambda \ \boldsymbol{X}\ =$
lyche-numerical-linear-algebra-F03645	281	■ 0.999	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A} \boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A} \boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $	$\boxed{\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A} \boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ }$. Since $\ \boldsymbol{X}\ \neq 0$ we obtain $ \lambda \leq \ \boldsymbol{A}\ $.
lyche-numerical-linear-algebra-F03646	281	■ 0.999	$\ \boldsymbol{X}\ \neq 0$	$\ \boldsymbol{X}\ \neq 0$	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A} \boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $. Since $\ \boldsymbol{X}\ \neq 0$ we obtain $ \lambda \leq \ \boldsymbol{A}\ $.
lyche-numerical-linear-algebra-F03647	281	■ 0.999	$ \lambda \leq \ \boldsymbol{A}\ $	$ \lambda \leq \ \boldsymbol{A}\ $	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A} \boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $. Since $\ \boldsymbol{X}\ \neq 0$ we obtain $\boxed{ \lambda \leq \ \boldsymbol{A}\ }$.
lyche-numerical-linear-algebra-F03648	281	■ 1.000	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ \leq \rho(\boldsymbol{A}) + \epsilon$	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ \leq \rho(\boldsymbol{A}) + \epsilon$	that $\boxed{\rho(\boldsymbol{A}) < \ \boldsymbol{A}\ < \rho(\boldsymbol{A}) + \epsilon}$.
lyche-numerical-linear-algebra-F03649	281	■ 1.000	$\boldsymbol{R} = [r_{ij}]$	$\boldsymbol{R} = [r_{ij}]$	there is a unitary matrix $\boldsymbol{U}$ and an upper triangular matrix $\boldsymbol{R} = [r_{ij}]$ such that
lyche-numerical-linear-algebra-F03650	281	■ 1.000	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{R}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{R}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{R}$. For $t > 0$ we define $\boldsymbol{D}_t := \text{diag}(t, t^2, \dots, t^n) \in \mathbb{R}^{n \times n}$, and note that
lyche-numerical-linear-algebra-F03651	281	■ 1.000	$\boldsymbol{D}_t := \text{diag}(t, t^2, \dots, t^n) \in \mathbb{R}^{n \times n}$	$\boldsymbol{D}_t := \text{diag}(t, t^2, \dots, t^n) \in \mathbb{R}^{n \times n}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{R}$. For $t > 0$ we define $\boxed{\boldsymbol{D}_t := \text{diag}(t, t^2, \dots, t^n) \in \mathbb{R}^{n \times n}}$, and note that
lyche-numerical-linear-algebra-F03652	281	■ 0.718	(i, j)	(i, j)	the (i, j) element in $\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1}$ is given by $t^{i-j} r_{ij}$ for all i, j . For $n = 3$
lyche-numerical-linear-algebra-F03653	281	■ 0.718	$\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1}$	$\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1}$	the (i, j) element in $\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1}$ is given by $t^{i-j} r_{ij}$ for all i, j . For $n = 3$

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lyche-numerical-linear-algebra-281-F03654	281	0.718	$t^{i-j} r_{ij}$	$t^{i-j} r_{ij}$	the (i, j) element in $D_t R D_t^{-1}$ is given by $t^{i-j} r_{ij}$ for all i, j . For $n = 3$
lyche-numerical-linear-algebra-281-F03655	281	0.877	$\ \boldsymbol{B}\ _t := \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1$	$\ \boldsymbol{B}\ _t := \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1$	$\ \boldsymbol{B}\ _t := \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1$. We leave it as an exercise to show that $\ \cdot\ _t$ is a
lyche-numerical-linear-algebra-281-F03656	281	0.877	$\left\ \boldsymbol{B} \right\ _t$	$\ \boldsymbol{B}\ _t$	$\ \boldsymbol{B}\ _t := \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1$. We leave it as an exercise to show that $\ \cdot\ _t$ is a
lyche-numerical-linear-algebra-281-F03657	281	1.000	$\ \boldsymbol{B}\ := \ \boldsymbol{B}\ _t$	$\ \boldsymbol{B}\ := \ \boldsymbol{B}\ _t$	consistent matrix norm on $\mathbb{C}^{n \times n}$. We define $\ \boldsymbol{B}\ := \ \boldsymbol{B}\ _t$, where t is chosen so
lyche-numerical-linear-algebra-282-F03658	282	1.000	λ^k	λ^k	then λ^k is an eigenvalue of $\boldsymbol{A}^k$ for any $k \in \mathbb{N}$. By Theorem 12.11 we then obtain
lyche-numerical-linear-algebra-282-F03659	282	1.000	$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) \leq \ \boldsymbol{A}^k\ $	$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) \leq \ \boldsymbol{A}^k\ $	$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) < \ \boldsymbol{A}^k\ $ for any $k \in \mathbb{N}$ so that $\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ ^{1/k}$. Let $\epsilon > 0$ and
lyche-numerical-linear-algebra-282-F03660	282	1.000	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}^k\ ^{1/k}$	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}^k\ ^{1/k}$	$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) \leq \ \boldsymbol{A}^k\ $ for any $k \in \mathbb{N}$ so that $\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ ^{1/k}$. Let $\epsilon > 0$ and
lyche-numerical-linear-algebra-282-F03661	282	1.000	$\boldsymbol{B} := (\rho(\boldsymbol{A}) + \epsilon)^{-1} \boldsymbol{A}$	$\boldsymbol{B} := (\rho(\boldsymbol{A}) + \epsilon)^{-1} \boldsymbol{A}$	consider the matrix $\boldsymbol{B} := (\rho(\boldsymbol{A}) + \epsilon)^{-1} \boldsymbol{A}$. Then $\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A}) + \epsilon) < 1$
lyche-numerical-linear-algebra-282-F03662	282	1.000	$\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A}) + \epsilon) < 1$	$\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A}) + \epsilon) < 1$	consider the matrix $\boldsymbol{B} := (\rho(\boldsymbol{A}) + \epsilon)^{-1} \boldsymbol{A}$. Then $\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A}) + \epsilon) < 1$
lyche-numerical-linear-algebra-282-F03663	282	1.000	$\ \boldsymbol{B}^k\ \rightarrow 0$	$\ \boldsymbol{B}^k\ \rightarrow 0$	and $\ \boldsymbol{B}^k\ \rightarrow 0$ by Theorem 12.10 as $k \rightarrow \infty$. Choose $N \in \mathbb{N}$ such that $\ \boldsymbol{B}^k\ < 1$
lyche-numerical-linear-algebra-282-F03664	282	1.000	$k \rightarrow \infty$	$k \rightarrow \infty$	and $\ \boldsymbol{B}^k\ \rightarrow 0$ by Theorem 12.10 as $k \rightarrow \infty$. Choose $N \in \mathbb{N}$ such that $\ \boldsymbol{B}^k\ < 1$
lyche-numerical-linear-algebra-282-F03665	282	1.000	$\ \boldsymbol{B}^k\ < 1$	$\ \boldsymbol{B}^k\ < 1$	and $\ \boldsymbol{B}^k\ \rightarrow 0$ by Theorem 12.10 as $k \rightarrow \infty$. Choose $N \in \mathbb{N}$ such that $\ \boldsymbol{B}^k\ < 1$
lyche-numerical-linear-algebra-282-F03666	282	1.000	$k \geq N$	$k \geq N$	for all $k \geq N$. Then for $k \geq N$
lyche-numerical-linear-algebra-282-F03667	282	1.000	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}^k\ ^{1/k} \leq \rho(\boldsymbol{A}) + \epsilon$	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}^k\ ^{1/k} \leq \rho(\boldsymbol{A}) + \epsilon$	We have shown that $\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}^k\ ^{1/k} \leq \rho(\boldsymbol{A}) + \epsilon$ for $k \geq N$. Since ϵ is arbitrary
lyche-numerical-linear-algebra-282-F03668	282	1.000	$\sum_{k=0}^{\infty} \boldsymbol{A}_k$	$\sum_{k=0}^{\infty} \boldsymbol{A}_k$	Consider an infinite series $\sum_{k=0}^{\infty} \boldsymbol{A}_k$ of matrices in $\mathbb{C}^{n \times n}$. We say that the series
lyche-numerical-linear-algebra-282-F03669	282	1.000	$\{\boldsymbol{S}_m\}$	$\{\boldsymbol{S}_m\}$	converges if the sequence of partial sums $\{\boldsymbol{S}_m\}$ given by $\boldsymbol{S}_m = \sum_{k=0}^m \boldsymbol{A}_k$ converges.
lyche-numerical-linear-algebra-282-F03670	282	1.000	$\boldsymbol{S}_m = \sum_{k=0}^m \boldsymbol{A}_k$	$\boldsymbol{S}_m = \sum_{k=0}^m \boldsymbol{A}_k$	converges if the sequence of partial sums $\{\boldsymbol{S}_m\}$ given by $\boldsymbol{S}_m = \sum_{k=0}^m \boldsymbol{A}_k$ converges.
lyche-numerical-linear-algebra-282-F03671	282	1.000	$\ \boldsymbol{S}_l - \boldsymbol{S}_m\ < \epsilon$	$\ \boldsymbol{S}_l - \boldsymbol{S}_m\ < \epsilon$	there exists an integer N so that $\ \boldsymbol{S}_l - \boldsymbol{S}_m\ < \epsilon$ for all $l > m \geq N$.
lyche-numerical-linear-algebra-282-F03672	282	1.000	$l > m \geq N$	$l > m \geq N$	there exists an integer N so that $\ \boldsymbol{S}_l - \boldsymbol{S}_m\ < \epsilon$ for all $l > m \geq N$.
lyche-numerical-linear-algebra-282-F03673	282	1.000	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	1. The series $\sum_{k=0}^{\infty} \boldsymbol{B}^k$ converges if and only if $\rho(\boldsymbol{B}) < 1$.
lyche-numerical-linear-algebra-282-F03674	282	1.000	$\rho(\boldsymbol{B}) < 1$	$\rho(\boldsymbol{B}) < 1$	1. The series $\sum_{k=0}^{\infty} \boldsymbol{B}^k$ converges if and only if $\rho(\boldsymbol{B}) < 1$.
lyche-numerical-linear-algebra-282-F03675	282	1.000	$(\boldsymbol{I} - \boldsymbol{B})$	$(\boldsymbol{I} - \boldsymbol{B})$	2. If $\rho(\boldsymbol{B}) < 1$ then $(\boldsymbol{I} - \boldsymbol{B})$ is nonsingular and $(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$.
lyche-numerical-linear-algebra-282-F03676	282	1.000	$(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$	$(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$	2. If $\rho(\boldsymbol{B}) < 1$ then $(\boldsymbol{I} - \boldsymbol{B})$ is nonsingular and $(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$.
lyche-numerical-linear-algebra-282-F03677	282	1.000	$\ \boldsymbol{B}\ < 1$	$\ \boldsymbol{B}\ < 1$	3. If $\ \boldsymbol{B}\ < 1$ for some consistent matrix norm $\ \cdot\ $ on $\mathbb{C}^{n \times n}$ then
lyche-numerical-linear-algebra-283-F03678	283	1.000	$\boldsymbol{S}_m := \sum_{k=0}^m \boldsymbol{B}^k$	$\boldsymbol{S}_m := \sum_{k=0}^m \boldsymbol{B}^k$	1. Suppose $\rho(\boldsymbol{B}) < 1$. We show that $\boldsymbol{S}_m := \sum_{k=0}^m \boldsymbol{B}^k$ is a Cauchy sequence and

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lyche-numerical-linear-algebra-283-F03679	283	1.000	$l > m$	$l > m$	norm $\ \cdot\ $ on $\mathbb{C}^{n \times n}$ such that $\ B\ < 1$. Then for $l > m$
lyche-numerical-linear-algebra-283-F03680	283	1.000	$\frac{\ B\ ^{N+1}}{1 - \ B\ } < \epsilon$	$\frac{\ B\ ^{N+1}}{1 - \ B\ } < \epsilon$	But then $\{S_m\}$ is a Cauchy sequence provided N is such that $\frac{\ B\ ^{N+1}}{1 - \ B\ } < \epsilon$.
lyche-numerical-linear-algebra-283-F03681	283	0.876	$S_m x =$	$S_m x =$	Conversely, suppose (λ, x) is an eigenpair for B with $ \lambda \geq 1$. We find $S_m x =$
lyche-numerical-linear-algebra-283-F03682	283	1.000	$\sum_{k=0}^m B^k x = \left(\sum_{k=0}^m \lambda^k \right) x$	$\sum_{k=0}^m B^k x = \left(\sum_{k=0}^m \lambda^k \right) x$	$\sum_{k=0}^m B^k x = \left(\sum_{k=0}^m \lambda^k \right) x$. Since λ^k does not tend to zero the series $\sum_{k=0}^{\infty} \lambda^k$ is
lyche-numerical-linear-algebra-283-F03683	283	1.000	$\sum_{k=0}^{\infty} \lambda^k$	$\sum_{k=0}^{\infty} \lambda^k$	$\sum_{k=0}^m B^k x = \left(\sum_{k=0}^m \lambda^k \right) x$. Since λ^k does not tend to zero the series $\sum_{k=0}^{\infty} \lambda^k$ is
lyche-numerical-linear-algebra-283-F03684	283	1.000	$\left\{ S_m x \right\}$	$\{S_m x\}$	not convergent and therefore $\{S_m x\}$ and hence $\{S_m\}$ does not converge.
lyche-numerical-linear-algebra-283-F03685	283	1.000	$B^{m+1} \rightarrow 0$	$B^{m+1} \rightarrow 0$	Since $\rho(B) < 1$ we conclude that $B^{m+1} \rightarrow 0$ and hence taking limits in (12.36)
lyche-numerical-linear-algebra-283-F03686	283	0.947	$\left(\sum_{k=0}^{\infty} B^k \right) (I - B) = I$	$\left(\sum_{k=0}^{\infty} B^k \right) (I - B) = I$	we obtain $\left(\sum_{k=0}^{\infty} B^k \right) (I - B) = I$ which completes the proof of 2.
lyche-numerical-linear-algebra-283-F03687	283	0.999	$\left\ (I - B)^{-1} \right\ = \left\ \sum_{k=0}^{\infty} B^k \right\ \leq \sum_{k=0}^{\infty} \ B\ ^k = \frac{1}{1 - \ B\ }$	$\ (I - B)^{-1}\ = \left\ \sum_{k=0}^{\infty} B^k \right\ \leq \sum_{k=0}^{\infty} \ B\ ^k = \frac{1}{1 - \ B\ }$	3. By 2: $\ (I - B)^{-1}\ = \left\ \sum_{k=0}^{\infty} B^k \right\ \leq \sum_{k=0}^{\infty} \ B\ ^k = \frac{1}{1 - \ B\ }$.
lyche-numerical-linear-algebra-283-F03689	283	0.546	$G_J v = \mu v$	$G_J v = \mu v$	write these equations using the matrix formulation $TV + VT = h^2 F$. If $G_J v = \mu v$
lyche-numerical-linear-algebra-283-F03690	283	1.000	$v := \operatorname{vec}(V) \in \mathbb{R}^{m^2}$	$v := \operatorname{vec}(V) \in \mathbb{R}^{m^2}$	where $v := \operatorname{vec}(V) \in \mathbb{R}^{m^2}$ and $v_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-283-F03691	283	1.000	$v_{i,j} = 0$	$v_{i,j} = 0$	where $v := \operatorname{vec}(V) \in \mathbb{R}^{m^2}$ and $v_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-283-F03692	283	1.000	$i \in \{0, m+1\}$	$i \in \{0, m+1\}$	where $v := \operatorname{vec}(V) \in \mathbb{R}^{m^2}$ and $v_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-283-F03693	283	1.000	$j \in \{0, m+1\}$	$j \in \{0, m+1\}$	where $v := \operatorname{vec}(V) \in \mathbb{R}^{m^2}$ and $v_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-283-F03694	283	0.588	(λ, w)	(λ, w)	Suppose (λ, w) is an eigenpair for G_ω . By (12.21) $(I - \omega L)^{-1}(\omega R + (1 -$
lyche-numerical-linear-algebra-283-F03695	283	0.588	$(12.21)(I - \omega L)^{-1}(\omega R + (1 -$	$(12.21)(I - \omega L)^{-1}(\omega R + (1 -$	Suppose (λ, w) is an eigenpair for G_ω . By (12.21) $(I - \omega L)^{-1}(\omega R + (1 -$
lyche-numerical-linear-algebra-283-F03696	283	1.000	$\omega) I) w = \lambda w$	$\omega) I) w = \lambda w$	$\omega) I) w = \lambda w$ or
lyche-numerical-linear-algebra-284-F03697	284	1.000	$w = \operatorname{vec}(W)$	$w = \operatorname{vec}(W)$	Let $w = \operatorname{vec}(W)$, where $W \in \mathbb{C}^{m \times m}$. Then (12.38) can be written
lyche-numerical-linear-algebra-284-F03698	284	1.000	$W \in \mathbb{C}^{m \times m}$	$W \in \mathbb{C}^{m \times m}$	Let $w = \operatorname{vec}(W)$, where $W \in \mathbb{C}^{m \times m}$. Then (12.38) can be written
lyche-numerical-linear-algebra-284-F03699	284	1.000	$w_{i,j} = 0$	$w_{i,j} = 0$	where $w_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-284-F03700	284	0.430	(μ, v)	(μ, v)	Proof Suppose (λ, w) is an eigenpair for G_ω . We claim that (μ, v) is an eigenpair
lyche-numerical-linear-algebra-284-F03701	284	1.000	$v = (V)$	$v = (V)$	for G_J , where μ is given by (12.40) and $v = (V)$ with $v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$.
lyche-numerical-linear-algebra-284-F03702	284	1.000	$v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$	$v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$	for G_J , where μ is given by (12.40) and $v = (V)$ with $v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$.
lyche-numerical-linear-algebra-284-F03703	284	1.000	$w_{i,j}$	$w_{i,j}$	Indeed, replacing $w_{i,j}$ by $\lambda^{(i+j)/2} v_{i,j}$ in (12.39) and cancelling the common factor

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lyche-numerical-linear-algebra-284 F03704	284	■ 1.000	$\lambda^{(i+j) / 2} v_{i, j}$	$\lambda^{(i+j)/2} v_{i,j}$	Indeed, replacing $w_{i,j}$ by $\lambda^{(i+j)/2} v_{i,j}$ in (12.39) and cancelling the common factor
lyche-numerical-linear-algebra-284 F03705	284	■ 1.000	$\lambda^{(i+j) / 2}$	$\lambda^{(i+j)/2}$	$\lambda^{(i+j)/2}$ we obtain
lyche-numerical-linear-algebra-284 F03706	284	■ 0.999	$\boldsymbol{v} =: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W} = \operatorname{vec}(\boldsymbol{W})$	$\boldsymbol{v} =: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W} = \operatorname{vec}(\boldsymbol{W})$	(12.41). We define as before $\boldsymbol{v} =: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W} = \operatorname{vec}(\boldsymbol{W})$ with $w_{i,j} := \lambda^{(i+j)/2} v_{i,j}$.
lyche-numerical-linear-algebra-284 F03707	284	■ 0.999	$w_{i, j} := \lambda^{(i+j) / 2} v_{i, j}$	$w_{i,j} := \lambda^{(i+j)/2} v_{i,j}$	(12.41). We define as before $\boldsymbol{v} =: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W} = \operatorname{vec}(\boldsymbol{W})$ with $w_{i,j} := \lambda^{(i+j)/2} v_{i,j}$.
lyche-numerical-linear-algebra-284 F03708	284	■ 1.000	$\lambda^{-(i+j) / 2}$	$\lambda^{-(i+j)/2}$	Inserting this in (12.37) and canceling $\lambda^{-(i+j)/2}$ we obtain
lyche-numerical-linear-algebra-284 F03709	284	■ 1.000	$\omega \lambda^{1 / 2}$	$\omega \lambda^{1/2}$	Multiplying by $\omega \lambda^{1/2}$ we obtain
lyche-numerical-linear-algebra-284 F03710	284	■ 0.952	$\omega \mu \lambda^{1 / 2} = \lambda + \omega - 1$	$\omega \mu \lambda^{1/2} = \lambda + \omega - 1$	Thus, if $\omega \mu \lambda^{1/2} = \lambda + \omega - 1$ then by (12.39) $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$.
lyche-numerical-linear-algebra-284 F03711	284	■ 0.952	$\lambda, \boldsymbol{w}$	$\lambda, \boldsymbol{w}$	Thus, if $\omega \mu \lambda^{1/2} = \lambda + \omega - 1$ then by (12.39) $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$.
lyche-numerical-linear-algebra-285 F03712	285	■ 1.000	$\rho(\boldsymbol{G}_\omega) = \lambda(\mu) $	$\rho(\boldsymbol{G}_\omega) = \lambda(\mu) $	that $\rho(\boldsymbol{G}_\omega) = \lambda(\mu) $, where $\lambda(\mu)$ is an eigenvalue of $\boldsymbol{G}_\omega$ satisfying (12.41) for
lyche-numerical-linear-algebra-285 F03713	285	■ 1.000	$\lambda(\mu)$	$\lambda(\mu)$	that $\rho(\boldsymbol{G}_\omega) = \lambda(\mu) $, where $\lambda(\mu)$ is an eigenvalue of $\boldsymbol{G}_\omega$ satisfying (12.41) for
lyche-numerical-linear-algebra-285 F03714	285	■ 1.000	$\frac{1}{2} \cos (j \pi h) + \frac{1}{2} \cos (k \pi h)$	$\frac{1}{2} \cos(j\pi h) + \frac{1}{2} \cos(k\pi h)$	some eigenvalue μ of $\boldsymbol{G}_J$. The eigenvalues of $\boldsymbol{G}_J$ are $\frac{1}{2} \cos(j\pi h) + \frac{1}{2} \cos(k\pi h)$,
lyche-numerical-linear-algebra-285 F03715	285	■ 1.000	$-\mu$	$-\mu$	$j, k = 1, \dots, m$, so μ is real and both μ and $-\mu$ are eigenvalues. Thus, to compute
lyche-numerical-linear-algebra-285 F03716	285	■ 1.000	$\lambda = \lambda(\mu)$	$\lambda = \lambda(\mu)$	(12.41) for $\lambda = \lambda(\mu)$ gives
lyche-numerical-linear-algebra-285 F03717	285	■ 0.998	$d(0)=4>0$	$d(0) = 4 > 0$	Moreover $d(0) = 4 > 0$ and $d(2) = 4\mu^2 - 4 < 0$. As a function of ω , $\lambda(\mu)$ changes
lyche-numerical-linear-algebra-285 F03718	285	■ 0.998	$d(2)=4 \mu^2-4<0$	$d(2) = 4\mu^2 - 4 < 0$	Moreover $d(0) = 4 > 0$ and $d(2) = 4\mu^2 - 4 < 0$. As a function of ω , $\lambda(\mu)$ changes
lyche-numerical-linear-algebra-285 F03719	285	■ 0.998	$\omega, \lambda(\mu)$	$\omega, \lambda(\mu)$	Moreover $d(0) = 4 > 0$ and $d(2) = 4\mu^2 - 4 < 0$. As a function of ω , $\lambda(\mu)$ changes
lyche-numerical-linear-algebra-285 F03720	285	■ 0.618	$d(\omega)=0$	$d(\omega) = 0$	from real to complex when $d(\omega) = 0$. The root in $(0, 2)$ is
lyche-numerical-linear-algebra-285 F03721	285	■ 1.000	$\tilde{\omega}(\mu)$	$\tilde{\omega}(\mu)$	Both $\lambda(\mu)$ and $\tilde{\omega}(\mu)$ are strictly increasing as functions of μ . It follows that $ \lambda(\mu) $
lyche-numerical-linear-algebra-285 F03722	285	■ 1.000	$ \lambda(\mu) $	$ \lambda(\mu) $	Both $\lambda(\mu)$ and $\tilde{\omega}(\mu)$ are strictly increasing as functions of μ . It follows that $ \lambda(\mu) $
lyche-numerical-linear-algebra-285 F03723	285	■ 1.000	$\mu=\rho(\boldsymbol{G}_J)=: \beta$	$\mu = \rho(\boldsymbol{G}_J) =: \beta$	is maximized for $\mu = \rho(\boldsymbol{G}_J) =: \beta$ and for this value of μ we obtain (12.27) for
lyche-numerical-linear-algebra-285 F03724	285	■ 0.995	$0<\omega \leq \tilde{\omega}(\beta)=\omega^*$	$0 < \omega \leq \tilde{\omega}(\beta) = \omega^*$	$0 < \omega < \tilde{\omega}(\beta) = \omega^*$.
lyche-numerical-linear-algebra-285 F03725	285	■ 1.000	$\rho(\boldsymbol{G}_\omega)=\omega-1$	$\rho(\boldsymbol{G}_\omega) = \omega - 1$	Evidently $\rho(\boldsymbol{G}_\omega) = \omega - 1$ is strictly increasing in $\omega^* < \omega < 2$. Equation (12.29)
lyche-numerical-linear-algebra-285 F03726	285	■ 1.000	$\omega^*<\omega<2$	$\omega^* < \omega < 2$	Evidently $\rho(\boldsymbol{G}_\omega) = \omega - 1$ is strictly increasing in $\omega^* < \omega < 2$. Equation (12.29)
lyche-numerical-linear-algebra-285 F03727	285	■ 1.000	$0<\omega<\omega^*$	$0 < \omega < \omega^*$	will follow if we can show that $\rho(\boldsymbol{G}_\omega)$ is strictly decreasing in $0 < \omega < \omega^*$. By
lyche-numerical-linear-algebra-285 F03728	285	■ 1.000	$\beta^2\left(\omega^2\beta^2-4\omega+4\right)<\left(2-\omega\beta^2\right)^2$	$\beta^2\left(\omega^2\beta^2-4\omega+4\right)<\left(2-\omega\beta^2\right)^2$	Since $\beta^2(\omega^2\beta^2 - 4\omega + 4) < (2 - \omega\beta^2)^2$ the numerator is negative and the strict
lyche-numerical-linear-algebra-286 F03729	286	■ 1.000	$a_{i i}=d \neq 0$	$a_{ii} = d \neq 0$	Exercise 12.1 (Richardson and Jacobi) Show that if $a_{ii} = d \neq 0$ for all i then

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lyche-numerical-linear-algebra-286-F03730	286	■ 1.000	$\alpha:=1 / d$	$\alpha := 1/d$	Richardson's method with $\alpha := 1/d$ is the same as Jacobi's method.
lyche-numerical-linear-algebra-286-F03731	286	■ 1.000	$u_{\{j\}}$	u_j	all eigenvalues λ_j of A have positive real parts u_j for $j = 1, \dots, n$ and that α is
lyche-numerical-linear-algebra-286-F03732	286	■ 0.766	$0<\alpha<\min _{\{j\}}\left(2 u_{\{j\}} /\left \lambda_{\{j\}}\right ^2\right)$	$0 < \alpha < \min_j \left(2u_j/ \lambda_j ^2\right)$	real. Show that the R method converges if and only if $0 < \alpha < \min_j (2u_j/ \lambda_j ^2)$.
lyche-numerical-linear-algebra-286-F03733	286	■ 1.000	$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	method and Gauss-Seidel's method diverge for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.
lyche-numerical-linear-algebra-286-F03734	286	■ 0.945	$\boldsymbol{A}:=\left[\begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array}\right] \in \mathbb{R}^{2 \times 2}$	$A := \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$	and only if Jacobi converges for a 2 by 2 matrix $A := \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$.
lyche-numerical-linear-algebra-286-F03735	286	■ 1.000	$\left[\begin{array}{ccc} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{array}\right]$	$\begin{bmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{bmatrix}$	values) that the matrix $\begin{bmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{bmatrix}$ is positive definite for $-1/2 < a < 1$. Thus, GS
lyche-numerical-linear-algebra-286-F03736	286	■ 1.000	$-1 / 2<a<1$	$-1/2 < a < 1$	values) that the matrix $\begin{bmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{bmatrix}$ is positive definite for $-1/2 < a < 1$. Thus, GS
lyche-numerical-linear-algebra-286-F03737	286	■ 1.000	$1 / 2<a<1$	$1/2 < a < 1$	$1/2 < a < 1$.
lyche-numerical-linear-algebra-286-F03738	286	■ 0.995	$\boldsymbol{G}_{\{1\}}$	G_1	Exercise 12.6 (Example: GS Diverges, J Converges) Let G_J and G_1 be the
lyche-numerical-linear-algebra-286-F03739	286	■ 0.924	$\boldsymbol{A}:=\left[\begin{array}{ccc} 1 & 0 & 1 / 2 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{array}\right] .1$	$A := \begin{bmatrix} 1 & 0 & 1/2 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} .1$	$A := \begin{bmatrix} 1 & 0 & 1/2 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} .1$
lyche-numerical-linear-algebra-286-F03740	286	■ 0.996	$\boldsymbol{G}_{\{1\}}:=\left[\begin{array}{ccc} 0 & 0 & -1 / 2 \\ 0 & 0 & 1 / 2 \\ 0 & 0 & -1 \end{array}\right]$	$G_1 := \begin{bmatrix} 0 & 0 & -1/2 \\ 0 & 0 & 1/2 \\ 0 & 0 & -1 \end{bmatrix}$	a) Show that $G_1 := \begin{bmatrix} 0 & 0 & -1/2 \\ 0 & 0 & 1/2 \\ 0 & 0 & -1 \end{bmatrix}$ and conclude that GS diverges.
lyche-numerical-linear-algebra-286-F03741	286	■ 1.000	$p(\lambda):=\operatorname{det}(\lambda \boldsymbol{I}-\boldsymbol{G}_{\{J\}})=\lambda^3+\frac{1}{2} \lambda+\frac{1}{2}$	$p(\lambda) := \det (\lambda I - G_J) = \lambda^3 + \frac{1}{2}\lambda + \frac{1}{2}$	b) Show that $p(\lambda) := \det(\lambda I - G_J) = \lambda^3 + \frac{1}{5}\lambda + \frac{1}{5}$.
lyche-numerical-linear-algebra-286-F03742	286	■ 0.997	$p(\lambda) \neq 0$	$p(\lambda) \neq 0$	c) Show that if $ \lambda \geq 1$ then $p(\lambda) \neq 0$. Conclude that J converges.
lyche-numerical-linear-algebra-286-F03743	286	■ 1.000	$\left a_{i i}\right >\sum_{j \neq i}\left a_{i j}\right $	$ a_{ii} > \sum_{j \neq i} a_{ij} $	method converges if $ a_{ii} > \sum_{j \neq i} a_{ij} $ for $i = 1, \dots, n$.
lyche-numerical-linear-algebra-287-F03744	287	■ 1.000	$r:=\max _{\{i\}} r_{\{i\}}<1$	$r := \max_i r_i < 1$	GS method. Suppose $r := \max_i r_i < 1$, where $r_i = \sum_{j \neq i} \frac{ a_{ij} }{ a_{ii} }$. Show using
lyche-numerical-linear-algebra-287-F03745	287	■ 1.000	$r_{\{i\}}=\sum_{j \neq i} \frac{\left a_{i j}\right }{\left a_{i i}\right }$	$r_i = \sum_{j \neq i} \frac{ a_{ij} }{ a_{ii} }$	GS method. Suppose $r := \max_i r_i < 1$, where $r_i = \sum_{j \neq i} \frac{ a_{ij} }{ a_{ii} }$. Show using
lyche-numerical-linear-algebra-287-F03746	287	■ 1.000	$\left \boldsymbol{\epsilon}_{k+1}(\{j\})\right \leq r\left\ \boldsymbol{\epsilon}_k\right\ _{\infty}$	$ \epsilon_{k+1}(j) \leq r \ \epsilon_k\ _\infty$	induction on i that $ \epsilon_{k+1}(j) \leq r \ \epsilon_k\ _\infty$ for $j = 1, \dots, i$. Conclude that Gauss-
lyche-numerical-linear-algebra-287-F03747	287	■ 1.000	$j=1, \ldots, i$	$j = 1, \dots, i$	induction on i that $ \epsilon_{k+1}(j) \leq r \ \epsilon_k\ _\infty$ for $j = 1, \dots, i$. Conclude that Gauss-
lyche-numerical-linear-algebra-287-F03748	287	■ 1.000	$a \in$	$a \in$	Exercise 12.9 (Convergence Example for Fix Point Iteration) Consider for $a \in$
lyche-numerical-linear-algebra-287-F03749	287	■ 1.000	$\boldsymbol{x}_{\{0\}}=\mathbf{0}$	$x_0 = \mathbf{0}$	Starting with $x_0 = \mathbf{0}$ show by induction
lyche-numerical-linear-algebra-287-F03750	287	■ 1.000	$\boldsymbol{x}=[1,1]^T$	$x = [1, 1]^T$	and conclude that the iteration converges to the fixed-point $x = [1, 1]^T$ for $ a < 1$
lyche-numerical-linear-algebra-287-F03751	287	■ 1.000	$ a <1$	$ a < 1$	and conclude that the iteration converges to the fixed-point $x = [1, 1]^T$ for $ a < 1$
lyche-numerical-linear-algebra-287-F03752	287	■ 1.000	$ a >1$	$ a > 1$	and diverges for $a > 1$. Show that $\rho(G) = 1 - \eta$ with $\eta = 1 - a $. Compute the

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lyche-numerical-linear-algebra-F03753	287	1.000	$\eta=1- a $	$\eta = 1 - a $	and diverges for $ a > 1$. Show that $\rho(\boldsymbol{G}) = 1 - \eta$ with $\eta = 1 - a $. Compute the
lyche-numerical-linear-algebra-F03754	287	0.999	$a=0.9$	$a = 0.9$	estimate (12.31) for the rate of convergence for $a = 0.9$ and $s = 16$ and compare
lyche-numerical-linear-algebra-F03755	287	0.999	$s=16$	$s = 16$	estimate (12.31) for the rate of convergence for $a = 0.9$ and $s = 16$ and compare
lyche-numerical-linear-algebra-F03756	287	1.000	$ a ^k \leq 10^{-16}$	$ a ^k \leq 10^{-16}$	with the true number of iterations determined from $ a ^k < 10^{-16}$.
lyche-numerical-linear-algebra-F03757	287	1.000	$\rho(\boldsymbol{G}_J)=1/2$	$\rho(\boldsymbol{G}_J) = 1/2$	in Example 12.2. Show that $\rho(\boldsymbol{G}_J) = 1/2$. Then show that $\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - 2^{-k}$
lyche-numerical-linear-algebra-F03758	287	1.000	$\boldsymbol{x}_k(1)=\boldsymbol{x}_k(2)=1-2^{-k}$	$\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - 2^{-k}$	in Example 12.2. Show that $\rho(\boldsymbol{G}_J) = 1/2$. Then show that $\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - 2^{-k}$
lyche-numerical-linear-algebra-F03759	287	1.000	$\boldsymbol{A}=\boldsymbol{I}-\boldsymbol{L}-\boldsymbol{L}^T$	$\boldsymbol{A} = \boldsymbol{I} - \boldsymbol{L} - \boldsymbol{L}^T$	be written $\boldsymbol{A} = \boldsymbol{I} - \boldsymbol{L} - \boldsymbol{L}^T$, where $\boldsymbol{L}$ is lower triangular with zero's on the diagonal,
lyche-numerical-linear-algebra-F03760	287	1.000	$l_{i,j}=0$	$l_{i,j} = 0$	$l_{i,j} = 0$, when $j \geq i$. The method is defined by
lyche-numerical-linear-algebra-F03761	287	1.000	$j \geq i$	$j \geq i$	$l_{i,j} = 0$, when $j \geq i$. The method is defined by
lyche-numerical-linear-algebra-F03762	287	1.000	$\boldsymbol{N}$	$\boldsymbol{N}$	where $\boldsymbol{M}$ and $\boldsymbol{N}$ are given by the splitting
lyche-numerical-linear-algebra-F03763	287	1.000	$\boldsymbol{M}^{-1}\boldsymbol{N}$	$\boldsymbol{M}^{-1}\boldsymbol{N}$	a) Let $\boldsymbol{x} \neq \mathbf{0}$ be an eigenvector of $\boldsymbol{M}^{-1}\boldsymbol{N}$ with eigenvalue λ . Show that
lyche-numerical-linear-algebra-F03764	288	1.000	$\boldsymbol{x}=[x(1), x(2), \ldots, x(n)]^T$	$\boldsymbol{x} = [x(1), x(2), \ldots, x(n)]^T$	1. Choose $\boldsymbol{x} = [x(1), x(2), \ldots, x(n)]^T$.
lyche-numerical-linear-algebra-F03765	288	0.996	$i=1,2, \ldots, n-1, n, n, n-1, n-2, \ldots, 1$	$i = 1, 2, \ldots, n-1, n, n, n-1, n-2, \ldots, 1$	for $i = 1, 2, \ldots, n-1, n, n, n-1, n-2, \ldots, 1$
lyche-numerical-linear-algebra-F03766	288	0.999	$\boldsymbol{b}:=[1,9,-2]^T$	$\boldsymbol{b} := [1, 9, -2]^T$	and $\boldsymbol{b} := [1, 9, -2]^T$.
lyche-numerical-linear-algebra-F03767	288	1.000	$\boldsymbol{x}_0=[1,1,1]^t$	$\boldsymbol{x}_0 = [1, 1, 1]^t$	a) With $\boldsymbol{x}_0 = [1, 1, 1]^t$, carry out one iteration of the Gauss-Seidel method to find
lyche-numerical-linear-algebra-F03768	288	1.000	$\boldsymbol{x}_1 \in \mathbb{R}^3$	$\boldsymbol{x}_1 \in \mathbb{R}^3$	$\boldsymbol{x}_1 \in \mathbb{R}^3$.
lyche-numerical-linear-algebra-F03769	288	1.000	$\boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{E} \in \mathbb{C}^{n \times n}$. Show that $\boldsymbol{A} + \boldsymbol{E}$ is nonsingular if $\rho(\boldsymbol{A}^{-1}\boldsymbol{E}) < 1$.
lyche-numerical-linear-algebra-F03770	288	1.000	$\rho(\boldsymbol{A}^{-1}\boldsymbol{E})<1$	$\rho(\boldsymbol{A}^{-1}\boldsymbol{E}) < 1$	$\boldsymbol{E} \in \mathbb{C}^{n \times n}$. Show that $\boldsymbol{A} + \boldsymbol{E}$ is nonsingular if $\rho(\boldsymbol{A}^{-1}\boldsymbol{E}) < 1$.
lyche-numerical-linear-algebra-F03771	289	0.998	$\lim_{k \rightarrow \infty} \ \boldsymbol{A}^k\ ^{1/k} = \rho(\boldsymbol{A})$	$\lim_{k \rightarrow \infty} \ \boldsymbol{A}^k\ ^{1/k} = \rho(\boldsymbol{A})$	Exercise 12.15 (Slow Spectral Radius Convergence) The convergence $\lim_{k \rightarrow \infty} \ \boldsymbol{A}^k\ ^{1/k} = \rho(\boldsymbol{A})$ can be quite slow. Consider
lyche-numerical-linear-algebra-F03772	289	1.000	$ \lambda = \rho(\boldsymbol{A}) < 1$	$ \lambda = \rho(\boldsymbol{A}) < 1$	If $ \lambda = \rho(\boldsymbol{A}) < 1$ then $\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$ for any $a \in \mathbb{R}$. We show below that the
lyche-numerical-linear-algebra-F03773	289	1.000	$(1, n)$	$(1, n)$	$(1, n)$ element of $\boldsymbol{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$ for $k \geq n-1$.
lyche-numerical-linear-algebra-F03775	289	1.000	$f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$	$f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$	$(1, n)$ element of $\boldsymbol{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$ for $k \geq n-1$.
lyche-numerical-linear-algebra-F03776	289	1.000	$k \geq n-1$	$k \geq n-1$	$(1, n)$ element of $\boldsymbol{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$ for $k \geq n-1$.
lyche-numerical-linear-algebra-F03777	289	1.000	$n=5$	$n = 5$	a) Pick an n , e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9$, $a = 10$, and
lyche-numerical-linear-algebra-F03778	289	1.000	$f(k)$	$f(k)$	a) Pick an n , e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9$, $a = 10$, and

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lyche-numerical-linear-algebra- F03779	289	1.000	$\lambda=0.9, a=10$	$\lambda = 0.9, a = 10$	a) Pick an n , e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9, a = 10$, and
lyche-numerical-linear-algebra- F03780	289	1.000	$n-1 \leq k \leq 200$	$n - 1 \leq k \leq 200$	$n - 1 \leq k \leq 200$. Your program should also compute $\max_k f(k)$. Use your
lyche-numerical-linear-algebra- F03781	289	1.000	$\max_k f(k)$	$\max_k f(k)$	$n - 1 \leq k \leq 200$. Your program should also compute $\max_k f(k)$. Use your
lyche-numerical-linear-algebra- F03782	289	1.000	$f(k) < 10^{-8}$	$f(k) < 10^{-8}$	program to determine how large k must be before $f(k) < 10^{-8}$.
lyche-numerical-linear-algebra- F03783	289	1.000	$\text{boldsymbol{E}} := (\text{boldsymbol{A}} - \lambda \text{boldsymbol{I}}) / a$	$E := (A - \lambda I)/a$	b) We can determine the elements of A^k explicitly for any k . Let $E := (A - \lambda I)/a$.
lyche-numerical-linear-algebra- F03784	289	1.000	$\text{boldsymbol{E}}^k = \begin{bmatrix} \mathbf{0} & I_{n-k} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$E^k = \begin{bmatrix} \mathbf{0} & I_{n-k} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	Show by induction that $E^k = \begin{bmatrix} \mathbf{0} & I_{n-k} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$ for $1 \leq k \leq n - 1$ and that $E^n = \mathbf{0}$.
lyche-numerical-linear-algebra- F03785	289	1.000	$\text{boldsymbol{E}}^n = \mathbf{0}$	$E^n = \mathbf{0}$	Show by induction that $E^k = \begin{bmatrix} \mathbf{0} & I_{n-k} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$ for $1 \leq k \leq n - 1$ and that $E^n = \mathbf{0}$.
lyche-numerical-linear-algebra- F03786	289	0.997	$\text{boldsymbol{A}}^k = (a \text{boldsymbol{E}} + \lambda \text{boldsymbol{I}})^k = \sum_{j=0}^{\min\{k, n-1\}} \binom{k}{j} a^j \lambda^{k-j} \text{boldsymbol{E}}^j$	$A^k = (aE + \lambda I)^k = \sum_{j=0}^{\min\{k, n-1\}} \binom{k}{j} a^j \lambda^{k-j} E^j$	c) We have $A^k = (aE + \lambda I)^k = \sum_{j=0}^{\min\{k, n-1\}} \binom{k}{j} a^j \lambda^{k-j} E^j$ and conclude that the
lyche-numerical-linear-algebra- F03787	289	1.000	$\text{boldsymbol{A}}^n \rightarrow \mathbf{0}$	$A^n \rightarrow \mathbf{0}$	• Give a necessary and sufficient condition that $A^n \rightarrow \mathbf{0}$.
lyche-numerical-linear-algebra- F03788	289	1.000	$\ \text{boldsymbol{A}}\ < \rho(\text{boldsymbol{A}})$	$\ A\ < \rho(A)$	• Is there a matrix norm $\ \cdot\ $ consistent on $\mathbb{C}^{n \times n}$ such that $\ A\ < \rho(A)$?
lyche-numerical-linear-algebra- F03789	290	0.969	$\kappa := \frac{\lambda_{\max}}{\lambda_{\min}}$	$\kappa := \frac{\lambda_{\max}}{\lambda_{\min}}$	condition number of A is given by $\kappa := \frac{\lambda_{\max}}{\lambda_{\min}}$, where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest
lyche-numerical-linear-algebra- F03790	290	1.000	$\langle \text{boldsymbol{x}}, \text{boldsymbol{y}} \rangle = \text{boldsymbol{x}}^T \text{boldsymbol{y}}$	$\langle x, y \rangle = x^T y$	on $\mathbb{R}^n$, the usual inner product $\langle x, y \rangle = x^T y$ with the associated Euclidian norm
lyche-numerical-linear-algebra- F03791	290	1.000	$\ \text{boldsymbol{x}}\ _2 = \sqrt{\text{boldsymbol{x}}^T \text{boldsymbol{x}}}$	$\ x\ _2 = \sqrt{x^T x}$	$\ x\ _2 = \sqrt{x^T x}$, and the A-inner product and the corresponding A-norm given by
lyche-numerical-linear-algebra- F03792	291	1.000	$\text{boldsymbol{x}}, \text{boldsymbol{y}}, \text{boldsymbol{z}} \in$	$x, y, z \in$	We note that the A -inner product is an inner product on $\mathbb{R}^n$. Indeed, for any $x, y, z \in$
lyche-numerical-linear-algebra- F03793	291	1.000	$\langle \text{boldsymbol{x}}, \text{boldsymbol{x}} \rangle_{\text{boldsymbol{A}}} = \text{boldsymbol{x}}^T \text{boldsymbol{A}} \text{boldsymbol{x}} \geq 0$	$\langle x, x \rangle_A = x^T A x \geq 0$	1. $\langle x, x \rangle_A = x^T A x > 0$ and $\langle x, x \rangle_A = 0$ if and only if $x = \mathbf{0}$, since A is positive
lyche-numerical-linear-algebra- F03794	291	1.000	$\langle \text{boldsymbol{x}}, \text{boldsymbol{x}} \rangle_{\text{boldsymbol{A}}} = 0$	$\langle x, x \rangle_A = 0$	1. $\langle x, x \rangle_A = x^T A x \geq 0$ and $\langle x, x \rangle_A = 0$ if and only if $x = \mathbf{0}$, since A is positive
lyche-numerical-linear-algebra- F03795	291	0.983	$\langle \text{boldsymbol{x}}, \text{boldsymbol{y}} \rangle_{\text{boldsymbol{A}}} := \text{boldsymbol{x}}^T \text{boldsymbol{A}} \text{boldsymbol{y}} = (\text{boldsymbol{x}}^T \text{boldsymbol{A}} \text{boldsymbol{y}})^T = \text{boldsymbol{y}}^T \text{boldsymbol{A}}^T \text{boldsymbol{x}} = \text{boldsymbol{y}}^T \text{boldsymbol{A}} \text{boldsymbol{x}} = \langle \text{boldsymbol{y}}, \text{boldsymbol{x}} \rangle_{\text{boldsymbol{A}}}$	$\langle x, y \rangle_A := x^T A y = (x^T A y)^T = y^T A^T x = y^T A x = \langle y, x \rangle_A$	2. $\langle x, y \rangle_A := x^T A y = (x^T A y)^T = y^T A^T x = y^T A x = \langle y, x \rangle_A$ by symmetry of
lyche-numerical-linear-algebra- F03796	291	1.000	$\langle \text{boldsymbol{x}} + \text{boldsymbol{y}}, \text{boldsymbol{z}} \rangle_{\text{boldsymbol{A}}} := (\text{boldsymbol{x}} + \text{boldsymbol{y}})^T \text{boldsymbol{A}} \text{boldsymbol{z}} = \text{boldsymbol{x}}^T \text{boldsymbol{A}} \text{boldsymbol{z}} + \text{boldsymbol{y}}^T \text{boldsymbol{A}} \text{boldsymbol{z}} = \langle \text{boldsymbol{x}}, \text{boldsymbol{z}} \rangle_{\text{boldsymbol{A}}} + \langle \text{boldsymbol{y}}, \text{boldsymbol{z}} \rangle_{\text{boldsymbol{A}}}$	$\langle x + y, z \rangle_A := (x + y)^T A z = x^T A z + y^T A z = \langle x, z \rangle_A + \langle y, z \rangle_A$	3. $\langle x + y, z \rangle_A := x^T A z + y^T A z = \langle x, z \rangle_A + \langle y, z \rangle_A$, true for any A .
lyche-numerical-linear-algebra- F03797	291	1.000	$\text{boldsymbol{A}} \in \mathbb{R}^{n \times n}, \text{boldsymbol{b}} \in \mathbb{R}^n$	$A \in \mathbb{R}^{n \times n}, b \in \mathbb{R}^n$	Consider for a positive definite $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$ the quadratic
lyche-numerical-linear-algebra- F03798	291	1.000	$c \in \mathbb{R}$	$c \in \mathbb{R}$	Consider for a positive definite $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$ the quadratic

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lyche-numerical-linear-algebra-F03799	291	1.000	$Q: \mathbb{R}^n \rightarrow \mathbb{R}$	$Q: \mathbb{R}^n \rightarrow \mathbb{R}$	function $Q: \mathbb{R}^n \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra-F03800	291	1.000	$\mathbf{y}, \mathbf{h} \in \mathbb{R}^n$	$\mathbf{y}, \mathbf{h} \in \mathbb{R}^n$	The following expansion will be used repeatedly. For $\mathbf{y}, \mathbf{h} \in \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}$
lyche-numerical-linear-algebra-F03801	291	1.000	$\varepsilon \in \mathbb{R}$	$\varepsilon \in \mathbb{R}$	The following expansion will be used repeatedly. For $\mathbf{y}, \mathbf{h} \in \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}$
lyche-numerical-linear-algebra-F03802	291	0.996	$\mathbf{r}(\mathbf{y}) := \mathbf{b} - \mathbf{A}\mathbf{y}$	$\mathbf{r}(\mathbf{y}) := \mathbf{b} - \mathbf{A}\mathbf{y}$	if and only if $\mathbf{A}\mathbf{x} = \mathbf{b}$. Moreover, the residual $\mathbf{r}(\mathbf{y}) := \mathbf{b} - \mathbf{A}\mathbf{y}$ for any $\mathbf{y} \in \mathbb{R}^n$ is
lyche-numerical-linear-algebra-F03803	291	1.000	$\mathbf{r}(\mathbf{y}) = -\nabla Q(\mathbf{y})$	$\mathbf{r}(\mathbf{y}) = -\nabla Q(\mathbf{y})$	equal to the negative gradient, i.e., $\mathbf{r}(\mathbf{y}) = -\nabla Q(\mathbf{y})$, where $\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$.
lyche-numerical-linear-algebra-F03804	291	1.000	$\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$	$\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$	equal to the negative gradient, i.e., $\mathbf{r}(\mathbf{y}) = -\nabla Q(\mathbf{y})$, where $\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$.
lyche-numerical-linear-algebra-F03805	291	0.998	$\mathbf{y} = \mathbf{x}, \varepsilon = 1$	$\mathbf{y} = \mathbf{x}, \varepsilon = 1$	Proof If $\mathbf{y} = \mathbf{x}$, $\varepsilon = 1$, and $\mathbf{A}\mathbf{x} = \mathbf{b}$, then (13.5) simplifies to $Q(\mathbf{x} + \mathbf{h}) =$
lyche-numerical-linear-algebra-F03806	291	0.998	$Q(\mathbf{x} + \mathbf{h}) =$	$Q(\mathbf{x} + \mathbf{h}) =$	Proof If $\mathbf{y} = \mathbf{x}$, $\varepsilon = 1$, and $\mathbf{A}\mathbf{x} = \mathbf{b}$, then (13.5) simplifies to $Q(\mathbf{x} + \mathbf{h}) =$
lyche-numerical-linear-algebra-F03807	291	1.000	$Q(\mathbf{x}) + \frac{1}{2} \mathbf{h}^T \mathbf{A} \mathbf{h}$	$Q(\mathbf{x}) + \frac{1}{2} \mathbf{h}^T \mathbf{A} \mathbf{h}$	$Q(\mathbf{x}) + \frac{1}{2} \mathbf{h}^T \mathbf{A} \mathbf{h}$, and since $\mathbf{A}$ is positive definite $Q(\mathbf{x} + \mathbf{h}) > Q(\mathbf{x})$ for all nonzero
lyche-numerical-linear-algebra-F03808	291	1.000	$Q(\mathbf{x} + \mathbf{h}) > Q(\mathbf{x})$	$Q(\mathbf{x} + \mathbf{h}) > Q(\mathbf{x})$	$Q(\mathbf{x}) + \frac{1}{2} \mathbf{h}^T \mathbf{A} \mathbf{h}$, and since $\mathbf{A}$ is positive definite $Q(\mathbf{x} + \mathbf{h}) > Q(\mathbf{x})$ for all nonzero
lyche-numerical-linear-algebra-F03809	291	0.982	$\mathbf{h} \in \mathbb{R}^n$	$\mathbf{h} \in \mathbb{R}^n$	$\mathbf{h} \in \mathbb{R}^n$. It follows that $\mathbf{x}$ is the unique minimum of Q . Conversely, if $\mathbf{A}\mathbf{x} \neq \mathbf{b}$ and
lyche-numerical-linear-algebra-F03810	291	0.982	$\mathbf{A}\mathbf{x} \neq \mathbf{b}$	$\mathbf{A}\mathbf{x} \neq \mathbf{b}$	$\mathbf{h} \in \mathbb{R}^n$. It follows that $\mathbf{x}$ is the unique minimum of Q . Conversely, if $\mathbf{A}\mathbf{x} \neq \mathbf{b}$ and
lyche-numerical-linear-algebra-F03811	292	1.000	$\mathbf{h} := \mathbf{r}(\mathbf{x})$	$\mathbf{h} := \mathbf{r}(\mathbf{x})$	$\mathbf{h} := \mathbf{r}(\mathbf{x})$, then by (13.5), $Q(\mathbf{x} + \varepsilon \mathbf{h}) - Q(\mathbf{x}) = -\varepsilon(\mathbf{h}^T \mathbf{r}(\mathbf{x}) - \frac{1}{2} \varepsilon \mathbf{h}^T \mathbf{A} \mathbf{h}) < 0$ for
lyche-numerical-linear-algebra-F03812	292	1.000	$Q(\mathbf{x} + \varepsilon \mathbf{h}) - Q(\mathbf{x}) = -\varepsilon \left(\mathbf{h}^T \mathbf{r}(\mathbf{x}) - \frac{1}{2} \varepsilon \mathbf{h}^T \mathbf{A} \mathbf{h} \right) < 0$	$Q(\mathbf{x} + \varepsilon \mathbf{h}) - Q(\mathbf{x}) = -\varepsilon \left(\mathbf{h}^T \mathbf{r}(\mathbf{x}) - \frac{1}{2} \varepsilon \mathbf{h}^T \mathbf{A} \mathbf{h} \right) < 0$	$\mathbf{h} := \mathbf{r}(\mathbf{x})$, then by (13.5), $Q(\mathbf{x} + \varepsilon \mathbf{h}) - Q(\mathbf{x}) = -\varepsilon(\mathbf{h}^T \mathbf{r}(\mathbf{x}) - \frac{1}{2} \varepsilon \mathbf{h}^T \mathbf{A} \mathbf{h}) < 0$ for
lyche-numerical-linear-algebra-F03813	292	1.000	$\mathbf{x}_0 \in \mathbb{R}^n$	$\mathbf{x}_0 \in \mathbb{R}^n$	1. Choose $\mathbf{x}_0 \in \mathbb{R}^n$.
lyche-numerical-linear-algebra-F03814	292	1.000	α_k	α_k	Choose a “step length” α_k ,
lyche-numerical-linear-algebra-F03815	292	1.000	$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$	$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$	Compute $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$.
lyche-numerical-linear-algebra-F03816	293	0.999	$Q(\mathbf{x}_{k+1})$	$Q(\mathbf{x}_{k+1})$	For a fixed direction $\mathbf{p}_k$ we say that α_k is optimal if $Q(\mathbf{x}_{k+1})$ is as small as
lyche-numerical-linear-algebra-F03817	293	1.000	$Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = Q(\mathbf{x}_k) - \alpha \mathbf{p}_k^T \mathbf{r}_k + \frac{1}{2} \alpha^2 \mathbf{p}_k^T \mathbf{A} \mathbf{p}_k$	$Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = Q(\mathbf{x}_k) - \alpha \mathbf{p}_k^T \mathbf{r}_k + \frac{1}{2} \alpha^2 \mathbf{p}_k^T \mathbf{A} \mathbf{p}_k$	By (13.5) we have $Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = Q(\mathbf{x}_k) - \alpha \mathbf{p}_k^T \mathbf{r}_k + \frac{1}{2} \alpha^2 \mathbf{p}_k^T \mathbf{A} \mathbf{p}_k$, where $\mathbf{r}_k :=$
lyche-numerical-linear-algebra-F03818	293	1.000	$\mathbf{r}_k :=$	$\mathbf{r}_k :=$	By (13.5) we have $Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = Q(\mathbf{x}_k) - \alpha \mathbf{p}_k^T \mathbf{r}_k + \frac{1}{2} \alpha^2 \mathbf{p}_k^T \mathbf{A} \mathbf{p}_k$, where $\mathbf{r}_k :=$
lyche-numerical-linear-algebra-F03819	293	1.000	$\mathbf{b} - \mathbf{A}\mathbf{x}_k$	$\mathbf{b} - \mathbf{A}\mathbf{x}_k$	$\mathbf{b} - \mathbf{A}\mathbf{x}_k$. Since $\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k \geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial \alpha} Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = 0$.
lyche-numerical-linear-algebra-F03820	293	1.000	$\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k \geq 0$	$\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k \geq 0$	$\mathbf{b} - \mathbf{A}\mathbf{x}_k$. Since $\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k \geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial \alpha} Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = 0$.
lyche-numerical-linear-algebra-F03821	293	1.000	$\frac{\partial}{\partial \alpha} Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = 0$	$\frac{\partial}{\partial \alpha} Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = 0$	$\mathbf{b} - \mathbf{A}\mathbf{x}_k$. Since $\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k \geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial \alpha} Q(\mathbf{x}_k + \alpha \mathbf{p}_k) = 0$.

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lyche-numerical-linear-algebra-293 F03822	293	1.000	$\boldsymbol{p}_k=\boldsymbol{r}_k$	$\boldsymbol{p}_k = \boldsymbol{r}_k$	choose $\boldsymbol{p}_k = \boldsymbol{r}_k$ the negative gradient, and the optimal α_k . Starting from $\boldsymbol{x}_0$ we
lyche-numerical-linear-algebra-293 F03823	293	1.000	$k=0,1,2 \ldots$	$k = 0, 1, 2 \dots$	compute for $k = 0, 1, 2 \dots$
lyche-numerical-linear-algebra-293 F03824	293	1.000	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{r}_k)^T \boldsymbol{r}_k = 0$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k = (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{r}_k)^T \boldsymbol{r}_k = 0$	are orthogonal. Indeed, by (13.10), $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k = (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{r}_k)^T \boldsymbol{r}_k = 0$ since $\alpha_k =$
lyche-numerical-linear-algebra-293 F03825	293	1.000	$\alpha_k =$	$\alpha_k =$	are orthogonal. Indeed, by (13.10), $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k = (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{r}_k)^T \boldsymbol{r}_k = 0$ since $\alpha_k =$
lyche-numerical-linear-algebra-293 F03826	293	0.887	$\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$	$\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$	$\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$ and $\boldsymbol{A}$ is symmetric.
lyche-numerical-linear-algebra-293 F03827	293	1.000	$\boldsymbol{x}_0 = [-1, -1/2]^T$	$\boldsymbol{x}_0 = [-1, -1/2]^T$	Starting with $\boldsymbol{x}_0 = [-1, -1/2]^T$ and $\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$ we find
lyche-numerical-linear-algebra-293 F03828	293	1.000	$\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$	$\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$	Starting with $\boldsymbol{x}_0 = [-1, -1/2]^T$ and $\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$ we find
lyche-numerical-linear-algebra-294 F03829	294	1.000	$k \geq 1$	$k \geq 1$	and in general for $k \geq 1$
lyche-numerical-linear-algebra-294 F03830	294	1.000	$\alpha_k = 1/2$	$\alpha_k = 1/2$	Since $\alpha_k = 1/2$ is constant for all k the methods of Richardson, Jacobi and steepest
lyche-numerical-linear-algebra-294 F03831	294	1.000	$\ \boldsymbol{x}_{j+1}\ _2 / \ \boldsymbol{x}_j\ = \ \boldsymbol{r}_{j+1}\ _2 / \ \boldsymbol{r}_j\ _2 = 1/2$	$\ \boldsymbol{x}_{j+1}\ _2 / \ \boldsymbol{x}_j\ = \ \boldsymbol{r}_{j+1}\ _2 / \ \boldsymbol{r}_j\ _2 = 1/2$	of convergence is determined from $\ \boldsymbol{x}_{j+1}\ _2 / \ \boldsymbol{x}_j\ = \ \boldsymbol{r}_{j+1}\ _2 / \ \boldsymbol{r}_j\ _2 = 1/2$ for
lyche-numerical-linear-algebra-294 F03832	294	1.000	$\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0$	$\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0$	A-orthogonal search directions i.e., $\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0$ for all $i \neq j$.
lyche-numerical-linear-algebra-294 F03833	294	1.000	$\boldsymbol{r}_0 =$	$\boldsymbol{r}_0 =$	As in the steepest descent method we choose a starting vector $\boldsymbol{x}_0 \in \mathbb{R}^n$. If $\boldsymbol{r}_0 =$
lyche-numerical-linear-algebra-294 F03834	294	1.000	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0 = \mathbf{0}$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0 = \mathbf{0}$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0 = \mathbf{0}$ then $\boldsymbol{x}_0$ is the exact solution and we are finished, otherwise we initially
lyche-numerical-linear-algebra-294 F03835	294	1.000	$\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$	$\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$	make a steepest descent step. It follows that $\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$ and $\boldsymbol{p}_0 := \boldsymbol{r}_0$.
lyche-numerical-linear-algebra-294 F03836	294	1.000	$\boldsymbol{p}_0 := \boldsymbol{r}_0$	$\boldsymbol{p}_0 := \boldsymbol{r}_0$	make a steepest descent step. It follows that $\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$ and $\boldsymbol{p}_0 := \boldsymbol{r}_0$.
lyche-numerical-linear-algebra-294 F03837	294	1.000	$j \geq 0$	$j \geq 0$	For the general case we define for $j \geq 0$
lyche-numerical-linear-algebra-294 F03838	294	1.000	$\boldsymbol{p}_j$	$\boldsymbol{p}_j$	1. $\boldsymbol{p}_j$ is computed by the Gram-Schmidt orthogonalization process applied to the
lyche-numerical-linear-algebra-294 F03839	294	0.999	$\boldsymbol{r}_0, \dots, \boldsymbol{r}_j$	$\boldsymbol{r}_0, \dots, \boldsymbol{r}_j$	residuals $\boldsymbol{r}_0, \dots, \boldsymbol{r}_j$ using the $\boldsymbol{A}$ -inner product. The search directions are there-
lyche-numerical-linear-algebra-294 F03840	294	1.000	α_j	α_j	3. It can be shown that the step length α_j is optimal for all j (cf. Exercise 13.7).
lyche-numerical-linear-algebra-295 F03841	295	1.000	$\boldsymbol{x}_j$	$\boldsymbol{x}_j$	$\boldsymbol{x}_j$ is well defined, $\boldsymbol{r}_j \neq 0$, and $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ for $i, j = 0, 1, \dots, k, i \neq j$. Then $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-algebra-295 F03842	295	1.000	$\boldsymbol{r}_j \neq 0$	$\boldsymbol{r}_j \neq 0$	$\boldsymbol{x}_j$ is well defined, $\boldsymbol{r}_j \neq 0$, and $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ for $i, j = 0, 1, \dots, k, i \neq j$. Then $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-algebra-295 F03843	295	1.000	$\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$	$\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$	$\boldsymbol{x}_j$ is well defined, $\boldsymbol{r}_j \neq 0$, and $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ for $i, j = 0, 1, \dots, k, i \neq j$. Then $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-algebra-295 F03844	295	1.000	$i, j=0,1, \ldots, k, i \neq j$	$i, j = 0, 1, \dots, k, i \neq j$	$\boldsymbol{x}_j$ is well defined, $\boldsymbol{r}_j \neq 0$, and $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ for $i, j = 0, 1, \dots, k, i \neq j$. Then $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-algebra-295 F03845	295	1.000	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$	is well defined and $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$ for $j = 0, 1, \dots, k$.
lyche-numerical-linear-algebra-295 F03846	295	1.000	$j=0,1, \ldots, k$	$j = 0, 1, \dots, k$	is well defined and $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$ for $j = 0, 1, \dots, k$.

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lyche-numerical-linear-algebra-F03847	295	1.000	$\boldsymbol{r}_j$	$\boldsymbol{r}_j$	Proof Since the residuals $\boldsymbol{r}_j$ are orthogonal and nonzero for $j \leq k$, they are linearly
lyche-numerical-linear-algebra-F03848	295	1.000	$j \leq k$	$j \leq k$	Proof Since the residuals $\boldsymbol{r}_j$ are orthogonal and nonzero for $j \leq k$, they are linearly
lyche-numerical-linear-algebra-F03849	295	1.000	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-algebra-F03850	295	1.000	$i < k$	$i < k$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-algebra-F03851	295	1.000	$\boldsymbol{r}_{k+1}$	$\boldsymbol{r}_{k+1}$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-algebra-F03852	295	1.000	$j < k$	$j < k$	orthogonality for $j < k$ and by the definition of α_k for $j = k$. This completes
lyche-numerical-linear-algebra-F03853	295	1.000	$\operatorname{dim} \mathbb{R}^n = n$	$\dim \mathbb{R}^n = n$	orthogonal and therefore linearly independent if they are nonzero. Since $\dim \mathbb{R}^n = n$
lyche-numerical-linear-algebra-F03854	295	1.000	$\boldsymbol{r}_0, \dots, \boldsymbol{r}_n$	$\boldsymbol{r}_0, \dots, \boldsymbol{r}_n$	the $n + 1$ residuals $\boldsymbol{r}_0, \dots, \boldsymbol{r}_n$ cannot all be nonzero and we must have $\boldsymbol{r}_k = 0$ for
lyche-numerical-linear-algebra-F03855	295	1.000	$\boldsymbol{r}_k = 0$	$\boldsymbol{r}_k = 0$	the $n + 1$ residuals $\boldsymbol{r}_0, \dots, \boldsymbol{r}_n$ cannot all be nonzero and we must have $\boldsymbol{r}_k = 0$ for
lyche-numerical-linear-algebra-F03856	295	1.000	$j = k + 1$	$j = k + 1$	With $j = k + 1$ (13.11) therefore takes the simple form $\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$ and
lyche-numerical-linear-algebra-F03857	295	1.000	$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$	$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$	With $j = k + 1$ (13.11) therefore takes the simple form $\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$ and
lyche-numerical-linear-algebra-F03858	295	1.000	$\boldsymbol{x}_0, \boldsymbol{p}_0 = \boldsymbol{r}_0 =$	$\boldsymbol{x}_0, \boldsymbol{p}_0 = \boldsymbol{r}_0 =$	To summarize, in the conjugate gradient method we start with $\boldsymbol{x}_0, \boldsymbol{p}_0 = \boldsymbol{r}_0 =$
lyche-numerical-linear-algebra-F03859	295	1.000	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$ and then generate a sequence of vectors $\{\boldsymbol{x}_k\}$ as follows:
lyche-numerical-linear-algebra-F03860	296	1.000	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$	itive definite linear system $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Starting as in Example 13.1 with
lyche-numerical-linear-algebra-F03861	296	1.000	$\boldsymbol{x}_0 = \begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$	$\boldsymbol{x}_0 = \begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$	$\boldsymbol{x}_0 = \begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$ we find $\boldsymbol{p}_0 = \boldsymbol{r}_0 = \begin{bmatrix} 3/2 \\ 0 \end{bmatrix}$ and then
lyche-numerical-linear-algebra-F03862	296	1.000	$\boldsymbol{p}_0 = \boldsymbol{r}_0 = \begin{bmatrix} 3/2 \\ 0 \end{bmatrix}$	$\boldsymbol{p}_0 = \boldsymbol{r}_0 = \begin{bmatrix} 3/2 \\ 0 \end{bmatrix}$	$\boldsymbol{x}_0 = \begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$ we find $\boldsymbol{p}_0 = \boldsymbol{r}_0 = \begin{bmatrix} 3/2 \\ 0 \end{bmatrix}$ and then
lyche-numerical-linear-algebra-F03863	296	1.000	$\boldsymbol{x}_2$	$\boldsymbol{x}_2$	Thus $\boldsymbol{x}_2$ is the exact solution as illustrated in the right part of Fig. 13.1.
lyche-numerical-linear-algebra-F03864	296	0.969	$\ \boldsymbol{r}_k\ _2 / \ \boldsymbol{b}\ _2 \leq$	$\ \boldsymbol{r}_k\ _2 / \ \boldsymbol{b}\ _2 \leq$	starting vector for the iteration. The iteration is stopped when $\ \boldsymbol{r}_k\ _2 / \ \boldsymbol{b}\ _2 \leq \text{tol}$ or
lyche-numerical-linear-algebra-F03865	297	0.803	$k >$	$k >$	$k > \text{itmax}$. K is the number of iterations used:
lyche-numerical-linear-algebra-F03866	297	0.996	$(\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p})$	$(\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p})$	1. one matrix times vector ($\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p}$),
lyche-numerical-linear-algebra-F03867	297	1.000	$(\boldsymbol{p}^T \boldsymbol{t})$	$(\boldsymbol{p}^T \boldsymbol{t})$	2. two inner products ($(\boldsymbol{p}^T \boldsymbol{t})$ and $\boldsymbol{r}^T \boldsymbol{r}$),
lyche-numerical-linear-algebra-F03868	297	1.000	$(\boldsymbol{r}^T \boldsymbol{r})$	$(\boldsymbol{r}^T \boldsymbol{r})$	2. two inner products ($(\boldsymbol{p}^T \boldsymbol{t})$ and $\boldsymbol{r}^T \boldsymbol{r}$),
lyche-numerical-linear-algebra-F03869	297	0.758	$\boldsymbol{x} = \boldsymbol{x} + \alpha \boldsymbol{p}, \boldsymbol{r} = \boldsymbol{r} - \alpha \boldsymbol{t}$	$\boldsymbol{x} = \boldsymbol{x} + \alpha \boldsymbol{p}, \boldsymbol{r} = \boldsymbol{r} - \alpha \boldsymbol{t}$	3. three vector-plus-scalar-times-vector ($\boldsymbol{x} = \boldsymbol{x} + \alpha \boldsymbol{p}, \boldsymbol{r} = \boldsymbol{r} - \alpha \boldsymbol{t}$) and $\boldsymbol{p} = \boldsymbol{r} +$
lyche-numerical-linear-algebra-F03870	297	0.758	$\boldsymbol{p} = \boldsymbol{r} +$	$\boldsymbol{p} = \boldsymbol{r} +$	3. three vector-plus-scalar-times-vector ($\boldsymbol{x} = \boldsymbol{x} + \alpha \boldsymbol{p}, \boldsymbol{r} = \boldsymbol{r} - \alpha \boldsymbol{t}$ and $\boldsymbol{p} = \boldsymbol{r} +$
lyche-numerical-linear-algebra-F03871	297	1.000	$\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p}$	$\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p}$	The dominating part is the computation of $\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p}$.
lyche-numerical-linear-algebra-F03872	297	1.000	$\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$	$\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$	Kronecker sum $\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$, where $\boldsymbol{T}_1 := \text{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$

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lyche-numerical-linear-algebra-F03873	297	■ 1.000	$\backslash\boldsymbol{T}_1:=\operatorname{tridiag}_m(a, d, a)$ $\backslash\operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$	$T_1 := \operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$	Kronecker sum $T_2 := T_1 \otimes I + I \otimes T_1$, where $T_1 := \operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$
lyche-numerical-linear-algebra-F03874	297	■ 1.000	$\backslash\boldsymbol{f}=[1, \ldots, 1]^T \in \mathbb{R}^n$	$\boldsymbol{f} = [1, \dots, 1]^T \in \mathbb{R}^n$	(cf. Theorem 10.2). We set $h = 1/(m + 1)$ and $\boldsymbol{f} = [1, \dots, 1]^T \in \mathbb{R}^n$
lyche-numerical-linear-algebra-F03875	297	■ 1.000	$a=1 / 9, d=5 / 18$	$a = 1/9, d = 5/18$	1. $a = 1/9, d = 5/18$, the Averaging matrix.
lyche-numerical-linear-algebra-F03876	297	■ 1.000	$a=-1, d=2$	$a = -1, d = 2$	2. $a = -1, d = 2$, the Poisson matrix.
lyche-numerical-linear-algebra-F03877	297	■ 1.000	$O(5 \ n)$	$O(5n)$	Note that for our test problems T_2 only has $O(5n)$ nonzero elements. Therefore,
lyche-numerical-linear-algebra-F03878	297	■ 1.000	$\backslash\boldsymbol{t}$	$\boldsymbol{t}$	taking advantage of the sparseness of T_2 we can compute $\boldsymbol{t}$ in Algorithm 13.1
lyche-numerical-linear-algebra-F03879	298	■ 1.000	$\backslash\boldsymbol{V}, \backslash\boldsymbol{R}, \backslash\boldsymbol{P}, \backslash\boldsymbol{B}, \backslash\boldsymbol{T} \in$ $\backslash\boldsymbol{B}, \backslash\boldsymbol{T} \in$	$\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$	tageous to use a matrix equation formulation. We define matrices $\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$
lyche-numerical-linear-algebra-F03880	298	■ 1.000	$\backslash\boldsymbol{x}=\operatorname{vec}(\backslash\boldsymbol{V}),$ $\backslash\boldsymbol{r}=\operatorname{vec}(\backslash\boldsymbol{R}),$ $\backslash\boldsymbol{p}=\operatorname{vec}(\backslash\boldsymbol{P}),$ $\backslash\boldsymbol{t}=\operatorname{vec}(\backslash\boldsymbol{T})$	$\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V}), \boldsymbol{r} = \operatorname{vec}(\boldsymbol{R}), \boldsymbol{p} = \operatorname{vec}(\boldsymbol{P}), \boldsymbol{t} = \operatorname{vec}(\boldsymbol{T})$	$\mathbb{R}^{m \times m}$ by $\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V}), \boldsymbol{r} = \operatorname{vec}(\boldsymbol{R}), \boldsymbol{p} = \operatorname{vec}(\boldsymbol{P}), \boldsymbol{t} = \operatorname{vec}(\boldsymbol{T})$, and $h^2 \boldsymbol{f} = \operatorname{vec}(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F03881	298	■ 1.000	$h^2 \ \backslash\boldsymbol{f}=\operatorname{vec}(\backslash\boldsymbol{B})$	$h^2 \boldsymbol{f} = \operatorname{vec}(\boldsymbol{B})$	$\mathbb{R}^{m \times m}$ by $\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V}), \boldsymbol{r} = \operatorname{vec}(\boldsymbol{R}), \boldsymbol{p} = \operatorname{vec}(\boldsymbol{P}), \boldsymbol{t} = \operatorname{vec}(\boldsymbol{T})$, and $h^2 \boldsymbol{f} = \operatorname{vec}(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F03882	298	■ 1.000	$\backslash\boldsymbol{T}_2 \ \backslash\boldsymbol{x}=h^2 \ \backslash\boldsymbol{f}$ $\Longleftarrow \backslash\boldsymbol{T}_1 \ \backslash\boldsymbol{V}+$ $\backslash\boldsymbol{V} \ \backslash\boldsymbol{T}_1=\backslash\boldsymbol{B}$	$T_2 \boldsymbol{x} = h^2 \boldsymbol{f} \iff T_1 \boldsymbol{V} + \boldsymbol{V} T_1 = \boldsymbol{B}$	Then $T_2 \boldsymbol{x} = h^2 \boldsymbol{f} \iff T_1 \boldsymbol{V} + \boldsymbol{V} T_1 = \boldsymbol{B}$, and $\boldsymbol{t} = T_2 \boldsymbol{p} \iff \boldsymbol{T} = T_1 \boldsymbol{P} + \boldsymbol{P} T_1$.
lyche-numerical-linear-algebra-F03883	298	■ 1.000	$\backslash\boldsymbol{t}=\backslash\boldsymbol{T}_2 \ \backslash\boldsymbol{p}$ $\Longleftarrow \backslash\boldsymbol{T}=\backslash\boldsymbol{T}_1 \ \backslash\boldsymbol{P}+$ $\backslash\boldsymbol{P} \ \backslash\boldsymbol{T}_1$	$\boldsymbol{t} = T_2 \boldsymbol{p} \iff \boldsymbol{T} = T_1 \boldsymbol{P} + \boldsymbol{P} T_1$	Then $T_2 \boldsymbol{x} = h^2 \boldsymbol{f} \iff T_1 \boldsymbol{V} + \boldsymbol{V} T_1 = \boldsymbol{B}$, and $\boldsymbol{t} = T_2 \boldsymbol{p} \iff \boldsymbol{T} = T_1 \boldsymbol{P} + \boldsymbol{P} T_1$.
lyche-numerical-linear-algebra-F03884	298	■ 0.958	$t o l=10^{-8}$	$tol = 10^{-8}$	For both the averaging- and Poisson matrix we use $tol = 10^{-8}$.
lyche-numerical-linear-algebra-F03885	298	■ 1.000	$K / \sqrt{n}$	$K/\sqrt{n}$	of iterations, and $K/\sqrt{n}$.
lyche-numerical-linear-algebra-F03886	299	■ 1.000	$\ \backslash\boldsymbol{x}\ _{\boldsymbol{A}}:=\sqrt{\backslash\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}$	$\ \boldsymbol{x}\ _A := \sqrt{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}$	Recall that the $\boldsymbol{A}$ -norm of a vector $\boldsymbol{x} \in \mathbb{R}^n$ is given by $\ \boldsymbol{x}\ _A := \sqrt{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}$. The
lyche-numerical-linear-algebra-F03887	299	■ 0.995	$\kappa=\operatorname{cond}_2(\backslash\boldsymbol{A}):=\lambda_{\text {max }} / \lambda_{\text {min }}$	$\kappa = \operatorname{cond}_2(\boldsymbol{A}) := \lambda_{\max} / \lambda_{\min}$	Here $\kappa = \operatorname{cond}_2(\boldsymbol{A}) := \lambda_{\max} / \lambda_{\min}$ is the spectral condition number of $\boldsymbol{A}$, where
lyche-numerical-linear-algebra-F03888	299	■ 1.000	$\frac{\kappa-1}{\kappa+1}<1$	$\frac{\kappa - 1}{\kappa + 1} < 1$	1. Since $\frac{\kappa-1}{\kappa+1} < 1$ the steepest descent method always converges for a positive
lyche-numerical-linear-algebra-F03889	299	■ 1.000	$\frac{\kappa-1}{\kappa+1}$	$\frac{\kappa - 1}{\kappa + 1}$	definite matrix. The convergence can be slow when $\frac{\kappa-1}{\kappa+1}$ is close to one, and this
lyche-numerical-linear-algebra-F03890	300	■ 1.000	$d=5 / 18, a=1 / 9$	$d = 5/18, a = 1/9$	For the averaging problem given by $d = 5/18, a = 1/9$, the largest and smallest
lyche-numerical-linear-algebra-F03891	300	■ 0.999	$\lambda_{\text {max }}=\frac{5}{9}+\frac{4}{9} \cos (\pi h)$	$\lambda_{\max} = \frac{5}{9} + \frac{4}{9} \cos(\pi h)$	eigenvalue of T_2 are given by $\lambda_{\max} = \frac{5}{9} + \frac{4}{9} \cos(\pi h)$ and $\lambda_{\min} = \frac{5}{9} - \frac{4}{9} \cos(\pi h)$.
lyche-numerical-linear-algebra-F03892	300	■ 0.999	$\lambda_{\text {min }}=\frac{5}{9}-\frac{4}{9} \cos (\pi h)$	$\lambda_{\min} = \frac{5}{9} - \frac{4}{9} \cos(\pi h)$	eigenvalue of T_2 are given by $\lambda_{\max} = \frac{5}{9} + \frac{4}{9} \cos(\pi h)$ and $\lambda_{\min} = \frac{5}{9} - \frac{4}{9} \cos(\pi h)$.
lyche-numerical-linear-algebra-F03893	300	■ 1.000	$\mathbb{W}_0=\{\mathbf{0}\}$	$\mathbb{W}_0 = \{\mathbf{0}\}$	The Krylov spaces are defined by $\mathbb{W}_0 = \{\mathbf{0}\}$ and
lyche-numerical-linear-algebra-F03894	301	■ 1.000	$\operatorname{dim}(\mathbb{W}_k) \leq k$	$\dim(\mathbb{W}_k) \leq k$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$.
lyche-numerical-linear-algebra-F03895	301	■ 1.000	$\backslash\boldsymbol{v} \in \mathbb{W}_k$	$\boldsymbol{v} \in \mathbb{W}_k$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$.
lyche-numerical-linear-algebra-F03896	301	■ 1.000	$\boldsymbol{A} \backslash\boldsymbol{v} \in \mathbb{W}_{k+1}$	$\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$.
lyche-numerical-linear-algebra-F03897	301	■ 1.000	$\backslash\boldsymbol{p}_0=\backslash\boldsymbol{r}_0$	$\boldsymbol{p}_0 = \boldsymbol{r}_0$	Proof Equation (13.22) clearly holds for $k = 0$ since $\boldsymbol{p}_0 = \boldsymbol{r}_0$. Suppose it holds

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lyche-numerical-linear-algebra-F03898	301	1.000	$\boldsymbol{r}_{k+1}=\boldsymbol{r}_k-\alpha_k\boldsymbol{A}\boldsymbol{p}_k\in\mathbb{W}_{k+2},\boldsymbol{p}_{k+1}=\boldsymbol{r}_{k+1}+\beta_k\boldsymbol{p}_k\in\mathbb{W}_{k+2}$	$\boldsymbol{r}_{k+1}=\boldsymbol{r}_k-\alpha_k\boldsymbol{A}\boldsymbol{p}_k\in\mathbb{W}_{k+2},\boldsymbol{p}_{k+1}=\boldsymbol{r}_{k+1}+\beta_k\boldsymbol{p}_k\in\mathbb{W}_{k+2}$	for some $k\geq 0$. Then $\boldsymbol{r}_{k+1}=\boldsymbol{r}_k-\alpha_k\boldsymbol{A}\boldsymbol{p}_k\in\mathbb{W}_{k+2},\boldsymbol{p}_{k+1}=\boldsymbol{r}_{k+1}+\beta_k\boldsymbol{p}_k\in\mathbb{W}_{k+2}$
lyche-numerical-linear-algebra-F03899	301	0.995	$\mathbb{W}_{k+2}$	$\mathbb{W}_{k+2}$	$\mathbb{W}_{k+2}$ and $\boldsymbol{x}_{k+1}-\boldsymbol{x}_0\stackrel{(13.12)}{=}\boldsymbol{x}_k-\boldsymbol{x}_0+\alpha_k\boldsymbol{p}_k\in\mathbb{W}_{k+1}$. Thus (13.22) follows by
lyche-numerical-linear-algebra-F03900	301	0.995	$\boldsymbol{x}_{k+1}-\boldsymbol{x}_0\stackrel{(13.12)}{=}\boldsymbol{x}_k-\boldsymbol{x}_0+\alpha_k\boldsymbol{p}_k\in\mathbb{W}_{k+1}$	$\boldsymbol{x}_{k+1}-\boldsymbol{x}_0\stackrel{(13.12)}{=}\boldsymbol{x}_k-\boldsymbol{x}_0+\alpha_k\boldsymbol{p}_k\in\mathbb{W}_{k+1}$	$\mathbb{W}_{k+2}$ and $\boldsymbol{x}_{k+1}-\boldsymbol{x}_0\stackrel{(13.12)}{=}\boldsymbol{x}_k-\boldsymbol{x}_0+\alpha_k\boldsymbol{p}_k\in\mathbb{W}_{k+1}$. Thus (13.22) follows by
lyche-numerical-linear-algebra-F03901	301	0.984	$\boldsymbol{w}\in\mathbb{W}_k$	$\boldsymbol{w}\in\mathbb{W}_k$	induction. The equation (13.23) follows since any $\boldsymbol{w}\in\mathbb{W}_k$ is a linear combination
lyche-numerical-linear-algebra-F03902	301	1.000	$\{\boldsymbol{r}_0,\boldsymbol{r}_1,\dots,\boldsymbol{r}_{k-1}\}$	$\{\boldsymbol{r}_0,\boldsymbol{r}_1,\dots,\boldsymbol{r}_{k-1}\}$	of $\{\boldsymbol{r}_0,\boldsymbol{r}_1,\dots,\boldsymbol{r}_{k-1}\}$ and also $\{\boldsymbol{p}_0,\boldsymbol{p}_1,\dots,\boldsymbol{p}_{k-1}\}$.
lyche-numerical-linear-algebra-F03903	301	1.000	$\{\boldsymbol{p}_0,\boldsymbol{p}_1,\dots,\boldsymbol{p}_{k-1}\}$	$\{\boldsymbol{p}_0,\boldsymbol{p}_1,\dots,\boldsymbol{p}_{k-1}\}$	of $\{\boldsymbol{r}_0,\boldsymbol{r}_1,\dots,\boldsymbol{r}_{k-1}\}$ and also $\{\boldsymbol{p}_0,\boldsymbol{p}_1,\dots,\boldsymbol{p}_{k-1}\}$.
lyche-numerical-linear-algebra-F03904	301	0.995	$\boldsymbol{u}:=\boldsymbol{x}_k-\boldsymbol{x}_0-\boldsymbol{w}$	$\boldsymbol{u}:=\boldsymbol{x}_k-\boldsymbol{x}_0-\boldsymbol{w}$	Proof Fix k , let $\boldsymbol{w}\in\mathbb{W}_k$ and $\boldsymbol{u}:=\boldsymbol{x}_k-\boldsymbol{x}_0-\boldsymbol{w}$. By (13.22) $\boldsymbol{u}\in\mathbb{W}_k$ and then (13.23)
lyche-numerical-linear-algebra-F03905	301	0.995	$\boldsymbol{u}\in\mathbb{W}_k$	$\boldsymbol{u}\in\mathbb{W}_k$	Proof Fix k , let $\boldsymbol{w}\in\mathbb{W}_k$ and $\boldsymbol{u}:=\boldsymbol{x}_k-\boldsymbol{x}_0-\boldsymbol{w}$. By (13.22) $\boldsymbol{u}\in\mathbb{W}_k$ and then (13.23)
lyche-numerical-linear-algebra-F03906	301	1.000	$\langle\boldsymbol{x}-\boldsymbol{x}_k,\boldsymbol{u}\rangle=\boldsymbol{r}_k^T\boldsymbol{u}=0$	$\langle\boldsymbol{x}-\boldsymbol{x}_k,\boldsymbol{u}\rangle=\boldsymbol{r}_k^T\boldsymbol{u}=0$	implies that $\langle\boldsymbol{x}-\boldsymbol{x}_k,\boldsymbol{u}\rangle=\boldsymbol{r}_k^T\boldsymbol{u}=0$. Using Corollary 5.2 we obtain
lyche-numerical-linear-algebra-F03907	301	1.000	$\boldsymbol{u}=\mathbf{0}$	$\boldsymbol{u}=\mathbf{0}$	with equality for $\boldsymbol{u}=\mathbf{0}$.
lyche-numerical-linear-algebra-F03908	301	1.000	$\boldsymbol{x}_0\neq\mathbf{0}$	$\boldsymbol{x}_0\neq\mathbf{0}$	solution $\boldsymbol{x}$ in the $\boldsymbol{A}$ -norm. More generally, if $\boldsymbol{x}_0\neq\mathbf{0}$ then $\boldsymbol{x}-\boldsymbol{x}_k=(\boldsymbol{x}-\boldsymbol{x}_0)-$
lyche-numerical-linear-algebra-F03909	301	1.000	$\boldsymbol{x}-\boldsymbol{x}_k=(\boldsymbol{x}-\boldsymbol{x}_0)-$	$\boldsymbol{x}-\boldsymbol{x}_k=(\boldsymbol{x}-\boldsymbol{x}_0)-$	solution $\boldsymbol{x}$ in the $\boldsymbol{A}$ -norm. More generally, if $\boldsymbol{x}_0\neq\mathbf{0}$ then $\boldsymbol{x}-\boldsymbol{x}_k=(\boldsymbol{x}-\boldsymbol{x}_0)-$
lyche-numerical-linear-algebra-F03910	301	1.000	$(\boldsymbol{x}_k-\boldsymbol{x}_0)$	$(\boldsymbol{x}_k-\boldsymbol{x}_0)$	$(\boldsymbol{x}_k-\boldsymbol{x}_0)$ and $\boldsymbol{x}_k-\boldsymbol{x}_0$ is the element in $\mathbb{W}_k$ that is closest to $\boldsymbol{x}-\boldsymbol{x}_0$ in the $\boldsymbol{A}$ -norm.
lyche-numerical-linear-algebra-F03911	301	1.000	$\boldsymbol{x}_k-\boldsymbol{x}_0$	$\boldsymbol{x}_k-\boldsymbol{x}_0$	$(\boldsymbol{x}_k-\boldsymbol{x}_0)$ and $\boldsymbol{x}_k-\boldsymbol{x}_0$ is the element in $\mathbb{W}_k$ that is closest to $\boldsymbol{x}-\boldsymbol{x}_0$ in the $\boldsymbol{A}$ -norm.
lyche-numerical-linear-algebra-F03912	301	1.000	$p(t):=\sum_{j=0}^ma_jt^j$	$p(t):=\sum_{j=0}^ma_jt^j$	Recall that to each polynomial $p(t):=\sum_{j=0}^ma_jt^j$ there corresponds a matrix
lyche-numerical-linear-algebra-F03913	301	0.808	$p(\boldsymbol{A}):=a_0\boldsymbol{I}+a_1\boldsymbol{A}+\dots+a_m\boldsymbol{A}^m$	$p(\boldsymbol{A}):=a_0\boldsymbol{I}+a_1\boldsymbol{A}+\dots+a_m\boldsymbol{A}^m$	polynomial $p(\boldsymbol{A}):=a_0\boldsymbol{I}+a_1\boldsymbol{A}+\dots+a_m\boldsymbol{A}^m$. Moreover, if $(\lambda_j,\boldsymbol{u}_j)$ are eigenpairs
lyche-numerical-linear-algebra-F03914	301	0.979	$(p(\lambda_j),\boldsymbol{u}_j)$	$(p(\lambda_j),\boldsymbol{u}_j)$	of $\boldsymbol{A}$ then $(p(\lambda_j),\boldsymbol{u}_j)$ are eigenpairs of $p(\boldsymbol{A})$ for $j=1,\dots,n$.
lyche-numerical-linear-algebra-F03915	301	0.979	$p(\boldsymbol{A})$	$p(\boldsymbol{A})$	of $\boldsymbol{A}$ then $(p(\lambda_j),\boldsymbol{u}_j)$ are eigenpairs of $p(\boldsymbol{A})$ for $j=1,\dots,n$.
lyche-numerical-linear-algebra-F03916	301	0.991	$(\lambda_j,\boldsymbol{u}_j),j=1,2,\dots,n$	$(\lambda_j,\boldsymbol{u}_j),j=1,2,\dots,n$	$\mathbb{R}^{n\times n}$ is positive definite with orthonormal eigenpairs $(\lambda_j,\boldsymbol{u}_j),j=1,2,\dots,n$,
lyche-numerical-linear-algebra-F03917	301	1.000	$\boldsymbol{r}_0:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_0$	$\boldsymbol{r}_0:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_0$	and let $\boldsymbol{r}_0:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_0$ for some $\boldsymbol{x}_0\in\mathbb{R}^n$. To each $\boldsymbol{w}\in\mathbb{W}_k$ there corresponds
lyche-numerical-linear-algebra-F03918	302	1.000	$P(t):=\sum_{j=0}^{k-1}a_jt^j$	$P(t):=\sum_{j=0}^{k-1}a_jt^j$	a polynomial $P(t):=\sum_{j=0}^{k-1}a_jt^j$ such that $\boldsymbol{w}=P(\boldsymbol{A})\boldsymbol{r}_0$. Moreover, if $\boldsymbol{r}_0=$
lyche-numerical-linear-algebra-F03919	302	1.000	$\boldsymbol{w}=P(\boldsymbol{A})\boldsymbol{r}_0$	$\boldsymbol{w}=P(\boldsymbol{A})\boldsymbol{r}_0$	a polynomial $P(t):=\sum_{j=0}^{k-1}a_jt^j$ such that $\boldsymbol{w}=P(\boldsymbol{A})\boldsymbol{r}_0$. Moreover, if $\boldsymbol{r}_0=$
lyche-numerical-linear-algebra-F03920	302	1.000	$\sum_{j=1}^n\sigma_j\boldsymbol{u}_j$	$\sum_{j=1}^n\sigma_j\boldsymbol{u}_j$	$\sum_{j=1}^n\sigma_j\boldsymbol{u}_j$ then
lyche-numerical-linear-algebra-F03921	302	1.000	$\boldsymbol{w}=a_0\boldsymbol{r}_0+a_1\boldsymbol{A}\boldsymbol{r}_0+\dots+a_{k-1}\boldsymbol{A}^{k-1}\boldsymbol{r}_0$	$\boldsymbol{w}=a_0\boldsymbol{r}_0+a_1\boldsymbol{A}\boldsymbol{r}_0+\dots+a_{k-1}\boldsymbol{A}^{k-1}\boldsymbol{r}_0$	Proof If $\boldsymbol{w}\in\mathbb{W}_k$ then $\boldsymbol{w}=a_0\boldsymbol{r}_0+a_1\boldsymbol{A}\boldsymbol{r}_0+\dots+a_{k-1}\boldsymbol{A}^{k-1}\boldsymbol{r}_0$ for some scalars

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lyche-numerical-linear-algebra- F03922	302	1.000	$a_{\{0\}}, \ldots, a_{\{k-1\}}$	$a_0, \dots, a_{k-1}$	$a_0, \dots, a_{k-1}$. But then $w = P(A)r_0$. We find $x - x_0 - P(A)r_0 = A^{-1}(r_0 -$
lyche-numerical-linear-algebra- F03923	302	1.000	$\boldsymbol{x}-\boldsymbol{x}_{\{0\}}-P(\boldsymbol{A})\boldsymbol{r}_0=\boldsymbol{A}^{-1}\left(\boldsymbol{r}_0-\boldsymbol{x}_{\{0\}}\right)$	$x - x_0 - P(A)r_0 = A^{-1}(r_0 -$	$a_0, \dots, a_{k-1}$. But then $w = P(A)r_0$. We find $x - x_0 - P(A)r_0 = A^{-1}(r_0 -$
lyche-numerical-linear-algebra- F03924	302	1.000	$P(\boldsymbol{A})\boldsymbol{r}_0=\boldsymbol{A}^{-1}Q(\boldsymbol{A})\boldsymbol{r}_0$	$AP(A)r_0 = A^{-1}Q(A)r_0$	$AP(A)r_0 = A^{-1}Q(A)r_0$ and $A(x - x_0 - P(A)r_0) = Q(A)r_0$. Therefore,
lyche-numerical-linear-algebra- F03925	302	1.000	$\boldsymbol{A}\left(\boldsymbol{x}-\boldsymbol{x}_{\{0\}}-P(\boldsymbol{A})\boldsymbol{r}_0\right)=Q(\boldsymbol{A})\boldsymbol{r}_0$	$A(x - x_0 - P(A)r_0) = Q(A)r_0$	$AP(A)r_0 = A^{-1}Q(A)r_0$ and $A(x - x_0 - P(A)r_0) = Q(A)r_0$. Therefore,
lyche-numerical-linear-algebra- F03926	302	1.000	$\boldsymbol{r}_{\{0\}}$	r_0	Using the eigenvector expansion for r_0 we obtain
lyche-numerical-linear-algebra- F03927	303	0.978	$0<a<b$	$0 < a < b$	$0 < a < b$ is an interval containing all the eigenvalues of A . Then in the conjugate
lyche-numerical-linear-algebra- F03928	303	0.978	A	A	$0 < a < b$ is an interval containing all the eigenvalues of A . Then in the conjugate
lyche-numerical-linear-algebra- F03929	303	1.000	$\Pi_{\{k\}}$	Π_k	where Π_k denotes the class of univariate polynomials of degree $\leq k$ with real
lyche-numerical-linear-algebra- F03930	303	1.000	$\leq k$	$\leq k$	where Π_k denotes the class of univariate polynomials of degree $\leq k$ with real
lyche-numerical-linear-algebra- F03931	303	0.998	$Q(t)=1$	$Q(t) = 1$	Proof By (13.25) with $Q(t) = 1$ (corresponding to $P(A) = 0$) we find $\ x -$
lyche-numerical-linear-algebra- F03932	303	0.998	$P(\boldsymbol{A})=\mathbf{0}$	$P(A) = 0$	Proof By (13.25) with $Q(t) = 1$ (corresponding to $P(A) = 0$) we find $\ x -$
lyche-numerical-linear-algebra- F03933	303	0.998	$\ \boldsymbol{x}-$	$\ x -$	Proof By (13.25) with $Q(t) = 1$ (corresponding to $P(A) = 0$) we find $\ x -$
lyche-numerical-linear-algebra- F03934	303	0.969	$\ \boldsymbol{x}_{\{0\}}\ _{\boldsymbol{A}}^2=\sum_{j=1}^n\frac{\sigma_j^2}{\lambda_j}$	$x_0\ _A^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}$	$x_0\ _A^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}$. Therefore, by the best approximation property Theorem 13.4
lyche-numerical-linear-algebra- F03935	303	1.000	$Q \in \Pi_{\{k\}}$	$Q \in \Pi_k$	where $Q \in \Pi_k$ and $Q(0) = 1$. Minimizing over such polynomials Q and taking
lyche-numerical-linear-algebra- F03936	303	1.000	$Q(0)=1$	$Q(0) = 1$	where $Q \in \Pi_k$ and $Q(0) = 1$. Minimizing over such polynomials Q and taking
lyche-numerical-linear-algebra- F03937	303	0.999	$\kappa=\lambda_{\text{max}} / \lambda_{\text{min}}$	$\kappa = \lambda_{\max} / \lambda_{\min}$	where $\kappa = \lambda_{\max} / \lambda_{\min}$ is the spectral condition number of A . Ignoring the second
lyche-numerical-linear-algebra- F03938	303	1.000	$y=1 / x$	$y = 1/x$	$y = 1/x$, using $2^k/k! = 2, k = 1, 2$, and $2^k/k! < 2$ for $k > 2$, we find
lyche-numerical-linear-algebra- F03939	303	1.000	$2^{\{k\}} / k!=2, k=1,2$	$2^k/k! = 2, k = 1, 2$	$y = 1/x$, using $2^k/k! = 2, k = 1, 2$, and $2^k/k! < 2$ for $k > 2$, we find
lyche-numerical-linear-algebra- F03940	303	1.000	$2^{\{k\}} / k!<2$	$2^k/k! < 2$	$y = 1/x$, using $2^k/k! = 2, k = 1, 2$, and $2^k/k! < 2$ for $k > 2$, we find
lyche-numerical-linear-algebra- F03941	303	1.000	$k>2$	$k > 2$	$y = 1/x$, using $2^k/k! = 2, k = 1, 2$, and $2^k/k! < 2$ for $k > 2$, we find
lyche-numerical-linear-algebra- F03942	304	1.000	$c \notin[a, b]$	$c \notin [a, b]$	$c \notin [a, b]$ and $k \in \mathbb{N}$. Consider the set $\mathcal{S}_k$ of all polynomials Q of degree $\leq k$ such
lyche-numerical-linear-algebra- F03943	304	1.000	$\mathcal{S}_{\{k\}}$	$\mathcal{S}_k$	$c \notin [a, b]$ and $k \in \mathbb{N}$. Consider the set $\mathcal{S}_k$ of all polynomials Q of degree $\leq k$ such
lyche-numerical-linear-algebra- F03944	304	1.000	$Q(c)=1$	$Q(c) = 1$	that $Q(c) = 1$. For any continuous function f on $[a, b]$ we define
lyche-numerical-linear-algebra- F03945	304	1.000	$Q^* \in \mathcal{S}_{\{k\}}$	$Q^* \in \mathcal{S}_k$	We want to find a polynomial $Q^* \in \mathcal{S}_k$ such that
lyche-numerical-linear-algebra- F03946	304	1.000	$Q^{\{*\}}$	Q^*	We will show that Q^* is uniquely given as a suitably shifted and normalized
lyche-numerical-linear-algebra- F03947	304	1.000	$T_{\{n\}}$	T_n	version of the Chebyshev polynomial . The Chebyshev polynomial T_n of degree n

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lyche-numerical-linear-algebra- F03948	304	0.999	$T_{\{0\}}(t)=1$	$T_0(t) = 1$	starting with $T_0(t) = 1$ and $T_1(t) = t$. Thus $T_2(t) = 2t^2 - 1$, $T_3(t) = 4t^3 - 3t$ etc.
lyche-numerical-linear-algebra- F03949	304	0.999	$T_{\{1\}}(t)=t$	$T_1(t) = t$	starting with $T_0(t) = 1$ and $T_1(t) = t$. Thus $T_2(t) = 2t^2 - 1$, $T_3(t) = 4t^3 - 3t$ etc.
lyche-numerical-linear-algebra- F03950	304	0.999	$T_{\{2\}}(t)=2\ t^{\{2\}}-1$, $T_{\{3\}}(t)=4\ t^{\{3\}}-3\ t$	$T_2(t) = 2t^2 - 1, T_3(t) = 4t^3 - 3t$	starting with $T_0(t) = 1$ and $T_1(t) = t$. Thus $T_2(t) = 2t^2 - 1, T_3(t) = 4t^3 - 3t$ etc.
lyche-numerical-linear-algebra- F03951	304	1.000	$T_{\{n\}}(t)=\cos\ (n\ \arccos\ t)$	$T_n(t) = \cos(n\ \arccos t)$	1. $T_n(t) = \cos(n\ \arccos t)$ for $t \in [-1, 1]$,
lyche-numerical-linear-algebra- F03952	304	1.000	$t\ \text{in}[-1,1]$	$t \in [-1, 1]$	1. $T_n(t) = \cos(n\ \arccos t)$ for $t \in [-1, 1]$,
lyche-numerical-linear-algebra- F03953	304	1.000	$T_{\{n\}}(t)=\frac{1}{2}\left[\left(t+\sqrt{t^2-1}\right)^n+\left(t+\sqrt{t^2-1}\right)^{-n}\right]$	$T_n(t) = \frac{1}{2} \left[\left(t + \sqrt{t^2 - 1} \right)^n + \left(t + \sqrt{t^2 - 1} \right)^{-n} \right]$	2. $T_n(t) = \frac{1}{2} \left[\left(t + \sqrt{t^2 - 1} \right)^n + \left(t + \sqrt{t^2 - 1} \right)^{-n} \right]$ for $ t \geq 1$.
lyche-numerical-linear-algebra- F03954	304	1.000	$ t \ \geq\ 1$	$ t \geq 1$	2. $T_n(t) = \frac{1}{2} \left[\left(t + \sqrt{t^2 - 1} \right)^n + \left(t + \sqrt{t^2 - 1} \right)^{-n} \right]$ for $ t \geq 1$.
lyche-numerical-linear-algebra- F03955	304	1.000	$P_{\{n\}}(t)=\cos\ (n\ \arccos\ t)$	$P_n(t) = \cos(n\ \arccos t)$	1. With $P_n(t) = \cos(n\ \arccos t)$ we have $P_n(t) = \cos n\phi$, where $t = \cos \phi$.
lyche-numerical-linear-algebra- F03956	304	1.000	$P_{\{n\}}(t)=\cos\ n\ \phi$	$P_n(t) = \cos n\phi$	1. With $P_n(t) = \cos(n\ \arccos t)$ we have $P_n(t) = \cos n\phi$, where $t = \cos \phi$.
lyche-numerical-linear-algebra- F03957	304	1.000	$t=\cos\ \phi$	$t = \cos \phi$	1. With $P_n(t) = \cos(n\ \arccos t)$ we have $P_n(t) = \cos n\phi$, where $t = \cos \phi$.
lyche-numerical-linear-algebra- F03958	304	1.000	$P_{\{n\}}$	P_n	and it follows that P_n satisfies the same recurrence relation as T_n . Since $P_0 = T_0$
lyche-numerical-linear-algebra- F03959	304	1.000	$P_{\{0\}}=T_{\{0\}}$	$P_0 = T_0$	and it follows that P_n satisfies the same recurrence relation as T_n . Since $P_0 = T_0$
lyche-numerical-linear-algebra- F03960	304	1.000	$P_{\{1\}}=T_{\{1\}}$	$P_1 = T_1$	and $P_1 = T_1$ we have $P_n = T_n$ for all $n \geq 0$.
lyche-numerical-linear-algebra- F03961	304	1.000	$P_{\{n\}}=T_{\{n\}}$	$P_n = T_n$	and $P_1 = T_1$ we have $P_n = T_n$ for all $n \geq 0$.
lyche-numerical-linear-algebra- F03962	304	1.000	$x_{\{n\}}:=T_{\{n\}}(t)$	$x_n := T_n(t)$	2. Fix t with $ t \geq 1$ and let $x_n := T_n(t)$ for $n \geq 0$. The recurrence relation for the
lyche-numerical-linear-algebra- F03963	305	0.711	$x_{\{n\}}=z^{\{n\}}$	$x_n = z^n$	To solve this difference equation we insert $x_n = z^n$ into (13.31) and obtain $z^{n+1} -$
lyche-numerical-linear-algebra- F03964	305	0.711	$z^{\{n+1\}} -$	$z^{n+1} -$	To solve this difference equation we insert $x_n = z^n$ into (13.31) and obtain $z^{n+1} -$
lyche-numerical-linear-algebra- F03965	305	1.000	$2\ t\ z^{\{n\}}+z^{\{n-1\}}=0$	$2tz^n + z^{n-1} = 0$	$2tz^n + z^{n-1} = 0$ or $z^2 - 2tz + 1 = 0$. The roots of this equation are
lyche-numerical-linear-algebra- F03966	305	1.000	$z^{\{2\}}-2\ t\ z+1=0$	$z^2 - 2tz + 1 = 0$	$2tz^n + z^{n-1} = 0$ or $z^2 - 2tz + 1 = 0$. The roots of this equation are
lyche-numerical-linear-algebra- F03967	305	1.000	$z_{\{1\}}^{\{n\}},\ z_{\{2\}}^{\{n\}}$	z_1^n, z_2^n	Now z_1^n, z_2^n and more generally $c_1 z_1^n + c_2 z_2^n$ are solutions of (13.31) for any
lyche-numerical-linear-algebra- F03968	305	1.000	$c_{\{1\}}\ z_{\{1\}}^{\{n\}}+c_{\{2\}}\ z_{\{2\}}^{\{n\}}$	$c_1 z_1^n + c_2 z_2^n$	Now z_1^n, z_2^n and more generally $c_1 z_1^n + c_2 z_2^n$ are solutions of (13.31) for any
lyche-numerical-linear-algebra- F03969	305	1.000	$c_{\{1\}}$	c_1	constants c_1 and c_2 . We find these constants from the initial conditions $x_0 =$
lyche-numerical-linear-algebra- F03970	305	1.000	$c_{\{2\}}$	c_2	constants c_1 and c_2 . We find these constants from the initial conditions $x_0 =$
lyche-numerical-linear-algebra- F03971	305	1.000	$x_{\{0\}}=$	$x_0 =$	constants c_1 and c_2 . We find these constants from the initial conditions $x_0 =$
lyche-numerical-linear-algebra- F03972	305	1.000	$c_{\{1\}}+c_{\{2\}}=1$	$c_1 + c_2 = 1$	$c_1 + c_2 = 1$ and $x_1 = c_1 z_1 + c_2 z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is
lyche-numerical-linear-algebra- F03973	305	1.000	$x_{\{1\}}=c_{\{1\}}\ z_{\{1\}}+c_{\{2\}}\ z_{\{2\}}=t$	$x_1 = c_1 z_1 + c_2 z_2 = t$	$c_1 + c_2 = 1$ and $x_1 = c_1 z_1 + c_2 z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is
lyche-numerical-linear-algebra- F03974	305	1.000	$z_{\{1\}}+z_{\{2\}}=2\ t$	$z_1 + z_2 = 2t$	$c_1 + c_2 = 1$ and $x_1 = c_1 z_1 + c_2 z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is

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lyche-numerical-linear-algebra- F03975	305	■ 1.000	$c_{\{1\}}=c_{\{2\}}=\frac{1}{2}$	$c_1 = c_2 = \frac{1}{2}$	
lyche-numerical-linear-algebra- F03976	305	■ 1.000	$a < b, c \notin [a, b]$	$a < b, c \notin [a, b]$	Theorem 13.6 (A Minimal Norm Problem) Suppose $a < b, c \notin [a, b]$ and $k \in$
lyche-numerical-linear-algebra- F03977	305	■ 1.000	$k \in$	$k \in$	Theorem 13.6 (A Minimal Norm Problem) Suppose $a < b, c \notin [a, b]$ and $k \in$
lyche-numerical-linear-algebra- F03978	305	■ 1.000	$\mathbb{N}$	$\mathbb{N}$	$\mathbb{N}$. If $Q \in \mathcal{S}_k$ and $Q \neq Q^*$ then $\ Q\ _\infty > \ Q^*\ _\infty$.
lyche-numerical-linear-algebra- F03979	305	■ 1.000	$Q \in \mathcal{S}_k$	$Q \in \mathcal{S}_k$	$\mathbb{N}$. If $Q \in \mathcal{S}_k$ and $Q \neq Q^*$ then $\ Q\ _\infty > \ Q^*\ _\infty$.
lyche-numerical-linear-algebra- F03980	305	■ 1.000	$Q \neq Q^*$	$Q \neq Q^*$	$\mathbb{N}$. If $Q \in \mathcal{S}_k$ and $Q \neq Q^*$ then $\ Q\ _\infty > \ Q^*\ _\infty$.
lyche-numerical-linear-algebra- F03981	305	■ 1.000	$\ Q\ _\infty > \ Q^*\ _\infty$	$\ Q\ _\infty > \ Q^*\ _\infty$	$\mathbb{N}$. If $Q \in \mathcal{S}_k$ and $Q \neq Q^*$ then $\ Q\ _\infty > \ Q^*\ _\infty$.
lyche-numerical-linear-algebra- F03982	305	■ 1.000	$p(z)=p'(z)=0$	$p(z) = p'(z) = 0$	$p(z) = p'(z) = 0$, we say that p has a double zero at z . Counting such a zero as
lyche-numerical-linear-algebra- F03983	305	■ 1.000	$ Q^* $	$ Q^* $	$ Q^* $ takes on its maximum $1/ T_k(u(c)) $ at the $k+1$ points $\mu_0, \dots, \mu_k$ in $[a, b]$
lyche-numerical-linear-algebra- F03984	305	■ 1.000	$1/ T_k(u(c)) $	$1/ T_k(u(c)) $	$ Q^* $ takes on its maximum $1/ T_k(u(c)) $ at the $k+1$ points $\mu_0, \dots, \mu_k$ in $[a, b]$
lyche-numerical-linear-algebra- F03985	305	■ 1.000	$\mu_0, \dots, \mu_k$	$\mu_0, \dots, \mu_k$	$ Q^* $ takes on its maximum $1/ T_k(u(c)) $ at the $k+1$ points $\mu_0, \dots, \mu_k$ in $[a, b]$
lyche-numerical-linear-algebra- F03986	305	■ 1.000	$u(\mu_i) = \cos(i\pi/k)$	$u(\mu_i) = \cos(i\pi/k)$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \dots, k$. Suppose $Q \in \mathcal{S}_k$ and that
lyche-numerical-linear-algebra- F03987	305	■ 1.000	$i=0, 1, \dots, k$	$i = 0, 1, \dots, k$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \dots, k$. Suppose $Q \in \mathcal{S}_k$ and that
lyche-numerical-linear-algebra- F03988	305	■ 1.000	$Q \in \mathcal{S}_k$	$Q \in \mathcal{S}_k$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \dots, k$. Suppose $Q \in \mathcal{S}_k$ and that
lyche-numerical-linear-algebra- F03989	305	■ 1.000	$\ Q\ _\infty \leq \ Q^*\ _\infty$	$\ Q\ _\infty \leq \ Q^*\ _\infty$	$\ Q\ _\infty \leq \ Q^*\ _\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show
lyche-numerical-linear-algebra- F03990	305	■ 1.000	$Q \equiv Q^*$	$Q \equiv Q^*$	$\ Q\ _\infty \leq \ Q^*\ _\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show
lyche-numerical-linear-algebra- F03991	305	■ 1.000	$f \equiv Q - Q^*$	$f \equiv Q - Q^*$	$\ Q\ _\infty \leq \ Q^*\ _\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show
lyche-numerical-linear-algebra- F03992	305	■ 1.000	$f(c)=0$	$f(c) = 0$	$f(c) = 0$, this means that $f \equiv 0$ or equivalently $Q \equiv Q^*$.
lyche-numerical-linear-algebra- F03993	305	■ 1.000	$f \equiv 0$	$f \equiv 0$	$f(c) = 0$, this means that $f \equiv 0$ or equivalently $Q \equiv Q^*$.
lyche-numerical-linear-algebra- F03994	305	■ 1.000	$I_j = [\mu_{j-1}, \mu_j]$	$I_j = [\mu_{j-1}, \mu_j]$	Consider $I_j = [\mu_{j-1}, \mu_j]$ for a fixed j . Let
lyche-numerical-linear-algebra- F03995	305	■ 1.000	$\sigma_j \leq 0$	$\sigma_j \leq 0$	We have $\sigma_j \leq 0$. For if say $Q^*(\mu_j) > 0$ then
lyche-numerical-linear-algebra- F03996	305	■ 1.000	$Q^*(\mu_j) > 0$	$Q^*(\mu_j) > 0$	We have $\sigma_j \leq 0$. For if say $Q^*(\mu_j) > 0$ then
lyche-numerical-linear-algebra- F03997	305	■ 1.000	$f(\mu_j) \leq 0$	$f(\mu_j) \leq 0$	so that $f(\mu_j) \leq 0$. Moreover,
lyche-numerical-linear-algebra- F03998	305	■ 1.000	$f(\mu_{j-1}) \geq 0$	$f(\mu_{j-1}) \geq 0$	Thus $f(\mu_{j-1}) \geq 0$ and it follows that $\sigma_j \leq 0$. Similarly, $\sigma_j \leq 0$ if $Q^*(\mu_j) < 0$.
lyche-numerical-linear-algebra- F03999	305	■ 1.000	$Q^*(\mu_j) < 0$	$Q^*(\mu_j) < 0$	Thus $f(\mu_{j-1}) \geq 0$ and it follows that $\sigma_j \leq 0$. Similarly, $\sigma_j \leq 0$ if $Q^*(\mu_j) < 0$.
lyche-numerical-linear-algebra- F04000	305	■ 1.000	$\sigma_j < 0, f$	$\sigma_j < 0, f$	If $\sigma_j < 0$, f must have a zero in I_j since it is continuous. Suppose $\sigma_j = 0$.
lyche-numerical-linear-algebra- F04001	305	■ 1.000	I_j	I_j	If $\sigma_j < 0$, f must have a zero in I_j since it is continuous. Suppose $\sigma_j = 0$.

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lyche-numerical-linear-algebra F04002	305	■ 1.000	$\sigma_j=0$	$\sigma_j = 0$	If $\sigma_j < 0$, f must have a zero in I_j since it is continuous. Suppose $\sigma_j = 0$.
lyche-numerical-linear-algebra F04003	305	■ 1.000	$f(\mu_{j-1})=0$	$f(\mu_{j-1}) = 0$	Then $f(\mu_{j-1}) = 0$ or $f(\mu_j) = 0$. If $f(\mu_j) = 0$ then $Q(\mu_j) = Q^*(\mu_j)$. But then
lyche-numerical-linear-algebra F04004	305	■ 1.000	$f(\mu_j)=0$	$f(\mu_j) = 0$	Then $f(\mu_{j-1}) = 0$ or $f(\mu_j) = 0$. If $f(\mu_j) = 0$ then $Q(\mu_j) = Q^*(\mu_j)$. But then
lyche-numerical-linear-algebra F04005	305	■ 1.000	$Q(\mu_j)=Q^*(\mu_j)$	$Q(\mu_j) = Q^*(\mu_j)$	Then $f(\mu_{j-1}) = 0$ or $f(\mu_j) = 0$. If $f(\mu_j) = 0$ then $Q(\mu_j) = Q^*(\mu_j)$. But then
lyche-numerical-linear-algebra F04006	305	■ 0.998	$\mu_j \in (a, b)$	$\mu_j \in (a, b)$	μ_j is a maximum or minimum both for Q and Q^* . If $\mu_j \in (a, b)$ then $Q'(\mu_j) =$
lyche-numerical-linear-algebra F04007	305	■ 0.998	$Q'(\mu_j)=$	$Q'(\mu_j) =$	μ_j is a maximum or minimum both for Q and Q^* . If $\mu_j \in (a, b)$ then $Q'(\mu_j) =$
lyche-numerical-linear-algebra F04008	306	■ 1.000	$Q^*(\mu_j)=0$	$Q^{*'}(\mu_j) = 0$	$Q^{*'}(\mu_j) = 0$. Thus $f(\mu_j) = f'(\mu_j) = 0$, and f has a double zero at μ_j . We can
lyche-numerical-linear-algebra F04009	306	■ 1.000	$f(\mu_j)=f'(\mu_j)=0$	$f(\mu_j) = f'(\mu_j) = 0$	$Q^{*'}(\mu_j) = 0$. Thus $f(\mu_j) = f'(\mu_j) = 0$, and f has a double zero at μ_j . We can
lyche-numerical-linear-algebra F04010	306	■ 1.000	I_{j+1}	I_{j+1}	count this as one zero for I_j and one for I_{j+1} . If $\mu_j = b$, we still have a zero in I_j .
lyche-numerical-linear-algebra F04011	306	■ 1.000	$\mu_j=b$	$\mu_j = b$	count this as one zero for I_j and one for I_{j+1} . If $\mu_j = b$, we still have a zero in I_j .
lyche-numerical-linear-algebra F04012	306	■ 1.000	μ_{j-1}	μ_{j-1}	Similarly, if $f(\mu_{j-1}) = 0$, a double zero of f at μ_{j-1} appears if $\mu_{j-1} \in (a, b)$. We
lyche-numerical-linear-algebra F04013	306	■ 1.000	$\mu_{j-1} \in (a, b)$	$\mu_{j-1} \in (a, b)$	Similarly, if $f(\mu_{j-1}) = 0$, a double zero of f at μ_{j-1} appears if $\mu_{j-1} \in (a, b)$. We
lyche-numerical-linear-algebra F04014	306	■ 1.000	I_{j-1}	I_{j-1}	count this as one zero for I_{j-1} and one for I_j .
lyche-numerical-linear-algebra F04015	306	■ 1.000	$I_j, j=$	$I_j, j =$	In this way we associate one zero of f for each of the k intervals $I_j, j =$
lyche-numerical-linear-algebra F04016	306	■ 0.999	$1, 2, \ldots, k$	$1, 2, \ldots, k$	$1, 2, \ldots, k$. We conclude that f has at least k zeros in $[a, b]$.
lyche-numerical-linear-algebra F04017	306	■ 1.000	$c=0$	$c = 0$	Theorem 13.6 with a , and b , the smallest and largest eigenvalue of A , and $c = 0$
lyche-numerical-linear-algebra F04018	306	■ 1.000	$t=(b+a)/(b-a)$	$t = (b + a)/(b - a)$	Moreover with $t = (b + a)/(b - a)$ we have
lyche-numerical-linear-algebra F04019	307	■ 1.000	$M:=\lambda_{\text{max}}$	$M := \lambda_{\max}$	where $M := \lambda_{\max}$ and $m := \lambda_{\min}$ are the largest and smallest eigenvalue of A ,
lyche-numerical-linear-algebra F04020	307	■ 1.000	$m:=\lambda_{\text{min}}$	$m := \lambda_{\min}$	where $M := \lambda_{\max}$ and $m := \lambda_{\min}$ are the largest and smallest eigenvalue of A ,
lyche-numerical-linear-algebra F04021	307	■ 0.997	$(\lambda_j^{-1}, \boldsymbol{u}_j)$	$(\lambda_j^{-1}, \boldsymbol{u}_j)$	Proof If $(\lambda_j, \boldsymbol{u}_j)$ are orthonormal eigenpairs of A then $(\lambda_j^{-1}, \boldsymbol{u}_j)$ are eigenpairs
lyche-numerical-linear-algebra F04022	307	■ 1.000	$\boldsymbol{A}^{-1}, j=1, \ldots, n$	$\boldsymbol{A}^{-1}, j = 1, \ldots, n$	for $\boldsymbol{A}^{-1}, j = 1, \ldots, n$. Let $\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{u}_j$ be the corresponding eigenvector
lyche-numerical-linear-algebra F04023	307	■ 1.000	$\boldsymbol{y}=\sum_{j=1}^n c_j \boldsymbol{u}_j$	$\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{u}_j$	for $\boldsymbol{A}^{-1}, j = 1, \ldots, n$. Let $\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{u}_j$ be the corresponding eigenvector
lyche-numerical-linear-algebra F04024	307	■ 1.000	$f:[m c, M c] \rightarrow \mathbb{R}$	$f : [mc, Mc] \rightarrow \mathbb{R}$	where $f : [mc, Mc] \rightarrow \mathbb{R}$ is given by $f(x) := x + 1/x$. By (13.38)
lyche-numerical-linear-algebra F04025	307	■ 1.000	$f(x):=x+1/x$	$f(x) := x + 1/x$	where $f : [mc, Mc] \rightarrow \mathbb{R}$ is given by $f(x) := x + 1/x$. By (13.38)
lyche-numerical-linear-algebra F04026	307	■ 1.000	$f \in C^2$	$f \in C^2$	Since $f \in C^2$ and f'' is positive it follows from Lemma 8.2 that f is a convex
lyche-numerical-linear-algebra F04027	307	■ 1.000	$f^{\prime \prime}$	f''	Since $f \in C^2$ and f'' is positive it follows from Lemma 8.2 that f is a convex

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lyche-numerical-linear-algebra- F04028	308	1.000	$c:=1/\sqrt{mM}$	$c := 1/\sqrt{mM}$	Choosing $c := 1/\sqrt{mM}$ we find $f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$.
lyche-numerical-linear-algebra- F04029	308	1.000	$f(m\ c)=f(M\ c)=\sqrt{\frac{M}{m}}+\sqrt{\frac{m}{M}}=\sqrt{\frac{M+m}{m\ M}}$	$f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$	Choosing $c := 1/\sqrt{mM}$ we find $f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$.
lyche-numerical-linear-algebra- F04030	308	1.000	$\boldsymbol{\epsilon}_j:=\boldsymbol{x}-\boldsymbol{x}_j, \ j=0,1,\ldots$	$\boldsymbol{\epsilon}_j := \boldsymbol{x} - \boldsymbol{x}_j, j = 0, 1, \dots$	<i>Proof of (13.19)</i> Let $\boldsymbol{\epsilon}_j := \boldsymbol{x} - \boldsymbol{x}_j, j = 0, 1, \dots$, where $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. It is enough to
lyche-numerical-linear-algebra- F04031	308	1.000	$\left \boldsymbol{\epsilon}_k\right _{\boldsymbol{A}}\leq\left(\frac{\kappa-1}{\kappa+1}\right)\left \boldsymbol{\epsilon}_{k-1}\right \leq\cdots\leq\left(\frac{\kappa-1}{\kappa+1}\right)^k\left \boldsymbol{\epsilon}_0\right $	$\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}} \leq \left(\frac{\kappa-1}{\kappa+1}\right) \ \boldsymbol{\epsilon}_{k-1}\ \leq \dots \leq \left(\frac{\kappa-1}{\kappa+1}\right)^k \ \boldsymbol{\epsilon}_0\ $	for then $\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}} \leq \left(\frac{\kappa-1}{\kappa+1}\right) \ \boldsymbol{\epsilon}_{k-1}\ \leq \dots \leq \left(\frac{\kappa-1}{\kappa+1}\right)^k \ \boldsymbol{\epsilon}_0\ $. It follows from (13.8) that
lyche-numerical-linear-algebra- F04032	309	1.000	$\boldsymbol{r}_{m+1}=\mathbf{0}$	$\boldsymbol{r}_{m+1} = \mathbf{0}$	gradient method m be the smallest integer such that $\boldsymbol{r}_{m+1} = \mathbf{0}$. For $k \leq m$ we
lyche-numerical-linear-algebra- F04033	309	1.000	$k \leq m$	$k \leq m$	gradient method m be the smallest integer such that $\boldsymbol{r}_{m+1} = \mathbf{0}$. For $k \leq m$ we
lyche-numerical-linear-algebra- F04034	309	1.000	$\left \boldsymbol{\epsilon}_{k+1}\right _2<\left \boldsymbol{\epsilon}_k\right _2$	$\ \boldsymbol{\epsilon}_{k+1}\ _2 < \ \boldsymbol{\epsilon}_k\ _2$	have $\ \boldsymbol{\epsilon}_{k+1}\ _2 < \ \boldsymbol{\epsilon}_k\ _2$. More precisely,
lyche-numerical-linear-algebra- F04035	309	0.983	$\boldsymbol{\epsilon}_j=\boldsymbol{x}-\boldsymbol{x}_j$	$\boldsymbol{\epsilon}_j = \boldsymbol{x} - \boldsymbol{x}_j$	where $\boldsymbol{\epsilon}_j = \boldsymbol{x} - \boldsymbol{x}_j$ and $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.
lyche-numerical-linear-algebra- F04036	309	1.000	$j \leq m$	$j \leq m$	Proof For $j \leq m$
lyche-numerical-linear-algebra- F04037	309	1.000	$\beta_{i-1} \cdots \beta_k=\left(\boldsymbol{r}_i^T \boldsymbol{r}_i\right) /\left(\boldsymbol{r}_k^T \boldsymbol{r}_k\right)$	$\beta_{i-1} \cdots \beta_k = \left(\boldsymbol{r}_i^T \boldsymbol{r}_i\right) / \left(\boldsymbol{r}_k^T \boldsymbol{r}_k\right)$	and since $\beta_{i-1} \cdots \beta_k = \left(\boldsymbol{r}_i^T \boldsymbol{r}_i\right) / \left(\boldsymbol{r}_k^T \boldsymbol{r}_k\right)$ we find
lyche-numerical-linear-algebra- F04038	310	1.000	$\operatorname{cond}_2(\boldsymbol{A})$	$\operatorname{cond}_2(\boldsymbol{A})$	For problems $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ of size n , where both n and $\operatorname{cond}_2(\boldsymbol{A})$ are large, it is
lyche-numerical-linear-algebra- F04039	310	1.000	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{B} \boldsymbol{b}$	$\boldsymbol{B}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{B}\boldsymbol{b}$	equivalent system $\boldsymbol{B}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{B}\boldsymbol{b}$, where $\boldsymbol{B}$ is nonsingular and $\operatorname{cond}_2(\boldsymbol{B}\boldsymbol{A})$ is smaller
lyche-numerical-linear-algebra- F04040	310	1.000	$\operatorname{cond}_2(\boldsymbol{B} \boldsymbol{A})$	$\operatorname{cond}_2(\boldsymbol{B}\boldsymbol{A})$	equivalent system $\boldsymbol{B}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{B}\boldsymbol{b}$, where $\boldsymbol{B}$ is nonsingular and $\operatorname{cond}_2(\boldsymbol{B}\boldsymbol{A})$ is smaller
lyche-numerical-linear-algebra- F04041	310	0.578	$\boldsymbol{B}=\boldsymbol{M}^{-1}$	$\boldsymbol{B} = \boldsymbol{M}^{-1}$	$\boldsymbol{B} = \boldsymbol{M}^{-1}$. We cannot use CG on $\boldsymbol{B}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{B}\boldsymbol{b}$ directly since $\boldsymbol{B}\boldsymbol{A}$ in general is not
lyche-numerical-linear-algebra- F04042	310	1.000	$\boldsymbol{B}=\boldsymbol{C}^T \boldsymbol{C}$	$\boldsymbol{B} = \boldsymbol{C}^T \boldsymbol{C}$	such that $\boldsymbol{B} = \boldsymbol{C}^T \boldsymbol{C}$. ($\boldsymbol{C}$ is only needed for the derivation and will not appear in the
lyche-numerical-linear-algebra- F04043	310	1.000	$\boldsymbol{y}:=\boldsymbol{C}^{-T} \boldsymbol{x}$	$\boldsymbol{y} := \boldsymbol{C}^{-T} \boldsymbol{x}$	where $\boldsymbol{y} := \boldsymbol{C}^{-T} \boldsymbol{x}$. We have 3 linear systems
lyche-numerical-linear-algebra- F04044	311	1.000	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T$	$\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T$	being positive definite the matrix $\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T$ is similar to $\boldsymbol{B}\boldsymbol{A}$. Indeed,
lyche-numerical-linear-algebra- F04045	311	1.000	$\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right) \boldsymbol{y}=\boldsymbol{C} \boldsymbol{b}$	$\left(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T\right) \boldsymbol{y} = \boldsymbol{C}\boldsymbol{b}$	We apply the conjugate gradient method to $\left(\boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T\right) \boldsymbol{y} = \boldsymbol{C}\boldsymbol{b}$. Denoting the
lyche-numerical-linear-algebra- F04046	311	0.997	$\boldsymbol{q}_k$	$\boldsymbol{q}_k$	search direction by $\boldsymbol{q}_k$ and the residual by $\boldsymbol{z}_k := \boldsymbol{C}\boldsymbol{b} - \boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T \boldsymbol{y}_k$ we obtain the
lyche-numerical-linear-algebra- F04047	311	0.997	$\boldsymbol{z}_k:=\boldsymbol{C} \boldsymbol{b}-\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \boldsymbol{y}_k$	$\boldsymbol{z}_k := \boldsymbol{C}\boldsymbol{b} - \boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T \boldsymbol{y}_k$	search direction by $\boldsymbol{q}_k$ and the residual by $\boldsymbol{z}_k := \boldsymbol{C}\boldsymbol{b} - \boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T \boldsymbol{y}_k$ we obtain the
lyche-numerical-linear-algebra- F04048	311	0.998	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}, \boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k$	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$	Here $\boldsymbol{x}_k$ will be an approximation to the solution $\boldsymbol{x}$ of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$ is
lyche-numerical-linear-algebra- F04049	311	0.573	$\boldsymbol{s}_k=\boldsymbol{C}^T \boldsymbol{z}_k=\boldsymbol{C}^T\left(\boldsymbol{C}-\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right) \boldsymbol{y}_k=\boldsymbol{B} \boldsymbol{b}-$	$\boldsymbol{s}_k = \boldsymbol{C}^T \boldsymbol{z}_k = \boldsymbol{C}^T \left(\boldsymbol{C} - \boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T\right) \boldsymbol{y}_k = \boldsymbol{B}\boldsymbol{b} -$	the residual in the original system, and $\boldsymbol{s}_k = \boldsymbol{C}^T \boldsymbol{z}_k = \boldsymbol{C}^T \left(\boldsymbol{C} - \boldsymbol{C}\boldsymbol{A}\boldsymbol{C}^T\right) \boldsymbol{y}_k = \boldsymbol{B}\boldsymbol{b} -$

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lyche-numerical-linear-algebra- F04050	311	■ 1.000	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}_k$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}_k$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}_k$ is the residual in the preconditioned system. If we set $\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$, $\boldsymbol{p}_0 =$
lyche-numerical-linear-algebra- F04051	311	■ 1.000	$\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$, $\boldsymbol{p}_0 =$	$\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$, $\boldsymbol{p}_0 =$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}_k$ is the residual in the preconditioned system. If we set $\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$, $\boldsymbol{p}_0 =$
lyche-numerical-linear-algebra- F04052	311	■ 0.835	$\boldsymbol{s}_0 = \boldsymbol{B} \boldsymbol{r}_0$	$\boldsymbol{s}_0 = \boldsymbol{B} \boldsymbol{r}_0$	$\boldsymbol{s}_0 = \boldsymbol{B} \boldsymbol{r}_0$, we obtain the following preconditioned conjugate gradient algorithm for
lyche-numerical-linear-algebra- F04053	311	■ 0.995	$\boldsymbol{x} (= \boldsymbol{x}_0)$	$\boldsymbol{x} (= \boldsymbol{x}_0)$	and $\boldsymbol{x} (= \boldsymbol{x}_0)$ is the starting iteration.
lyche-numerical-linear-algebra- F04054	312	■ 1.000	ρ	ρ	Apart from the calculation of ρ this algorithm is quite similar to Algorithm 13.1.
lyche-numerical-linear-algebra- F04055	312	■ 1.000	$\boldsymbol{w} = \boldsymbol{B} * \boldsymbol{t}$	$\boldsymbol{w} = \boldsymbol{B} * \boldsymbol{t}$	The main additional work is contained in $\boldsymbol{w} = \boldsymbol{B} * \boldsymbol{t}$. We'll discuss this further in
lyche-numerical-linear-algebra- F04056	312	■ 1.000	$\boldsymbol{y}_k$	$\boldsymbol{y}_k$	where $\boldsymbol{y}_k$ is the conjugate gradient approximation to the solution $\boldsymbol{y}$ of (13.46) and κ
lyche-numerical-linear-algebra- F04057	313	■ 0.998	$\boldsymbol{B} \boldsymbol{x}$	$\boldsymbol{B} \boldsymbol{x}$	2. The evaluation of $\boldsymbol{B} \boldsymbol{x}$ for a given vector $\boldsymbol{x}$ should not be expensive in storage and
lyche-numerical-linear-algebra- F04058	313	■ 1.000	$u = u(\boldsymbol{x}, \boldsymbol{y})$	$u = u(\boldsymbol{x}, \boldsymbol{y})$	and c are given and we seek a function $u = u(\boldsymbol{x}, \boldsymbol{y})$ such that (13.52) holds. We
lyche-numerical-linear-algebra- F04059	313	■ 1.000	$c(\boldsymbol{x}, \boldsymbol{y}) > 0$	$c(\boldsymbol{x}, \boldsymbol{y}) > 0$	assume that c and f are defined and continuous on Ω and that $c(\boldsymbol{x}, \boldsymbol{y}) > 0$ for
lyche-numerical-linear-algebra- F04060	313	■ 0.991	$(\boldsymbol{x}, \boldsymbol{y}) \in \Omega$	$(\boldsymbol{x}, \boldsymbol{y}) \in \Omega$	all $(\boldsymbol{x}, \boldsymbol{y}) \in \Omega$. The problem (13.52) reduces to the Poisson problem (10.1) in the
lyche-numerical-linear-algebra- F04061	313	■ 1.000	$c(\boldsymbol{x}, \boldsymbol{y}) = 1$	$c(\boldsymbol{x}, \boldsymbol{y}) = 1$	special case where $c(\boldsymbol{x}, \boldsymbol{y}) = 1$ for $(\boldsymbol{x}, \boldsymbol{y}) \in \Omega$.
lyche-numerical-linear-algebra- F04062	313	■ 1.000	$m \in \mathbb{N}$	$m \in \mathbb{N}$	To solve (13.52) numerically, we choose $m \in \mathbb{N}$, set $h := 1/(m + 1)$, and define
lyche-numerical-linear-algebra- F04063	313	■ 1.000	$v_{j,k} \approx u(\boldsymbol{x}_j, \boldsymbol{y}_k)$	$v_{j,k} \approx u(\boldsymbol{x}_j, \boldsymbol{y}_k)$	We compute approximations $v_{j,k} \approx u(\boldsymbol{x}_j, \boldsymbol{y}_k)$ on a grid of points
lyche-numerical-linear-algebra- F04064	314	■ 1.000	f, g	f, g	using a finite difference method. For univariate functions f, g we approximate
lyche-numerical-linear-algebra- F04065	314	■ 1.000	$f_{j,k} := f(\boldsymbol{x}_j, \boldsymbol{y}_k)$, $(d\boldsymbol{v})_{j,k} := (d_1 \boldsymbol{v})_{j,k} + (d_2 \boldsymbol{v})_{j,k}$	$f_{j,k} := f(\boldsymbol{x}_j, \boldsymbol{y}_k)$, $(d\boldsymbol{v})_{j,k} := (d_1 \boldsymbol{v})_{j,k} + (d_2 \boldsymbol{v})_{j,k}$	where $f_{j,k} := f(\boldsymbol{x}_j, \boldsymbol{y}_k)$, $(d\boldsymbol{v})_{j,k} := (d_1 \boldsymbol{v})_{j,k} + (d_2 \boldsymbol{v})_{j,k}$,
lyche-numerical-linear-algebra- F04066	314	■ 1.000	$c_{p,q} = c(\boldsymbol{p} h, \boldsymbol{q} h)$	$c_{p,q} = c(\boldsymbol{p} h, \boldsymbol{q} h)$	and where $c_{p,q} = c(\boldsymbol{p} h, \boldsymbol{q} h)$ for $\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}$. The equation (13.53) can be written in
lyche-numerical-linear-algebra- F04067	314	■ 1.000	$\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}$	$\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}$	and where $c_{p,q} = c(\boldsymbol{p} h, \boldsymbol{q} h)$ for $\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}$. The equation (13.53) can be written in
lyche-numerical-linear-algebra- F04068	314	■ 0.997	$h^2 L_h \boldsymbol{v} = h^2 \boldsymbol{F}$	$h^2 L_h \boldsymbol{v} = h^2 \boldsymbol{F}$	as unknowns. The system $h^2 L_h \boldsymbol{v} = h^2 \boldsymbol{F}$ can be written in standard form $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$
lyche-numerical-linear-algebra- F04069	315	■ 1.000	$\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$	$\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$	Lemma 13.6 (Discrete Inner Product) If $\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$ and $v_{j,k} = w_{j,k} = 0$ for
lyche-numerical-linear-algebra- F04070	315	■ 1.000	$v_{j,k} = w_{j,k} = 0$	$v_{j,k} = w_{j,k} = 0$	Lemma 13.6 (Discrete Inner Product) If $\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$ and $v_{j,k} = w_{j,k} = 0$ for
lyche-numerical-linear-algebra- F04071	315	■ 1.000	$(j, k) \in \partial I_m$	$(j, k) \in \partial I_m$	$(j, k) \in \partial I_m$, then
lyche-numerical-linear-algebra- F04072	315	■ 0.994	$m \in \mathbb{N}$, $a_i, b_i, c_i \in \mathbb{R}$	$m \in \mathbb{N}$, $a_i, b_i, c_i \in \mathbb{R}$	Proof If $m \in \mathbb{N}$, $a_i, b_i, c_i \in \mathbb{R}$ for $i = 0, \dots, m$ and $b_0 = c_0 = b_{m+1} = c_{m+1} = 0$
lyche-numerical-linear-algebra- F04073	315	■ 0.994	$i = 0, \dots, m$	$i = 0, \dots, m$	Proof If $m \in \mathbb{N}$, $a_i, b_i, c_i \in \mathbb{R}$ for $i = 0, \dots, m$ and $b_0 = c_0 = b_{m+1} = c_{m+1} = 0$
lyche-numerical-linear-algebra- F04074	315	■ 0.994	$b_0 = c_0 = b_{m+1} = c_{m+1} = 0$	$b_0 = c_0 = b_{m+1} = c_{m+1} = 0$	Proof If $m \in \mathbb{N}$, $a_i, b_i, c_i \in \mathbb{R}$ for $i = 0, \dots, m$ and $b_0 = c_0 = b_{m+1} = c_{m+1} = 0$
lyche-numerical-linear-algebra- F04075	315	■ 1.000	$(d_1 \boldsymbol{v})_{j,k} w_{j,k}$	$(d_1 \boldsymbol{v})_{j,k} w_{j,k}$	and the right hand side of (13.60) follows. We apply (13.60) to $(d_1 \boldsymbol{v})_{j,k} w_{j,k}$ and
lyche-numerical-linear-algebra- F04076	315	■ 1.000	$(d_2 \boldsymbol{v})_{j,k} w_{j,k}$	$(d_2 \boldsymbol{v})_{j,k} w_{j,k}$	$(d_2 \boldsymbol{v})_{j,k} w_{j,k}$ given by (13.55) and (13.59) follows.

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lyche-numerical-linear-algebra- F04077	315	1.000	$\left\langle L_h v, \boldsymbol{W} \right\rangle = \left\langle \boldsymbol{W}, L_h v \right\rangle$	$\langle L_h v, \boldsymbol{W} \rangle = \langle \boldsymbol{W}, L_h v \rangle$	Proof By (13.59) $\langle L_h v, \boldsymbol{W} \rangle = \langle \boldsymbol{W}, L_h v \rangle$ and symmetry follows. We take $\boldsymbol{W} =$
lyche-numerical-linear-algebra- F04078	315	1.000	$\boldsymbol{W} =$	$\boldsymbol{W} =$	Proof By (13.59) $\langle L_h v, \boldsymbol{W} \rangle = \langle \boldsymbol{W}, L_h v \rangle$ and symmetry follows. We take $\boldsymbol{W} =$
lyche-numerical-linear-algebra- F04079	315	1.000	$c_{j+\frac{1}{2},k}$	$c_{j+\frac{1}{2},k}$	$\boldsymbol{V}$ and obtain quadratic factors in (13.59). Since $c_{j+\frac{1}{2},k}$ and $c_{j,k+\frac{1}{2}}$ correspond to
lyche-numerical-linear-algebra- F04080	315	1.000	$c_{j,k+\frac{1}{2}}$	$c_{j,k+\frac{1}{2}}$	$\boldsymbol{V}$ and obtain quadratic factors in (13.59). Since $c_{j+\frac{1}{2},k}$ and $c_{j,k+\frac{1}{2}}$ correspond to
lyche-numerical-linear-algebra- F04081	315	1.000	$\left\langle L_h v, \boldsymbol{V} \right\rangle \geq 0$	$\langle L_h v, \boldsymbol{V} \rangle \geq 0$	and $\langle L_h v, \boldsymbol{V} \rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m \times m}$. If $\langle L_h v, \boldsymbol{V} \rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra- F04082	315	1.000	$\boldsymbol{V} \in \mathbb{R}^{m \times m}$	$\boldsymbol{V} \in \mathbb{R}^{m \times m}$	and $\langle L_h v, \boldsymbol{V} \rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m \times m}$. If $\langle L_h v, \boldsymbol{V} \rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra- F04083	315	1.000	$\left\langle L_h v, \boldsymbol{V} \right\rangle = 0$	$\langle L_h v, \boldsymbol{V} \rangle = 0$	and $\langle L_h v, \boldsymbol{V} \rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m \times m}$. If $\langle L_h v, \boldsymbol{V} \rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra- F04084	315	1.000	$v_{j,k+1} = v_{j,k}$	$v_{j,k+1} = v_{j,k}$	must be zero, and $v_{j,k+1} = v_{j,k}$ for $k = 0, 1, \dots, m$ and $j = 1, \dots, m$. Now
lyche-numerical-linear-algebra- F04085	315	1.000	$k=0,1,\ldots,m$	$k = 0, 1, \dots, m$	must be zero, and $v_{j,k+1} = v_{j,k}$ for $k = 0, 1, \dots, m$ and $j = 1, \dots, m$. Now
lyche-numerical-linear-algebra- F04086	315	1.000	$v_{j,0} = v_{j,m+1} = 0$	$v_{j,0} = v_{j,m+1} = 0$	$v_{j,0} = v_{j,m+1} = 0$ implies that $\boldsymbol{V} = \mathbf{0}$. It follows that the linear system (13.56) is
lyche-numerical-linear-algebra- F04087	315	1.000	$\boldsymbol{V} = \mathbf{0}$	$\boldsymbol{V} = \mathbf{0}$	$v_{j,0} = v_{j,m+1} = 0$ implies that $\boldsymbol{V} = \mathbf{0}$. It follows that the linear system (13.56) is
lyche-numerical-linear-algebra- F04088	316	1.000	$\boldsymbol{b} \in \mathbb{R}^n$	$\boldsymbol{b} \in \mathbb{R}^n$	Consider solving $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, where $\boldsymbol{A}$ is given by (13.57) and $\boldsymbol{b} \in \mathbb{R}^n$. Since $\boldsymbol{A}$ is
lyche-numerical-linear-algebra- F04089	316	1.000	$\boldsymbol{x} \in \mathbb{R}^n$	$\boldsymbol{x} \in \mathbb{R}^n$	positive definite it is nonsingular and the system has a unique solution $\boldsymbol{x} \in \mathbb{R}^n$.
lyche-numerical-linear-algebra- F04090	316	1.000	$m = \sqrt{n}$	$m = \sqrt{n}$	to find $\boldsymbol{x}$. Since the bandwidth of $\boldsymbol{A}$ is $m = \sqrt{n}$ both of these methods require $O(n^2)$
lyche-numerical-linear-algebra- F04091	316	1.000	$c(x, y) \equiv 1$	$c(x, y) \equiv 1$	If we choose $c(x, y) \equiv 1$ in (13.52), we get the Poisson problem. With this in
lyche-numerical-linear-algebra- F04092	316	1.000	$\boldsymbol{A}_p$	$\boldsymbol{A}_p$	mind, we may think of the coefficient matrix $\boldsymbol{A}_p$ arising from the discretization of
lyche-numerical-linear-algebra- F04093	316	1.000	$\boldsymbol{B} = \boldsymbol{A}_p^{-1}$	$\boldsymbol{B} = \boldsymbol{A}_p^{-1}$	$\boldsymbol{B} = \boldsymbol{A}_p^{-1}$, the inverse of the discrete Poisson matrix as a preconditioner for the
lyche-numerical-linear-algebra- F04094	316	0.928	$\boldsymbol{w} = \boldsymbol{B}\boldsymbol{t}$	$\boldsymbol{w} = \boldsymbol{B}\boldsymbol{t}$	Consider Algorithm 13.3. With this preconditioner the calculation $\boldsymbol{w} = \boldsymbol{B}\boldsymbol{t}$ takes
lyche-numerical-linear-algebra- F04095	316	1.000	$\boldsymbol{A}_p \boldsymbol{w}_k = \boldsymbol{t}_k$	$\boldsymbol{A}_p \boldsymbol{w}_k = \boldsymbol{t}_k$	the form $\boldsymbol{A}_p \boldsymbol{w}_k = \boldsymbol{t}_k$.
lyche-numerical-linear-algebra- F04096	316	1.000	$\boldsymbol{A}_p \boldsymbol{w} = \boldsymbol{t}$	$\boldsymbol{A}_p \boldsymbol{w} = \boldsymbol{t}$	This method can be utilized to solve $\boldsymbol{A}_p \boldsymbol{w} = \boldsymbol{t}$.
lyche-numerical-linear-algebra- F04097	316	1.000	$\boldsymbol{x}_0 = 0$	$\boldsymbol{x}_0 = 0$	lem (13.52). We used $\boldsymbol{x}_0 = 0$ and $\epsilon = 10^{-8}$. The results are shown in Table 13.3.
lyche-numerical-linear-algebra- F04098	316	1.000	$\epsilon = 10^{-8}$	$\epsilon = 10^{-8}$	lem (13.52). We used $\boldsymbol{x}_0 = 0$ and $\epsilon = 10^{-8}$. The results are shown in Table 13.3.
lyche-numerical-linear-algebra- F04099	316	1.000	$O(n \log_2 n)$	$O(n \log_2 n)$	without preconditioning to $O(n^{3/2})$ or $O(n \log_2 n)$ with preconditioning. This is not
lyche-numerical-linear-algebra- F04101	317	1.000	$0 < c_0 \leq$	$0 < c_0 \leq$	Theorem 13.12 (Eigenvalues of Preconditioned Matrix) Suppose $0 < c_0 \leq$
lyche-numerical-linear-algebra- F04102	317	0.973	$c(x, y) \leq c_1$	$c(x, y) \leq c_1$	$c(x, y) \leq c_1$ for all $(x, y) \in [0, 1]^2$. For the eigenvalues of the matrix $\boldsymbol{BA} = \boldsymbol{A}_p^{-1} \boldsymbol{A}$
lyche-numerical-linear-algebra- F04103	317	0.973	$(x, y) \in [0, 1]^2$	$(x, y) \in [0, 1]^2$	$c(x, y) \leq c_1$ for all $(x, y) \in [0, 1]^2$. For the eigenvalues of the matrix $\boldsymbol{BA} = \boldsymbol{A}_p^{-1} \boldsymbol{A}$
lyche-numerical-linear-algebra- F04104	317	0.973	$\boldsymbol{BA} = \boldsymbol{A}_p^{-1} \boldsymbol{A}$	$\boldsymbol{BA} = \boldsymbol{A}_p^{-1} \boldsymbol{A}$	$c(x, y) \leq c_1$ for all $(x, y) \in [0, 1]^2$. For the eigenvalues of the matrix $\boldsymbol{BA} = \boldsymbol{A}_p^{-1} \boldsymbol{A}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04105	317	1.000	$\boldsymbol{A}_{-p}^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$	$\boldsymbol{A}_p^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$	Proof Suppose $\boldsymbol{A}_p^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ for some $\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{A}_p \boldsymbol{x}$.
lyche-numerical-linear-algebra F04106	317	1.000	$\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$	$\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$	Proof Suppose $\boldsymbol{A}_p^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ for some $\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{A}_p \boldsymbol{x}$.
lyche-numerical-linear-algebra F04107	317	1.000	$\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{A}_p \boldsymbol{x}$	$\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{A}_p \boldsymbol{x}$	Proof Suppose $\boldsymbol{A}_p^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ for some $\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{A}_p \boldsymbol{x}$.
lyche-numerical-linear-algebra F04108	317	1.000	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}$	We computed $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}$ in (13.59) and we obtain $\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x}$ by setting all the c 's there
lyche-numerical-linear-algebra F04109	317	1.000	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x}$	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x}$	We computed $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}$ in (13.59) and we obtain $\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x}$ by setting all the c 's there
lyche-numerical-linear-algebra F04110	317	0.997	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x} > 0$	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x} > 0$	Thus $\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x} > 0$ and bounding all the c 's in (13.59) from below by c_0 and above
lyche-numerical-linear-algebra F04111	317	0.997	$c_{\{0\}}$	c_0	Thus $\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x} > 0$ and bounding all the c 's in (13.59) from below by c_0 and above
lyche-numerical-linear-algebra F04112	317	1.000	$c_{\{0\}} \leq \lambda \leq c_{\{1\}}$	$c_0 \leq \lambda \leq c_1$	which implies that $c_0 < \lambda < c_1$ for all eigenvalues λ of $\boldsymbol{B} \boldsymbol{A} = \boldsymbol{A}_p^{-1} \boldsymbol{A}$.
lyche-numerical-linear-algebra F04113	317	0.998	$c(x, y) = e^{-x+y}$	$c(x, y) = e^{-x+y}$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$
lyche-numerical-linear-algebra F04114	317	0.998	$c_{\{0\}} = e^{-2}$	$c_0 = e^{-2}$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$
lyche-numerical-linear-algebra F04115	317	0.998	$c_{\{1\}} = 1$	$c_1 = 1$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$
lyche-numerical-linear-algebra F04116	317	0.998	$\kappa \leq e^2 \approx$	$\kappa \leq e^2 \approx$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa < e^2 \approx$
lyche-numerical-linear-algebra F04117	317	1.000	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^T$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^T$	Exercise 13.2 (Paraboloid) Let $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^T$ be the spectral decomposition of $\boldsymbol{A}$,
lyche-numerical-linear-algebra F04118	318	0.966	$\boldsymbol{v} = [v_1, \dots, v_n]^T := \boldsymbol{U}^T \boldsymbol{y}$	$\boldsymbol{v} = [v_1, \dots, v_n]^T := \boldsymbol{U}^T \boldsymbol{y}$	$\boldsymbol{v} = [v_1, \dots, v_n]^T := \boldsymbol{U}^T \boldsymbol{y}$, and set $\boldsymbol{c} := \boldsymbol{U}^T \boldsymbol{b} = [c_1, \dots, c_n]^T$. Show that
lyche-numerical-linear-algebra F04119	318	0.966	$\boldsymbol{c} := \boldsymbol{U}^T \boldsymbol{b} = [c_1, \dots, c_n]^T$	$\boldsymbol{c} := \boldsymbol{U}^T \boldsymbol{b} = [c_1, \dots, c_n]^T$	$\boldsymbol{v} = [v_1, \dots, v_n]^T := \boldsymbol{U}^T \boldsymbol{y}$, and set $\boldsymbol{c} := \boldsymbol{U}^T \boldsymbol{b} = [c_1, \dots, c_n]^T$. Show that
lyche-numerical-linear-algebra F04120	318	0.999	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{r}_k$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{r}_k$	the iteration is $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{r}_k$, where $\boldsymbol{r}_k$ is the residual, $\boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$, and
lyche-numerical-linear-algebra F04121	318	0.999	$\boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$	$\boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$	the iteration is $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{r}_k$, where $\boldsymbol{r}_k$ is the residual, $\boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$, and
lyche-numerical-linear-algebra F04122	318	0.973	$\alpha_k = \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$	$\alpha_k = \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$	$\alpha_k = \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$.
lyche-numerical-linear-algebra F04123	318	0.927	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \boldsymbol{b} = [1 \ 1]^T$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \boldsymbol{b} = [1 \ 1]^T$	a) Compute $\boldsymbol{x}_1$ if $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \boldsymbol{b} = [1 \ 1]^T$ and $\boldsymbol{x}_0 = \mathbf{0}$.
lyche-numerical-linear-algebra F04124	318	1.000	$\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}$	$\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}$	b) If the k -th error, $\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}$, is an eigenvector of $\boldsymbol{A}$, what can you say about
lyche-numerical-linear-algebra F04125	318	0.907	$\boldsymbol{x}_1 = \left(\frac{\boldsymbol{b}^T \boldsymbol{b}}{\boldsymbol{b}^T \boldsymbol{A} \boldsymbol{b}} \right) \boldsymbol{b}$	$\boldsymbol{x}_1 = \left(\frac{\boldsymbol{b}^T \boldsymbol{b}}{\boldsymbol{b}^T \boldsymbol{A} \boldsymbol{b}} \right) \boldsymbol{b}$	conjugate gradient method when $\boldsymbol{x}_0 = \mathbf{0}$. (Answer: $\boldsymbol{x}_1 = \left(\frac{\boldsymbol{b}^T \boldsymbol{b}}{\boldsymbol{b}^T \boldsymbol{A} \boldsymbol{b}} \right) \boldsymbol{b}$.)
lyche-numerical-linear-algebra F04126	318	0.990	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	system $\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$ starting with $\boldsymbol{y}_0 = \mathbf{0}$. ³
lyche-numerical-linear-algebra F04127	318	0.990	$\boldsymbol{y}_0 = \mathbf{0}$. ³	$\boldsymbol{y}_0 = \mathbf{0}$. ³	system $\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$ starting with $\boldsymbol{y}_0 = \mathbf{0}$. ³

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04128	318	1.000	$\boldsymbol{p}_k = \boldsymbol{r}_k + \sum_{j=0}^{k-1} a_{k,j} \boldsymbol{r}_j$	$\boldsymbol{p}_k = \boldsymbol{r}_k + \sum_{j=0}^{k-1} a_{k,j} \boldsymbol{r}_j$	² Hint: use induction on k to show that $\boldsymbol{p}_k = \boldsymbol{r}_k + \sum_{j=0}^{k-1} a_{k,j} \boldsymbol{r}_j$ for some constants $a_{k,j}$.
lyche-numerical-linear-algebra F04129	318	0.881	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0$	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0$	³ Hint: The conjugate gradient method for $\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0$ can be written $\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k \boldsymbol{q}_k$, $\gamma_k :=$
lyche-numerical-linear-algebra F04130	318	0.881	$\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k \boldsymbol{q}_k$, $\gamma_k :=$	$\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k \boldsymbol{q}_k$, $\gamma_k :=$	³ Hint: The conjugate gradient method for $\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0$ can be written $\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k \boldsymbol{q}_k$, $\gamma_k :=$
lyche-numerical-linear-algebra F04131	318	0.888	$\frac{\boldsymbol{s}_k^T \boldsymbol{s}_k}{\boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k}$, $\boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k \boldsymbol{A} \boldsymbol{q}_k$, $\boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k \boldsymbol{q}_k$, $\delta_k := \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{s}_k}$	$\frac{\boldsymbol{s}_k^T \boldsymbol{s}_k}{\boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k}$, $\boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k \boldsymbol{A} \boldsymbol{q}_k$, $\boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k \boldsymbol{q}_k$, $\delta_k := \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{s}_k}$	$\frac{\boldsymbol{s}_k^T \boldsymbol{s}_k}{\boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k}$, $\boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k \boldsymbol{A} \boldsymbol{q}_k$, $\boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k \boldsymbol{q}_k$, $\delta_k := \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{s}_k}$. Show that $\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$,
lyche-numerical-linear-algebra F04132	318	0.888	$\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$	$\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$	$\frac{\boldsymbol{s}_k^T \boldsymbol{s}_k}{\boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k}$, $\boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k \boldsymbol{A} \boldsymbol{q}_k$, $\boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k \boldsymbol{q}_k$, $\delta_k := \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{s}_k}$. Show that $\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$,
lyche-numerical-linear-algebra F04133	318	0.977	$\boldsymbol{s}_k = \boldsymbol{r}_k$	$\boldsymbol{s}_k = \boldsymbol{r}_k$	$\boldsymbol{s}_k = \boldsymbol{r}_k$, and $\boldsymbol{q}_k = \boldsymbol{p}_k$, for $k = 0, 1, 2, \dots$
lyche-numerical-linear-algebra F04134	318	0.977	$\boldsymbol{q}_k = \boldsymbol{p}_k$	$\boldsymbol{q}_k = \boldsymbol{p}_k$	$\boldsymbol{s}_k = \boldsymbol{r}_k$, and $\boldsymbol{q}_k = \boldsymbol{p}_k$, for $k = 0, 1, 2, \dots$
lyche-numerical-linear-algebra F04135	319	1.000	$n = 100, 400, 1600, 2500$	$n = 100, 400, 1600, 2500$	to test for say $n = 100, 400, 1600, 2500$, and tabulate K/n instead of $K/\sqrt{n}$ in the
lyche-numerical-linear-algebra F04136	319	1.000	K / n	K/n	to test for say $n = 100, 400, 1600, 2500$, and tabulate K/n instead of $K/\sqrt{n}$ in the
lyche-numerical-linear-algebra F04137	319	1.000	$\boldsymbol{A} \in \mathbb{R}^{m,n}$, $\boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m,n}$, $\boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m,n}$, $\boldsymbol{b} \in \mathbb{R}^m$ and $\boldsymbol{A}^T \boldsymbol{A}$ is positive definite. ⁴ Explain why only the following
lyche-numerical-linear-algebra F04138	319	0.996	$\boldsymbol{r} = \boldsymbol{A}'(\boldsymbol{b} - \boldsymbol{A}^* \boldsymbol{x})$; $\boldsymbol{p} = \boldsymbol{r}$	$\boldsymbol{r} = \boldsymbol{A}'(\boldsymbol{b} - \boldsymbol{A}^* \boldsymbol{x})$; $\boldsymbol{p} = \boldsymbol{r}$	1. $\boldsymbol{r} = \boldsymbol{A}'(\boldsymbol{b} - \boldsymbol{A}^* \boldsymbol{x})$; $\boldsymbol{p} = \boldsymbol{r}$;
lyche-numerical-linear-algebra F04139	319	0.926	$\boldsymbol{r} = \boldsymbol{r} - \boldsymbol{a}^* \boldsymbol{A}^* \boldsymbol{t}$	$\boldsymbol{r} = \boldsymbol{r} - \boldsymbol{a}^* \boldsymbol{A}^* \boldsymbol{t}$	3. $\boldsymbol{r} = \boldsymbol{r} - \boldsymbol{a}^* \boldsymbol{A}^* \boldsymbol{t}$
lyche-numerical-linear-algebra F04140	319	1.000	$\text{cond}_2(\boldsymbol{A})^2$	$\text{cond}_2(\boldsymbol{A})^2$	Note that the condition number of the normal equations is $\text{cond}_2(\boldsymbol{A})^2$, the square of
lyche-numerical-linear-algebra F04141	319	1.000	$\{\boldsymbol{v}_i\}_{i=1}^k$	$\{\boldsymbol{v}_i\}_{i=1}^k$	a) Let $\{\boldsymbol{v}_i\}_{i=1}^k$ be a set of linearly independent vectors in $\mathbb{R}^n$, and let $\langle \cdot, \cdot \rangle$ be an
lyche-numerical-linear-algebra F04142	319	1.000	$n_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$	$n_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$	inner product in $\mathbb{R}^n$. Explain that the $k \times k$ -matrix $\boldsymbol{N}$ with entries $n_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$
lyche-numerical-linear-algebra F04143	319	0.997	$\mathbb{W} \subset \mathbb{R}^n$	$\mathbb{W} \subset \mathbb{R}^n$	b) Let $\mathbb{W} \subset \mathbb{R}^n$ be any linear subspace. Show that there is one and only one vector
lyche-numerical-linear-algebra F04144	319	1.000	$\hat{\boldsymbol{x}} \in \mathbb{W}$	$\hat{\boldsymbol{x}} \in \mathbb{W}$	$\hat{\boldsymbol{x}} \in \mathbb{W}$ so that
lyche-numerical-linear-algebra F04145	319	1.000	$\hat{\boldsymbol{x}}$	$\hat{\boldsymbol{x}}$	and that $\hat{\boldsymbol{x}}$ satisfies
lyche-numerical-linear-algebra F04146	319	1.000	$\mathbb{W}$	$\mathbb{W}$	space $\mathbb{W}$ is taken to be the Krylov space
lyche-numerical-linear-algebra F04147	320	1.000	$\boldsymbol{x}_k \in \mathbb{W}_k$	$\boldsymbol{x}_k \in \mathbb{W}_k$	$\boldsymbol{x}_k$. Assume that $\boldsymbol{x}_k \in \mathbb{W}_k$ is already determined. In addition, assume that we
lyche-numerical-linear-algebra F04148	320	1.000	$\boldsymbol{p}_k \in \mathbb{W}_{k+1}$	$\boldsymbol{p}_k \in \mathbb{W}_{k+1}$	already have computed a “search direction” $\boldsymbol{p}_k \in \mathbb{W}_{k+1}$ such that $\ \boldsymbol{A} \boldsymbol{p}_k\ _2 =$
lyche-numerical-linear-algebra F04149	320	1.000	$\ \boldsymbol{A} \boldsymbol{p}_k\ _2 =$	$\ \boldsymbol{A} \boldsymbol{p}_k\ _2 =$	already have computed a “search direction” $\boldsymbol{p}_k \in \mathbb{W}_{k+1}$ such that $\ \boldsymbol{A} \boldsymbol{p}_k\ _2 =$
lyche-numerical-linear-algebra F04150	320	1.000	$\ \boldsymbol{p}_k\ _{\boldsymbol{A}} = 1$	$\ \boldsymbol{p}_k\ _{\boldsymbol{A}} = 1$	$\ \boldsymbol{p}_k\ _{\boldsymbol{A}} = 1$, and such that

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lyche-numerical-linear-algebra-F04151	320	1.000	$\alpha_k \in \mathbb{R}$	$\alpha_k \in \mathbb{R}$	Show that $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$ for a suitable $\alpha_k \in \mathbb{R}$, and express α_k in terms of
lyche-numerical-linear-algebra-F04152	320	1.000	$\mathbf{p}_{k-2}, \mathbf{p}_{k-1}$	$\mathbf{p}_{k-2}, \mathbf{p}_{k-1}$	further that the vectors $\mathbf{p}_{k-2}, \mathbf{p}_{k-1}$, and $\mathbf{p}_k$ are already known with properties as
lyche-numerical-linear-algebra-F04153	320	1.000	$k \leq 3$	$k \leq 3$	a) Determine the vectors defining the Krylov spaces for $k \leq 3$ taking as initial
lyche-numerical-linear-algebra-F04154	320	1.000	$\left[\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b} \right] = \left[\begin{array}{rrr} 4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4 \end{array} \right]$	$[\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}] = \begin{bmatrix} 4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4 \end{bmatrix}$	approximation $\mathbf{x} = \mathbf{0}$. Answer: $[\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}] = \begin{bmatrix} 4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4 \end{bmatrix}$.
lyche-numerical-linear-algebra-F04155	321	0.983	$\operatorname{dim}(\mathbb{W}_k) = k$	$\dim(\mathbb{W}_k) = k$	• $\dim(\mathbb{W}_k) = k$ for $k = 0, 1, 2, 3$.
lyche-numerical-linear-algebra-F04156	321	0.983	$k=0,1,2,3$	$k = 0, 1, 2, 3$	• $\dim(\mathbb{W}_k) = k$ for $k = 0, 1, 2, 3$.
lyche-numerical-linear-algebra-F04157	321	0.987	$\mathbf{x}_3$	$\mathbf{x}_3$	• $\mathbf{x}_3$ is the exact solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.
lyche-numerical-linear-algebra-F04158	321	1.000	$\mathbf{r}_0, \dots, \mathbf{r}_{k-1}$	$\mathbf{r}_0, \dots, \mathbf{r}_{k-1}$	• $\mathbf{r}_0, \dots, \mathbf{r}_{k-1}$ is an orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra-F04159	321	1.000	$k=1,2,3$	$k = 1, 2, 3$	• $\mathbf{r}_0, \dots, \mathbf{r}_{k-1}$ is an orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra-F04160	321	1.000	$\mathbf{p}_0, \dots, \mathbf{p}_{k-1}$	$\mathbf{p}_0, \dots, \mathbf{p}_{k-1}$	• $\mathbf{p}_0, \dots, \mathbf{p}_{k-1}$ is an $\mathbf{A}$ -orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra-F04161	321	1.000	$\ \mathbf{r}_k\ $	$\{\ \mathbf{r}_k\ \}$	• $\{\ \mathbf{r}_k\ \}$ is monotonically decreasing.
lyche-numerical-linear-algebra-F04162	321	0.991	$\ \mathbf{x}_k - \mathbf{x}\ $	$\{\ \mathbf{x}_k - \mathbf{x}\ \}$	• $\{\ \mathbf{x}_k - \mathbf{x}\ \}$ is monotonically decreasing.
lyche-numerical-linear-algebra-F04163	321	1.000	$\mathbf{B}^T = -\mathbf{B}$	$\mathbf{B}^T = -\mathbf{B}$	Let $\mathbf{B} \in \mathbb{R}^{n \times n}$ be an antisymmetric matrix, i.e., $\mathbf{B}^T = -\mathbf{B}$, and let $\mathbf{A} := \mathbf{I} - \mathbf{B}$,
lyche-numerical-linear-algebra-F04164	321	1.000	$\mathbf{A} := \mathbf{I} - \mathbf{B}$	$\mathbf{A} := \mathbf{I} - \mathbf{B}$	Let $\mathbf{B} \in \mathbb{R}^{n \times n}$ be an antisymmetric matrix, i.e., $\mathbf{B}^T = -\mathbf{B}$, and let $\mathbf{A} := \mathbf{I} - \mathbf{B}$.
lyche-numerical-linear-algebra-F04165	321	0.998	$\ \mathbf{A}\mathbf{x}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{B}\mathbf{x}\ _2^2$	$\ \mathbf{A}\mathbf{x}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{B}\mathbf{x}\ _2^2$	b) Show that $\ \mathbf{A}\mathbf{x}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{B}\mathbf{x}\ _2^2$ and that $\ \mathbf{A}\ _2 = \sqrt{1 + \ \mathbf{B}\ _2^2}$.
lyche-numerical-linear-algebra-F04166	321	0.998	$\ \mathbf{A}\ _2 = \sqrt{1 + \ \mathbf{B}\ _2^2}$	$\ \mathbf{A}\ _2 = \sqrt{1 + \ \mathbf{B}\ _2^2}$	b) Show that $\ \mathbf{A}\mathbf{x}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{B}\mathbf{x}\ _2^2$ and that $\ \mathbf{A}\ _2 = \sqrt{1 + \ \mathbf{B}\ _2^2}$.
lyche-numerical-linear-algebra-F04167	321	1.000	$\ \mathbf{A}\ _2 \leq 1$	$\ \mathbf{A}\ _2 \leq 1$	and $\ \mathbf{A}\ _2 \leq 1$.
lyche-numerical-linear-algebra-F04168	321	1.000	$1 \leq k \leq n, \mathcal{W} = \operatorname{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$	$1 \leq k \leq n, \mathcal{W} = \operatorname{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$	d) Let $\mathcal{W} = \operatorname{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$ a k -dimensional subspace of $\mathbb{R}^n$ and
lyche-numerical-linear-algebra-F04169	321	1.000	$\mathbf{x} \in \mathcal{W}$	$\mathbf{x} \in \mathcal{W}$	$\mathbf{b} \in \mathbb{R}^n$. Show that if $\mathbf{x} \in \mathcal{W}$ is such that
lyche-numerical-linear-algebra-F04170	321	0.835	$\ \mathbf{x}\ _2 \leq \ \mathbf{b}\ _2$	$\ \mathbf{x}\ _2 \leq \ \mathbf{b}\ _2$	then $\ \mathbf{x}\ _2 \leq \ \mathbf{b}\ _2$.
lyche-numerical-linear-algebra-F04171	322	1.000	$\mathbf{x} := \sum_{j=1}^k x_j \mathbf{w}_j$	$\mathbf{x} := \sum_{j=1}^k x_j \mathbf{w}_j$	With $\mathbf{x} := \sum_{j=1}^k x_j \mathbf{w}_j$ the problem (13.65) is equivalent to finding real
lyche-numerical-linear-algebra-F04172	322	1.000	$x_1, \dots, x_k$	$x_1, \dots, x_k$	numbers $x_1, \dots, x_k$ solving the linear system
lyche-numerical-linear-algebra-F04173	322	0.996	$\mathbf{x}^* := \mathbf{A}^{-1}\mathbf{b}$	$\mathbf{x}^* := \mathbf{A}^{-1}\mathbf{b}$	e) Let $\mathbf{x}^* := \mathbf{A}^{-1}\mathbf{b}$. Show that
lyche-numerical-linear-algebra-F04174	322	1.000	$\mathbf{x}_k \in \mathcal{W}_k$	$\mathbf{x}_k \in \mathcal{W}_k$	For $k = 1, 2, \dots, n$ we define $\mathbf{x}_k \in \mathcal{W}_k$ by

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lyche-numerical-linear-algebra-F04202	324	1.000	$f:[a, b] \rightarrow \mathbb{R}$	$f:[a, b] \rightarrow \mathbb{R}$	Exercise 13.16 (Maximum of a Convex Function) Show that if $f:[a, b] \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra-F04203	324	1.000	$\max_{a \leq x \leq b} f(x) \leq \max\{f(a), f(b)\}$	$\max_{a \leq x \leq b} f(x) \leq \max\{f(a), f(b)\}$	is convex then $\max_{a \leq x \leq b} f(x) \leq \max\{f(a), f(b)\}$.
lyche-numerical-linear-algebra-F04204	326	1.000	$\mathbf{A} \mathbf{e}_i = a_{ii} \mathbf{e}_i$	$\mathbf{A} \mathbf{e}_i = a_{ii} \mathbf{e}_i$	Diagonal Matrices The eigenpairs are easily determined. Since $\mathbf{A} \mathbf{e}_i = a_{ii} \mathbf{e}_i$
lyche-numerical-linear-algebra-F04205	326	0.965	$(\lambda_i, \mathbf{e}_i)$	$(\lambda_i, \mathbf{e}_i)$	the eigenpairs are $(\lambda_i, \mathbf{e}_i)$, where $\lambda_i = a_{ii}$ for $i = 1, \dots, n$. Moreover, the
lyche-numerical-linear-algebra-F04206	326	0.965	$\lambda_i = a_{ii}$	$\lambda_i = a_{ii}$	the eigenpairs are $(\lambda_i, \mathbf{e}_i)$, where $\lambda_i = a_{ii}$ for $i = 1, \dots, n$. Moreover, the
lyche-numerical-linear-algebra-F04207	326	1.000	$\det(\mathbf{A} - \lambda \mathbf{I}) = \prod_{i=1}^n (a_{ii} - \lambda)$	$\det(\mathbf{A} - \lambda \mathbf{I}) = \prod_{i=1}^n (a_{ii} - \lambda)$	the eigenvalues Since $\det(\mathbf{A} - \lambda \mathbf{I}) = \prod_{i=1}^n (a_{ii} - \lambda)$ the eigenvalues are $\lambda_i = a_{ii}$
lyche-numerical-linear-algebra-F04208	326	1.000	$\mathbf{A}_i \mathbf{X}_i = \mathbf{X}_i \mathbf{D}_i$	$\mathbf{A}_i \mathbf{X}_i = \mathbf{X}_i \mathbf{D}_i$	Here the eigenpair problem reduces to r smaller problems. Let $\mathbf{A}_i \mathbf{X}_i = \mathbf{X}_i \mathbf{D}_i$
lyche-numerical-linear-algebra-F04209	326	1.000	$\mathbf{A}_i$	$\mathbf{A}_i$	define the eigenpairs of $\mathbf{A}_i$ for $i = 1, \dots, r$ and let $\mathbf{X} := \text{diag}(\mathbf{X}_1, \dots, \mathbf{X}_r)$,
lyche-numerical-linear-algebra-F04210	326	1.000	$\mathbf{X} := \text{diag}(\mathbf{X}_1, \dots, \mathbf{X}_r)$	$\mathbf{X} := \text{diag}(\mathbf{X}_1, \dots, \mathbf{X}_r)$	define the eigenpairs of $\mathbf{A}_i$ for $i = 1, \dots, r$ and let $\mathbf{X} := \text{diag}(\mathbf{X}_1, \dots, \mathbf{X}_r)$,
lyche-numerical-linear-algebra-F04211	326	1.000	$\mathbf{D} := \text{diag}(\mathbf{D}_1, \dots, \mathbf{D}_r)$	$\mathbf{D} := \text{diag}(\mathbf{D}_1, \dots, \mathbf{D}_r)$	$\mathbf{D} := \text{diag}(\mathbf{D}_1, \dots, \mathbf{D}_r)$. Then the eigenpairs for $\mathbf{A}$ are given by
lyche-numerical-linear-algebra-F04212	327	0.988	$\mathbf{A}_{11}, \mathbf{A}_{22}, \dots, \mathbf{A}_{rr}$	$\mathbf{A}_{11}, \mathbf{A}_{22}, \dots, \mathbf{A}_{rr}$	$\mathbf{A}_{11}, \mathbf{A}_{22}, \dots, \mathbf{A}_{rr}$ be the diagonal blocks of $\mathbf{A}$. By Property 8. of determinants
lyche-numerical-linear-algebra-F04213	327	1.000	$\pi_{\mathbf{A}}(\lambda)$	$\pi_{\mathbf{A}}(\lambda)$	a small change in one of the coefficients of $\pi_{\mathbf{A}}(\lambda)$ can lead to a large change in the
lyche-numerical-linear-algebra-F04214	327	0.405	$\pi_{\mathbf{A}}(\lambda) = \lambda^{16}$	$\pi_{\mathbf{A}}(\lambda) = \lambda^{16}$	roots of the polynomial. For example, if $\pi_{\mathbf{A}}(\lambda) = \lambda^{16}$ and $q(\lambda) = \lambda^{16} - 10^{-16}$ then
lyche-numerical-linear-algebra-F04215	327	0.405	$q(\lambda) = \lambda^{16} - 10^{-16}$	$q(\lambda) = \lambda^{16} - 10^{-16}$	roots of the polynomial. For example, if $\pi_{\mathbf{A}}(\lambda) = \lambda^{16}$ and $q(\lambda) = \lambda^{16} - 10^{-16}$ then
lyche-numerical-linear-algebra-F04216	327	0.999	$\lambda_j = 10^{-1} e^{2\pi i j / 16}$	$\lambda_j = 10^{-1} e^{2\pi i j / 16}$	the roots of $\pi_{\mathbf{A}}$ are all equal to zero, while the roots of q are $\lambda_j = 10^{-1} e^{2\pi i j / 16}$,
lyche-numerical-linear-algebra-F04217	327	1.000	$j=1, \dots, 16$	$j = 1, \dots, 16$	$j = 1, \dots, 16$. The roots of q have absolute value 0.1 and a perturbation in one
lyche-numerical-linear-algebra-F04218	327	1.000	10^{-16}	10^{-16}	of the polynomial coefficients of magnitude 10^{-16} has led to an error in the roots
lyche-numerical-linear-algebra-F04219	328	1.000	$R \cap C$	$R \cap C$	Then any eigenvalue of $\mathbf{A}$ lies in $R \cap C$ where $R = R_1 \cup R_2 \cup \dots \cup R_n$ and
lyche-numerical-linear-algebra-F04220	328	1.000	$R = R_1 \cup R_2 \cup \dots \cup R_n$	$R = R_1 \cup R_2 \cup \dots \cup R_n$	Then any eigenvalue of $\mathbf{A}$ lies in $R \cap C$ where $R = R_1 \cup R_2 \cup \dots \cup R_n$ and
lyche-numerical-linear-algebra-F04221	328	1.000	$C = C_1 \cup C_2 \cup \dots \cup C_n$	$C = C_1 \cup C_2 \cup \dots \cup C_n$	$C = C_1 \cup C_2 \cup \dots \cup C_n$.
lyche-numerical-linear-algebra-F04222	328	0.621	$\lambda \in R_i$	$\lambda \in R_i$	Proof Suppose $(\lambda, \mathbf{x})$ is an eigenpair for $\mathbf{A}$. We claim that $\lambda \in R_i$, where i is such
lyche-numerical-linear-algebra-F04223	328	0.997	$ x_i = \ \mathbf{x}\ _{\infty}$	$ x_i = \ \mathbf{x}\ _{\infty}$	that $ x_i = \ \mathbf{x}\ _{\infty}$. Indeed, $\mathbf{A} \mathbf{x} = \lambda \mathbf{x}$ implies that $\sum_j a_{ij} x_j = \lambda x_i$ or $(\lambda - a_{ii}) x_i =$
lyche-numerical-linear-algebra-F04224	328	0.997	$\sum_j a_{ij} x_j = \lambda x_i$	$\sum_j a_{ij} x_j = \lambda x_i$	that $ x_i = \ \mathbf{x}\ _{\infty}$. Indeed, $\mathbf{A} \mathbf{x} = \lambda \mathbf{x}$ implies that $\sum_j a_{ij} x_j = \lambda x_i$ or $(\lambda - a_{ii}) x_i =$
lyche-numerical-linear-algebra-F04225	328	0.997	$(\lambda - a_{ii}) x_i =$	$(\lambda - a_{ii}) x_i =$	that $ x_i = \ \mathbf{x}\ _{\infty}$. Indeed, $\mathbf{A} \mathbf{x} = \lambda \mathbf{x}$ implies that $\sum_j a_{ij} x_j = \lambda x_i$ or $(\lambda - a_{ii}) x_i =$
lyche-numerical-linear-algebra-F04226	328	1.000	$\sum_{j \neq i} a_{ij} x_j$	$\sum_{j \neq i} a_{ij} x_j$	$\sum_{j \neq i} a_{ij} x_j$. Dividing by x_i and taking absolute values we find
lyche-numerical-linear-algebra-F04227	328	1.000	$ x_j / x_i \leq 1$	$ x_j / x_i \leq 1$	since $ x_j / x_i \leq 1$ for all j . Thus $\lambda \in R_i$.

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lyche-numerical-linear-algebra F04228	328	1.000	$C_{\{j\}}$	C_j	But these are the column disks C_j of A . Hence $\lambda \in C_j$ for some j .
lyche-numerical-linear-algebra F04229	328	1.000	$\lambda \in C_{\{j\}}$	$\lambda \in C_j$	But these are the column disks C_j of A . Hence $\lambda \in C_j$ for some j .
lyche-numerical-linear-algebra F04230	328	1.000	$r_{\{i\}}$	r_i	with center at a_{ii} and radius r_i , c.f. Fig. 14.1. R_i is called a (Gerschgorin) row disk.
lyche-numerical-linear-algebra F04231	328	1.000	$R_{\{1\}}, \ldots, R_{\{n\}}$	$R_1, \dots, R_n$	An eigenvalue λ lies in the union of the row disks $R_1, \dots, R_n$ and also in the
lyche-numerical-linear-algebra F04232	328	1.000	$C_{\{1\}}, \ldots, C_{\{n\}}$	$C_1, \dots, C_n$	union of the column disks $C_1, \dots, C_n$. If A is Hermitian then $R_i = C_i$ for $i =$
lyche-numerical-linear-algebra F04233	328	1.000	$R_{\{i\}}=C_{\{i\}}$	$R_i = C_i$	union of the column disks $C_1, \dots, C_n$. If A is Hermitian then $R_i = C_i$ for $i =$
lyche-numerical-linear-algebra F04234	328	1.000	$R_{\{1\}}=R_{\{m\}}=\{z \in \mathbb{R}: z-2 \leq 1\}$	$R_1 = R_m = \{z \in \mathbb{R} : z - 2 \leq 1\}$	eigenvalues are real. We find $R_1 = R_m = \{z \in \mathbb{R} : z - 2 \leq 1\}$ and
lyche-numerical-linear-algebra F04235	328	1.000	$\lambda \in [0, 4]$	$\lambda \in [0, 4]$	We conclude that $\lambda \in [0, 4]$ for any eigenvalue λ of T . To check this, we recall that
lyche-numerical-linear-algebra F04236	329	1.000	$4\left[\sin \frac{\pi}{2(m+1)}\right]^2$	$4 \left[\sin \frac{\pi}{2(m+1)} \right]^2$	When m is large the smallest eigenvalue $4 \left[\sin \frac{\pi}{2(m+1)} \right]^2$ is very close to zero and
lyche-numerical-linear-algebra F04237	329	0.969	$4\left[\sin \frac{m\pi}{2(m+1)}\right]^2$	$4 \left[\sin \frac{m\pi}{2(m+1)} \right]^2$	the largest eigenvalue $4 \left[\sin \frac{m\pi}{2(m+1)} \right]^2$ is very close to 4. Thus Gerschgorin's theorem
lyche-numerical-linear-algebra F04238	329	1.000	$\boldsymbol{A}(0)=\boldsymbol{D}$	$\boldsymbol{A}(0) = \boldsymbol{D}$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t , every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra F04239	329	1.000	$\boldsymbol{A}(1)=\boldsymbol{A}$	$\boldsymbol{A}(1) = \boldsymbol{A}$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t , every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra F04240	329	1.000	$\boldsymbol{A}(t)$	$\boldsymbol{A}(t)$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t , every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra F04241	329	1.000	$R_{\{i\}}(t)$	$R_i(t)$	row disks $R_i(t)$ of $\boldsymbol{A}(t)$ have radius proportional to t , indeed
lyche-numerical-linear-algebra F04242	329	1.000	$0 \leq t_1 < t_2 \leq 1$	$0 \leq t_1 < t_2 \leq 1$	Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of
lyche-numerical-linear-algebra F04243	329	1.000	$R_{\{i\}}\left(t_1\right) \subset R_{\{i\}}\left(t_2\right)$	$R_i\left(t_1\right) \subset R_i\left(t_2\right)$	Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of
lyche-numerical-linear-algebra F04244	329	1.000	$R_{\{i\}}(1)$	$R_i(1)$	Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of
lyche-numerical-linear-algebra F04245	329	0.992	$\bigcup_{k=1}^p R_{\{i_k\}}(1)$	$\bigcup_{k=1}^p R_{i_k}(1)$	$\boldsymbol{A}$ for all i . Suppose $\bigcup_{k=1}^p R_{i_k}(1)$ are disjoint from the other disks of $\boldsymbol{A}$ and set
lyche-numerical-linear-algebra F04246	329	0.992	$R^{\{p\}}(t):=\bigcup_{k=1}^p R_{\{i_k\}}(t)$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$ for $t \in [0, 1]$. Now $R^p(0)$ contains only the p eigenvalues
lyche-numerical-linear-algebra F04247	329	0.992	$t \in [0, 1]$	$t \in [0, 1]$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$ for $t \in [0, 1]$. Now $R^p(0)$ contains only the p eigenvalues
lyche-numerical-linear-algebra F04248	329	0.992	$R^{\{p\}}(0)$	$R^p(0)$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$ for $t \in [0, 1]$. Now $R^p(0)$ contains only the p eigenvalues
lyche-numerical-linear-algebra F04249	329	1.000	$a_{\{i_1\}, i_{\{1\}}}, \ldots, a_{\{i_p\}, i_{\{p\}}}$	$a_{i_1, i_1}, \dots, a_{i_p, i_p}$	$a_{i_1, i_1}, \dots, a_{i_p, i_p}$ of $\boldsymbol{A}(0) = \boldsymbol{D}$. As t increases from zero to one the set $R^p(t)$ is
lyche-numerical-linear-algebra F04250	329	1.000	$R^{\{p\}}(t)$	$R^p(t)$	$a_{i_1, i_1}, \dots, a_{i_p, i_p}$ of $\boldsymbol{A}(0) = \boldsymbol{D}$. As t increases from zero to one the set $R^p(t)$ is
lyche-numerical-linear-algebra F04251	329	1.000	$R^{\{p\}}(1)$	$R^p(1)$	cannot loose or gain eigenvalues. It follows that $R^p(1)$ must contain p eigenvalues
lyche-numerical-linear-algebra F04252	329	1.000	$\boldsymbol{A}=\left[\begin{array}{ccc} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{bmatrix}$	<i>Example 14.2</i> Consider the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{bmatrix}$, where $ \epsilon_i \leq 10^{-15}$ all i . By

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lyche-numerical-linear-algebra-F04253	329	1.000	$\left \epsilon_i\right \leq 10^{-15}$	$ \epsilon_i \leq 10^{-15}$	<i>Example 14.2</i> Consider the matrix $A = \begin{bmatrix} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{bmatrix}$, where $ \epsilon_i \leq 10^{-15}$ all i . By
lyche-numerical-linear-algebra-F04254	329	1.000	$\lambda_1, \lambda_2, \lambda_3$	$\lambda_1, \lambda_2, \lambda_3$	Corollary 14.1 the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of A are distinct and satisfy $ \lambda_j - j \leq$
lyche-numerical-linear-algebra-F04255	329	1.000	$\left \lambda_j - j\right \leq$	$ \lambda_j - j \leq$	Corollary 14.1 the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of A are distinct and satisfy $ \lambda_j - j \leq$
lyche-numerical-linear-algebra-F04256	329	1.000	2×10^{-15}	2×10^{-15}	2×10^{-15} for $j = 1, 2, 3$.
lyche-numerical-linear-algebra-F04257	329	1.000	$j=1,2,3$	$j = 1, 2, 3$	2×10^{-15} for $j = 1, 2, 3$.
lyche-numerical-linear-algebra-F04258	329	0.979	$\boldsymbol{A}+$	$\boldsymbol{A}+$	we think of $\boldsymbol{E}$ as a perturbation of $\boldsymbol{A}$. By how much do the eigenvalues of $\boldsymbol{A}$ and $\boldsymbol{A}+$
lyche-numerical-linear-algebra-F04259	329	1.000	$\boldsymbol{A}_{\{0\}}:=\mathbf{0}$	$\boldsymbol{A}_0 := \mathbf{0}$	We illustrate this by considering two examples. Suppose $\boldsymbol{A}_0 := \mathbf{0}$ is the zero
lyche-numerical-linear-algebra-F04260	329	0.998	$\lambda \in \sigma\left(\boldsymbol{A}_{\{0\}}+\boldsymbol{E}\right)=\sigma(\boldsymbol{E})$	$\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$	matrix. If $\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$, then $ \lambda \leq \ \boldsymbol{E}\ _\infty$ by Theorem 12.11, and any
lyche-numerical-linear-algebra-F04261	329	0.998	$ \lambda \leq\ \boldsymbol{E}\ _{\infty}$	$ \lambda \leq \ \boldsymbol{E}\ _\infty$	matrix. If $\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$, then $ \lambda \leq \ \boldsymbol{E}\ _\infty$ by Theorem 12.11, and any
lyche-numerical-linear-algebra-F04262	330	0.998	$\boldsymbol{A}_{\{0\}}$	$\boldsymbol{A}_0$	zero eigenvalue of $\boldsymbol{A}_0$ is perturbed by at most $\ \boldsymbol{E}\ _\infty$. On the other hand consider
lyche-numerical-linear-algebra-F04263	330	0.998	$\ \boldsymbol{E}\ _{\infty}$	$\ \boldsymbol{E}\ _\infty$	zero eigenvalue of $\boldsymbol{A}_0$ is perturbed by at most $\ \boldsymbol{E}\ _\infty$. On the other hand consider
lyche-numerical-linear-algebra-F04264	330	1.000	$\boldsymbol{A}_{\{1\}}+\boldsymbol{E}$	$\boldsymbol{A}_1 + \boldsymbol{E}$	The characteristic polynomial of $\boldsymbol{A}_1 + \boldsymbol{E}$ is $\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$, and the zero
lyche-numerical-linear-algebra-F04265	330	1.000	$\pi(\lambda):=(-1)^n\left(\lambda^n-\epsilon\right)$	$\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$	The characteristic polynomial of $\boldsymbol{A}_1 + \boldsymbol{E}$ is $\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$, and the zero
lyche-numerical-linear-algebra-F04266	330	1.000	$ \lambda =\ \boldsymbol{E}\ _{\infty}^{1 / n}$	$ \lambda = \ \boldsymbol{E}\ _\infty^{1/n}$	eigenvalues of $\boldsymbol{A}_1$ are perturbed by the amount $ \lambda = \ \boldsymbol{E}\ _\infty^{1/n}$. Thus, for $n = 16$, a
lyche-numerical-linear-algebra-F04267	330	1.000	$n=16$	$n = 16$	eigenvalues of $\boldsymbol{A}_1$ are perturbed by the amount $ \lambda = \ \boldsymbol{E}\ _\infty^{1/n}$. Thus, for $n = 16$, a
lyche-numerical-linear-algebra-F04268	330	0.978	$\epsilon=10^{-16}$	$\epsilon = 10^{-16}$	perturbation of say $\epsilon = 10^{-16}$ gives a change in eigenvalue of 0.1.
lyche-numerical-linear-algebra-F04269	330	1.000	$\ \boldsymbol{E}\ _{\infty}^{1 / n}$	$\ \boldsymbol{E}\ _\infty^{1/n}$	The following theorem shows that a dependence $\ \boldsymbol{E}\ _\infty^{1/n}$ is the worst that can
lyche-numerical-linear-algebra-F04270	330	1.000	$\mu \in$	$\mu \in$	Theorem 14.2 (Elsner’s Theorem (1985)) Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$. To every $\mu \in$
lyche-numerical-linear-algebra-F04271	330	1.000	$\sigma(\boldsymbol{A}+\boldsymbol{E})$	$\sigma(\boldsymbol{A} + \boldsymbol{E})$	$\sigma(\boldsymbol{A} + \boldsymbol{E})$ there is a $\lambda \in \sigma(\boldsymbol{A})$ such that
lyche-numerical-linear-algebra-F04272	330	1.000	$\left\ \boldsymbol{u}_{\{1\}}\right\ _{\{2\}}=1$	$\ \boldsymbol{u}_1\ _2 = 1$	μ . Let $\boldsymbol{u}_1$ with $\ \boldsymbol{u}_1\ _2 = 1$ be an eigenvector corresponding to μ , and extend $\boldsymbol{u}_1$ to
lyche-numerical-linear-algebra-F04273	330	1.000	$\ \boldsymbol{E}\ _2^{1 / n}$	$\ \boldsymbol{E}\ _2^{1/n}$	elements of the matrix. The factor $\ \boldsymbol{E}\ _2^{1/n}$ shows that this dependence is almost,
lyche-numerical-linear-algebra-F04274	330	1.000	$\left[\begin{array}{cc} 1 & 1 \\ \epsilon & 1 \end{array}\right]$	$\begin{bmatrix} 1 & 1 \\ \epsilon & 1 \end{bmatrix}$	but not quite, differentiable. As an example, the eigenvalues of the matrix $\begin{bmatrix} 1 & 1 \\ \epsilon & 1 \end{bmatrix}$ are
lyche-numerical-linear-algebra-F04275	330	1.000	$1 \pm \sqrt{\epsilon}$	$1 \pm \sqrt{\epsilon}$	$1 \pm \sqrt{\epsilon}$ and these expressions are not differentiable at $\epsilon = 0$.
lyche-numerical-linear-algebra-F04276	330	1.000	$\epsilon=0$	$\epsilon = 0$	$1 \pm \sqrt{\epsilon}$ and these expressions are not differentiable at $\epsilon = 0$.
lyche-numerical-linear-algebra-F04277	331	1.000	$1 / n$	$1/n$	nondefective matrices we can get rid of the annoying exponent $1/n$ in $\ \boldsymbol{E}\ _2$
lyche-numerical-linear-algebra-F04278	331	1.000	$\ \boldsymbol{E}\ _2$	$\ \boldsymbol{E}\ _2$	nondefective matrices we can get rid of the annoying exponent $1/n$ in $\ \boldsymbol{E}\ _2$

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lyche-numerical-linear-algebra-331-F04279	331	1.000	$\mu \in \mathbb{C}$	$\mu \in \mathbb{C}$	any $\mu \in \mathbb{C}$ and $\mathbf{x} \in \mathbb{C}^n$ with $\ \mathbf{x}\ _p = 1$ we can find an eigenvalue λ of $\mathbf{A}$ such that
lyche-numerical-linear-algebra-331-F04280	331	0.999	$\mathbf{r} := \mathbf{Ax} - \mu \mathbf{x}$	$\mathbf{r} := \mathbf{Ax} - \mu \mathbf{x}$	where $\mathbf{r} := \mathbf{Ax} - \mu \mathbf{x}$ and $K_p(\mathbf{X}) := \ \mathbf{X}\ _p \ \mathbf{X}^{-1}\ _p$. If for some $\mathbf{E} \in \mathbb{C}^{n \times n}$ it holds
lyche-numerical-linear-algebra-331-F04281	331	0.999	$K_p(\mathbf{X}) := \ \mathbf{X}\ _p \ \mathbf{X}^{-1}\ _p$	$K_p(\mathbf{X}) := \ \mathbf{X}\ _p \ \mathbf{X}^{-1}\ _p$	where $\mathbf{r} := \mathbf{Ax} - \mu \mathbf{x}$ and $K_p(\mathbf{X}) := \ \mathbf{X}\ _p \ \mathbf{X}^{-1}\ _p$. If for some $\mathbf{E} \in \mathbb{C}^{n \times n}$ it holds
lyche-numerical-linear-algebra-331-F04282	331	0.902	$(\mu, \mathbf{x})$	$(\mu, \mathbf{x})$	that $(\mu, \mathbf{x})$ is an eigenpair for $\mathbf{A} + \mathbf{E}$, then we can find an eigenvalue λ of $\mathbf{A}$ such
lyche-numerical-linear-algebra-331-F04283	331	1.000	$\mu \in \sigma(\mathbf{A})$	$\mu \in \sigma(\mathbf{A})$	Proof If $\mu \in \sigma(\mathbf{A})$ then we can take $\lambda = \mu$ and (14.2), (14.3) hold trivially. So
lyche-numerical-linear-algebra-331-F04284	331	1.000	$\lambda = \mu$	$\lambda = \mu$	Proof If $\mu \in \sigma(\mathbf{A})$ then we can take $\lambda = \mu$ and (14.2), (14.3) hold trivially. So
lyche-numerical-linear-algebra-331-F04285	331	1.000	$\mu \notin \sigma(\mathbf{A})$	$\mu \notin \sigma(\mathbf{A})$	assume $\mu \notin \sigma(\mathbf{A})$. Since $\mathbf{A}$ is nondefective it can be diagonalized, we have $\mathbf{A} =$
lyche-numerical-linear-algebra-331-F04286	331	1.000	$\mathbf{XDX}^{-1}$	$\mathbf{XDX}^{-1}$	$\mathbf{XDX}^{-1}$, where $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $(\lambda_j, \mathbf{x}_j)$ are the eigenpairs of $\mathbf{A}$ for $j =$
lyche-numerical-linear-algebra-331-F04287	331	1.000	$(\lambda_j, \mathbf{x}_j)$	$(\lambda_j, \mathbf{x}_j)$	$\mathbf{XDX}^{-1}$, where $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $(\lambda_j, \mathbf{x}_j)$ are the eigenpairs of $\mathbf{A}$ for $j =$
lyche-numerical-linear-algebra-331-F04288	331	1.000	$\mathbf{D}_1 := \mathbf{D} - \mu \mathbf{I}$	$\mathbf{D}_1 := \mathbf{D} - \mu \mathbf{I}$	$1, \dots, n$. Define $\mathbf{D}_1 := \mathbf{D} - \mu \mathbf{I}$. Then $\mathbf{D}_1^{-1} = \text{diag}((\lambda_1 - \mu)^{-1}, \dots, (\lambda_n - \mu)^{-1})$
lyche-numerical-linear-algebra-331-F04289	331	1.000	$\mathbf{D}_1^{-1} = \text{diag}((\lambda_1 - \mu)^{-1}, \dots, (\lambda_n - \mu)^{-1})$	$\mathbf{D}_1^{-1} = \text{diag}((\lambda_1 - \mu)^{-1}, \dots, (\lambda_n - \mu)^{-1})$	$1, \dots, n$. Define $\mathbf{D}_1 := \mathbf{D} - \mu \mathbf{I}$. Then $\mathbf{D}_1^{-1} = \text{diag}((\lambda_1 - \mu)^{-1}, \dots, (\lambda_n - \mu)^{-1})$
lyche-numerical-linear-algebra-331-F04290	331	0.658	$(\mathbf{A} + \mathbf{E})\mathbf{x} = \mu \mathbf{x}$	$(\mathbf{A} + \mathbf{E})\mathbf{x} = \mu \mathbf{x}$	But then (14.2) follows. If $(\mathbf{A} + \mathbf{E})\mathbf{x} = \mu \mathbf{x}$, then $\mathbf{0} = \mathbf{Ax} - \mu \mathbf{x} + \mathbf{Ex} = \mathbf{r} + \mathbf{Ex}$.
lyche-numerical-linear-algebra-331-F04291	331	0.658	$\mathbf{0} = \mathbf{Ax} - \mu \mathbf{x} + \mathbf{Ex} = \mathbf{r} + \mathbf{Ex}$	$\mathbf{0} = \mathbf{Ax} - \mu \mathbf{x} + \mathbf{Ex} = \mathbf{r} + \mathbf{Ex}$	But then (14.2) follows. If $(\mathbf{A} + \mathbf{E})\mathbf{x} = \mu \mathbf{x}$ then $\mathbf{0} = \mathbf{Ax} - \mu \mathbf{x} + \mathbf{Ex} = \mathbf{r} + \mathbf{Ex}$.
lyche-numerical-linear-algebra-331-F04292	331	0.954	$\ \mathbf{r}\ _p = \ \mathbf{Ex}\ _p \leq \ \mathbf{E}\ _p$	$\ \mathbf{r}\ _p = \ \mathbf{Ex}\ _p \leq \ \mathbf{E}\ _p$	But then $\ \mathbf{r}\ _p = \ \mathbf{Ex}\ _p \leq \ \mathbf{E}\ _p$. Inserting this in (14.2) proves (14.3).
lyche-numerical-linear-algebra-331-F04293	331	1.000	$K_p(\mathbf{X})$	$K_p(\mathbf{X})$	be magnified by at most $K_p(\mathbf{X})$, the condition number of the eigenvector matrix
lyche-numerical-linear-algebra-332-F04294	332	0.593	$K_2(\mathbf{X}) = 1$	$K_2(\mathbf{X}) = 1$	we restrict attention to the 2-norm then $K_2(\mathbf{X}) = 1$ and (14.3) implies the following
lyche-numerical-linear-algebra-332-F04295	332	1.000	$ \lambda - \mu \leq \ \mathbf{E}\ _2$	$ \lambda - \mu \leq \ \mathbf{E}\ _2$	eigenvalue λ of $\mathbf{A}$ such that $ \lambda - \mu \leq \ \mathbf{E}\ _2$.
lyche-numerical-linear-algebra-332-F04296	332	0.995	$\mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$	$\mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$	Lemma 14.1 (p-Norm of a Diagonal Matrix) If $\mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a
lyche-numerical-linear-algebra-332-F04297	332	1.000	$\ \mathbf{A}\ _p = \rho(\mathbf{A})$	$\ \mathbf{A}\ _p = \rho(\mathbf{A})$	diagonal matrix then $\ \mathbf{A}\ _p = \rho(\mathbf{A})$ for $1 \leq p \leq \infty$.
lyche-numerical-linear-algebra-332-F04298	332	1.000	$p < \infty$	$p < \infty$	Proof For $p = \infty$ the proof is left as an exercise. For any $\mathbf{x} \in \mathbb{C}^n$ and $p < \infty$ we
lyche-numerical-linear-algebra-332-F04299	332	0.874	$\ \mathbf{A}\ _p = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\ \mathbf{Ax}\ _p}{\ \mathbf{x}\ _p} \leq \rho(\mathbf{A})$	$\ \mathbf{A}\ _p = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\ \mathbf{Ax}\ _p}{\ \mathbf{x}\ _p} \leq \rho(\mathbf{A})$	Thus $\ \mathbf{A}\ _p = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\ \mathbf{Ax}\ _p}{\ \mathbf{x}\ _p} \leq \rho(\mathbf{A})$. But from Theorem 12.11 we have $\rho(\mathbf{A}) \leq$
lyche-numerical-linear-algebra-332-F04300	332	0.784	$\mu, \mathbf{x}$	$\mu, \mathbf{x}$	where $\mathbf{r} := \mathbf{Ax} - \mu \mathbf{x}$. If for some $\mathbf{E} \in \mathbb{C}^{n \times n}$ it holds that $(\mu, \mathbf{x})$ is an eigenpair for
lyche-numerical-linear-algebra-333-F04301	333	1.000	$\mathbf{B} := \mu \mathbf{A}^{-1}$	$\mathbf{B} := \mu \mathbf{A}^{-1}$	and (14.4) follows from (14.2). To prove (14.5) we define the matrices $\mathbf{B} := \mu \mathbf{A}^{-1}$
lyche-numerical-linear-algebra-333-F04302	333	0.858	$\mathbf{F} := -\mathbf{A}^{-1} \mathbf{E}$	$\mathbf{F} := -\mathbf{A}^{-1} \mathbf{E}$	and $\mathbf{F} := -\mathbf{A}^{-1} \mathbf{E}$. If $(\lambda_j, \mathbf{x})$ are the eigenpairs for $\mathbf{A}$ then $(\frac{\mu}{\lambda_j}, \mathbf{x})$ are the eigenpairs
lyche-numerical-linear-algebra-333-F04303	333	0.858	$(\lambda_j, \mathbf{x})$	$(\lambda_j, \mathbf{x})$	and $\mathbf{F} := -\mathbf{A}^{-1} \mathbf{E}$. If $(\lambda_j, \mathbf{x})$ are the eigenpairs for $\mathbf{A}$ then $(\frac{\mu}{\lambda_j}, \mathbf{x})$ are the eigenpairs

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lyche-numerical-linear-algebra F04304	333	0.858	$\left(\frac{\mu}{\lambda_j}, \boldsymbol{x}\right)$	$\left(\frac{\mu}{\lambda_j}, \boldsymbol{x}\right)$	and $\boldsymbol{F} := -\boldsymbol{A}^{-1}\boldsymbol{E}$. If $(\lambda_j, \boldsymbol{x})$ are the eigenpairs for $\boldsymbol{A}$ then $(\frac{\mu}{\lambda_j}, \boldsymbol{x})$ are the eigenpairs
lyche-numerical-linear-algebra F04305	333	0.999	$(1, \boldsymbol{x})$	$(1, \boldsymbol{x})$	Thus $(1, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{B} + \boldsymbol{F}$. Applying Theorem 14.3 to this eigenvalue
lyche-numerical-linear-algebra F04306	333	0.999	$\boldsymbol{B} + \boldsymbol{F}$	$\boldsymbol{B} + \boldsymbol{F}$	Thus $(1, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{B} + \boldsymbol{F}$. Applying Theorem 14.3 to this eigenvalue
lyche-numerical-linear-algebra F04307	333	1.000	$\left \frac{\mu}{\lambda} - 1\right \leq K_p(\boldsymbol{X})\ \boldsymbol{F}\ _p = K_p(\boldsymbol{X})\ \boldsymbol{A}^{-1}\boldsymbol{E}\ _p$	$\left \frac{\mu}{\lambda} - 1\right \leq K_p(\boldsymbol{X})\ \boldsymbol{F}\ _p = K_p(\boldsymbol{X})\ \boldsymbol{A}^{-1}\boldsymbol{E}\ _p$	we can find $\lambda \in \sigma(\bar{\boldsymbol{A}})$ such that $\left \frac{\mu}{\lambda} - 1\right \leq K_p(\boldsymbol{X})\ \boldsymbol{F}\ _p = K_p(\boldsymbol{X})\ \boldsymbol{A}^{-1}\boldsymbol{E}\ _p$ which
lyche-numerical-linear-algebra F04308	333	1.000	$1, 2, \ldots, i-2, i=3, 4, \ldots, n$	$1, 2, \ldots, i-2, i=3, 4, \ldots, n$	$1, 2, \ldots, i-2, i=3, 4, \ldots, n$. We will reduce $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ to upper Hessenberg form
lyche-numerical-linear-algebra F04309	333	1.000	$\boldsymbol{A}_1 = \boldsymbol{A}$	$\boldsymbol{A}_1 = \boldsymbol{A}$	by unitary similarity transformations. Let $\boldsymbol{A}_1 = \boldsymbol{A}$ and define $\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k$ for
lyche-numerical-linear-algebra F04310	333	1.000	$\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k$	$\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k$	by unitary similarity transformations. Let $\boldsymbol{A}_1 = \boldsymbol{A}$ and define $\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k$ for
lyche-numerical-linear-algebra F04311	333	1.000	$k=1, 2, \ldots, n-2$	$k=1, 2, \ldots, n-2$	$k=1, 2, \ldots, n-2$. Here $\boldsymbol{H}_k$ is a Householder transformation chosen to introduce
lyche-numerical-linear-algebra F04312	333	1.000	$\boldsymbol{H}_k$	$\boldsymbol{H}_k$	$k=1, 2, \ldots, n-2$. Here $\boldsymbol{H}_k$ is a Householder transformation chosen to introduce
lyche-numerical-linear-algebra F04313	333	1.000	$\boldsymbol{A}_{n-1}$	$\boldsymbol{A}_{n-1}$	$\boldsymbol{A}_{n-1}$ will be upper Hessenberg. Householder transformations were used in Chap. 5
lyche-numerical-linear-algebra F04314	333	1.000	$\boldsymbol{A}_k^* = \boldsymbol{A}_k$	$\boldsymbol{A}_k^* = \boldsymbol{A}_k$	if $\boldsymbol{A}_k^* = \boldsymbol{A}_k$ then
lyche-numerical-linear-algebra F04315	334	1.000	$\boldsymbol{B}_k \in \mathbb{C}^{k, k}$	$\boldsymbol{B}_k \in \mathbb{C}^{k, k}$	Suppose $\boldsymbol{B}_k \in \mathbb{C}^{k, k}$ is upper Hessenberg, and the first $k-1$ columns of $\boldsymbol{D}_k \in$
lyche-numerical-linear-algebra F04316	334	1.000	$\boldsymbol{D}_k \in$	$\boldsymbol{D}_k \in$	Suppose $\boldsymbol{B}_k \in \mathbb{C}^{k, k}$ is upper Hessenberg, and the first $k-1$ columns of $\boldsymbol{D}_k \in$
lyche-numerical-linear-algebra F04317	334	0.999	$\mathbb{C}^{n-k, k}$	$\mathbb{C}^{n-k, k}$	$\mathbb{C}^{n-k, k}$ are zero, i.e. $\boldsymbol{D}_k = [\boldsymbol{0}, \boldsymbol{0}, \ldots, \boldsymbol{0}, d_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$ be a
lyche-numerical-linear-algebra F04318	334	0.999	$\boldsymbol{D}_k = [\boldsymbol{0}, \boldsymbol{0}, \ldots, \boldsymbol{0}, d_k]$	$\boldsymbol{D}_k = [\boldsymbol{0}, \boldsymbol{0}, \ldots, \boldsymbol{0}, d_k]$	$\mathbb{C}^{n-k, k}$ are zero, i.e. $\boldsymbol{D}_k = [\boldsymbol{0}, \boldsymbol{0}, \ldots, \boldsymbol{0}, d_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$ be a
lyche-numerical-linear-algebra F04319	334	0.999	$\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$	$\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$	$\mathbb{C}^{n-k, k}$ are zero, i.e. $\boldsymbol{D}_k = [\boldsymbol{0}, \boldsymbol{0}, \ldots, \boldsymbol{0}, d_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$ be a
lyche-numerical-linear-algebra F04320	334	0.999	$\boldsymbol{V}_k \boldsymbol{d}_k = \alpha_k \boldsymbol{e}_1$	$\boldsymbol{V}_k \boldsymbol{d}_k = \alpha_k \boldsymbol{e}_1$	Householder transformation such that $\boldsymbol{V}_k \boldsymbol{d}_k = \alpha_k \boldsymbol{e}_1$. Define
lyche-numerical-linear-algebra F04321	334	1.000	$\boldsymbol{V}_k \boldsymbol{D}_k = [\boldsymbol{V}_k \boldsymbol{0}, \ldots, \boldsymbol{V}_k \boldsymbol{0}, \boldsymbol{V}_k \boldsymbol{d}_k] = (\boldsymbol{0}, \ldots, \boldsymbol{0}, \alpha_k \boldsymbol{e}_1)$	$\boldsymbol{V}_k \boldsymbol{D}_k = [\boldsymbol{V}_k \boldsymbol{0}, \ldots, \boldsymbol{V}_k \boldsymbol{0}, \boldsymbol{V}_k \boldsymbol{d}_k] = (\boldsymbol{0}, \ldots, \boldsymbol{0}, \alpha_k \boldsymbol{e}_1)$	Now $\boldsymbol{V}_k \boldsymbol{D}_k = [\boldsymbol{V}_k \boldsymbol{0}, \ldots, \boldsymbol{V}_k \boldsymbol{0}, \boldsymbol{V}_k \boldsymbol{d}_k] = (\boldsymbol{0}, \ldots, \boldsymbol{0}, \alpha_k \boldsymbol{e}_1)$. Moreover, the matrix
lyche-numerical-linear-algebra F04322	334	1.000	$(k+1) \times (k+1)$	$(k+1) \times (k+1)$	$\boldsymbol{B}_k$ is not affected by the $\boldsymbol{H}_k$ transformation. Therefore the upper left $(k+1) \times (k+1)$
lyche-numerical-linear-algebra F04323	334	1.000	$\boldsymbol{v}_k$	$\boldsymbol{v}_k$	To find $\boldsymbol{A}_{k+1}$ we use Algorithm 5.1 to find $\boldsymbol{v}_k$ and α_k . We store $\boldsymbol{v}_k$ in the k th
lyche-numerical-linear-algebra F04324	334	1.000	$\boldsymbol{L}(k+1:n, k) = \boldsymbol{v}_k$	$\boldsymbol{L}(k+1:n, k) = \boldsymbol{v}_k$	column of a matrix $\boldsymbol{L}$ as $\boldsymbol{L}(k+1:n, k) = \boldsymbol{v}_k$. This leads to the following algorithm
lyche-numerical-linear-algebra F04325	334	0.994	$\boldsymbol{B} \cdot \boldsymbol{B}$	$\boldsymbol{B} \cdot \boldsymbol{B}$	transformations. The algorithm returns the reduced matrix $\boldsymbol{B}$. $\boldsymbol{B}$ is tridiagonal if $\boldsymbol{A}$
lyche-numerical-linear-algebra F04326	334	1.000	$\boldsymbol{B} = \boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$	$\boldsymbol{B} = \boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$	unitary matrix $\boldsymbol{Q}$ such that $\boldsymbol{B} = \boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$. Algorithm 5.1 is used in each step of the
lyche-numerical-linear-algebra F04327	335	1.000	$\boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$	$\boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$	$\boldsymbol{Q}$ is orthogonal and $\boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$ is upper Hessenberg. We need to compute the product

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lyche-numerical-linear-algebra- F04328	335	0.950	$\boldsymbol{Q}=\boldsymbol{H}_1\boldsymbol{H}_2\cdots\boldsymbol{H}_{n-2}$	$\boldsymbol{Q} = \boldsymbol{H}_1\boldsymbol{H}_2\cdots\boldsymbol{H}_{n-2}$	$\boldsymbol{Q} = \boldsymbol{H}_1\boldsymbol{H}_2\cdots\boldsymbol{H}_{n-2}$, where $\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix}$ and $\boldsymbol{v}_k \in \mathbb{R}^{n-k}$. Since $\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-algebra- F04329	335	0.950	$\boldsymbol{H}_k=\left[\begin{array}{cc}\boldsymbol{I}&\mathbf{0}\\\mathbf{0}&\boldsymbol{I}-\boldsymbol{v}_k\boldsymbol{v}_k^T\end{array}\right]$	$\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix}$	$\boldsymbol{Q} = \boldsymbol{H}_1\boldsymbol{H}_2\cdots\boldsymbol{H}_{n-2}$, where $\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix}$ and $\boldsymbol{v}_k \in \mathbb{R}^{n-k}$. Since $\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-algebra- F04330	335	0.950	$\boldsymbol{v}_k\in\mathbb{R}^{n-k}$	$\boldsymbol{v}_k \in \mathbb{R}^{n-k}$	$\boldsymbol{Q} = \boldsymbol{H}_1\boldsymbol{H}_2\cdots\boldsymbol{H}_{n-2}$, where $\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix}$ and $\boldsymbol{v}_k \in \mathbb{R}^{n-k}$. Since $\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-algebra- F04331	335	0.950	$\boldsymbol{v}_1\in\mathbb{R}^{n-1}$	$\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$	$\boldsymbol{Q} = \boldsymbol{H}_1\boldsymbol{H}_2\cdots\boldsymbol{H}_{n-2}$, where $\boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{bmatrix}$ and $\boldsymbol{v}_k \in \mathbb{R}^{n-k}$. Since $\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-algebra- F04332	335	1.000	$\boldsymbol{v}_{n-2}\in\mathbb{R}^2$	$\boldsymbol{v}_{n-2} \in \mathbb{R}^2$	and $\boldsymbol{v}_{n-2} \in \mathbb{R}^2$ it is most economical to assemble the product from right to left. We
lyche-numerical-linear-algebra- F04333	335	1.000	$\boldsymbol{Q}_{k+1}$	$\boldsymbol{Q}_{k+1}$	Suppose $\boldsymbol{Q}_{k+1}$ has the form $\begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix}$, where $\boldsymbol{U}_k \in \mathbb{R}^{n-k,n-k}$. Then
lyche-numerical-linear-algebra- F04334	335	1.000	$\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{U}_k\end{array}\right]$	$\begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix}$	Suppose $\boldsymbol{Q}_{k+1}$ has the form $\begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix}$, where $\boldsymbol{U}_k \in \mathbb{R}^{n-k,n-k}$. Then
lyche-numerical-linear-algebra- F04335	335	1.000	$\boldsymbol{U}_k\in\mathbb{R}^{n-k,n-k}$	$\boldsymbol{U}_k \in \mathbb{R}^{n-k,n-k}$	Suppose $\boldsymbol{Q}_{k+1}$ has the form $\begin{bmatrix} \boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k \end{bmatrix}$, where $\boldsymbol{U}_k \in \mathbb{R}^{n-k,n-k}$. Then
lyche-numerical-linear-algebra- F04336	335	1.000	$\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_n$	$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$	Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. In this section
lyche-numerical-linear-algebra- F04337	335	1.000	$1\leq m\leq n$	$1 \leq m \leq n$	some $1 \leq m \leq n$. Using Householder similarity transformations as outlined in the
lyche-numerical-linear-algebra- F04338	336	1.000	$c_i=0$	$c_i = 0$	Suppose one of the off-diagonal elements is equal to zero, say $c_i = 0$. We then have
lyche-numerical-linear-algebra- F04339	336	1.000	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{A}_1&\mathbf{0}\\\mathbf{0}&\boldsymbol{A}_2\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_1 & \mathbf{0} \\ \mathbf{0} & \boldsymbol{A}_2 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_1 & \mathbf{0} \\ \mathbf{0} & \boldsymbol{A}_2 \end{bmatrix}$, where
lyche-numerical-linear-algebra- F04340	336	1.000	$c_i\neq 0$	$c_i \neq 0$	so that $\boldsymbol{A}$ is irreducible, i.e., $c_i \neq 0$ for $i = 1, \dots, n-1$.
lyche-numerical-linear-algebra- F04341	336	1.000	$x\in\mathbb{R}$	$x \in \mathbb{R}$	for $x \in \mathbb{R}$ the polynomial $p_k(x) := \det(x\boldsymbol{I}_k - \boldsymbol{A}_k)$ for $k = 1, \dots, n$, where $\boldsymbol{A}_k$
lyche-numerical-linear-algebra- F04342	336	1.000	$p_k(x):=\operatorname{det}\left(x\boldsymbol{I}_k-\boldsymbol{A}_k\right)$	$p_k(x) := \det(x\boldsymbol{I}_k - \boldsymbol{A}_k)$	for $x \in \mathbb{R}$ the polynomial $p_k(x) := \det(x\boldsymbol{I}_k - \boldsymbol{A}_k)$ for $k = 1, \dots, n$, where $\boldsymbol{A}_k$
lyche-numerical-linear-algebra- F04343	336	1.000	p_n	p_n	The eigenvalues of $\boldsymbol{A}$ are the roots of the polynomial p_n . Using the last column to
lyche-numerical-linear-algebra- F04344	336	1.000	$k\geq 2$	$k \geq 2$	expand for $k \geq 2$ the determinant $p_{k+1}(x)$ we find
lyche-numerical-linear-algebra- F04345	336	1.000	$p_{k+1}(x)$	$p_{k+1}(x)$	expand for $k \geq 2$ the determinant $p_{k+1}(x)$ we find
lyche-numerical-linear-algebra- F04346	336	0.990	$p_1(x)=x-d_1$	$p_1(x) = x - d_1$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-algebra- F04347	336	0.990	$p_2(x)=\left(x-d_2\right)\left(x-d_1\right)-c_1^2$	$p_2(x) = (x - d_2)(x - d_1) - c_1^2$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-algebra- F04348	336	0.990	$k=0,1$	$k = 0, 1$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-algebra- F04349	336	1.000	$p_{-1}(x)=0$	$p_{-1}(x) = 0$	if we define $p_{-1}(x) = 0$ and $p_0(x) = 1$. For M sufficiently large we have
lyche-numerical-linear-algebra- F04350	336	1.000	$p_0(x)=1$	$p_0(x) = 1$	if we define $p_{-1}(x) = 0$ and $p_0(x) = 1$. For M sufficiently large we have

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lyche-numerical-linear-algebra-F04351	336	1.000	$y_{\{1\}} \in (-M, d_{\{1\}})$	$y_1 \in (-M, d_1)$	Since p_2 is continuous there are $y_1 \in (-M, d_1)$ and $y_2 \in (d_1, M)$ such that
lyche-numerical-linear-algebra-F04352	336	1.000	$y_{\{2\}} \in (d_{\{1\}}, M)$	$y_2 \in (d_1, M)$	Since p_2 is continuous there are $y_1 \in (-M, d_1)$ and $y_2 \in (d_1, M)$ such that
lyche-numerical-linear-algebra-F04353	336	1.000	$p_{\{2\}}(y_{\{1\}}) = p_{\{2\}}(y_{\{2\}}) = 0$	$p_2(y_1) = p_2(y_2) = 0$	$p_2(y_1) = p_2(y_2) = 0$. It follows that the root d_1 of p_1 separates the roots of p_2 , so
lyche-numerical-linear-algebra-F04354	336	1.000	$y_{\{1\}}$	y_1	y_1 and y_2 must be distinct. Consider next
lyche-numerical-linear-algebra-F04355	336	1.000	$y_{\{2\}}$	y_2	y_1 and y_2 must be distinct. Consider next
lyche-numerical-linear-algebra-F04356	337	1.000	$y_{\{1\}} < d_{\{1\}} < y_{\{2\}}$	$y_1 < d_1 < y_2$	Since $y_1 < d_1 < y_2$ we have for M sufficiently large
lyche-numerical-linear-algebra-F04357	337	1.000	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}$	x_1, x_2, x_3	Thus the roots x_1, x_2, x_3 of p_3 are separated by the roots y_1, y_2 of p_2 . In the general
lyche-numerical-linear-algebra-F04358	337	1.000	$p_{\{3\}}$	p_3	Thus the roots x_1, x_2, x_3 of p_3 are separated by the roots y_1, y_2 of p_2 . In the general
lyche-numerical-linear-algebra-F04359	337	1.000	$z_{\{1\}}, \ldots, z_{\{k-1\}}$	$z_1, \dots, z_{k-1}$	case suppose for $k \geq 2$ that the roots $z_1, \dots, z_{k-1}$ of p_{k-1} separate the roots
lyche-numerical-linear-algebra-F04360	337	1.000	$p_{\{k-1\}}$	p_{k-1}	case suppose for $k \geq 2$ that the roots $z_1, \dots, z_{k-1}$ of p_{k-1} separate the roots
lyche-numerical-linear-algebra-F04361	337	1.000	$y_{\{1\}}, \ldots, y_{\{k\}}$	$y_1, \dots, y_k$	$y_1, \dots, y_k$ of p_k . Choose M so that $y_0 := -M < y_1, y_{k+1} := M > y_k$. Then
lyche-numerical-linear-algebra-F04362	337	1.000	$p_{\{k\}}$	p_k	$y_1, \dots, y_k$ of p_k . Choose M so that $y_0 := -M < y_1, y_{k+1} := M > y_k$. Then
lyche-numerical-linear-algebra-F04363	337	1.000	$y_{\{0\}} := -M < y_{\{1\}}, y_{\{k+1\}} := M > y_{\{k\}}$	$y_0 := -M < y_1, y_{k+1} := M > y_k$	$y_1, \dots, y_k$ of p_k . Choose M so that $y_0 := -M < y_1, y_{k+1} := M > y_k$. Then
lyche-numerical-linear-algebra-F04364	337	1.000	$j=0, k+1$	$j = 0, k + 1$	This holds for $j = 0, k + 1$, and for $j = 1, \dots, k$ since
lyche-numerical-linear-algebra-F04365	337	0.832	$x_{\{1\}}, \ldots, x_{\{k+1\}}$	$x_1, \dots, x_{k+1}$	It follows that the roots $x_1, \dots, x_{k+1}$ are separated by the roots $y_1, \dots, y_k$ of p_k and
lyche-numerical-linear-algebra-F04366	337	1.000	$\boldsymbol{A} = \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E}$	$\boldsymbol{A} = \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E}$	We say that two matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$ are congruent if $\boldsymbol{A} = \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E}$ for some
lyche-numerical-linear-algebra-F04367	337	1.000	$\boldsymbol{D}, \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$	$\boldsymbol{D}, \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$	congruent and similar to a diagonal matrix $\boldsymbol{D}, \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$ where $\boldsymbol{U}$ is unitary. The
lyche-numerical-linear-algebra-F04368	337	1.000	$\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$	$\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$	eigenvalues of $\boldsymbol{A}$ are the diagonal elements of $\boldsymbol{D}$. Let $\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$ and $v(\boldsymbol{A})$ denote
lyche-numerical-linear-algebra-F04369	337	1.000	$v(\boldsymbol{A})$	$v(\boldsymbol{A})$	eigenvalues of $\boldsymbol{A}$ are the diagonal elements of $\boldsymbol{D}$. Let $\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$ and $v(\boldsymbol{A})$ denote
lyche-numerical-linear-algebra-F04370	337	1.000	$\pi(\boldsymbol{A}) + \zeta(\boldsymbol{A}) + v(\boldsymbol{A}) = n$	$\pi(\boldsymbol{A}) + \zeta(\boldsymbol{A}) + v(\boldsymbol{A}) = n$	all eigenvalues are real and $\pi(\boldsymbol{A}) + \zeta(\boldsymbol{A}) + v(\boldsymbol{A}) = n$.
lyche-numerical-linear-algebra-F04371	337	1.000	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{B}), \zeta(\boldsymbol{A}) = \zeta(\boldsymbol{B})$	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{B}), \zeta(\boldsymbol{A}) = \zeta(\boldsymbol{B})$	congruent then $\pi(\boldsymbol{A}) = \pi(\boldsymbol{B}), \zeta(\boldsymbol{A}) = \zeta(\boldsymbol{B})$ and $v(\boldsymbol{A}) = v(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F04372	337	1.000	$v(\boldsymbol{A}) = v(\boldsymbol{B})$	$v(\boldsymbol{A}) = v(\boldsymbol{B})$	congruent then $\pi(\boldsymbol{A}) = \pi(\boldsymbol{B}), \zeta(\boldsymbol{A}) = \zeta(\boldsymbol{B})$ and $v(\boldsymbol{A}) = v(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F04373	337	1.000	$\pi(\boldsymbol{A}) = k$	$\pi(\boldsymbol{A}) = k$	are diagonal matrices. Suppose $\pi(\boldsymbol{A}) = k$ and $\pi(\boldsymbol{B}) = m < k$. We shall show that
lyche-numerical-linear-algebra-F04374	337	1.000	$\pi(\boldsymbol{B}) = m < k$	$\pi(\boldsymbol{B}) = m < k$	are diagonal matrices. Suppose $\pi(\boldsymbol{A}) = k$ and $\pi(\boldsymbol{B}) = m < k$. We shall show that
lyche-numerical-linear-algebra-F04375	337	1.000	$\boldsymbol{E}_1$	$\boldsymbol{E}_1$	this leads to a contradiction. Let $\boldsymbol{E}_1$ be the upper left $m \times k$ corner of $\boldsymbol{E}$. Since
lyche-numerical-linear-algebra-F04376	337	1.000	$m \times k$	$m \times k$	this leads to a contradiction. Let $\boldsymbol{E}_1$ be the upper left $m \times k$ corner of $\boldsymbol{E}$. Since
lyche-numerical-linear-algebra-F04377	337	1.000	$m < k$	$m < k$	$m < k$, we can find a nonzero x such that $\boldsymbol{E}_1 x = \mathbf{0}$ (cf. Lemma 1.3). Let $y^T =$

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lyche-numerical-linear-algebra-F04378	337	1.000	$\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$	$m < k$, we can find a nonzero $\boldsymbol{x}$ such that $\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$ (cf. Lemma 1.3). Let $\boldsymbol{y}^T =$
lyche-numerical-linear-algebra-F04379	337	1.000	$\boldsymbol{y}^T =$	$\boldsymbol{y}^T =$	$m < k$, we can find a nonzero $\boldsymbol{x}$ such that $\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$ (cf. Lemma 1.3). Let $\boldsymbol{y}^T =$
lyche-numerical-linear-algebra-F04380	337	0.996	$\left[\boldsymbol{x}^T, \mathbf{0}^T \right] \in \mathbb{C}^n$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$, and $\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$. Then $z_i = 0$ for $i = 1, 2, \dots, m$.
lyche-numerical-linear-algebra-F04381	337	0.996	$\boldsymbol{z} = \left[z_1, \dots, z_n \right]^T = \boldsymbol{E} \boldsymbol{y}$	$\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$, and $\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$. Then $z_i = 0$ for $i = 1, 2, \dots, m$.
lyche-numerical-linear-algebra-F04382	337	0.996	$z_i = 0$	$z_i = 0$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$, and $\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$. Then $z_i = 0$ for $i = 1, 2, \dots, m$.
lyche-numerical-linear-algebra-F04383	337	0.996	$i=1,2, \dots, m$	$i = 1, 2, \dots, m$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$, and $\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$. Then $z_i = 0$ for $i = 1, 2, \dots, m$.
lyche-numerical-linear-algebra-F04384	337	1.000	$\mu_i \leq 0$	$\mu_i \leq 0$	$\mu_i \leq 0$ for $i \geq m + 1$ then
lyche-numerical-linear-algebra-F04385	337	1.000	$i \geq m+1$	$i \geq m + 1$	$\mu_i \leq 0$ for $i \geq m + 1$ then
lyche-numerical-linear-algebra-F04386	338	1.000	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$	We conclude that $\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$ if $\boldsymbol{A}$ and $\boldsymbol{B}$ are diagonal. Moreover, $v(\boldsymbol{A}) =$
lyche-numerical-linear-algebra-F04387	338	1.000	$v(\boldsymbol{A}) =$	$v(\boldsymbol{A}) =$	We conclude that $\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$ if $\boldsymbol{A}$ and $\boldsymbol{B}$ are diagonal. Moreover, $v(\boldsymbol{A}) =$
lyche-numerical-linear-algebra-F04388	338	1.000	$\pi(-\boldsymbol{A}) = \pi(-\boldsymbol{B}) = v(\boldsymbol{B})$	$\pi(-\boldsymbol{A}) = \pi(-\boldsymbol{B}) = v(\boldsymbol{B})$	$\pi(-\boldsymbol{A}) = \pi(-\boldsymbol{B}) = v(\boldsymbol{B})$ and $\zeta(\boldsymbol{A}) = n - \pi(\boldsymbol{A}) - v(\boldsymbol{A}) = n - \pi(\boldsymbol{B}) - v(\boldsymbol{B}) =$
lyche-numerical-linear-algebra-F04389	338	1.000	$\zeta(\boldsymbol{A}) = n - \pi(\boldsymbol{A}) - v(\boldsymbol{A}) = n - \pi(\boldsymbol{B}) - v(\boldsymbol{B}) =$	$\zeta(\boldsymbol{A}) = n - \pi(\boldsymbol{A}) - v(\boldsymbol{A}) = n - \pi(\boldsymbol{B}) - v(\boldsymbol{B}) =$	$\pi(-\boldsymbol{A}) = \pi(-\boldsymbol{B}) = v(\boldsymbol{B})$ and $\zeta(\boldsymbol{A}) = n - \pi(\boldsymbol{A}) - v(\boldsymbol{A}) = n - \pi(\boldsymbol{B}) - v(\boldsymbol{B}) =$
lyche-numerical-linear-algebra-F04390	338	1.000	$\zeta(\boldsymbol{B})$	$\zeta(\boldsymbol{B})$	$\zeta(\boldsymbol{B})$. This completes the proof for diagonal matrices.
lyche-numerical-linear-algebra-F04391	338	1.000	$\boldsymbol{U}_1^* \boldsymbol{A} \boldsymbol{U}_1 = \boldsymbol{D}_1$	$\boldsymbol{U}_1^* \boldsymbol{A} \boldsymbol{U}_1 = \boldsymbol{D}_1$	Let in the general case $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$ be unitary matrices such that $\boldsymbol{U}_1^* \boldsymbol{A} \boldsymbol{U}_1 = \boldsymbol{D}_1$
lyche-numerical-linear-algebra-F04392	338	0.999	$\boldsymbol{U}_2^* \boldsymbol{B} \boldsymbol{U}_2 = \boldsymbol{D}_2$	$\boldsymbol{U}_2^* \boldsymbol{B} \boldsymbol{U}_2 = \boldsymbol{D}_2$	and $\boldsymbol{U}_2^* \boldsymbol{B} \boldsymbol{U}_2 = \boldsymbol{D}_2$ where $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ are diagonal matrices. Since $\boldsymbol{A} = \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E}$,
lyche-numerical-linear-algebra-F04393	338	0.999	$\boldsymbol{D}_1 = \boldsymbol{F}^* \boldsymbol{D}_2 \boldsymbol{F}$	$\boldsymbol{D}_1 = \boldsymbol{F}^* \boldsymbol{D}_2 \boldsymbol{F}$	we find $\boldsymbol{D}_1 = \boldsymbol{F}^* \boldsymbol{D}_2 \boldsymbol{F}$ where $\boldsymbol{F} = \boldsymbol{U}_2^* \boldsymbol{E} \boldsymbol{U}_1$ is nonsingular. Thus $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$
lyche-numerical-linear-algebra-F04394	338	0.999	$\boldsymbol{F} = \boldsymbol{U}_2^* \boldsymbol{E} \boldsymbol{U}_1$	$\boldsymbol{F} = \boldsymbol{U}_2^* \boldsymbol{E} \boldsymbol{U}_1$	we find $\boldsymbol{D}_1 = \boldsymbol{F}^* \boldsymbol{D}_2 \boldsymbol{F}$ where $\boldsymbol{F} = \boldsymbol{U}_2^* \boldsymbol{E} \boldsymbol{U}_1$ is nonsingular. Thus $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$
lyche-numerical-linear-algebra-F04395	338	1.000	$\boldsymbol{D}_1, \boldsymbol{B}$	$\boldsymbol{D}_1, \boldsymbol{B}$	are congruent diagonal matrices. But since $\boldsymbol{A}$ and $\boldsymbol{D}_1, \boldsymbol{B}$ and $\boldsymbol{D}_2$ have the same
lyche-numerical-linear-algebra-F04396	338	1.000	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{D}_1) = \pi(\boldsymbol{D}_2) = \pi(\boldsymbol{B})$	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{D}_1) = \pi(\boldsymbol{D}_2) = \pi(\boldsymbol{B})$	eigenvalues, we find $\pi(\boldsymbol{A}) = \pi(\boldsymbol{D}_1) = \pi(\boldsymbol{D}_2) = \pi(\boldsymbol{B})$. Similar results hold for ζ
lyche-numerical-linear-algebra-F04397	338	1.000	ζ	ζ	eigenvalues, we find $\pi(\boldsymbol{A}) = \pi(\boldsymbol{D}_1) = \pi(\boldsymbol{D}_2) = \pi(\boldsymbol{B})$. Similar results hold for ζ
lyche-numerical-linear-algebra-F04398	338	1.000	$\boldsymbol{A} = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$ is symmetric and that $\alpha \in \mathbb{R}$ is such that $\boldsymbol{A} - \alpha \boldsymbol{I}$
lyche-numerical-linear-algebra-F04399	338	1.000	$\alpha \in \mathbb{R}$	$\alpha \in \mathbb{R}$	$\boldsymbol{A} = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$ is symmetric and that $\alpha \in \mathbb{R}$ is such that $\boldsymbol{A} - \alpha \boldsymbol{I}$
lyche-numerical-linear-algebra-F04400	338	1.000	$\boldsymbol{A} - \alpha \boldsymbol{I}$	$\boldsymbol{A} - \alpha \boldsymbol{I}$	$\boldsymbol{A} = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$ is symmetric and that $\alpha \in \mathbb{R}$ is such that $\boldsymbol{A} - \alpha \boldsymbol{I}$
lyche-numerical-linear-algebra-F04401	338	0.999	$\boldsymbol{A} - \alpha \boldsymbol{I} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^T$	$\boldsymbol{A} - \alpha \boldsymbol{I} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^T$	has an symmetric LU factorization, i.e. $\boldsymbol{A} - \alpha \boldsymbol{I} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^T$ where $\boldsymbol{L}$ is unit lower
lyche-numerical-linear-algebra-F04402	338	1.000	$d_1(\alpha), \dots, d_n(\alpha)$	$d_1(\alpha), \dots, d_n(\alpha)$	$d_1(\alpha), \dots, d_n(\alpha)$ in $\boldsymbol{D}$ can be computed recursively as follows
lyche-numerical-linear-algebra-F04403	338	1.000	$\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^T$	$\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^T$	elements in $\boldsymbol{D}$ in an $\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^T$ factorization we see that the formulas in (14.8) follows
lyche-numerical-linear-algebra-F04404	338	1.000	$v(\boldsymbol{A} - \alpha \boldsymbol{I}) = v(\boldsymbol{D})$	$v(\boldsymbol{A} - \alpha \boldsymbol{I}) = v(\boldsymbol{D})$	By the previous theorem $v(\boldsymbol{A} - \alpha \boldsymbol{I}) = v(\boldsymbol{D})$, the number of negative diagonal

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lyche-numerical-linear-algebra-F04405	338	■ 0.988	$(\boldsymbol{A} - \alpha \boldsymbol{I}) \boldsymbol{x} = (\lambda - \alpha) \boldsymbol{x}$	$(\boldsymbol{A} - \alpha \boldsymbol{I}) \boldsymbol{x} = (\lambda - \alpha) \boldsymbol{x}$	elements in $\boldsymbol{D}$. If $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ then $(\boldsymbol{A} - \alpha \boldsymbol{I}) \boldsymbol{x} = (\lambda - \alpha) \boldsymbol{x}$, and $\lambda - \alpha$ is an eigenvalue
lyche-numerical-linear-algebra-F04406	338	■ 0.988	$\lambda - \alpha$	$\lambda - \alpha$	elements in $\boldsymbol{D}$. If $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ then $(\boldsymbol{A} - \alpha \boldsymbol{I}) \boldsymbol{x} = (\lambda - \alpha) \boldsymbol{x}$, and $\lambda - \alpha$ is an eigenvalue
lyche-numerical-linear-algebra-F04407	338	■ 1.000	$v(\boldsymbol{A} - \alpha \boldsymbol{I})$	$v(\boldsymbol{A} - \alpha \boldsymbol{I})$	of $\boldsymbol{A} - \alpha \boldsymbol{I}$. But then $v(\boldsymbol{A} - \alpha \boldsymbol{I})$ equals the number of eigenvalues of $\boldsymbol{A}$ which are
lyche-numerical-linear-algebra-F04408	338	■ 1.000	$\lambda_1 \geq \lambda_2 \geq$	$\lambda_1 \geq \lambda_2 \geq$	Corollary 14.2 can be used to determine the m th eigenvalue of $\boldsymbol{A}$, where $\lambda_1 \geq \lambda_2 \geq$
lyche-numerical-linear-algebra-F04409	338	■ 0.999	$\cdots \geq \lambda_n$	$\cdots \geq \lambda_n$	$\cdots \geq \lambda_n$. Using Gerschgorin's theorem we first find an interval $[a, b]$, such that
lyche-numerical-linear-algebra-F04410	338	■ 0.999	(a, b)	(a, b)	(a, b) contains the eigenvalues of $\boldsymbol{A}$. Let for $x \in [a, b]$
lyche-numerical-linear-algebra-F04411	339	■ 1.000	$\rho(a) =$	$\rho(a) =$	be the number of eigenvalues of $\boldsymbol{A}$ which are strictly greater than x . Clearly $\rho(a) =$
lyche-numerical-linear-algebra-F04412	339	■ 1.000	$n, \rho(b) = 0$	$n, \rho(b) = 0$	$n, \rho(b) = 0$. Choosing a tolerance ϵ and using bisection we proceed as follows:
lyche-numerical-linear-algebra-F04413	339	■ 1.000	$\left\{ \left[a_j, b_j \right] \right\}$	$\{[a_j, b_j]\}$	We generate a sequence $\{[a_j, b_j]\}$ of intervals, each containing λ_m and $b_j - a_j =$
lyche-numerical-linear-algebra-F04414	339	■ 1.000	$b_j - a_j =$	$b_j - a_j =$	We generate a sequence $\{[a_j, b_j]\}$ of intervals, each containing λ_m and $b_j - a_j =$
lyche-numerical-linear-algebra-F04415	339	■ 1.000	$2^{-j}(b - a)$	$2^{-j}(b - a)$	$2^{-j}(b - a)$.
lyche-numerical-linear-algebra-F04416	339	■ 1.000	$d_k(\alpha)$	$d_k(\alpha)$	As it stands this method will fail if in (14.8) one of the $d_k(\alpha)$ is zero. One
lyche-numerical-linear-algebra-F04417	339	■ 1.000	$\delta_k =$	$\delta_k =$	possibility is to replace such a $d_k(\alpha)$ by a suitable small number, say $\delta_k =$
lyche-numerical-linear-algebra-F04418	339	■ 1.000	$c_k \epsilon_M$	$c_k \epsilon_M$	$c_k \epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra-F04419	339	■ 1.000	ϵ_M	ϵ_M	$c_k \epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra-F04420	339	■ 1.000	2×10^{-16}	2×10^{-16}	$c_k \epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra-F04421	339	■ 1.000	$ d_k(\alpha) < \delta_k $	$ d_k(\alpha) < \delta_k $	replacement is done if $ d_k(\alpha) < \delta_k $.
lyche-numerical-linear-algebra-F04422	340	■ 0.998	$\boldsymbol{A} (a_{i,i} >$	$\boldsymbol{A} (a_{i,i} >$	Gershgorin's circle theorem that a strictly diagonally dominant matrix $\boldsymbol{A} (a_{i,i} >$
lyche-numerical-linear-algebra-F04423	340	■ 1.000	$\sum_{j \neq i} a_{i,j} $	$\sum_{j \neq i} a_{i,j} $	$\sum_{j \neq i} a_{i,j} $ for all i) is nonsingular.
lyche-numerical-linear-algebra-F04424	340	■ 1.000	$R := R_1 \cup \cdots \cup R_n$	$R := R_1 \cup \cdots \cup R_n$	$\boldsymbol{A} \in \mathbb{R}^{n,n}$ lie inside $R \cap C$, where $R := R_1 \cup \cdots \cup R_n$ is the union of the row disks
lyche-numerical-linear-algebra-F04425	340	■ 1.000	$C = C_1 \cup \cdots \cup C_n$	$C = C_1 \cup \cdots \cup C_n$	R_i of $\boldsymbol{A}$, and $C := C_1 \cup \cdots \cup C_n$ is the union of the column disks C_j . You do not
lyche-numerical-linear-algebra-F04426	340	■ 0.987	$\boldsymbol{s} = [s_1, \dots, s_n] \in \mathbb{R}^n$	$\boldsymbol{s} = [s_1, \dots, s_n] \in \mathbb{R}^n$	computes the centres $\boldsymbol{s} = [s_1, \dots, s_n] \in \mathbb{R}^n$ of the row and column disks, and their
lyche-numerical-linear-algebra-F04427	340	■ 1.000	$\boldsymbol{r} = [r_1, \dots, r_n] \in \mathbb{R}^n$	$\boldsymbol{r} = [r_1, \dots, r_n] \in \mathbb{R}^n$	radii $\boldsymbol{r} = [r_1, \dots, r_n] \in \mathbb{R}^n$ and $\boldsymbol{c} = [c_1, \dots, c_n] \in \mathbb{R}^n$, respectively.
lyche-numerical-linear-algebra-F04428	340	■ 1.000	$\boldsymbol{c} = [c_1, \dots, c_n] \in \mathbb{R}^n$	$\boldsymbol{c} = [c_1, \dots, c_n] \in \mathbb{R}^n$	radii $\boldsymbol{r} = [r_1, \dots, r_n] \in \mathbb{R}^n$ and $\boldsymbol{c} = [c_1, \dots, c_n] \in \mathbb{R}^n$, respectively.
lyche-numerical-linear-algebra-F04429	340	■ 1.000	$\boldsymbol{A}(t_2)$	$\boldsymbol{A}(t_2)$	$0 \leq t_1 < t_2 \leq 1$ and that μ is an eigenvalue of $\boldsymbol{A}(t_2)$. Show, using Theorem 14.2
lyche-numerical-linear-algebra-F04430	340	■ 1.000	$\boldsymbol{A} = \boldsymbol{A}(t_1)$	$\boldsymbol{A} = \boldsymbol{A}(t_1)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that
lyche-numerical-linear-algebra-F04431	340	■ 1.000	$\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$	$\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that

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lyche-numerical-linear-algebra-F04432	340	■ 1.000	$\boldsymbol{A} \left(t_1 \right)$	$\boldsymbol{A} (t_1)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that
lyche-numerical-linear-algebra-F04433	340	■ 1.000	$\ \boldsymbol{A}\ _{\infty} =$	$\ \boldsymbol{A}\ _{\infty} =$	Exercise 14.6 (∞-Norm of a Diagonal Matrix) Give a direct proof that $\ \boldsymbol{A}\ _{\infty} =$
lyche-numerical-linear-algebra-F04434	340	■ 1.000	$\left[a_{kj} \right], \boldsymbol{E} = \left[e_{kj} \right]$	$[a_{kj}], \boldsymbol{E} = [e_{kj}]$	$[a_{kj}], \boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra-F04435	340	■ 1.000	$\boldsymbol{B} = \left[b_{kj} \right]$	$\boldsymbol{B} = [b_{kj}]$	$[a_{kj}], \boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra-F04436	340	■ 1.000	$\mathbb{R}^{n,n}$	$\mathbb{R}^{n,n}$	$[a_{kj}], \boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra-F04437	341	■ 1.000	$\boldsymbol{B} = \boldsymbol{A} + \boldsymbol{E}$	$\boldsymbol{B} = \boldsymbol{A} + \boldsymbol{E}$	and $\boldsymbol{B} = \boldsymbol{A} + \boldsymbol{E}$, where $0 < \epsilon < 1$. Thus for $n = 4$,
lyche-numerical-linear-algebra-F04438	341	■ 1.000	$\ \boldsymbol{A}\ _2 = \ \boldsymbol{B}\ _2 = 1$	$\ \boldsymbol{A}\ _2 = \ \boldsymbol{B}\ _2 = 1$	b) Show that $\ \boldsymbol{A}\ _2 = \ \boldsymbol{B}\ _2 =$ for arbitrary $n \in \mathbb{N}$.
lyche-numerical-linear-algebra-F04439	341	■ 1.000	$\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$	$\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$	c) Recall Elsner's Theorem (Theorem 14.2). Let $\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$ be given by (14.10).
lyche-numerical-linear-algebra-F04440	341	■ 1.000	$\mu = \epsilon^{1/n}$	$\mu = \epsilon^{1/n}$	$\mu = \epsilon^{1/n}$ of $\boldsymbol{B}$? How sharp is this upper bound?
lyche-numerical-linear-algebra-F04441	341	■ 1.000	$\frac{10}{3}n^3 = 5G_n$	$\frac{10}{3}n^3 = 5G_n$	Show that the number of arithmetic operations for Algorithm 14.1 is $\frac{10}{3}n^3 = 5G_n$.
lyche-numerical-linear-algebra-F04442	341	■ 1.000	$\frac{4}{3}n^3 = 2G_n$	$\frac{4}{3}n^3 = 2G_n$	of arithmetic operations required by Algorithm 14.2 is $\frac{4}{3}n^3 = 2G_n$.
lyche-numerical-linear-algebra-F04443	341	■ 1.000	$\boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k$	$\boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k$	compute $\boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k$ where $\boldsymbol{E}_k$ is symmetric. Dropping subscripts we have to compute
lyche-numerical-linear-algebra-F04444	341	■ 1.000	$\boldsymbol{E}_k$	$\boldsymbol{E}_k$	compute $\boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k$ where $\boldsymbol{E}_k$ is symmetric. Dropping subscripts we have to compute
lyche-numerical-linear-algebra-F04445	341	■ 1.000	$\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{E} (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T)$	$\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{E} (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T)$	a product of the form $\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{E} (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T)$. Let $\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}$, $\beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$ and
lyche-numerical-linear-algebra-F04446	341	■ 1.000	$\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}, \beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$	$\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}, \beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$	a product of the form $\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{E} (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T)$. Let $\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}$, $\beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$ and
lyche-numerical-linear-algebra-F04447	341	■ 0.999	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$. Show that $\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$. Since $\boldsymbol{G}$ is symmetric, only the sub- or
lyche-numerical-linear-algebra-F04448	341	■ 0.999	$\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$	$\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$. Show that $\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$. Since $\boldsymbol{G}$ is symmetric, only the sub- or
lyche-numerical-linear-algebra-F04449	341	■ 1.000	$O(4n^3/3)$	$O(4n^3/3)$	be shown that we need $O(4n^3/3)$ operations to tridiagonalize a symmetric matrix
lyche-numerical-linear-algebra-F04450	342	■ 0.999	d_k	d_k	a) Let d_k be the diagonal elements of $\boldsymbol{D}$ in an LDL* factorization of $\boldsymbol{A}_n$. Show that
lyche-numerical-linear-algebra-F04451	342	■ 0.999	$\boldsymbol{A}_n$	$\boldsymbol{A}_n$	a) Let d_k be the diagonal elements of $\boldsymbol{D}$ in an LDL* factorization of $\boldsymbol{A}_n$. Show that
lyche-numerical-linear-algebra-F04452	342	■ 1.000	$5 + \sqrt{24} < d_k \leq 10, k = 1, 2, \dots, n$	$5 + \sqrt{24} < d_k \leq 10, k = 1, 2, \dots, n$	$5 + \sqrt{24} < d_k \leq 10, k = 1, 2, \dots, n$.
lyche-numerical-linear-algebra-F04453	342	■ 1.000	$D_n := \det(\boldsymbol{A}_n) > (5 + \sqrt{24})^n$	$D_n := \det(\boldsymbol{A}_n) > (5 + \sqrt{24})^n$	b) Show that $D_n := \det(\boldsymbol{A}_n) > (5 + \sqrt{24})^n$. Give $n_0 \in \mathbb{N}$ such that your computer
lyche-numerical-linear-algebra-F04454	342	■ 1.000	$n_0 \in \mathbb{N}$	$n_0 \in \mathbb{N}$	b) Show that $D_n := \det(\boldsymbol{A}_n) > (5 + \sqrt{24})^n$. Give $n_0 \in \mathbb{N}$ such that your computer
lyche-numerical-linear-algebra-F04455	342	■ 1.000	D_{n_0}	D_{n_0}	gives an overflow when D_{n_0} is computed in floating point arithmetic.
lyche-numerical-linear-algebra-F04456	342	■ 0.935	$\boldsymbol{B} = \boldsymbol{U}^T \boldsymbol{D} \boldsymbol{U}$	$\boldsymbol{B} = \boldsymbol{U}^T \boldsymbol{D} \boldsymbol{U}$	where $\boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{B}$ is symmetric positive definite. Then $\boldsymbol{B} = \boldsymbol{U}^T \boldsymbol{D} \boldsymbol{U}$ where
lyche-numerical-linear-algebra-F04457	342	■ 1.000	$\hat{\boldsymbol{A}} = \boldsymbol{D}^{-1/2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^T \boldsymbol{D}^{-1/2}$	$\hat{\boldsymbol{A}} = \boldsymbol{D}^{-1/2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^T \boldsymbol{D}^{-1/2}$	$\hat{\boldsymbol{A}} = \boldsymbol{D}^{-1/2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^T \boldsymbol{D}^{-1/2}$ where

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lyche-numerical-linear-algebra- F04458	342	1.000	$\hat{\mathbf{A}}$	$\hat{\mathbf{A}}$	a) Show that $\hat{\mathbf{A}}$ is symmetric.
lyche-numerical-linear-algebra- F04459	342	1.000	$\hat{\mathbf{A}} = \hat{\mathbf{U}}^T \hat{\mathbf{D}} \hat{\mathbf{U}}$	$\hat{\mathbf{A}} = \hat{\mathbf{U}}^T \hat{\mathbf{D}} \hat{\mathbf{U}}$	Let $\hat{\mathbf{A}} = \hat{\mathbf{U}}^T \hat{\mathbf{D}} \hat{\mathbf{U}}$ where $\hat{\mathbf{U}}$ is orthonormal and $\hat{\mathbf{D}}$ is diagonal. Set $\mathbf{E} =$
lyche-numerical-linear-algebra- F04460	342	1.000	$\hat{\mathbf{D}}$	$\hat{\mathbf{D}}$	Let $\hat{\mathbf{A}} = \hat{\mathbf{U}}^T \hat{\mathbf{D}} \hat{\mathbf{U}}$ where $\hat{\mathbf{U}}$ is orthonormal and $\hat{\mathbf{D}}$ is diagonal. Set $\mathbf{E} =$
lyche-numerical-linear-algebra- F04461	342	1.000	$\mathbf{E} =$	$\mathbf{E} =$	Let $\hat{\mathbf{A}} = \hat{\mathbf{U}}^T \hat{\mathbf{D}} \hat{\mathbf{U}}$ where $\hat{\mathbf{U}}$ is orthonormal and $\hat{\mathbf{D}}$ is diagonal. Set $\mathbf{E} =$
lyche-numerical-linear-algebra- F04462	342	1.000	$\mathbf{U}^T \mathbf{D}^{-1/2} \hat{\mathbf{U}}^T$	$\mathbf{U}^T \mathbf{D}^{-1/2} \hat{\mathbf{U}}^T$	$\mathbf{U}^T \mathbf{D}^{-1/2} \hat{\mathbf{U}}^T$.
lyche-numerical-linear-algebra- F04463	342	0.999	$\mathbf{E}^T \mathbf{A} \mathbf{E} = \hat{\mathbf{D}}, \mathbf{E}^T \mathbf{B} \mathbf{E} = \mathbf{I}$	$\mathbf{E}^T \mathbf{A} \mathbf{E} = \hat{\mathbf{D}}, \mathbf{E}^T \mathbf{B} \mathbf{E} = \mathbf{I}$	b) Show that $\mathbf{E}$ is nonsingular and that $\mathbf{E}^T \mathbf{A} \mathbf{E} = \hat{\mathbf{D}}, \mathbf{E}^T \mathbf{B} \mathbf{E} = \mathbf{I}$.
lyche-numerical-linear-algebra- F04464	342	1.000	$\mathbf{A} = \text{tridiag}(\mathbf{c}, \mathbf{d}, \mathbf{c})$	$\mathbf{A} = \text{tridiag}(\mathbf{c}, \mathbf{d}, \mathbf{c})$	Exercise 14.14 (Program Code for One Eigenvalue) Suppose $\mathbf{A} = \text{tridiag}(\mathbf{c}, \mathbf{d}, \mathbf{c})$
lyche-numerical-linear-algebra- F04465	342	1.000	$d_1, \dots, d_n$	$d_1, \dots, d_n$	is symmetric and tridiagonal with elements $d_1, \dots, d_n$ on the diagonal and
lyche-numerical-linear-algebra- F04466	342	1.000	$c_1, \dots, c_{n-1}$	$c_1, \dots, c_{n-1}$	$c_1, \dots, c_{n-1}$ on the neighboring subdiagonals. Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be the
lyche-numerical-linear-algebra- F04467	342	0.998	$k = \text{counting}(\mathbf{c}, \mathbf{d}, \mathbf{x})$	$k = \text{counting}(\mathbf{c}, \mathbf{d}, \mathbf{x})$	a) Write a function $k = \text{counting}(\mathbf{c}, \mathbf{d}, \mathbf{x})$ which for given $\mathbf{x}$ counts the
lyche-numerical-linear-algebra- F04468	342	1.000	$d_j(\mathbf{x})$	$d_j(\mathbf{x})$	above if one of the $d_j(\mathbf{x})$ is close to zero.
lyche-numerical-linear-algebra- F04469	342	0.907	$(a, b]$	$(a, b]$	interval $(a, b]$ containing all eigenvalues of $\mathbf{A}$ and then generates a sequence
lyche-numerical-linear-algebra- F04470	342	0.901	$\{(a_j, b_j]\}$	$\{(a_j, b_j]\}$	$\{(a_j, b_j]\}$ of intervals each containing λ_m . Iterate until $b_j - a_j \leq (b - a)\epsilon_M$,
lyche-numerical-linear-algebra- F04471	342	0.901	$b_j - a_j \leq (b - a)\epsilon_M$	$b_j - a_j \leq (b - a)\epsilon_M$	$\{(a_j, b_j]\}$ of intervals each containing λ_m . Iterate until $b_j - a_j \leq (b - a)\epsilon_M$,
lyche-numerical-linear-algebra- F04472	342	1.000	$\epsilon_M \approx 2.22 \times 10^{-16}$	$\epsilon_M \approx 2.22 \times 10^{-16}$	where ϵ_M is MATLAB's machine epsilon eps. Typically $\epsilon_M \approx 2.22 \times 10^{-16}$.
lyche-numerical-linear-algebra- F04473	342	0.795	$\mathbf{T} := \text{tridiag}(-1, 2, -1)$	$\mathbf{T} := \text{tridiag}(-1, 2, -1)$	c) Test the program on $\mathbf{T} := \text{tridiag}(-1, 2, -1)$ of size 100. Compare the exact
lyche-numerical-linear-algebra- F04474	342	1.000	λ_5	λ_5	value of λ_5 with your result and the result obtained by using MATLAB's built-in
lyche-numerical-linear-algebra- F04475	343	1.000	$x \in \mathbb{C}$	$x \in \mathbb{C}$	is upper Hessenberg and $x \in \mathbb{C}$. We will study two algorithms to compute $f(x) =$
lyche-numerical-linear-algebra- F04476	343	1.000	$f(x) =$	$f(x) =$	is upper Hessenberg and $x \in \mathbb{C}$. We will study two algorithms to compute $f(x) =$
lyche-numerical-linear-algebra- F04477	343	1.000	$\det(\mathbf{A} - x\mathbf{I})$	$\det(\mathbf{A} - x\mathbf{I})$	$\det(\mathbf{A} - x\mathbf{I})$.
lyche-numerical-linear-algebra- F04478	343	1.000	$\mu \in \sigma(\mathbf{A} + \mathbf{E})$	$\mu \in \sigma(\mathbf{A} + \mathbf{E})$	14.7.1 Suppose $\mathbf{A}, \mathbf{E} \in \mathbb{C}^{n \times n}$. To every $\mu \in \sigma(\mathbf{A} + \mathbf{E})$ there is a $\lambda \in \sigma(\mathbf{A})$ which
lyche-numerical-linear-algebra- F04479	345	0.604	$2z_k/3^k$	$2z_k/3^k$	It follows that $2z_k/3^k$ converges to $[1, -1]$, an eigenvector corresponding to the
lyche-numerical-linear-algebra- F04480	345	0.604	$[1, -1]$	$[1, -1]$	It follows that $2z_k/3^k$ converges to $[1, -1]$, an eigenvector corresponding to the
lyche-numerical-linear-algebra- F04481	345	0.954	$\lambda = 3$	$\lambda = 3$	dominant eigenvalue $\lambda = 3$. The sequence of Rayleigh quotients $\{z_k^T \mathbf{A} z_k / z_k^T z_k\}$
lyche-numerical-linear-algebra- F04482	345	0.954	$\{z_k^T \mathbf{A} z_k / z_k^T z_k\}$	$\{z_k^T \mathbf{A} z_k / z_k^T z_k\}$	dominant eigenvalue $\lambda = 3$. The sequence of Rayleigh quotients $\{z_k^T \mathbf{A} z_k / z_k^T z_k\}$
lyche-numerical-linear-algebra- F04483	345	0.628	z_0	z_0	To understand better what happens we expand z_0 in terms of the eigenvectors

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lyche-numerical-linear-algebra-F04484	345	1.000	$\left(\lambda_j^k, \boldsymbol{v}_j\right), j=1,2$	$(\lambda_j^k, \boldsymbol{v}_j), j = 1, 2$	Since $\boldsymbol{A}^k$ has eigenpairs $(\lambda_j^k, \boldsymbol{v}_j), j = 1, 2$ we find
lyche-numerical-linear-algebra-F04485	345	0.749	$3^{-k} z_k = c_1 \boldsymbol{v}_1 + 3^{-k} c_2 \boldsymbol{v}_2 \rightarrow c_1 \boldsymbol{v}_1$	$3^{-k} z_k = c_1 \boldsymbol{v}_1 + 3^{-k} c_2 \boldsymbol{v}_2 \rightarrow c_1 \boldsymbol{v}_1$	Thus $3^{-k} z_k = c_1 \boldsymbol{v}_1 + 3^{-k} c_2 \boldsymbol{v}_2 \rightarrow c_1 \boldsymbol{v}_1$. Since $c_1 \neq 0$ we obtain convergence to the
lyche-numerical-linear-algebra-F04486	345	0.749	$c_1 \neq 0$	$c_1 \neq 0$	Thus $3^{-k} z_k = c_1 \boldsymbol{v}_1 + 3^{-k} c_2 \boldsymbol{v}_2 \rightarrow c_1 \boldsymbol{v}_1$. Since $c_1 \neq 0$ we obtain convergence to the
lyche-numerical-linear-algebra-F04487	345	1.000	$\left(\lambda_j, \boldsymbol{v}_j\right), j=1, \dots, n$	$(\lambda_j, \boldsymbol{v}_j), j = 1, \dots, n$	Let $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ have eigenpairs $(\lambda_j, \boldsymbol{v}_j), j = 1, \dots, n$ with $ \lambda_1 > \lambda_2 \geq \dots \geq$
lyche-numerical-linear-algebra-F04488	345	1.000	$ \lambda_1 > \lambda_2 \geq \dots \geq$	$ \lambda_1 > \lambda_2 \geq \dots \geq$	Let $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ have eigenpairs $(\lambda_j, \boldsymbol{v}_j), j = 1, \dots, n$ with $ \lambda_1 > \lambda_2 \geq \dots \geq$
lyche-numerical-linear-algebra-F04489	345	1.000	$ \lambda_n $	$ \lambda_n $	$ \lambda_n $.
lyche-numerical-linear-algebra-F04490	345	0.905	$z_0 \in \mathbb{C}^n$	$z_0 \in \mathbb{C}^n$	Given $z_0 \in \mathbb{C}^n$ we assume that
lyche-numerical-linear-algebra-F04491	345	0.999	z_0	z_0	multiplicity one. The second assumption says that z_0 has a component in the
lyche-numerical-linear-algebra-F04492	345	0.522	$z_0 = c_1 \boldsymbol{v}_1 + c_2 \boldsymbol{v}_2 + \dots + c_n \boldsymbol{v}_n$	$z_0 = c_1 \boldsymbol{v}_1 + c_2 \boldsymbol{v}_2 + \dots + c_n \boldsymbol{v}_n$	To see what happens let $z_0 = c_1 \boldsymbol{v}_1 + c_2 \boldsymbol{v}_2 + \dots + c_n \boldsymbol{v}_n$, where by assumption
lyche-numerical-linear-algebra-F04493	345	1.000	$\boldsymbol{A}^k \boldsymbol{v}_j = \lambda_j^k \boldsymbol{v}_j$	$\boldsymbol{A}^k \boldsymbol{v}_j = \lambda_j^k \boldsymbol{v}_j$	(ii) of (15.2) we have $c_1 \neq 0$. Since $\boldsymbol{A}^k \boldsymbol{v}_j = \lambda_j^k \boldsymbol{v}_j$ for all j we see that
lyche-numerical-linear-algebra-F04494	345	1.000	λ_1^k	λ_1^k	Dividing by λ_1^k we find
lyche-numerical-linear-algebra-F04495	346	0.772	$(\lambda_j / \lambda_1)^k \rightarrow 0$	$(\lambda_j / \lambda_1)^k \rightarrow 0$	Assumption (i) of (15.2) implies that $(\lambda_j / \lambda_1)^k \rightarrow 0$ as $k \rightarrow \infty$ for all $j \geq 2$ and
lyche-numerical-linear-algebra-F04496	346	0.534	(i)	(i)	matrices as long as (i) and (ii) of (15.2) hold, see for example page 58 of [18].
lyche-numerical-linear-algebra-F04497	346	0.989	z_k	z_k	In practice we need to scale the iterates z_k somehow and we normally do not
lyche-numerical-linear-algebra-F04498	346	0.766	$\boldsymbol{x}_0 = z_0 / \ z_0\ $	$\boldsymbol{x}_0 = z_0 / \ z_0\ $	know λ_1 . Instead we choose a norm on $\mathbb{C}^n$, set $\boldsymbol{x}_0 = z_0 / \ z_0\ $ and generate for
lyche-numerical-linear-algebra-F04499	346	1.000	$k=1,2, \dots$	$k = 1, 2, \dots$	$k = 1, 2, \dots$ unit vectors as follows:
lyche-numerical-linear-algebra-F04500	346	1.000	$\lambda_1 > 0$	$\lambda_1 > 0$	In particular, if $\lambda_1 > 0$ and $c_1 > 0$ then the sequence $\{\boldsymbol{x}_k\}$ will converge to the
lyche-numerical-linear-algebra-F04501	346	1.000	$c_1 > 0$	$c_1 > 0$	In particular, if $\lambda_1 > 0$ and $c_1 > 0$ then the sequence $\{\boldsymbol{x}_k\}$ will converge to the
lyche-numerical-linear-algebra-F04502	346	1.000	$\boldsymbol{u}_1 := \boldsymbol{v}_1 / \ \boldsymbol{v}_1\ $	$\boldsymbol{u}_1 := \boldsymbol{v}_1 / \ \boldsymbol{v}_1\ $	eigenvector $\boldsymbol{u}_1 := \boldsymbol{v}_1 / \ \boldsymbol{v}_1\ $ of unit length.
lyche-numerical-linear-algebra-F04503	346	0.980	$\boldsymbol{x}_k = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $	$\boldsymbol{x}_k = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $	Proof By induction on k it follows that $\boldsymbol{x}_k = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $ for all $k \geq 0$, where $\boldsymbol{z}_k =$
lyche-numerical-linear-algebra-F04504	346	0.980	$\boldsymbol{z}_k =$	$\boldsymbol{z}_k =$	Proof By induction on k it follows that $\boldsymbol{x}_k = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $ for all $k \geq 0$, where $\boldsymbol{z}_k =$
lyche-numerical-linear-algebra-F04505	346	0.999	$\boldsymbol{A}^k \boldsymbol{z}_0$	$\boldsymbol{A}^k \boldsymbol{z}_0$	$\boldsymbol{A}^k \boldsymbol{z}_0$. Indeed, this holds for $k = 1$, and if it holds for $k - 1$ then $\boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} =$
lyche-numerical-linear-algebra-F04506	346	0.999	$\boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} =$	$\boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} =$	$\boldsymbol{A}^k \boldsymbol{z}_0$. Indeed, this holds for $k = 1$, and if it holds for $k - 1$ then $\boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} =$
lyche-numerical-linear-algebra-F04507	346	1.000	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $ and $\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\) (\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $.
lyche-numerical-linear-algebra-F04508	346	1.000	$\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\) (\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $	$\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\) (\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $ and $\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\) (\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $.

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lyche-numerical-linear-algebra F04509	346	1.000	$r(\lambda) := \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}$	$r(\lambda) := \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}$	residual $\boldsymbol{r}(\lambda) := \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}$.
lyche-numerical-linear-algebra F04510	346	1.000	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \rightarrow \mathbb{R}$ be given by $\rho(\lambda) = \ \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\ _2$. Then ρ is
lyche-numerical-linear-algebra F04511	346	1.000	$\rho : \mathbb{C} \rightarrow \mathbb{R}$	$\rho : \mathbb{C} \rightarrow \mathbb{R}$	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \rightarrow \mathbb{R}$ be given by $\rho(\lambda) = \ \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\ _2$. Then ρ is
lyche-numerical-linear-algebra F04512	346	1.000	$\rho(\lambda) = \ \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\ _2$	$\rho(\lambda) = \ \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\ _2$	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \rightarrow \mathbb{R}$ be given by $\rho(\lambda) = \ \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\ _2$. Then ρ is
lyche-numerical-linear-algebra F04513	346	0.906	$\lambda := \frac{\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^* \boldsymbol{u}}$	$\lambda := \frac{\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^* \boldsymbol{u}}$	minimized when $\lambda := \frac{\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^* \boldsymbol{u}}$, the Rayleigh quotient for $\boldsymbol{A}$.
lyche-numerical-linear-algebra F04514	347	1.000	$\{\boldsymbol{u}, \boldsymbol{U}\}$	$\{\boldsymbol{u}, \boldsymbol{U}\}$	Proof Assume $\boldsymbol{u}^* \boldsymbol{u} = 1$ and extend $\boldsymbol{u}$ to an orthonormal basis $\{\boldsymbol{u}, \boldsymbol{U}\}$ for $\mathbb{C}^n$. Then
lyche-numerical-linear-algebra F04515	347	1.000	$\boldsymbol{U}^* \boldsymbol{u} = \mathbf{0}$	$\boldsymbol{U}^* \boldsymbol{u} = \mathbf{0}$	$\boldsymbol{U}^* \boldsymbol{u} = \mathbf{0}$ and
lyche-numerical-linear-algebra F04516	347	1.000	$\lambda = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}$	$\lambda = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}$	and ρ has a global minimum at $\lambda = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}$.
lyche-numerical-linear-algebra F04517	347	0.909	$(\mu, \boldsymbol{u})$	$(\mu, \boldsymbol{u})$	$(\mu, \boldsymbol{u})$ is an approximate eigenpair with $\ \boldsymbol{u}\ _2 = 1$, then by (14.4) we can find an
lyche-numerical-linear-algebra F04518	347	0.909	$\ \boldsymbol{u}\ _2 = 1$	$\ \boldsymbol{u}\ _2 = 1$	$(\mu, \boldsymbol{u})$ is an approximate eigenpair with $\ \boldsymbol{u}\ _2 = 1$, then by (14.4) we can find an
lyche-numerical-linear-algebra F04519	347	0.860	$l, \boldsymbol{x}$	$l, \boldsymbol{x}$	estimate for the eigenvalue is used to compute a dominant eigenpair $(l, \boldsymbol{x})$ of $\boldsymbol{A}$
lyche-numerical-linear-algebra F04520	347	0.767	$\ \boldsymbol{A}\boldsymbol{x} - l\boldsymbol{x}\ _2 / \ \boldsymbol{A}\ _F < tol$	$\ \boldsymbol{A}\boldsymbol{x} - l\boldsymbol{x}\ _2 / \ \boldsymbol{A}\ _F < tol$	$\ \boldsymbol{A}\boldsymbol{x} - l\boldsymbol{x}\ _2 / \ \boldsymbol{A}\ _F < tol$. If no such eigenpair is found in K iterations the value
lyche-numerical-linear-algebra F04521	347	0.877	$= K + 1$	$= K + 1$	$it \equiv K + 1$ is returned.
lyche-numerical-linear-algebra F04522	348	0.900	$\boldsymbol{z} = [0.6602, 0.3420]$	$\boldsymbol{z} = [0.6602, 0.3420]$	In each case we start with the random vector $\boldsymbol{z} \equiv [0.6602, 0.3420]$ and $tol = 10^{-6}$.
lyche-numerical-linear-algebra F04523	348	0.900	$= 10^{-6}$	$= 10^{-6}$	In each case we start with the random vector $\boldsymbol{z} = [0.6602, 0.3420]$ and $tol = 10^{-6}$.
lyche-numerical-linear-algebra F04524	348	1.000	$\frac{ \lambda_2 }{ \lambda_1 }$	$\frac{ \lambda_2 }{ \lambda_1 }$	convergence depends on the ratio $\frac{ \lambda_2 }{ \lambda_1 }$. If this ratio is small then the convergence
lyche-numerical-linear-algebra F04525	348	1.000	$\lambda_1 = 5.3723$	$\lambda_1 = 5.3723$	$\boldsymbol{A}_1$ are $\lambda_1 = 5.3723$ and $\lambda_2 = -0.3723$ giving a quite small ratio of 0.07 and
lyche-numerical-linear-algebra F04526	348	1.000	$\lambda_2 = -0.3723$	$\lambda_2 = -0.3723$	$\boldsymbol{A}_1$ are $\lambda_1 = 5.3723$ and $\lambda_2 = -0.3723$ giving a quite small ratio of 0.07 and
lyche-numerical-linear-algebra F04527	348	1.000	$\lambda_2 = 1.9$	$\lambda_2 = 1.9$	$\lambda_2 \equiv 1.9$ and the corresponding ratio is 0.95 resulting in slow convergence.
lyche-numerical-linear-algebra F04528	348	1.000	$\boldsymbol{A} - s\boldsymbol{I}$	$\boldsymbol{A} - s\boldsymbol{I}$	choose a number s and apply the power method to the matrix $\boldsymbol{A} - s\boldsymbol{I}$. The number
lyche-numerical-linear-algebra F04529	348	1.000	$\lambda - s$	$\lambda - s$	s is called a shift since it shifts an eigenvalue λ of $\boldsymbol{A}$ to $\lambda - s$ of $\boldsymbol{A} - s\boldsymbol{I}$. Sometimes
lyche-numerical-linear-algebra F04530	348	1.000	$(\boldsymbol{A} - s\boldsymbol{I})^{-1}$	$(\boldsymbol{A} - s\boldsymbol{I})^{-1}$	method to the inverse matrix $(\boldsymbol{A} - s\boldsymbol{I})^{-1}$, where s is a shift. If $\boldsymbol{A}$ has eigenvalues
lyche-numerical-linear-algebra F04531	348	1.000	$\lim_{s \rightarrow \lambda_1} \mu_1(s) = \infty$	$\lim_{s \rightarrow \lambda_1} \mu_1(s) = \infty$	Suppose λ_1 is a simple eigenvalue of $\boldsymbol{A}$. Then $\lim_{s \rightarrow \lambda_1} \mu_1(s) = \infty$, while
lyche-numerical-linear-algebra F04532	348	1.000	$\lim_{s \rightarrow \lambda_1} \mu_j(s) = (\lambda_j - \lambda_1)^{-1} < \infty$	$\lim_{s \rightarrow \lambda_1} \mu_j(s) = (\lambda_j - \lambda_1)^{-1} < \infty$	$\lim_{s \rightarrow \lambda_1} \mu_j(s) = (\lambda_j - \lambda_1)^{-1} < \infty$ for $j = 2, \dots, n$. Hence, by choosing s
lyche-numerical-linear-algebra F04533	348	1.000	$j = 2, \dots, n$	$j = 2, \dots, n$	$\lim_{s \rightarrow \lambda_1} \mu_j(s) = (\lambda_j - \lambda_1)^{-1} < \infty$ for $j = 2, \dots, n$. Hence, by choosing s
lyche-numerical-linear-algebra F04534	349	0.987	s_{k-1}	s_{k-1}	previous Rayleigh quotient s_{k-1} as the current shift. In each iteration we need to
lyche-numerical-linear-algebra F04535	349	1.000	$\boldsymbol{A}\boldsymbol{x}_k$	$\boldsymbol{A}\boldsymbol{x}_k$	We can avoid the calculation of $\boldsymbol{A}\boldsymbol{x}_k$ in (iii) and (iv). Let

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lyche-numerical-linear-algebra-F04536	350	■ 0.913	i i	ii	this error disappears when we normalize y_k in (ii). Miraculously, the normalized
lyche-numerical-linear-algebra-F04537	350	■ 0.995	(s, \boldsymbol{x})	$(s, \boldsymbol{x})$	Given an approximation $(s, \boldsymbol{x})$ to an eigenpair $(\lambda, \boldsymbol{v})$ of a matrix $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. The
lyche-numerical-linear-algebra-F04538	350	■ 0.995	(\lambda, \boldsymbol{v})	$(\lambda, \boldsymbol{v})$	Given an approximation $(s, \boldsymbol{x})$ to an eigenpair $(\lambda, \boldsymbol{v})$ of a matrix $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. The
lyche-numerical-linear-algebra-F04539	350	■ 1.000	n r	nr	one Rayleigh quotient iteration. The length nr of the new residual is also returned.
lyche-numerical-linear-algebra-F04540	350	■ 1.000	\lambda_1=(5-\sqrt{33}) / 2 \approx -0.37	$\lambda_1 = (5 - \sqrt{33})/2 \approx -0.37$	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is $\lambda_1 = (5 - \sqrt{33})/2 \approx -0.37$. Starting with $\boldsymbol{x} = [1, 1]^T$ and $s =$
lyche-numerical-linear-algebra-F04541	350	■ 1.000	s=	$s =$	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is $\lambda_1 = (5 - \sqrt{33})/2 \approx -0.37$. Starting with $\boldsymbol{x} = [1, 1]^T$ and $s =$
lyche-numerical-linear-algebra-F04545	351	■ 0.960	\boldsymbol{Q} \in \mathbb{C}^{n \times n}	$\boldsymbol{Q} \in \mathbb{C}^{n \times n}$	same. If $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ is a QR factorization then $\boldsymbol{Q} \in \mathbb{C}^{n \times n}$ is unitary, $\boldsymbol{Q}^* \boldsymbol{Q} = \boldsymbol{I}$ and
lyche-numerical-linear-algebra-F04546	351	■ 1.000	\boldsymbol{R} \in \mathbb{C}^{n \times n}	$\boldsymbol{R} \in \mathbb{C}^{n \times n}$	$\boldsymbol{R} \in \mathbb{C}^{n \times n}$ is upper triangular.
lyche-numerical-linear-algebra-F04547	351	■ 1.000	\boldsymbol{R}_k \boldsymbol{Q}_k	$\boldsymbol{R}_k \boldsymbol{Q}_k$	The determination of the QR factorization of $\boldsymbol{A}_k$ and the computation of $\boldsymbol{R}_k \boldsymbol{Q}_k$
lyche-numerical-linear-algebra-F04548	351	■ 1.000	\boldsymbol{R}_k = \boldsymbol{Q}_k^* \boldsymbol{A}_k	$\boldsymbol{R}_k = \boldsymbol{Q}_k^* \boldsymbol{A}_k$	point, since $\boldsymbol{R}_k = \boldsymbol{Q}_k^* \boldsymbol{A}_k$ we find
lyche-numerical-linear-algebra-F04549	352	■ 1.000	\boldsymbol{A}_{10}	$\boldsymbol{A}_{10}$	$\boldsymbol{A}_{10}$ is almost diagonal and contains approximations to the eigenvalues $\lambda_1 = 3$ and
lyche-numerical-linear-algebra-F04550	352	■ 1.000	\lambda_1=3	$\lambda_1 = 3$	$\boldsymbol{A}_{10}$ is almost diagonal and contains approximations to the eigenvalues $\lambda_1 = 3$ and
lyche-numerical-linear-algebra-F04551	352	■ 0.999	\lambda_2=1	$\lambda_2 = 1$	$\lambda_2 = 1$ on the diagonal.
lyche-numerical-linear-algebra-F04552	352	■ 1.000	\lambda_1, \dots, \lambda_4	$\lambda_1, \dots, \lambda_4$	This matrix is almost quasi-triangular and estimates for the eigenvalues $\lambda_1, \dots, \lambda_4$
lyche-numerical-linear-algebra-F04553	352	■ 0.835	\boldsymbol{A}_{14}	$\boldsymbol{A}_{14}$	of $\boldsymbol{A}$ can now easily be determined from the diagonal blocks of $\boldsymbol{A}_{14}$. The 1×1
lyche-numerical-linear-algebra-F04554	352	■ 1.000	\lambda_1 \approx 2.323	$\lambda_1 \approx 2.323$	blocks give us two real eigenvalues $\lambda_1 \approx 2.323$ and $\lambda_4 \approx 0.2275$. The middle
lyche-numerical-linear-algebra-F04555	352	■ 1.000	\lambda_4 \approx 0.2275	$\lambda_4 \approx 0.2275$	blocks give us two real eigenvalues $\lambda_1 \approx 2.323$ and $\lambda_4 \approx 0.2275$. The middle
lyche-numerical-linear-algebra-F04556	352	■ 0.813	\lambda_2 \approx 0.0914+0.4586 i	$\lambda_2 \approx 0.0914 + 0.4586i$	2×2 block has complex eigenvalues resulting in $\lambda_2 \approx 0.0914 + 0.4586i$ and $\lambda_3 \approx$
lyche-numerical-linear-algebra-F04557	352	■ 0.813	\lambda_3 \approx	$\lambda_3 \approx$	2×2 block has complex eigenvalues resulting in $\lambda_2 \approx 0.0914 + 0.4586i$ and $\lambda_3 \approx$
lyche-numerical-linear-algebra-F04558	352	■ 1.000	\left(\boldsymbol{A}_k\right)_k	$(\boldsymbol{A}_k)_k$	$(\boldsymbol{A}_k)_k$ converges to the triangular Schur form (Cf. Theorem 6.5) if all the eigenvalues
lyche-numerical-linear-algebra-F04559	352	■ 0.674	k=1,2,3 \ldots	$k = 1, 2, 3 \dots$	Theorem 15.3 (QR and Power) For $k = 1, 2, 3, \dots$, the QR factorization of $\boldsymbol{A}^k$ is
lyche-numerical-linear-algebra-F04560	352	■ 0.674	$\boldsymbol{A}^k$	$\boldsymbol{A}^k$	Theorem 15.3 (QR and Power) For $k = 1, 2, 3, \dots$, the QR factorization of $\boldsymbol{A}^k$ is
lyche-numerical-linear-algebra-F04561	352	■ 1.000	\boldsymbol{A}^k = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k	$\boldsymbol{A}^k = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k$	$\boldsymbol{A}^k = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k$, where
lyche-numerical-linear-algebra-F04562	352	■ 1.000	\boldsymbol{Q}_1, \dots, \boldsymbol{Q}_k, \boldsymbol{R}_1, \dots, \boldsymbol{R}_k	$\boldsymbol{Q}_1, \dots, \boldsymbol{Q}_k, \boldsymbol{R}_1, \dots, \boldsymbol{R}_k$	and $\boldsymbol{Q}_1, \dots, \boldsymbol{Q}_k, \boldsymbol{R}_1, \dots, \boldsymbol{R}_k$ are the matrices generated by the basic QR algo-
lyche-numerical-linear-algebra-F04563	353	■ 1.000	\tilde{\boldsymbol{Q}}_1 \tilde{\boldsymbol{R}}_1 = \boldsymbol{Q}_1 \boldsymbol{R}_1 = \boldsymbol{A}_1	$\tilde{\boldsymbol{Q}}_1 \tilde{\boldsymbol{R}}_1 = \boldsymbol{Q}_1 \boldsymbol{R}_1 = \boldsymbol{A}_1$	The proof is by induction on k . Clearly $\tilde{\boldsymbol{Q}}_1 \tilde{\boldsymbol{R}}_1 = \boldsymbol{Q}_1 \boldsymbol{R}_1 = \boldsymbol{A}_1$. Suppose
lyche-numerical-linear-algebra-F04564	353	■ 1.000	\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^{k-1}	$\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^{k-1}$	$\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^{k-1}$ for some $k \geq 2$. Since $\boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k$ and using (15.11)
lyche-numerical-linear-algebra-F04565	353	■ 1.000	\boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k	$\boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k$	$\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^{k-1}$ for some $k \geq 2$. Since $\boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k$ and using (15.11)

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lyche-numerical-linear-algebra- F04566	353	1.000	$\tilde{\boldsymbol{R}}_k$	$\tilde{\boldsymbol{R}}_k$	Since $\tilde{\boldsymbol{R}}_k$ is upper triangular, its first column is a multiple of $\boldsymbol{e}_1$ so that
lyche-numerical-linear-algebra- F04567	353	1.000	$\left\ \tilde{\boldsymbol{q}}_1^{(k)}\right\ _2=1$	$\left\ \tilde{\boldsymbol{q}}_1^{(k)}\right\ _2=1$	Since $\ \tilde{\boldsymbol{q}}_1^{(k)}\ _2 = 1$ the first column of $\tilde{\boldsymbol{Q}}_k$ is the result of applying the normalized
lyche-numerical-linear-algebra- F04568	353	1.000	$\tilde{\boldsymbol{Q}}_k$	$\tilde{\boldsymbol{Q}}_k$	Since $\ \tilde{\boldsymbol{q}}_1^{(k)}\ _2 = 1$ the first column of $\tilde{\boldsymbol{Q}}_k$ is the result of applying the normalized
lyche-numerical-linear-algebra- F04569	353	1.000	$\boldsymbol{x}_0=\boldsymbol{e}_1$	$\boldsymbol{x}_0=\boldsymbol{e}_1$	power iteration (15.6) to the starting vector $\boldsymbol{x}_0=\boldsymbol{e}_1$. If this iteration converges we
lyche-numerical-linear-algebra- F04570	353	1.000	$\lambda_1\boldsymbol{e}_1$	$\lambda_1\boldsymbol{e}_1$	It can be shown that the first column of $\boldsymbol{A}_k$ must then converge to $\lambda_1\boldsymbol{e}_1$, where λ_1 is a
lyche-numerical-linear-algebra- F04571	353	1.000	$\left(\tilde{\boldsymbol{Q}}_k^*\boldsymbol{A}\tilde{\boldsymbol{Q}}_k\right)_k$	$\left(\tilde{\boldsymbol{Q}}_k^*\boldsymbol{A}\tilde{\boldsymbol{Q}}_k\right)_k$	Indeed, what is observed in practice is that the sequence $(\tilde{\boldsymbol{Q}}_k^*\boldsymbol{A}\tilde{\boldsymbol{Q}}_k)_k$ converges to a
lyche-numerical-linear-algebra- F04572	353	1.000	$\boldsymbol{H}_1$	$\boldsymbol{H}_1$	an initial reduction of $\boldsymbol{A}$ to upper Hessenberg form $\boldsymbol{H}_1$ using Algorithm 14.1, the
lyche-numerical-linear-algebra- F04573	353	1.000	$\boldsymbol{P}_{i,i+1}, i=1,\ldots,n-1$	$\boldsymbol{P}_{i,i+1}, i=1,\ldots,n-1$	determine plane rotations $\boldsymbol{P}_{i,i+1}, i=1,\ldots,n-1$ so that $\boldsymbol{P}_{n-1,n}\cdots\boldsymbol{P}_{1,2}\boldsymbol{H}_1=\boldsymbol{R}_1$
lyche-numerical-linear-algebra- F04574	353	1.000	$\boldsymbol{P}_{n-1,n}\cdots\boldsymbol{P}_{1,2}\boldsymbol{H}_1=\boldsymbol{R}_1$	$\boldsymbol{P}_{n-1,n}\cdots\boldsymbol{P}_{1,2}\boldsymbol{H}_1=\boldsymbol{R}_1$	determine plane rotations $\boldsymbol{P}_{i,i+1}, i=1,\ldots,n-1$ so that $\boldsymbol{P}_{n-1,n}\cdots\boldsymbol{P}_{1,2}\boldsymbol{H}_1=\boldsymbol{R}_1$
lyche-numerical-linear-algebra- F04575	353	1.000	$\boldsymbol{H}_1=\boldsymbol{Q}_1\boldsymbol{R}_1$	$\boldsymbol{H}_1=\boldsymbol{Q}_1\boldsymbol{R}_1$	is upper triangular. The details were described in Sect. 5.6. Thus $\boldsymbol{H}_1=\boldsymbol{Q}_1\boldsymbol{R}_1$,
lyche-numerical-linear-algebra- F04576	353	0.999	$\boldsymbol{Q}_1=\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$	$\boldsymbol{Q}_1=\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$	where $\boldsymbol{Q}_1=\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$ is a QR factorization of $\boldsymbol{H}_1$. To finish the QR step we
lyche-numerical-linear-algebra- F04577	353	0.996	$\boldsymbol{R}_1\boldsymbol{Q}_1=\boldsymbol{R}_1\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$	$\boldsymbol{R}_1\boldsymbol{Q}_1=\boldsymbol{R}_1\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$	compute $\boldsymbol{R}_1\boldsymbol{Q}_1=\boldsymbol{R}_1\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$. This postmultiplication step is illustrated by
lyche-numerical-linear-algebra- F04578	353	0.999	$(i+1,i)$	$(i+1,i)$	The postmultiplication by $\boldsymbol{P}_{i,i+1}$ introduces a nonzero in position $(i+1,i)$
lyche-numerical-linear-algebra- F04579	354	0.993	$\boldsymbol{R}\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$	$\boldsymbol{R}\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$	matrix $\boldsymbol{R}\boldsymbol{P}_{1,2}^*\cdots\boldsymbol{P}_{n-1,n}^*$ is upper Hessenberg and a QR step leaves the Hessenberg
lyche-numerical-linear-algebra- F04580	354	1.000	$a_{i+1,i}$	$a_{i+1,i}$	If a subdiagonal element $a_{i+1,i}$ of an upper Hessenberg matrix $\boldsymbol{A}$ is equal to zero,
lyche-numerical-linear-algebra- F04581	354	1.000	$\boldsymbol{A}(1:i,1:i)$	$\boldsymbol{A}(1:i,1:i)$	matrices $\boldsymbol{A}(1:i,1:i)$ and $\boldsymbol{A}(i+1:n,i+1:n)$. Thus, if during the iteration
lyche-numerical-linear-algebra- F04582	354	1.000	$\boldsymbol{A}(i+1:n,i+1:n)$	$\boldsymbol{A}(i+1:n,i+1:n)$	matrices $\boldsymbol{A}(1:i,1:i)$ and $\boldsymbol{A}(i+1:n,i+1:n)$. Thus, if during the iteration
lyche-numerical-linear-algebra- F04583	354	1.000	$\left a_{i+1,i}^{(k)}\right \leq \epsilon$	$\left a_{i+1,i}^{(k)}\right \leq \epsilon$	To see what effect this can have on the eigenvalues of $\boldsymbol{A}$ suppose $ a_{i+1,i}^{(k)} \leq \epsilon$. Let
lyche-numerical-linear-algebra- F04584	354	0.992	$\hat{\boldsymbol{A}}_k:=\boldsymbol{A}_k-\boldsymbol{a}_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T$	$\hat{\boldsymbol{A}}_k:=\boldsymbol{A}_k-\boldsymbol{a}_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T$	$\hat{\boldsymbol{A}}_k:=\boldsymbol{A}_k-\boldsymbol{a}_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T$ be the matrix obtained from $\boldsymbol{A}_k$ by setting the $(i+1,i)$
lyche-numerical-linear-algebra- F04585	354	1.000	$\boldsymbol{A}_k=\tilde{\boldsymbol{Q}}_{k-1}^*\boldsymbol{A}\tilde{\boldsymbol{Q}}_{k-1}$	$\boldsymbol{A}_k=\tilde{\boldsymbol{Q}}_{k-1}^*\boldsymbol{A}\tilde{\boldsymbol{Q}}_{k-1}$	element equal to zero. Since $\boldsymbol{A}_k=\tilde{\boldsymbol{Q}}_{k-1}^*\boldsymbol{A}\tilde{\boldsymbol{Q}}_{k-1}$ we have
lyche-numerical-linear-algebra- F04586	354	1.000	$\tilde{\boldsymbol{Q}}_{k-1}$	$\tilde{\boldsymbol{Q}}_{k-1}$	Since $\tilde{\boldsymbol{Q}}_{k-1}$ is unitary, $\ \boldsymbol{E}\ _F=\left\ \boldsymbol{a}_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\right\ _F=\left a_{i+1,i}^{(k)}\right \leq \epsilon$ and setting
lyche-numerical-linear-algebra- F04587	354	1.000	$\left\ \boldsymbol{E}\right\ _F=\left\ \boldsymbol{a}_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\right\ _F=\left a_{i+1,i}^{(k)}\right \leq \epsilon$	$\left\ \boldsymbol{E}\right\ _F=\left\ \boldsymbol{a}_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\right\ _F=\left a_{i+1,i}^{(k)}\right \leq \epsilon$	Since $\tilde{\boldsymbol{Q}}_{k-1}$ is unitary, $\ \boldsymbol{E}\ _F=\left\ \boldsymbol{a}_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\right\ _F=\left a_{i+1,i}^{(k)}\right \leq \epsilon$ and setting
lyche-numerical-linear-algebra- F04588	354	1.000	$a_{i+1,i}^{(k)}=0$	$a_{i+1,i}^{(k)}=0$	$a_{i+1,i}^{(k)}=0$ amounts to a perturbation in the original $\boldsymbol{A}$ of at most ϵ . For how to
lyche-numerical-linear-algebra- F04589	355	1.000	$\boldsymbol{R}_k=\boldsymbol{Q}_k^*(\boldsymbol{A}_k-s_k\boldsymbol{I})$	$\boldsymbol{R}_k=\boldsymbol{Q}_k^*(\boldsymbol{A}_k-s_k\boldsymbol{I})$	Since $\boldsymbol{R}_k=\boldsymbol{Q}_k^*(\boldsymbol{A}_k-s_k\boldsymbol{I})$ we find
lyche-numerical-linear-algebra- F04590	355	1.000	$\boldsymbol{Q}_k$	$\boldsymbol{Q}_k$	Theorem 15.3 and also the inverse power method. In fact the last column of $\boldsymbol{Q}_k$

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lyche-numerical-linear-algebra- F04591	355	■ 0.998	$\backslash\boldsymbol{A}-s_{\{k\}} \backslash\boldsymbol{I}=\backslash\boldsymbol{Q}_{\{k\}} \backslash\boldsymbol{R}_{\{k\}}$	$A-s_k I=Q_k R_k$	since $A-s_k I=Q_k R_k$ we have $(A-s_k I)^*=R_k^* Q_k^*$ and $(A-s_k I)^* Q_k=R_k^*$.
lyche-numerical-linear-algebra- F04592	355	■ 0.998	$\backslash\left(\backslash\boldsymbol{A}-s_{\{k\}} \backslash\boldsymbol{I}\right)^{*}=\backslash\boldsymbol{R}_{\{k\}}^* \backslash\boldsymbol{Q}_{\{k\}}^*$	$(A-s_k I)^*=R_k^* Q_k^*$	since $A-s_k I=Q_k R_k$ we have $(A-s_k I)^*=R_k^* Q_k^*$ and $(A-s_k I)^* Q_k=R_k^*$.
lyche-numerical-linear-algebra- F04593	355	■ 0.998	$\backslash\left(\backslash\boldsymbol{A}-s_{\{k\}} \backslash\boldsymbol{I}\right)^{*} \backslash\boldsymbol{Q}_{\{k\}}=\backslash\boldsymbol{R}_{\{k\}}^*$	$(A-s_k I)^* Q_k=R_k^*$	since $A-s_k I=Q_k R_k$ we have $(A-s_k I)^*=R_k^* Q_k^*$ and $(A-s_k I)^* Q_k=R_k^*$.
lyche-numerical-linear-algebra- F04594	355	■ 1.000	$\backslash\boldsymbol{R}_{\{k\}}^*$	R_k^*	Thus, since R_k^* is lower triangular with n, n element $\bar{r}_{nn}^{(k)}$ we find $(A-s_k I)^* Q_k e_n =$
lyche-numerical-linear-algebra- F04595	355	■ 1.000	n, n	n, n	Thus, since R_k^* is lower triangular with n, n element $\bar{r}_{nn}^{(k)}$ we find $(A-s_k I)^* Q_k e_n =$
lyche-numerical-linear-algebra- F04596	355	■ 1.000	$\backslash\bar{r}_{\{n\}}^{\{k\}}$	$\bar{r}_{nn}^{(k)}$	Thus, since R_k^* is lower triangular with n, n element $\bar{r}_{nn}^{(k)}$ we find $(A-s_k I)^* Q_k e_n =$
lyche-numerical-linear-algebra- F04597	355	■ 1.000	$\backslash\left(\backslash\boldsymbol{A}-s_{\{k\}} \backslash\boldsymbol{I}\right)^{*} \backslash\boldsymbol{Q}_{\{k\}} \backslash\boldsymbol{e}_{\{n\}}=$	$(A-s_k I)^* Q_k e_n =$	Thus, since R_k^* is lower triangular with n, n element $\bar{r}_{nn}^{(k)}$ we find $(A-s_k I)^* Q_k e_n =$
lyche-numerical-linear-algebra- F04598	355	■ 1.000	$\backslash\boldsymbol{R}_{\{k\}}^* \backslash\boldsymbol{e}_{\{n\}}=\backslash\bar{r}_{\{n\}}^{\{k\}} \backslash\boldsymbol{e}_{\{n\}}$	$R_k^* e_n = \bar{r}_{nn}^{(k)} e_n$	$R_k^* e_n = \bar{r}_{nn}^{(k)} e_n$ from which the conclusion follows.
lyche-numerical-linear-algebra- F04599	355	■ 1.000	$s_{\{k\}}:=\backslash\boldsymbol{e}_{\{n\}}^T \backslash\boldsymbol{A}_{\{k\}} \backslash\boldsymbol{e}_{\{n\}}$	$s_k := e_n^T A_k e_n$	The shift $s_k := e_n^T A_k e_n$ is called the Rayleigh quotient shift , while the
lyche-numerical-linear-algebra- F04600	355	■ 0.999	$\backslash\boldsymbol{A}:=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right]$	$A := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	is given by $A := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. We obtain $A_k = A$ for all $k \in \mathbb{N}$. For convergence of the
lyche-numerical-linear-algebra- F04601	355	■ 0.999	$\backslash\boldsymbol{A}_{\{k\}}=\backslash\boldsymbol{A}$	$A_k = A$	is given by $A := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. We obtain $A_k = A$ for all $k \in \mathbb{N}$. For convergence of the
lyche-numerical-linear-algebra- F04602	355	■ 1.000	$\backslash\boldsymbol{A} \backslash\boldsymbol{u}-\backslash\lambda \backslash\boldsymbol{u}$	$Au - \lambda u$	Exercise 15.1 (Orthogonal Vectors) Show that u and $Au - \lambda u$ are orthogonal
lyche-numerical-linear-algebra- F04603	355	■ 1.000	$\backslash\lambda=\frac{\backslash\boldsymbol{u}^* \backslash\boldsymbol{A} \backslash\boldsymbol{u}}{\backslash\boldsymbol{u}^* \backslash\boldsymbol{u}}$	$\lambda = \frac{u^* Au}{u^* u}$	when $\lambda = \frac{u^* Au}{u^* u}$.
lyche-numerical-linear-algebra- F04604	358	■ 1.000	$f: \backslash\mathbb{R}^n \rightarrow \backslash\mathbb{R}$	$f: \mathbb{R}^n \rightarrow \mathbb{R}$	For any sufficiently differentiable $f: \mathbb{R}^n \rightarrow \mathbb{R}$ we recall that the partial derivative
lyche-numerical-linear-algebra- F04605	358	■ 1.000	$\backslash\nabla f(\backslash\boldsymbol{x}) \in \backslash\mathbb{R}^n$	$\nabla f(\boldsymbol{x}) \in \mathbb{R}^n$	$\nabla f(\boldsymbol{x}) \in \mathbb{R}^n$, and the hessian $Hf = \nabla \nabla^T f(\boldsymbol{x}) \in \mathbb{R}^{n \times n}$ of f by
lyche-numerical-linear-algebra- F04606	358	■ 1.000	$\backslash\boldsymbol{H} f=\backslash\nabla \backslash\nabla^T f(\backslash\boldsymbol{x}) \in \backslash\mathbb{R}^n \times \mathbb{R}^n$	$Hf = \nabla \nabla^T f(\boldsymbol{x}) \in \mathbb{R}^{n \times n}$	$\nabla f(\boldsymbol{x}) \in \mathbb{R}^n$, and the hessian $Hf = \nabla \nabla^T f(\boldsymbol{x}) \in \mathbb{R}^{n \times n}$ of f by
lyche-numerical-linear-algebra- F04607	358	■ 1.000	$\backslash\nabla^T f:=\left(\backslash\nabla f\right)^T$	$\nabla^T f := (\nabla f)^T$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla \nabla^T$ and $\nabla^T \nabla$ are
lyche-numerical-linear-algebra- F04608	358	■ 1.000	$\backslash\nabla \backslash\nabla^T$	$\nabla \nabla^T$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla \nabla^T$ and $\nabla^T \nabla$ are
lyche-numerical-linear-algebra- F04609	358	■ 1.000	$\backslash\nabla^T \backslash\nabla$	$\nabla^T \nabla$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla \nabla^T$ and $\nabla^T \nabla$ are
lyche-numerical-linear-algebra- F04610	358	■ 1.000	$\backslash\nabla^T \backslash\nabla f=D_1^2 f+\cdots+D_n^2 f=: \backslash\nabla^2$	$\nabla^T \nabla f = D_1^2 f + \cdots + D_n^2 f =: \nabla^2$	quite different. Indeed, $\nabla^T \nabla f = D_1^2 f + \cdots + D_n^2 f =: \nabla^2$ the Laplacian of f ,
lyche-numerical-linear-algebra- F04611	358	■ 1.000	$f, g: \backslash\mathbb{R}^n \rightarrow \backslash\mathbb{R}$	$f, g: \mathbb{R}^n \rightarrow \mathbb{R}$	Lemma 16.1 (Product Rules) For $f, g: \mathbb{R}^n \rightarrow \mathbb{R}$ we have the product rules
lyche-numerical-linear-algebra- F04612	358	■ 0.982	$\backslash\nabla(f g)=\backslash\nabla f g+g \backslash\nabla f, \quad \backslash\nabla^T(f g)=f \backslash\nabla^T g+g \backslash\nabla^T f$	$\nabla(fg) = f \nabla g + g \nabla f, \quad \nabla^T(fg) = f \nabla^T g + g \nabla^T f$	1. $\nabla(fg) = f \nabla g + g \nabla f, \quad \nabla^T(fg) = f \nabla^T g + g \nabla^T f$.
lyche-numerical-linear-algebra- F04613	358	■ 1.000	$\backslash\nabla \backslash\nabla^T(f g)=\backslash\nabla f \backslash\nabla^T g+\backslash\nabla g \backslash\nabla^T f+f \backslash\nabla \backslash\nabla^T g+g \backslash\nabla \backslash\nabla^T f$	$\nabla \nabla^T(fg) = \nabla f \nabla^T g + \nabla g \nabla^T f + f \nabla \nabla^T g + g \nabla \nabla^T f$	2. $\nabla \nabla^T(fg) = \nabla f \nabla^T g + \nabla g \nabla^T f + f \nabla \nabla^T g + g \nabla \nabla^T f$.
lyche-numerical-linear-algebra- F04614	358	■ 1.000	$\backslash\nabla^2(f g)=2 \backslash\nabla^T f \backslash\nabla g+f \backslash\nabla^2 g+g \backslash\nabla^2 f$	$\nabla^2(fg) = 2 \nabla^T f \nabla g + f \nabla^2 g + g \nabla^2 f$	3. $\nabla^2(fg) = 2 \nabla^T f \nabla g + f \nabla^2 g + g \nabla^2 f$.
lyche-numerical-linear-algebra- F04615	359	■ 1.000	$\backslash\boldsymbol{f}=\left[f_1, \ldots f_m\right]^T: \backslash\mathbb{R}^n \rightarrow \backslash\mathbb{R}^m$	$\boldsymbol{f} = [f_1, \dots f_m]^T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	We define the Jacobian of a vector function $\boldsymbol{f} = [f_1, \dots f_m]^T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ as
lyche-numerical-linear-algebra- F04616	359	■ 1.000	$f(\backslash\boldsymbol{x})=f(x, y)=x^2-x y+y^2$	$f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$	As an example, if $f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$ and $\boldsymbol{g}(x, y) := [f(x, y), x - y]^T$

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lyche-numerical-linear-algebra- F04617	359	1.000	$\boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y}) := [f(\boldsymbol{x}, \boldsymbol{y}), \boldsymbol{x} - \boldsymbol{y}]^T$	$\boldsymbol{g}(x, y) := [f(x, y), x - y]^T$	As an example, if $f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$ and $\boldsymbol{g}(x, y) := [f(x, y), x - y]^T$
lyche-numerical-linear-algebra- F04618	359	1.000	$f \in C^2(\Omega)$	$f \in C^2(\Omega)$	Lemma 16.2 (Second Order Taylor Expansion) Suppose $f \in C^2(\Omega)$, where
lyche-numerical-linear-algebra- F04619	359	0.633	$\Omega \in \mathbb{R}^n$	$\Omega \in \mathbb{R}^n$	$\Omega \in \mathbb{R}^n$ contains two points $\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$, such that the line segment $L := \{\boldsymbol{x} + t\boldsymbol{h} :$
lyche-numerical-linear-algebra- F04620	359	0.633	$\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$	$\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$	$\Omega \in \mathbb{R}^n$ contains two points $\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$, such that the line segment $L := \{\boldsymbol{x} + t\boldsymbol{h} :$
lyche-numerical-linear-algebra- F04621	359	0.633	$L := \{\boldsymbol{x} + t\boldsymbol{h} :$	$L := \{\boldsymbol{x} + t\boldsymbol{h} :$	$\Omega \in \mathbb{R}^n$ contains two points $\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$, such that the line segment $L := \{\boldsymbol{x} + t\boldsymbol{h} :$
lyche-numerical-linear-algebra- F04622	359	1.000	$t \in (0, 1) \subset \Omega$	$t \in (0, 1) \subset \Omega$	$t \in (0, 1) \subset \Omega$. Then
lyche-numerical-linear-algebra- F04623	359	1.000	$g : [0, 1] \rightarrow \mathbb{R}$	$g : [0, 1] \rightarrow \mathbb{R}$	Proof Let $g : [0, 1] \rightarrow \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra- F04624	359	1.000	$g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$	$g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$	Proof Let $g : [0, 1] \rightarrow \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra- F04625	359	1.000	$g \in C^2[0, 1]$	$g \in C^2[0, 1]$	Proof Let $g : [0, 1] \rightarrow \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra- F04626	359	1.000	$\boldsymbol{c} = \boldsymbol{x} + u\boldsymbol{h}$	$\boldsymbol{c} = \boldsymbol{x} + u\boldsymbol{h}$	we obtain (16.2) with $\boldsymbol{c} = \boldsymbol{x} + u\boldsymbol{h}$.
lyche-numerical-linear-algebra- F04627	360	1.000	$m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in \mathbb{R}^{m \times n}, \boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^m$	$m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in \mathbb{R}^{m \times n}, \boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^m$	Lemma 16.3 (Functions Involving Matrices) For any $m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in \mathbb{R}^{m \times n}$, and $\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^m$ we have
lyche-numerical-linear-algebra- F04628	360	1.000	$\nabla(\boldsymbol{y}^T \boldsymbol{C}) = \nabla^T(\boldsymbol{C} \boldsymbol{x}) = \boldsymbol{C}$	$\nabla(\boldsymbol{y}^T \boldsymbol{C}) = \nabla^T(\boldsymbol{C} \boldsymbol{x}) = \boldsymbol{C}$	1. $\nabla(\boldsymbol{y}^T \boldsymbol{C}) = \nabla^T(\boldsymbol{C} \boldsymbol{x}) = \boldsymbol{C}$,
lyche-numerical-linear-algebra- F04629	360	1.000	$\nabla(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = (\boldsymbol{B} + \boldsymbol{B}^T) \boldsymbol{x}, \nabla^T(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = \boldsymbol{x}^T (\boldsymbol{B} + \boldsymbol{B}^T)$	$\nabla(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = (\boldsymbol{B} + \boldsymbol{B}^T) \boldsymbol{x}, \nabla^T(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = \boldsymbol{x}^T (\boldsymbol{B} + \boldsymbol{B}^T)$	2. $\nabla(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = (\boldsymbol{B} + \boldsymbol{B}^T) \boldsymbol{x}, \nabla^T(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = \boldsymbol{x}^T (\boldsymbol{B} + \boldsymbol{B}^T)$,
lyche-numerical-linear-algebra- F04630	360	1.000	$\nabla \nabla^T(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = \boldsymbol{B} + \boldsymbol{B}^T$	$\nabla \nabla^T(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = \boldsymbol{B} + \boldsymbol{B}^T$	3. $\nabla \nabla^T(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}) = \boldsymbol{B} + \boldsymbol{B}^T$.
lyche-numerical-linear-algebra- F04631	360	1.000	$D_i(\boldsymbol{y}^T \boldsymbol{C}) = \lim_{h \rightarrow 0} \frac{1}{h} ((\boldsymbol{y} + h\boldsymbol{e}_i)^T \boldsymbol{C} - \boldsymbol{y}^T \boldsymbol{C}) = \boldsymbol{e}_i^T \boldsymbol{C}$	$D_i(\boldsymbol{y}^T \boldsymbol{C}) = \lim_{h \rightarrow 0} \frac{1}{h} ((\boldsymbol{y} + h\boldsymbol{e}_i)^T \boldsymbol{C} - \boldsymbol{y}^T \boldsymbol{C}) = \boldsymbol{e}_i^T \boldsymbol{C}$	1. We find $D_i(\boldsymbol{y}^T \boldsymbol{C}) = \lim_{h \rightarrow 0} \frac{1}{h} ((\boldsymbol{y} + h\boldsymbol{e}_i)^T \boldsymbol{C} - \boldsymbol{y}^T \boldsymbol{C}) = \boldsymbol{e}_i^T \boldsymbol{C}$ and $D_i(\boldsymbol{C} \boldsymbol{x}) =$
lyche-numerical-linear-algebra- F04632	360	1.000	$D_i(\boldsymbol{C} \boldsymbol{x}) = \lim_{h \rightarrow 0} \frac{1}{h} (\boldsymbol{C}(\boldsymbol{x} + h\boldsymbol{e}_i) - \boldsymbol{C} \boldsymbol{x}) = \boldsymbol{C} \boldsymbol{e}_i$	$D_i(\boldsymbol{C} \boldsymbol{x}) = \lim_{h \rightarrow 0} \frac{1}{h} (\boldsymbol{C}(\boldsymbol{x} + h\boldsymbol{e}_i) - \boldsymbol{C} \boldsymbol{x}) = \boldsymbol{C} \boldsymbol{e}_i$	$\lim_{h \rightarrow 0} \frac{1}{h} (\boldsymbol{C}(\boldsymbol{x} + h\boldsymbol{e}_i) - \boldsymbol{C} \boldsymbol{x}) = \boldsymbol{C} \boldsymbol{e}_i$ and 1. follows.
lyche-numerical-linear-algebra- F04633	360	1.000	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$	1979-3, $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$ inequality, 119
lyche-numerical-linear-algebra- F04634	363	1.000	$\boldsymbol{T} \boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$, 79	$\boldsymbol{T} \boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$, 79	1981-3, solving $\boldsymbol{T} \boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$, 79
lyche-numerical-linear-algebra- F04635	365	0.995	$\boldsymbol{A}^* \boldsymbol{A}$, 86	$\boldsymbol{A}^* \boldsymbol{A}$, 86	$\boldsymbol{A}^* \boldsymbol{A}$, 86
lyche-numerical-linear-algebra- F04636	374	1.000	$\boldsymbol{X} \boldsymbol{X}$	$\boldsymbol{X} \boldsymbol{X}$	Engineering $\boldsymbol{X} \boldsymbol{X}$.